Person on the Edge: Lessons from Crosslinguistic Variation in Syntactic Person Restrictions

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This thesis presents the findings of a large-scale crosslinguistic survey of *person restrictions*—a phenomenon where the co-occurrence of multiple pronouns in a clause is allowed or disallowed depending on their person value. This is the broadest study of this phenomenon to date, spanning 106 languages from 26 families and 3 isolates. It examines both person restrictions holding between indirect and direct objects and those holding between subjects and objects and also includes languages with direct/inverse systems and person-based argument indexing, in addition to the traditional Person-Case Constraint. The study establishes a number of typological generalizations, including the properties that define the class of pronouns which are crosslinguistically subject to person restrictions and, more importantly, establishes two previously unnoticed typological gaps. The first typological gap concerns the interaction between the strength of the person restriction—how many pronoun combinations are disallowed—and the domain of the person restrictions—which arguments in the clause are affected by it. The second typological gap concerns the direction of the restriction—which of the two pronouns in a restricted pair is affected by the restriction.

A new analysis of person restrictions is proposed where the pronouns subject to person restrictions start the syntactic derivation unspecified for a person value, getting valued during the derivation. Only phase heads—the heads that determine the points of spell-out in a syntactic derivation—may provide a value to such pronouns by way of a strictly local syntactic operation: Agree.

It is shown that the proposed analysis of person restrictions derives the existence of the two typological gaps established by the survey and also explains why certain patterns of person restrictions are much less frequent than others crosslinguistically. All this is shown to follow from independently needed assumptions concerning syntactic domains, argument

structure, syntactic movement, and the timing of grammatical operations. Finally, the special syntactic status of person features, which is responsible for the existence of person restrictions, is tied to the semantic properties that set person apart from other categories such as number or gender which do not constrain the distribution of pronouns in the way person does.

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APPROVAL PAGE

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Person on the Edge: Lessons from Crosslinguistic Variation in Syntactic Person Restrictions

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Za mamo.

Some notes on the revised version

What you're reading is a revised version of my PhD thesis. The original was filed on August 22nd 2019 and is at the moment of writing these notes on embargo. Since I filed the thesis, I've gotten many requests to share it and just as many inquiries into why there is an embargo. This is my attempt to provide some context. There's a sensitive personal reason behind the series of events that led to the embargo, so I'll try to provide context without going too much into that, focusing mostly on what is and isn't in the two versions of the thesis, and why.

The euphemistic short version of the story is that I was busy making other plans and then life happened. The blunt short version is that a month before the hard deadline for filing, I had to deal with the sudden and unexpected passing of my mother. As a result, the filing had to be rushed, so there was no time to go over the thesis to check for typos and consistency across examples, glosses, and translations. Additionally, the acknowledgments section, list of tables, list of abbreviations and acronyms, and a separate conclusion chapter were cut to save time. Most importantly, Chapters 5 and 6 were not revised to the full extent that was planned after my defense. The idea was to put the thesis on embargo, and complete the version I originally planned to file when things would have settled a bit. But do things ever truly settle, even a bit? With my first student ready to file her PhD thesis, it just didn't feel right anymore for mine to not be publicly available. It was time.

The revised version has all of what I noted above was missing in the filed version. In addition, there are some—mostly minor—revisions throughout the thesis: corrected typos, minor style corrections, more consistent glossing conventions and translations of examples, and corrected references. That also means that minor changes in page numbers, footnote numbers, and example numbers were unavoidable in a lot of cases.

The only content change in the thesis concerns a single data point from my crosslinguistic survey of person restrictions: the Chukotko-Kamchatkan language Koryak. At the time of my defense, I was exclusively using Bernard Comrie's work from the 1980's as a source on the language, which didn't contain a full paradigm for agreement with objects in double object ditransitives. Luckily, I was made aware of Rafael Abramovitz's work on the language soon after, although I was not able to make the necessary changes to the thesis before the filing. In short, the only difference in the revised version is that instead of discussing a single language with a certain type of person restriction pattern, I now discuss two from the same language family. This does not change any of my crosslinguistic generalizations, but the relevant tables and discussions have been edited to include Koryak as one of the two languages, alongside Alutor, that have that particular person restriction pattern.

Related to this last note, I had to employ a lot of restraint to keep the revised version rooted in August 2019, ignoring many works that have been published since (I wanted this to be a "director's cut" and not a "special edition" . . .). This means that any relevant research that came out after August 2019 is not referenced, and any works that I had access to in manuscript form at the time, but have since been published, are cited and engaged with as if only the pre-published versions exist.

Once the embargo on the original version expires, you can play a fun game of spot the difference with the two versions. However, I ask anyone who wants to engage with or cite my thesis, to please reference the page and example numbers from the revised version.

Adrian Stegovec
Leipzig, 23rd July 2025

Acknowledgments

Based on what I've seen in other dissertation acknowledgments, this is where you first thank your committee. But I can't help but first acknowledge the teachers who set me on the path of eventually becoming a linguist. I am going from memory, but I'm pretty sure that I took my first linguistics classes with them in the following order: Janez Orešnik, Tatjana Marvin, Marija Golden, Sašo Živanović, Jonathan Kaye, and Milena Milojević Sheppard. On top of that, the Slovenian linguistics community at large was always a very welcoming group. Regardless of the venue or situation, I could speak my mind and feel at home already as an undergrad. The names that immediately come to mind as representative of this community are: Lanko Marušič, Petra Mišmaš, and Rok Žaucer. Finally, I have to single out Tatjana and Sašo as the advisors for my BA theses and my mentors. They encouraged me to give my first conference presentations and convinced me to apply to PhD programs abroad.

Another community that deserves a special mention here is the EGG Summer School. I attended my first EGG in Debrecen, Hungary in 2008 based on a recommendation from Sašo, and then attended it every summer until 2014: Poznań, Poland (2009), Constanța, Romania (2010), České Budějovice, Czechia (2011), Wrocław, Poland (2012, 2013), and finally Debrecen again to go full circle. I can't name all the teachers and students who made the EGG such a great (learning) experience for me, so I hope no one will feel left out if I only list the schools I attended. But since this is an opportunity to record things for posterity: the first "traditional" EGG Cocktail Party was organized by Andy Murphy and me in 2014.

When it came to my transition to grad school, Tatjana and Sašo couldn't have entrusted me to a better team. Here I want to thank in particular my dissertation committee of Željko Bošković, Jonathan Bobaljik, Magdalena Kaufmann, and Susi Wurmbrand. They were all key figures, first as teachers and then as committee members, in shaping my PhD experience.

I'm struggling to find things to say about Željko as an advisor that haven't been said before in countless other acknowledgments. So I'll just say: It's true! The dedication, the support, the wisdom, the right advice at exactly the right time. All of it. It's all true.

Jonathan was instrumental in helping me scale up my project on person restrictions to the typological study presented in this dissertation, especially during my independent study with him. He is a seemingly endless source of relevant linguistic facts and always knows how to give you a solid reality check when your analyses start running away from you.

I don't think I would have gotten into semantics if it were not for Magda. I appreciate the research we did on Slovenian imperatives, and her guidance when it was time to do my

own semantics research, including parts of Chapter 5 of this thesis. Working with her is like having access to a living encyclopedia of semantics.

Susi was my first syntax teacher at UConn. It was a challenging class, and I definitely struggled at times, but I also loved every minute of it. When it came to my research, I could always rely on Susi's straight talk feedback. She held us all up to a high standard, and made sure that we always worked out all the details of our analyses, and I really appreciate that.

In addition to my committee, I also want to thank the entire UConn linguistics faculty. I always appreciated the willingness of all the faculty members to give feedback, advice, or help out with various aspects of life in academia. While I did not take classes with all of them, they have all contributed to my education in some way: Andrea Calabrese, Marie Coppola, Jon Gajewski, Harry van der Hulst, Stefan Kaufmann, Diane Lillo-Martin, William Snyder, and Jon Sprouse. A special thanks also goes to Tamara Cohen. She manages to make administrative work look easy (and don't even for a moment think it actually is) and she is probably the reason why I was able to deal with my fear of/aversion to bureaucracy.

I was lucky to have overlapped with some amazing visiting faculty during my time as a student at UConn: Norbert Corver, Jairo Nunes, Ian Roberts, and Mamoru Saito. I gained a lot from their classes, lecture series, and direct feedback on my research. I can safely say that they all, in one way or another, shaped the research on display in my thesis.

Even with the amazing team of professors I just listed, I would be lying if I said that my first year at UConn wasn't challenging. Suddenly I was in a different country, surrounded by strangers, and a very different environment than what I was used to. I really appreciated that I didn't have to go through all of this alone, sharing my struggles with my cohort: Karina Bertolino, Ryosuke Hattori, Emma Nguyen, Jayeon Park, and Roberto Petrosino.

On top of that, the whole community of UConn linguistics PhD students was and still is a tight-knit, friendly, and stimulating bunch. During my open house visit to UConn, I was struck by how relaxed everyone was, how I immediately felt like a part of the group, and how relaxed the interactions between students and faculty were. It was a major factor in my decision to choose UConn. Over the years, I have gotten so much useful feedback, so many fun moments, and so much support and good advice from virtually all other students I overlapped with, but I want to single out especially: Neda Todorović, Gísli Rúnar Harðarson, Aida Talić, Troy Messick, Zheng Shen, Koji Shimamura, Marcin Dadan, Renato Lacerda, Yuta Sakamoto, Abbie Thornton, YongSuk Yoo, Christos Christopoulos, Hiromune Oda, Pietro Cerrone, Yoshiki Fujiwara, Lily Kwok, Shuyan Wang, Muyi Yang, Teruyuki Mizuno, Ivana Jovović, and Si Kai Lee. Thank you all!

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cover each others weak spots and enhance our strengths. I don't have to stress as much with choosing the right words, knowing that you know what I really want to say better than I do (too bad that we can't collaborate on these acknowledgments).

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And finally, a giant thank you to Paula! You're the first person I ask for an opinion on anything, you're the best person to just exist next to, and there's so many other nice things I want to say, but I'm just not good enough of a writer to say them in a nice way. So here's an anecdote instead. At some point in this thesis, you'll see the phrase *"Mind the gap!"* I asked Paula for a quote that sums up a chapter of my thesis, and 1.5 seconds later I had it, and it was just right. Also, thank you for introducing me to Stamppot. While he "moved in" only after my defense, his presence made an immediate impact.

Contents

List of Tables

List of Abbreviations and Acronyms

1P	first person
1	first person (in glosses/abbreviated)
2P	second person
2	second person (in glosses/abreviated)
3P	third person
3	third person (in glosses/abbreviated)
3’	third person obviative
A	nomimal class A (Kiowa-Tanoan)
ABL	ablative case
ABS	absolutive case
ACC	accusative case
ACT	active voice
AF	Agent-Focus (Mayan)
AG	Agent argument
AI+O	Animate Intransitive + Object (Algonquian)
ALL	allative case
AN	animate
ANML	animal
AP	antipassive voice
APPL	applicative marking
ApplP	applicative phrase

ASP	aspect marker
AspP	aspect phrase
AUX	auxiliary verb
B	nominal class B (Kiowa-Tanoan)
BEN	benefactive marking
C	nominal class C (Kiowa-Tanoan)
CAUS	causative marking
CMPL	completive aspect
COMP	complementizer
CONJ	conjunct (Algonquian; agreement paradigm)
CP	complementizer phrase
DAT	dative case
DEB	debitive mood
DEF	definite
DIR	direct marking
DO	direct object
DP	determiner phrase
DU	dual number
DUR	durative aspect
EA	external argument
EMPH	emphatic marker, emphasizer
ERG	ergative case
EXPER.PAST	experienced past tense/evidential
FACT	factive mood/evidential
FAM	familiar individual
F	feminine gender
FOC	focus marker

FUT	future tense
FV	final vowel (Bantu verbal suffix)
GEN	genitive case
GL	Goal argument
HAB	habitual aspect
HUM	human
IA	internal argument
IC	initial change (Algonquian; clause type affix)
IMPF	imperfect aspect
INAN	inanimate
INCL	inclusive (first person)
INCMPL	incompletive aspect
IND	indicative mood
INDEF	indefinite
INF	infinitive verb
INSTR	instrumental case/marking
INTR	intransitivity marking
INV	inverse marking
IO	indirect object
IRR	irrealis mood
LCA	Linear Correspondence Axiom
LF	Logical Form
M	masculine gender
NEG	negative marker
N	neuter gender
NOM	nominative case

NP	noun phrase
NPAST	non-past tense
NSPEC	non-specific
O	object
OBL	oblique case
OBV	obviative
PASS	passive voice
PAST	past tense
PCC	Person-Case Constraint
PERF	perfective aspect
PF	Photetic Form
PL	plural number
POT	potential mood
PP	prepositional phrase
PRC	present contunuous verb stem (Cherokee)
PRED	predicate marker
PRES	present tense
pron	pronoun
PROX	proximate
PUNC	punctual aspect
REFL	reflexive
RESP	respected individual
REV	reversative, reversive
SG	singular number
SPEC	specific
SU	subject
SUBR	subordinator, subordinate
SUP	superessive case

TH	Theme argument
TOP	topic marker
TR	transitivity marking
UG	Universal Grammar
VE	vegetation gender
VEN	venitive
VoiceP	voice phrase
VP	verb phrase
*v*P	“little *v*” phrase
X	unspecificed argument
Z	zoic gender

When my love
Stands next to your love
I can't compare love
When it's not love

—Talking Heads, "Love → Building on Fire"

1

Introduction

The generative enterprise established in the wake of Chomsky 1957, 1965 is sometimes characterized by non-practitioners as dealing exclusively in highly abstract analyses of syntactic phenomena within a limited empirical domain—a few hand-picked well-studied languages, at best. This is meant to contrast with the typological work in the tradition of Greenberg (1957, 1963) and Comrie (1981), concerned with hard facts collected through large-scale crosslinguistic surveys. But there is no real basis for this dichotomy. While it might be that the number of languages discussed in an average generative work is smaller than in an average typological one, one cannot say that the issue of crosslinguistic variation is something the generative enterprise is not concerned with. This is perhaps most evident with the *Principles and Parameters* approach to language (Chomsky 1981; Chomsky and Lasnik 1993), where the issue of crosslinguistic variation is placed front and center.

Although the theoretical backdrop is different, the empirical question driving the Principles and Parameters approach is completely in line with what Greenberg (1957, 1963) set out to do: identifying linguistic universals—those grammatical traits that all languages share, but also those traits that no language has—obscured by the veil of crosslinguistic variation. What the Principles and Parameters approach offers in addition is a neat answer to the question of why such universals exist. It hypothesizes an initial state of a child's language faculty or *Universal Grammar (UG)*, which is the same for all human beings and contains "clues" that allow any child to acquire any natural human language, but also parameters to be shaped by experience, which is what gives rise to crosslinguistic differences—crucially, all within the bounds of UG. In other words, which languages are possible and which languages are impossible is shaped by the mechanisms first language acquisition. Typologists and generativists thus care for the same empirical problem, generativists only differ in the theoretical means by which we seek to solve it. This should have no bearing on the methods we use. In fact, there are many advantages generative theoretical assumptions can bring to large-scale typological work (see Baker and McCloskey 2007; Baker 2010; Cinque 2007, 2013a, 2013b for general discussion of these points). The complication is only that as generativists we are sometimes not willing to sacrifice careful abstract analyses of individual linguistic phenomena for a larger linguistic sample, and that is good, but it also means we have to find a way to find balance between them, which is hard to do.

But a number of us nonetheless do it. The emerging *generative typology* approach that rides the fine line between in-depth abstract studies of individual languages and large-scale crosslinguistic surveys has already made several important contributions. To list a few areas in syntax and morphology where theoretical advances have been made within this approach:

word and morpheme order patterns (Cinque 1999, 2005, 2013b; Julien 2002; Radkevich 2010; Abels and Neeleman 2012; Biberauer, Holmberg, and Roberts 2014; Tvica 2017; i.a.), lexical categories (Baker 2003b), agreement and concord (Baker 2008b; Bobaljik 2008), case marking (Baker 2015; Baker and Bobaljik 2017), and suppletion and syntcretism patterns (Bobaljik 2007, 2012; Caha 2009; Radkevich 2010; Moskal 2015a, 2015b; Harbour 2016; Smith et al. 2018; i.a.).

One unique advantage of generative typology is the identification of non-obvious implicational relationships between phenomena guided by theoretical predictions. Two notable examples of this kind are Mark Baker's work on polysynthesis (Baker 1996), which explores how morphological argument indexing (i.e. *head-marking*; Nichols 1986, 1992) relates to incorporation, word order, and binding phenomena, and Željko Bošković's work on the NP/DP parameter (Bošković 2008, 2012), which explores how the presence of lexicalized definite articles in a language relates to a number of, at first glance, unrelated syntactic phenomena, such as left branch extraction, the possibility of scrambling, clitic doubling, and negative raising. Regardless of whether these generalizations will stand the test of time, or perhaps turn out to be only strong crosslinguistic tendencies, they would have been missed if not for the theoretical assumptions that made us look for them in the first place.

This thesis follows this tradition. It does, however, depart from the works listed above in at least one significant way: the phenomena that have so far received the generative typology treatment concern either universal or fairly transparent attributes of languages, for example the basic word order, the case marking pattern, the functional category inventory of a language. Person restrictions, the phenomenon I will look at, is very different: these restrictions are not found in every language and even in the languages where they are found, they are limited to only a small subset of syntactic contexts.

The rationale behind focusing on a phenomenon of this type from a typological perspective is the following: if the UG hypothesis outlined above is correct, UG should have no blind spots. All linguistic phenomena, no matter how rare or peculiar, should be explainable by using theoretical mechanisms that are not language specific. This means that the properties of person restrictions should also fall out from the universal architecture of our language faculty. Furthermore, if the number of operations driving syntax is very limited, as assumed in *Minimalism* (Chomsky 1993, 1995), the full range of empirically observable syntactic phenomena must be the result of the same small number of operations interacting with each other. By that rationale, there should be a lot of overlap in variation across different phenomena. We expect to find a large number of implicational universals, and person restrictions specifically are expected to not be isolated from this. As we will see, this is exactly what we find when we give this phenomenon the typological treatment. What might appear to be a rather quirky surface fact of some languages actually turns out to have implications for theories of syntactic operations, syntactic domains, pronoun typology, and how syntax interacts with its interfaces—both on the sound and the meaning side.

1.1 Person restrictions

There are languages that display a peculiar restriction on the distribution of personal pronouns: a personal pronoun that can normally express the full range of person values (i.e. it can be first, second, or third person) can only occur with a restricted set of person values in the context of another personal pronoun in a particular syntactic domain. I refer to all phenomena that fit this description with the cover term *person restrictions*.

Perhaps the most famous case of a person restriction occurs with clitic pronouns in double object constructions—the *Person-Case Constraint (PCC)* (Bonet 1994) (sometimes also called the **me lui Constraint*; Bonet 1991). This restriction has been the subject of discussion in generative linguistics from very early on, already being noted in the seminal works on Romance clitics of Perlmutter (1968, 1970, 1971) and Kayne (1975).[1] The person restriction is illustrated with French examples in (1) and (2) (the types of pronouns and related elements that are typically sensitive to person restrictions are highlighted by the use of bold font in the relevant examples throughout the thesis).[2,3] The syntactic context for the restriction in French are ditransitive clauses where both objects are realized as clitic pronouns. The grammatical combinations of the two pronouns are presented in (1), where we see that when the two object clitics co-occur, the *indirect object (IO)* clitic pronoun can express the full range of person values: *third (3P)*, *second (2P)*, or *first person (1P)*.

(1) French [*Romance*]:

a. Marie **le** **lui** donnera. — 3.IO + 3.DO
   Marie 3.M.DO 3.DAT give.FUT.3
   'Marie will give **it** to **him**.' (Perlmutter 1970, 214)

b. Philippe **te** **le** présentera. — 2.IO + 3.DO
   Philippe 2.IO 3.M.ACC introduce.FUT.3
   'Philippe will introduce **him** to **you**.' (Postal 1990, 119)

c. Jean **me** **le** donnera. — 1.IO + 3.DO
   Jean 1.IO 3.M.ACC give.FUT.3
   'Jean will give **it** to **me**.' (Kayne 1969, 40)

However, in the same context, the *direct object (DO)* clitic pronoun cannot have any person value but 3P. This is illustrated by the ungrammatical examples in (2); no matter what the person of the IO clitic is, the DO clitic that co-occurs with it cannot be 1P or 2P.[4]

1. The original empirical observation goes back at least to Meyer-Lübke (1899).

2. All examples are glossed using Leipzig glossing rules. Unmarked number, case, and tense are left out unless relevant, while caseless/case-syncretic pronouns are glossed according to their grammatical role.

3. All ditransitive constructions are translated into English as prepositional datives, where the PCC is never observed. Since some English speakers exhibit the PCC (Bonet 1991; Richards 2008a) or find the relevant examples marked, I want to avoid using ungrammatical or unnatural sounding translations.

4. Although the ungrammaticality of the relevant examples is technically already noted in Perlmutter 1968, 194–5n14, the first proper discussion is in Perlmutter 1970, where the restriction is stated as:

(i) If the direct object of *recomendar* ['recommend'] is first or second person, use of the clitic form of the indirect object results in an ungrammatical sentence. (Perlmutter 1970, 231)

With the footnote: "[(i)] *certainly applies to a wider class of cases than just objects of* recomendar*, but the problem of determining the range of cases to which it applies does not concern us here.*" (Perlmutter 1970, 231n37). The discussion is repeated in Chapter 2 of Perlmutter 1971, based on the earlier paper.

(2) a. *Ton cousin **me** **lui** a recommandé. — ✗3.IO + 1.DO
your cousin 1.DO 3.DAT has.3 recommended
'Your cousin has recommended **me** to **him/her**.' (Perlmutter 1970, 232)

b. *Philippe **te** **lui** présentera. — ✗3.IO + 2.DO
Philippe 2.DO 3.DAT introduce.FUT.3
'Philippe will introduce **you** to **him**.' (Postal 1990, 119)

c. *Jean **me** **te** présente. — ✗1.IO + 2.DO
Jean 1.IO 2.DO introduce.3
'Jean introduces **you** to **me**.' (Laenzlinger 1993, 242)

d. *Il **te** **m'** envoya. — ✗2.IO + 1.DO
he 2.IO 1.DO sent.3
'He sent **me** to **you**.' (Nicol 2005, 142)

The descriptive generalization for this particular person restriction can be stated as in (3).[5]

(3) *Person-Case Constraint*
If IO and DO pronouns co-occur, the DO must be 3P.

A curious property of person restrictions is that they only apply in very specific syntactic contexts and only with some types of pronouns. For example, the object clitic combination in (4a) is ungrammatical in French. But the same pronoun combination can be expressed with a prepositional dative construction as in (4b).

(4) a. *Lucille **te** **leur** présentera. — ✗3.IO + 2.DO
Lucille 2.DO 3PL.DAT introduce.FUT.3

b. Lucille **te** présentera à elles. — 3.IO + 2.DO
Lucille 2.DO introduce.FUT.3 to them
'Lucille will introduce **you** to **them**.' (Řezáč 2011, 93)

5. The description of the person restriction I use here is based on Bonet's:

(i) *The *me lui/I-II Constraint*
a. In a combination of a direct object and an indirect object, the direct object has to be third person
b. Both the direct object and the indirect object are phonologically weak (Bonet 1991, 177)

Bonet later updated this to (ii) (where DAT and ACC-3rd are clitic/agreement elements related to arguments):

(ii) *Person-Case Constraint:* If DAT then ACC-3rd (Bonet 1994, 36)

Person restrictions thus do not result in the complete impossibility of expressing specific person combinations of multiple pronouns—the restrictions are specific to particular syntactic contexts and configurations. There is almost always a way in a given language with such restrictions to express the banned pronoun combinations via an alternative construction or by changing the type of one of the relevant pronouns.

Importantly, the complementary distribution between different realizations of a banned person combination is not necessarily between a double object construction and prepositional dative construction. For example, in Greek, which also displays the PCC, clitic pronouns can be the only objects in a clause or they can double noun phrases or prosodically independent *strong pronouns*, as in the double object construction in (5a).[6] However, it is impossible to have a double object construction where only the DO is a clitic—as in the French example in (4b) above—even when this would avoid a banned clitic combination, as shown in (5b).

(5) Greek [*Hellenic*]:

a. (**Tu**) (**to**) edhose *tu Petru* *to vivlio* i Maria.
3.M.GEN 3.N.ACC gave.3 the Petros.GEN the book.ACC the Maria.NOM
'Maria gave Peter [**him**] the book [**it**].' (Anagnostopoulou 2003, 63)

b. ***Me** sistisan { *ekinu* / *aftu* }. $\boxed{✗\,3.\text{IO} + 1.\text{DO}}$
1.ACC introduce.3PL { him / him.GEN }
'They introduced **me** to *him*.' (Anagnostopoulou 2003, 312)

However, the person restriction can be avoided in the reverse case, where the IO is expressed as a strong pronoun without a clitic double and the DO can be a clitic, as seen with the contrast between (6a), where the person restriction applies, and (6b), where it is voided.

(6) a. *Tha **tu** **se** stilune $\boxed{✗\,3.\text{IO} + 2.\text{DO}}$
FUT 3.M.GEN 2.ACC send.3PL
'They will send **you** to **him**.' (Anagnostopoulou 2005, 202)

6. The italic font is not meant to signal emphasis. It is only used to highlight strong pronouns (as opposed to clitic or weak ones) and noun phrases/strong pronouns doubled by clitics or agreement affixes. Since there are no unmarked English equivalents to clitic/agreement doubling, either the clitic pronoun (or agreement affix expressed as a pronoun) or its double is in square brackets in relevant translations throughout the thesis.

b. Tha **tu** stilune *esena* $\boxed{\text{3.IO + 2.DO}}$
FUT 3.M.GEN send.3PL you.ACC
'They will send *you* to **him**.' (Anagnostopoulou 2003, 253)

Note that in the case of the Greek person restriction, what makes the restriction go away is making the restricted pronoun (the DO) not a clitic but an undoubled strong pronoun, whereas in French it was the other pronoun (the IO) which had to be realized differently: as a strong pronoun in a prepositional phrase. However, in both cases the restriction goes away by creating a context where one of the objects is not a clitic. Even beyond the French and Greek cases, person restrictions do not mean the complete impossibility of expressing certain person values on the two arguments, but merely the need to employ alternative grammatical means to do so. As we will see later on, the type of the alternative means employed can sometimes even obscure that we are in fact dealing with a person restriction, especially given the many forms that person restrictions can take crosslinguistically.

Finally, consider how peculiar it is that only the grammatical category of person should be singled out in this way, as opposed to say number or gender. Person, number, gender are all grammatical categories generally assumed to be encoded on syntactic elements by way of formal features. Person, number and gender are commonly grouped together as a natural class: *φ-features* (Chomsky 1981). Considering this, person restrictions can be thought of as a ban on specific values of person features in the relevant syntactic contexts. For example, if the contrast between 1/2P and 3P features is encoded via a [±participant] feature, where [+participant] corresponds to 1/2P and [−participant] to 3P, the PCC can be characterized as a ban against the DO having a [+participant] feature in the context of an IO.

However, if person features can be singled out syntactically in this way, there is no a priory reason for not expecting the same to be a possibility for other φ-features. In analogy to the PCC, one might then expect a syntactic restriction on number, such as (7), which we could characterize as a ban on the DO having the feature [−singular] in the context of an IO.

(7) *Number-Case Constraint (unattested)*
If IO and DO pronouns co-occur, the DO cannot be plural.

However, such syntactic number restrictions are unattested in the world's languages (see also Nevins 2011). Similarly, restrictions on gender features are also unattested; there is no *Gender-Case Constraint* that would target masculine, feminine, or neuter pronouns.

This remains one of the main questions concerning person restrictions: why are there no comparable syntactic restrictions on other φ-features like number or gender? If person restrictions were limited to French or Romance languages only, then they could be dismissed as just a quirk or irregularity and an areal phenomenon. However, such restrictions are found across numerous unrelated language families of very different typological profiles, as we will see in the course of this thesis. This makes it quite difficult to entertain the idea that the singling out of person by these restrictions is merely a coincidence.

1.2 Crosslinguistic variation in person restrictions

When we expand the empirical domain beyond Romance, it soon becomes clear that person restrictions are found in virtually all language families (see e.g. Bonet 1991; Haspelmath 2004a; Baker 2008b; Řezáč 2011; Anagnostopoulou 2017). Additionally, person restrictions go way beyond the classic PCC case. As discussed below, they can affect arguments other than those involved in the classic PCC. Furthermore, even the classic PCC restriction can vary in terms of the excluded combinations of pronouns.

The basic point of variation observed in relation to person restrictions is in their *strength*. For example, although speakers of Spanish can have the same restriction as we saw in French—dubbed the *STRONG version* of the PCC, some speakers permit more combinations of clitic pronouns when they co-occur in double object ditransitives (Perlmutter 1970, 1971). In addition to clitic object pronoun combinations with a 3P DO, such speakers also permit combinations where both the IO and the DO clitic are 1P or 2P, like the example in (138a). However, just as in French, combinations of a 3P IO and a 1P or 2P DO are still ungrammatical for such speakers, as illustrated in (8b) and (8c) respectively.

(8) Spanish [*Romance*]:

a. **Te** **me** recomendaron. $\boxed{\text{1/2.IO + 2/1.DO}}$
2.IO/DO 1.DO/IO recommended.3PL
'They recommended **me** to **you**.'
'They recommended **you** to **me**.' (Perlmutter 1970:231)

b. ***Me** **le** recomendaron. $\boxed{\text{✗ 3.IO + 1.DO}}$
1.DO 3.DAT recommended.3PL
'They recommended **me** to **him/her**.'

c. ***Te** **le** recomendaron. $\boxed{\text{✗ 3.IO + 2.DO}}$
2.DO 3.DAT recommended.3PL
'They recommended **you** to **him/her**.' (Perlmutter 1970, 230)

This restriction pattern is the so-called *WEAK version* of the PCC (or just the *WEAK PCC* for short), which conforms to the descriptive generalization in (9).[7]

(9) *Person-Case Constraint (WEAK version)*
If IO and DO pronouns co-occur, and IO is 3P, the DO must be 3P.

Interestingly, as is the case in Spanish, the variation between STRONG and WEAK is even more generally notoriously fine-grained. It is often the case that speakers who are nominally of the same dialect can differ in terms of displaying either the STRONG or the WEAK PCC.

There is another key way in which person restrictions can vary, namely they can vary in terms of which arguments the restricted pronoun pair corresponds to—the *domain* of the

7. The description I use is again based on Bonet's:

(i) *The *me lui Constraint (WEAK VERSION)*
a. In a combination of a direct object and an indirect object, if there is one third person, it has to be the direct object
b. Both the direct object and the indirect object are phonologically weak (Bonet 1991, 181–2)

Bonet later updated this to (ii) (where DAT and ACC-3rd are clitic/agreement elements related to arguments):

(ii) *Weak Person-Case Constraint:* If DAT-3rd then ACC-3rd (Bonet 1994, 41) person restriction. Above we have seen cases of the PCC, which affects two object pronouns, but the same type of restriction can also be found with pairs of subject and object pronouns. An example of this type of restriction is found in Tundra Nenets (Dalrymple and Nikolaeva 2011, 134; Nikolaeva 2014, 202), where in an active transitive clause, the subject can be any person, while the object can only be 3P, if expressed via the marker traditionally analyzed as object agreement. The grammatical subject-object combinations are shown in (10).[8,9]

(10) Tundra Nenets [*Samoyedic*]:

a. *Wanya s'ita* ladə° **-Ø** **-da** — 3.SU + 3.O
John he.ACC hit -(3).O -3.SU
'**He**$_i$ [John] hit **him**$_k$.' (Dalrymple and Nikolaeva 2011, 133)

b. *pidər° Wera-m* lad°ə **-Ø** **-r°** — 2.SU + 3.O
you Wera.ACC hit -(3).O -2.SU
'**You** hit **him** [Wera].' (Nikolaeva 2014, 386)

c. *Wera-h ŋəno-m-ta* sulor-p'i **-Ø** **-wə** -s'° — 1.SU + 3.O
Wera.GEN boat-ACC-3 fix-DUR -(3).O -1.SU -PAST
'**I** fixed **it** [Wera's boat].' (Nikolaeva 2014, 145)

As we can see in (11), the object marker cannot cross-reference a 1P or 2P object (even when the 1/2P object pronoun it is meant to cross-reference is overt).

(11) a. **Wanya s'iqm'i* ladə° **-Ø** **-da** — ✗ 3.SU + 1.O
Wanya I.ACC hit -(1).O -3.SU
'**He** [Wanya] hit **me**.'

b. **Wanya s'it°* ladə° **-Ø** **-da** — ✗ 3.SU + 2.O
Wanya you.ACC hit -(2).O -3.SU
'**He** [Wanya] hit **you**.' (Dalrymple and Nikolaeva 2011, 134)

c. **man'° s'it°* xada-wənta **-Ø** **-w°** — ✗ 1.SU + 2.O
I you.ACC kill-DEB -2.(O) -1.SU
'**I** must kill **you**.' (Nikolaeva 2014, 106)

8. In many languages, some pronouns or agreement morphemes are not expressed overtly (common with 3P ones in particular). As the underlying presence of such morphemes is key for illustrating and discussing person restrictions, I represent them with 'Ø', where their position is inferred from the overt parts of the paradigm.

9. All sourced examples use the transcription from the original unless I provide examples of the same language from sources that use different transcription conventions. In that case, I generally use the transcription conventions of the more recent work for the sake of consistency.

Notice that the pattern is abstractly identical to the PCC, we only have to substitute the IO with the subject and the DO with any type of object in the description. Both are restrictions that force a pronominal/agreement element to be 3P in the presence of another element of the same type, which also means that the Tundra Nenets person restriction is a STRONG restriction specifically, and can be described as in (12).

(12) *Subject-object person restriction (STRONG version)*
If subject and object pronouns co-occur, the object must be 3P.

Another parallelism that we find with subject-object person restrictions is that they show the same exact variation in restriction strength as the PCC; that is, there exist STRONG and WEAK versions of such restrictions (we will see examples of this in Chapter 2).

Additionally, even the strategies that can void the subject-object person restrictions are of the same type as the ones that we have seen above in relation to the PCC, for example: realizing only one of the pertinent arguments as a clitic/argument marker. We can see an example of this from Tundra Nenets in (13): when the objects are 1/2P, the argument cross-referencing must employ only subject and no object markers.

(13) a. *Wanya s'iqm'i* ladə° **-Ø** — 3.SU (1.O)
Wanya I.ACC hit -3.SU
'**He** [Wanya] hit *me*.'

b. *Wanya s'it°* ladə° **-Ø** — 3.SU (2.O)
Wanya you.ACC hit -3.SU
'**He** [Wanya] hit *you*.' (Dalrymple and Nikolaeva 2011, 134)

c. *man'° s'it°* xada-wənta **-d°m** — 1.SU (2.O)
I you.ACC kill-DEB -1.SU
'**I** must kill *you*.' (Nikolaeva 2014, 106)

We can see in (13) that the subject markers are different from the ones in (10)–(11); they are markers otherwise used with intransitive verbs (Nikolaeva 2014, 78–80), this reveals the absence of object markers even when the expected forms would have to be null (as in this case). That the presence of the object marker restricts what the person of the object can be is also indicative of it not being a regular agreement marker, from which we would expect to only co-vary with the person features of the object and not to restrict them.

We will see eventually that there is even more crosslinguistic variation in the domain of person restrictions, especially when it comes to interactions with other language specific properties (which I look at briefly below). However, this illustration of the variation in the strength and domain of the restrictions will suffice for now to illustrate that the problem of analyzing person restrictions is actually two-fold: we need to explain how they arise in the first place as well as how the variation in person restriction patterns can arise.

1.3 What drives person restrictions?

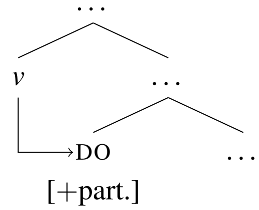
Person restrictions have received a number of analyses over the years. They were initially suggested by Perlmutter (1970, 1971) to be surface constraints on co-occurring morphemes—essentially: with a ditransitive verb, if the IO is a clitic and the DO is a clitic, then the DO cannot be 1/2P (see footnote 4). Note though that this amounts to restating the descriptive generalization as a constraint. In addition, it was not meant to be a universal constraint.

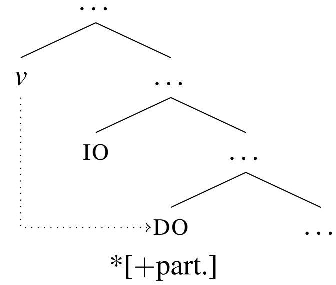
In contrast, many later analyses invoke universal person/topicality hierarchies in the style of Silverstein (1976). In such analyses, the restriction generally amounts to a prohibition of specific interactions between a person hierarchy, where 1P and 2P outrank 3P, and a thematic hierarchy, where Goal outranks Theme: e.g. an argument cannot outrank on the person scale a co-occurring argument that is higher on the thematic scale. That means the Theme cannot be 1/2P if the Goal is 3P. This approach is not tied to a particular framework; the basic idea has been implemented, for example, in Relational Grammar (Rosen 1990), Optimality Theory (Aissen 1999b), and in functionalist terms (Haspelmath 2004a).

Probably the first generative studies that proposed a universal treatment of the PCC were conducted by Bonet (1991, 1994). Her analysis of the PCC relies on a universal morphological constraint amounting to a ban against 1/2P clitic pronoun arguments occurring in the context of oblique case clitic pronoun arguments. More recently though, starting with Albizu (1997a), there has been a shift in generative circles towards syntactic analyses.

1.3.1 Person restrictions and Agree

The current mainstream syntactic approach to person restrictions attributes them to a syntactic intervention effect (Anagnostopoulou 1999, 2003, 2005; Béjar and Řezáč 2003; i.a.). More specifically, the idea is that person restrictions arise in cases when two or more arguments compete to establish a syntactic dependency with a single functional category. The assumption is that only one of them may do so: the argument structurally closest to the head of the functional category. For example, as illustrated in (14), a dependency between the functional head *v* and the DO is possible because no other argument intervenes between them. In contrast, as shown in (15), the same type of dependency between *v* and DO is blocked because of the IO argument that is structurally closer to *v* than the DO.

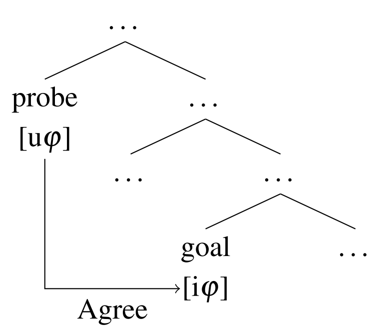
(14)


(15)


The basic idea is that successfully entering into a syntactic dependency with *v* is what licenses the DO pronoun to be 1/2P; in feature terms, it licenses the presence of a [+participant] feature on the DO. Without the dependency between *v* and the DO, the [+participant] feature is not allowed on the DO, and the pronoun can only be 3P, which is [−participant].

The relevant syntactic dependency assumed in the works cited above is *Agree* (Chomsky 2000, 2001), which is a dependency assumed to result, among other things, in morphological agreement between a verb and its arguments. Arguments bear semantically *interpretable* φ-features ([iφ]) specified with their own values, while agreement targets are functional syntactic heads with matching *uninterpretable* φ-features ([uφ]) which enter the syntactic derivation unvalued. In the original version of Agree proposed by Chomsky (2000), unvalued uninterpretable φ-features act as *probes* looking for a *goal*, which must be a valued interpretable φ-feature. Upon a successful probing, Agree can be established between the two as in (16) and the uninterpretable features acquire the value of the interpretable features.

(16)



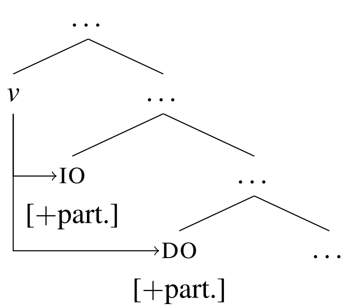
What intervention approaches to person restrictions add on top of this is the assumption that there is an asymmetry between the conditions for licensing 1/2P arguments and 3P arguments: 1/2P pronouns require Agree with person features, while 3P pronouns do not. However, there have been numerous proposals regarding how to implement this; to list just a few: (i) differences in the φ-feature make-up of pronouns affect the ability to license arguments via Case checking (Anagnostopoulou 2003, 2005; Adger and Harbour 2007), (ii) person features on arguments require special licensing via Agree in addition to Case licensing of the argument itself (Béjar and Řezáč 2003; Charnavel and Mateu

2015b; Pancheva and Zubizarreta 2018; i.a.), (iii) person features can only enter Agree in a Spec-Head configuration with the probe (Baker 2008b, 2011), and (iv) person features interact with Agree differently than other φ-features (Nevins 2011).

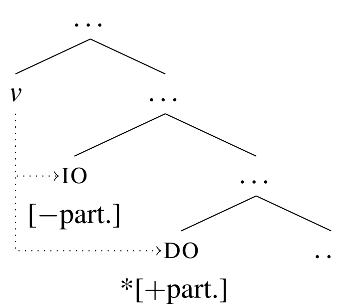
However, as I noted above, there is also a lot of crosslinguistic variation in person restrictions. For example, there exist WEAK restrictions, where both pronouns can be 1/2P, and restrictions between subject and object pronouns—these can not be derived with derivations like the one illustrated in (15) above.

The mainstream approach to variation in the case of different restriction strengths is to parameterize Agree itself, so that multiple pronouns can be licensed depending on their specific person value. The original proposal of this kind was by Anagnostopoulou (2005): in languages with a WEAK person restriction, a probe may establish Agree with more than one 1/2P goal, as in (17), but a 3P goal still blocks Agree with a lower goal, as in (18).

(17)



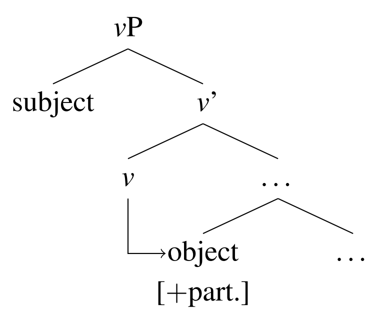
(18)



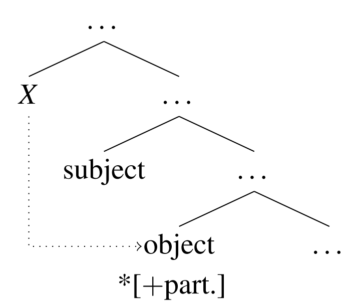
But since Anagnostopoulou's proposal there have been many versions of the basic idea, all with slightly different assumptions regarding how fine-grained the parameterization of Agree can be (see Nevins 2007, 2011; Béjar and Řezáč 2009; Deal 2015; Coon and Keine 2018; i.a.). These differences are usually motivated by the crosslinguistic variation in person restrictions. Note though that these modifications all involve a departure from the original conception of Agree, which is blind to specific values of features and only distinguishes between unvalued probe features and valued goal features. This is therefore a place where we need to exercise great care; we need to make sure that the additional machinery we introduce is not unnecessarily complicating the system or making it too powerful—allowing it to generate even the types of person restrictions that are never attested.

Another key issue relating to intervention analyses, and one that has not received much attention, is what kinds of functional elements can function as probes that license 1/2P arguments. For example, in the case of person restrictions between a subject and an object, the relevant probe cannot be *v* given the standard intervention approach. This is because *v* is assumed to be the head that introduces the subject in active sentences (Chomsky 2000). As shown in (19), the subject cannot block Agree between *v* and the object, so the object can be 1/2P. What is needed to derive this type of person restrictions is a probe higher in the structure of the clause than both the subject and the object. As seen in (20), in that case the subject will block Agree between the probe *X* and the object, restricting the latter to 3P.

(19)



(20)



The question is what determines that in a language with a subject-object person restriction *X*, and not another head, can be the probe that licenses a 1/2P subject? For example, if there was another probe *Y* between the subject and the object, then *X* and *Y* could license a 1/2P subject and object respectively and the person restriction would go away. Why is is that not all languages simply default to such configurations that yield no person restrictions? Related to this, can any functional category in the clausal spine in principle be such a probe?

These are the types of questions that we may only start answering once we have a better idea of what the actual limits of variation are in the domain of person restrictions. It may be that we have only scratched the surface of the possible restriction patterns, but it may also turn out that the current approaches potentially allow the derivation of person restriction patterns that we never find in the world's languages.

1.3.2 Person restrictions and the typology of pronouns

Another factor that must be considered in any theory of person restrictions is which syntactic elements can be affected by them. So far, all the examples of person restrictions we looked at involved either clitic pronouns or similar phonologically weak elements—in the case of the subject and object markers in Tundra Nenets. The type of the co-occurring pronouns does appear to be crucial for whether or not person restrictions can arise.

A very nice illustration of this can be made with Modern Standard Arabic, where object clitics can co-occur in a ditransitive, as in (21a), and they are subject to the PCC, as (21b) shows. Crucially, the restriction on the DO goes away as soon as we change it from a clitic to a prosodically independent pronoun—also known as a strong pronoun, as in (21c).

(21) Modern Standard Arabic [*Semitic*]:

a. ʔaᶜṭay-tu **-ka** **-hu** — 2.IO + 3.DO
gave-1 -2.IO -3.M.DO
'I gave **him** to **you**.'

b. *ʔaᶜṭaa **-hu** **-ka** l-ʔustaaḏ-u — ✗ 3.IO + 2.DO
gave.3 -3.IO -2.DO the-teacher-NOM
'The teacher gave **you** to **him**.'

c. ʔaᶜṭaa **-hu** l-ʔustaaḏ-u *ʔiyya-ka* — 3.IO + 2.DO
gave.3 -3.IO the-teacher-NOM you-ACC
'The teacher gave *you* to **him**.' (Fassi-Fehri 1993, 104–5)

What makes this case different from the French one earlier (cf. (4)) is that voiding the person restriction does not require a change in the type of ditransitive construction—here it is a double object construction in both the restricted case in (21b) and the unrestricted case in (21c). And unlike in Greek (cf. (5)–(6)), there is no clitic doubling in Modern Standard Arabic, so clitic and strong pronouns are always in complementary distribution.

Person restrictions appear to exclusively affect prosodically deficient or weak pronouns and similar elements—for which I will use the cover term deficient pronouns for the time being. The affinity of deficient pronouns to person restrictions was already discussed by Perlmutter (1970, 1971) and Bonet (1991, 1994). However, what is interesting is that not all such pronouns are affected by person restrictions. This is nicely illustrated by comparing the Slovenian example in (22) with the Polish one in (23). The languages have similar deficient pronoun systems, but only the Slovenian ones are subject to person restrictions.

(22) Slovenian [*Slavic*]:
*Mama **mu** **te** bo predstavila. ✗3.IO + 2.DO
mom 3.M.DAT 2.ACC will introduce
'Mom will introduce **you** to **him**.' (Stegovec 2015b, 108)

(23) Polish [*Slavic*]:
Jak **mu** **cię** nieprzytomną dostarczą, ... 3.IO + 2.DO
when 3.M.DAT 2.ACC unconscious deliver ...
'When they deliver **you** to **him** unconscious, ...' (Bondaruk 2012, 68)

Similar contrasts can even be found within one language. For example, as noted by Cardinaletti and Starke (1994, 1999), the PCC applies between two object clitics in Italian, as in (24a), but with a plural IO *weak pronoun* (as opposed to a clitic pronoun in the technical sense of Cardinaletti and Starke 1994, 1999) the PCC is not observed, as shown in (24b).

(24) Italian [*Romance*]:
a. *Gianni **mi** **gli** ha presentato. ✗3.IO + 1.DO
Gianni 1.DO 3.M.DAT has presented
'Gianni has presented **me** to **him**.'
b. Gianni **mi** ha presentato **loro**. 3.IO + 1.DO
Gianni 1.DO has presented 3.PL.DAT
'Gianni has presented **me** to **them**.' (Cardinaletti and Starke 1994, 63)

However, it is not even always the case that weak pronouns (as opposed to clitic ones) never give rise to person restrictions. As discussed by Anagnostopoulou (2008), many Germanic languages show such restrictions with pronouns that are classified as weak rather than clitic.

Finding what determines the right natural class of pronouns in terms of sensitivity to person restrictions is therefore not straightforward, but still something that must be determined if we hope to find the right analysis of person restrictions. Because of this, one of the goals of this thesis is also to determine what defines the natural class of pronouns which are crosslinguistically affected by person restrictions.

1.3.3 Person restrictions and language acquisition

It is important that, whatever theory of person restrictions we propose, it can also explain how children acquiring a language with person restrictions can converge on the target restriction pattern given only the language information they receive as an input—this is, oddly, a point that is pretty much overlooked by recent studies of the phenomenon.

The main issue we have to consider is essentially a version of *Baker's paradox* (Baker 1979).[10] Children do not systematically receive any information on which sentences are ungrammatical in their target language—the input only contains utterances that are grammatical in the language. However, given a particular phenomenon, some grammars may generate a set of expressions that are in a subset relation to the expressions generated by another grammar, so how does the child determine based only on positive evidence what is the target grammar?

Abstractly speaking, a grammar α may generate the strings *AB* and *BC* while another grammar β may generate the strings *AB*, *BC*, as well as *AC* and *DC*. Since the set of strings generated by α is properly contained in the set generated by β, the child does not have sufficient information upon hearing only the strings *AB* and *BC* to determine what the target grammar is. If the target grammar is α, how is the child to know that the strings *AC* and *DC*

10. Part of the more general 'logical problem of language acquisition' (Baker 1979; Pinker 1979, 1989; Baker and McCarthy 1981; Hornstein and Lightfoot 1981; i.a.): children acquire language despite an inadequate linguistic input. See also *Poverty of the Stimulus* (Chomsky 1980) and *Plato's Problem* (Chomsky 1986).

are not going to appear in the input at any future point? Similarly, if the target grammar is β, how is the child to know that the strings in *AC* and *DC* were "accidentally" absent in the input up to this point? Wrong inferences about what the target grammar is should then be numerous in such cases. So how do children learners avoid or overcome such errors in the absence of negative evidence that would tell them which forms are ungrammatical?

This is exactly the problem with acquiring a language with a person restriction. The child learning the language has to identify which combinations of pronouns are ungrammatical based only on the grammatical combinations encountered in the input. Suppose the child is acquiring a language with a STRONG version of the PCC. Upon first hearing that language's equivalent of *'They showed **you him**'*, the child only knows that a combination of a 2P IO and a 3P DO is grammatical. But that information is also compatible with the target grammar having a WEAK version of the PCC or having no restriction at all; see (25).

(25) *Grammatical person combinations per restriction (IO + DO):*

a. STRONG: 1 + 3, **2 + 3**, 3 + 3
b. WEAK: 1 + 3, **2 + 3**, 3 + 3, 1 + 2, 2 + 1
c. none: 1 + 3, **2 + 3**, 3 + 3, 1 + 2, 2 + 1, 3 + 2, 3 + 1

Additionally, a child may hear a pronoun combination that is ungrammatical in the double object frame in the context of a different construction where it is grammatical, like *'They showed **you** to **him**'*. The child cannot generalize based on this example what the grammatical pronoun combinations are in double object constructions.

One can also expect the positive evidence itself to be scarce, especially in the case of the PCC. Double object constructions with two object pronouns are not all that common in adult speech, and those with two 1/2P objects—required to tease apart WEAK from STRONG restrictions—are even rarer even in languages without the relevant person restriction (see e.g. Haspelmath 2004a, 32–6 for discussion based around a corpus study of German).

Although the present work is not an acquisition study per se, the acquisition problem sketched out above is going to be a key consideration in the analysis of person restrictions I will propose down the line. It will be crucial that the key evidence required for the child to acquire a person restriction does not come from the person combinations of pronouns that are attested in the input—which, as just discussed, would be insufficient, but from independent grammatical cues relating to the pronouns.

1.4 Why the typological treatment?

At the beginning of this chapter I outlined some general arguments in favor of generative typology studies. In this section I want to motivate such a study specifically in relation to person restrictions. Additionally, I will provide some illustration of what kind of phenomena, aside from the classic cases of person restrictions seen above, will be examined in this study.

Past studies of person restrictions have focused empirically either on in-depth studies of individual languages and groups of related languages (e.g. Romance), or on identifying crosslinguistic variation based on limited comparisons between individual, sometimes unrelated, languages. These studies have led to much progress in our understanding of person restrictions (see Anagnostopoulou 2017 for an overview and references). As noted above, Bonet (1991) and Haspelmath (2004a) helped establish the universality of the PCC—not in the sense that all languages exhibit it, but in the sense that it is not merely an areal or genetically restricted phenomenon. Albizu (1997b) and Baker (2008b, 2011) have reached similar conclusions about person restrictions more generally, also including cases of person restrictions between subject and object pronouns. Another important contribution was identifying the existence of different strengths and patterns of person restrictions (Bonet 1991; Anagnostopoulou 2005; Nevins 2007; Pancheva and Zubizarreta 2018; i.a.). However, there has not yet been a systematic typological exploration of the limits of crosslinguistic variation in the domain of person restrictions. This is in part due to very practical reasons: generally reference grammars, which are the primary tool for large-scale typological work, are not explicit about the pronoun combinations allowed in a given language.

The most extensive and systematic survey so far is Albizu's (1997), with 45 languages spanning 19 families and 3 isolates.[11] Albizu's language sample, although quite extensive, was not compiled and examined with the intention to identify the limits of crosslinguistic variation in the domain of person restrictions, but mainly to examine the common properties of the person restrictions found in unrelated languages.[12] What has been missing in work on person restrictions so far is a systematic study of typological gaps—the patterns which are never attested in the world's languages. If we are to have an account that derives all and only the patterns that are attested, we need to first determine what those are.

The main parameters of variation, as we saw above, are: (i) in the strength of the restriction, or how many and which person combinations of pronouns restricted, and (ii) in the domain of the restriction, or which are the pairs of arguments where the restriction is observed—co-occurring subject and object pronouns or co-occurring object pronouns. The possibility of interaction between these two parameters of variation has been left pretty much unexplored in previous studies. In order to understand how the variation comes about in the first place, we need to know whether these parameters are independent from each other or not. Recall that in intervention analyses of person restrictions the variation in restriction strength is attributed to crosslinguistic differences in the Agree operation, while the variation in the domain of the restriction is attributed to variation in the identity of the functional category that is the initiator of Agree. In principle, variation in the operation

11. There are typological works dealing with person, like Cysouw 2003 and Siewierska 2004, with larger language samples. However, they look at person as a grammatical category more broadly, only sometimes touching upon phenomena that I classify here as person restrictions; they do not systematically discuss them in a way that could help us address the theoretical questions outlined in the introduction.

12. Another drawback of Albizu 1997b is that the work is not widely accessible. Because of that I am deeply grateful to Pablo Albizu for having shared his work with me.

and the category that initiates it can be independent from each other, so finding unattested patterns in relation to the two parameters would already be an interesting result.

In the present work I tackle these issues directly by means of a broad typological study of the phenomenon, which is the largest so far with 106 languages of varying typological profiles, spanning 26 language families and 3 isolates. In examining these languages I focus in particular on the interactions of multiple parameters of variation, including new parameters of variation, and crucially on identifying previously unnoticed typological gaps. The hope is that identifying these and providing an explanation for them will give as a more restrictive theory of person restrictions and hopefully bring us closer to answering the question why person restrictions exist in the first place.

In addition to this, the size and the type of the language sample means that the survey will look at a number of different types of person restrictions, many of which have previously not been analyzed as such at all. These will be just as crucial in meeting the goals outlined above as the more well known cases of person restrictions. In order to get a feel for the diversity of the phenomena to be discussed, let us take a look at a few of the relevant cases.

All the cases of person restrictions discussed so far showed a systematic asymmetry in always restricting the same argument in a given pronoun pair. With the PCC, the restricted pronoun is always the DO and not the IO, and in a subject-object equivalent of the restriction, the restricted pronoun is always the object and not the subject. However, there are also examples of restrictions where this is not the case. For example, in Slovenian, there are two possible orders with object clitics in double object ditransitives. When the IO clitic precedes the DO clitic, the latter is restricted, as illustrated in (26); this is compatible with the canonical version of the PCC we saw examples of so far. In contrast, when the DO clitic precedes the IO clitic, as in (27), the restriction applies to the IO (Stegovec 2015b, 2019).

(26) Slovenian [*Slavic*]:

a. Mama **ti** **ga** bo predstavila. $\boxed{\text{2.IO + 3.DO}}$
mom 2.DAT 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him** to **you**.'

b. *Mama **mu** **te** bo predstavila. $\boxed{\text{✗ 3.IO + 2.DO}}$
mom 3.M.DAT 2.ACC FUT.3 introduce.F
'Mom will introduce **you** to **him**.'

(27) a. Mama **te** **mu** bo predstavila. $\boxed{\text{2.DO + 3.IO}}$
mom 2.ACC 3.M.DAT FUT.3 introduce.F
'Mom will introduce **you** to **him**.'

b. *Mama **ga** **ti** bo predstavila. $\boxed{\text{✗ 3.DO + 2.IO}}$
mom 3.M.ACC 2.DAT FUT.3 introduce.F
'Mom will introduce **him** to **you**.' (Stegovec 2015b, 108–9)

If we take the restrictions observed with each clitic order to just be two aspects of a single restriction, the emerging pattern is one where it is consistently the second clitic in the cluster which is person restricted. In other words, the restriction pattern is one where the order of clitics is affected by the person value of each clitic.

Direct/inverse systems are another manifestation of person restrictions. A direct/inverse system can be characterized as a system of *argument indexing* (i.e. marking the presence of an argument via agreement or pronominal markers on the verb) which is constrained by the person values of the arguments. Specifically, the indexing alternates between: (i) a *direct marking* paradigm, which is the baseline paradigm used with argument combinations compliant with the person restriction, and (ii) an *inverse marking* paradigm, which is used with argument combinations violating the person restriction. This is illustrated with the examples from Mapudungun in (28). When a subject and object are both 3P in Mapudungun, they can be indexed on the verb via markers from the direct paradigm, as shown in (28a). However, if the object is 1/2P, indexing with markers from the same paradigm is ungrammatical, as shown in (28b) for a 1P object.[13] Instead, the inverse paradigm, characterized by a dedicated

13. Many sources used for the survey do not provide the actual ungrammatical examples, but only observe the existence of the restriction. For exposition purposes, I construct, wherever possible, these examples based on independent information provided in the source and also note where the information is given in the citation.

inverse morpheme and the subject marking occurring in dative form, must be used with banned person combinations, as shown in (28c).

(28) Mapudungun [*Araucanian*]:

a. mütrüm **-fi** **-y** — $\boxed{\text{3.SU + 3.O}}$
call -3.O -3.SU
'**He**$_i$ called **him**$_k$.'

b. *mütrüm **-n** **-y** — $\boxed{\text{✗ 3.SU + 1.O}}$
call -1.O -3.SU

c. mütrüm-e **-n** **-ew** — $\boxed{\text{3.SU + 1.O}}$
call-INV -1.O -3.SU.DAT
'**He** called **me**.' (Smeets 2008, 153–65, 367)

Note that the distribution of the inverse follows a pattern we have already seen: the use of an alternative construction can realize otherwise banned person combinations. However, unlike the previous cases, like the relevant use of prepositional datives in French (cf. (4)), the inverse construction does not have a function outside realizing banned pronoun combinations.

Finally, there are person restrictions which combine some properties of both previous cases: (i) the restriction does not always apply to the same argument (like in Slovenian) and (ii) the indexing of arguments (i.e. the morphological registering of their presence) is sensitive to their person value (like in Mapudungun). Namely, in some languages not all arguments can be indexed on the verb and which arguments can be is determined based solely on their person value. Not only is the indexing based on person, but the unindexed argument can be additionally restricted in terms of person value. A case like this is found in Maasai, where the subject-object indexing is of the direct/inverse type and only one object in a double object construction may be indexed. Crucially, the indexed object may be either the IO or DO, and the choice is based on their person value. For example, in (29) we see that given a 2P and a 3P object, the 2P one is indexed whether it is the IO or DO (note that the indexing may in some cases constitute of only an inverse marker in Maasai).

(29) Maasai/Maa [*Eastern Nilotic*]:

a. **kɨ́-** ɨshɔ(r) ɔl-payíán $\boxed{\text{2.IO/DO (3.DO/IO)}}$
INV.3>2- give M-man.ACC
'**They** will give **you** to the man.'
'**They** will give the man to **you**.'

b. ***ɜ́-** ishɔ *nɨnyɜ́* iyíé $\boxed{\text{✗3.IO/DO (2.DO/IO)}}$
3(>3)- give him/her.ACC you.ACC
'**They** will give *you* to **him/her**.'
'**They** will give **him/her** to *you*.' (Lamoureaux 2004, 14–20)

More generally, any 1/2P object is picked for indexing over a 3P object in double object constructions; it is never an option for unindexed objects to be anything other than 3P (Lamoureaux 2004, 20). In other words, the indexed object can have any person value regardless of its grammatical role, whereas the unindexed object must be 3P.

Despite the differences between these phenomena, we will see when we look at them more closely below that they are abstractly the same type of person restriction. Furthermore, only when we put all of the restriction types together can we get a fuller picture of what is crosslinguistically possible and what is impossible in the realm of person restrictions.

1.5 Thesis outline

In the following, I present the outline of the thesis. Broadly speaking, Chapter 2 focuses on establishing the empirical basis for the theory of person restrictions developed in Chapters 3 and 4. Chapter 5 explores the connection between the syntax and semantics of person as a driving force behind person restrictions. Finally, Chapter 6 looks at miscellaneous implications and extensions of the proposed theory and offers some concluding remarks.

Chapter 2

In this chapter I present the empirical core of the study, which is a typological survey of 106 languages from 26 language families and 3 isolates. I first establish the syntactic nature of person restrictions, showing also that all the different manifestations of person restrictions share the same syntactic character. Following that, I look at the properties of the pronouns that crosslinguistically show sensitivity to person restrictions, establishing a correlation between prosodic strength and sensitivity to person restrictions—only phonologically weak pronouns show sensitivity to person restrictions—but also a correlation between binding behavior and sensitivity to person restrictions—pronouns which are sensitive to person restrictions can appear in more binding contexts than those which are not.

The maim contribution of this chapter, however, is the establishment of two typological gaps. The first concerns languages where the restrictions on subject-object pairs (abbreviated as EA-IA restrictions; explained in the chapter) coexist with the restrictions on object pairs (abbreviated as IA-IA restrictions; explained in the chapter). It turns out, that in such cases the strength of each restriction is not independent from the other. Their strength is determined according to the following descriptive generalization:

(30) *Strength Generalization*
If a language has person restrictions in both the EA-IA and IA-IA domain, the IA-IA person restriction cannot be weaker than the EA-IA person restriction.

Additionally, it turns out that languages where the EA-IA restriction is weaker than the IA-IA restriction are much more common than any other attested combinations of the two.

The second generalization concerns which of the two pronouns in a pair may be restricted, or the *direction* of the restriction. Recall that the canonical person restrictions involve cases where given an IO and DO, the DO is restricted (PCC), or given a subject and an object, the object is restricted. There are also cases like Slovenian or Maasai, where either the IO or DO is restricted, and similar patterns also exist for subject-object restrictions. This cases can be thought of as combining a canonical restriction (on the DO/object) with a reverse restriction (on the IO/subject). There are, however, no languages with only reverse restrictions. In other words, the following holds:

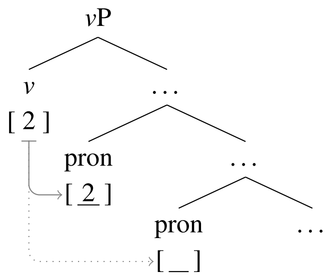
(31) *Direction Generalization*
If a language has a *reverse* person restriction in the EA-IA or IA-IA domain, it must also have a *canonical* counterpart of the same person restriction.

Chapter 3

In this chapter, I propose a new analysis of person restrictions based on the empirical insights from Chapter 2. The basic idea follows the logic of the syntactic intervention approaches outlined in Section 1.3.1, but with a twist. It is not the pronouns valuing the person features of the relevant functional head via Agree that allows them to be 1/2P, it is the person features of the pronoun that are valued by the functional head via Agree—the person features present on the functional head directly determine what the person value of the pronoun can be, they do not license a particular person value of the pronoun. The idea is based on the *minimal pronoun* approach to bound pronouns of Kratzer (2009) and thus relates to the special behavior of the relevant class of pronouns in relation to binding phenomena.

Person restrictions arise in the proposed system because the functional heads that can value such minimal pronouns are very limited. Specifically, I argue that valued person features are only found on *phase heads*—the syntactic heads, such as *v*, that determine the point at which syntactic structure is sent to the interfaces (Chomsky 2000). Because Agree can only be established between a probe and the structurally closest matching goal, the presence of multiple minimal pronoun in a given phase, such as *v*P, means that not all minimal pronouns in *v*'s domain can enter into an Agree relation with *v*. As illustrated in (32), a *v* phase head may only establish Agree with the structurally closest minimal pronoun, which is sufficient to value that pronoun's person features, but at the same time prevents the same for the minimal pronoun further away from *v*.

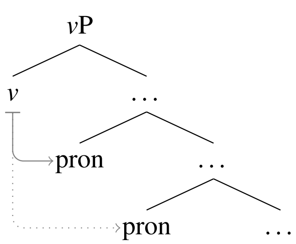
(32)



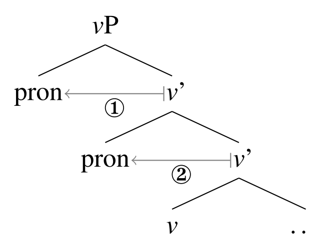
I argue that because *v* cannot establish Agree with the lower minimal pronoun and the pronoun consequently cannot be person values, the pronoun can only get a default 3P value.

I further argue that the derivation in (32) is not the only way in which minimal pronouns can be valued. In particular, the minimal pronouns can also move to the specifier of *v*P to be valued. This is actually what drives the STRONG VS. WEAK PCC variation in the proposed analysis: when both pronouns remain in-situ, as in (33), the intervention effects yields the STRONG PCC, while when both pronouns move to Spec*v*P to be valued, as in (34), the order in which the two pronouns must be valued yields the WEAK PCC.

(33) STRONG:



(34) WEAK:



I will argue that the two options are tied to a difference in the internal structure of the minimal pronouns, so the variation is essentially lexical in nature and does not require us to invoke any parameters on the Agree operation, which has important ramifications not only for person restrictions, but for theories of Agree and language variation in general.

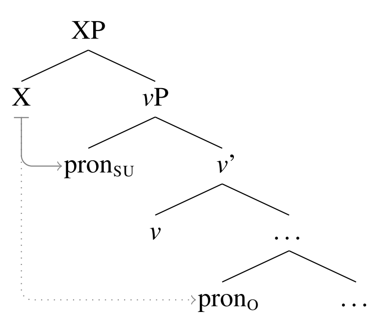
Additionally, I show that this approach can straightforwardly also derive the restriction patterns found in Slovenian or Maasai (see Section 1.4), which are problematic for other intervention analyses of person restrictions.

Chapter 4

In this chapter I show that the analysis of person restrictions proposed in Chapter 3 actually can actually provide an explanation for both the *Strength Generalization* and the *Direction Generalization*; in fact, they can be deduced from the basic assumptions of the proposed analysis in addition to independently proposed assumptions concerning phases, argument structure, movement operations, and the ordering of grammatical operations.

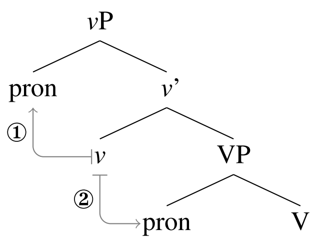
In the case of the *Strength Generalization*, I first show that the analysis proposed in Chapter 3 in relation to the PCC specifically can also be straightforwardly extended to person restrictions between subject and object pronouns. In the case of STRONG restrictions of this type, I argue that these arise in contexts where the phase head is not *v*, but a head higher in the clause (a possibility proposed independently by Gallego 2005; den Dikken 2007; Wurmbrand 2013, 2017b; Bošković 2013, 2014; Despić 2015; i.a.). As illustrated in (35), this yields the same intervention configuration with the phase head (X) and the subject and object minimal pronouns as we saw above with the STRONG PCC derivation.

(35) STRONG:



Alternatively, when the phase is *v*P, the subject of an active transitive will be introduced into the derivation at the edge of the phase (see Chomsky 2000). I argue that when the subject and object in such cases are minimal pronouns, the subject can be person valued by *v* first, followed by *v* also valuing the object, as illustrated in (36).

(36) WEAK:



Although this derivation is at first glance quite different from the one we saw in (34), where both pronouns are valued in the specifier of *v*P, I will show that they are equivalent in terms of the person restriction they yield. Namely, the restriction is a WEAK one in both cases as the result of the structurally higher pronoun always being valued by *v* before the lower one.

Having established that the proposed analysis can derive STRONG and WEAK person restrictions in both argument domains, I show that the proposed analysis can also derive all and only the attested patterns of person restrictions coexisting in both argument domains. In fact, I show also that no matter how we manipulate the variables of the analysis, it can never derive the unattested patterns, thus deducing the *Strength Generalization*.

After that, I shift the discussion to the *Direction Generalization*. I propose that all reverse person restrictions—where the restriction applies to the IO or the subject—are derived via scrambling (Saito and Hoji 1983; Saito 1985) of one pronoun over the other. Because scrambling is not driven by the checking or valuation of any features (Saito and Fukui 1998), it can occur before any person valuation and reverse the structural configuration of the pronouns in relation to the probe, and because such movement is entirely optional, there can never be a language with only a reverse person restriction in either of the argument domains.

Finally, I examine some more fine-grained person restriction patterns where 1P and 2P pronouns do not behave as a natural class, identifying also some patterns of this type that have not been noted by previous studies. I then propose an account of these more fine-grained restrictions which employs the mechanism of *feature inheritance* (Chomsky 2004, 2005, 2008; Ouali 2006), where a phase head may transfer its features to the head of the projection it takes as a complement. I then show, crucially, that in spite of the introduction of this additional mechanism the deductions of the *Strength Generalization* and the *Direction Generalization* are maintained.

Chapter 5

In this chapter, I explore a possible cause for the special status of person features in the syntax. I first show that there actually are other features that constrain the syntactic distribution of pronouns in the same way as person does. With the inclusion of person, all the features that behave this way are pertaining to *indexical expressions* (Kaplan 1978, 1989).

I suggest, based on this insight, that valued versions of such features must exist on phase heads as a type of "escape hatch" that ensures the interpretation of minimal pronouns in the semantics. The proposed idea is that the semantics of person, and other indexicals, does not directly constrain the syntax of person—which would violate the autonomy of syntax—but that without those features being there in the syntax, the minimal pronouns would simply not be able to receive a denotation in the semantics.

Chapter 6

The final chapter of the thesis plays two roles: a discussion of some further implications of the analysis proposed in Chapters 3 and 4 and a conclusion to the thesis.

The first part concerns implications regarding the acquisition of person restrictions with particular reference to pronoun typology. I suggest that person restrictions are acquired through the help from cues concerning phenomena independent from them, such as binding, but also that the morphological and prosodic properties of the pronouns themselves might offer important cues regarding the person restriction pattern in the language.

The second part concerns implications regarding the relationship between person restrictions and phases. I show how certain extraction and argument indexing asymmetries that co-vary with person restrictions can be derived in the analysis of person restrictions proposed in the earlier chapters.

Chapter 7

The final chapter serves as a conclusion to the thesis where I offer concluding remarks on the nature of person restrictions, some speculations on the reason for their existence, what they can tell us about the nature of syntactic derivations, and lastly discuss how some of the proposals made in the thesis ar meant to bridge the gap between formal analyses of person restrictions and functionalist ones.

There is a crack, a crack in everything
That's how the light gets in
—Leonard Cohen, "Anthem"

2

A typological survey of person restrictions

In this chapter, I present and discuss the results of the largest typological survey of person restrictions so far. The goals of the survey are to identify the shared properties of all person restrictions as well as the points of crosslinguistic variation, including the limits of the crosslinguistic variation in this domain. In the case of the later, the main points of variation that I will look at are: (i) the domain of person restrictions—the identity of the arguments that take part in the restriction, (ii) the strength of person restrictions—how many and which person combinations are excluded by the person restrictions, and (iii) the direction of person restrictions—which of the co-occurring arguments does the person restriction apply to.

The chapter starts with an overview of the language sample (Section 2.1) and a discussion establishing the syntactic character of person restrictions (Section 2.2). I then proceed in Section 2.3 to broaden the empirical scope of the study, showing that a number of person-sensitive grammatical phenomena that appear to be very different on the surface are in fact all the same phenomenon; namely, the same phenomenon as the prototypical cases of person restrictions introduced in Chapter 1. Following that, I show that the types of pronouns that are crosslinguistically sensitive to person restrictions constitute a very specific natural class, showing special behavior with respect to independent syntactic phenomena (Section 2.4). Finally, based on the parameters of variation established in Sections 2.2 and 2.3, I establish two previously unnoticed typological gaps and examine their theoretical implications (Section 2.5).

2.1 Language sample

The language sample used for this survey spans 106 languages from 26 families and 3 isolates.[1] The languages are listed in (1); references will be provided when individual examples are discussed below and the full list of sources is provided in the Appendix.

(1) *List of languages and language families included in the survey*:

[1] Indo-European:

- 1 French
- 2 Spanish
- 3 Catalan
- 4 Italian
- 5 Romanian
- 6 German
- 7 Swiss German
- 8 Zürich German
- 9 Dutch
- 10 Swedish
- 11 English

1. This is an expansion of the two previous largest samples, from Albizu 1997b and Haspelmath 2004a. It includes almost all of the 64 languages discussed in these two studies, which were re-examined for the current study, occasionally identifying some overlooked person restrictions. At the time of writing, I have yet to examine the relevant data from Kera [*Chadic*], Noon [*Cangin*], Kabardian [*Northwest Caucasian*], and Lakhota [*Siouan*] (the last three being languages Haspelmath reports as having no person restrictions).

- 12 Icelandic
- 13 Faroese
- 14 Slovenian
- 15 Bosnian-Croatian-Serbian
- 16 Czech
- 17 Polish
- 18 Bulgarian
- 19 Macedonian
- 20 Greek
- 21 Albanian
- 22 Kurdish
- 23 Pashto
- 24 Iron Ossetic
- 25 Digor Ossetic
- 26 Kashmiri

[2] Uralic:

- 27 Finnish
- 28 Hungarian
- 29 Eastern Mansi
- 30 Khanty/Ostyak
- 31 Tundra Nenets

[3] Afro-Asiatic:

- 32 Modern Standard Arabic
- 33 Classical Arabic
- 34 Cairene Arabic
- 35 Maltese
- 36 Senaya
- 37 Christian Barwar
- 38 Telkepe
- 39 Migama
- 40 Baraïn

[4] Nilo-Saharan:

- 41 Maasai/Maa

[5] Niger-Congo:

- 42 Sambaa
- 43 Haya
- 44 Swahili
- 45 Nyaturu/(Ki)Rimi
- 46 Limbum

[6] Kartvelian:

- 47 Georgian

[7] Northwest Caucasian:

- 48 Abhkaz

[8] Sino-Tibetan:

- 49 Hakha Chin
- 50 Chepang
- 51 Jyarong
- 52 Nocte
- 53 Tangut

[9] Austronesian:

- 54 Kambera
- 55 Manam
- 56 Tagalog

[10] Sepik-Ramu:

- 57 Yimas
- 58 Manambu

[11] Toricelli/Monumbo:

- 59 Monumbo

[12] Pama-Nyungan:

- 60 Djaru
- 61 Warlpiri

[13] Chukotko-Kamchatkan:

- 62 Chukchi
- 63 Koryak
- 64 Alutor
- 65 Itelmen

[14] Penutian:

66 Sahaptin
67 Takelma

[15] Algic:

68 Algonquin
69 Arapaho
70 Blackfoot
71 Plains Cree
72 Delaware
73 Meskwaki/Fox
74 Mi'kmaq
75 Maniwaki
76 Ojibwe
77 Passamaquoddy
78 Potawatomi

[16] Kiowa-Tanoan:

79 Southern Tiwa
80 Picurís
81 Arizona Tewa
82 Kiowa

[17] Iroquoian:

83 Mohawk
84 Cherokee

[18] Uto-Aztecan:

85 Tetelcingo Nahuatl
86 Classical Nahuatl
87 O'odham

[19] Mayan:

88 Tzotzil
89 Kaqchikel

[20] Oto-Manguean:

90 Oaxaca Zapotec

[21] Quechuan:

91 Quechua

[22] Salish:

92 Bella Coola
93 Clallam
94 Lummi
95 Halkomelem
96 Squamish
97 Lushootseed

[23] Dené-Yeniseian:

98 Koyukon
99 Navajo

[24] Eskimo-Aleut:

100 Labrador Inuttut
101 Inuktitut

[25] Araucanian:

102 Mapudungun

[26] Tupian:

103 Paraguayan Guaraní

isolates:

104 Basque
105 Zuni
106 Kutenai

Note that the survey includes languages from many distinct language families and with sometimes very different typological profiles. This not only ensures that we can identify commonalities across distinct language families, the linguistic diversity also allows us to examine person restrictions that appear to be quite distinct from each other on the surface; for example, we will be looking at: the classic Person-Case Constraint (PCC) between object pronouns, parallel restrictions between subject and object pronouns, person-based restrictions on pronoun order and argument indexing, and direct/inverse systems (we saw some illustrations of these phenomena in Chapter 1). Importantly, many of these were not previously considered from this perspective—that is, as being in any way related to the PCC. However, we will see below that grouping them together is not arbitrary, and that all these phenomena in fact pattern in the same way in terms of what governs their occurrence and crosslinguistic variation. Another important feature of the language sample is that multiple languages from the same family are included whenever possible, allowing for the study of minimal pairs consisting of closely related languages. Finally, the survey also includes languages without person restrictions. Comparisons between languages where one of them shows a person restriction and the other one does not are—especially in the case of comparisons between closely related languages—a particularly useful tool in seeking to identify what the source of person restrictions is.

Although the current study looks at more languages than a typical generative study, the basic methodology is still the same. The goal is to not just count the instances of particular grammatical traits or features (which would not be useful anyway, as the sample is not large enough to make meaningful statistical generalizations), but to do an in-depth formal study of person restrictions, using the data from the surveyed languages as the empirical base. In the course of this study, we will often zoom in on a few genetically and areally unrelated languages from the sample, examining not only the person restriction but also other grammatical factors that might help identify possible correlations. This approach will allow us to examine these languages in enough detail to make generalizations that go beyond mere descriptions of superficial traits (see Baker and McCloskey 2007; Baker 2010 on this strategy for doing typology, which they dub the “Middle way”). At the same time, we will also often compare closely related languages that differ in minimal ways. This micro-comparative approach is the best approximation we have in typological work to a controlled experiment (see Kayne 2005b, 8): it is obviously impossible to manipulate a language to see if changing one of its traits or features causes changes in any other area of the language, but by examining languages that differ from one another in very limited ways, we can hope to come very close to the desired effect. Since studies that look at both macro and micro-variation in this way are not that common in generative syntax, the discussion below can—in addition to being a study of person restrictions—also be taken as a working demonstration of the hybrid macro/micro-comparative approach.

2.2 The syntactic character of person restrictions

Many early generative treatments of person restrictions (e.g. Perlmutter 1971; Bonet 1991, 1994) characterize them either as a surface constraint on morpheme sequences or a morphological constraint on feature combinations. More recently though, there has been a shift towards syntactic treatments (Albizu 1997a; Anagnostopoulou 2003, 2005; Béjar and Řezáč 2003; i.a.). In this section, I offer a new argument for a syntactic approach, based on an overlooked person restriction pattern.

Let us start by considering the classic case of a person restriction: the Person-Case Constraint (PCC), which is described again in (2) and illustrated by the Greek sentences in (3)–(4). In Greek double object ditransitives, both objects can be expressed as clitic pronouns, where the IO clitic bears genitive case and the DO clitic bears accusative case. While the IO clitic can be either 3P, as in (3a), or 1/2P, as in (3b), the DO clitic can only be 3P in the context of the IO clitic. That means that all double object clitic combinations with a 1P or 2P DO are ungrammatical, as the examples in (4) illustrate.

(2) *PCC (STRONG version)*
If IO and DO pronouns co-occur, the DO must be 3P.

(3) Greek [*Hellenic*]:

a. **Tu** **tin** sistisa — $\boxed{\text{3.IO + 3.DO}}$
3.M.GEN 3.F.ACC introduced.1
'I introduced **her** to **him**.' (Anagnostopoulou 2003, 200)

b. Tha { **mu** / **su** } **ton** stilune — $\boxed{\text{1/2.IO + 3.DO}}$
FUT { 1.GEN / 2.GEN } 3.M.ACC send.3PL
'They will send **him** to **me/you**.'

(4) a. *Tha **mu** **se** stilune — $\boxed{✗\ \text{1.IO + 2.DO}}$
FUT 1.GEN 2.ACC send.3PL
'They will send **you** to **me**.'

b. *Tha **su** **me** sistisune — $\boxed{✗\ \text{2.IO + 1.DO}}$
FUT 2.GEN 1.ACC introduce.3PL
'They will introduce **me** to **you**.'

c. *Tha **tu** { **me** / **se** } stilune — $\boxed{✗\ \text{3.IO + 1/2.DO}}$
FUT 3.M.GEN { 1.ACC / 2.ACC } send.3PL
'They will send **me/you** to **him**.' (Anagnostopoulou 2003, 252; 2005, 202)

Consider now the potential ways of capturing the person restriction. For example, we can take the key factor distinguishing the restricted clitic from the unrestricted one to be: (i) the difference in case morphology (genitive case vs. accusative case), (ii) the difference in thematic roles (Goal vs. Theme), or (iii) the difference in their syntactic position; e.g. one clitic being placed lower in the syntactic structure of the clause than the other one.[2] In the first two options, the relevant factors concern grammatical properties inherent to each clitic (i.e. factors which are independent of their structural position). In the last option, on the other hand, those kind of factors do not play a direct role, since the person restriction is captured in purely positional terms (inherent properties of the pronouns can only have an indirect role if they determine or constrain the syntactic position of the pronouns).

All three types of approaches have in fact been proposed. Bonet (1991, 1994) offers an analysis of the PCC in terms of a morphological constraint on feature combinations between

2. The one option I do not consider here is an analysis in terms of surface order templates (see Perlmutter 1968, 1970, 1971). I discuss the issues that arise with that type of an analysis in detail in Section 2.5.3.

co-occurring clitics: a 1/2P clitic pronoun (i.e. one with a [+participant] feature) is banned in clitic clusters where another clitic pronoun bears an *oblique case*:[3]

(5) *Morphological constraint:*
*[+participant]$_{\text{CL}}$ / [[OBLIQUE]$_{\text{CL}}$ ___]

Another approach attributes restrictions like the PCC to interactions between extrinsic scales that rank grammatical attributes (Rosen 1990; Aissen 1999b; Haspelmath 2004a). This usually involves a thematic scale, where Goal arguments outrank Theme arguments (Goal > Theme), and a person scale, where 1/2P outranks 3P (1/2 > 3). The idea is that when arguments associate with their respective values on both scales, certain alignments are not tolerated; e.g. in a ditransitive, the Theme argument, which is low on the thematic scale, may not associate with 1/2P, which is high on the person scale, as illustrated in (6).

(6) *Scale misalignment:*

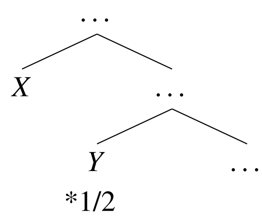
Goal	>	Theme
	✗	
1/2	>	3

Lastly, there are syntactic approaches to the PCC , which can differ greatly in terms of the specific mechanism that drives the person restriction, but the basic intuition behind all of them is that the context for the restriction should be stated in terms of differences in the syntactic position of the two pronouns. Specifically, the restriction applies to the structurally lower pronoun of the two: given co-occurring pronouns X and Y, where X asymmetrically c-commands Y, Y cannot be 1/2P. This is illustrated in (7).[4]

3. In contrast to the traditional use of the term *oblique case*, which groups together all cases excluding nominative case, Bonet (1991, 58) uses the term to group together all cases excluding nominative and accusative as *structural cases*, which means: genitive, dative, locative, partitive, and ablative, among others.

4. I assume (here and throughout) a fairly standard version of c-command, where Z c-commands Y iff Z does not dominate Y and Y does not dominate Z, and every category that dominates Z also dominates Y. Z then *asymmetrically* c-commands Y iff Z c-commands Y and Y does not c-command Z.

(7) *Syntactic approach:*



With respect to the canonical PCC pattern we saw in (3), all three approaches can capture why the person restriction applies to the DO. In the case of the morphological approach, this is so because the DO clitic occurs in the context of an oblique (e.g. dative or genitive) IO clitic. In the case of the thematic/person scale approach, this is so because the DO realizes a Theme argument in the context of a Goal argument. Finally, in the case of the syntactic approach, the IO clitic preceding the DO clitic can be taken to directly reflect their underlying syntactic positions: IO precedes DO because IO asymmetrically c-commands DO in the syntax—along the lines of Kayne's (1994) *Linear Correspondence Axiom (LCA)*, where asymmetrical c-command between two elements is directly mapped to linear precedence.

However, we will see in the next section that these different approaches can be teased apart when we look beyond the canonical PCC pattern. In particular, we will see that stating the restriction context in terms of morphological case or thematic roles will not work in the case of the person restriction attested in Slovenian (as well as some other languages from the survey, discussed in Section 2.5.3), which is sensitive to the position of of the two object pronouns, but not their case of thematic role.

2.2.1 A purely positional person restriction

In Slovenian double object ditransitives, both objects can be clitic pronouns just like in Greek. Clitic pronouns are second position clitics in Slovenian, which means they normally cluster in the second clausal position, and IO and DO clitics respectively bear dative and accusative case in standard double object ditransitives. As illustrated in examples in (8)–(9), when an IO clitic occurs before a DO clitic we get a person restriction pattern exactly like the one we saw in Greek: the IO clitic can be either 3P, as in (8a), or 1/2P, as in (8b), but the DO clitic can only be 3P, as shown by the ungrammaticality of the examples in (9).[5]

(8) Slovenian [*Slavic*]:

a. Mama **mu** **ga** bo predstavila. $\boxed{\text{3.IO + 3.DO}}$
mom 3.M.DAT 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him**$_i$ to **him**$_k$.'

b. Mama { **mi** / **ti** } **ga** bo predstavila. $\boxed{\text{1/2.IO + 3.DO}}$
mom { 1.DAT / 2.DAT } 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him** to **me/you**.'

(9) a. *Mama **mi** **te** bo predstavila. $\boxed{\text{✗ 1.IO + 2.DO}}$
mom 1.DAT 2.ACC FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

b. *Mama **ti** **me** bo predstavila. $\boxed{\text{✗ 2.IO + 1.DO}}$
mom 2.DAT 1.ACC FUT.3 introduce.F
'Mom will introduce **me** to **you**.'

c. *Mama **mu** { **me** / **te** } bo predstavila. $\boxed{\text{✗ 3.IO + 1/2.DO}}$
mom 3.M.DAT { 1.ACC / 2.ACC } FUT.3 introduce.F
'Mom will introduce **me/you** to **him**.' (Stegovec 2015b, 108–9)

What we see in (8)–(9) is a canonical PCC pattern, just like the one in Greek. But Slovenian in addition exhibits another kind of pattern. As seen in (10)–(11), IO and DO clitics may also appear in the reverse order from the one we saw in (8)–(9). Crucially, when the DO clitic precedes the IO clitic, the person restriction pattern also changes: it is now the DO clitic that can be either 3P, as in (10a), or 1/2P, as in (10b), whereas the IO clitic can only be 3P, as illustrated by ungrammaticality of the examples in (11).

(10) a. Mama **ga** **mu** bo predstavila. $\boxed{\text{3.DO + 3.IO}}$
mom 3.M.ACC 3.M.DAT FUT.3 introduce.F
'Mom will introduce **him**$_i$ to **him**$_k$.'

5. Just as in Spanish, discussed in Section 1.2 above, there is speaker variation in Slovenian regarding whether pairs of two 1/2P object clitics are grammatical, with the rest of the restriction pattern staying the same; this is the STRONG vs. WEAK PCC distinction. The variation affects the restriction seen in (8)–(9) as well as the one we will see in (10)–(11) below, which will be an important point later.

b. Mama { **me** / **te** } **mu** bo predstavila. [1/2.DO + 3.IO]
mom { 1.ACC / 2.ACC } 3.M.DAT FUT.3 introduce.F
'Mom will introduce **me/you** to **him**.'

(11) a. *Mama **me** **ti** bo predstavila. [✗1.DO + 2.IO]
mom 1.ACC 2.DAT FUT.3 introduce.F
'Mom will introduce **me** to **you**.'

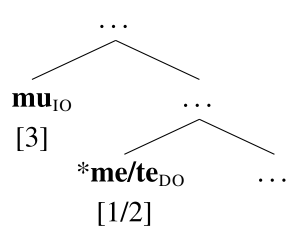
b. *Mama **te** **mi** bo predstavila. [✗2.DO + 1.IO]
mom 2.ACC 1.DAT FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

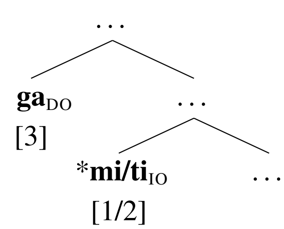
c. *Mama **ga** { **mi** / **ti** } bo predstavila. [✗3.DO + 1/2.IO]
mom 3.M.ACC { 1.DAT / 2.DAT } FUT.3 introduce.F
'Mom will introduce **him** to **me/you**.' (Stegovec 2015b, 108–9)

In terms of which of the two objects (IO or DO) is restricted to a 3P value, the pattern illustrated in (10)–(11) is a mirror image of the one we saw in (8)–(9) and in Greek above. The IO and DO person combinations that are ungrammatical when the IO clitic precedes the DO clitic, like the combination of a 3P IO and a 1/2P DO, are grammatical with the reversed clitic order. Likewise, clitic combinations that are grammatical in (8), like the combination of a 1/2P IO and a 3P DO, are ungrammatical with the reversed clitic order. Because of this restriction reversal, I will refer to the pattern in (10)–(11) as the *reverse PCC*, to distinguish it from the *canonical PCC* pattern in (8)–(9).

Consider then what the reverse PCC means for the morphological and thematic scale analyses. Recall that it is crucial for these analyses that there is an asymmetry regarding the grammatical properties of the pronouns involved in the restriction. With the morphological approach, the oblique case clitic is what causes the restriction on the other clitic, while with the thematic scale approach, the restriction crucially targets the Theme clitic due to it being lower on the thematic scale than the Goal. None of this works with the reverse PCC. It is the clitic with accusative (structural) case that is the context for the person restriction which applies to the clitic with dative (oblique) case. Similarly, the restricted clitic is a Goal, which is ranked higher on the thematic scale than the Theme.

The problem does not arise in the syntactic approach, based purely on differences in the syntactic placement of objects. As long as the clitic that is lower in the structure is the one being restricted, we do not have to concern ourselves with the inherent grammatical properties associated with each of the two clitics. The canonical PCC, where the DO must be 3P, arises when the IO clitic precedes the DO clitic because the former asymmetrically c-command the latter, as illustrated in (12) (corresponding to (9c)). But when the clitic order is reversed, the DO clitic asymmetrically c-commands the IO clitic, as illustrated in (13) (corresponding to (11c)), resulting in a reverse PCC pattern where the IO must be 3P.

(12)


(13)


The syntactic approach thus has a clear advantage in that it is the only one that can capture both the canonical and the reverse PCC pattern. Crucially, this complete independence from case and thematic roles can only be established based on cases like Slovenian, where the order of the clitic pronouns is freer. In most other languages, which were historically the basis for examining PCC effects, the clitic order is much more rigid and the mapping from case or thematic roles to the clitic order for a given clitic pair is pretty much one-to-one, obscuring the role of syntax in the person restriction.

One could raise an objection here that the person restriction in Slovenian simply has a different grammatical source and the similarity to the canonical PCC is only coincidental. To make such an argument we would have to ignore how closely the two patterns mirror each other. But there is an additional, much stronger, argument to be made for a unified analysis of the two, which is the fact that the canonical and reverse restrictions pattern exactly the same when it comes to the crosslinguistic variation in the banned person combinations.

As we saw in Chapter 1, the PCC varies crosslinguistically in terms of which combinations of person values may co-occur on the affected pronouns, where the main distinction is between STRONG VS. WEAK restrictions; using the terminology popularized by Bonet (1991, 1994) and Anagnostopoulou (2005). What we saw above in Greek was the STRONG version of the PCC, where the DO pronoun must always be 3P when the IO pronoun is present. The WEAK version of the PCC, as the name suggests, is a comparatively weaker restriction, where the the DO does not necessarily have to be 3P when the IO is also present—the DO only has to be 3P when the IO is also 3P. The descriptive generalization for this version of the PCC is given in (14). We can see a WEAK version of the PCC in the Classical Arabic examples in (15)–(16).[6] Notice that unlike with the STRONG restriction in Greek, combinations of two 1/2P clitics are allowed in Classical Arabic, as (15d) shows, whereas a 1/2P DO clitic is still ungrammatical in the context of a 3P IO clitic, as in (16).

(14) *PCC (WEAK version)*
If IO and DO pronouns co-occur, and the IO is 3P, the DO must be 3P.

(15) Classical Arabic [*Semitic*]:[7]

a. ʔaʕtˀaː **-haː -hʊ** — 3.IO + 3.DO
gave.3 -3.F.IO -3.M.DO
'He gave **him/it** to **her**.'

b. ʔaʕtˀaː **-niː -hɪ** — 1.IO + 3.DO
gave.3 -1.IO -3.M.DO
'He gave **him/it** to **me**.'

c. ʔaʕtˀaː **-ka -hu** — 2.IO + 3.DO
gave.3 -2.IO -3.M.DO
'He gave **him/it** to **you**.'

d. ʔaʕtˀaː **-niː -ka** — 1.IO + 2.DO
gave.3 -1.IO -2.DO
'He gave **you** to **me**.'

6. There is an additional restriction in Classical Arabic banning combinations of 2P IO and 1P DO clitics. I set aside this restriction until Section 4.3, where it will be discussed in detail (see also footnote 17).

7. Walkow's (2014) paradigm is based on Sībawayh's grammar from the late 8th century (1881 critical edition), the seminal book on Classical Arabic grammar.

(16) *ʔaʕtˁaː **-huː** { **-niː** / **-ka** } ✗ 3.IO + 1/2.DO
gave.3 -3.M.IO { -1.DO / -2.DO }
'He gave **me/you** to **him**.' (Walkow 2014, 139–40)

Variation between STRONG and WEAK versions of the PCC is very common even within the same language or dialect. Vast speaker variation is famously found in this domain in Spanish (Perlmutter 1971; Ormazabal and Romero 2007), Italian (Seuren 1976; Cardinaletti 2008), and Catalan (Bonet 1991; Walkow 2012). Crucially, the same variation is also found in Slovenian. What I presented above was the STRONG version of the restriction, but some speakers (*Slovenian B*) have the WEAK version of the same restriction. As can be seen in (17), when the IO clitic precedes the DO clitic in Slovenian B, the banned person combinations match those we have seen above in Classical Arabic. The only ungrammatical combinations are those where a 3P IO precedes a 1/2P DO, as in (18).

(17) Slovenian B [*Slavic*]:

a. Mama **mu** **ga** bo predstavila. 3.IO + 3.DO
mom 3.M.DAT 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him**$_i$ to **him**$_k$.'

b. Mama { **mi** / **ti** } **ga** bo predstavila. 1/2.IO + 3.DO
mom { 1.DAT / 2.DAT } 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him** to **me/you**.'

c. Mama **mi** **te** bo predstavila. 1.IO + 2.DO
mom 1.DAT 2.ACC FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

(18) *Mama **mu** { **me** / **te** } bo predstavila. ✗ 3.IO + 1/2.DO
mom 3.M.DAT { 1.ACC / 2.ACC } FUT.3 introduce.F
'Mom will introduce **me/you** to **him**.' (Stegovec 2015b, 108–9)

Most importantly, with the reversed order of object clitics, where the DO clitic precedes the IO clitic, the resulting reverse PCC pattern is also WEAK for Slovenian B speakers, as shown in (19)–(20). The roles of IO and DO are reversed, and the only ungrammatical combinations are those with a 3P.DO and a 1/2P.IO, as shown in (20).[8]

(19) a. Mama **ga** **mu** bo predstavila. $\boxed{\text{3.DO + 3.IO}}$
mom 3.M.ACC 3.M.DAT FUT.3 introduce.F
'Mom will introduce **him**$_i$ to **him**$_k$.'

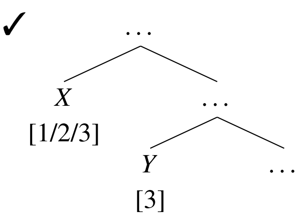
b. Mama { **me** / **te** } **mu** bo predstavila. $\boxed{\text{1/2.DO + 3.IO}}$
mom { 1.ACC / 2.ACC } 3.M.DAT FUT.3 introduce.F
'Mom will introduce **me/you** to **him**.'

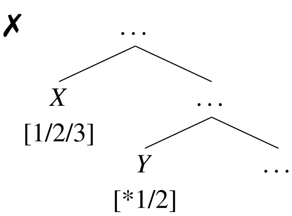
c. Mama **te** **mi** bo predstavila. $\boxed{\text{2.DO + 1.IO}}$
mom 2.ACC 1.DAT FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

(20) *Mama **ga** { **mi** / **ti** } bo predstavila. $\boxed{\text{✗ 3.DO + 1/2.IO}}$
mom 3.M.DAT { 1.ACC / 2.ACC } FUT.3 introduce.F
'Mom will introduce **him** to **me/you**.' (Stegovec 2015b, 108–9)

If the canonical PCC we observe in Greek and Classical Arabic and the combined canonical + reverse PCC we observe in Slovenian are the same grammatical phenomenon underlyingly, we can easily attribute the variation to the same STRONG/WEAK parameter. It is fairly straightforward to restate the STRONG and WEAK versions of both PCC patterns in structural terms. If, as suggested above, the order of the clitics is a direct reflection of the underlying syntax—namely, the first clitic asymmetrically c-commands the second one—all STRONG restriction patterns conform to (21), and all WEAK restriction patterns conform to (22).

(21) *STRONG person restriction*
If X asymmetrically c-commands Y, then Y must be 3P.

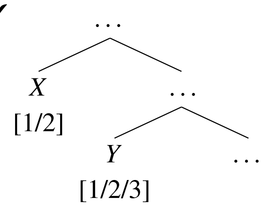
a. ✓


b. ✗


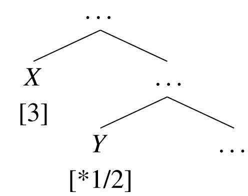
8. There is an additional restriction in Slovenian B with respect to combinations of 1P and 2P clitics. However, there is good reason to believe that this additional restriction is not syntactic, as it is sensitive to semantic perspective shifts like *interrogative flip* (see Stegovec 2019 as well as Section 5.4 for discussion).

(22) *WEAK person restriction*
If X asymmetrically c-commands Y, and X is 3P, then Y must be 3P.

a. ✓



b. ✗



If the person restrictions in Greek or Classical Arabic and Slovenian were not the same phenomenon underneath, then the the fact that the restriction patterns are virtually identical and even behave the same in terms of the STRONG vs. WEAK variation would have to be attributed to pure coincidence. As we will see in Sections 2.3 and 2.5, where the empirical domain is extended beyond the basic cases of person restrictions discussed here, the parallel is not limited to the canonical and reverse PCC patterns. This makes it even harder to dismiss the parallels as coincidental, strongly suggesting that we are dealing with a striking crosslinguistic constant in need of an explanation.

Of course, it is always a possibility that we are dealing with a coincidental parallelism. It would therefore be good to have additional evidence internal to the prototypical PCC cases (i.e. those without a canonical/reverse alternation) showing that these are syntactic in nature. Such evidence indeed exists and I will review some of it next.

2.2.2 More arguments for the syntactic nature of the PCC

A nice piece of evidence for the syntactic character of the PCC comes from Basque, where co-occurring dative and absolutive clitics show a STRONG person restriction in double object ditransitives and Agent-less transitives like the experiencer dative construction in (23a) (Bonet 1991, 1994; Albizu 1997a). Contrast the former with (23b), a motion verb construction with a dative goal; (23a) and (23b) crucially have an identical configuration of pronominal clitics, but only the experiencer dative construction shows the person restriction.

(23) Basque:

a. **Ni* *Miren-i* gusta-tzen **na-** tzai-Ø **-o**. ✗3.DAT ≫ 1.ABS
me.ABS Miren-DAT like-IMPF 1.ABS- AUX-ABS -3.DAT
'**He** [Miren] likes **me**.'

b. *Ni* *Peru-ri* hurbildu **na-** tzai-Ø **-o**. 1.ABS ≫ 3.DAT
me.ABS Peru-DAT approach 1.ABS- AUX-ABS -3.DAT
'**I** approached **it** [Peru].' (Albizu 1997a, 9–10)

Řezáč (2008b, 70–9) shows through a number of syntactic tests that the two constructions, although showing identical morphology on the auxiliary, differ in their underlying syntax: in (23a) the dative argument is base generated in a position asymmetrically c-commanding the absolutive argument (indicated as '≫'), while in (23b) the absolutive argument is the one asymmetrically c-commanding the dative argument in their base positions. It is thus the underlying syntactic structure that determines whether or not the absolutive clitic is person restricted—the restriction cannot be solely tied to the case the two clitics express.

Similar evidence is also found across Romance with contrasts between selected clitics and non-selected clitics like ethical datives (Perlmutter 1971; Bonet 1991, 1994; Albizu 1997a). For example, Catalan disallows co-occurring 1P DO and 3P IO clitics; see (24a).[9] However, the same surface clitic cluster is permitted when the 1P clitic in such a clitic combination is construed as an ethical dative, as in (24b).

(24) Catalan [*Romance*]:

a. **A en Josep*, **me** **li** va recomanar *la Mireia*. ✗3.IO ≫ 1.DO
to the Josep 1.DO 3.DAT recommended.3 the Mireia
'She [Mireia] recommended **me** to **him** [Josep].'

b. **Me** **li** van dir que havia suspès l'examen. 1.DAT ≫ 3.IO
1.DAT 3.DAT AUX.3PL say.INF that had.3 failed the.exam
'They told **him** (on **me**) that he had failed the exam.' (Bonet 1991, 178–9)

9. Note that, as discussed by Bonet (1991, 1995), the underlying asymmetric c-command relation between the objects (here: IO ≫ DO) is not reflected in the clitic order in Catalan. Bonet argues that the surface clitic order in Catalan is derived from the underlying syntax via a low level PF reordering and can be very idiosyncratic based on the clitics involved (see also footnote 10 on French).

Ethical datives (and other non-selected datives) consistently show syntactic behavior distinct from argument clitics. Specifically, they show behavior which indicates that such clitics are introduced in the structure outside the domain where argument clitics are introduced (see, among others, Rouveret and Vergnaud 1980; Authier and Reed 1992; Řezáč and Jouitteau 2008 on French; Borer and Grodzinsky 1986 on Hebrew; and Bošković 2001, 61–2 on Bosnian-Croatian-Serbian, where it is specifically shown that ethical dative clitics are introduced higher than argument dative clitics). Just as in the Basque case, we see here that syntax is what matters for the restriction, not surface considerations.

The final example of this kind comes from French (Postal 1990), where speakers who permit clitic climbing in contexts like (25b) differentiate between identical clitic clusters in a regular double object ditransitive, where a DO clitic is person restricted in the presence of a dative IO clitic, as in (25a),[10] and those in (25b) where the lower dative climbs into the matrix clause and the DO is not restricted.

(25) French [*Romance*]:

a. *Marcel **vous** **lui** recommandera. — $\boxed{\text{✗ 3.IO} \gg \text{2.DO}}$
   Lucille 2PL.DO 3.DAT recommend.FUT.3
   'Marcel will recommend **you** to **him/her**.' (Postal 1990, 104)

b. Je **vous**$_i$ **lui**$_j$ crois t_i [infidèle t_j]. — $\boxed{\text{2.DO} \gg \text{3.DAT}}$
   I 2PL.DO 3.DAT believe unfaithful
   'I believe **you** to be unfaithful to **him/her**.' (Postal 1990, 177)

The person restriction is observed here only when the DO clitic starts structurally lower than the dative clitic—the surface clitic sequence is irrelevant, consistent with the above cases.

10. Like in Catalan, it is standardly assumed that the surface clitic order does not reflect the underlying syntactic positions of the object clitics. In fact, the clitic surface order is rather idiosyncratic in French; e.g. with pairs of 3P object clitics DO precedes IO, but when IO is 1/2P, it must precede DO (Kayne 1975; Nicol 2005). Nonetheless, there is syntactic evidence for a uniform IO $\gg$ DO syntactic base. For example, floating quantifiers floated by an IO clitic have to precede those floated by a DO clitic even when the quantifiers follow participles—even when the surface clitic order is reversed (Kayne 1975; Cinque 1999; Bošković 2004a). This indicates that the IO $\gg$ DO configuration is the only option below the lowest derived position of the verb. The surface clitic order must therefore be established later, either via syntactic movement of the DO clitic after the point in the derivation where person restrictions occur or via a low level PF reordering, which is the usual assumption in the literature (Bonet 1991, 1995; Béjar and Řezáč 2003; Řezáč 2008b).

A possible objection to the syntactic take on the examples in (23)–(25) could be that the contrasted surface-identical clitic clusters might still have different sets of underlying morphosyntactic features, making the clitic clusters only phonologically, but not morphologically, equivalent. In fact, this is what Bonet (1991, 1994) proposes in response to the Catalan examples in (24). In order to exclude clitic pairs with ethical datives, Bonet modifies her morphological constraint (cf. (5) above) to target only clitics with the feature [ARGUMENT]; i.e. a clitic with the specification [ARGUMENT; +participant] cannot co-occur with a clitic with the specification [ARGUMENT; OBLIQUE].

Introducing the [ARGUMENT] feature will, however, not work for the Basque and French cases, where the unrestricted clitic pairs involve two argument clitics just like their restricted counterparts (see also Albizu 1997a; Řezáč 2008b). In order to maintain a morphological analysis for all the relevant cases, one would have to somehow exclude the unlikely natural class of absolutive subject clitics (cf. (23)), ethical dative clitics (cf. (24)), and climbed dative clitics (cf. (25)) in terms of features—which cannot be features encoding (non-)argument status or case features. Such a characterization would be really strange. In fact, as far as I am aware, there is no independent motivation for using such a feature-based characterization. In contrast, as we saw above, the syntactic characterization is corroborated by independent syntactic evidence in all three cases.

Furthermore, when we look beyond the classic cases of the PCC, like those in Romance or Greek, where an oblique case clitic co-occurs with structural case clitic, there is really no crosslinguistic consistency in terms of the case or even thematic roles of the affected pronouns. For example, many Bantu languages show PCC effects with object markers even though IO and DO markers and independent (pro)nouns are consistently not distinguished in terms of case (see Ormazabal and Romero 2007; Riedel 2009; I will discuss some Bantu examples in Sections 2.4.3 and 2.5.4). At the other extreme we have cases where the PCC occurs across different constructions with considerably different case combinations on the pronouns. In Warlpiri [*Pama-Nyungan*], ditransitive constructions show the STRONG person restriction with both dative-absolutive and dative-dative object pairs (Hale 1973b, 1983; Simpson 1991). The pattern with verbs that require dative-absolutive marking on the objects is shown in (26), where we can see that the DO clitic can never be 1/2P in the presence of an IO clitic (note that only 3P singular clitics overtly express dative case in Warlpiri, which is similar to Catalan and French, where dative forms exist only for 3P clitics).

(26) Warlpiri [*Pama-Nyungan*]:

a.

Punta-rni	kapirna	**-ngku**	**-Ø**.	2.IO ≫ 3.DO
take.away-NPAST	FUT.1	-2.IO	-3.DO	

‘I will take **him/her/it** away from **you**.’ (Hale 1983, 19)

b.

*Ngarrka-ngku	kapi	**-ji**	**-rla**	punta-mi.	✗ 3.IO ≫ 1.DO
man-ERG	FUT.3	-1.DO	-3.DAT	take.away-NPAST	

‘The man will take **me** away from **him**.’ (Simpson 1991, 339)

c.

*Wati-ngki	ka	**-ju**	**-ngku**	punta-rni.	✗ 1/2.IO ≫ 2/1.DO
man-ERG	PRES.3	-1.IO/DO	-2.DO/IO	take.away-NPAST	

‘He is taking **you/me** away from **me/you**.’ (Simpson 1991, 149)

But Warlpiri also has verbs that assign dative case to the DO, such as *warri* ‘seek’ in (27a). When a dative IO argument is added to such verbs, as in (27b), both clitics can bear dative case (in sequences of two dative clitics, the second one realizes as *-jinta*; Hale 1973b, 336, this is similar to how 3P IO clitics in Spanish realize as *se*; Perlmutter 1968, 1970, 1971).

(27) a.

Ngarrka-ngku	ka	**-rla**	*karli-ki*	warri-rni.
Man-ERG	PRES.3	-3.DAT	boomerang-DAT	seek-NONPAST

‘A man is looking for **it** [a boomerang].’ (Simpson 1991, 326)

b.

Ngajulu-rlu	karna	**-rla**	**-jinta**	*karli-ki*	warri-rni	*ngarrka-ku*.
I-ERG	PRES.1	-3.DO.DAT	-3.IO.DAT	boomerang-DAT	seek-NPAST	man-DAT

‘I’m looking for **it** [a boomerang] for **him** [a man].’ (Hale 1973b, 336)

Crucially, while the IO clitic in such constructions can also be 1P or 2P, as in (28a), the DO clitic is subject to a STRONG person restriction and can only be 3P, as shown in (28b).[11]

11. While neither clitic in (28b) overtly expresses dative case marking—since Warlpiri 1/2P object clitics never do—we know because of (27b) that both IO and DO must underlyingly have case features; see also Bonet 1991 regarding morphologically unrealized oblique/dative case features in Catalan 1/2P clitics.

(28) a. Ngajulu-rlu karna **-ngku -rla** *karli-ki* warri-rni *nyuntu-ku*.
I-ERG PRES.1 -2.IO -3.DO.DAT boomerang-DAT seek-NPAST you-DAT
'I'm looking for **it** [a boomerang] for **you**.'
(Laughren and Hoogenraad 1996, 115)

b. *Ngarrka-ngku lpa **-ju -ngku** *nyuntu-ku* warru-rnu *ngaju-ku*.
man-ERG PAST.3 -1.IO -2.DO you-DAT seek-PAST me-DAT
'A man is looking for **you** for **me**.' (Hale 1973b, 335)

Ossetic [*Eastern Iranian*] has an even more elaborate pattern, where a dative pronominal clitic can co-occur with pronominal clitics that bear a number of different cases (allative, superessive, or accusative), and nonetheless person restrictions are constant throughout those combinations (Erschler 2014, 2015).[12] With the examples in (29) from the Iron dialect of Ossetic, we even see a person restriction arising in a construction where neither of the clitics is dative: the clitic pair in question involves a superessive (SUP) clitic and an ablative (ABL) clitic (a pair that in Iron Ossetic yields a STRONG restriction; see Erschler 2014, 2015).

(29) Iron Ossetic [*Eastern Iranian*]:

a. ɐmɐ=**myl**=**šɐ** Mɐdinɐ jettɐmɐ niči ɐwwɐndy. $\boxed{\text{1.SUP} \gg \text{3.ABL}}$
and=1.SUP=3PL.ABL Madina besides nobody believes
'No one of **them** believes **me**, but Madina.'

b. *ɐmɐ=**jyl**=**nɐ** Mɐdinɐ jettɐmɐ niči ɐwwɐndy. $\boxed{\text{✗ 3.SUP} \gg \text{1.ABL}}$
and=3.SUP=1PL.ABL Madina besides nobody believes
'No one of **us** believes **them**, but Madina.' (Erschler 2014, 6)

Another restricted clitic combination in Iron Ossetic where neither clitic is dative is that of an allative and ablative clitic. The ablative clitic is restricted in such pairs in exactly the same way as in superessive-ablative pairs (Erschler 2015, 8). We thus see here a double dissociation between case and person restrictions: person restrictions are observed with different cases of the restricted clitic and with different cases of the unrestricted clitic.

12. In fact, some combinations may yield a STRONG restriction while others yield a WEAK restriction (there is also considerable dialectal variation between the Iron and Digor varieties; see Erschler 2014, 2015 for details). Since the variation in case is tied to syntactic differences, with the pronouns presumably occupying different syntactic positions, the variation in restriction patterns can be taken as further support for a syntactic analysis; more specifically, the analysis introduced in Chapters 3 and 4, where different syntactic positions of pronouns is what yields the STRONG vs. WEAK variation.

In addition to resisting a characterization in terms of case, the Ossetic person restrictions are also problematic for a characterization in terms of thematic roles (crucial for the scale approach; cf. (6) above), since the clitic pairs do not all conform to the Goal-Theme pattern and the restricted clitic is not necessarily the Theme. In fact, the (pro)nouns not marked with dative or accusative are typically non-core arguments. For example, ablative can mark the cause/source/accompanying circumstances of an action. Finally, Erschler (2015, 8) reports that some speakers of both Iron and Digor dialects of Ossetic can void person restrictions by changing the order of the clitics—suggesting that the restriction is, just as we saw for Slovenian, sensitive to the position of the clitics and not their inherent properties.

The case and the thematic role status of the pronouns subject to the PCC is thus not consistent crosslinguistically—sometimes not even language internally. So what unifies all the relevant contexts? Given how diverse they are, seeking to unify them in terms of shared inherent grammatical properties of the pronoun pairs runs the risk of positing properties tantamount to restating the *syntactic contexts in which person restrictions are observed*. If what we want is a theory with explanatory power, we need something better. Fortunately, the syntactic interpretation of person restrictions is more promising in this regard: a characterization in terms of asymmetric c-command is not only consistent with all the cases seen above, there is generally evidence for it independent from the person restriction itself. What remains to be explained then is why the structural configuration of the pronouns should matter for which person value they can have.

At this point, we have sufficient motivation for seeking a unified analysis of the PCC in purely syntactic terms, one that would derive the person restriction in languages with only the canonical PCC as well as the person restriction in languages, like Slovenian, with both the canonical and the reverse PCC. This is, in fact, the approach I take in Stegovec 2019, where I offer such an analysis which unifies the two types of languages. But we can actually do better than that. There are many person sensitive phenomena that superficially appear quite different from the PCC, either in the identity of the affected arguments, the morphological correlates of the restriction, or other grammatical consequences. There is a tendency in many works on person restrictions to push these aside as either completely unrelated to the PCC or as related but subject to different parameters of variation. I want to argue that all these other phenomena are actually the same phenomenon as the PCC and should therefore be accounted by the same analysis as the PCC. So before we can provide a syntactic analysis of person restrictions, let us extend the empirical domain of this study by looking beyond the cases of person restrictions that are usually discussed.

2.3 Looking beyond the standard cases

The PCC, as we saw, affects combinations of internal argument pronouns, which typically function as the IO and DO in the clause. However, there also exist person restrictions where the restriction affects combinations of external argument and internal argument pronouns. The parallelisms between these two types of restrictions are often dismissed or overlooked in the literature. For instance, Řezáč (2011) suggests that the PCC only exists in the STRONG version, attributing the apparent WEAK version to speakers interpreting the IO pronoun as an ethical dative in combinations of two 1/2P objects, suspending the person restriction for those combinations (as we saw above, ethical datives can be exempt from person restrictions). Řezáč acknowledges the existence of WEAK person restrictions only for combinations of subject and object pronouns—which suggests that the person restrictions in the two domains are fundamentally different for Řezáč. In contrast, Baker (2008b, 2011) does treat person restrictions in both domains on a par, but he assumes only STRONG restrictions are person restrictions in the sense discussed here, suggesting that WEAK restrictions are an altogether different type of phenomenon.

The few notable exceptions that acknowledge both the STRONG vs. WEAK variation and assume a common source for restrictions between object pairs and subject-object pairs are Albizu (1997b), Nevins (2011), and Pancheva and Zubizarreta (2018) (the latter indirectly by extending their earlier analysis of subject-object restrictions from Zubizarretta and Pancheva 2017 to ditransitives). The current crosslinguistic survey, based on a much larger language sample, confirms the presence of the STRONG vs. WEAK distinction across both argument contexts and across a number of phenomena seemingly distinct from the PCC. This suggests that all the different types of person restrictions are underlyingly the same phenomenon.

From here on, I divide person restrictions broadly according to the argument pairs they affect: (i) person restrictions on external-internal argument combinations or *EA-IA restrictions* for short, and (ii) person restrictions on internal-internal argument combinations or *IA-IA restrictions* for short. The classification of person restrictions along the lines of internal and external argument pairs anticipates the discussion in Section 2.5.1, where I motivate this classification as opposed to the more standard one in terms of grammatical roles. In discussing IA-IA restrictions, I will be describing the two affected arguments consistently as Goal (abbreviated as GL) and Theme (abbreviated as TH), even though IA-IA restrictions extend beyond combinations of those internal arguments. For example, IA-IA restrictions include the person restriction we saw above in Basque between a dative Experiencer and absolutive Theme (Bonet 1991, 1994; Albizu 1997a; cf. (23)), which arise in transitive clauses without an external (Agent) argument. The characterization in terms of Goal and Theme is used for expository purposes; I will provide more precise characterizations whenever those are relevant. Similarly, I will not be distinguishing different internal arguments in EA-IA restrictions. Typically, such restrictions do not differentiate between different internal arguments, so I will only disambiguate them when relevant.

2.3.1 Person restrictions on external-internal argument pairs

Let us first remind ourselves of the STRONG vs. WEAK variation in the case of the canonical PCC. The STRONG version of the PCC, recast here as (30) in terms of thematic roles, is illustrated with the repeated Greek examples in (31)–(32), whereas the WEAK version of the PCC, recast here as (33) in terms of thematic roles, is illustrated with the repeated Classical Arabic examples in (34)–(35).

(30) *STRONG IA-IA person restriction (PCC)*
If Goal and Theme pronouns co-occur, the Theme must be 3P.

(31) Greek [*Hellenic*]: **STRONG IA-IA**

a. **Tu** **tin** sistisa — 3.GL ≫ 3.TH
3.M.GEN 3.F.ACC introduced.1
'I introduced **her** to **him**.' (Anagnostopoulou 2003, 200)

b. Tha { **mu** / **su** } **ton** stilune — 1/2.GL ≫ 3.TH
FUT { 1.GEN / 2.GEN } 3.M.ACC send.3PL
'They will send **him** to **me/you**.'

(32) a. *Tha **mu** **se** stilune — ✗ 1.GL ≫ 2.TH
FUT 1.GEN 2.ACC send.3PL
'They will send **you** to **me**.'

b. *Tha **su** **me** sistisune — ✗ 2.GL ≫ 1.TH
FUT 2.GEN 1.ACC introduce.3PL
'They will introduce **me** to **you**.'

c. *Tha **tu** { **me** / **se** } stilune — ✗ 3.GL ≫ 1/2.TH
FUT 3.M.GEN { 1.ACC / 2.ACC } send.3PL
'They will send **me/you** to **him**.' (Anagnostopoulou 2003, 252; 2005, 202)

(33) *WEAK IA-IA person restriction (PCC)*
If Goal and Theme pronouns co-occur, and the Goal is 3P, the Theme must be 3P.

(34) Classical Arabic [*Semitic*]: **WEAK IA-IA**

a. ʔaʕtˀaː **-haː** **-hʊ** — 3.GL ≫ 3.TH
gave.3 -3.F.IO -3.M.DO
'He gave **him/it** to **her**.'

b. ʔaʕtˀaː **-niː** **-hɪ** — 1.GL ≫ 3.TH
gave.3 -1.IO -3.M.DO
'He gave **him/it** to **me**.'

c. ʔaʕtˀaː **-ka** **-hu** — $\boxed{\text{2.GL} \gg \text{3.TH}}$
gave.3 -2.IO -3.M.DO
'He gave **him/it** to **you**.'

d. ʔaʕtˀaː **-niː** **-ka** — $\boxed{\text{1.GL} \gg \text{2.TH}}$
gave.3 -1.IO -2.DO
'He gave **you** to **me**.'

(35) *ʔaʕtˀaː **-huː** { **-niː** / **-ka** } — $\boxed{✗\,\text{3.GL} \gg \text{1/2.TH}}$
gave.3 -3.M.IO { -1.DO / -2.DO }
'He gave **me/you** to **him**.' (Walkow 2014, 139–40)

Recall that the difference between the two PCC versions is in the combinations of the two 1/2P pronouns, which the STRONG version of the restriction disallows and the WEAK version allows. Both versions, however, disallow a 3P Goal co-occurring with a 1/2P Theme.

Although there are a number of languages with a lot of fine speaker variation concerning the STRONG VS. WEAK distinction in the IA-IA domain (see Sections 1.2 and 2.2), there are also languages and language families where WEAK restrictions are more robustly attested. For example, both Classical Arabic (Walkow 2014) and Modern Standard Arabic (Fassi-Fehri 1988, 1993) are only reported to show WEAK IA-IA restrictions, Riedel (2009) notes that within the Bantu family WEAK IA-IA restrictions are predominant, and Anagnostopoulou (2008) notes that WEAK IA-IA restrictions are also prevalent in Germanic, where almost all the languages with a person restriction display this pattern. This in itself may argue against Řezáč's (2011) suggestion that such restrictions are merely an extra-grammatical effect.

Person restrictions in the EA-IA domain crucially also display the same STRONG VS. WEAK variation. In the case of EA-IA restrictions, the STRONG restriction takes the form described in (36). Note that the only difference in relation with the STRONG IA-IA restriction (cf. (30)) is that instead of the Theme pronoun being restricted in the context of the Goal pronoun, the IA pronoun is restricted in the context of an EA pronoun.

(36) *STRONG EA-IA person restriction*
If EA and IA pronouns co-occur, the IA pronoun must be 3P.

An example of this restriction is found in Arizona Tewa (Kroskrity 1977, 1985). Like other Kiowa-Tanoan languages, Arizona Tewa encodes arguments on the verb via a prefix pronoun marker—generally treated as a portmanteau morpheme that co-varies with the subject, the direct object, and (if present) the indirect object, but, as discussed by Adger and Harbour (2007) and Adger, Harbour, and Watkins (2009), the Kiowa-Tanoan prefix markers can be decomposed into individual pronoun markers upon careful morphological and phonological analysis. In Arizona Tewa, the prefixes are available in active transitive clauses only for cases where the internal argument is 3P, as in (37) (this holds for both objects in ditransitives). The pronoun markers are *pronominal arguments* in the sense of Jelinek (1984); they are the real arguments of the verb, as opposed to the NPs/DPs they co-vary with, and thus obligatory. So if the EA is encoded by the prefix, the IA cannot be 1/2P, as shown in (38).

(37) Arizona Tewa [*Kiowa-Tanoan*]: **STRONG EA-IA**
sen { **dó-** / **ná:-** / **mán-** } mun. 1/2/3.EA + 3.IA
man { 1>3- / 2>3- / 3>3- } see
'**I/You/He** saw **him** [the man].'

(38) a. **na:* *ų* **???-** mun. ✗ 1.EA + 2.IA
I you 1>2- see
'**I** saw **you**'

b. **ų* *na:* **???-** mun. ✗ 2.EA + 1.IA
you I 2>1- see
'**You** saw **me**.'

c. **sen* **???-** mun. ✗ 3.EA + 1/2.IA
man 3>1/2- see
'**He** [the man] saw **me/you**.' (Kroskrity 1977, 86, 169, 171)

The '???' represents the ungrammatical prefix markers, since the forms themselves are never attested.[13] The independent pronouns in (38a) and (38b) are only used to disambiguate the examples; as the prefixes are the real arguments, the independent pronouns are optional.

13. One could reconstruct them through comparative historical analysis and morphological decomposition of the grammatical prefixes, but the hypothetical form of the banned prefixes is really irrelevant for the task at hand, so I will use '???' for the ungrammatical prefixes in Kiowa-Tanoan and other similar cases.

It might seem odd to be unable to express sentences as basic as those given in the translations of (38), but the ungrammaticality here means only that the meaning of these sentences is impossible to express in Arizona Tewa with the particular grammatical means used in (38). Recall from Chapter 1 that in the case of the PCC, the impossible argument combinations can still be expressed as prepositional ditransitives. Likewise, the person combinations in (38) are only ungrammatical as active sentences in Arizona Tewa; they can be alternatively realized as passives where the independent pronoun/noun phrase associated with the EA argument, if present, bears oblique morphology, as shown in (39).

(39) a. *ų̨* na:n-di **wí-** tay. $\boxed{\text{2.IA.SU + 1.EA.OBL}}$
you we-OBL PASS(1>2)- know
'**You** are known (or recognized) by us.'

b. *na:* { sen-en-di / ų̨-di } **dí-** k^{w}ek^{hw}ę́di. $\boxed{\text{1.IA.SU + 3/2.EA.OBL}}$
I { man-PL-OBL / you-OBL } PASS(3/2>1)- shoot
'**I** was shot by the man/by you.' (Kroskrity 1985, 311)

In fact, outside the sentences where all arguments are 3P, active and passive sentences in Arizona Tewa are in complementary distribution with respect to the person values of the arguments (Kroskrity 1977, 1985): 1/2P EAs require active voice and 1/2P IAs require passive voice. As discussed by Perlmutter (1971), Postal (1990), Bonet (1991, 1994), and Řezáč (2011), a comparable complementary distribution can also be found with IA-IA restrictions. In the French examples from in Chapter 1, repeated here in (40), we see that the person restriction forces the use of a prepositional dative construction. However, it is also the case that the use of the prepositional dative construction is ungrammatical when the two objects are unstressed/unfocused, as shown in (41).

(40) a. *Lucille **te** **leur** présentera. $\boxed{\text{✗ 3.IO + 2.DO}}$
Lucille 2.DO 3PL.DAT introduce.FUT.3

b. Lucille **te** présentera à elles. $\boxed{\text{3.IO + 2.DO}}$
Lucille 2.DO introduce.FUT.3 to them
'Lucille will introduce **you** to **them**.'

(41) a. *Lucille **la** présentera à elles. ✗ 3.IO + 3.DO
Lucille 3.F.DO introduce.FUT.3 to them

b. Lucille **la** présentera { à ELLES / aux filles }. 3.IO + 3.DO
Lucille 3.F.DO introduce.FUT.3 { to them.FOC / to.the girls }
'Lucille will introduce **you** to THEM/the girls.' (Řezáč 2011, 93)

This means that outside of cases where one of the object is stressed/focused, the use of a prepositional dative is only licensed if using the same two pronouns in a double object ditransitive would yield an ungrammatical result due to the person restriction.

Let us turn now to WEAK EA-IA restrictions, which can be described as in (42). Again, note that the only difference with respect to the WEAK IA-IA restriction (cf. (33)) is in the thematic roles of the two pronouns (EA substitutes for the Goal, IA for the Theme).

(42) *WEAK EA-IA person restriction*
If EA and IA pronouns co-occur, and the EA is 3P, the IA pronoun must be 3P.

Without leaving Kiowa-Tanoan, we can find WEAK versions of the EA-IA restrictions we saw in Arizona Tewa, such as the restriction pattern in Southern Tiwa (Allen and Frantz 1978; Rosen 1990), which will be discussed in detail later. The same kind of pattern is also found among many Salish languages, including Lummi (Jelinek and Demers 1983), where in an active sentence, the IA cannot be 1/2P whenever the EA is 3P. This is illustrated by the Lummi examples in (43)–(44).

(43) Lummi [*Salish*]: **WEAK EA-IA:**

a. x̣či-t **-Ø** { **-sən** / **-sxʷ** / **-s** } 1/2/3.EA + 3.IA
know-TR -3.O { -1.SU / -2.SU / -3.SU.ERG }
'**I/you/he** knows **him**.'

b. x̣či-t **-oŋəs -sən** 1.EA + 2.IA
know-TR -1/2.O -1.SU
'**I** know **you**.'

c. x̣či-t **-oŋəs -sxʷ** 2.EA + 1.IA
know-TR -1/2.O -2.SU
'**You** know **me**.'

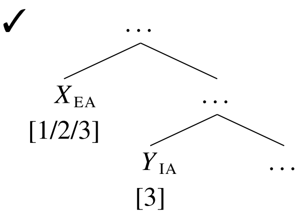
(44) *x̣či-t **-oŋəs -əs** ✗ 3.EA + 1/2.IA
know-TR -1/2.O -3.SU.ERG
'**He** knows **you/me**.' (Jelinek and Demers 1983, 168)

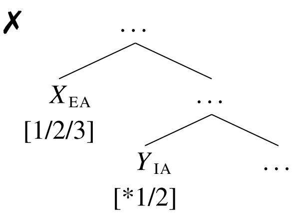
Just as in Arizona Tewa, the banned combinations can only be expressed with a passive, as shown in (45). In fact, the person value of the EA and IA is essentially what determines the voice of a clause in Lummi: if both EA and IA are 3P, then both voices are available, if the EA is 1/2P, then active voice is required, and if the IA is 1/2P, then passive voice is required.

(45) a. x̣či-t-ŋ **-sən** 1.IA.SU + 3.EA
know-TR-PASS -1.SU
'**I** am known (by him/someone).'
b. x̣či-t-ŋ **-sx^{w}** 2.IA.SU + 3.EA
know-TR-PASS -2.SU
'**You** are known (by him/someone).' (Jelinek and Demers 1983, 168)

We have seen that the STRONG vs. WEAK variation exists for person restrictions in the EA-IA domain as well as the IA-IA domain. More importantly, given that the base generated position of the EA is generally assumed to asymmetrically c-command the base positions of all the IAs, both patterns conform to the syntactic characterization of person restrictions introduced in Section 2.2. The STRONG EA-IA restriction can be described in terms of (46), while the WEAK EA-IA restriction can be described in terms of (47).

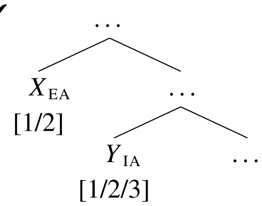
(46) *STRONG person restriction*
If X asymmetrically c-commands Y, then Y must be 3P.

a. ✓


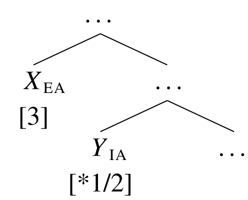
b. ✗


(47) *WEAK person restriction*
If X asymmetrically c-commands Y, and X is 3P, then Y must be 3P.

a. ✓



b. ✗



2.3.2 Direct/inverse systems

In addition to the types of person restrictions we saw in Arizona Tewa and Lummi above, there also exist other kinds of restrictions in the EA-IA domain—usually classified as *direct/inverse systems*. A direct/inverse system can be characterized as a system of argument indexing (i.e. marking the presence of an argument via agreement or pronominal markers on the verb) which is constrained by the person values of the arguments. That is, some person combinations are realized with the baseline transitive marking, or *direct* marking, while others require *inverse* marking, which either involves a dedicated inverse morpheme on the verb or the use of an entirely different pronominal marker paradigm. Crucially for our present purposes, the person combinations that require the inverse paradigm crosslinguistically correspond to either the STRONG or WEAK EA-IA restriction pattern.

A STRONG version of a direct/inverse system is found in Mapudungun (Zúñiga 2006; Smeets 2008). As shown in (48)–(49), its baseline direct paradigm is limited to the person combinations grammatical with STRONG EA-IA restrictions, like the one in Arizona Tewa.

(48) Mapudungun [*Araucanian*]: **STRONG EA-IA**
mütrüm **-fi** { **-n** / **-mi** / **-y** } 1/2/3.EA + 3.IA
call -3.O.(IND) { -1.SU / -2.SU / -3.SU }
'**I/you/he** called **him**.'

(49) a. *mütrüm -y**-u** **-n** ✗ 1.EA + 2.IA
call -IND-(1>)2.O -1.SU
'**I** called **you**.'

b. *mütrüm **-n** **-mi** ✗2.EA + 1.IA
call -1.O.(IND) -2.SU
'**You** called **me**.'

c. *mütrüm { **-n** / -y-**m** } **-y** ✗3.EA + 1/2.IA
call { -1.O.(IND) / -IND-(3>)2.O } -3.SU
'**He** called **me/you**.' (Smeets 2008, 153–65, 367)

The person combinations ungrammatical in the direct paradigm must be expressed with the inverse construction, which involves the addition of an inverse morpheme close to the verbal root and a change in the form of the subject pronominal marker, which Smeets (2008) analyzes as the dative form. This alternative construction is illustrated by the examples in (50), with the inverse morpheme *-e* underlined.

(50) a. mütrüm-e -y-**u** **-Ø** 1.EA + 2.IA
call-INV -IND-(1>)2.O -1.SU.DAT
'**I** called **you**.'

b. mütrüm-e -y-**m** **-ew** 3.EA + 2.IA
call-INV -IND-(3>)2.O -3.SU.DAT
'**He** called **you**.'

c. mütrüm-e **-n** **-Ø** 2.EA + 1.IA
call-INV -1.O.(IND) -2.SU.DAT
'**You** called **me**.'

d. mütrüm-e **-n** **-ew** 3.EA + 1.IA
call-INV -1.O.(IND) -3.SU.DAT
'**He** called **me**.' (Smeets 2008, 367)

In contrast to Mapudungun, the Cherokee (Scancarelli 1987; Montgomery-Anderson 2008) direct/inverse system patterns like the WEAK EA-IA person restriction we saw in Lummi. As shown in (51), the direct paradigm spans all person combinations except for 3P EAs co-occurring with a 1/2P IAs (cf. (52)).[14] Note that, like in Arizona Tewa, transitive pronominal prefixes in Cherokee are traditionally analyzed as portmanteau markers (although further morphological decomposition appears to be possible for most of the prefixes).

14. The plural 3P examples are used in (52) and (53) below due to the more transparent inverse morpheme; with singular 3P EAs, inverse morphology constitutes of null subject marking and the use of an alternative pronominal marker paradigm (see Scancarelli 1987; Dukes 1996; Montgomery-Anderson 2008 for discussion).

(51) Cherokee [*Iroquoian*]: **WEAK EA-IA**

a. { **jii-** / **hii-** / **a-** } ahyvhthéesk-óʔi 1/2/3.EA + 3.IA
{ 1>3.AN- / 2>3.AN- / 3(>3)- } kick:INCMPL-HAB
'**I/you/he** kick **him**.' (Montgomery-Anderson 2008, 180–3)

b. **kvv-** kahthiíya 1.EA + 2.IA
1>2- wait.for:PRC
'**I** am waiting for **you**.'

c. **ski-** kahthiíya 2.EA + 1.IA
2>1- wait.for:PRC
'**You** are waiting for **me**.' (Montgomery-Anderson 2008, 198)

(52) *{ **ki-** / **ja-** } kooh-vv́ʔi ✗ 3.EA + 1/2.IA
{ (3PL>)1- / (3PL>)2- } see:CMPL-EXPER.PAST
'**They** saw **me/you**.' (Montgomery-Anderson 2008, 191, 193)

The inverse counterparts to (52) are given in (53). The inverse for this combination of arguments is characterized by the addition of the inverse prefix *kee-* (underlined).

(53) a. kvv- **ki-** kooh-vv́ʔi 3.EA + 1.IA
INV.PL- (3>)1- see:CMPL-EXPER.PAST
'**They** saw **me**.'

b. kee- **ja-** kooh-vv́ʔi 3.EA + 2.IA
INV.PL- (3>)2- see:CMPL-EXPER.PAST
'**They** saw **you**.' (Montgomery-Anderson 2008, 191, 193)

Direct/inverse systems thus operate along the same STRONG/WEAK split we saw with the plain examples of EA-IA restrictions, which suggests the possibility of a unification. But before we examine that possibility, based on additional parallels with order person restrictions, let us consider the complicated status of direct/inverse systems.

In her typological overview of direct/inverse systems, Klaiman (1991, 1992) interestingly treats the EA-IA person restrictions of Arizona Tewa and Lummi as direct/inverse systems. This is due to the person-based complementary distribution of active and passive voice in those languages (see discussion above); this characterization treats active voice as the direct and passive voice as the inverse construction. But if a person-based complementary distribution between two constructions is sufficient to categorize something as a direct/inverse system, then virtually all the examples of person restrictions from the current survey, including the canonical PCC, can be categorized as direct/inverse systems; this is actually the path taken by Haspelmath (2007), who treats double object ditransitives as the direct and prepositional ditransitives as the inverse constructions in the case of the PCC. Alternatively, we could limit direct/inverse systems to only those person-based alternations closer to the "prototypical" direct/inverse system of Algonquian languages. However, even the Algonquian system does not seem to be as clear-cut as it is often presented (see Zúñiga 2006; Oxford 2014, 2019); in fact, the Algonquian system can be described as a combination of a person-based argument indexing system and a direct/inverse system (see Section 2.5.3). Similar cases are found in Trans-Himalayan (DeLancey 1981, 2017, 2018), where individual languages gravitate either towards person-based argument indexing or direct/inverse marking with a dedicated inverse morpheme, but very few are clear-cut cases of either. Lastly, there are also languages where what have previously been classified as dedicated inverse morphemes are actually morphemes with a much broader distribution (see e.g. Oxford 2014, 2019 on Algonquian, and Bobaljik to appear on Chukotko-Kamchatkan).

Going into more details about the intricacies of direct/inverse systems would take us to far astray, so I direct the reader to Givón (1994) and Jacques and Antonov (2014) for more comprehensive overviews. What I hope that the brief discussion above nonetheless conveys is that a uniform classification of direct/inverse systems is quite illusive. But that is actually not a problem in our case: no matter what we call the cases where alternative grammatical means are used to avoid ungrammaticality due to a specific person combination of arguments, the fact is that the alternative means are employed *because* of a person restriction in the baseline construction. In that sense, direct/inverse systems are for all intents and purposes like the EA-IA restrictions discussed in relation to Arizona Tewa and Southern Tiwa, and they can therefore be characterized in exactly the same syntactic terms (cf. (46)–(47) above).

2.3.3 Person-based argument indexing

The final phenomenon that can be integrated with those discussed so far are person-based argument indexing systems. In many languages, only some arguments can be indexed on the verb; i.e. their existence is registered by agreement, pronominal markers, or some other type of verbal morphology. In some of those languages, the choice of which argument will be indexed is determined solely by its person value and the arguments left unindexed are restricted in terms of the person values they can have. Importantly for us, such restrictions also conform to the STRONG vs. WEAK variation. For example, a STRONG person restriction on argument indexing in the IA-IA domain is described in (54).

(54) *STRONG IA-IA person restriction on argument indexing*
If only one of two IA is indexed, the other IA must be 3P.

This type of restriction is observed in Maasai (also Maa) (Lamoureaux 2004). In Maasai, only one of the two objects can be indexed on the verb, and due to the direct/inverse system in the EA-IA domain, the indexing may amount to an inverse marker being used based on the person of the subject and the relevant object. We can see an example of this in (55a), where the 2P object is identified based on the inverse marker (3.EA $\gg$ 2.IA being one of the contexts for inverse marking). The sentence can either be interpreted as having a 2P Goal and a 3P Theme ('the man') or a 2P Theme and 3P Goal, since the indexing of the object is blind to it being the Goal or Theme. Crucially, the object indexing pattern is not random, as the 3P object can never be indexed in such cases, as shown in (55b) (if allowed, this possibility would yield a direct marking pattern). In fact, given a 1/2P object and a 3P object, it is always the 1/2P one that is indexed in Maasai (Lamoureaux 2004, 20).

(55) Maasai/Maa [*Eastern Nilotic*]:

a. **kɨ́-** ɨshɔ(r) ɔl-payíán — 2.IO/DO + (3.DO/IO)
INV.3>2- give M-man.ACC
'**They** will give **you** to the man.'
'**They** will give the man to **you**.'

b. ***ɔ́-** ɨshɔ *nɨnyɔ́* iyíé — ✗ 3.IO/DO + (2.DO/IO)
3(>3)- give him/her.ACC you.ACC
'**They** will give *you* to **him/her**.'
'**They** will give **him/her** to *you*.' (Lamoureaux 2004, 14–20)

Furthermore, what we observe in the case of two 1/2P objects reveals that we are dealing here with a person restriction: such combinations cannot be expressed via a regular double object ditransitive. As shown in (56), such combinations of objects require the suffixation of a *venitive* morpheme on the verb—a morpheme otherwise used to express 'motion towards'.

(56) a. ***áa-** ɨshɔr *nánú* iyíé — ✗ 1.GL/TH + (2.TH/GL)
INV(3>1)- give me.ACC you.ACC
'**They** will give **me** to you.'
'**They** will give you to **me**.'

b. **áa-** ɨshɔr-ʉ̀(n) — 1.GL/TH + (2.TH/GL)
INV(3>1)- give-VEN
'**They** will give **me** (towards you).'
'**They** will give (you) towards **me**.' (Lamoureaux 2004, 20)

As with other person restrictions before, we find a complementary distribution between the baseline construction used with the grammatical person combinations and the alternative construction used with the ungrammatical ones. More importantly, since at most one object can be 1/2P, the person restriction corresponds to a STRONG one. This classification will become clearer when we compare the Maasai pattern with a minimally different WEAK one:

(57) *WEAK IA-IA person restriction on argument indexing*
If only one of two IA is indexed, and that IA is 3P, the other IA must be 3P.

This type of restriction is attested in Alutor (Savvina and Mel'čuk 1978; Mel'čuk 1988),[15] where only one object can be indexed in double object ditransitives (limited to the verb *jəl* 'give (as a wife)'), and this can be either the Goal or the Theme. Unlike in Maasai, combinations of two 1/2P objects are grammatical, even though only one of them can be indexed—but just like in Maasai, the indexing interacts with an additional EA-IA restriction.

Because of this EA-IA restriction (discussed in more detail in Section 4.1.5) transitives with a 3P EA and a 1/2P IA, for example, require a special inverse marking paradigm to index the arguments.[16] In double object ditransitives, either one of the IAs can be indexed this way; e.g. the 1P Goal is indexed in (128c), and the 1P Theme is indexed (58b).

(58) Alutor [*Chukotkan*]: **WEAK IA-IA**

a. əlləɣ-a ina-jəl -i *ɣəmək-əŋ* ɣəttə — 1.GL + (2.TH)
father-ERG INV(1.IO)-give -3.SU me-DAT you.ABS
'**He** [father] gave you to **me** (as a wife).'

b. əlləɣ-a ina-jəl -i ɣənək-əŋ *ɣəmmə* — 1.TH + (2.GL)
father-ERG INV(1.DO)-give -3.SU you-DAT me.ABS
'**He** [father] gave **me** to you (as a wife).' (Mel'čuk 1988, 295)

Note though that the unindexed IAs in (58) are, unlike in Maasai above, permitted to be 2P. This shows that we are not dealing with a STRONG person restriction here.

It is not only with two 1/2P IAs that the indexing is insensitive to the Goal/Theme distinction. Whenever one IA is 1/2P and the other is 3P, the 1/2P one is indexed whether it is the Goal or the Theme. We see in (128a) that the 1P Goal is indexed when the Theme is 3P, and in (59b), where the Goal is 3P and the Theme 1P, it is the Theme which is indexed; note also that in both cases we get the same marking on the verb as in (58) above.

15. Just like Classical Arabic (see footnote 6), Alutor has an additional restriction differentiating 1P and 2P objects, which I discuss in Section 4.3 in the context of a more general discussion of these types of restrictions.

16. I use the term "inverse marking" here following the analysis of Chukotko-Kamchatkan person restrictions of Comrie (1979b, 1980), although this classification has been since reassessed by Halle and Hale (1997), Bobaljik and Branigan (2006), and Bobaljik (to appear). The classification of the marking that surfaces in inverse person combination contexts, however, does not matter for the issue at hand, but see Sections 4.1.5 and 4.3 for a more detailed discussion of the Alutor person restrictions in both EA-IA and IA-IA domains.

(59) a. əlləɣ-a <u>ina-</u> jəl -i *ɣəmək-əŋ* şininkin ŋavakək — 1.GL + (3.TH)
father-ERG INV(1.IO)- give -3.SU me-DAT his.ABS daughter.ABS
'**He** [father] gave his daughter to **me** (as a wife).'

b. əlləɣ-a <u>ina-</u> jəl -i ənək-əŋ *ɣəmmə* — 1.TH + (3.GL)
father-ERG INV(1.DO)- give -3.SU him-DAT me.ABS
'**He** [father] gave **me** (as a wife) to him.' (Mel'čuk 1988, 295)

But note that this does not mean that Alutor is devoid of IA-IA person restrictions. Since the choice of the IA to be indexed on the verb is determined by their person value whenever at least one of the IAs is 1/2P, this means that a 3P will never be indexed in such cases. In other words, whenever a 3P IA is indexed on the verb in a double object construction, the other IA cannot be 1/2P, which makes the Alutor indexing patterns a WEAK IA-IA restriction. We see an example of this in (129): the 3P Theme cannot be indexed when the Goal is 1P, and the 3P Goal cannot be indexed when the Theme is 1P.

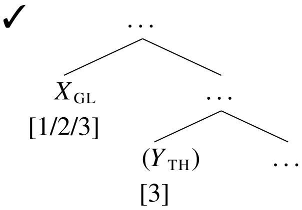
(60) a. *əlləɣ-a **Ø-** jəl **-nin** ɣəmək-əŋ *şininkin ŋavakək* — ✗3.TH + (1.GL)
father-ERG 3.SU- give -3>3.DO me-DAT his.ABS daughter.ABS
'**He** [father] gave **her** [his daughter] to me (as a wife).'

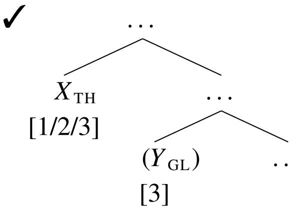
b. *əlləɣ-a **Ø-** jəl **-nin** *ənək-əŋ* ɣəmmə — ✗3.GL + (1.TH)
father-ERG 3.SU- give -3>3.IO him-DAT me.ABS
'**He** [father] gave me (as a wife) to **him**.' (Mel'čuk 1988, 295)

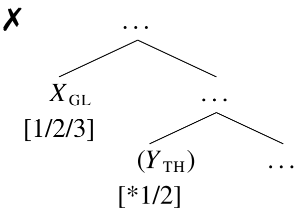
Note that the Maasai and Alutor restriction patterns are effectively canonical + reverse PCC patterns. Recall that in Slovenian, the canonical PCC, where the Theme is restricted, occurs when the Goal precedes Theme and the reverse PCC, where the Goal is restricted, occurs when the Theme precedes the Goal. In the case of argument indexing restrictions, the canonical PCC occurs when the Goal is indexed and the reverse PCC occurs when the Theme is indexed. Such person-based argument restrictions are therefore also problematic for analyses of person restrictions that rely on inherent properties of the two pronouns. In contrast, they can be straightforwardly characterized in syntactic terms along the same lines as the canonical + reverse PCC in Slovenian (see Section 2.2).

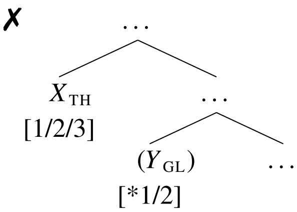
We can take the argument indexing options in languages like Maasai and Alutor to reflect a structural difference in terms of the c-command relations between the relevant pronoun markers. If the pronoun/agreement markers are the real arguments in these languages (see Jelinek 1984; Section 2.4.1), then the indexing pattern can be characterized as: only the structurally highest IA marker is morphologically realized. This way the STRONG and WEAK versions of the restriction can be characterized uniformly as shown in (61) and (62) respectively (the arguments in parentheses are the unrealized/unindexed ones).

(61) *STRONG person restriction*
If X asymmetrically c-commands Y, then Y must be 3P.

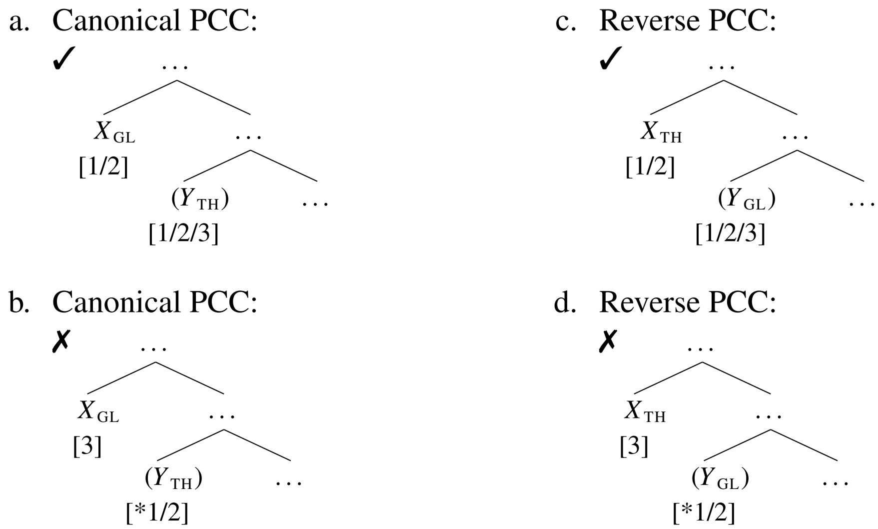
a. Canonical PCC:
✓


c. Reverse PCC:
✓


b. Canonical PCC:
✗


d. Reverse PCC:
✗


(62) *WEAK person restriction*
If X asymmetrically c-commands Y, and X is 3P, then Y must be 3P.



Although the canonical + reverse patterns we saw with Slovenian in Section 2.2 and here with Maasai and Alutor all involve IA-IA person restrictions, the possibility of canonical + reverse patterns exists also in the EA-IA domain (see Section 2.5.3), where we crucially also see restrictions that conform either to the STRONG or the WEAK restriction format.

2.3.4 Summary of restriction types

It is very striking that despite how different the person restrictions can appear on the surface, all of them conform to the same abstract STRONG and WEAK patterns. All the different patterns are summarized in Table 2.1 in both their STRONG and WEAK versions.

Table 2.1: Summary of different person restriction types

IA-IA restrictions (including the PCC)				
STRONG:	3.GL + 3.TH	1/2.GL + 3.TH	***1/2.GL + 2/1.TH**	***3.GL + 1/2.TH**
WEAK:	3.GL + 3.TH	1/2.GL + 3.TH	1/2.GL + 2/1.TH	***3.GL + 1/2.TH**
EA-IA restrictions (including direct/inverse systems)				
STRONG:	3.EA + 3.IA	1/2.EA + 3.IA	***1/2.EA + 2/1.IA**	***3.EA + 1/2.IA**
WEAK:	3.EA + 3.IA	1/2.EA + 3.IA	1/2.EA + 2/1.IA	***3.EA + 1/2.IA**
Canonical + reverse IA-IA restriction				
STRONG:	3.IA + 3.IA	1/2.IA + 3.IA	***1/2.IA + 2/1.IA**	***3.IA + 1/2.IA**
WEAK:	3.IA + 3.IA	1/2.IA + 3.IA	1/2.IA + 2/1.IA	***3.IA + 1/2.IA**
Canonical + reverse IA-IA argument indexing restriction				
STRONG:	3.IA + (3.IA)	1/2.IA + (3.IA)	***1/2.IA + (2/1.IA)**	***3.IA + (1/2.IA)**
WEAK:	3.IA + (3.IA)	1/2.IA + (3.IA)	1/2.IA + (2/1.IA)	***3.IA + (1/2.IA)**
Overarching abstract pattern				
STRONG:	3 $\gg$ 3	1/2 $\gg$ 3	***1/2 $\gg$ 2/1**	***3 $\gg$ 1/2**
WEAK:	3 $\gg$ 3	1/2 $\gg$ 3	1/2 $\gg$ 2/1	***3 $\gg$ 1/2**

In fact, the uniformity of the patterns extends to all of the person restrictions identified by the crosslinguistic survey, which are all either of the STRONG or WEAK type.[17] What is also important is that all the person restriction patterns are compatible with a syntactic characterization (see the summary of the patterns in Table 2.1): given two pronouns X and Y, where X asymmetrically c-commands Y, X can have any person value while Y cannot.

Furthermore, the crosslinguistic parameters of variation that we have established in the last two sections—that is: STRONG vs. WEAK and canonical vs. reverse—will provide the basis for identifying two new typological gaps in Section 2.5. But before discussing those, I

17. There is more fine-grained internal variation within the WEAK patterns, as noted, among others, also by Nevins (2007) and Pancheva and Zubizarreta (2018). However, for ease of exposition, I postpone the discussion of that variation until Section 4.3, after I introduce the analysis of person restrictions. Crucially, the finer-grained variation is attested for all the types of person restrictions discussed above as well (see Section 4.3 also regarding some exceptionally rare restriction patterns of this type).

will consider another interesting property concerning person restrictions, namely, that they crucially only arise with very specific types of personal pronouns.

2.4 Person restrictions and pronoun type

It has been recognized already since the very first formal treatments of person restrictions that not all pronouns are equal in their sensitivity to them. A good illustration of this is Bonet's (1991) characterization of the canonical PCC (known at that time as **me lui Constraint*), given in (63) (emphasis mine).

(63) *The *me lui Constraint*

a. In a combination of a direct object and an indirect object, the direct object has to be third person

b. Both the direct object and the indirect object are *phonologically weak*

(Bonet 1991, 177)

What is crucial in (63) is that the constraint is only applicable to *phonologically weak* elements, which is meant to cover clitic and other weak pronouns as well as agreement markers—they are all prosodically deficient elements that must prosodically attach to other words. The need for limiting person restrictions to phonologically weak pronouns can be illustrated with Slovenian. Recall that in Slovenian, the second object clitic in the clitic cluster is person restricted, as seen in (64a). The restriction crucially does not occur when the second object pronoun is a strong pronoun, as in (64b), or when both object pronouns are strong, as in (64c).

(64) Slovenian [*Slavic*]:

a. *Mama **mu** **te** bo predstavila. ✗3.GL ≫ 2.TH
mom 3.M.DAT 2.ACC will introduce

b. Mama **mu** bo predstavila *tebe*. 3.GL ≫ 2.TH
mom 3.M.DAT will introduce you.ACC

c. Mama bo predstavila *njemu* *tebe*. $\boxed{3.\textsc{gl} \gg 2.\textsc{th}}$
mom will introduce him.DAT you.ACC
'Mom will introduce **you** to **him**.'

Similar asymmetries can also be found in all languages with person restrictions and weak/strong pronoun oppositions. However, despite the wide recognition of this asymmetry, there is very little agreement in the literature on person restrictions as to what is behind it and what is the precise nature of the pronouns sensitive to person restrictions.

Here are some examples in addition to Bonet (1991). For example, it has been argued that person restrictions are characteristic of affixal elements (Perlmutter 1971; Zwicky 1977; Zwicky and Pullum 1983), where the reasoning is that person restrictions are all surface ordering restrictions and affixes crosslinguistically show much less freedom in ordering than clitics or other phonologically weak elements. We saw in the previous section that it is difficult to maintain such a view in the face of the irrelevance of surface forms for many person restrictions. A radically different view has been proposed by Béjar and Řezáč (2003) and Anagnostopoulou (2003, 2005, 2008), who argue that essentially all pronouns can in principle be sensitive to person restrictions. Cases where strong pronouns are exempt from person restrictions are attributed to the presence of additional non-pronominal structure (e.g. overt or null prepositions) (Béjar and Řezáč 2003) or to them not being required to move to the same syntactic position as the affected pronouns (Anagnostopoulou 2003). We will, however, see later in this section that the putative cases of restrictions with strong pronouns, crucial for these proposals, are actually radically different from all other person restrictions, strongly suggesting a different source. Finally, there are analyses that limit the restrictions to clitic-doubling configurations specifically (Nevins 2011; Coon and Keine 2018). I briefly discuss this line of approaches at the end of this section. Notice that the above proposals may require mutually exclusive assumptions regarding what is the natural class of affected pronouns. Consequently, a particular theory of person restrictions could often lead researchers to reclassify as affixes those elements that have been identified as clitics by unrelated linguistic factors and vice versa.

There is also a conceptual issue at play here. If person restrictions are at their core a syntactic phenomenon, as I argued in the previous section, why should the phonological status of pronouns play a role? This is especially troubling for *Minimalist* approaches to syntax (Chomsky 1993, 1995, 2000, 2001) and realizational approaches to morphology (*Distributed Morphology* and its kin; Bonet 1991; Noyer 1992; Halle and Marantz 1993, 1994; Harley and Noyer 1999), where syntax feeds fully independent semantic/logical (*Logical Form*; LF) and morphological/phonological (*Photetic Form*; PF) modules that do not feed back into syntax or directly interact with it—that is, they branch off at spell-out in concordance with the *Y-model* of grammar. Therefore, it should be impossible for any phonological trait of a pronoun to have direct consequences for its behavior in syntax.

Bonet (1991), in fact, already anticipated the issue. Although she placed person restrictions in the morphological component of grammar, she defined the context where they apply in syntactic terms: person restrictions occur between pronominal elements syntactically adjoined to the same *INFL* head—INFL being the locus of clausal agreement (also I or T). This was framed within the approach to Romance clitics of Kayne (1990, 1991), in which clitics are base generated in argument positions and move to adjoin to INFL. The fact that these pronouns are also phonologically weak can then be linked to them being heads rather than maximal projections (I entertain a similar idea in Section 3.2). Even if we were to accept that the pronouns affected by person restrictions crosslinguistically all move to adjoin to INFL (which is hard to motivate),[18] the issue noted in Section 2.2 would nonetheless remain: the morphological constraint proposed by Bonet (1991) is not

18. Consider for example the case of Slovenian clitic pronouns, which are second position clitics. Slovenian clitic pronouns are sensitive to person restrictions, but their placement would be quite clearly impossible to attribute to the clitics consistently residing in INFL (see Bošković 2001 for discussion). In fact, no one has even ever entertained such an account for second position clitics, due to its rather obvious implausibility.

transparently reflected in all the crosslinguistically attested person restrictions. Furthermore, since the morphosyntactic features of the subject also come to be located on INFL, we need something in addition to explain why some languages do and other languages do not have person restrictions involving non-oblique subjects. This is not to say that other analyses have a better grasp on how to characterize the relationship between being phonologically weak and showing person restrictions in the syntax. As we saw above, there is very little consensus on how to unify the class of pronouns that crosslinguistically show sensitivity to person restrictions.

In the remainder of this section I will discuss the insights the current crosslinguistic survey provides with respect to this issue. I will show that the relationship between phonological weakness and sensitivity to person restrictions is a one-way correlation: although all pronouns affected by person restrictions are phonologically weak elements, not all phonologically weak pronouns show person restrictions. In addition, I will argue that the relevant pronoun class is most clearly identifiable due to it being singled out by a syntactic phenomenon seemingly independent from person restrictions; namely, binding. Finally, I will briefly discuss why the correlation between phonological form and syntactic behavior does not contradict the autonomy of syntax.

2.4.1 A correlation with binding

The reviewed language sample identifies two groups of pronouns sensitive to person restrictions: (i) *pronominal arguments* in languages of the *head-marking* type (Nichols 1986, 1992; Jelinek 1984, 2006; Baker 1988, 1996), that is, languages that primarily use verbal morphology to express linguistic relationships between the arguments of the verb, with the pronominal markers serving as the real arguments as opposed to DPs/NPs/strong pronouns; and (ii) *clitic/weak pronouns* serving as the unmarked objects in the language, which can either double DPs/NPs/strong pronouns or must be in complementary distribution with them (depending on the language). At first glance it seems that the only unifying trait between the two groups is their status as phonologically weak—they are not free standing pronouns, but prosodically dependent on appropriate hosts. However, there is a more important syntactic connection between them which will be identified below.

Crucially, the prosodic character of the pronouns cannot be the only relevant factor. This can be illustrated with Polish, which has a weak/strong object pronoun opposition (Franks and King 2000, 149–152), but as shown in (65), co-occurring weak pronouns do not yield a person restriction (see Cetnarowska 2003; Migdalski 2006; Bondaruk 2012).[19]

(65) Polish [*Slavic*]:
Jak **mu** **cię** nieprzytomną dostarczą, ... $\boxed{\text{3.GL} \gg \text{2.TH}}$
when 3.M.DAT 2.ACC unconscious deliver ...
'When they deliver **you** to **him** unconscious, ...' (Bondaruk 2012, 68)

Similar, although less systematic, cases like this are also found in Germanic with the languages that show a weak/strong object pronoun split. As noted by Anagnostopoulou (2008), speakers of such languages vary significantly in terms of accepting pronoun combinations expected to yield person restrictions even though the pronouns are weak for all of them.

2.4.1.1 Pronominal argument languages

A much tighter relationship between person restrictions and an independent syntactic factor is observed when we look at binding. In *pronominal argument languages* specifically (Jelinek 1984, 2006; Baker 1988, 1996), reflexivization is not achieved, like it is in English, through the use of (strong) reflexive pronouns (see Baker 1988, 1996 also regarding additional binding idiosyncrasies). Instead, reflexive constructions require either intransitivization,

19. Pancheva and Zubizarreta (2018) use an example equivalent to (65) to suggest that Polish might have a type of WEAK restriction that bans only $3 \gg 1$ and $2 \gg 1$ pronoun combinations (see Section 4.3). However, not only does Polish allow $3/2 \gg 1$ combinations, colloquial Polish also lost the weak/strong pronoun distinction for 1P accusative pronouns, so we would not expect a restriction with these combinations in the first place.

special reflexive verb forms, or dummy nouns (usually body-part nouns), and sometimes even different combinations of all three strategies.

Consider Lummi, which we saw has a WEAK EA-IA restriction. In Lummi, reflexive constructions involve intransitivization and reflexive morphology on the verb; compare the transitive example (66a) with the reflexive one in (66b) (this strategy is used more generally across Salish; Jelinek and Demers 1994; Déchaine and Wiltschko 2017). Note that the pronominal marking in (66b) matches that of the intransitive (67a) and transitive passive (67b); Lummi has ergative alignment for 3P markers, and all three show no ergative marking.

(66) Lummi [*Salish*]:

a. nəp-t **-s** **-Ø** Transitive
advise-TR -3.SU.ERG -3.O.ABS
'**He** advised **him**.'

b. leŋ-t-oŋət **-Ø** Reflexive
see-TR-REFL -3.SU.ABS
'**He** looks at himself.' (Jelinek and Demers 1994, 704–5)

(67) a. čey **-Ø** Intransitive
work -3.SU.ABS
'**He** works.' (Jelinek and Demers 1994, 699)

b. 'əy'-t-ŋ **-Ø** Passive
good-TR-PASS -3.SU.ABS
'**It** has improved.' (Jelinek and Demers 1994, 711)

A similar case is found in Blackfoot (and Algonquian more generally), another language with a WEAK restriction between subject and object markers. In Blackfoot, reflexivization involves adding a dedicated reflexive morpheme that changes the pronominal marking from transitive to intransitive; as shown in (68). The intransitive character of the agreement can be seen by comparing (68) with the plain intransitive construction in (69a) and the unspecified agent constriction in (69b).[20]

20. The unspecified agent construction in (69b) is sometimes considered a plain passive, although this is somewhat controversial (see Frantz 1976; Dahlstrom 1991; Bliss 2013; Lochbihler 2013; Oxford 2014; Zúñiga 2016 for discussion). What matters here is that these constructions pattern closely to reflexives when it comes to argument indexing; the translation in (69b) is changed slightly from the original to highlight the parallel.

(68) Blackfoot [*Algonquian*]:

a. i-sspómmo **-yii** **-wa** *aakííkoan-a póós-i* Transitive
PAST-help.TR.AN -3>3' -3(SU) girl-3 cat-3'
'**She** [the girl] helped **it** [that cat].' (Frantz 1991, 52)

b. *om-a iimitaa-wa* siiksip-o:hsi **-wa** Reflexive
that-3 dog-3 PAST.bite.TR.AN-REFL.INTR.AN -3(SU)
'**He** [the dog] bit himself.' (Frantz 1971, 53)

(69) a. *om-a ninaa-wa* iyimmi **-wa** Intransitive
that-3 man-3 laugh.INTR.AN -3(SU)
'**He** [the man] laughed. (Frantz 1971, 33)

b. *ann-a pookáá-wa* ii-hts-íísit-o-wa **-wa** ann-i pokón-i Unspec. agent
that-3 child-3 IC-INSTR-hit-TR.AN-X>3 -3(SU) that-3' ball-3'
'**He** [the child] was hit with the ball (by someone).' (Bliss 2013, 48)

An interesting case with respect to reflexive marking is Southern Tiwa, which (at least superficially) utilizes both intransitivization with a reflexive marker for some subjects and plain intransitivization for others. The first option is illustrated in (70)–(71), although for this particular case the transitive and intransitive paradigm are morphologically indistinguishable—they are distinguished, for example, in the case of *i-bi-* '3B-3B' vs. *i-be-* '3B-REFL' vs. *i-* '3B' (see full paradigms in Rosen 1990, 673, 691).[21]

(70) Southern Tiwa [*Kiowa-Tanoan*]:

a. **Ø-** mũ-ban Transitive
3>3- see-PAST
'**He/She** saw **her/him**.' (Allen and Frantz 1986, 394)

b. *'uide* **Ø-** be-mũ-ban Reflexive
child.A 3- REFL-see-PAST
'**He** [the child] bit himself.' (Rosen 1990, 691)

(71) a. *seuanide* **Ø-** mi-ban Intransitive
man 3- go-PAST
'**He** [the man] went.' (Allen and Frantz 1986, 392)

b. *liorade* **Ø-** mũ-che-ban seuanide-ba Passive
child 3- see-PASS-PAST man-INSTR
'**She** [the lady] was seen by the man.' (Allen and Frantz 1978, 12)

21. 3P markers in Kiowa-Tanoan languages fit in three classes (A, B, C) corresponding essentially to different number values; they are not simply marked as singular, dual, and plural because of the phenomenon of *inverse number*, an intricate system of number marking found in Kiowa-Tanoan (see Noyer 1992; Harbour 2006).

The second option is shown in (72)–(73), where the pronominal marking on the verb with a reflexive interpretation is indistinguishable from that of an intransitive.

(72) Southern Tiwa [*Kiowa-Tanoan*]:

a. *seuanide* **ti-** mũ-ban $\boxed{\text{Transitive}}$
man 1>3- see-PAST
'**I** saw **him** [the man].' (Rosen 1990, 683)

b. **te-** mũ-ban $\boxed{\text{Reflexive}}$
1>REFL- see-PAST
'**I** saw myself.' (Rosen 1990, 691)

(73) a. **te-** mi-ban (eswela-'ay) $\boxed{\text{Intransitive}}$
1- go-PAST school-to
'**I** went (to school).' (Rosen 1990, 672)

b. seuanide-ba **te-** mũ-che-ban $\boxed{\text{Passive}}$
man-INSTR 1- see-PASS-PAST
'**I** was seen by the man.' (Rosen 1990, 697)

Interestingly, Southern Tiwa has yet another reflexivization strategy employed only in cases where there are two internal arguments. This strategy does not involve intransitivization. Instead, it has been argued to involve incorporation of a dummy noun ('self') historically related to the reflexive marker (see Rosen 1990, 690–5 for details). Like Lummi and Blackfoot, Southern Tiwa is also a language with an EA-IA person restriction.

The take away message from pronominal argument languages is that there is a tight relationship between the pronominal markers and reflexives, manifested mainly via their complementary distribution. Of course, the details concerning their complementary distribution are not always the same, but nonetheless informative. As we saw in Section 2.3, intransitivization (e.g. passivization) is also employed in these languages to void person restrictions. In cases where the complementary distribution is not the result of intransitivization, but merely the absence of pronominal marking for the reflexive, the complementary distribution of pronominal marking and reflexives can be tied to the so-called *Anaphor-*

Agreement Effect (Rizzi 1990a; Woolford 1999);[22] that is, the crosslinguistic generalization that anaphoric elements, like reflexives, cannot be agreement controllers—and by extension indexed by pronominal arguments (assuming pronominal arguments effectively replace traditional agreement; Jelinek 1984; Jelinek and Demers 1994; Baker 1988, 1996). In fact, this relationship between agreement and binding will be crucial in the analysis of person restrictions in Chapter 3. Finally, even reflexivization via dummy noun incorporation generally shows this type of complementary distribution between pronominal marking and reflexive marking—due to the complementary distribution of agreement and incorporation (Baker 1988, 1996).[23]

In addition to being singled out in terms of binding strategies, pronominal argument/head-marking languages also seem to be the only languages that exhibit EA-IA restrictions. In other words, person restrictions of this type are only found in languages where the primary way of realizing the verb's arguments, including both the EA and IAs, is via agreement/pronoun markers on the verb—typically accompanied by some or all of the syntactic criteria for pronominal arguments established by Jelinek (1984) and Jelinek and Demers (1994) and the related criteria for polysynthesis established by Baker (1988, 1996), such as virtually free order of phrases in a clause and free omission of discourse salient DPs/NPs and independent pronouns (Lummi, Blackfoot, and Southern Tiwa are all prototypical cases discussed by Jelinek and Baker). However, this is only a one-way correlation, as there are pronominal argument languages without person restrictions or with only an IA-IA restriction.

22. Yuan (2018) offers a particularly good illustration of the difference between intransitivization and the Anaphor-Agreement Effect in her discussion of apparent antipassives in Inuktitut.

23. The "generally" here is due to Southern Tiwa, an outlier in that it shows agreement with incorporated nouns—although the constraints on incorporation in the language are quite idiosyncratic independently of this (see Rosen 1990). On a related note, if Déchaine and Wiltschko (2017) are correct, Halkomelem reflexivization (which patterns with Lummi and Salish more broadly) is actually a form of verb-noun compounding and thus similar to the Southern Tiwa strategy of reflexivization via incorporation. Different reflexivization strategies are often hard to distinguish, possibly because of the grammaticalization path from reflexive pronouns and dummy nouns to verbal reflexive morphology and finally intransitivization (see Faltz 1977, 1985). See also Déchaine and Wiltschko 2017 for a proposal how to derive the different reflexivization strategies.

2.4.1.2 Clitic/weak object pronoun languages

The other class of pronouns that shows sensitivity to person restrictions can be described pre-theoretically as consisting of: (i) clitic/weak object pronouns in complementary distribution with strong pronouns and DPs/NPs (e.g. as in French or Slovenian), and (ii) clitic/weak object pronouns that can double co-occurring strong pronouns and DPs/NPs (e.g. as in Spanish or Catalan). We are faced here with a rather heterogeneous group of pronoun types, whose only apparent unifying feature is their phonological character as weak elements. However, it has been observed that the pronouns in (i) and (ii) are sensitive not only to person restrictions, but also a number of binding restrictions that set them apart from their strong counterparts. For example, when they co-occur they exhibit (logophoric) binding restrictions similar to person restrictions, as shown in (74); the DO clitic can be bound by the matrix subject only when the animate IO is not clitic doubled (the observation of this phenomenon for Spanish and Catalan was originally made by Roca 1992).

(74) Spanish [*Romance*]:

a. Mateo$_i$ piensa que **lo**$_i$ entregaste a la policá.
Mateo thinks that 3.DO.ACC handed.2 to the police
'Mateo thinks that you handed **him** over to the police.'

b. Mateo$_i$ piensa que **se** **lo**$_{*i}$ entregaste *a la policá*.
Mateo thinks that 3.IO.DAT 3.DO.ACC handed.2 to the police
'Mateo thinks that you handed **him** over to **them** [the police].'

(Ormazabal and Romero 2007, 327)

The connection between these facts and person restrictions has been discussed in detail for Catalan, Spanish (Ormazabal and Romero 2007), and French (Charnavel and Mateu 2015a). More general discussions are also provided by Bhatt and Šimík (2009) and Pancheva and Zubizarreta (2018) (apart from Spanish, Catalan, and French, Bhatt and Šimík also discuss data from Bosnian-Serbian-Croatian, Czech, and Slovenian, while Pancheva and Zubizarreta also discuss Bulgarian). I will come back to this binding restriction in Chapter 5. For now, I want to focus instead on a particularly informative minimal pair with respect to the connection between binding and person restrictions: Slovenian and Polish.

Remember that in Slovenian 3P IO clitic cannot precede a 2P DO clitic (cf. (75)), while in Polish, the same combination of weak object pronouns is grammatical (cf. (76)). Polish is a language where phonologically weak pronouns are not affected by person restrictions.

(75) Slovenian [*Slavic*]:
*Mama **mu** **te** bo predstavila. ✗3.GL ≫ 2.TH
mom 3.M.DAT 2.ACC will introduce
'Mom will introduce **you** to **him**.' [repeated from (64a)]

(76) Polish [*Slavic*]:
Jak **mu** **cię** nieprzytomną dostarczą, ... 3.GL ≫ 2.TH
when 3.M.DAT 2.ACC unconscious deliver ...
'When they deliver **you** to **him** unconscious, ...' [repeated from (65)]

The clitic/weak pronouns in Slovenian and Polish also differ regarding at least two binding phenomena. The first involves the so-called *Montalbetti Effect* (Montalbetti 1984): in many languages, null and weak pronouns can be bound variables whereas their strong counterparts cannot be. Montalbetti originally observed this for Spanish and Italian, while Despić (2011) discusses it for Bosnian-Croatian-Serbian; these are all languages that also show person restrictions. A version of the effect also exists in pronominal argument languages: pronominal arguments can be bound by indefinite quantifiers (Jelinek and Demers 1994, 731; Baker 1996, 60–1), whereas their strong counterparts can only be referential.[24]

Focusing on the Slovenian vs. Polish comparison, the Montalbetti Effect crucially holds for Slovenian, where clitic pronouns must be chosen over strong pronouns for bound variable readings, as shown in (77); while the clitic can be either bound or referential, the strong pronoun can only be referential.

24. Although this might not be true for all pronominal argument languages; see Déchaine and Wiltschko 2002 on Shuswap [*Salish*] strong pronouns, which can be bound variables.

(77) Slovenian [*Slavic*]:
Nihče$_i$ noče, da { **ga**$_{i,k}$ / *njega*$_{k,*i}$ } povabim.
no.one.NOM NEG.want.3 that { 3.M.ACC / him.ACC } invite.1
'No one$_i$ wants that I invite **him**$_{i,k}$.'

In Polish, on the other hand, strong and weak pronouns in the same contexts allow both bound variable and referential readings, as shown in (78).

(78) Polish [*Slavic*]:
Każdy chce że-by { **go**$_{i,k}$ / *jego*$_{i,k}$ } zaprosili.
everybody wants that-SBJV { 3.M.ACC / him.ACC } invited.PL
'Everybody$_i$ wants that they invite **him**$_{i,k}$.' (A. Pietraszko; M. Dadan, p.c.)

The other binding phenomenon relevant for the difference between Slovenian and Polish weak pronouns is the availability of *strict identity* and *sloppy identity* readings. Generally, personal pronouns can only receive strict identity readings in the relevant contexts (e.g. *"John scratched* ***his arm*** *and Mary scratched* ***it*** *too"* is incompatible with a reading where Mary scratched her, i.e. Mary's, arm). However, Slovenian is among the Slavic languages where clitic pronouns specifically may receive both strict and sloppy identity readings (see Runić 2013a, 2014; Franks 2013; Bošković 2018 also for Bosnian-Croatian-Serbian); as illustrated in (79a), where the object clitic in the second conjunct can refer either to Pero's father (strict) or Maja's father (sloppy).[25] Strong pronouns, conversely, retain the familiar ban on sloppy readings, and can only receive a strict reading, as in (79b), where the object pronoun in the second conjunct must refer to Pero's father.

25. Interestingly, while null subjects in Slovenian pattern with clitic pronouns in relation to the Montalbetti Effect, null subjects cannot get a sloppy reading in contexts comparable to (79) (same as in Bosnian-Croatian-Serbian; Runić 2014; Bošković 2018). This suggests that there must be a further underlying difference between the two types, which might also be why *pro* does not yield person restrictions when co-occurring with other deficient pronouns. I will suggest two possible explanations for what distinguishes *pro* (and more generally subject pronouns outside pronominal argument languages) from clitic/weak object pronouns in Section 3.3.

(79) Slovenian [*Slavic*]:

a. Pero ceni svojega očeta, in Maja **ga** tudi ceni Strict/Sloppy
Pero values self's.M.ACC father.ACC and Maja 3.M.ACC also values
'Pero$_i$ values his$_i$ father and Maja$_k$ values {**his$_i$ father / her$_k$ father**} too.'

b. Pero ceni svojega očeta, in Maja *njega* tudi ceni Strict
Pero values self's.M.ACC father.ACC and Maja him.ACC also values
'Pero$_i$ values his$_i$ father and Maja$_k$ values **his$_i$ father** too.'

Polish here differs from Slovenian again. Both weak and strong pronouns can only receive a strict identity reading in the same context, as is shown by the examples in (80).

(80) Polish [*Slavic*]:

a. Piotr docenia swojego ojca, i Marta też **go** docenia. Strict
Piotr values self's.M.ACC father.ACC and Marta also 3.M.ACC values.
'Piotr$_i$ values his$_i$ father and Marta$_k$ values **his$_i$ father** too.'

b. Piotr docenia swojego ojca, i Marta też *jego* docenia. Strict
Piotr values self's.M.ACC father.ACC and Marta also 3.M.ACC values.
'Piotr$_i$ values his$_i$ father and Marta$_k$ values **his$_i$ father** too.'

(A. Pietraszko; M. Dadan, p.c.)

The two closely related languages thus systematically differ in how their phonologically weak pronouns pattern in relation to their strong counterparts across different phenomena. In Slovenian, the phonologically weak clitic pronouns are sensitive to person restrictions, can be bound variables, and may have both strict and sloppy identity readings, while their strong counterparts are free from person restrictions, resist being bound variables, and may only get strict identity readings. In Polish, conversely, both the phonologically weak and the strong pronouns pattern with the Slovenian strong pronouns.

Whatever the underlying difference between a Slovenian-type weak pronoun and a Polish-type weak pronoun is, it has ramifications for both person restrictions and binding. Crucially, this does not necessarily mean that all weak pronouns sensitive to person restrictions will show exactly the same binding behavior that we saw in Slovenian. In fact, we know this not to be true in the case of the availability of sloppy readings: there are languages with person restrictions that do not permit sloppy readings with any pronouns, including Bulgarian, Macedonian, and seemingly all of Romance (Runić 2014; Bošković 2018).[26]

Similarly to the differences in the reflexivization strategies we saw with pronominal argument languages, there is variation regarding which binding phenomena will correlate with person restrictions in the case of clitic/weak object pronoun languages. Of course, the size of the language sample again makes it impossible to check the behavior of pronouns in a range of different binding contexts for all the relevant languages, but the binding restriction examples noted above in addition to the Slovenian vs. Polish minimal pair are strongly suggestive of a link between binding and person restrictions. The above discussion implies that in order for the person restriction to arise there needs to be a binding asymmetry between the pronoun types to begin with, which is consistent with Polish, where there is no binding asymmetry of this type and where person restrictions do not arise despite the existence of a phonologically weak pronoun series. At any rate, the relevant correlation between person restrictions and pronoun type can be stated as: if a language has person restrictions and two types of pronouns, then the person restriction will arise with the type that allows more bound readings or appears in more bound contexts. Note, however, that the correlation between person restrictions and binding is a one-way implication, just as the correlation between person restrictions and being a phonologically weak pronoun.

There are thus two major correlations between pronoun type and person restrictions: (i) person restrictions only affect phonologically weak pronouns, and (ii) weak pronouns sensi-

26. Runić (2014) and Bošković (2018) tie the impossibility of sloppy readings here to Bulgarian, Macedonian, and Romance having definite articles—unlike Slovenian and Bosnian-Croatian-Serbian. However, as Yuan (2018) observes, Italian and Romanian allow sloppy readings with a right context (Runić 2014 herself notes the speaker variation in relation to this for Italian, but not for Romanian), whereas French consistently does not. A more fine-grained explanation thus seems to be needed, and a good starting point for one might be exploring the differences in the distribution of articles within Romance. As Longobardi (1994) notes, there are several contexts in Italian where bare nouns without an article are allowed, in contrast to French. However, exploring this in any more detail would go beyond the present goals.

tive to person restrictions always allow more bound readings than their unrestricted strong counterparts. Additionally, there is a generalization specific to EA-IA person restrictions.

2.4.2 A further note on the PF-to-syntax relationship

Although we saw that the prosodic status of pronouns alone cannot be the source of person restrictions, it should still worry us, if we care about the autonomy of syntax, that there is an apparent correlation between the prosodic status of pronouns and person restrictions, even though it is only a one directional one. Let us start tackling this issue by first considering a possible way to state the difference between cases where weak pronouns show a contrast with strong pronouns in relation to binding and person restrictions and those cases where they behave uniformly with respect to both phenomena. While this will not yet provide us with a solution, it will be helpful in better illustrating the nature of the problem.

The issue can be illustrated by utilizing the idea that different pronoun types correspond to different levels of structural complexity, following, among others, Cardinaletti and Starke (1994, 1999) and Déchaine and Wiltschko (2002). The basic idea behind these proposals is that the differences in syntactic, semantic, and morpho-phonological properties between different types of pronouns can be reduced to differences in their internal structure, where weak pronouns correspond to a less complex structure and strong pronouns to a more complex structure. However, the relationship between meaning, syntactic function, and morphological form is not necessarily completely isomorphic. As discussed below, there can in principle be mismatches between form and syntactic function or meaning and syntactic function (see also Despić 2011 for related discussion).

In light of the line of research noted above, one way to characterize the asymmetry between Slovenian-type weak and strong pronouns is to posit additional structure in strong pronouns whose reflex is phonological strength and more rigidity with respect to binding—

also related to resistance to person restrictions. Anticipating the possibility of mismatches between phonological strength and the binding rigidity/person restriction resistance, I attribute the former to Σ, whose reflex at PF is prosodic independence (cf. Cardinaletti and Starke 1994, 1999), and the latter to D, whose reflex at LF is referential independence (the label 'D' is used for expository purposes; I leave open the head's precise identity at this point). The assumption for a Slovenian-type language is then that Σ and D obligatorily cluster in pronouns. In the case of bound variable contexts, this results in pronouns (whose root is represented as 'pron') that lack Σ+D realizing as phonologically weak and capable of receiving a bound variable interpretation, as illustrated in (81a), and pronouns with Σ+D realizing as phonologically strong and incapable of a bound variable reading due to D inducing a referential reading of the pronoun, as illustrated abstractly in (81b).

(81) a. No one$_i$ wants that I invite [pron$_{i,k}$] (pron $\Leftrightarrow$ **ga**)
b. No one$_i$ wants that I invite [Σ+D$_k$ [pron$_{k,*i}$]] (Σ+D+pron $\Leftrightarrow$ *njega*)

Contrast this with the Polish case, where the weak/strong distinction has no effect on binding. This can be interpreted, given present assumptions, as Σ and D occurring independently of each other. The lack of correlation between phonological strength and binding behavior can be attributed to the D being null (and without other phonological/prosodic effects), and Σ alone having the reflex of phonological strength (the lack of syntactic/semantic contribution from Σ does not have to be total, it just cannot have any consequences for binding/person restrictions). Weak pronoun forms (cf. (82)) and strong pronoun forms (cf. (83)) then realize either D-less or D-full structures. So speakers can always parse the surface form as D-less when a bound variable interpretation is needed.

(82) a. No one$_i$ wants that I invite [pron$_{i,k}$] (pron $\Leftrightarrow$ **go**)
b. No one$_i$ wants that I invite [D$_k$ [pron$_{k,*i}$]] (D $\Leftrightarrow$ Ø, pron $\Leftrightarrow$ **go**)

(83) a. No one$_i$ wants that I invite [Σ [pron$_{i,k}$]] (Σ+pron $\Leftrightarrow$ *jego*)
b. No one$_i$ wants that I invite [D$_k$ [Σ [pron$_{k,*i}$]]] (D $\Leftrightarrow$ Ø, Σ+pron $\Leftrightarrow$ *jego*)

The same logic can be applied to person restrictions. Suppose that the referentiality of D is also what allows the pronoun to have any person value, even in contexts where a person restriction would otherwise result (a technical implementation is given in Chapter 3). Then, in a language like Slovenian, where Σ and D always co-occur, strong pronouns are correctly predicted to resist person restrictions, as illustrated in (84); D's ability to isolate pronouns from person restrictions is indicated by a person subscript matching that of the pronoun.

(84) a. *He showed [pron_3] [pron_{*2}] ($\text{pron}_3 \Leftrightarrow$ **mu**, $\text{pron}_2 \Leftrightarrow$ **te**)
b. He showed [pron_3] [$\Sigma+\text{D}_2$ [pron_2]] ($\text{pron}_3 \Leftrightarrow$ **mu**, $\Sigma+\text{D}+\text{pron}_2 \Leftrightarrow$ *tebe*)

In Polish, conversely, both strong and weak pronoun forms can realize either D-less and D-full structures, which means that speakers always have a grammatical parse for each of the surface forms in person restriction contexts, as illustrated in (85).

(85) a. *He showed [pron_3] [pron_{*2}] ($\text{pron}_3 \Leftrightarrow$ **mu**, $\text{pron}_2 \Leftrightarrow$ **cię**)
b. He showed [pron_3] [D_2 [pron_2]] ($\text{pron}_3 \Leftrightarrow$ **mu**, $\text{D}+\text{pron}_2 \Leftrightarrow$ **cię**)
c. *He showed [pron_3] [Σ [pron_{*2}]] ($\text{pron}_3 \Leftrightarrow$ **mu**, $\Sigma+\text{pron}_2 \Leftrightarrow$ *ciebie*)
d. He showed [pron_3] [D_2 [Σ [pron_2]]] ($\text{pron}_3 \Leftrightarrow$ **mu**, $\text{D}+\Sigma+\text{pron}_2 \Leftrightarrow$ *ciebie*)

Since Σ and D can be independent, it should also be possible that a language could have pronouns distinguished only in the presence or absence of Σ, in which case, both weak and strong pronouns should be sensitive to person restrictions, as in (86). This possibility, however, is unattested; that is, it seems that it is not realized.

(86) a. *He showed [pron_3] [pron_{*2}] ($\text{pron}_3 \Leftrightarrow$ **'im**, $\text{pron}_2 \Leftrightarrow$ **you**)
b. *He showed [pron_3] [Σ [pron_{*2}]] ($\text{pron}_3 \Leftrightarrow$ **'im**, $\Sigma+\text{pron}_2 \Leftrightarrow$ *you*)

The issue here is that even though the difference between weak pronouns sensitive to person restrictions and those insensitive to them can be represented in a syntactic way, this preliminary analysis does not rule out the option of strong pronouns being sensitive to person restrictions. A solution to this problem will be offered in Section 3.2 of Chapter 3, where the prosodic strength or weakness will not be tied directly to the presence or absence of a specific syntactic head (Σ), but to a particular set of syntax-prosody mapping principles.

2.4.3 Person restrictions with strong pronouns?

Before we can look for an explanation of why strong pronouns are not affected by person restrictions, we have to look at the cases that have been suggested to show otherwise. The most famous putative example of this is found in Icelandic (Sigurðsson 1990–1991, 1996; Taraldsen 1995; Anagnostopoulou 2003; Béjar and Řezáč 2003; Holmberg and Hróarsdóttir 2004; Sigurðsson and Holmberg 2008). Another example involves restrictions in copular constructions with two strong nominative pronouns in Polish (Bondaruk 2012), German (Heycock 2012; Coon, Keine, and Wagner 2017; Coon and Keine 2018) and Slovenian. I will argue below that while these types of restrictions do arise in similar syntactic contexts as person restrictions, they are radically different in character, as indicated by the strategies that can be used to void them and the fact that they can also affect features other than person.

2.4.3.1 The Icelandic nominative object restriction

In Icelandic quirky dative subject constructions, the verb can agree with a nominative object instead of a subject. This is shown in (87a,b), where the verb shows plural agreement with the 3P.PL object (noun phrase or pronoun). Interestingly, in such configurations, the verb cannot agree with a 1P or 2P person object, yielding an ungrammatical sentence when the nominative object is a strong 1P pronoun, as in (87c), or a strong 2P pronoun, as in (87d).

(87) Icelandic [*Germanic*]:

a. Henni líkuð**u** ekki *Þessar athugasemdir.* $\boxed{\text{DAT} \gg \text{3PL.NOM}}$
she.DAT liked.3PL not these.NOM comments.NOM
'She did not like these comments.' (Sigurðsson 1996, 22)

b. Henni líkuðu *Þeir*. $\boxed{\text{DAT} \gg \text{3PL.NOM}}$
she.DAT liked.3PL they.NOM
'She liked them.' (G. R. Harðarson, p.c.)

c. *Henni líkuð**um** *við*. $\boxed{\text{✗ DAT} \gg \text{1PL.NOM}}$
she.DAT liked.1PL we
'She liked us.'

d. *Henni leidduð**uð** *Þið*. $\boxed{\text{✗ DAT} \gg \text{2PL.NOM}}$
she.DAT liked.2PL you.NOM
'She liked y'all.' (Sigurðsson 1996, 27)

As noted by Anagnostopoulou (2003) and Béjar and Řezáč (2003), the syntactic context where the Icelandic restriction occurs is strikingly similar to the canonical case of the PCC, with a dative argument asymmetrically c-commanding an argument with structural case (cf. also the person restriction in Basque from Section 2.2).

However, the Icelandic restriction is also different in a number of ways. We can see that the restriction actually concerns the agreement on the verb and not the nominative pronoun with ECM-like constructions like (88a), where the quirky subject matrix verb optionally allows agreement with the downstairs nominative argument. Note that in such constructions the restriction does not occur when the verb shows default agreement, as shown in (88b).

(88) a. Mér { virt**i**st / virt**u**st } [*Þær* vera duglegar].
me.DAT { seemed.3SG / seemed.3PL } they to.be industrious.PL
'I found them to be industrious.' (Sigurðsson 1996, 23)

b. Henni { virt**i**st / *virt**um**st } [*við* vera duglegar].
her.DAT { seemed.3SG / seemed.1PL } we to.be industrious.PL
'She found us to be industrious.' (Sigurðsson 1996, 36)

Even more tellingly, the use of 1P and 2P objects becomes tolerated by speakers when the agreement on the verb is syncretic for 1P, 2P, and 3P, as (89) shows.

(89) ?Henni leidd**i**st { *ég* / *Þú* }.
her.DAT bored.at.1/2/3 { I / you.NOM }
'She was bored by me/you.' (Sigurðsson 1996, 29)

As we will see below, syncretism does not void true person restrictions, which is expected if the latter are in fact purely syntactic in nature. This also suggests that at least some aspect of the Icelandic restriction actually concerns the morphological realization of agreement on the verb and not the person features on the pronoun.

A case of agreement syncretism is also trivially found with non agreeing forms such as infinitives, and as (90) illustrates, 1P and 2P objects are also tolerated in those cases.[27]

(90) ?Hún vonaðist auðvitað til [að leiðast { *við* / *Þið* / *Þeir* } ekki mikið].
she hoped of.course for to find.boring.INF { we / you.NOM / they } not much
'She of course hoped not to find us/you/them very boring.'
(Sigurðsson and Holmberg 2008, 271)

Sigurðsson and Holmberg (2008) note that this is only possible with some verbs. For example, 'like' (*líka*) as opposed to 'be bored by' (*leiðast*) does not tolerate 1/2P objects even in these contexts. And there is a good reason for that, as 'like' generally does not take human objects, whereas 'be bored by' does, as shown in (91). Since 1/2P pronouns must take human referents, the difference between the two verbs is not coincidental.

(91) a. Mér lík**ar** *bókin*. $\boxed{\text{DAT} \gg [-\text{human}]}$
I.DAT like.3SG book.the.NOM
'I like the book.'

b. *Mér lík**ar** *stúlkan*. $\boxed{\text{✗ DAT} \gg [+\text{human}]}$
I.DAT like.3SG girl.the.NOM
'I like the girl.'

c. Mér leidd**i**st *stúlkan*. $\boxed{\text{DAT} \gg [+\text{human}]}$
I.DAT was.bored.with.3SG girl.the.NOM
'The girl bores me.' (Taraldsen 1994, 44–5)

Although there is disagreement with respect to the source of this additional restriction (see Taraldsen 1994; Maling and Jónsson 1995 for different proposals), it seems to target a specific lexical class of verbs—not a specific argument structure. This additional restriction

27. However, there seems to be speaker variation with respect to this; Bobaljik (2008, 318–9n27) notes that Höskuldur Thráinsson (p.c. to Bobaljik) still finds 1/2P nominative objects ungrammatical in such cases.

is often overlooked or not controlled for by researchers arguing for a unification of the restrictions on Icelandic nominative objects with person restrictions.

The point here is that the Icelandic restriction, although arising in a similar syntactic context, shows sensitivity to factors that true person restrictions do not (see below). I suggest therefore that it is an altogether different phenomenon: a complex interaction between restrictions on the realization of verbal agreement in contexts with more than one agreement controller, as suggested by the sensitivity of the restriction to syncretic agreement forms and lack of agreement (see Burzio 2000; Schütze 2003; Boeckx 2008; Ussery 2013; Stegovec 2016; Ackema and Neeleman 2018; Coon and Keine 2018 for different implementations).[28] The additional selectional restriction of some verbs for non-human objects (Taraldsen 1994; Maling and Jónsson 1995) is also unlike what we observe in person restrictions, where the key factors are the structural configurations of the arguments and the type of pronouns involved. For example, it is not the case with the PCC in Greek that a 1/2P Theme object is simply impossible with some ditransitive verbs, it is specifically impossible when the Theme is a clitic asymmetrically c-commanded by a Goal clitic—when the Theme is a strong pronoun, there is no restriction (see Anagnostopoulou 2003, 253; Section 1.1).

The sensitivity to syncretism is an especially important distinction between the two phenomena. As discussed by Adger and Harbour (2007), based primarily on data from Kiowa [*Kiowa-Tanoan*], true person restrictions crucially do not show sensitivity to syncretic forms. In Kiowa, the person restriction is a STRONG one between internal arguments, so the DO must be 3P in the presence of an IO. All the pronominal markers are fused into one verbal prefix (although morphological decomposition is possible in most cases). Crucially,

28. An indirect argument for the restriction being about agreement and not the pronoun can be made by way of comparison with quirky subject constructions in Faroese (Schütze 2003) and Finnish (Kiparsky 2001; Řezáč 2006). In the closely related Faroese, there is no agreement with the object in such constrictions because the object is always accusative and there is no agreement with accusative arguments in the language; crucially the object can always be 1/2P. In Finish, the case on the object alternates between nominative, if the object is 3P, and accusative, if the object is 1/2P; again there is no restriction because there is no agreement with accusative arguments. The minimal difference in case/agreement thus correlates with the absence of the restriction.

plural animate IOs are never morphologically realized within the prefix, as shown in (92a). One could imagine in this case that since the IO is never expressed, there should be no restriction on the DO. But this is not what is observed, as (92b) shows. Despite the identical form of the prefix, the DO can only be 1/2P if the IO is realized as a PP, as shown in (92c); note that PPs are crucially never registered by the prefix in Kiowa.

(92) Kiowa [*Kiowa-Tanoan*]:

a. **Gya-** kɔ́ɔ́gya. — $\boxed{\text{3PL.IO} \gg \text{3.DO}}$
1(>3AN.PL)>3- got
'I got **it** (for **them**).'

b. **Kɔ́ígú* **em-** pɔɔhíítɔɔ. — $\boxed{\text{✗ 3PL.IO} \gg \text{2.DO}}$
Kiowas 1(>3AN.PL)>2- bring.FUT
'I will take **you** to **them**, the Kiowas.'

c. Kɔ́ígú-em **em-** pɔɔhíítɔɔ. — $\boxed{\text{2.DO} + \text{PP}}$
Kiowas-LOC 1>2- bring.FUT
'I will take **you** to the Kiowas.' (Adger and Harbour 2007, 10–1)

A similar case can be made based on Swahili data [*Bantu*] (Riedel 2009). In Swahili, only the IO is registered by an object marker on the verb in double object constructions (cf. (93a), (94)), but the DO is nonetheless restricted in person according to a WEAK person restriction: if the IO is 3P, the DO must also be 3P, as is illustrated by the ungrammatical (93b).[29]

(93) Swahili [*Bantu*]:

a. N-li- **ku-** onyesha *Juma*. — $\boxed{\text{2.IO} \gg \text{3.DO}}$
1.SU-PAST- 2.O show Juma
'I showed **him** [Juma] to **you**.'
'*I showed **you** to **him** [Juma].'

b. *N-li- **mw-** onyesha *Juma wewe*. — $\boxed{\text{✗ 3.IO} \gg \text{2.DO}}$
1.SU-PAST- 3.O show Juma you
'I showed **you** to **him** [Juma].'
'I showed **him** [Juma] to **you**.'

29. The translation for (93b) is changed from the original in Riedel 2009; Swahili consistently doubles the IO with the object marker on the verb and not the DO, as the original translation suggests. Based on Riedel's prose accompanying the example, the translation for this example is most likely a typo.

(94) a. A-li- **ni-** onyesha *wewe*. $\boxed{1.\text{IO} \gg 2.\text{DO}}$
3.SU-PAST- 1.O show you
'He showed **you** to **me**.'

b. A-li- **ku-** onyesha *mimi*. $\boxed{2.\text{IO} \gg 1.\text{DO}}$
3.SU-PAST- 2.O show me
'He showed **me** to **you**.' (Riedel 2009, 151–2)

Similarly to the Kiowa case, the DO can be 1/2P in the context of a 3P IO only if one of the objects is a PP, which is never registered by an object marker (Riedel 2009, 152).

In both the Kiowa and Swahili cases, we see examples where the pronominal marking is syncretic by virtue of not being overtly expressed. This shows a clear contrast with the Icelandic restriction, where syncretic agreement morphology can void the restriction. Additionally, such cases are also problematic for analyses that ascribe person restrictions to a failure of multiple clitic doubling, such as Nevins (2011) and Coon and Keine (2018). That is to say, if the restriction arises because certain person combinations cause a failure of clitic doubling, then cases where there is no multiple clitic doubling to begin with should not give rise to person restrictions. What would have to be posited in examples like Kiowa or Swahili is some sort of covert clitic doubling, but at least Coon and Keine (2018) assume that lack of an overt exponent is actually one of the ways to void a person restriction.

Another possibility for Swahili at least, is that the person restriction actually applies to the strong 1/2P DO pronouns, which is actually what Riedel (2009) suggests. This would make the Swahili person restriction like what the Icelandic restriction has been argued to be. However, what this analysis would have to downplay is that the strategy for voiding the person restriction—making one of the objects a PP—also in general blocks objects from being doubled by an object marker on the verb. Furthermore, in other Bantu languages, as Riedel (2009) also notes, person restrictions can be observed between two visible co-occurring object markers and can be voided with the same strategy. If we want to maintain a unified analysis of person restrictions across Bantu, then we would have to say that Swahili only differs in having a null DO marker in double object constructions.

2.4.3.2 Restrictions in copular constructions

Moving to the other case where person restrictions have been claimed to arise with strong pronouns, it has been observed for Polish (Bondaruk 2012) and German (Heycock 2012; Coon, Keine, and Wagner 2017; Coon and Keine 2018) that in copular constructions where both DPs/NPs bear nominative case, and are because of that potential agreement controllers, restrictions on agreement similar to person restrictions arise. In German, for example, given a 1/2P DP and a 3P DP, a copula must always agree with the 1/2P DP, as shown in (95)–(96) (in the boxes summarizing the examples, the agreement controller is in bold).

(95) German [*Germanic*]:

a. *Du* **bist** das Problem.
you are.2 the problem
$\boxed{\mathbf{2} \gg 3}$

b. *Du **ist** *das Problem*.
you is the problem
'You are the problem.'
$\boxed{✗\, 2 \gg \mathbf{3}}$

(96) a. Das Problem **bist** *du*.
the problem are.2 you
$\boxed{3 \gg \mathbf{2}}$

b. **Das Problem* **ist** du.
the problem is you
'The problem is you.' (Coon, Keine, and Wagner 2017, 205)
$\boxed{✗\, \mathbf{3} \gg 2}$

When the copular construction takes the form of a so-called assumed identity sentence (Heycock 2012), the first DP is the presumed subject of the construction. While such sentences might require the appropriate context to be semantically felicitous (e.g. one person playing another in a game, play, or movie), the two DPs may be pronouns of different person values. In such cases, the object pronoun cannot be 1/2P if the subject that agrees with the copula is 3P, as illustrated in (97). Crucially, such restrictions do not only operate over different person values, as the object may not be plural in such constructions when the subject that agrees with the copula is singular, as the examples in (98) show.

(97) a. *Ich* **bin** er. $\boxed{\mathbf{1} \gg 3}$
I am he
'I am him.'

b. **Er* **ist** ich. $\boxed{\text{✗}\,\mathbf{3} \gg 1}$
he is I
'He is me.'

(98) a. *Sie* **sind** er. $\boxed{\mathbf{PL} \gg \text{SG}}$
they are he
'They are him.'

b. **Er* **ist** sie. $\boxed{\text{✗}\,\mathbf{SG} \gg \text{PL}}$
he is they
'He is they.'

(Coon, Keine, and Wagner 2017, 206)

A similar restriction arises in Slovenian, although it patterns even less with true person restrictions, suggesting that perhaps the restrictions on copular agreement are not as closely related to true person restrictions as Coon, Keine, and Wagner (2017) and Coon and Keine (2018) suggest. The basic restriction in assumed identity sentences is given in (99)–(100). It can be described as follows: if one pronoun is 1/2P or non-singular and the other is 3P or singular, then the former controls agreement with the copula, unless the later is the subject.

(99) <u>Slovenian [*Slavic*]:</u>

a. *On* **bo** jaz. $\boxed{\mathbf{3} \gg 1}$
he FUT.3 I
'He will be me.'

b. *Jaz **bo** *on*. $\boxed{\text{✗}\,1 \gg \mathbf{3}}$
I FUT.3 he
'I will be him.'

c. On bo**m** *jaz*. $\boxed{3 \gg \mathbf{1}}$
he FUT.1 I
'He will be me.'

d. *Jaz* bo**m** on. $\boxed{\mathbf{1} \gg 3}$
I FUT.1 he
'I will be him.'

(100) a. *On* **bo** onadva. $\boxed{\mathbf{SG} \gg \text{DU}}$
he FUT.3 they.DU
'He will be the two of them.'

b. *Onadva **bo** *on*. $\boxed{\text{✗}\,\text{DU} \gg \mathbf{SG}}$
they FUT.3 he
'The two of them will be him.'

c. On bo**sta** *onadva*. $\boxed{\text{SG} \gg \mathbf{DU}}$
he FUT.3DU they.DU
'He will be the two of them.'

d. *Onadva* bo**sta** on. $\boxed{\mathbf{DU} \gg \text{SG}}$
they.DU FUT.3DU he
'The two of them will be him.'

The key difference from the German pattern is that being the linearly first pronoun (possibly analyzed as the subject) can override the requirement that a 1/2P or non-singular pronoun be the controller. In other words, the agreement controller is picked based on two conditions: (i) agree with the most specified value (assuming that 1/2P is more specified than 3P, and non-singular more specified than singular), or (ii) agree with the closest nominative element. When neither is met, the result is ungrammatical. Additionally, when both pronouns are 1/2P and condition (i) cannot be met by either pronoun, condition (ii) is the only option, as illustrated by the examples in (101), where the copula always agrees with the first pronoun.

(101) a. *Jaz* bo**m** ti. $\boxed{\mathbf{1} \gg 2}$
I FUT.1 you
'I will be you.'

b. *Ti bo**m** *jaz*. $\boxed{✗\, 2 \gg \mathbf{1}}$
you FUT.1 I
'You will be me.'

c. *Ti* bo**š** jaz. $\boxed{\mathbf{2} \gg 1}$
you FUT.2 I
'You will be me.'

d. *Jaz bo**š** *ti*. $\boxed{✗\, 1 \gg \mathbf{2}}$
I FUT.2 you
'I will be you.'

We can see at this point the observed restriction is starting to look less and less like a WEAK person restriction. The second pronoun *can* be 1/2P if the first one is 3P, as long as the 3P one agrees with the copula. Similarly, the second pronoun *cannot* be 1/2P if the first one is also 1/2P person, as long as the first pronoun is the one agreeing with the copula.

Finally, just like the Icelandic restriction, the Slovenian restriction in copular constructions is voided with syncretic agreement forms. In Slovenian, 2P and 3P agreement morphology is always syncretic in the dual. Thus, in (102), with dual 3P and 2P pronouns, there is no observed restriction regardless of the order of the pronouns.

(102) a. Onadva bo**sta** vidva. $\boxed{3\text{DU} \gg 2\text{DU}}$
they.DU FUT.2/3 you.DU
'The two of them will be the two of you.'

b. Vidva bo**sta** onadva. $\boxed{2\text{DU} \gg 3\text{DU}}$
you.DU FUT.2/3 them.DU
'The two of you will be the two of them.'

Coon and Keine (2018) observe a similar effect in German, where subjunctive agreement is syncretic between 3P and 1P. They also observe that the restriction disappears with non agreeing forms like infinitives—like in Icelandic (in Slovenian such examples cannot be constructed with infinitives due to independent complications). I take these similarities between the copular clause restrictions and the agreement restrictions in Icelandic, in addition to the copular restrictions also applying to number, to indicate that we are dealing here with an altogether different kind of phenomenon than the true person restrictions observed in combinations of weak pronouns (see Coon and Keine 2018 for a similar point).

2.4.4 Summary of pronoun type discussion

To sum up, not all phonologically weak pronouns or agreement markers are subject to person restrictions, but all those that are, are phonologically weak. Furthermore, sensitivity to person restrictions correlates with specific behavior with a range of binding phenomena, suggesting a connection between person restrictions and the referential status of the restricted elements. Based on this, I will refer to all elements subject to person restrictions as pronouns from now on, regardless of their realization as affixes, clitics, or other weak forms.

What requires an explanation then is what kind of syntactic representation for the pronouns can simultaneously be responsible for the special behavior with respect to these seemingly unrelated phenomena and why this syntactic property always correlates with a phonologically weak status of the pronouns. An analysis of the relationship between person restrictions, binding, and phonologically weak pronouns will be given in Chapter 3.

2.5 Two crosslinguistic gaps and their implications

So far we have established that by looking across all the different incarnations of person restrictions, we find a number of crosslinguistic commonalities concerning their format and the elements affected by them. One such commonality is the existence of the STRONG vs. WEAK restriction variation. While this does highlight the fact that person restrictions vary from language to language, it is also a striking revelation that the same variation is present in a number of phenomena that can superficially look very different; for example, canonical person restriction cases, direct/inverse systems, and person-based argument indexing. In this section, we take an additional step further and show that the variation is more limited than one might expect; that is, more limited than what should be logically possible.

As noted in Chapter 1, a lot of recent work on person restrictions has focused on deriving the attested crosslinguistic variation. It is of course important to have an analysis which can account for all the variation, but we must also be careful not to introduce theoretical possibilities that over-generate with respect to the attested patterns (i.e. derive unattested patterns in addition to the attested ones). The goal of this section is to establish, in a broad sense, what kind of theories of crosslinguistic variation we should not pursue in the area of person restrictions. I again take a comparative approach; I will first establish two crosslinguistic gaps and demonstrate that the existing analyses do not provide us with a way to explain the existence of the gaps in a principled way.

Recall that STRONG and WEAK variants of person restrictions are attested in both the EA-IA and IA-IA domains. When a language has either just an EA-IA or an IA-IA restriction, the person restriction may conform to either format. However, in some languages, EA-IA and IA-IA restrictions can exist side-by-side. In those cases, not all logically attested combinations of STRONG and WEAK restrictions are attested. In particular, as will be established below, the crosslinguistically attested combinations of EA-IA and IA-IA restrictions in terms of their strength conform to the new generalization in (103).

(103) *Strength Generalization*
If a language has person restrictions in both the EA-IA and IA-IA domain, the IA-IA person restriction cannot be weaker than the EA-IA person restriction.

What this means is that there are no languages that can have an EA-IA restriction that is STRONG and an IA-IA restriction that is WEAK.

In addition to the strength gap described by the *Strength Generalization*, there exists another crosslinguistic gap which concerns the *direction* of the person restriction; namely, which of the two pronouns in the affected pair is the one to which the person restriction applies. Recall that we have seen above both languages where the person restriction always has the canonical form, like the PCC in Greek, where given a Goal pronoun and a Theme pronoun, the Theme is restricted, and languages where the canonical restriction exists alongside a corresponding reverse restriction, like the reverse PCC in Slovenian, Maasai, or Alutor, where given a Goal pronoun and a Theme pronoun, the Goal is restricted. Curiously, as will be established below, reverse person restrictions are limited only to such cases where they coexist with their canonical counterpart, revealing the following new generalization:

(104) *Direction Generalization*
If a language has a *reverse* person restriction in the EA-IA or IA-IA domain, it must also have a *canonical* counterpart of the same person restriction.

In other words, no language can only have a reverse person restriction either in the EA-IA or IA-IA domain, which means that there are no languages that would consistently (i.e. only) restrict the EA in the presence of an IA or the Goal in the presence of a Theme.

In the rest of this section I first establish the existence of the two crosslinguistic gaps and then discuss the implications these gaps have for the analysis of person restrictions.

2.5.1 The strength gap

Earlier, in Sections 2.2 and 2.3, we saw that EA-IA and IA-IA restrictions can both be either STRONG or WEAK. But we have not considered cases where EA-IA and IA-IA restrictions coexist in one language. In fact, Arizona Tewa, a language where we only looked at the EA-IA restriction, also has a person restriction in the IA-IA domain (Kroskrity 1977, 1985):

outside of passives and prepositional datives, whenever there is an EA, the IA must be 3P (applies to the Goal in ditransitives), and whenever there is a Goal, the Theme must be 3P. That is, Arizona Tewa has a STRONG restriction in both domains.

The related language Southern Tiwa also has both an EA-IA and an IA-IA restriction (Allen and Frantz 1978; Rosen 1990), but with one major difference: the EA-IA restriction is a WEAK one. The pattern is illustrated by the examples in (105); note that just as with other Kiowa-Tanoan languages, the pronominal marking is all encoded in one pronominal prefix.

(105) Southern Tiwa [*Kiowa-Tanoan*]: **WEAK EA-IA**

a. { **ti-** / **a-** / **Ø-** } mũ-ban. 1/2/3.EA ≫ 3.IA
{ 1>3- / 2>3- / 3>3- } see-PAST
'**I/You/He** saw **him**.'

b. **i-** mũ-ban. 1.EA ≫ 2.IA
1>2- see-PAST
'**I** saw **you**'

c. **bey-** mũ-ban. 2.EA ≫ 1.IA
2>1- see-PAST
'**You** saw **me**.'

(106) *seuan **???-** mũ-ban. ✗3.EA ≫ 1/2.IA
man.A 3>1/2- see-PAST
'**He** [the man] saw **me/you**.' (Allen and Frantz 1978, 11–2)

Just as in Arizona Tewa, the banned person combinations can be realized in Southern Tiwa via a passive construction, as shown in (107) (cf. (106)). The main difference is that the passive must be used to void the person restriction with *fewer* person combinations.

(107) a. seuanide-ba **te-** mũ-che-ban. 1.IA.SU + 3.EA
man.A-INSTR 1.SU- see-PASS-PAST
'**I** was seen by the man.'

b. seuanide-ba **a-** mũ-che-ban. 2.IA.SU + 3.EA
man.A-INSTR 2.SU- see-PASS-PAST
'**You** were seen by the man.' (Allen and Frantz 1978, 11–2)

In contrast, the IA-IA restriction in Southern Tiwa is a STRONG restriction just like in Arizona Tewa. Whenever a Goal argument is present, and it is not expressed as a PP, the Theme must be 3P. The basic restriction pattern in ditransitives is illustrated in (108)–(109). Note that the provided examples are not sufficient to show that we are truly dealing with a STRONG restriction here; we will examine additional examples required to show that below.[30]

(108) Southern Tiwa [*Kiowa-Tanoan*]: **STRONG IA-IA**

a. 'uide **tow-** wia-ban. 1.EA ≫ 3.GL ≫ 3.TH
   child 1>3>3PL- give-PAST
   '**I** gave **them** to **him/her** [the child].'

b. **ow-** wia-ban. 2.EA ≫ 3.GL ≫ 3.TH
   2>3>3PL- give-PAST
   '**You** gave **them** to **him/her**.'

c. **kam-** musa- wia-ban. 1.EA ≫ 2.GL ≫ 3.TH
   1>2>3PL- cat(PL)- give-PAST
   '**I** gave **them** [the cats] to **you**.'

d. **bow-** wia-ban. 2.EA ≫ 1.GL ≫ 3.TH
   2>1>3PL- give-PAST
   '**You** gave **them** to **me**.'

(109) a. ***???-** wia-ban. ✗1.EA ≫ 3.GL ≫ 2.TH
   1>3>2- give-PAST
   '**I** gave **you** to **him/her/them**.'

b. ***???-** wia-ban. ✗2.EA ≫ 3.GL ≫ 1.TH
   2>3>1- give-PAST
   '**You** gave **me** to **him/her/them**.' (Rosen 1990, 672–3, 677)

Importantly, this restriction pattern, with a WEAK EA-IA restriction and a STRONG IA-IA restriction, is not limited to Southern Tiwa. But before we can look at why the Southern Tiwa pattern is important, let us consider where it stands in relation to all other crosslinguistically attested EA-IA and IA-IA restrictions along the STRONG/WEAK parameter. The distribution of the different restriction strengths of the person restrictions across the languages in the survey is summarized in Table 2.2; from the 106 languages in the language sample there

30. Another complication is the **ergative-dative constraint* (Rosen 1990), which bans ergative subjects and IOs from co-occurring. The ban occurs regardless of the person of IO and DO—it is only indirectly connected to person in that ergative markers are limited to 3P. It is possible, if both ergative and dative are actually inherent cases (see Woolford 1997, 2006), that the ban actually targets co-occurring inherent case-marked pronouns. In any case, it does not seem to be a person restriction per se.

are 114 data points concerning the STRONG or WEAK status of the restrictions due to the cases where there is speaker variation language-internally (e.g. in Spanish, Catalan, Italian, or Slovenian, as discussed in Section 2.2).[31] The languages in the table are first categorized based on the strength of the EA-IA restriction and secondly by the IA-IA restriction. The options for each domain are: ∅ (no restriction), WEAK, or STRONG.

Table 2.2: Languages according to restriction strength in EA-IA and IA-IA domains

	Total: **114**								
EA-IA	∅	WEAK	STRONG	∅	WEAK	STRONG	∅	WEAK	STRONG
IA-IA	STRONG	STRONG	STRONG	WEAK	WEAK	WEAK	∅	∅	∅
	Spanish	Maasai	Tangut	Spanish’	Alutor		German’	Hungarian	Kurdish
	French	Chukchi	Senaya	Catalan’	Koryak		Dutch’	Chepang*	Mupudungun*
	Catalan	Cherokee	Telkepe	Italian’			Swedish’	Jyarong	Eastern Mansi
	Italian	Sahaptin	Arizona Tewa	Romanian			English’	Nocte	Tundra Nenets
	English	Algonquin	Oaxaca Zapotec	German			Icelandic	Pashto	Christian Barwar
	Slovenian	Blackfoot		Zürich German			Polish	Kashmiri*	Kaqchikel
	Bulgarian	Plains Cree		Swiss German			Khanty	Bella Coola	Labrador Inuttut*
	Macedonian	Delaware		Dutch			Abkhaz*	Clallam	
	Greek	Meskwaki/Fox		Swedish			Tagalog*	Lummi	
	Albanian	Mi’kmaq		Slovenian’			Manambu*	Halkomelem	
	Iron Ossetic	Ojibwe		B-C-S			Itelmen*	Squamish	
	Basque	Passamaquoddy		Czech			Class. Nahuatl*	Arapaho	
	Cairene Arabic	Southern Tiwa		Bulgarian’			Lushootseed	Zuni	
	Maltese	Picuris		Macedonian’			Kutenai		
	Migama	Quechua		Iron Ossetic’			Koyukon		
	Baraïn	Paraguayan Guaraní		Digor Ossetic			Navajo		
	Sambaa			M. S. Arabic			Inuktitut*		
	Nyaturu			Class. Arabic					
	Georgian			Sambaa’					
	Kambera			Haya					
	Manam			Swahili					
	Yimas			Limbum					
	Monumbo			Lai					
	Warlpiri			Djaru					
	Mohawk*								
	Takelma								
	Kiowa								
	Tetelcingo Nahuatl								
	O’odham								
	Tzotzil								
	30	**16**	**5**	**24**	**2**	**0**	**17**	**13**	**7**

31. Conventions used in the table: prime (’) marks languages present in two groups due to speaker variation or variation in strength across different constructions; an asterisk (*) marks languages reported with conflicting patterns in the literature or confounds suggesting that the reported pattern might not be accurate (see Appendix). Crucially, the different options do not invalidate the generalizations discussed in Sections 2.5.1 and 2.5.3.

The person restrictions attested with the languages in Table 2.2 include all the different types discussed in Sections 2.2 and 2.3; in the case of coexisting EA-IA and IA-IA restrictions, the two restrictions may even be of different types. For example, Maasai [*Nilo-Saharan*] has a canonical WEAK EA-IA restriction that takes the form of a direct/inverse system and a canonical + reverse STRONG IA-IA restriction that takes the form of a person-based argument indexing restriction (see Lamoureaux 2004; Section 2.3). Most Algonquian languages, like Ojibwe/Nishnaabemwin, on the other hand, have a WEAK EA-IA restriction that shows both properties of a canonical + reverse restriction and a direct/inverse system (see Section 2.5.3) alongside a canonical STRONG IA-IA restriction. But the restrictions in the two argument domains can also be exactly of the same type: for example, in Telkepe [*Neo-Aramaic*] (Kalin 2014b) both the EA-IA and the IA-IA restriction are canonical and STRONG, while in Tangut [*Quiangic*] (Kepping 1979) both the EA-IA and the IA-IA restriction are canonical + reverse and take the form of person-based argument indexing.

Despite of all the variation, the possibilities for co-occurring EA-IA and IA-IA restrictions are limited. The Southern Tiwa pattern we saw above, where the EA-IA restriction is WEAK and the IA-IA restriction is STRONG (henceforth WEAK-STRONG), is actually by far the most common combination crosslinguistically. The WEAK-STRONG combination is also attested with a number of different combinations of restriction types; two cases that we saw above are Ojibwe and Maasai, where the WEAK EA-IA restriction is part of a direct/inverse system and the STRONG IA-IA restriction is just a plain person restriction. The ubiquitous nature of this combination in spite of the variation in the type of the two restrictions, provides additional support for all the types being the same phenomenon underneath. The other attested combinations involve two coexisting restrictions of the same strength, such as in Telkepe and Tangut (see above), where both restrictions are STRONG (henceforth STRONG-STRONG), or Alutor [*Chukotkan*] (Mel'čuk 1988), where both restrictions are WEAK (henceforth WEAK-WEAK). Note that both of these restriction combinations, especially WEAK-WEAK, are extremely rare. But most importantly, there is a combination that is completely unattested: a pattern where the EA-IA restriction is STRONG and the IA-IA restriction is WEAK (henceforth STRONG-WEAK). All the crosslinguistic possibilities and the attested number of languages for each combination are summarized in Table 2.3.

Table 2.3: Number of languages with restrictions in both EA-IA and IA-IA domains

IA-IA \ EA-IA	WEAK	STRONG
WEAK	2	**0**
STRONG	16	5

The generalization describing the possible and impossible languages with respect to the parameters on restriction strength and affected arguments can therefore be stated as:

(110) *Strength Generalization*
If a language has person restrictions in both the EA-IA and IA-IA domain, the IA-IA person restriction cannot be weaker than the EA-IA person restriction.[32]

It should be acknowledged that the language sample, although relatively large, does not allow us to make claims about the statistical significance of the typological gap. However, as will will see in the later chapters, the attested restriction patterns have independent grammatical correlates; a closer inspection of how EA-IA and IA-IA restrictions interact with independent phenomena both in isolation and when they co-exist strongly suggests that the

32. Note that I have so far only distinguished between STRONG and WEAK restrictions; as will be discussed in Section 4.3, there is additional variation within the WEAK class—the label WEAK then covers all cases where the restricted pronoun is not always 3P, but its person is restricted based on the person of the other pronoun. The *Strength Generalization* can then be interpreted in two ways: (i) the IA-IA restriction cannot be weaker than the EA-IA restriction, or (ii) the IA-IA restriction cannot be WEAK if the EA-IA restriction is STRONG. The only case for which this distinction matters are WEAK-WEAK languages, and there are only two languages of this type in my sample: Alutor and Koryak [*Chukotko-Kamchatkan*]. As we will see in Sections 4.1.5 and 4.3, there are two ways to analyze their pattern, and neither makes Alutor and Koryak counterexamples to (i) or (ii).

generalization is valid. Pending further motivation to be provided below, I will treat the *Strength Generalization* as valid and consider its implications.

Additionally, I will treat differences in the number of attested patterns as a tentative approximation of the actual differences in frequency despite the small size of the sample. Languages with only an IA-IA restriction are easily found, and so are languages with a WEAK EA-IA restriction, or for that matter a WEAK-STRONG restriction combination. In contrast, finding languages with a STRONG EA-IA restriction, a STRONG-STRONG, or a WEAK-WEAK combination is rather difficult.

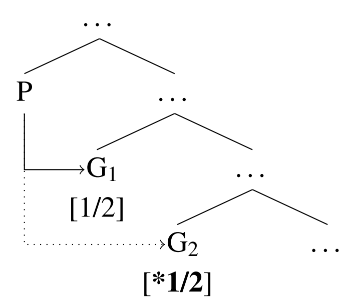
2.5.2 Implications of the strength gap

Having identified the curious crosslinguistic gap, we should consider its theoretical implications by looking at whether the existing analyses of person restrictions have a way of explaining its existence. Due to the arguments for the fundamentally syntactic character of person restrictions provided in Section 2.2, I will limit myself only to the current mainstream syntactic approach, although similar issues arise for other analyses as well. The mainstream approach to person restrictions involves reducing them to a syntactic intervention effect (see Anagnostopoulou 2003, 2005; Béjar and Řezáč 2003, 2009; Nevins 2007, 2011; Adger and Harbour 2007; i.a.). Although individual analyses differ regarding the specific implementation of how and why the person restriction arises, their common core is that the restriction is tied to a failure of establishing an *Agree* relation (Chomsky 2000) between a verbal/inflectional syntactic head and multiple arguments within its syntactic domain.

2.5.2.1 The intervention approach to person restrictions

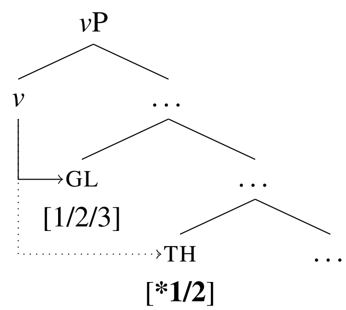
For presentation purposes, I will illustrate the principles behind the intervention approach to person restrictions with a bare-bones prototypical version of the approach. Crucially, the differences between individual intervention analyses currently on the market do not have a bearing on the points made below. The basic idea is that the successful establishment of an Agree relation between a probe (P) and a goal (G) is a prerequisite for the goal to be able to have 1/2P features. Since Agree is assumed to apply only between P and the closest matching G it c-commands, a configuration with one P and two Gs, like (111), will ensure that G_1 structurally intervenes between P and G_2.

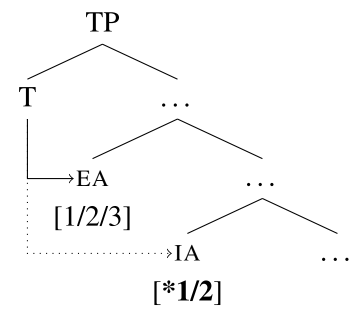
(111)



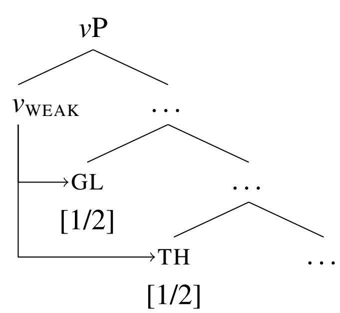
Since a successful Agree relation with P is a prerequisite for an argument to bear 1/2P features, G_2 will be restricted to 3P (due to the blocking effect of G_1), thus yielding a STRONG person restriction pattern between G_1 and G_2. A nice result of the intervention approach is that it offers a straightforward explanation for why person restrictions consistently target the structurally lowest pronoun in the pair.

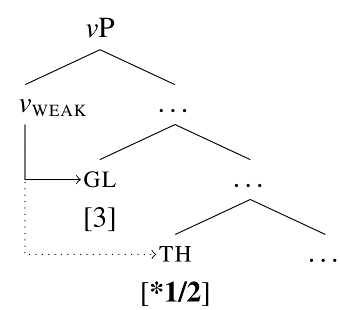
However, in order to derive the variation in the domain of the person restriction (EA-IA vs. IA-IA), we need to assume that different syntactic heads can play the role of the relevant probe. In the case of IA-IA restrictions, the probe is most commonly assumed to be the *v* head (Chomsky 2000), associated with the introduction of the EA and case assignment to the IAs. Since this head is assumed to asymmetrically c-command both the Goal and Theme in ditransitives, it fits the role of the probe responsible for IA-IA restrictions (see Anagnostopoulou 2003, 2005; Béjar and Řezáč 2003), which can be derived as illustrated in (112). Turning to the case of EA-IA restrictions, here we need a probe that asymmetrically c-commands both the EA and IA, with EA intervening between it and the IA. A good candidate in this respect is the T head (also I or INFL), typically associated with subject agreement and nominative case assignment (Chomsky 2000). The person restriction can in this case be derived as shown in (113) (see Nevins 2011; Kalin and van Urk 2015 for approaches where T is assumed to be responsible for EA-IA person restrictions).

(112)


(113)


Finally, and most importantly for the issue at hand, we need a way to derive the STRONG VS. WEAK restriction variation. The standard way to achieve this involves parameterizing the probe so that with the "STRONG setting" it behaves like we described above, while with the "WEAK setting" it exceptionally allows Agree between the probe and both arguments—but only when both of them are 1/2P, as shown in (114) (see Anagnostopoulou 2005; Nevins 2007, 2011). The "WEAK setting" requires the adoption of some version of *Multiple Agree* (Hiraiwa 2001, 2005), but constraining it in a way that it will be blocked when the closest goal is 3P, as shown in (115), making regular Agree the only option in that case.

(114)


(115)


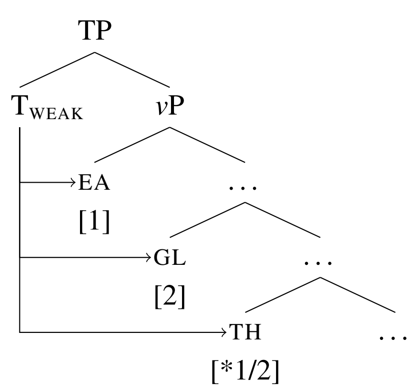
We now have the sufficient analytic tools to explore the implications of the *Strength Generalization* for syntactic intervention approaches to person restrictions.

Consider the possibilities at our disposal for varying restriction strength. We can vary the identity of the relevant probe, which is required to get both EA-IA and IA-IA restrictions, and we can vary the ability of the probe to establish Agree with multiple arguments, which is required to get STRONG and WEAK person restrictions. For languages with both EA-IA and IA-IA restrictions we therefore have two options: (i) relegating all the Agree-licensing responsibilities to T, or (ii) splitting the Agree-licensing responsibilities between T and *v*. Let us look at how option (i) fairs in relation to the *Strength Generalization* first.

2.5.2.2 One parameter for both domains

Assuming that the probe on T can probe all of the arguments it c-commands, which in a ditransitive includes the EA and two IAs (see Nevins 2011 who proposes that all three are accessible because the IAs move to the edge of *v*P), then we can derive STRONG-STRONG and WEAK-WEAK combinations by parameterizing T as either a "STRONG" or a "WEAK" probe respectively. A complication arises though if we try to derive a WEAK-STRONG combination. We can in principle set T to be a "WEAK" probe, deriving a WEAK restriction in the EA-IA domain, but in order to restrict the Theme to 3P we need a mechanism independent of Agree, since, as shown in (116), a "WEAK" probe must establish Agree with all 1/2P goals in its domain—only a 3P goal will induce an intervention effect (see above).

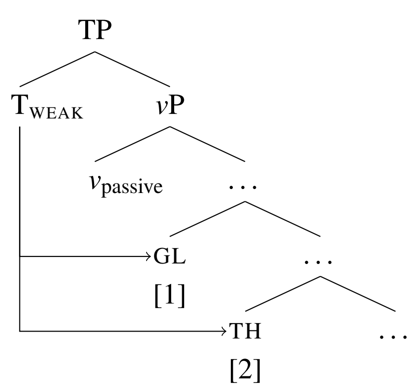
(116)



One option to salvage this approach is to stipulate that in (116) only the Theme is inaccessible to the probe, but this comes dangerously close to just restating the facts. Remember that WEAK-STRONG is by far the most common combination crosslinguistically, so we would want a principled explanation for this contextual inaccessibility, if it is at all possible. Another option would be to say that the additional restriction is not a person restriction per se, but a binding restriction. Since in (116) we have three 1/2P pronouns, two will inevitably co-refer and induce a violation of *Binding Principle B* (Chomsky 1981; cf. '**I* introduced *me* to you'); as long as the two topmost pronouns have different person values they will be unaffected by the restriction, but a third 1/2P pronoun will yield an ungrammatical result.[33]

In this type of analysis, where the Theme cannot be 1/2P due to Principle B because the two other arguments are already 1P and 2P respectively, the restriction on the Theme is predicted to only apply with three arguments. If we were to remove an argument out of the picture, the restriction should change. For example, if we were to passivize the clause and remove the EA, the IA-IA restriction should become a WEAK one, as shown in (117).

(117)



This is, however, not what we observe in the cases where we can check this prediction. In addition, addressing the prediction in question will also resolve a question we left open in

33. A possible workaround for the binding violation would be to make one of the objects plural, which should at least marginally improve the grammaticality of such cases (cf. '?*I* introduced *us* to you' in English). It is hard to test this prediction on a typological study of the scale of the current one; however, none of the sources I used indicate that this is a strategy that can void the person restriction on the Theme. We will also see a better argument against the analysis under consideration next.

relation to Southern Tiwa back in Section 2.5.1: how can we be sure the IA-IA restriction is really a STRONG one? Remember that we only provided a few ditransitive examples to illustrate the STRONG IA-IA restriction in Southern Tiwa. Due to the WEAK EA-IA restriction, we cannot check combinations like *3.EA $\gg$ 1.GL $\gg$ 2.TH, since these would be excluded due to having a 3.EA co-occurring with a 1/2P.IA. Additionally, checking combinations like *1.EA $\gg$ 2.GL $\gg$ 1.TH, which are tolerated by the WEAK EA-IA restriction, we encounter the Principle B complication raised above. Luckily, Southern Tiwa has a workaround for *3.EA $\gg$ 1/2.IA combinations: using a passive construction where the 3P EA is either implicit or realized as a non-argumental oblique, as in (118a). Crucially, in such ditransitive passives, the Theme remains restricted to 3P even when the EA is 1/2P, as shown in (118b), which is a combination allowed with WEAK restrictions. This means that the IA-IA restriction remains STRONG in passives, counter to the predictions of the Principle B analysis.

(118) Southern Tiwa [*Kiowa-Tanoan*]:

a. liorade-ba **in-** khwian- wia-che-ban
lady-INSTR 1>3- dog- give-PASS-PAST
'**I** was given **him** [the dog] by the lady.'
$\boxed{\text{1.GL.SU} \gg \text{3.TH.O} + \text{PP.3.EA}}$

b. *liorade-ba **???-** wia-che-ban
lady-INSTR 1>2- give-PASS-PAST
'**I** was given **you** by the lady.'
$\boxed{✗\ \text{1.GL.SU} \gg \text{2.TH.O} + \text{PP.3.EA}}$

(Allen and Frantz 1978, 13–6)

We can also test this prediction of the Principle B analysis in another Southern Tiwa construction. Some transitive verbs in the language yield Agent-less quirky subject constructions, using the same pronominal marking paradigm as ditransitive passives (Rosen 1990, 673–4). With these constructions, the person restriction is always STRONG: the quirky Goal subject can be any person, as in (119a), where the subject can be 1P, 2P, or 3P, while the Theme object must be 3P. As illustrated in (119b), the Theme cannot be 1/2P in such constrictions; the restriction is only lifted if the Goal is a dative PP, as in (119c).

(119) a. { **im-** / **kam-** / **am-** } seuan-wan-ban — 1/2/3.GL.SU ≫ 3.TH.O
{ 1>3PL- / 2>3PL- / 3>3PL- } man-come-PAST
'**He** [the man] came to **me/you/him**.'

b. *???- wan-ban — ✗ 1.GL.SU ≫ 2.TH.O
1>2- come-PAST
'**You** came to **me**.'

c. **a-** wan-ban na-'ay — 2.TH.SU + PP.1.GL
2- come-PAST me-to
'**You** came to **me**.' (Rosen 1990, 673–4)

We find similar cases in Algonquian languages, which almost exclusively have a WEAK-STRONG restriction combination. In Algonquian, some verbs—although formally transitive—surface with intransitive morphology (the construction is usually referred to as *Animate Intransitive* + *Object* or AI+O; Goddard 1979). In such constructions, the object is always 3P (Goddard 1979, 37), which makes the person restriction a STRONG one. This is shown in (120a) with examples from Central Ojibwe, using the verb *gagweji* (roughly 'sell'), which in this construction can only take one object (the Goal can only be implicit). Contrast the first three examples with the regular transitive (120d), where the object can be 1/2P.

(120) Ojibwe [*Algonquian*]:

a. mii wi pii *niwi* gaa-ggweji-daawe **-d** — 3 ≫ 3.IA
EMPH that time he(O) IC.PAST-try-sell -3.SU
'That's when **he** tried to sell **him** (to someone).'

b. *mii wi pii { *gii* / *nii* } gaa-ggweji-daawe **-d** — ✗ 3 ≫ 1/2.IA
EMPH that time { me(O) / you(O) } IC.PAST-try-sell -3.SU
'That's when **he** tried to sell **me/you** (to someone).'

c. *mii wi pii *nii* gaa-ggweji-daawe **-yaanh** — ✗ 2 ≫ 1.IA
EMPH that time you(O) IC.PAST-try-sell -1.SU
'That's when **I** tried to sell **you** (to someone).'

d. mii wi pii gaa-ndom-n **-yaanh** — 1.EA ≫ 2.IA
EMPH that time IC.PAST-shoot-2>1 -1.SU
"That's when **I** called **you**? (Rhodes 1990b, 407–8)

Note that the translations are somewhat misleading here in that they suggest a regular transitive interpretation. But as Dahlstrom (2013) observes, the verb class associated with

AI+O constructions mainly consists of verbs where the IA is not directly affected by the argument translated as the EA, suggesting that these are close to Agent-less quirky subject constructions just like the Southern Tiwa examples above. This is further supported by the fact that AI+O constructions resist passivization (Rhodes 1990b; but see Quinn 2006 for some exceptional cases). This would then mean that the difference between (120d) and the other examples is due to the former being an EA-IA transitive and the latter being effectively IA-IA transitives. Additionally, even though I have not been able to systematically check this for all Algonquian languages with the WEAK-STRONG combination, unspecified agent ditransitives (which are the closest Algonquian equivalent to passives) appear to retain the STRONG restriction on the Theme,[34] paralleling the Southern Tiwa examples in (118).

What we can see then is that, while relegating all the 1/2P-licensing to T can derive EA-IA and IA-IA restrictions of equal strength, the ways in which we could extend this approach to WEAK-STRONG languages are either ad hoc or make wrong predictions about the behavior of person restrictions in Agent-less transitive constructions.

Additionally, we saw that in languages with a WEAK-STRONG restriction combination the two restrictions track the thematic roles rather than the grammatical roles of the pronouns. The STRONG restriction between objects of a ditransitive remains STRONG under passivization, in which case it applies between a Goal subject and an Theme object. This contrasts with the person restriction between EA subjects and IA objects of an active (di)transitive clause, which is always WEAK. Given this insensitivity to grammatical role changes, it should now make more sense why I chose to characterize the domains of person restrictions in terms of thematic roles (EA-IA, IA-IA) rather than grammatical roles (SU-O, IO-DO).

However, this should not be taken to mean that person restrictions are always unaffected by changes in grammatical roles. As we saw on many occasions above, passivization can void EA-IA restrictions—by either dropping the EA completely or expressing it as an oblique

34. This is based on the fact that all such constructions in my sources always have 3P Theme arguments.

phrase/PP not registered by a pronoun marker. In addition, passivization may also void IA-IA restrictions under the right conditions. Namely, if the change in grammatical role is also accompanied by a change in pronoun type. For example, in Italian an IA-IA restriction occurs between two object clitics, as shown in (121a), but in a corresponding passive like (121b) there is no restriction. This is because the Theme is a canonical subject in the passive, which in Italian makes it either a full DP, an emphatic strong pronoun, or a *pro* (null subject), none of which take part in person restrictions (see Section 2.4).

(121) Italian [*Romance*]:

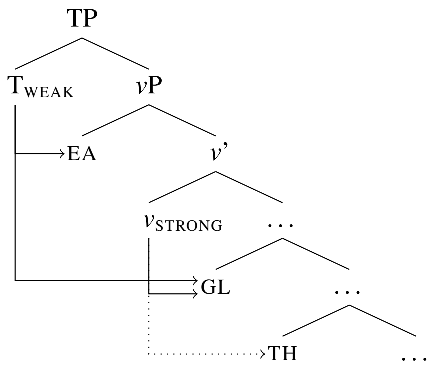
a. ***Ti** **gli** presenterá (Pietro). ✗3.GL ≫ 2.TH
2.DO 3.M.DAT introduce.FUT.3 Pietro
'Pietro/**He** will introduce **you** to **him**.'

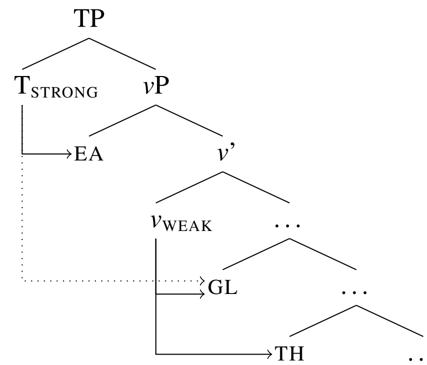
b. **Gli** sarai presentato (da Pietro). 3.GL ≫ 2.TH
3.M.DAT be.FUT.2 introduced.M by Pietro
'**You** will be introduced to **him** (by Pietro).' (P. Cerrone, p.c.)

The correct generalization is then: provided that the change in grammatical role does not bring with it a change to a pronoun type which does not take part in person restrictions, the strength of person restrictions is not affected by changes in grammatical role. The fact that the strength of person restrictions tracks thematic roles rather than grammatical roles will be important for developing a new analysis of EA-IA restrictions in Chapter 4.

2.5.2.3 One parameter for each domain

Returning to the discussion of the *Strength Generalization* and its implications for existing analyses, we can now turn to the other hypothetical analysis of coexisting EA-IA and IA-IA restrictions, where both T and *v* are probes that can license 1/2P pronouns via Agree. Because of languages where EA-IA and IA-IA restrictions exist in isolation—in either STRONG or WEAK form—it must be allowed that the T and *v* probes can each be parameterized to yield either a STRONG or a WEAK restriction; recall that T is tied to EA-IA restrictions and *v* to IA-IA restrictions. It then stands to reason that in languages with coexisting EA-IA and IA-IA restrictions, T and *v* are permitted the same parameter settings. But that also makes it possible to derive both the attested WEAK-STRONG combination, as illustrated in (122), as well as the unattested STRONG-WEAK combination, as illustrated in (123).[35]

(122)


(123)


Of course, one could impose restrictions on the parametric options for specifically the cases when both probes are required for 1/2P-licensing, but these would not be motivated by anything else other than deriving the absence of STRONG-WEAK languages, or at least I am not aware of any principled reason to impose such restrictions.

2.5.2.4 *Summing up the issue*

We can summarize the issue presented by the *Strength Generalization* in the following way. Relegating all the probing to T under-generates with respect to the attested patterns. This is because, while this approach can generate the attested STRONG-STRONG and WEAK-WEAK combinations, it cannot generate the also attested WEAK-STRONG combination. Conversely,

35. In both derivations, there is an independent issue for approaches where Agree licenses 1/2P pronouns (Anagnostopoulou 2003, 2005; Béjar and Řezáč 2003; Adger and Harbour 2007; i.a.). Goals are Agreed with more than once, which is excluded by the *Activity Condition* (Chomsky 2000, 2001; Řezáč 2003; i.a.): a goal that is Agreed with becomes inaccessible for later Agree operations. And even if this assumption is abandoned, T_{WEAK} in (122) will negate the limited licensing possibilities of v_{STRONG} (licensing a 1/2P TH that *v* can not), and in (123) v_{WEAK} will similarly negate the limited licensing possibilities of T_{STRONG} (licensing a 1/2P GL that T can not). In order to derive distinct EA-IA and IA-IA restrictions via parameterized/relativized probes on T and *v*, one has to assume person restrictions arise because some requirement of the probe is not met (Nevins 2007, 2011; Coon and Keine 2018; i.a.) rather than because some requirement of the goals is not met.

splitting the probing duties between T and *v* over-generates with respect to the attested patterns. This is because this approach can generate all four logically possible combinations, including the unattested STRONG-WEAK combination.

Additionally, neither approach can reflect the differences in the frequency of attested patterns. Consider first the approach with one parameter for both domains. Even if we could make the IA-IA restriction somehow exceptional only in WEAK-STRONG languages, the one parameter approach is tailor made for deriving STRONG-STRONG and WEAK-WEAK restriction combinations, which are much rarer than the WEAK-STRONG combination that must be given an exceptional status. A different problem arises with the two parameter approach. Even if we could somehow specifically exclude the parameter settings that yield the STRONG-WEAK combination, all three remaining combinations should be equally likely to be found, given the independence of the restrictions in the two domains.

To be fair, these issues are not limited to syntactic intervention approaches: imposing a single parameter on both EA-IA and IA-IA restrictions will always leave us with no way of deriving mixed strength combinations, and imposing two completely independent parameters for the EA-IA and IA-IA restrictions will always leave us with no way of deriving the gap. The same holds for the frequency issue: we either expect a bias towards same strength combinations, or no bias at all. What we need is an analysis that will allow us to vary the restriction strength freely for each domain in isolation, but make the strength of the EA-IA restriction limit the options for the strength of the IA-IA restriction when they coexist.

It should also be noted that deriving the STRONG vs. WEAK variation in the way illustrated above involves a parameterization of the Agree operation itself, which is one of the core computational mechanisms of grammar (Chomsky 2000, 2001). It is commonly assumed that allowing the parameterization of the computational mechanisms of grammar leads to problems, among other things, in relation to the learnability of such parameters and optimal language design (Borer 1984; Chomsky 1995, 2005; Biberauer et al. 2014). Because of this it is often suggested that we should favor theories of language variation where at least the core grammatical operations, such as Agree, are crosslinguistically uniform and where most, if not all, parameters of variation are not the result of differences in the computational system of language but results from differences in the lexical properties of the syntactic elements involved; the view Baker (2008a) dubs the *Borer-Chomsky Conjecture* based on the proposals of Borer (1984) and Chomsky (1995) along these lines. Thus, although the empirical problem with the STRONG vs. WEAK parameterization noted above should be our primary concern, we should also consider that it would be conceptually more appealing to develop an alternative which does not require Agree to be crosslinguistically parameterized.

In spite of the issues with the mainstream approach to person restrictions we saw above, I will argue in Chapter 3 that an analysis in terms of syntactic intervention, similar to the one outlined here, is actually on the right track. The main difference between the analysis outlined in this section and the one proposed in Chapter 3, which will allow us to deduce the *Strength Generalization*, is that in the latter the STRONG vs. WEAK variation does not result from parameterizing the probe but from the structural configuration of the affected arguments in relation to the probe. We will see that this alternative analysis not only derives the generalization, it also makes the WEAK-STRONG combination fall out as the default option, which is a welcome result given that the WEAK-STRONG combination seems to be the combination that languages with both EA-IA and IA-IA restrictions gravitate towards.

2.5.3 The direction gap

In addition to the STRONG vs.WEAK variation, there was another key point of variation discussed in Sections 2.2 and 2.3. As discussed in those sections, there are canonical person restrictions: in the case of an EA-IA restriction the IA is restricted and in the case of an IA-IA restriction the Theme is restricted. However, there are also reverse person restrictions: in the case of an EA-IA restriction, the EA is restricted and in the case of an IA-IA restriction, the Goal is restricted. We have seen, for example, that in the case of the Slovenian IA-IA restriction, both the canonical and the reverse versions of the person restriction are attested. Recall that, due to the two possible orders of the object clitics, the person restriction may either be canonical, where the Theme clitic is restricted, as illustrated in (124), or it can be a reverse restriction, where the Goal clitic is restricted, as illustrated in (125).

(124) Slovenian [*Slavic*]:

a. Mama **ti** **ga** bo predstavila.
   mom 2.DAT 3.M.ACC will introduce
   'Mom will introduce **him** to **you**.' $\boxed{2.\text{GL} \gg 3.\text{TH}}$

b. *Mama **mu** **te** bo predstavila.
   mom 3.M.DAT 2.ACC will introduce
   'Mom will introduce **you** to **him**.' $\boxed{✗\,3.\text{GL} \gg 2.\text{TH}}$

(125) a. Mama **te** **mu** bo predstavila.
   mom 2.ACC 3.M.DAT will introduce
   'Mom will introduce **you** to **him**.' $\boxed{2.\text{TH} \gg 3.\text{GL}}$

b. *Mama **ga** **ti** bo predstavila.
   mom 3.M.ACC 2.DAT will introduce
   'Mom will introduce **him** to **you**.' $\boxed{✗\,3.\text{TH} \gg 2.\text{GL}}$

(Stegovec 2015b, 108–9)

The crucial factor in determining the restriction pattern is the order of the object clitics, which, as we can see with (124) vs. (125), is Goal before Theme in the case of the canonical restriction and Theme before Goal in the case of the reverse restriction. We took the order of the two clitics to be a reflection of the underlying syntax, so that the first clitic always asymmetrically c-commands the second one. This was one of the key points in motivating the syntactic status of person restrictions.

The Slovenian restriction was not the only case of a canonical + reverse pattern we saw. This kind of restriction pattern is also instantiated in person-based indexing restrictions: person restrictions that are sensitive only to which of the two arguments is indexed on the verb. One of the examples of this type of restriction we saw above was from Maasai. In a

Maasai double object ditransitive, only one object may be indexed on the verb, regardless of its thematic or grammatical role, as shown in (126a). Crucially, given a 1/2P and 3P object, it must be the 1/2P object that is indexed, as shown in (126b), double object ditransitives can never be construed as having two 1/2P objects, as shown in (126c).

(126) Maasai/Maa [*Eastern Nilotic*]:

a. **kɨ́-** ɨshɔ(r) ɔl-payián — 2.GL/TH ≫ (3.TH/GL)
INV(3>2)- give M-man.ACC
'**They** will give **you** to the man.'
'**They** will give the man to **you**.'

b. ***ɜ́-** ɨshɔ *nɨnyɜ́* iyíé — ✗ 3.GL/TH ≫ (2.TH/GL)
3(>3)- give him/her.ACC you.ACC
'**They** will give *you* to **him/her**.'
'**They** will give **him/her** to *you*.'

c. ***áa-** ɨshɔr *nánú* iyíé — ✗ 1.GL/TH ≫ (2.TH/GL)
INV(3>1)- give me.ACC you.ACC
'**They** will give **me** to *you*.'
'**They** will give *you* to **me**.' (Lamoureaux 2004, 14–20)

Although all the examples of canonical + reverse restriction patterns discussed so far involved ditransitives, and thus IA-IA restrictions, restrictions of this type are also attested in the EA-IA domain, and we will see some examples of that kind of restriction below.

The main purpose of this section is to establish a typological gap concerning the distribution of canonical and reverse person restrictions and to discuss its implications for analyses of person restrictions. Before establishing the existence of the gap, I will first show that canonical and reverse person restrictions are not limited to one particular argument domain. This will make the existence of the gap all the more striking, since it will be shown that it exists for both EA-IA and IA-IA restrictions.

In reviewing the implications of the gap, I will first consider a non-syntactic surface order alternative to canonical + reverse restriction patterns (in the spirit of Perlmutter 1968, 1970, 1971), showing that it is too unconstrained to derive only the attested restriction patterns. In addition, I will review some complex restriction patterns that at first glance seem to interact with the order of pronouns in different ways than what we have seen in Slovenian. I will show that those patterns, upon closer inspection, actually present further support for a syntactic approach. Finally, I will discuss the shortcomings of two existing syntactic approaches in relation to the canonical vs. reverse variation.

2.5.3.1 Reverse EA-IA *restrictions*

As a first step towards establishing the constraints on the occurrence of canonical and reverse restrictions let us first establish that reverse person restrictions occur also in the EA-IA domain. A notable example of an EA-IA equivalent of the canonical + reverse person restriction pattern in Slovenian can be found in virtually all Algonquian languages.

Algonquian languages are all pronominal argument languages, and one of their defining characteristics is that the placement and order of the pronoun markers is independent from their grammatical or thematic role (Hockett 1939, 1948, 1966; Goddard 1979; Oxford 2014, 2019). I will illustrate this and the EA-IA person restriction with the case of Nishnaabemwin, a dialect of Ojibwe (Valentine 2001). When the EA and IA are both 3P, the markers indexing discourse backgrounded entities (*proximate*) precede those indexing foregrounded entities (*obviative*) (a similar pattern is found in Slovenian with two 3P clitics, whose order is also governed by discourse-related factors; see Stegovec 2019, 16).

As shown in (127), the proximate plural 3P markers (*w-* '3P'; *-waa* 'PL') always precede the obviative singular 3P marker (*-n* '3P.OBV'). Their placement is independent from the EA/IA or subject/object status of the corresponding arguments; in (127a) EA markers precede IA markers, whereas in (127b) IA markers precede EA markers.

(127) Nishnaabemwin/Ojibwe [*Algonquian*]:

a. **w-** waabm -aa **-waa -n** $\boxed{3.\text{EA} \gg 3'.\text{IA}}$
3- see -EA>3 -PL -3.OBV
'**They** see **him**$'$.'

b. **w-** waabm -igo **-waa -n** $\boxed{3.\text{IA} \gg 3'.\text{EA}}$
3- see -INV.3>IA -PL -3.OBV
'**He**$'$ sees **them**.' (Valentine 2001, 265–75, 287)

In most cases, including (127), one can infer from the *theme sign* (*-aa* vs. *-igo*), often analyzed as a direct/inverse morpheme, which marker expresses the EA and which the IA.

In contrast, with mixed person arguments what determines the marker order is their person value: 1P and 2P markers always precede 3P ones. This is shown in (128a) for 1P and 3P and in (128b) for 2P and 3P. As we see in (128c), 1P and 2P markers also can co-occur, which shows that the person restriction in Nishnaabemwin is not a STRONG one.[36]

(128) a. **n-** waabm -aa **-g** $\boxed{1.\text{EA} \gg 3.\text{IA}}$
1- see -EA>3 -3PL
'**I** see **them**.'

b. **g-** waabm -aa **-g** $\boxed{2.\text{EA} \gg 3.\text{IA}}$
2- see -EA>3 -3PL
'**You** see **them**.'

c. **g-** waabm **-imin** $\boxed{2.\text{EA} \gg 1.\text{IA}}$
2- see -1PL
'**You** see **us**.' (Valentine 2001, 265–75, 287)

Crucially, when the EA is 3P and the IA is 1/2P, 3P markers cannot precede the 1/2P ones. This is illustrated in (129); the 1/2P marker is '???' and the plural marker optional because the form and appearance of the post-verbal markers is highly contextual and we cannot know for sure what the markers would be in the unattested configurations. Note also that the person restriction is independent of direct/inverse marking; see (129a) vs. (129b).

36. Although there is an asymmetry between 1P and 2P person markers: 2P (and 1P inclusive) markers always precede 1P ones—the famous Algonquian *Person Hierarchy* (Hockett 1939, 1948, 1966; Goddard 1979) See Section 4.3, where it is discussed in the context of other related 1P vs. 2P asymmetries.

(129) a. ***w-** waabm -aa (**-waa**) **-???** ✗3.EA ≫ 1/2.IA
3- see -EA>3 -PL -1/2

b. ***w-** waabm -igoo (**-waa**) **-???**
3- see -INV.3>IA -PL -1/2
'**They** see **me/you**.' (Valentine 2001, 265–75, 287)

These person combinations are only possible if the EA and IA markers switch place, as in (130), where the IA markers precede the EA markers. Other than the inverse morphology, the examples in (130) look just like their 1/2P EA with 3P IA counterparts in (128).

(130) a. **n-** waabm -igoo **-g** 1.IA ≫ 3.EA
1- see -INV.3>IA -3PL
'**They** see **me**.'

b. **g-** waabm -igoo **-g** 2.IA ≫ 3.EA
2- see -INV.3>IA -3PL
'**They** see **you**.' (Valentine 2001, 265–75, 287)

Notice that the Nishnaabemwin person restriction is in a way an EA-IA equivalent of the Slovenian IA-IA restriction pattern. The first pronoun marker (sometimes split into preverbal and post-verbal morphemes) can have any person value, while the second one is person restricted.[37] Since the first marker corresponds to either the EA or IA and the second one likewise to either the IA or EA, we get a pattern where, if EA is first, then IA is restricted (a canonical restriction), and if IA is first, then EA is restricted (a reverse restriction).

Aside from this type of canonical + reverse EA-IA restriction, we also find such restrictions in the form of person-base argument indexing. For example, an EA-IA equivalent of the Maasai IA-IA restriction is found in Kaqchikel (Preminger 2011, 2014). In Kaqchikel *Agent-Focus* constructions, used when the subject of a transitive verb undergoes Ā-movement (like focus or *wh*-movement), either the EA or IA may be indexed on the verb. Which argument is going to be indexed depends on person: given two arguments where one is 1/2P and the

37. Although, if Oxford (2014, 2019) is correct, the post verbal markers in Algonquian are object agreement rather than pronoun markers. That would make the pattern more like person-based indexing, which would not change things much since I will ultimately argue that the two types of restrictions are underlyingly the same.

other 3P, the former will be indexed regardless of it being the EA or an IA, as illustrated by (131) vs. (132). Additionally, all Agent-Focus constructions with two 1/2P arguments are ungrammatical regardless of which one is indexed, as shown in (133).

(131) Kaqchikel [*Quichean-Mamean*]:

a. ja *rat* x- **at-** ax-an *ri achin*
FOC you CMPL- 2- hear-AF the man
'It was **you** that heard the man.' [2.EA ≫ (3.IA)]

b. ja *ri achin* x- **at-** ax-an *rat*
FOC the man CMPL- 2- hear-AF you
'It was the man that heard **you**.' [2.IA ≫ (3.EA)] (Preminger 2014, 18)

(132) a. *ja *ri achin* x- **Ø-** ax-an *rat*
FOC the man CMPL- 3.ABS- hear-AF you
'It was **him** [the man] that heard you.' [✗ 3.EA ≫ (2.IA)]

b. *ja *rat* x- **Ø-** ax-an *ri achin*
FOC you CMPL- 3.ABS- hear-AF the man
'It was you that heard **him** [the man].' [✗ 3.IA ≫ (2.EA)] (Preminger 2014, 18, 22)

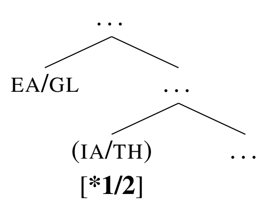
(133) a. *ja *yïn* x- { **in-** / **at-** } ax-an *rat*
FOC me CMPL- { 1- / 2- } hear-AF you
'It was **me** that heard **you**.' [✗ 1.EA ≫ (2.IA) / 2.IA ≫ (1.EA)]

b. *ja *rat* x- { **at-** / **in-** } ax-an *yïn*
FOC you CMPL- { 2- / 1- } hear-AF me
'It was **you** that heard **me**.' [✗ 2.EA ≫ (1.IA) / 1.IA ≫ (2.EA)] (Preminger 2014, 22)

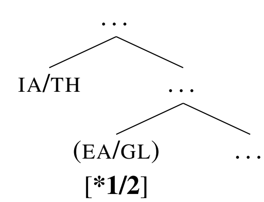
This is essentially the pattern we saw in Maasai. The only difference is that instead of arising in the IA-IA domain, it arises in the EA-IA domain between subject and object. When the EA is indexed on the verb, the IA is restricted (a canonical restriction), while when the IA is indexed on the verb, the EA is restricted (a reverse restriction).

There are therefore restrictions parallel to the canonical + reverse restrictions in the IA-IA domain also in the EA-IA domain. Additionally, there are cases where pronouns/markers are overt and their order affects the restriction pattern (like Nishnaabemwin/Ojibwe) as well as person-based argument indexing cases (like Kaqchikel), where the restricted argument is always the *unindexed* one. I suggest based on this parallelism that we are in fact dealing with the same type of restrictions, which we can, based on the discussion in Sections 2.2 and 2.3, characterize in the following way: canonical restriction patterns arise when the EA/Goal asymmetrically c-commands the IA/Theme, as illustrated in (134a) (parentheses indicate the unindexed argument in person-based indexing systems), while reverse restriction patterns arise when the IA/Theme asymmetrically c-commands the IA/Goal, as illustrated in (134b).

(134) a. Canonical restriction:



b. Reverse restriction:



2.5.3.2 Establishing the typological gap

Having established that canonical and reverse person restrictions exist in both argument domains, we can ask ourselves whether there are any restrictions on the occurrence of the two types of restrictions crosslinguistically. I present all the attested person restrictions with respect to direction (canonical, reverse, or canonical + reverse) identified by the crosslinguistic survey in Table 2.4 (the table excludes languages without person restrictions).[38] A language appears in two columns if it has both an EA-IA and an IA-IA restriction.

38. Just as in Table 2.2 above an asterisk (*) marks languages reported with conflicting patterns in the literature or confounds suggesting that the reported pattern might not be accurate (see Appendix).

Table 2.4: Languages according to restriction direction

110					
EA-IA			IA-IA		
canonical-only	canonical + reverse	reverse-only	canonical-only	canonical + reverse	reverse-only
Kurdish	Pashto		French	German*	
Kashmiri	Chepang		Spanish	Zürich German	
Hungarian	Tangut		Catalan	Slovenian	
Eastern Mansi	Arapaho		Italian	Czech*	
Tundra Nenets	Algonquin		Romanian	Maasai/Maa	
Senaya	Blackfoot		Swiss German	Haya	
Christian Barwar	Plains Cree		Dutch	Djaru	
Telkepe	Delaware		Swedish	Alutor	
Maasai/Maa	Maniwaki		English*	Koryak	
Jyarong	Ojibwe		B-C-S	Quechua	
Nocte	Passamaquoddy		Bulgarian		
Sahaptin	Potawatomi		Macedonian		
Mi'kmaq	Meskwaki/Fox		Greek		
Southern Tiwa	Kaqchikel		Albanian		
Picuris	Zuni		Iron Ossetic		
Arizona Tewa	Quechua		Digor Ossetic		
Clallam			Basque		
Lummi			M. S. Arabic		
Halkomelem			Classical Arabic		
Squamish			Cairene Arabic		
Mapudungun			Maltese		
Labrador Inuttut*			Senaya		
Oaxaca Zapotec			Telkepe		
Paraguayan Guaraní			Baraïn		
Cherokee			Migama		
			Limbum		
			Sambaa		
			Swahili		
			Nyaturu		
			Georgian		
			Hakha Chin		
			Kambera		
			Manam		
			Yimas		
			Warlpiri		
			Chukchi		
			Sahaptin		
			Takelma		
			Algonquin		
			Blackfoot		
			Delaware		
			Maniwaki		
			Mi'kmaq		
			Ojibwe		
			Passamaquoddy		
			Plains Cree		
			Potawatomi		
			Arizona Tewa		
			Kiowa		
			Picuris		
			Southern Tiwa		
			Tetelcingo Nahuatl		
			O'odham		
			Tzotzil		
			Oaxaca Zapotec		
			Paraguayan Guaraní		
			Cherokee		
			Mohawk		
25	**17**	**0**	**58**	**10**	**0**

The list of languages in the table again includes all types of person restrictions that we have established based on the survey, and reveals a notable typological gap in both the EA-IA domain and the IA-IA domain. The gap concerning the distribution of reverse person restrictions is captured by the following generalization:

(135) *Direction Generalization*
If a language has a *reverse* person restriction in the EA-IA or IA-IA domain, it must also have a *canonical* counterpart of the same person restriction.

In other words, canonical person restrictions can exist in isolation (i.e. not accompanied by a corresponding reverse pattern), but reverse person restrictions never exist in isolation. That means that no language consistently imposes a person restriction only on the IA in the context of EA-IA restrictions or only on the Goal in the context of IA-IA restrictions.

2.5.4 Implications of the direction gap

But what does the *Direction Generalization* mean for analyses of person restrictions? Recall that I have argued that canonical person restrictions arise in syntactic configurations like those illustrated in (136), where the EA or Goal asymmetrically c-command, the IA or Theme respectively, resulting in a person restriction on the IA or Theme.

(136) ✓Canonical restriction:
a. 2.EA/GL $\gg$ 3.IA/TH
b. 3.EA/GL $\gg$ ***2.IA/TH**

In contrast, canonical + result patterns result when we have two possible syntactic configurations: one just like the one in (136), shown again in (137a,b), and another syntactic configuration where the IA or Theme asymmetrically c-commands the EA or Goal respectively, resulting in a person restriction on the the EA or Goal, shown in (137c,d). This alternative structure can in principle arise either via base generation of the two arguments in different syntactic positions or via movement of one argument over the other (I return to these two options in Section 4.2, where I offer an analysis of the typological gap in question), the effect on the person restriction pattern is the same.

(137) ✓Canonical + reverse restriction:
a. 2.EA/GL $\gg$ 3.IA/TH
b. 3.EA/GL $\gg$ ***2.IA/TH**
c. 2.IA/TH $\gg$ 3.EA/GL
d. 3.IA/TH $\gg$ ***2.EA/GL**

But if this alternative structural configuration is a possibility with canonical + reverse patterns, why can it not also be a possibility elsewhere? Why can there be no language with just the structural options in (138)? Note that the options in (138) are the same as (137c,d) from the combined canonical + reverse restriction in (137) above.

(138) ✗Reverse restriction:
a. 2.IA/TH $\gg$ 3.EA/GL
b. 3.IA/TH $\gg$ ***2.EA/GL**

To my knowledge, no current syntactic analysis has a way to handle this issue. Although, admittedly, reverse person restrictions have not really been discussed by generative syntacticians (one exception being my previous work on Slovenian; Stegovec 2015b, 2016, 2019), and they have definitely not been discussed from a comparative typological perspective, so it is no surprise that the *Direction Generalization* remained under the radar.

However, as we have established in the previous sections, a syntactic approach most directly reflects the consistency of the person restriction patterns across different pairs of arguments, different morphological cases, and other surface factors that vary from language to language. Dismissing it due to the present lack of an explanation for the *Direction Generalization* would also mean abandoning the results we have gotten so far. The alternative, which is to treat the canonical/reverse alternation as a purely superficial phenomenon, has the same issue noted above in relation to the syntactic approach, plus additional ones, as I will show next. In contrast, we will see in Chapter 4 that a syntactic analysis that can derive the *Direction Generalization* is possible.

2.5.4.1 Deep or surface restrictions?

The Slovenian restriction pattern is superficially compatible with an analysis that makes no reference to syntactic structure, namely, a surface constraint on the order of clitics: 1/2P clitics must precede 3P clitics regardless of their respective case values. This can be implemented technically as a positive templatic constraint in the style of Perlmutter (1968, 1970, 1971). For example, in the case of the Slovenian (WEAK) person restriction, the template would be something like (139), where each slot can be filled by an object clitic according to person specified in the slot regardless of any other grammatical property; I use '»' to represent a linear precedence relation between syntactic elements. What is prohibited by this particular template is any linear sequence of pronouns that cannot fit the template, as illustrated in (140), where the 3P pronouns precede the 2P ones (see (124)–(125) above).

(139) a. [**ti** 1/2] » [1/2] » [**ga** 3] » [3] (**ti** ⇔ 2.DAT, **ga** ⇔ 3.M.ACC)
b. [**te** 1/2] » [1/2] » [**mu** 3] » [3] (**mu** ⇔ 3.M.DAT, **te** ⇔ 2.ACC)
c. [**mi** 1/2] » [**te** 1/2] » [3] » [3] (**mi** ⇔ 1.DAT, **te** ⇔ 2.ACC)

(140) a. ***ga** » [**ti** 1/2] » [1/2] » [3] » [3]
b. ***mu** » [**te** 1/2] » [1/2] » [3] » [3]

However, consider that this type of approach can in principle also derive canonical-only patterns. One way to do this is to superimpose additional constraints that reference grammatical role on top of the constraint that references person values. For example, a canonical-only restriction would result from: (i) 1/2P clitics must precede 3P clitics, and (ii) IO clitics must precede DO clitics. This would mean that a 1/2P DO clitic could not co-occur with a 3P

IO clitic, since the clitics could not be ordered with respect to both (i) and (ii). But notice that this approach can just as easily derive the unattested reverse-only restriction: (i) 3P clitics must precede 1/2P clitics, and (ii) IO clitics must precede DO clitics. In fact, surface order templates are almost completely unconstrained as a theoretical device: they can in principle reference any grammatical property and order any number of slots in any given way. For example, even by limiting the slots only to specific person values, we can still derive patterns like: 1P clitics must precede 3P clitics and 3P clitics must precede 2P clitics. Restriction patterns of this type are, to my knowledge, not attested.

Suppose that we try to constrain templates to only permit the options required to derive attested patterns of person-based pronoun order. As I show below, some of the complex pronoun order patterns of this kind would require us to posit the types of options that inevitably also permit deriving unattested reverse-only patterns. It can therefore be shown that even constrained versions of templates are still too unconstrained to properly handle the limited options exhibited by person restrictions. In contrast, a syntactic interpretation of the same facts requires a small number of parameters that can straightforwardly model the attested variation between closely related languages. While this will not yet provide us with a syntactic solution to the *Direction Generalization*, it will be useful later in Chapter 4, where it will be shown how a syntactic interpretation of the facts enables us to pinpoint the source of reverse person restrictions; i.e. what makes reverse person restrictions special.

Not all canonical + reverse restrictions are like the Slovenian case in relation to the surface order of pronouns. For example, in Haya double object ditransitives (Duranti 1979), the objects may be realized as two object markers on the verb. The object markers are not differentiated by case marking or in any other way, and the sequence of object markers may be interpreted as either the Goal preceding the Theme or the reverse, as shown in (141).

(141) Haya [*Bantu*]:
a-ka- **ba-** **mu-** leetela. 3.TH/GL » 3.GL/TH
3.SU-PAST- 3PL.O- 3.O- bring.APPL
'He brought **them** to **him**.'
'He brought **him** to **them**.' (Duranti 1979, 40)

This is reminiscent of Slovenian, where the object clitics come in two orders, except that in Haya the lack of morphological case distinctions yields complete Goal/Theme ambiguity. Another similarity with Slovenian is that the order of object markers appears to be constrained by their person, but the restriction appears to be exactly the mirror image of Slovenian. For instance, when a 1/2P and a 3P object marker co-occur, the 3P one must precede the 1/2P one regardless of which one corresponds to the Goal and which one to the Theme. This is illustrated with (142a) vs. (142b). Note that the restriction is a WEAK one, since combinations like (142c), where both objects are 1/2P, are allowed.

(142) a. a-ka- **mu-** { **n-** / **ku-** } deetela. 3.TH/GL » 1/2.GL/TH
3.SU-PAST- 3.O- { 1.O- / 2.O } bring.APPL
'He brought **him** to **me/you**.'
'He brought **me/you** to **him**.'

b. *a-ka- { **n-** / **ku-** } **mu-** leetela. ✗ 1/2.TH/GL » 3.GL/TH
3.SU-PAST- { 1.O- / 2.O- } 3.O- bring.APPL
'He brought **me/you** to **him**.'
'He brought **him** to **me/you**.'

c. a-ka- **ku-** **n-** deetela. 2.TH/GL » 1.GL/TH
3.SU-PAST- 2.O- 1.O- bring.APPL
'He brought **you** to **me**.'
'He brought **me** to **you**.' (Duranti 1979, 40)

In order to capture the Haya pattern in terms of a template, we would have to posit a template which is exactly the mirror version of the Slovenian one, with all the 3P slots preceding all the 1/2P slots, as in (143). This would derive the possibility of two co-occurring 3P object markers (see (143a)), two co-occurring 1/2P object markers (see (143b)), as well as a 3P object marker preceding a 1/2P one (see (143c)), and finally it would exclude the ungrammatical combinations with a 1/2P object marker preceding a 3P one (see (143d)).

(143) a. $\boxed{\textbf{ba}\ _3}$ » $\boxed{\textbf{mu}\ _3}$ » $\boxed{\ _{1/2}}$ » $\boxed{\ _{1/2}}$ (**ba** ⇔ 3PL.O, **mu** ⇔ 3.O)

b. $\boxed{\ _3}$ » $\boxed{\ _3}$ » $\boxed{\textbf{ku}\ _{1/2}}$ » $\boxed{\textbf{n}\ _{1/2}}$ (**ku** ⇔ 2.O, **n** ⇔ 1.O)

c. $\boxed{\textbf{mu}\ _3}$ » $\boxed{\ _3}$ » $\boxed{\textbf{ku}\ _{1/2}}$ » $\boxed{\ _{1/2}}$ (**mu** ⇔ 3.O, **ku** ⇔ 2.O)

d. ***ku** » $\boxed{\textbf{mu}\ _3}$ » $\boxed{\ _3}$ » $\boxed{\ _{1/2}}$ » $\boxed{\ _{1/2}}$

This analysis would be descriptively adequate, but it would also miss an important generalization concerning the Haya restriction in relation to the restriction in the related Sambaa.

Sambaa can also express both objects in a double object ditransitive as object markers on the verb, in which case these conform to a canonical WEAK IA-IA restriction (Riedel 2009). Part of the paradigm is given in (144); note also the consistent TH » GL marker order.

(144) Sambaa [*Bantu*]:

a. a-za- **m-** { **ni-** / **ku-** / **wa-** } onyesha $\boxed{\text{1/2/3.GL} \gg \text{3.TH}}$
3.SU-PERF- 3.DO- { 1.IO- / 2.IO- / 3PL.IO- } show
'He pointed **her** out to **me/you/them**.'

b. a-za- **ku-** **ni-** onyesha $\boxed{\text{1.GL} \gg \text{2.TH}}$
3.SU-PERF- 2.DO- 1.IO- show
'He pointed **you** out to **me**.'

c. a- za- **ni-** **ku-** onyesha $\boxed{\text{2.GL} \gg \text{1.TH}}$
3.SU- PERF- 1.DO- 2.IO- show
'He pointed **me** out to **you**.'

d. *a-za- { **ni-** / **ku-** } **mu-** onyesha $\boxed{\text{✗ 3.GL} \gg \text{1/2.TH}}$
3.SU-PERF- { 1.DO- / 2.DO- } 3.IO- show
'He pointed **me/you** out to **her**.' (Riedel 2009, 76, 140)

In fact, even in Haya there are certain combinations of object markers that rigidly surface with only one order. For example, when the Goal is 3P plural and the Theme is 3P singular, the only possible interpretation of the object marker sequence is TH » GL, as (145) shows.

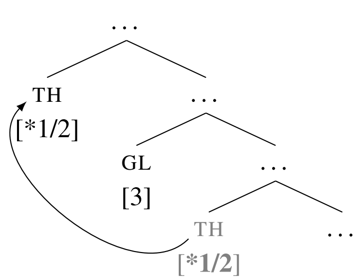
(145) Haya [*Bantu*]: $\boxed{\text{3.TH » 3.GL}}$

a-ka- **mu-** **ba-** leetela.
3.SU-PAST- 3.DO- 3PL.IO- bring.APPL
'He brought **him** to **them**.'
'*He brought **them** to **him**.' (Duranti 1979, 40)

If we were to analyze the Haya restriction as purely superficial, we would disregard the similarity with the Sambaa pattern. Note that both languages have a WEAK person restriction pattern and that the TH » GL order of object markers is the one that in both languages yields a canonical person restriction—a person restriction on the Theme object marker.

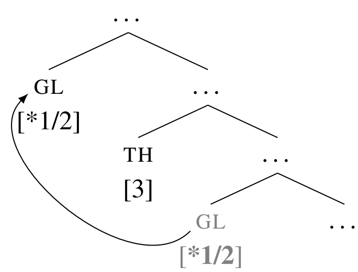
Anticipating the discussion in Chapter 4, suppose that there is a point in the syntactic derivation where person restrictions arise, and that only the c-command relations at that point in the derivation matter for determining whether the person restriction will be canonical or reverse. Any reordering of the pronouns *after* that point in the derivation will then not have any effect on the person restriction. If this approach is on the right track, then deriving the Sambaa pattern requires only that the Goal asymmetrically c-commands the Theme at the point of the derivation when the person restriction applies and that the Theme marker may cross the Goal marker at a later point in the derivation, as in (146). This would then also be the derivation yielding the TH » GL order and a canonical person restriction in Haya.

(146) Canonical restriction (TH » GL; Sambaa & Haya):



Conversely, the Haya GL » TH order, which yields the reverse person restriction, can be derived as in (147): the Theme asymmetrically c-commands the Goal at the point in the derivation when the person restriction occurs, which results in a person restriction on the Goal, but the Goal subsequently moves over the Theme to yield a GL » TH order.

(147) Reverse restriction (GL » TH; Haya only):



With this type of a syntactic approach (discussed in more detail in Chapter 4), we can reduce the difference between Haya and Sambaa to the former but not the latter allowing two structural configurations of the Goal and Theme at the point the person restriction arises, resulting in a canonical + reverse person restriction. In Sambaa, the Goal is consistently higher than the Theme at that point in the derivation, resulting in a consistent canonical restriction. What unifies the two cases is the obligatory reordering of the object markers *after* the relevant point in the derivation. Note that if we treated the Haya person restriction pattern as a surface phenomenon entirely unrelated to the Sambaa canonical person restriction, we would miss the similarity between the two patterns entirely. And given that the two languages are from the same language family, it is sensible to take an approach to the two patterns that reduces the differences between them to minimal points of variation.

Another restriction pattern that is very relevant for the present discussion is found with some speakers of Swiss German (Bonet 1991), who like in Slovenian and Haya, allow two alternative orders of clitic pronouns. Interestingly, this clitic order alternation interacts with the person restriction in an even more intricate way than what we have seen in Slovenian or Haya. The relevant Swiss German speakers exchibit a canonical IA-IA restriction with

GL » TH clitic orders, as seen with (148a) vs. (148d). However, with the alternative TH » GL clitic order, these speakers exhibit no person restrictions, as seen with (148b) vs. (148c).[39]

(148) Swiss German [*Germanic*]:

a. D' Maria zeigt **mir** **en**. — 1.GL » 3.TH
the Maria shows 1.DAT 3.M.ACC

b. D' Maria zeigt **en** **mir**. — 3.TH » 1.GL
the Maria shows 3.M.ACC 1.DAT
'Maria shows **him** to **me**.'

c. D' Maria zeigt **mi** **em**. — 1.TH » 3.GL
the Maria shows 1.ACC 3.M.DAT

d. *D' Maria zeigt **em** **mich**. — ✗ 3.GL » 1.TH
the Maria shows 3.M.DAT 1.ACC
'Maria shows **me** to **him**.' (Bonet 1991, 188n12)

If we wanted to capture this pattern in terms of a surface order template, pretty much the only way to do it would be what is shown in (149). The key to deriving the pattern this way lies in the 1/2P slots being interspersed between the 3P slots, where the former are specified only for person information and the latter are specified for both person and case values (grammatical or thematic roles would work just as well). Crucially, the first 3P slot must be specified for accusative case and the second one for dative case. This way, accusative 3P pronouns may either follow 1/2P pronouns (see (149a)) or precede them (see (149b)), but dative 3P pronouns may only follow them (see (149c) vs. (149d)).

(149)

a.		**mir** 1/2	»	**en** 3.ACC	»	1/2	»	3.DAT
b.		1/2	»	**en** 3.ACC	»	**mir** 1/2	»	3.DAT
c.		1/2	»	3.ACC	»	**mi** 1/2	»	**em** 3.DAT
d.	***em** »	1/2	»	3.ACC	»	**mi** 1/2	»	3.DAT

The analysis in (149) does capture the facts, but it faces a complication when it comes to the variation found within Swiss German. As Werner (1999) reports specifically for Zürich

39. A small number of consulted Slovenian speakers also had judgments similar to (148).

German speakers, their person restriction pattern, shown in (150)–(151),[40] looks exactly like the Slovenian canonical + reverse person restriction. The clitic pronouns occur either in the order GL » TH (cf. (150a)) or in the order TH » GL (cf. (151a)). With the former option, the person restriction is canonical, as shown in (151b), while with the latter option, the person restriction is the reverse one, as shown in (150b).

(150) Zürich German [*Germanic*]:

a. … das d' **mer** **en** halt morn bringsch. $\boxed{\text{1.GL » 3.TH}}$
that 2.NOM 1.DAT 3.M.ACC just tomorrow bring.2

b. *… das d' **en** **mer** halt morn bringsch. $\boxed{\text{✗3.TH » 1.GL}}$
that 2.NOM 3.M.ACC 1.DAT just tomorrow bring.2
'… that you just bring **him** to **me** tomorrow then.'

(151) a. De Max hät **mi** **em** voorgschtellt. $\boxed{\text{1.TH » 3.GL}}$
the Max has 1.ACC 3.M.DAT introduced.3

b. ??De Max hät **em** **mi** voorgschtellt. $\boxed{\text{✗3.GL » 1.TH}}$
the Max has 3.M.DAT 1.ACC introduced.3
'Max has introduced **me** to **him**.' (Werner 1999, 81)

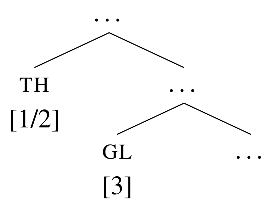
If we were to characterize this restriction in terms of templates, the differences with respect to the other Swiss German pattern would not be trivial. The Zürich German template would be like the Slovenian one (cf. (139)–(140)), with all 1/2P slots preceding the 3P slots and with neither referencing case. This would be quite a different template than the one presented above in (149), where the 3P needed to reference case and be interspersed between the 1/2P slots. It would be rather surprising that differences between speakers of what is nominally the same language would have to be attributed to such arbitrary differences.

The syntactic approach outlined above, on the other hand, offers a rather straightforward solution to the variation. Zürich German is just like Slovenian, in that it has a canonical + reverse restriction pattern, which is the result of two different structural configurations of the Goal and Theme at the point in the derivation when person restrictions apply. That

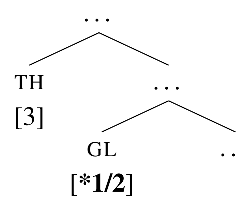
40. As far as I could gather, Werner (1999) uses '??' to mark unacceptability in (151b) because the speakers he consulted tend to replace the weak pronoun forms with strong ones to ameliorate the construction.

means that the TH » GL clitic order, like in Slovenian, always corresponds to the Theme clitic asymmetrically c-commanding the Goal clitic. As shown in (152) and (153), this results in a reverse pattern with a person restriction on the Goal.

(152) Reverse structure:

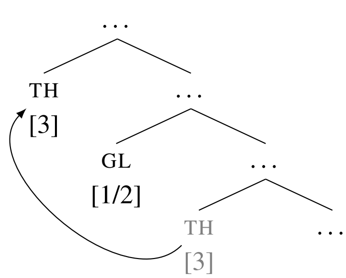


(153) Reverse structure:

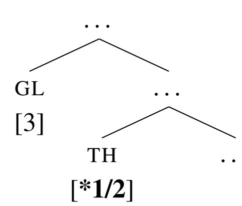


Deriving the other Swiss German pattern requires only one minor change: the option of another way to derive the TH » GL clitic order, in particular, movement of the Theme clitic over the Goal clitic after the point in the derivation when the person restriction applies, as shown in (154). Note that this is the same type of movement that was used to explain the TH » GL order of object markers in Sambaa and Haya above (cf. (146)), and one which crucially yields a canonical person restriction pattern.

(154) Canonical structure:



(155) Canonical structure:



The reason why the TH » GL order does not show any person restrictions in Swiss German is then because both (153) and (154) map to the same surface order of clitics. That means that even though (153) is ungrammatical, the derivation in (154) is grammatical with the same surface clitic string. The GL » TH clitic order, on the other hand, can only correspond to the structure in (155), so this clitic order will always yield a canonical person restriction.

Setting aside the issue of variation between closely related languages for a moment, consider again the three surface templates required to derive the Slovenian (and Zürich German) pattern, the Haya pattern, and the (other) Swiss German pattern, presented respectively in (156a), (156b), and (156c). The templates in (156a) and (156b) yield canonical + reverse patterns, which is possible because the slots reference only person information. In contrast, (156c) must add case information to the slots in order to yield only a canonical pattern.

(156) a. $\boxed{1/2} \gg \boxed{1/2} \gg \boxed{3} \gg \boxed{3}$ (canonical + reverse)

b. $\boxed{3} \gg \boxed{3} \gg \boxed{1/2} \gg \boxed{1/2}$ (canonical + reverse)

c. $\boxed{1/2} \gg \boxed{3.\textsc{acc}} \gg \boxed{1/2} \gg \boxed{3.\textsc{dat}}$ (canonical)

But once we allow for case to be referenced by templates, it is just as easy to reverse the case specification of the two 3P slots—a reverse case order would be independently needed to derive the clitic order of languages like Bosnian-Croatian-Serbian (Stjepanović 1999), where dative clitics consistently precede accusative ones. That means that by limiting ourselves only to the template options required to derive the attested clitic order patterns we are still able to construct a template like (157), which yields a reverse-only restriction pattern excluded by the *Direction Generalization*: the only person combination it disallows is the one where a 3P accusative pronoun (a Theme) precedes a 1/2P dative pronoun (a Goal), which means the dative Goal pronoun is the one being restricted.

(157) ✗ $\boxed{1/2} \gg \boxed{3.\textsc{dat}} \gg \boxed{1/2} \gg \boxed{3.\textsc{acc}}$ (reverse)

Therefore, a surface-oriented approach inevitably also generates crosslinguistically unattested patterns, even if we limit the templates to employ only the options needed independently to derive the crosslinguistically attested patterns. Additionally, a surface-oriented analysis would entirely miss the similarities between closely related patterns such as the ones in Haya and Sambaa or the two varieties of Swiss German. In contrast, a syntactic interpretation of the facts unifies all these patterns with all the other the restriction patterns we have encountered time and time again in the course of this chapter, and can model the differences between them as minimal differences in clitic reordering possibilities.

Based on the discussion above, I conjecture that all restrictions on clitic/weak pronoun order governed by person values are syntactic.[41] Additional evidence for their syntactic nature, like the cases examined in Section 2.2, may not be available for all the relevant languages in the survey, but as we will see in Chapter 4, only if all these restrictions are syntactic can we straightforwardly explain the lack of reverse-only restrictions.

2.5.4.2 Issues with non-structural asymmetries

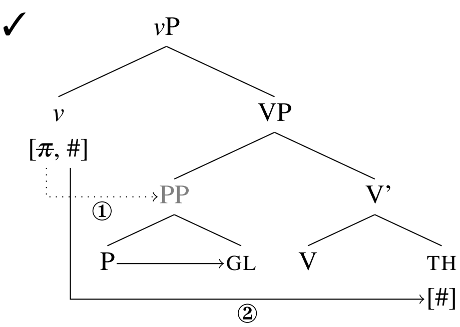
Although I think that a syntactic approach to canonical + reverse restrictions is the way to go, it is not the case that any syntactic analysis can derive such restriction patterns. Specifically, the majority of existing syntactic intervention analyses will have issues with deriving the reverse restriction in particular regardless of whether the asymmetric c-command relation between the arguments is reversed. That is mainly because most syntactic analyses still maintain a residue of the morphological approach to person restrictions (like Bonet 1991, 1994), in that it is crucial for such analyses that there is a case/feature asymmetry between the pronoun that is the structural intervener and the pronoun that is person restricted.

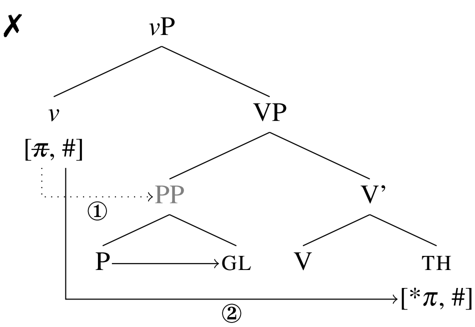
For example, Béjar and Řezáč (2003) argue that person restrictions are exclusive to syntactic contexts where a dative Goal (or other inherent case marked element) blocks the establishment of Agree (Chomsky 2000; see Section 2.5.1) between *v* and an accusative Theme (or other structural case marked element). They assume specifically that inherent case arguments are prepositional phrases (PPs), which *v* cannot establish Agree with, but which

41. This does not necessarily mean that all clitic order patterns are syntactic. In fact, whether or not clitic order phenomena that are insensitive to person values are syntactic or not has no bearing on my claims concerning person restrictions. However, Emonds (1975) has shown that any surface constraint can be restated as a ‘deep’ syntactic one. In relation to the latter, see e.g. Laenzlinger 1993, Kayne 1994, Ordóñez 2002, Cardinaletti 2008 and Ciucivara 2009 for purely syntactic approaches to clitic order.

crucially allow the pronoun within them to be 1/2P in any syntactic context—essentially because the preposition can establish Agree with the pronoun as part of inherent case assignment. Let us consider the consequences of this by looking at a simplified derivation of canonical STRONG person restrictions in the system of Béjar and Řezáč (2003).

Béjar and Řezáč assume that person and number features (represented as π and #) respectively) on *v* can probe independently of each other (following Taraldsen 1995), and that person crucially always probes first. As illustrated in (158), that means that when a PP Goal asymmetrically c-commands the Theme, the person probe on *v* cannot establish Agree with the PP, but *v* also cannot establish Agree with the Theme, because the Goal intervenes between them. However, the PP can be removed as an intervener after this step in the derivation (by moving above *v*P), which in turn allows the number probe on *v* to establish Agree with the number features on the Theme. The assumption here is that 3P corresponds to a total lack of person features, so all the features on the Theme can be Agreed with if it is 3P. This is not the case, however, if the Theme is 1/2P. In that case, illustrated in (159), Béjar and Řezáč propose that the failure of *v* to establish Agree with the Theme's person features is what causes the ungrammaticality of 1/2P Themes in such contexts.

(158) ✓


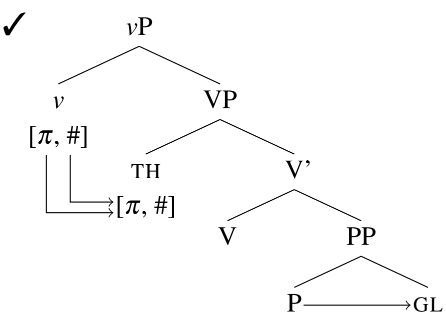
(159) ✗


The key point here is that the Goal does not interact syntactically with *v* at any point in the derivation. In fact, if it were accessible for Agree with *v*, the latter could establish Agree with the Goal both for person and number features, and not Agree with the Theme at all.

Recall also that the PP status is what allows the Goal to have any person value regardless of the fact that it can never enter into an Agree relation with *v*.

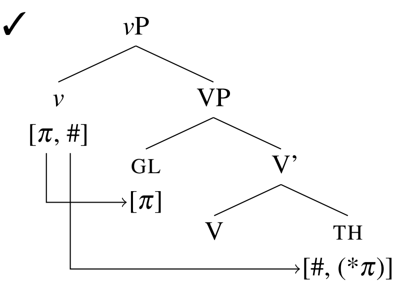
That is also where the problem arises with deriving reverse person restrictions. These must involve, in the IA-IA domain, configurations where the Theme asymmetrically c-commands the Goal. But note that the Theme does *not* in such cases prevent the Goal from entering into an Agree relation. As Goals are always PPs for Béjar and Řezáč (2003), the structure that we would expect to yield a reverse person restriction would look something like (160) (whether or not the arguments are base generated in these positions is not relevant for this point). The Theme is closer to *v* here than the Goal, so *v* can establish Agree with the Theme for both person and number features, which means the Theme can have any person value. But the Goal is a PP, so the fact that *v* cannot Agree with it does not matter—it can anyway have any person value without having to be a target for Agree with *v*.

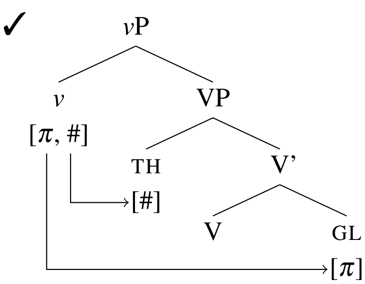
(160) ✓



This means that the system of Béjar and Řezáč (2003) cannot derive reverse person restrictions, due to crucially relying on the case asymmetry between the Goal and the Theme to derive the person restriction in the first place.

A similar problem arises for the analysis of Anagnostopoulou (2003, 2005), which also relies on split person and number probing, but assumes a slightly different asymmetry between the Goal and Theme. Anagnostopoulou's suggestion is that the Goal only has person features (or at least its number features are inaccessible for syntactic computation). As a result, it is not necessary to impose a specific order on the probing of the person and number features on *v*; the reason why *v* can only Agree with the Theme in number features is just because the Goal only intervenes for Agree in person features, as illustrated in (161). However, this introduces a different kind of a problem for reverse restrictions. Specifically, since Anagnostopoulou also assumes that 3P is the total lack of person features, a 3P Theme intervening between *v* and the Goal will only block Agree in number features. Thus, as shown in (162), *v* can still Agree with the Goal in person features, which allows the Goal to have any person value—incorrectly predicting the absence of a person restriction.

(161) ✓


(162) ✓


The derivation of reverse person restrictions is not problematic only for Béjar and Řezáč (2003) and Anagnostopoulou (2003, 2005). It is problematic for any syntactic approach that relies on non-structural asymmetries between the two pronouns involved in the restriction—not necessarily their case or person/number features. For example, in the case of Adger and Harbour (2007), the asymmetry that drives the restriction involves the specific heads that introduce the pronouns into the derivation: the Goal is introduced by an applicative head, while the Theme is introduced as the complement of the verb. In Stegovec 2019, I discuss, in relation to Slovenian in particular, why any analysis that is driven by non-structural asymmetries between the Goal and the Theme cannot derive reverse person restrictions, so I direct the reader to that work for a more comprehensive discussion. For now, what is important is that when we eventually set out to develop a new analysis of person restriction in Chapter 3, we will know what needs to be avoided. It will be crucial that the person restriction arises purely from the structural configuration of the two pronouns; inherent non-structural asymmetries between the relevant pronouns cannot play any role in it.

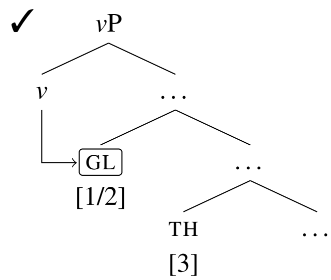
2.5.4.3 Issues with selectively blind probes

Before concluding, let us consider canonical + reverse restrictions in relation to a different kind of a syntactic approach. Specifically, an approach that assumes a uniform syntactic structure across canonical and reverse restrictions and instead relegates the source of variation between them to the domain of agreement. The discussion is largely based around Preminger's (2011, 2014) analysis of the Kaqchikel person-based indexing facts we have seen above, although it should apply more broadly to other cases. I will discuss the proposal in relation to IA-IA restrictions to keep things closer to the last two case studies.

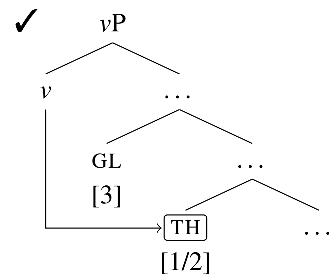
Preminger argues that person-based argument indexing does not depend on differences in syntactic structure. Instead, he proposes that such restrictions arise because Agree, assumed to be a prerequisite for argument indexing, can be relativized to specific values of morphosyntactic features, which in turn results in selective blindness to structural interveners (somewhat similarly to the "trigger-happy agreement" approach of Comrie 2003). The idea is essentially that if a probe (agreement target) is specified to look for a goal (agreement controller) with 1/2P features, a 3P goal structurally closer to the probe will be ignored and will not block the establishment of Agree between the probe and the 1/2P goal.

In a double object ditransitive, a *v* probe relativized this way will then be capable of establishing Agree with either the Goal, as in (163), or the Theme, as in (164), since they are both technically the closest goals of the kind relevant for the relativized probe. The indexing of the argument on the verb (indicated by the box around the relevant argument) is then solely a reflex of a successfully established Agree between *v* and an appropriate goal.

(163) Canonical:

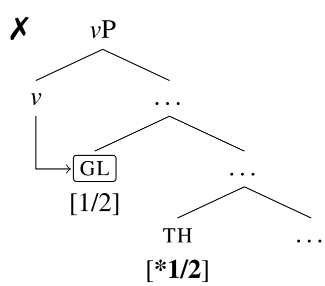


(164) Reverse:

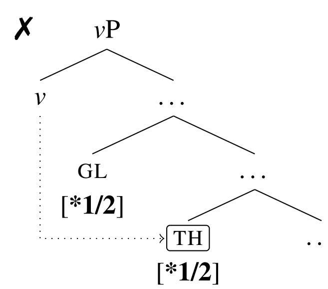


Conversely, if both objects are 1/2P, only the Goal will be accessible for Agree in (165). And since both goals have the 1/2P feature specification relevant to the probe, the Goal will also block Agree between *v* and the Theme, as in (166). Agree is then impossible with both 1/2P objects, which will yield an ungrammatical result following the approaches discussed in Section 2.5.1, where 1/2P features must be licensed via Agree.

(165) Canonical:



(166) Reverse:

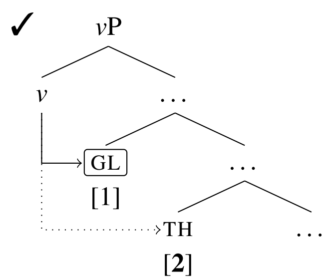


Under this approach there is a one-to-one correlation between indexing and the ability to be 1/2P, which fits a STRONG IA-IA restriction like Maasai, where the unindexed object is always 3P. However, what makes this approach well suited for STRONG restrictions actually makes it problematic for their WEAK counterparts, where unindexed objects do not have to be 3P, such as in Alutor (see Section 2.3.3).

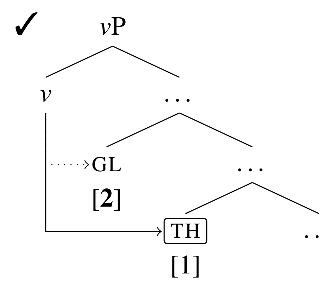
For the sake of completeness, I will also consider the additional 1P vs. 2P asymmetry found in Alutor (see footnote 15), even though I have yet to discuss such person restrictions in detail (they will be discussed in Section 4.3). However, to show that the issue is not due to the additional 1P vs. 2P asymmetry, I will also consider an alternative pattern that lacks it, and show that similar problems arise in relation to the selective probing approach.

Recall that in Alutor, when the two objects are 1/2P and 3P, the 1/2P one will be indexed whether it is the Goal or the Theme. Such cases can then be handled the same way as in (163)–(166) above. However, it is also the case that when the objects are 1P and 2P, the 1P one will be indexed whether it is the Goal or the Theme. In that case, the probe would have to be specified to pick a 1P goal and ignore a 2P goal, as illustrated in (167)–(168).

(167) Canonical:



(168) Reverse:

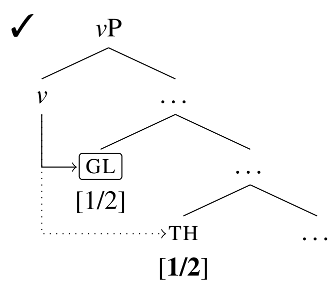


But there are two issues with this: (i) the 2P object is never in an Agree relation so its person feature is not licensed, and (ii) the probe must look for different features depending on the features of both arguments. What is meant with (ii) is that with a 1/2P goal and a 3P goal, both 1P and 2P features satisfy the probe, but with a 1P goal and a 2P goal, only 1P features satisfy the probe. The whole point of probing, which is to have a procedure that picks an appropriate goal upon reaching it without look-ahead, is lost, since the probe must know ahead of time what the two goals are in order to know what kind of goal will satisfy it. Also, to avoid issue (i), one would have to assume that with two 1/2P goals, the probe can establish Agree with both (i.e. Multiple Agree; see Section 2.5.1), but that only the 1P one is picked for indexing—sacrificing the one-to-one correlation between Agree and indexing.

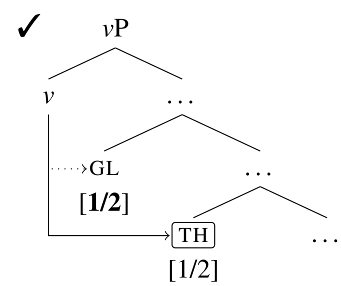
Consider now the same pattern but without the 1P vs. 2P asymmetry, where 1/2P objects are picked over 3P ones for indexing, and either of two 1/2P objects can be indexed—a pure

WEAK indexing restriction, which excludes only the option of indexing a 3P object when the other one is 1/2P. In this case, the problem with licensing the unindexed object is the same as above; as shown in (169)–(170), if one is to maintain a one-to-one correlation between Agree and indexing, one of two 1/2P objects is going to be left unlicensed.

(169) Canonical:



(170) Reverse:



If we managed to avoid this issue by employing Multiple Agree with both goals, then we would have to relegate the picking of the indexed argument to a mechanism independent of Agree. The procedure for would have to be something along the lines of: (a) if the objects are 1/2P and 3P, then Agree and index the 1/2P object, and (b) if the objects are both 1/2P, then Agree with both and index either one, but only one.

The point here is not that the selective probing approach cannot under any circumstances derive WEAK restrictions. The approach, as we have just seen, can be tweaked in a number of ways in order to derive them. But the aspects of the approach that are attractive in relation to STRONG restrictions are immediately lost as soon as we try to extend it to WEAK restrictions. The question here is whether the introduction of the extra theoretical machinery is worth the cost? And related to this, is the extra theoretical machinery just restating the observed facts? We should want to avoid a system that comes close to an unconstrained templatic approach, where any feature specification on the slots is legitimate, since this would not match the very limited variation we see crosslinguistically in the domain of person restrictions.

2.6 What should the alternative analysis be like?

What we should strive for is an analysis that captures all the generalizations with minimal means. In the next chapter I will offer a syntactic analysis which will be shown to better reflect the crosslinguistic generalizations established in this chapter. More precisely, the proposed analysis not only derives all the attested patterns, but it also makes the unattested patterns follow from independently required principles of grammar—it does this without introducing theoretical machinery specifically designed to exclude the gaps. As we will see, the proposed analysis incorporates the core insight of the previous syntactic analyses, namely, the idea that person restrictions arise in syntactic intervention contexts. But it also differs from other syntactic analyses in important respects that will ultimately enable us to explain the two typological gaps established in the last two sections, in addition to offering us a way to explain why some attested restriction patterns are rarer than others.

Turn it upside down

—Brian Eno and Peter Schmidt, *Oblique Strategies*

3

A new analysis of person restrictions

In the previous chapter we established that person restrictions are a syntactic phenomenon, but that nonetheless existing syntactic analyses fall short of deriving the attested crosslinguistic variation—in particular, they cannot explain the existence of two typological gaps identified by the crosslinguistic survey. In addition to the issues that arise in relation to the typological gaps, the existing analyses fail to draw a connection between person restrictions and pronoun type, which we established in Chapter 2 with respect to two separate domains: (i) the pronouns sensitive to person restrictions always also behave in a special way with respect to several binding phenomena (e.g. they cannot be anaphors, they must be used for bound variable readings, they can receive both strict and sloppy identity readings), and (ii) the pronouns sensitive to person restrictions are consistently phonologically weak.

Such correlations are unexpected with existing syntactic approaches: the Agree relation assumed to license 1/2P features can in principle arise with any type of a pronoun. A pronoun that is a goal for Agree can in principle be phonologically weak or strong, and there is no principled reason why only the 1/2P features on pronouns that are independently singled out by other syntactic phenomena, such as binding, should require special licensing via Agree. Of course, additional assumptions can be introduced to limit person restrictions to only some pronouns. For example, Anagnostopoulou (2008) constrains Agree with respect to pronoun type, and Preminger (2019) proposes a fine-tuning of the licensing requirement to apply only to goals that are "canonical agreement targets" (what counts as such a target could be defined in a way to only include some pronoun types). However, approaching the issue this way, by limiting person restrictions to a particular pronoun type by brute force, does not bring us any closer to explaining the correlation between pronoun type and person restrictions. Given the robustness of the correlation, we should seek an analysis where sensitivity to person restrictions is tied to independent properties of the relevant pronoun type in a principled way, which is the key aspect of the analysis I propose below.

In this chapter, I propose an analysis of person restrictions that follows the general logic of the accounts where person restrictions arise due to syntactic intervention. However, the analysis departs from the previous ones by placing all the burden of crosslinguistic variation in this domain on differences in the internal structure of the pronouns and differences in the syntactic position of the pronouns at the point when the syntactic dependencies driving the person restrictions are established. Regarding specifically the correlation with pronoun type, the proposed analysis ties person restrictions directly to a structural deficiency of the relevant pronouns. The particular structural deficiency is not only the source of person restrictions, it is also intrinsically tied to binding and prosodic deficiency.

Establishing the nature of the affected pronouns will thus be the first step of the analysis, which will be done in Section 3.1. This will be followed by a discussion of the syntactic contexts relevant for the emergence of person restrictions in Section 3.2, and a discussion of the link between the structural deficiency of pronouns and their prosodic deficiency in Section 3.3. In Section 3.4, I will discuss the nature of the grammatical operations that drive the person restrictions as well as certain economy principles that constrain them. Once we have established all the mechanisms behind the person restrictions, I will demonstrate how the proposed analysis derives the STRONG vs. WEAK variation in Section 3.5 as well as how it derives reverse person restrictions in Section 3.6. This will give us the basis for deriving the two typological gaps established in Chapter 2, which will be done in Chapter 4.

3.1 Minimal pronouns

The first step I take in establishing the new analysis is to basically turn things on their head. The existing syntactic analyses approach the restrictions on the distribution of specific person features by invoking the requirement that these features must enter into an Agree relation in order to be licensed—failing to do so causes the person restriction. In the analysis I propose, the relevant pronouns start the syntactic derivation without a person value and must be assigned one in the course of the derivation—the failure of the pronoun to be assigned a value is what is behind the person restriction. In fact, under this approach there is really no actual restriction on the pronouns having specific person values, there is only a restriction on whether or not the pronoun can be assigned a person value.

A pronoun lacking a person value is correlated to a type of syntactic deficiency that I elaborate on below. The basic idea comes from similar proposals in the domain of binding; specifically, the approaches to bound pronouns of Heim (2008), Kratzer (2009), Zanuttini, Pak, and Portner (2012), Wurmbrand (2017a), and Messick (2017). In these approaches, certain types of pronouns, usually called *minimal pronouns*, receive their values for person, number, and gender from an external source, which means they can start the syntactic derivation without being specified for those values. Minimal pronouns have been invoked to deal with a number of binding phenomena such as fake indexicals (Heim 2008; Kratzer 2009; Wurmbrand 2017a), agreement sensitive to shifting indexicals (Messick 2017), and semantics-morphosyntax mismatches found with imperative subjects (Zanuttini, Pak, and Portner 2012), as well as just bound pronouns more generally. The basic idea for bound pronouns is that the values of the person, number, and gender features of the pronouns come to match those of their antecedents because they actually receive those values from the antecedent (we will see how below). The reasoning behind adopting this possibility for pronouns comes directly from the observation made in Chapter 2 that the pronouns sensitive to person restrictions also show special behavior when it comes to binding.

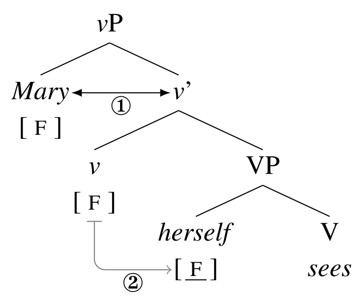
I want to argue that the pronouns sensitive to person restrictions are minimal pronouns due to a severe structural deficiency which is reflected in a subset of their φ-features being unspecified for a value. Specifically, I propose that such pronouns are special with regards to their person features, which begin the syntactic derivation unvalued and must receive a value from an external source. The unvalued person features are also meant to capture the fact that, in addition to being sensitive to person restrictions, the prosodically weak pronouns that fall into this group are always referentially dependent: they refer to some entity already salient in the discourse. This contrasts with strong pronouns, which are never affected by person restrictions, and which can be referentially independent; namely, they can introduce new referents in the discourse. The idea of relating referential dependence in pronouns to structural deficiency goes back to Cardinaletti and Starke (1994, 1999). The reasoning behind implementing referential dependence in terms of unvalued person features will be fully fleshed out in Chapter 5, where I motivate the connection between the two in semantic terms. For the task at hand, which is deriving the existence of person restrictions in the syntax, it will suffice to show the consequences of unvalued person for the syntactic behavior of the pronouns, as well as to establish how structural deficiency in pronouns relates to unvalued person features. I establish the last part in the next section, where I propose that there is a universal structural threshold for having the ability to bear valued person features.

If these deficient pronouns indeed begin the syntactic derivation without a person value as I am proposing, where do they receive it from? In contexts where the pronouns sensitive to person restrictions occur in isolation—that is, when such a pronoun is the only one in a clause—the relevant pronoun can have any person value. This means that such a pronoun must be able to freely receive the person values it can express from an external source. As I mentioned above, in minimal pronoun analyses of bound pronouns, these get their value from the antecedent that binds them (depending on the analysis, either directly or through mediating heads). But the issue is what happens to minimal pronouns when there is no antecedent, meaning, outside of binding contexts—in such cases, they must somehow still get a person value from somewhere.

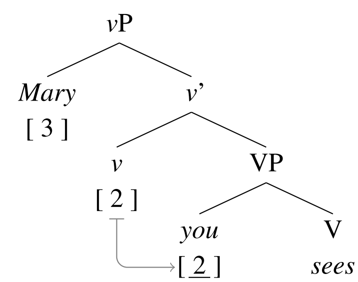
I suggest that outside of binding contexts, minimal pronouns receive a person value by entering into a syntactic dependency with functional syntactic heads that have valued person features. This means that, unlike what is traditionally assumed, elements other than nouns and pronouns may bear valued φ-features. This is not an entirely new idea. It has been suggested before that syntactic heads responsible for introducing specific arguments, such as *v*, which introduces the EA, can be specified with valued φ-features that consequently restrict what φ-features the argument they introduce may bear. For example, Adger and Harbour (2007) suggest that *v* heads and applicative heads respectively restrict EAs and Goals in this way. Similarly, Legate (2014) proposes an analysis of voice alternations in Acehnese [*Malayo-Polynesian*] in these terms, where specific types of *v* ('Voice' in her terms) may bear valued φ-features that restrict the EA to a particular person value. Although the details of how I implement this idea and the use I make of it differ significantly from the aforementioned works, we can see that allowing for valued φ-features to exist outside of arguments themselves is something that seems to be needed independently of the analysis of person restrictions proposed here.

The difference between feature valuation under binding and the alternative method of valuation under discussion here is illustrated in (1)–(2). The main difference is that with binding, shown in (1), φ-feature values come to be shared between the antecedent and the bound pronoun (details below), whereas with the alternative method, tentatively labeled ‘agreement’, shown in (2), the two arguments each have different feature values—and the object pronoun receives its person value from *v*.

(1) Binding (‘Mary sees **herself**’):



(2) Agreement (‘Mary sees **you**’):



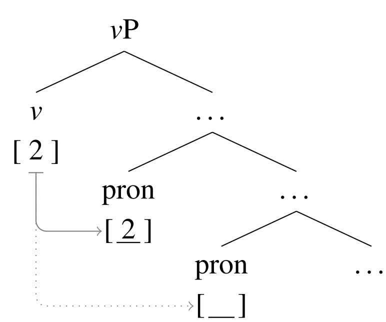
Previous works on minimal pronouns generally consider only valuation in binding contexts. The particular example in (1) illustrates the approach of Kratzer (2009). The binding relation between the antecedent (‘Mary’) and the bound reflexive pronoun (‘herself’) is mediated by *v*, which has valued φ-features that are licensed via Spec-Head agreement (‘①’) with the φ-features of ‘Mary’ (more precisely, the two sets of features become the same feature set via the *Predication* mechanism available in such contexts; see Kratzer 2009 for details). The valued features of *v* in turn value those of the minimal pronoun under binding (‘②’).[1] But once minimal pronouns are a possibility in binding contexts, we can ask the question:

1. In contrast, in the approach of Wurmbrand (2017a) the valuation occurs directly between the antecedent and the minimal pronoun. The choice between the two options has no bearing on the analysis of person restrictions I offer below, since what matters is that the methods of valuation in binding contexts and the methods of valuation outside of them are different and in complementary distribution.

what happens to them outside of those contexts? If minimal pronouns are provided by the grammar, how do they behave in contexts where they are not bound?

Recall from Section 2.4, that anaphors such as ‘herself’ in (1), are crosslinguistically in complementary distribution with agreement. This is the generalization referred to as the anaphor agreement effect (Rizzi 1990a; Woolford 1999). That means, for instance, that object anaphors are never controllers of object agreement. If agreement is the morphological reflex of a successful Agree operation (Chomsky 2000), then what this means is that valuation configurations like (1) must be in complementary distribution with Agree triggered by *v* (normally assumed to be the locus of object agreement). What I propose is that in the absence of binding, which can value minimal pronouns, such pronouns receive a value directly from *v*, either via Agree or other means that I will elaborate on below.

This idea might seem unorthodox, given that Agree is traditionally considered to involve arguments providing values for the φ-features on the functional head. But as we will see below, the reverse direction of valuation in this particular case does not require a departure from the standard approach to Agree. Pending the discussion in question, consider what this means for clauses where multiple minimal pronouns co-occur. Assuming two minimal pronouns are present in the Agree domain of *v*, only one will be the structurally closest one and therefore eligible for receiving a person value via Agree with *v*, as shown in (3). The inaccessibility of the lower pronoun to *v* and the consequent inability of the pronoun to receive a person value will be what drives person restrictions in the current analysis.

(3)


There is, of course, more to derivation of person restrictions than what the outline in (3) suggests. But before we can discuss the derivation in more precise terms, we need to establish some additional core assumptions of the analysis. The first thing to consider in this respect is which functional heads are allowed to bear valued person features.

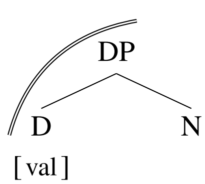
3.2 A structural threshold for person values

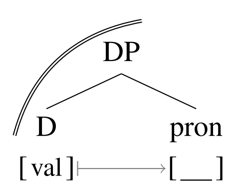
In the analyses of person restrictions discussed in Section 2.5.2, any functional head may in principle license 1/2P features via Agree, as long as it has unvalued φ-features that make it a probe. But, as we saw, this type of approach in conjunction with the possibility of parameterizing the probes to yield STRONG or WEAK person restrictions either under-generates or over-generates with respect to the attested patterns, depending on how many probes are assumed to be present in the clause.

In the current approach, what matters for the appearance of person restrictions is the identity of functional heads that may bear valued person features. Kratzer (2009) specifically suggests that, outside of arguments, only the phase heads *v* and C (Chomsky 2000) may bear valued φ-features. A phase head, as standardly understood, is a head that determines the edge of a cyclic syntactic domain, making that domain inaccessible for further syntactic operations (see, among others, Chomsky 2000, 2001 for different implementations of the basic idea). Taking this proposal as a starting point, I want to make a more general claim about the relation between phase heads and valued person features, but one that is also sufficiently restricted to not over-generate. Specifically, I propose that all phase heads, not just *v* and C, can introduce valued φ-features. This includes heads of prepositional phrases (PPs) (McGinnis 2001; Abels 2003a, 2003b; Řezáč 2008a; i.a.) and determiner phrases (DPs) (Adger 2003; Svenonius 2004; Bošković 2005, 2008; Despić 2011; i.a.), which have been argued to be phase heads as well. In fact, I am assuming that the extended projections of all major lexical categories, which includes verbs, nouns, prepositions, and adjectives, are phases, as argued by Bošković (2013, 2014). Lastly, I want to make an even stronger claim: not only can all phase heads have valued φ-features, but valued person features specifically can *only* be hosted by phase heads.

I follow the gist of the system of Bošković (2013, 2014), where all the extended projections of main categories are phases regardless of their size, but I depart from it in that I assume that pronouns specifically can either constitute a phase of not, depending on the amount of structure inside them. Namely, structurally deficient pronouns (in the sense of Cardinaletti and Starke 1994, 1999), which I assume are mainly composed of bits of inflectional material, do not have enough structure to project a phase boundary. In contrast, strong pronouns have more internal structure and can project a phase boundary. Pending the discussion of the structure of pronouns in Chapter 6, I will assume that strong pronouns are phases because they effectively have the same internal structure as noun phrases.

Let us consider the implications of this approach. In a language with determiners, the phase boundary for extended projections of nouns will fall on DP, as in (4), making D the phase head (phase boundaries are represented with a double line). In the current system, this means that D is the only head in the DP that may bear valued person features (also associated with referential independence, as suggested above). A strong pronoun has pretty much the same structure, as shown in (5). The main difference is that the pronominal DP takes a pronominal root (pron) as its complement. The pronominal root is identical to free standing minimal pronoun, except that here the D head can value its person features.

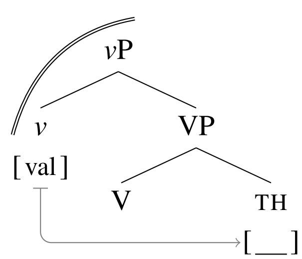
(4)


(5)


In a language without determiners, which have been argued to have less structure in the nominal domain (see Bošković 2012), the parallel would be the same, except that a different category would serve as the phase head (see Bošković 2015 for discussion). But given that for the issue at hand the NP/DP split does not seem to play a role I will set this variation aside and present all noun phrases and strong pronouns as DPs.

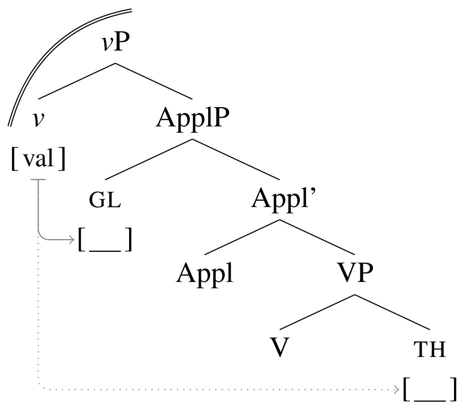
What is important here is that if a pronoun is structurally deficient and lacks the structure necessary to project its own phase boundary, it will not have the ability to bear valued person features and as a result will have to receive a value from a source outside of the pronoun. But this also means that if such valuation is blocked, the pronoun will not be able to receive a value, which will force it to get a default 3P value, thus yielding a person restriction (the details of how this happens will be discussed in Section 3.5).

We can now look at how this approach to the distribution of valued person features causes person restrictions to arise. Ignoring EAs for now, let us consider a scenario where a single object minimal pronoun occurs inside a *v*P phase, as in (6). In this case, we have a one-to-one ratio between the number of phase heads and minimal pronouns.

(6) 

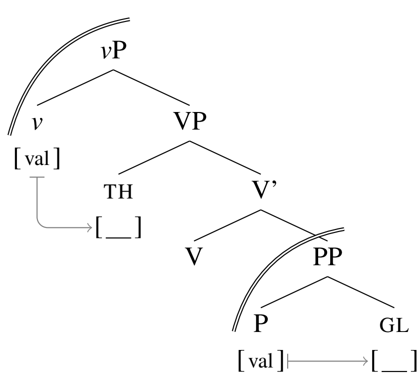
Given that there are no other pronouns inside *v*P that could block syntactic dependencies between *v* and the Theme pronoun, the minimal pronoun can receive any person value from *v*. Contrast this with a ditransitive clause, where a Goal argument is introduced between *v* and the Theme; I will be assuming, here and below, an applicative analysis of double object ditransitives (Marantz 1993; Pylkkänen 2002, 2008; Anagnostopoulou 2003; Georgala 2011), where the Goal is introduced in the specifier of an ApplP projection that dominates the VP containing the Theme. In this case, as illustrated in (7), a Goal minimal pronoun can establish a dependency with *v* without a problem. However, the Goal also structurally intervenes between *v* and the Theme, which will prevent the Theme minimal pronoun from receiving a person value and thus allowing it to only receive a default 3P value.

(7)

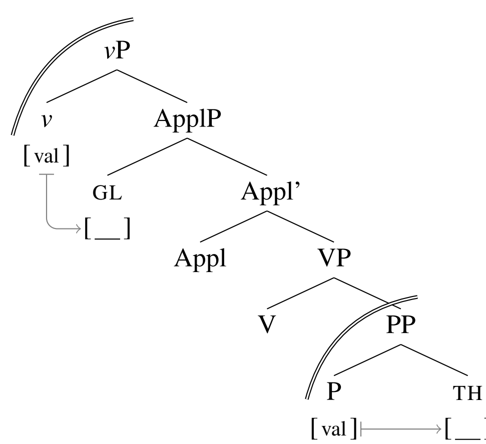


A structure like (7) represents the canonical syntactic context for IA-IA restrictions, discussed in detail in Section 3.5. In comparison, let us consider a prepositional dative, where the Goal is within a PP, which is a phase. As shown in (8), both Theme and Goal minimal pronouns have accessible sources of valued person features in such constructions: *v* can value the Theme, while P can value the Goal. The same is also true of ditransitives where the Theme is in a PP, as shown in (9): here *v* can value the Goal, whereas P can value the Theme.

(8)



(9)



The structures in (8)–(9) predict what we see crosslinguistically: person restrictions never arise with prepositional datives. Recall that in French, person restrictions arise with double object ditransitives, like in (10a), but not in prepositional dative ditransitives, as (10b) shows.

(10) <u>French [*Romance*]:</u>

a. *Lucille **te** **leur** présentera. — ✗3.GL ≫ 2.TH
   Lucille 2.DO 3PL.DAT introduce.FUT.3

b. Lucille **te** présentera à *elles*. — 2.TH ≫ PP.3.GL
   Lucille 2.DO introduce.FUT.3 to them
   'Lucille will introduce **you** to **them**.' (Řezáč 2011, 93)

But this is not only limited to PP that are Goals, as shown by the Spanish example in (11), where the 1P pronoun in the PP can either correspond to the Goal or to the Theme.

(11) <u>Spanish [*Romance*]:</u>

Te recomendaron a *mi*. — 2.TH/GL ≫ PP.1.GL/TH
2.DO/IO recommended.3PL to me
'They recommended **you** to **me**.'
'They recommended **me** to **you**.' (Bonet 1991, 203)

Although generally, like in the French and Spanish examples above, the pronouns in PPs are always strong pronouns (see Abels 2003a, 2003b for an analysis and discussion), there are also languages where the pronouns in PPs can be deficient pronouns; e.g. pronominal clitics appear on postpositions in O'odham [*Piman*] (Hale 1959; Zepeda 1983) and on prepositions in Zapotec [*Zapotecan*] (Operstein 2003; Foley, Kalivoda, and Toosarvandani to appear) and Telkepe [*Neo-Aramaic*] (Kalin 2014b).[2] But regardless of the status of the pronouns inside the PPs, they are never affected by person restrictions.

Since strong pronouns are also phases, they actually do not have to be inside a PP to obviate person restrictions. Recall that this is what we see in Slovenian, where as long as one object pronoun in a ditransitive is strong, no person restriction arises, as shown in (12).

2. Similar cases exist in Bosnian-Croatian-Serbian and Slovenian, although they are not as productive.

(12) Slovenian [*Slavic*]:

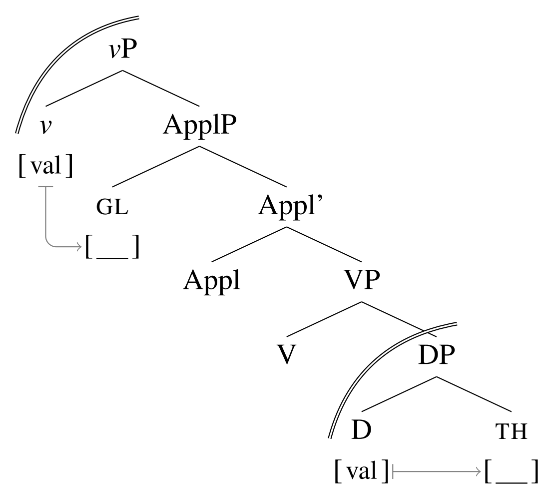
a. *Mama **mu** **te** bo predstavila.
mom 3.M.DAT 2.ACC will introduce
✗ 3.GL ≫ 2.TH

b. Mama **mu** bo predstavila *tebe*.
mom 3.M.DAT will introduce you.ACC
3.GL ≫ 2.TH

c. Mama bo predstavila *njemu* *tebe*.
mom will introduce him.DAT you.ACC
'Mom will introduce **you** to **him**.'
3.GL ≫ 2.TH

This is expected from the current analysis, where strong pronouns are phases. As illustrated in (13), even in double object ditransitives, where the Goal intervenes between *v* and the Theme, a strong pronoun Theme can get any person value from the D phase head.

(13)



There are two main takeaways from this section. The first one is that person restrictions can be directly tied to having a larger number of minimal pronouns than heads with valued person features—a direct consequence of phase heads being the only elements that may bear valued person features. The other takeaway is that pronouns may only be phases when they have sufficient internal structure, so pronouns that are too structurally deficient to be phases cannot have a person value and are thus predicted to be sensitive to person restrictions. As we will see next, we can directly tie the structural deficiency to the prosodic deficiency of the pronouns that are affected by person restrictions. Even better, the approach in terms of phases will allow us to explain why the person restrictions only arise with phonologically weak pronouns, but not all such pronouns are subject to person restrictions.

3.3 Linking structural and prosodic deficiency

Recall from Chapter 2 that there is a one way correlation between the prosodic/phonological character of pronouns and their sensitivity to person restrictions. That is, person restrictions only arise with phonologically weak pronouns, but there exist phonologically weak pronouns which are not affected by person restrictions. The prosodic deficiency/phonological weakness refers here to these pronouns always being prosodically dependent elements that require incorporation into larger prosodic units. If this prosodic deficiency were to be treated as independent of the structural deficiency of these pronouns—for example, by assuming that the morphological exponents of these pronouns are merely specified to be clitic elements in the lexicon independently of their underlying syntactic structure—then the connection between their syntactic and prosodic behavior would remain a mystery.

I instead assume an approach where the prosodic status of linguistic objects is determined by mapping syntactic units to prosodic units (Selkirk 1978; Selkirk 1980, 1984, 2011; Nespor and Vogel 1986; i.a.). Specifically, I take the approach of Cardinaletti and Starke (1994, 1999), who directly tie structural deficiency in pronouns to prosodic deficiency: deficient, but not strong, pronouns form a single prosodic word with adjacent lexical elements (i.e. their "hosts"), because such pronouns lack a particular syntactic projection.[3] In line with the present approach, where deficient pronouns are defined in terms of their inability to project a phase, I adopt the syntax-prosody mapping approach of Compton and Pittman (2010) and Fenger (to appear, in preparation), where prosodic words correspond to syntactic phases.

3. This is as precise as I will formulate what constitutes prosodically deficient behavior. But it is important to note that prosodic deficiency can manifest in a number of ways, the discussion of which would take us too far afield. See for example Selkirk 1996 regarding different types of clitic behavior.

3.3.1 Phases and their relation to prosodic (in)dependence

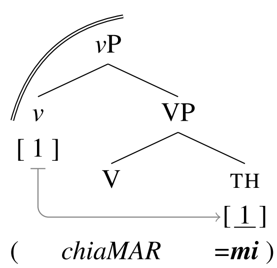
I illustrate the basic principle of the proposed analysis with the Italian minimal pair in (14). Italian object clitic pronouns, like the one in (14a), are sensitive to person restrictions (Seuren 1976; Cardinaletti and Starke 1994, 1999; Cardinaletti 2008) and form a single prosodic unit with the adjacent verb. In contrast, strong pronouns, like the one in (14b), are insensitive to person restrictions and are prosodically independent.

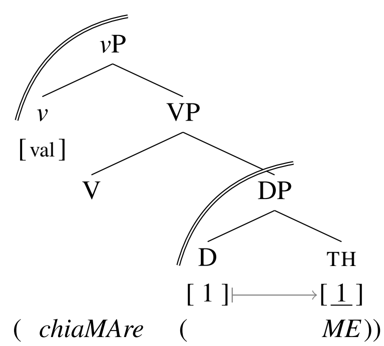
(14) Italian [*Romance*]:

a. Non chiamar **mi**!
   NEG call.INF 1.O
   'Don't call **me**!'

b. Non chiamare *me*!
   NEG call.INF me
   'Don't call me!' (P. Cerrone, p.c.)

The basic structures I propose for these examples and their mapping to prosody are illustrated respectively in (15) and (16), where parentheses indicate the boundaries of prosodic words and capital letters indicate stressed syllables. I discuss the structures in more detail below.

(15)


(16)


In (15), the pronoun does not project a phase, so it must receive a person value from the closest phase head: *v*. But this also means that the pronoun does not project its own prosodic word at PF, and must become part of the prosodic word projected by the verb; here I simplify this to correspond to the *v*P phase (see Fenger to appear, in preparation for discussion of how the domain of the verb's prosodic word can vary). In (16), on the other hand, the pronoun is a strong one, which means it does project its own phase. Consequently, the pronoun's D head may host valued person features, which allows the strong pronoun to have any person value independently of the features on *v*. Additionally, the strong pronoun corresponds to its own prosodic word at PF, which is reflected by the verb and pronoun in (16) each being their own domains for main word stress assignment.

However, as we also saw, not all prosodically deficient pronouns are sensitive to person restrictions; the particular example we discussed earlier in this regard was Polish, a language with weak object pronouns but no person restrictions. Interestingly, there is evidence that even in relation to prosodic deficiency the status of Polish weak pronouns is different from that of comparable pronouns in languages where such pronouns are sensitive to person restrictions. For example, clitic object pronouns in Bosnian-Croatian-Serbian show person restrictions (see Runić 2013b), and their prosodic deficiency is reflected in their status as second position clitics. As illustrated in (17a,b), such clitics require a host to their left, which must be the first prosodic constituent (a prosodic word or a phonological phrase) in the clause. They cannot, as in (17c), appear without such a host, or, as in (17d), following more than one prosodic word (where these words do not form a phonological phrase).

(17) Bosnian-Croatian-Serbian [*Slavic*]:

a. Petar **ti** **ga** daje.
   Peter 2.DAT 3.ACC giving.3

b. Daje **ti** **ga** Petar.
   giving.3 2.DAT 3.ACC Peter

c. ***Ti** **ga** Petar daje.
   2.DAT 3.ACC Peter giving.3

d. *Petar daje **ti** **ga**.
   Peter giving.3 2.DAT 3.ACC
   'Peter is giving **it** to **you**.' (Bošković 2001, 8-9, p.c.)

In contrast, Polish weak object pronouns are much less restricted in terms of their distribution. While they do require a preceding prosodic word to host them, see (18a) vs. (18b), they do not show the second position requirement. We can see this in (18c), where the weak object pronouns are preceded by two prosodic words.

(18) Polish [*Slavic*]:

a. Piotrek **ci** **go** dał.
Peter 2.DAT 3.ACC gave.M

b. ***Ci** **go** Piotrek dał.
2.DAT 3.ACC Peter gave.M

c. Piotrek dał **ci** **go**.
Peter gave.M 2.DAT 3.ACC
'Peter gave **it** to **you**.' (Bošković 2001, 169)

Franks (1998/2010) notes some other ways in which Polish weak pronouns are special: they do not have to cluster with other prosodically deficient elements like auxiliary clitics, and they do not even have to form clusters with other weak pronouns—as long as each pronoun has a prosodic host, they can be easily separated. What we see in Polish is that the weak pronouns have a PF requirement for prosodic support, but that is practically the only difference in their distribution with respect to noun phrases and strong pronouns. Based on this, Franks (1998/2010) suggests that Polish weak pronouns are only prosodically deficient, but not syntactically deficient. This mismatch between PF deficiency and syntactic deficiency can be encoded quite straightforwardly in the current approach, as will be demonstrated.

There is a clear contrast between prosodically dependent pronouns that impose a selectional requirement on their hosts and those that do not. On one hand, we have clitic object pronouns in Italian, which require specifically a verbal host, or clitic object pronouns in Bosnian-Croatian-Serbian, which have an adjacency requirement that is projected onto the host (Bošković 2001). On the other hand, we have the Polish weak object pronouns, which only have the (enclitic) requirement for some prosodic word to precede them at PF; its category is irrelevant, nor does it matter if the prosodic word is clause initial or not.

In relation to this split, I propose that pronouns that are deficient prosodically as the result of a syntactic deficiency can impose more restrictions on their hosts—a host of a particular syntactic category, or a host at the edge of a clause (more precisely, the edge of an *intonational phrase*); the idea is that the restrictions are essentially selectional requirements, meaning that they are at least partly syntactic (i.e. a reflex of their syntactic content; in other words, from their structural deficiency). In contrast, Polish-style weak pronouns involve pure prosodic incorporation: the pronoun only requires an adjacent prosodic word. Here I assume that phases, which are normally mapped onto prosodic words, can in some cases be ignored at PF (as discussed directly below). In this, I follow the analysis of periphrastic phenomena in the verbal domain by Fenger (to appear, in preparation).

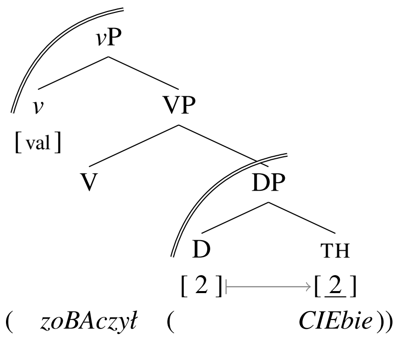
The basic idea in relation to Polish is that strong pronouns, like the one in (19b), and weak pronouns, like the one in (19a), are both phases in the syntax, but that in the latter case the phase is ignored for mapping to prosodic units at PF (recall that normally, there is correspondence between phases and prosodic words). As a result, the weak pronoun does not correspond to its own prosodic word, so it must incorporate into the preceding prosodic word. The difference between the two cases is illustrated in (20)–(21); for sake of comparison with the Italian case above, the preceding prosodic word corresponds to *v*P. (20) shows the derivation for a strong pronoun and (21) the derivation for a weak pronoun.[4]

(19) a. Zobaczył *ciebie*
saw.M you
'He saw you.'

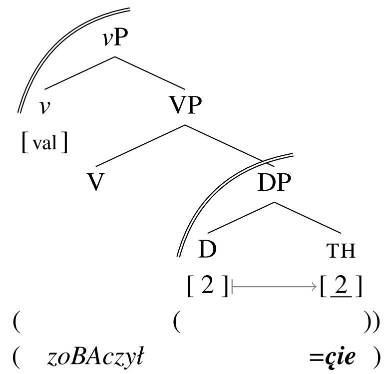
b. Zobaczył **cię**
saw.M 2.ACC
'He saw **you**.' (M. Dadan, p.c.)

4. Although, I will not go into the details of how this analysis interacts with different stress assignment rules (which can vary greatly from language to language), it is important to note that the analysis is applicable also to lexical stress. In the case of lexical stress, a lexically accented syllable is assigned stress in relation to other elements within the relevant prosodic word (see e.g. Kiparsky and Halle 1977 on the *Russian Stress Rule*), so a change in the size of the prosodic word will also affect stress in a lexical stress system.

(20)



(21)



What is important here is that both types of pronouns are phases, but the phase is reflected in the prosodic grouping only with the strong pronouns. In (21), the prosodic word that would normally correspond to the phase projected by the pronoun is ignored at PF. Both types of pronouns thus contain phase heads, which can bear valued person features an make the pronouns insensitive to person restrictions regardless of their prosodic status at PF.

Notice that this approach to syntax-PF mapping derives the one-way correlation between prosodic deficiency and sensitivity to person restrictions. A pronoun that is a phase and therefore insensitive to person restrictions will crosslinguistically have the possibility of either being prosodically independent, if the mapping from phase to prosodic word is always maintained at PF (as in Italian), or being prosodically dependent, if the mapping from phase to prosodic word can be ignored at PF (as in Polish). On the other hand, a structurally deficient pronoun, which is not a phase and therefore sensitive to person restrictions, can only be prosodically dependent. This is because undoing prosodic units at PF is possible, but it is not possible to create prosodic units at PF out of nothing, due to prosodic units always being established with reference to syntactic structure. The one-way correlation between prosodic deficiency and sensitivity to person restrictions is thus captured.

Additionally, the account offers a way to capture the differences in prosodic host selection between pronouns sensitive to person restrictions and those insensitive to them. Weak pronouns sensitive to person restrictions are always syntactically deficient and as a result impose more restrictions on the identity of the host—its category or its position in relation to other prosodic units. In contrast, weak pronouns insensitive to person restrictions are only prosodically deficient and as a result impose very simplistic conditions on their hosts—they have to be prosodic words (i.e. capable of bearing stress). Although this approach to the split between the two types of weak pronouns was established based on one minimal pair, it is possible to extend it to a whole other class of exceptional weak pronouns.

3.3.2 Why some subject pronouns are exceptional

Recall from Section 2.4 that outside pronominal argument languages, weak subject pronouns never take part in person restrictions. The most obvious case is *pro* in null subject languages (like Spanish or Italian), which is phonologically weak by virtue of being phonologically null, but also consistently never yields any person restrictions when co-occurring with weak object pronouns. This is surprising since *pro* has been argued to be of the same syntactic type as weak object pronouns (Cardinaletti and Starke 1994, 1999; Holmberg 2005).

The special status of *pro* in this respect is not an areally or generically restricted phenomenon. Null subject languages are found in Romance, with Spanish and Italian being prototypical cases (Perlmutter 1971; Chomsky 1981; Rizzi 1982, 1986a), in Slavic (Perlmutter 1971; Franks 1995), in many varieties of Arabic (Farghaly 1982; Jelinek 1983, 2002; Kenstowicz 1989), and in Bantu (Bresnan and McHombo 1987; Schneider-Zioga 2000, 2007; Baker 2003a). Note that most of the null subject languages in these language groups actually also show person restrictions, but the person restrictions in question are always between object pronouns—null subjects never take part. Crucially, this is not exclusive to *null* subject pronouns. For example, French weak subject pronouns (Kayne 1975; Rizzi 1986b) and subject clitics in Northern Italian dialects (Benincà 1983; Renzi and Vanelli 1983; Rizzi 1986b; Poletto 2000; Poletto and Tortora 2016) also do not show person-based co-occurrence restrictions with respect to clitic object pronouns. Therefore, the relevant distinction is not necessarily between null and overt weak pronominal elements.

Starting with the overt weak subject pronouns, we can find asymmetries reminiscent of those between second position clitic pronouns like those in Bosnian-Croatian-Serbian and Polish weak pronouns discussed above even language internally regarding weak subjects and deficient pronouns. In French, weak subject pronouns (sometimes categorized as subject clitics) show a much looser relationship with their host than object clitics; for example, in complex inversion contexts (Kayne 1972, 1975, 1983; Rizzi and Roberts 1989; Sportiche 1999; Roberts 2010b; i.a.), the object clitic fronts together with the verb, remaining a proclitic, while the weak subject pronoun changes its position in relation to the verb. We can see with the non-inversion case in (22a), that both subject and object deficient pronouns precede the host, while in (22b) only the object clitic precedes it—the weak subject pronoun attaches to the auxiliary as an enclitic.

(22) French [*Romance*]:

a. **Je** **t'** aime. [SU= [O=V]]
   1.SU 2.O love
   '**I** love **you**.'

b. **L'** as **-tu** vu? [[O=V] =SU]
   3.O have -2.SU see
   'Have **you** seen **it**?' (Roberts 2010b, 104–5)

In the current approach, this asymmetry can be modeled as a difference in the type of pronominal deficiency. The object clitics are just like the Italian ones discussed above—syntactically deficient pronouns that do not project a phase. The weak subject pronouns, on the other hand, project a phase and only become prosodically weak at PF, like Polish weak object pronouns. That is why they do not incorporate into the prosodic word of the verb early, but only after it has reached its final landing site. The idea is essentially an updated version of the very first analysis of French weak subject pronouns by Kayne (1972), where it was suggested that subject clitics attach to the verb at a later stage than object clitics, which would be PF in the current system. This was interestingly also given as one of the reasons why they do not take part in person restrictions.

French subject-object and object-object combinations of prosodically weak pronouns can even be homophonous, but differ in terms of person restrictions. The restriction only applies when both pronouns are object clitics, as in (23a). The homophonous sequence in (23b), in contrast, is grammatical. Crucially, even when one of the pronouns changes grammatical roles in a ditransitive passive, the restriction goes away, as in (23c) (cf. (23a)).

(23) French [*Romance*]:

a. *Paul **vous** **lui** présentera. — ✗3.GL ≫ 2.TH
   Paul 2PL.DO 3.DAT introduce.FUT.3
   'Paul will introduce **you** to **him**.'

b. **Vous** **lui** présenterez Paul. — 2.EA ≫ 3.GL
   2PL.SU 3.DAT introduce.FUT.2 Paul
   '**You** will introduce Paul to **him**.' (Kayne 1975, 87n18)

c. **Vous** **lui** serez présenté (par Philippe). — 3.GL ≫ 2.TH
   2PL.SU 3.DAT be.FUT.2PL introduced.2PL by Philippe
   '**You** will be introduced to **him** (by Philippe).' (Postal 1990, 123)

This data can be straightforwardly explained by the suggested analysis where weak subject pronouns in French actually only become deficient at PF like Polish weak object pronouns.[5]

In fact, the same analysis can also be extended to null subject languages like Spanish or Italian, where *pro* never takes part in person restrictions. The idea would be that *pro* is essentially a regular pronoun or NP/DP that gets elided at PF (see Holmberg 2005; Roberts 2007, 2010a; Richards 2016). In the current approach, this would mean that *pro* is a phase in the syntax that gets elided at PF (compatible with the approach of Bošković 2014,

5. One difference with Polish here is that nothing can intervene between a French weak subject pronoun and the verb (Kayne 1975, 84). However, this is not necessarily a selectional requirement, and it might just be a consequence of the syntactic closeness between the subject pronoun and the verb—-French has a rather rigid word order in contrast to Polish. Alternatively, the adjacency requirement could follow from Richards's (2016) idea that some EPP/pro-drop effects are prosodically determined (see also below).

where ellipsis is phase-constrained), correctly predicting its independence with respect to person valuation. The null nature of *pro* is not, under this analysis, the result of a syntactic deficiency, which fits nicely Richards's (2016) idea that the existence of null subjects in some languages might actually be tied to prosodic factors relating to the verb: it is the verb's prosodic requirement which forces the subject to be null, not a deficiency of the subject.

One complication here is that *pro* can get bound variable readings unlike its overt strong pronoun counterparts, i.e. the Montalbetti Effect—one of the asymmetries that was used in Section 2.4 to distinguish phonologically weak pronouns sensitive to person restrictions from those that are not. Importantly, even though the *pro* does conform to the Montalbetti Effect, it is not the case that there are no other binding asymmetries that single out *pro* with respect to clitic/weak object pronouns, at least in some languages. For example, although *pro* in Slovenian patterns with clitic pronouns in relation to the Montalbetti Effect, *pro* cannot get a sloppy reading in the relevant contexts (see Section 2.4.1.2), unlike clitic pronouns (the same is true for Bosnian-Croatian-Serbian; Runić 2014; Bošković 2018). I take this to indicate that while some cases of *pro* might be deficient enough to be subject to the Montalbetti effect (see Holmberg 2005 on different types of *pro*), no *pro* is deficient enough to not be a phase, which is what the binding asymmetry in Slovenian and the general insensitivity of *pro* to person restrictions is pointing towards. What remains to be determined in future work is how exactly this deficiency causes some binding asymmetries but not others.

The insensitivity of *pro* and some weak subject pronouns to person restrictions can thus be given a principled explanation: they are fundamentally different types of deficient pronouns, which retain their phase status in the syntax but only become deficient post-syntactically. This allows us to come back to an observation made in Section 2.4, namely that EA-IA restrictions are only found in pronominal argument languages. At that point this was established as a one-way correlation, since there exist pronominal languages with only IA-IA restrictions or no restrictions. What I want to suggest here is that the agreement/pronoun markers in such languages might not be a uniform class and can differ in ways relevant for person restrictions despite no discernible differences in the surface forms. Specifically, in pronominal argument languages with no EA-IA restrictions, at least part of the EA-IA marking paradigm involves true agreement morphology licensing a phasal *pro*, just like in Spanish or Italian, rather than a syntactically deficient minimal pronoun.

There is at least some preliminary evidence suggesting that this is correct. In her study of micro-variation in Inuit, Yuan (2018) observes that the absolutive object markers, despite being virtually identical in their surface forms, show distinct behavior in relation to a number syntactic and semantic phenomena, concluding that they are heterogeneous in terms of their underlying syntactic status: the Kalaallisut markers are true object agreement, Inuktitut markers are doubling object clitics, and Labrador Inuttut markers are non-doubling object clitics. Interestingly, Labrador Inuttut, where the markers are the most pronoun-like, is also the only variety that has been reported to show EA-IA person restrictions (Johns 1996; Johns and Kučerova 2017) (although the status of the restriction is somewhat controversial; see Compton 2018; Yuan 2018). The size of the current language sample unfortunately prevents us from running the necessary careful syntactic and semantic tests for all the relevant languages. However, doing this would be the natural next step to make. The prediction is clear here: when the markers display properties consistent with true agreement and traditional null subjects, we also expect there to be no EA-IA restrictions.

3.4 Only one way to *Agree*

So far, the aspects of the proposed analysis that we examined mainly concerned establishing the class of pronouns sensitive to person restrictions. Now we can finally look at the syntactic operations that manipulate syntactic elements and whose constrained application is to blame for certain pronouns not being able to have the full range of person values.

As I noted above, the valuation of minimal pronouns (outside of binding contexts) is contingent on Agree between the local phase head and the minimal pronoun. However, not all person restrictions are the same; there needs to be a way to vary the valuation of pronouns so that the outcome may be a STRONG or a WEAK restriction. As we saw in Chapters 1 and 2, existing syntactic analyses of the STRONG vs. WEAK variation parameterize Agree itself: whether Agree is possible only between one probe and one goal (yielding a STRONG restriction), or whether Agree is also possible between one probe and multiple goals given the right combination of goals (yielding a WEAK restriction). Recall that, for a number of reasons, including considerations of learnability and optimal language design, we should favor approaches to crosslinguistic variation where the core grammatical operations like Agree are crosslinguistically uniform. The alternative is that most, if not all, parameters of variation are not the result of crosslinguistic differences in the computational system of language but the result of differences in the lexical properties of the syntactic elements involved (Borer 1984; Chomsky 1995, 2005; Biberauer et al. 2014). Of course, it is always a possibility that language variation with respect to a particular linguistic phenomenon will resist such a characterization. However, as I argue below, not only can the variation in person restrictions be characterized in this way, such a characterization also offers us a solution to the issue of the typological gaps in this domain, which was raised in Section 2.5.

I argue that all the variation in person restrictions can be derived using a single universal version of Agree, constrained by universal economy conditions on syntactic derivations. All the crosslinguistic variation is due to lexical differences in the syntactic elements which I explore further in the subsequent chapters. The definition of the Agree operation that I will be assuming from now on is provided in (24) and follows the original conception of Agree by Chomsky (2000) quite closely. The key point is that probes only look for matching valued features—never specific values of features—and the domain of application for Agree is subject to strict locality conditions: a probe may never skip a matching goal.

(24) *Agree* can be established between an active probe and an accessible matching goal:

a. A goal G_1 is accessible to a probe P if the matching goal is in P's c-command domain and there is no matching goal G_2, such that G_2 asymmetrically c-commands G_1;

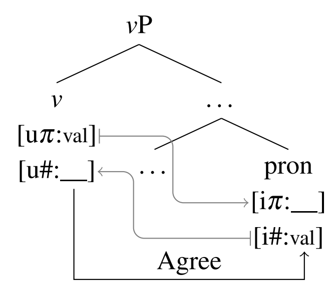
b. Unvalued features ([F:__]) are active probes, while valued features of the same feature type ([F:val]) are their matching goals.

I will be referring to the locality condition on Agree in (24a) as *Agree Closest* (adopting the terminology of Bošković 2003). My main departure from Chomsky (2000) is only in that I allow for the features that are interpretable at LF to be probes. In this I follow approaches to feature valuation in the spirit of Pesetsky and Torrego (2007) and Bošković (2007, 2011), where interpretable features ([iF]) may enter the derivation unvalued and uninterpretable features ([uF]) may enter the derivation valued. Specifically, I adopt Bošković's (2011) approach where all unvalued features are active probes—in other words, what drives Agree is valuation—and their goals are always matching valued features; in both cases, the features may be either interpretable or uninterpretable. This type of on approach to the relation between interpretability and valuation is needed in our case because of the existence of minimal pronouns. Already with minimal pronouns in binding configurations, discussed by Kratzer (2009), the minimal pronouns must have unvalued interpretable φ-features and the heads that bind them must have valued uninterpretable φ-features.

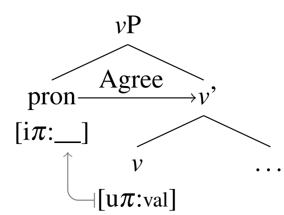
Recall from Section 3.2 that valued person features are limited to phase heads. What is crucial here is that only person features can have a valued status on phase heads outside of binding contexts, all other φ-features on the phase head must actually be unvalued. Similarly, minimal pronouns outside of binding contexts have unvalued person features whereas any other φ-features they have are valued. In Chapter 5, I offer an explanation for why it is specifically person features (and occasionally other features with similar semantics) that can be valued on phase heads and why the valuation status of the φ-features on minimal pronouns is flipped. Pending that discussion, what matters are the syntactic consequences of valued person being limited to phase heads.

Importantly, unvalued φ-features on phase heads—for example, number features (#-*features* or simply [#])—are active probes in the current system. As illustrated in (25), *v* can probe and establish Agree with a pronoun with matching valued #-features (recall that on minimal pronouns features other than person are valued), and the person features (π-*features* or simply [π]) on the pronoun can also be valued by *v* as a result of this Agree relation—we will see below why this is possible. However, in the current system the unvalued [π] features on minimal pronouns are also active probes, so they can establish Agree with valued [π] features they c-command; for example, the [π] features on a minimal pronoun may Agree with the valued [π] features of *v* when the pronoun is in the Spec of *v*P, as illustrated in (26).

(25) Agree ("long distance"):



(26) Agree ("Spec-Head"):



The two possible configurations in which person valuation of the minimal pronouns can take place will be crucial for deriving the STRONG/WEAK variation in the next section. Where we will see that the manner in which a minimal pronoun's person features end up being valued changes based on the lexical differences between the minimal pronouns across different languages. These lexical differences can affect the course of the syntactic derivation by way of interacting with universal derivational economy principles, which I review next.

It is commonly assumed that the application of syntactic operations is constrained by principles of derivational economy (see, among others, Chomsky 1991, 1993, 1995;

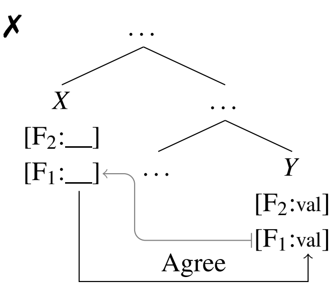
Bošković 1997b; and Bošković and Messick to appear for an overview). This means syntactic derivations are constrained so that the triggering of superfluous syntactic operations and introduction of superfluous syntactic structure is avoided. The economy principle that will be key in deriving different person restriction patterns is *Maximize Agree*, given in (27).

(27) *Maximize Agree*
If a probe matches a goal, all matching features on the probe and goal must be valued at that point in the derivation. (see also Řezáč 2004, 477)

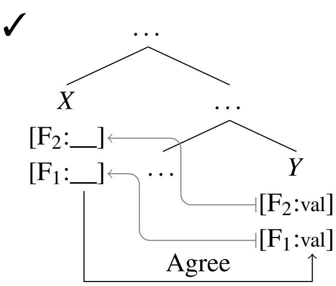
All unvalued features are probes in the current system, and a single syntactic head may host a number of features, which can be either valued or unvalued. For each of them to probe and establish Agree independently of each other would be quite a redundancy. However, if we treat Agree as an operation that applies between two syntactic heads, then Maximize Agree can ensure that once Agree is established between two heads, it does not need to be established again—no matter how many sets of matching features the probe and goal have.

The application of this principle is illustrated in (28)–(29). What Maximize Agree bans are derivations like (28), where the unvalued F_1 feature on the *X* head establishes Agree with its valued counterpart on the *Y* head and is valued to the exclusion of the other probe on the *X* head, the unvalued F_2, which then has to probe independently of F_1 to establish Agree and be valued. Instead, Maximize Agree enforces the derivation in (29), which involves the same exact features as (28), but where the Agree established between *X* and *Y* because of F_1 also enables F_2 to be valued without having to probe and establish Agree on its own.

(28) ✗

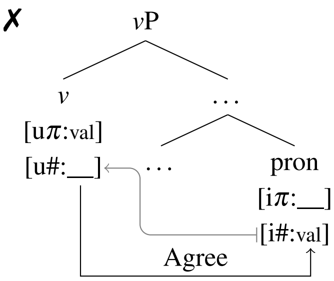


(29) ✓

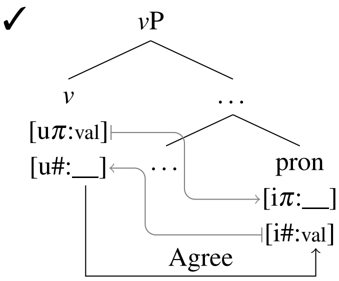


Note that Maximize Agree makes no reference to the direction of the multiple instances of valuation, and this comes into play with the cases where Agree is initiated between a phase head and a minimal pronoun. In this case, Agree is established as the result of the unvalued #-features on the phase head and the matching valued #-features on the minimal pronoun. But Maximize Agree disallows derivations like (30), where only the #-features are valued as part of this Agree relation. As a result, the π-features on the pronoun can also be valued by the π-features on the phase head as part of the same Agree relation, as shown in (31).

(30) ✗



(31) ✓

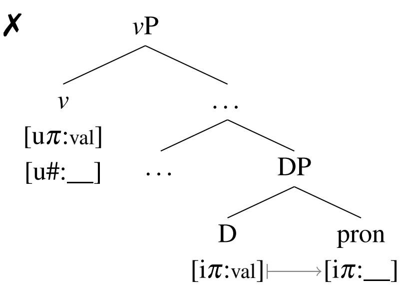


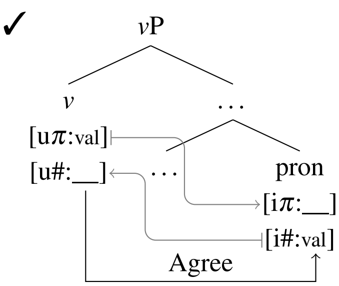
The situation here is different from (28)–(29), because the second active probe is not on the same head as the probe that initiates Agree, which will be a key point in the next section.

Whereas Maximize Agree is a principle of economy that constraints a grammatical operation, there are also principles of economy of representation that are relevant for person restrictions. In fact, even before derivational economy as a general concept was adopted, it has been acknowledged that syntactic structures of smaller structural complexity are preferred over those with a greater structural complexity—all other things being equal. Some proposals in this vein include: *Avoid Pronoun* (Chomsky 1981, 65), *The Minimal Structure Principle* (Law 1991), *Structural Economy Principle* (Safir 1993, 64), Speas (1994, 186–187), *Minimize Structure* (Cardinaletti and Starke 1994; 1999, 198), Chomsky (1995, 294), Bošković (1997b, 37–9), *Minimize Restrictors!* (Schlenker 2005, 391), and *Minimize DP!* (Patel-Grosz and Grosz 2017, 279). Given that the approach to pronoun typology I adopt already largely builds on the work of Cardinaletti and Starke (1994, 1999), I also adopt the economy principle in (32), which is based on theirs.

(32) *Minimize Structure*
Given two extended projections of the same lexical item: α and β, if α has fewer syntactic nodes than β, β is used iff α is independently ruled out.
(see also Cardinaletti and Starke 1999, 47)

As I discuss in more detail below in relation to person restrictions, the result of (32) is that strong pronouns, even though they are usually compatible with the same syntactic contexts that deficient (minimal) pronouns are compatible with, will be unavailable in the contexts where corresponding minimal pronouns are possible (cf. (33) vs. (34)). Consequently, strong pronouns are only possible in the syntactic contexts where the corresponding deficient pronouns yield an ungrammatical result, such as person restriction contexts.

(33) ✗


(34) ✓


We will see next that these universal economy principles can derive the variation in person restrictions by interacting with purely lexical differences between languages: most of the variation in person restrictions amounts to differences in the internal structure of deficient minimal pronouns and differences in the identity of the host of valued person features.

3.5 Deriving the STRONG vs. WEAK variation

With the essential theoretical assumptions in place, we can now look at how the proposed system derives person restrictions and the STRONG vs. WEAK variation in particular.

When it come to STRONG person restrictions, the derivation will not be that different from previous analyses. Although the source of the person restriction itself will be different—failure to value the person features of a minimal pronoun—the restriction will nonetheless arise due to the blocking of Agree between a functional head and a pronoun. Specifically, a STRONG restriction boils down to Agree Closest: the phase head may only Agree with the structurally closest minimal pronoun, which leaves all other minimal pronouns unable to receive a person value and consequently restricted to 3P.

The approach I take with WEAK person restrictions is different. Recall that the standard approach to WEAK restrictions involves relaxing the constraints on Agree in a way that allows a probe to establish Agree with multiple goals given specific person values on the goals; that is, with two 1/2P goals, but not otherwise. In contrast, I have proposed above that Agree is universally the same operation that involves a one-to-one relationship between a probe and a goal (i.e. a probe can only have one goal). The key in getting multiple minimal pronouns to be valued by the same head is in the pronouns themselves. The minimal pronouns can act as probes themselves due to their unvalued person features; that way multiple pronouns can stand in an Agree relation with the same phase head without the need for a special Multiple Agree operation. The basic idea, which I expand on below, is that WEAK person restrictions are essentially a *Superiority* effect (Chomsky 1973), in contrast to STRONG restrictions, which result directly from Agree Closest. The variation between STRONG and WEAK restrictions arises due to the valuation options available to the pronouns depending on where they are in the structure in relation to the most local phase head.

3.5.1 Deriving STRONG person restrictions

Although the analysis in terms of Agree Closest is applicable to both EA-IA and IA-IA STRONG person restrictions, I will focus here only on the IA-IA domain, specifically the

STRONG canonical PCC, to illustrate the basic principle behind STRONG restrictions. But before we look at the derivation itself, let us remind ourselves of what the STRONG canonical PCC pattern looks like. Remember that this is the restriction pattern we find, among other cases, in Greek double object ditransitives: the Goal clitic can have any person value, while the Theme clitic can only be 3P. The examples showing this are presented again in (35).

(35) Greek [*Hellenic*]:

a. Tha { **mu** / **su** / **tu** } **ton** stilune — 1/2/3.GL≫3.TH
FUT { 1.GEN / 2.GEN / 3.M.GEN } 3.M.ACC send.3PL
'They will send **him** to **me/you/him**.'

b. *Tha **mu** **se** stilune — ✗1.GL≫2.TH
FUT 1.GEN 2.ACC send.3PL
'They will send **you** to **me**.'

c. *Tha **tu** { **me** / **se** } stilune — ✗3.GL≫1/2.TH
FUT 3.M.GEN { 1.ACC / 2.ACC } send.3PL
'They will send **me/you** to **him**.' (Anagnostopoulou 2003, 200, 252; 2005, 202)

The STRONG canonical PCC is also found in languages like Slovenian, where it alternates with a corresponding reverse person restriction. In Slovenian, the canonical STRONG pattern is found when the Goal clitic precedes the Theme clitic, as shown again in (36).

(36) Slovenian [*Slavic*]:

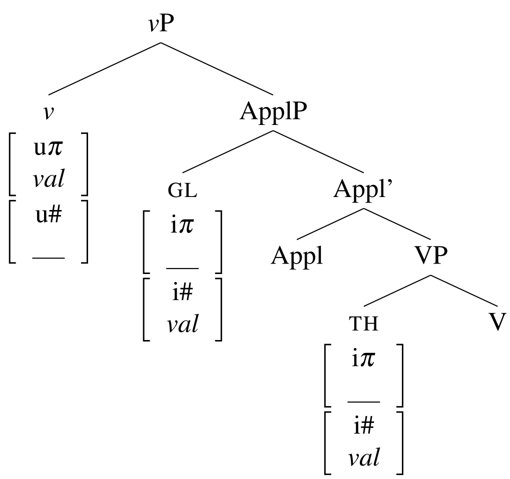
a. Mama { **mi** / **ti** / **mu** } **ga** bo predstavila. — 1/2/3.GL≫3.TH
mom { 1.DAT / 2.DAT / 3.M.DAT } 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him** to **me/you/him**.'

b. *Mama **mi** **te** bo predstavila. — ✗1.GL≫2.TH
mom 1.DAT 2.ACC FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

c. *Mama **mu** { **me** / **te** } bo predstavila. — ✗3.GL≫1/2.TH
mom 3.M.DAT { 1.ACC / 2.ACC } FUT.3 introduce.F
'Mom will introduce **me/you** to **him**.' (Stegovec 2015b, 108–9)

As noted in Section 3.2, I adopt an applicative analysis of double object ditransitives (Marantz 1993; Pylkkänen 2002, 2008; Anagnostopoulou 2003; Georgala 2011), where the base position of the Theme is as the complement of the verb inside VP, while the Goal resides in the specifier of the applicative phrase (ApplP), whose head takes the VP as its complement.[6] Importantly, the phase head *v* asymmetrically c-commands both arguments as ApplP is the complement of *v*P. The structure of a double object ditransitive with minimal pronouns as the Goal and Theme is shown in (37).[7] Note that *v* has a valued uninterpretable person feature by virtue of being a phase head, but it also has an unvalued uninterpretable number feature which is an active probe. The minimal pronouns on the other hand have unvalued interpretable person features and valued interpretable person features.

(37)



The structure in (37) is not meant to imply that *v* is always the only functional head that is also a probe. In fact, Appl may also be a probe, but since it is not a phase head it cannot bear valued person features, which is all that matters for what we are trying to explain here. Therefore, I will ignore the possibility of φ-feature probes on Appl for ease of exposition.

6. Different types of double object ditransitives may differ with respect to whether the Goal originates in SpecApplP or raises there from a position within VP but above the Theme (Georgala 2011). Since this has no bearing on my analysis of person restrictions, I consistently present the Goal as originating in SpecApplP.

7. The minimal pronouns are labeled according to their thematic role in all derivations from now on. This is based on the relationship between thematic roles and restriction strength established in Section 2.5.2.2. The labels are not meant to imply that pronoun type tracks thematic roles; as we also saw in Section 2.5.2.2, the type of a pronoun can change based on its grammatical role, but also case or number (see Cardinaletti and Starke 1994, 1999). Specific pronoun types can therefore have paradigms delimited by grammatical role, case, number: e.g. a Theme is a clitic object in an active context, but a *pro* subject in a passive one. This can be implemented as a filter excluding specific pronoun types in contexts for which they lack an appropriate form.

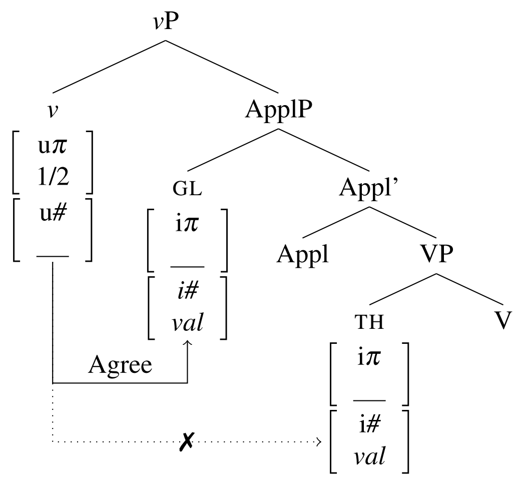
Since two 1/2P pronouns never co-occur with STRONG restriction patterns, we do not have to concern ourselves yet with combinations where the pronouns are 1P and 2P respectively—so I will utilize a simplified person feature system pending the discussion of WEAK person restrictions below. Thus, in a given derivation, a valued person feature on *v* may either have a positive participant specification, represented as shown in (38a), or have a bare 3P specification, as shown in (38b). The latter is crucially distinct from the unvalued status of the person features of a minimal pronoun, shown in (38c).

(38) a. $\underset{\begin{bmatrix} \text{u}\pi \\ 1/2 \end{bmatrix}}{v}$ (1/2P) b. $\underset{\begin{bmatrix} \text{u}\pi \end{bmatrix}}{v}$ (3P) c. $\underset{\begin{bmatrix} \text{i}\pi \\ \_\_ \end{bmatrix}}{\text{pron}}$ (unvalued)

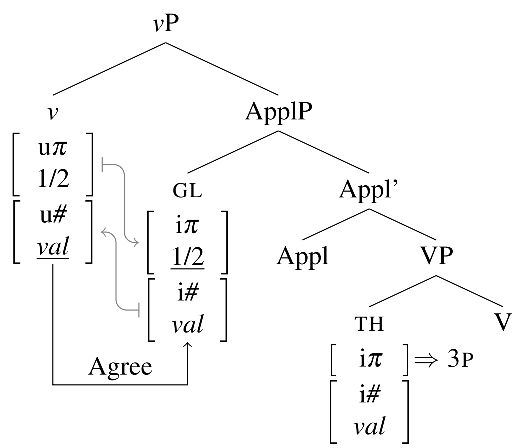
Recall from Section 3.2 that a failure to value the person feature of a minimal pronoun results in the pronoun getting a default 3P person value. The question here is why should such a failure of valuation limit the pronoun to 3P and not instead just prohibit the pronoun from appearing. I argue that 3P in fact corresponds to a person feature without a value. In this I follow Béjar and Řezáč (2009) and Preminger (2014) who assume that the default person value, meaning the value assigned upon a failed valuation (which I assume does not cause derivations to crash; see Preminger 2014), is no different from valued 3P. 3P is just a person feature with no positive speaker or participant specification, not to be confused with the lack of person features or an unvalued person feature. In this view, a person feature that was valued 3P and one that received a default 3P value are equivalent at the interfaces.

Let us now look at the derivation of a double object ditransitive with two minimal pronouns, starting with a *v* specified as 1/2P. Since *v* also has unvalued number features, those must probe for the closest matching valued features, which reside on the Goal. The [u#] probe on *v* and the valued [i#] feature on the Goal then initiate Agree between *v* and the Goal, as shown in (39a); since the Goal asymmetrically c-commands the Theme, Agree between *v* and the Theme is impossible due to Agree Closest. Once Agree is established, the valued [i#] on the Goal may value the [u#] on *v*, and due to the Maximize Agree principle (see (40)) all other matching features on *v* and the Goal may value each other. This means that, as shown in (39b), the person features on the Goal may also be valued as 1/2P by *v*.[8]

(39) a.



b.



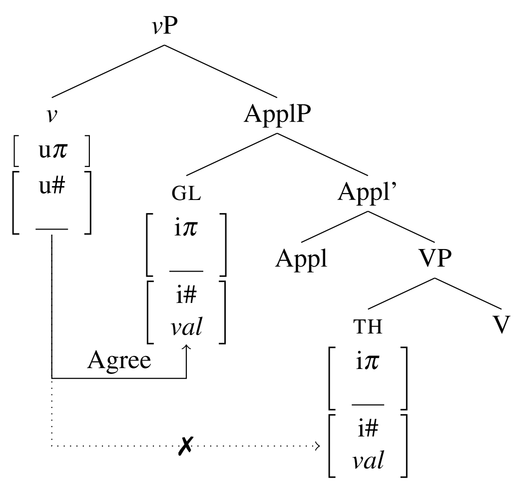
8. Note that φ-feature Agree is separated from Case in the current system (contra Chomsky 2000). This will be important for the derivation of reverse person restrictions in Section 3.6 (recall from Chapter 2 that relating person restrictions to Case causes a serious problem for deriving reverse restriction patterns). I assume the Goal in particular is assigned theta-related dative case by Appl, which also assigns the Goal its theta-role, under the local specifier-head configuration (Woolford 2006) (if we wished to keep this in line with a more traditional approach to inherent case, we could also assume that V raises to Appl to theta-license the Goal; see here Lasnik 1995). It is probably a good idea to keep Case and φ-feature Agree separated even independently of person restrictions. For example, Bošković (2007) discusses Blackfoot [*Algonquian*] examples from Frantz 1978 that show object agreement between a matrix verb and the first conjunct of the coordinated subject of the embedded finite clause (the fact that the controller of the object agreement is a conjunct excludes a movement analysis because of the coordinate structure constraint; Ross 1967). In such examples we see object agreement with an element that does not get accusative case from the matrix *v*, so Agree with *v* must be independent from accusative assignment (see Bošković 2007 for more relevant discussion and examples).

(40) *Maximize Agree*
If a probe matches a goal, all matching features on the probe and goal must be valued at that point in the derivation.

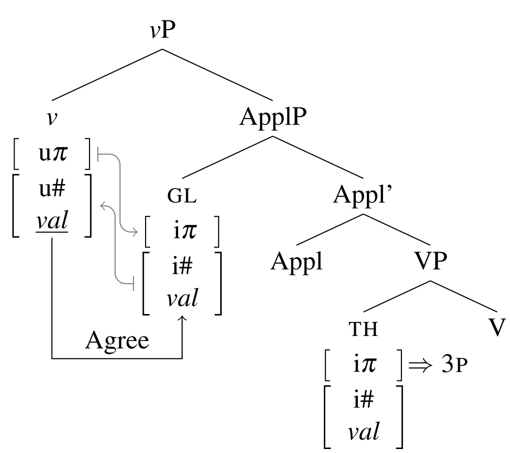
As the Theme cannot enter into Agree with *v* due to Agree Closest, its person features cannot be valued. Recall here that a person feature without a value corresponds to 3P, which means that since the Theme's [iπ] cannot be valued it must be spelled out as 3P at the interfaces. This is the source of the STRONG person restriction: no matter what happens, the Theme pronoun's [iπ] cannot be valued by *v*, so it will always be 3P whenever a Goal is present.

The derivation goes pretty much the same if the person feature on *v* is a bare 3P one. As shown in (41), the only difference is that when *v* values the Goal pronoun it has no participant features to share, so the Goal ends up being 3P.

(41) a.



b.



Since the Goal also intervenes between *v* and the Theme here, the latter also cannot be valued, spelling out as 3P. The person assigned to the Goal therefore has no bearing on the restriction; which is the result we want with respect to the STRONG restriction pattern.

Importantly, the analysis I just presented treats the Goal and the Theme exactly the same, aside from their position in the structure, which will be crucial later in in Section 3.6 with the derivation of the reverse version of the STRONG restriction.

3.5.2 Deriving WEAK person restrictions

Let us turn now to WEAK person restrictions, specifically the WEAK version of the canonical PCC, which will make it easier to highlight the difference between STRONG and WEAK PCC derivations. Remember that the main difference is that with WEAK restrictions the two pronouns may both be 1/2P; only combinations where a 3P pronoun asymmetrically c-commands a 1/2P pronoun are impossible. We can see this in the case of the canonical PCC with the Classical Arabic examples repeated in (42),[9] but also the Slovenian examples from dialect B repeated in (43), which show the person combinations that are available when the Goal clitic precedes the Theme clitic for the speakers that have a WEAK restriction.

(42) Classical Arabic [*Semitic*]:

a. ʔaʕtˀaː { **-niː** / **-ka** / **-haː** } **-hʊ** — 1/2/3.GL≫3.TH
gave.3 { -1.IO / -2.IO / -3.F.IO } -3.M.DO
'He gave **him/it** to **me/you/her**.'

b. ʔaʕtˀaː **-niː** **-ka** — 1.GL≫2.TH
gave.3 -1.IO -2.DO
'He gave **you** to **me**.'

c. *ʔaʕtˀaː **-huː** { **-niː** / **-ka** } — ✗3.GL≫1/2.TH
gave.3 -3.M.IO { -1.DO / -2.DO }
'He gave **me/you** to **him**.' (Walkow 2014, 139–40)

9. In order to condense the data, I ignore the vowel quality changes on the DO clitic, which changes depending on the preceding clitic; I only show the form that surfaces when the preceding clitic is the 3P.F **-haː**.

(43) Slovenian B [*Slavic*]:

a. Mama { **mi** / **ti** / **mu** } **ga** bo predstavila. $\boxed{\text{1/2/3.GL} \gg \text{3.TH}}$
mom { 1.DAT / 2.DAT / 3.M.DAT } 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him** to **me/you/him**.'

b. Mama **mi** **te** bo predstavila. $\boxed{\text{1.GL} \gg \text{2.TH}}$
mom 1.DAT 2.ACC FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

c. *Mama **mu** { **me** / **te** } bo predstavila. $\boxed{\text{✗3.GL} \gg \text{1/2.TH}}$
mom 3.M.DAT { 1.ACC / 2.ACC } FUT.3 introduce.F
'Mom will introduce **me/you** to **him**.' (Stegovec 2015b, 108–9)

Having refreshed our memory on the basic facts, consider now an alternative descriptive way to characterize the WEAK restriction: the lower pronoun can be 1/2P only when the higher pronoun is 1/2P as well. In the current system that means that the lower pronoun can be valued 1/2P only when the higher pronoun is valued 1/2P as well.

Described in these terms, the WEAK restriction has the signature of a Superiority effect (Chomsky 1973; Pesetsky 1982; Rudin 1988; Cheng and Demirdash 1990; Lasnik and Saito 1992; Bošković 1997a; i.a.). Superiority describes a class of displacement phenomena, where given two potential targets for syntactic movement, either only the structurally higher of the two can be targeted or the structurally lower of the two cannot be targeted to the exclusion of the higher one. We can characterize such cases abstractly as in (44).[10]

(44) *Superiority*
A syntactic operation cannot target Y without also targeting X if

(i) X and Y are of the same type relevant for the syntactic operation, and

(ii) X asymmetrically c-commands Y.

10. The original *Superiority Condition* is:

> No rule can involve X, Y in the structure
> $\ldots X \ldots [\ldots Z \ldots WYV \ldots] \ldots$
> where the rule applies ambiguously to Z and Y, and Z is superior to Y.
>
> The category A is "superior" to category B in the phrase marker if every major category dominating A dominates B as well but not conversely. (Chomsky 1973, 246)

Superiority is common with *wh*-extraction. The original cases discussed by Chomsky (1973) show that in English, given two *wh*-phrases, only the highest can undergo *wh*-movement, as seen in (45), where object *wh*-movement over the subject *wh*-word is banned.

(45) *John knows what$_j$ [who$_i$ saw t_j] (Chomsky 1973, 245)

Superiority is also observed when both *wh*-elements in a question undergo movement, such as in a language that has multiple *wh*-fronting like Bulgarian (Rudin 1988), where Superiority ensures the order of the fronted *wh*-words reflects their base order, as seen in (46), where the fronted subject *wh*-word must precede the fronted object *wh*-word.

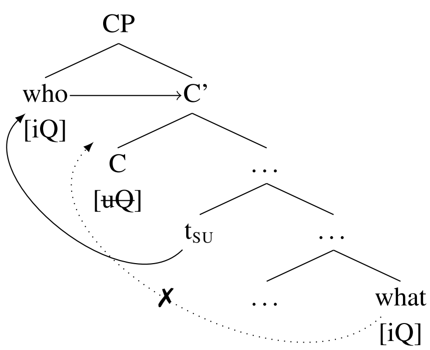
(46) Bulgarian [*Slavic*]:
a. Koj$_i$ kogo$_j$ [t_i vidjal t_j]?
who whom saw
b. *Kogo$_j$ koj$_i$ [t_i vidjal t_j]?
whom who saw
'Who saw whom?' (Bošković 1997a, 232)

The reason why the Bulgarian case is the result of Superiority as well as its connection to WEAK person restrictions becomes clearer if when we consider a common analysis of Superiority effects with *wh*-fronting. Superiority in wh-fronting can be captured straightforwardly with feature checking analyses (Richards 1997, 2001; Bošković 1999, 2007).

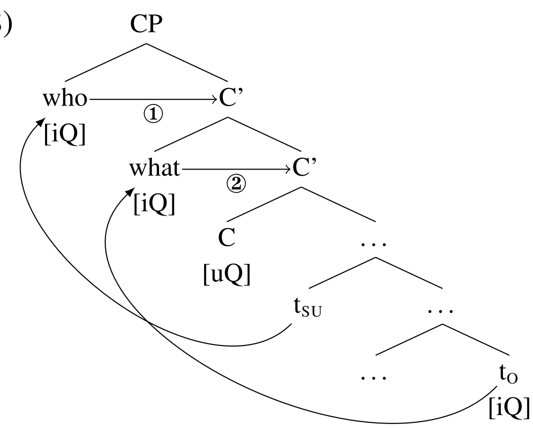
Consider, for example, the analysis of Bošković (2007), where the movement of *wh*-phrases to C is driven by the need of the moving element to check its interrogative/focus feature ([iQ]) against the matching feature on C ([uQ]). This approach can be made more in line with the present assumptions concerning Agree/valuation, by making the [iQ] features on *wh*-phrases unvalued (just like the person features on minimal pronouns) and the [uQ] features on C valued (just like the person features on *v*). Note that the movement of *wh*-phrases to SpecCP must, like all other movement, obey *Shortest Move* (Richards 1997, 2001; Bošković 1999), which means that, given two *wh*-phrases, only the closest one can move to

C. If then the [iQ] feature of the closest *wh*-phrase Agrees with and deactivates the [uQ] feature on C, no other *wh*-word can move to C, as illustrated in (47), which exactly describes the Superiority effect in English *wh*-fronting in (45). In order to derive Bulgarian-style multiple *wh*-fronting, the [uQ] feature on C must survive the fronting of of the *wh*-subject and Agree between [iQ] and [uQ]. This allows for the *wh*-object to also move to C to value its [iQ], *tucking in* under the *wh*-subject in SpecCP (Richards 1997, 2001), as illustrated in (48). The tucking in of the *wh*-object also follows here from Shortest Move.

(47)



(48)

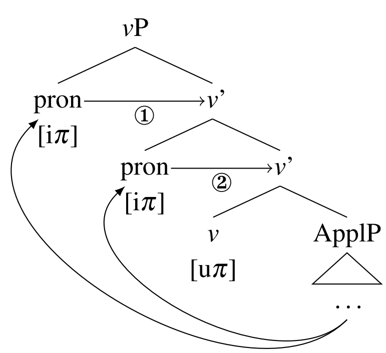


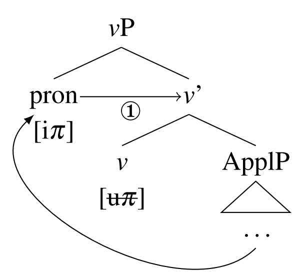
The *wh*-subject must thus front first—and its [iQ] is valued by the [uQ] on C first, followed by movement of the *wh*-object—whose [iQ] can then also be valued by [uQ] on C. The order in which the two movements take place is reflected in their order in SpecCP, with the *wh*-subject preceding the *wh*-object. Because of Shortest Move, this is the only order in which the two *wh*-phrases can move and consequently the only order in which they can appear. This is why Superiority in Bulgarian takes the form of order preservation.

The case of multiple *wh*-fronting reveals that, in addition to the prototypical cases of feature driven operations that result in the immediate deletion of the uninterpretable/unvalued feature (Chomsky 1995, 2000, 2001), there are also cases where the relevant feature is not immediately deleted and can trigger multiple operations of the same kind. This possibility is not reserved for multiple *wh*-fronting (see e.g. Demuth and Gruber 1995 on multiple instances of subject agreement in Bantu; Bobaljik 1995 on verb movement in Germanic; and Collins 1995; Ura 1996; Hiraiwa 2001, 2005 for a more general discussion of cases where one feature drives multiple instances of the same operation).

I argue that WEAK person restrictions are another phenomenon of this kind. Consider that the Bulgarian Superiority pattern is abstractly just like the WEAK person restriction pattern, if we substitute *wh*-fronting for pronouns being 1/2P: the lower pronoun can be valued as 1/2P iff the higher pronoun is also valued as 1/2P. I want to argue that this similarity is not coincidental, and that both phenomena involve the same kind of multiple valuation configuration driven by the valuation of features on the moving elements.

In a nutshell, I propose that multiple minimal pronouns can be valued 1/2P if they are all valued by moving to Spec*v*P, which is abstractly similar to Bulgarian *wh*-fronting. This movement is driven by the unvalued [iπ] feature on the pronouns, which is an active probe (see discussion below), and the multiple pronouns can all be valued because the valued [uπ] on *v* is not deleted after the first pronoun is valued, as shown in (49). Alternatively, the first pronoun that moves to Spec*v*P and gets valued may also delete *v*'s [uπ] feature, as in (50).

(49)


(50)


The outcome in (50) can crucially bleed person valuation for any subsequent pronouns that would move to Spec*v*P, yielding a mixed 1/2P and 3P combination where the 1/2P pronoun is always the highest pronoun—the one that moves to Spec*v*P first given Shortest Move.

We will see that these are the only two ways in which a minimal pronoun can be valued as 1/2P in a WEAK restriction language, which is why there are no combinations where the higher pronoun is 3P and the lower one is 1/2P. In addition to deriving why only these two valuation options are possible, I will also argue that the reason why the options in (49) and (50) are both available is due to the the person features on *v* having a special status, which is a consequence of the features being valued uninterpretable features: they *can* be, but *need not* be, deleted right after they value a matching unvalued feature (see also Bošković 2011).

Since we are going to be dealing with more than one 1/2P pronoun in a single derivation, we need to update the person feature system so that we can distinguish 1P and 2P pronouns while the features themselves still come from the same source. For this purpose I adopt a privative approach to person features, in particular an approach where person features stand in a dependency relation to each other (see e.g. Harley and Ritter 2002; McGinnis 2005; Béjar and Řezáč 2009). In the particular system I adopt, a bare [π] feature corresponds to 3P (see also Section 3.5.1), a [π] feature combined with a participant feature (represented as '2') corresponds to 2P, and a [π] feature combined with both a participant and a speaker feature (represented as '1') corresponds to 1P. Crucially, a participant feature may not occur in the absence of the root [π] feature, and likewise, an author feature may not occur in the absence of a participant feature; it is logically impossible to be a speaker who is also not a speech act participant. In other words, the speaker feature is dependent on the participant feature (represented as '2[1]'), and the participant feature is dependent on [π].

Although this feature system has several implications for feature valuation that I come back to below, the main point right now is that a phase head fully specified with all the person features, such as the one in (51a), will be capable of assigning two different values to two different pronouns: one of the pronouns receiving only the participant '2' feature and the other one receiving both the participant '2' and the speaker '1' feature. With all other specifications, like (51b) and (51c), the options are going to be more limited, as we will see.

(51) a. $\overset{v}{\begin{bmatrix} \mathrm{u}\pi \\ 2[1] \end{bmatrix}}$ (1P, 2P) b. $\overset{v}{\begin{bmatrix} \mathrm{u}\pi \\ 2 \end{bmatrix}}$ (2P) c. $\overset{v}{\begin{bmatrix} \mathrm{u}\pi \end{bmatrix}}$ (3P)

In relation to what allows multiple pronouns to stand in a valuation relation with a single head, my main point of departure with respect to the Multiple Agree approaches, like those of Anagnostopoulou (2005) and Nevins (2007, 2011), is in the source of the STRONG VS. WEAK parameterization. As I noted above, under the analysis presented here, the variation will be due to lexical differences, not the parameterization of core grammatical operations. In the current system, the parameterization can be attributed to a minor difference in the internal structure of the minimal pronouns. In STRONG PCC languages the minimal pronouns are effectively one feature bundle with all the φ-features on the same head, as shown in (52). On the other hand, in WEAK PCC languages the pronouns are more articulated in their internal structure, having individual φ-features reside on different heads, as shown in (53).

(52) $\overset{\text{pron}}{\begin{matrix} \begin{bmatrix} \mathrm{i}\pi \\ \_\_ \end{bmatrix} \\ \begin{bmatrix} \mathrm{i}\# \\ \mathit{val} \end{bmatrix} \end{matrix}}$

(53) $\overset{\text{pron}}{\begin{bmatrix} \mathrm{i}\pi \\ \_\_ \end{bmatrix} \quad \begin{bmatrix} \mathrm{i}\# \\ \mathit{val} \end{bmatrix}}$

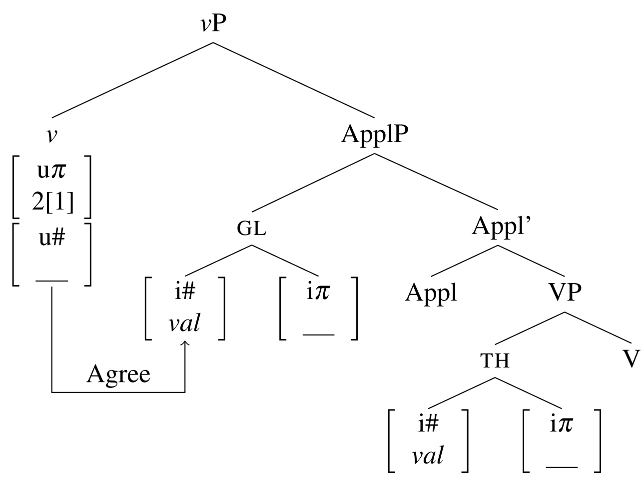
We will see, mainly in Chapter 6, that this difference in structure can be tied to certain morphological or other PF differences and the observation that the strength of the PCC effect decreases as the clitic/weak pronouns become less prosodically deficient (related also to the discussion in Section 3.3.1 above). At this point, what is important is that this minute difference in internal structure has a cascading effect on how the minimal pronouns interact with *v*. The differences are due to Agree being a relation established between two heads and the Maximize Agree principle, repeated in (54).

(54) *Maximize Agree*
If a probe matches a goal, all matching features on the probe and goal must be valued at that point in the derivation.

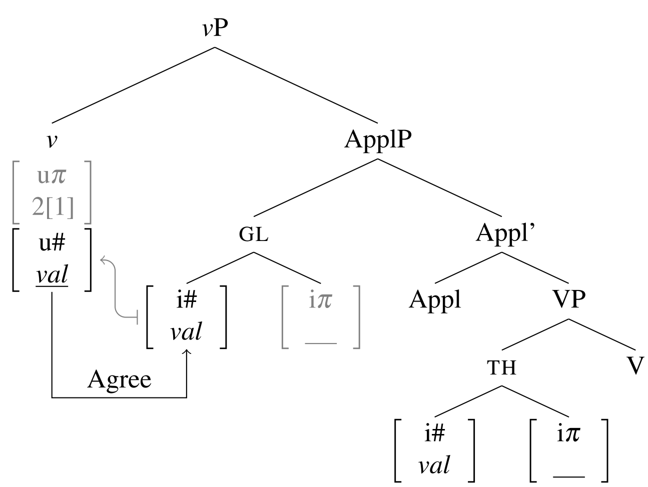
Since the pronoun with the richer internal structure in (53) has φ-features on different heads, Maximize Agree is not applicable and consequently each of the heads has to be valued separately. Let us consider the consequences of this in a derivation which is otherwise identical to the STRONG derivations discussed in the previous section.

We start off the derivation with a *v* fully specified with person features and an active [u#] probe, as shown in (55a). Agree is established between [u#] and the valued [i#] on the Goal, but due to [i#] and [iπ] being on separate heads, only [u#] can be valued, as shown in (55b).

(55) a.



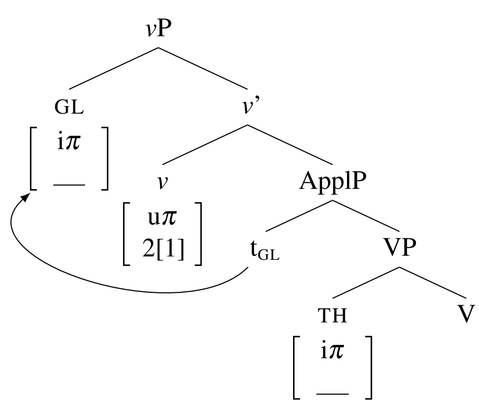
b.



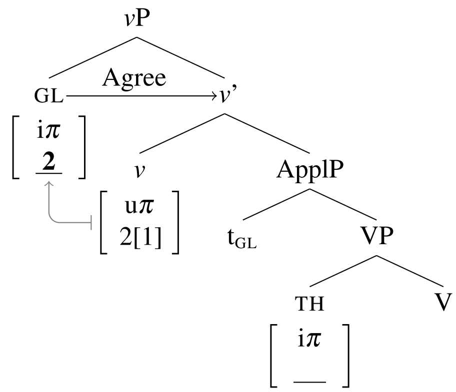
The [iπ] feature on the Goal crucially stays unvalued and therefore an active probe. But the [iπ] feature does not c-command a matching valued feature in its base position—the only valued person feature inside *v*P is on *v*. I assume here, following Bošković (2007), that unvalued features on arguments, like the [iπ] on the Goal pronoun, can trigger movement of the argument to a position where a matching valued feature can value them. In the case of (55b), this means that the Goal has to move to Spec*v*P so its unvalued [iπ] can be valued (see Bošković 2007, 2011 on other cases of movement driven by unvalued features on the argument; e.g. arguments moving so their unvalued Case feature can probe and be valued).

This step of the derivation is shown in (56a) (since they no longer matter for the derivation, I leave out the number features and Appl heads from here on): the Goal minimal pronoun is forced to move to Spec*v*P due to its unvalued [iπ] feature, an active probe, since the unvalued [iπ] feature does not c-command a matching valued goal in the Goal's base position. Once the Goal has moved to Spec*v*P, its unvalued [iπ] feature can probe and it finds the matching valued [uπ] feature on *v* (since the Goal is a specifier of *v*P, it c-commands *v*). After that, Agree can be established between the Goal's [iπ] feature and *v*'s [uπ] feature and the unvalued [iπ] feature can be valued by the [uπ] feature, which is shown in (56b).

(56) a.



b.

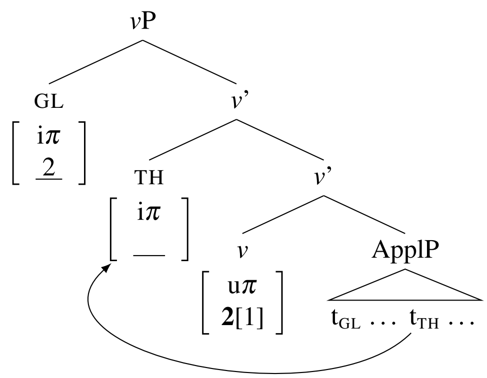


At this point, the feature geometry of person limits valuation to the participant ‘2’ feature because the speaker ‘1’ feature is dependent on the participant ‘2’ feature being present in the person feature set. This valuation procedure must be in place to prevent illicit feature sets to arise on the pronouns; a pronoun with a speaker but no participant feature would be a semantically malformed object. Only after all valuation involving participant features has taken place can valuation of the speaker feature occur.[11]

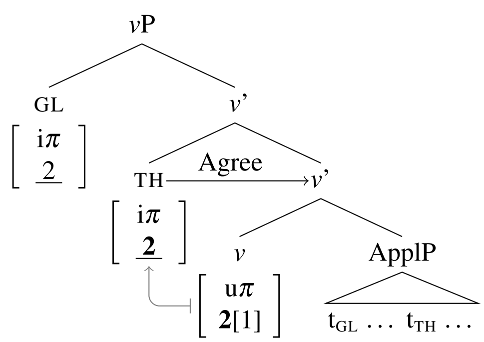
In this particular derivation, there is another candidate for person valuation: the Theme pronoun. As with the Goal pronoun before, the unvalued [iπ] on the Theme pronoun also does not c-command a matching goal in situ, which, following the system of Bošković (2007), triggers movement of the pronoun. The Theme pronoun must then move to Spec*v*P as well. It does so, as shown in (57a), by tucking in (Richards 1997, 2001) below the Goal (recall the multiple *wh*-fronting example in (48) above), where its unvalued [uπ] feature can also probe, establish Agree with the matching valued [uπ] feature on *v*, resulting in the pronoun being valued with the participant feature, as shown in (57b).

11. Note that this ordering of valuation steps is not in violation of Maximize Agree. Firstly, it is imposed by the feature geometry; failure to comply potentially yields an ungrammatical result. Secondly, Maximize Agree only pertains to different sets of unvalued features being valued simultaneously as part of the same Agree operation; it does not constrain the steps that make up the valuation of a single feature set.

(57) a.



b.

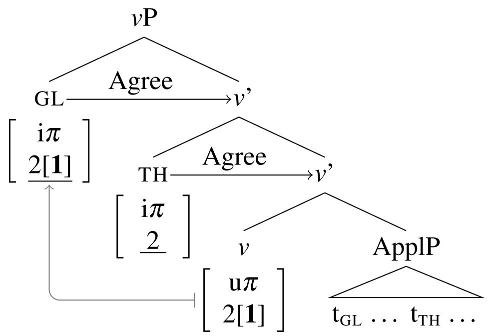


The tucking in of the Theme here also follows from Bošković's (2007) system, where each movement step driven by an unvalued feature on the moving element obeys the Shortest Move requirement: movement must target the closest head with a matching valued feature or, when there is no such head in the current phase, the phase head (in order for the moving element to escape being sent to spell-out; this is what makes successive cyclic movement possible in a system where movement is driven by features on the moving element).

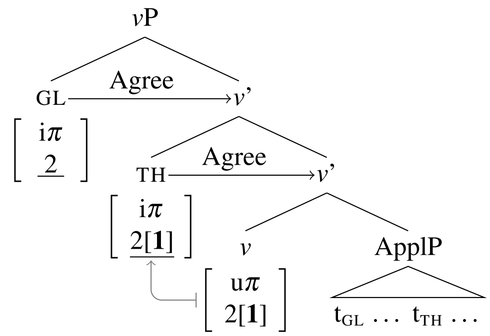
Both pronouns have a participant feature at this point, and both are in an Agree relation with *v*, so either the Goal (cf (58)) or the Theme (cf. (59)) can receive the speaker feature.[12]

12. The intuition here is similar to *The Conservation Law of Agree* of Hiraiwa (2005, 41): "Agree relations are unchanged and retained after Merge." In this case, the merger of the Theme does not undo the Agree relation the [iπ] of the Goal has already established with the [uπ] of *v*.

(58)



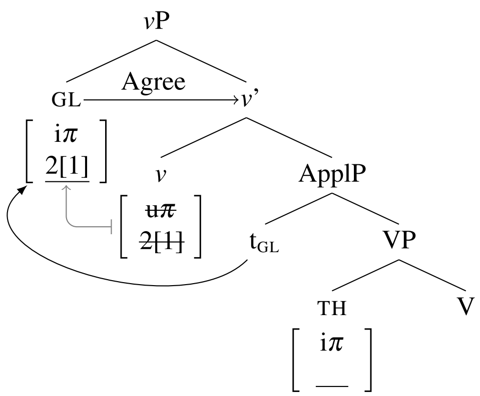
(59)



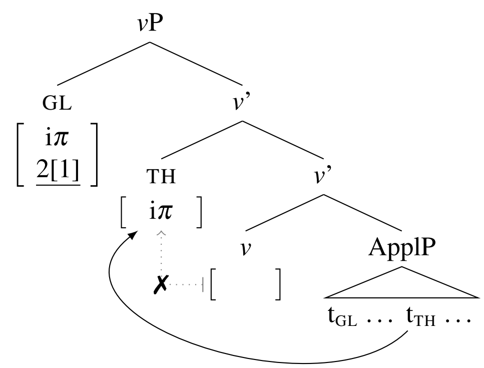
The option in (58) yields a 1P≫2P pronoun combination, whereas the option in (59) yields a 2P≫1P pronoun combination. Note that it is not possible for both pronouns to receive the speaker feature (unless it is possible in the language for one of them to have the status of a reflexive), as this would yield two 1P pronouns and therefore a Principle B violation.

The derivation we went though step by step in (55)–(59) is however only one of the two options that we get in a WEAK PCC language in the system developed here. This is the option that we get when the [uπ] feature on *v* is not deleted after it values the Goal pronoun. As I suggested above, the [uπ] may also get deleted immediately upon taking part in valuation. In that case, the derivation proceeds exactly the same way up until the point when the Goal pronoun moves to Spec*v*P to be valued. At that point, the [uπ] is immediately deleted after it values the [iπ] on the Goal 1P, as shown in (60a); if the [uπ] only had participant '2' features it would value the pronoun 2P. This means that when the Theme, which must move due to its unvalued [iπ] feature, also moves to Spec*v*P, there is no more valued [uπ] feature there, and the Theme can only receive a default 3P value, as shown in (60b). In short, the valuation of the Goal bleeds the valuation of the Theme.

(60) a.



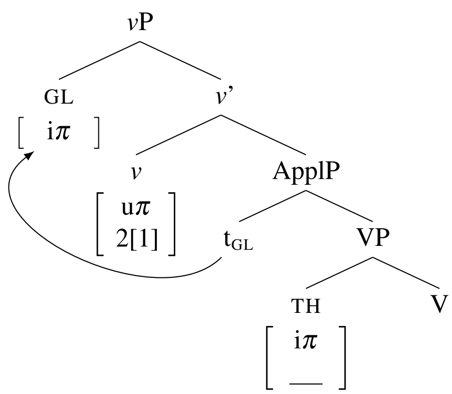
b.



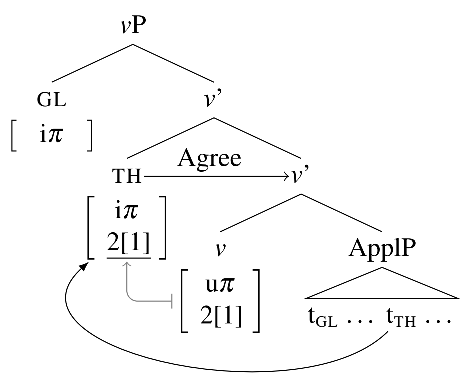
I propose that the reason why the [uπ] feature on *v* may either survive the first valuation or not is due to its status as a *valued* uninterpretable feature. Unlike *unvalued* uninterpretable features, which start the derivation in need of a value and no longer have a purpose immediately after they are valued, valued uninterpretable features already start the derivation this way. I suggest that what makes them special with respect to valuation and deletion is this: they *can* be, but *need not* be, deleted right after they value a matching unvalued feature (see also Bošković 2011 on the special status of valued uninterpretable features in this respect).

Crucially, the derivations illustrated above in (55)–(59) and (60) are the only derivations available when *v* bears a [uπ] specified with participant/speaker features. There is no way to derive pronoun combinations with a 3P Goal and a 1/2P Theme in this system (specifically with canonical person restrictions; see Section 3.6 on reverse restrictions). The impossibility of other valuation possibilities follows from minimal pronoun not being able to move to Spec*v*P and not receiving a 1/2P value when the [uπ] on *v* is specified with participant or participant and speaker features. An example of a valuation configuration impossible because of this is shown in (61a). Crucially, (61a) would have to be the first step in deriving a pronoun combination with a 3P Goal and a 1/2P Theme; the Goal minimal pronoun would have to move to Spec*v*P first and not be valued, while the Theme minimal pronoun would have to move after that, tuck in under the Goal, and be valued, as shown in (61b).

(61) a. ✗



b. ✗

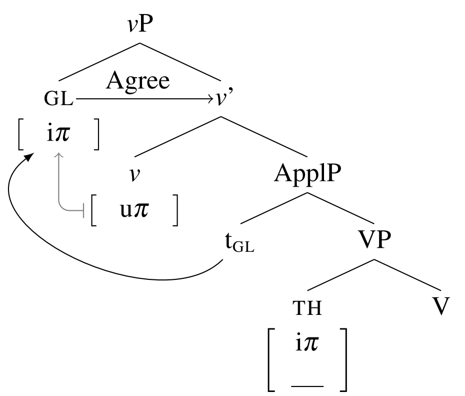


Essentially, the WEAK restriction arises because the only way to get the two pronouns to be 1/2P and 3P respectively is with the deletion of the [uπ] on *v* after the first pronoun moves to Spec*v*P. The [uπ] cannot withhold is participant/speaker features, because Agree is an obligatory operation that always takes place if the conditions for it are met. And in our case, a minimal pronoun moving to Spec*v*P is always an active probe. Therefore, if *v*'s [uπ] has participant/speaker features when the pronoun with an unvalued [iπ] moves to

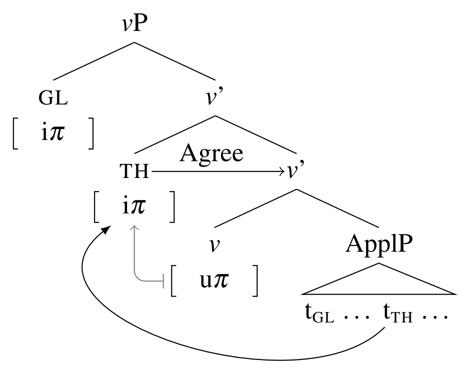
Spec*v*P—either because this is the first pronoun to move to Spec*v*P, or the features on *v* have not been deleted yet—it must be valued as 1/2P.

The only way for a Goal pronoun to be 3P in the current system is if *v* has a bare [uπ] set without any participant/speaker features. In that case, once the Goal pronoun moves to Spec*v*P to be valued, as shown in (62a), there are no participant/speaker features on *v* to value it. The same holds when the Theme subsequently moves to Spec*v*P in (62b).

(62) a.



b.

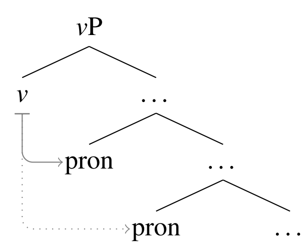


To sum up, the WEAK person restriction in double object ditransitives arises in multiple valuation configurations, where the two object pronouns move to the phase head *v* in order to receive a person value. They move there in the first place because their internal structure prevents them from being valued in situ, in contrast to what happens in STRONG restriction languages. Note that because the multi-valuation configuration arises as a cascade effect from a minute difference in the internal structure of the pronouns (which, as we will see in

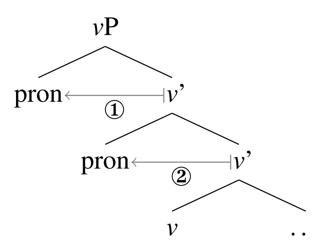
Chapter 6, can have consequences beyond its effect on person restrictions), we can derive the crosslinguistic variation in question without parameterizing core grammatical operations or invoking special versions of Agree.

The most important aspect of this approach to the STRONG vs. WEAK variation is that the two patterns are the result of the minimal pronouns being valued in different structural configurations with respect to the phase head with valued person features. The restriction is STRONG when, as shown in (63), the phase head asymmetrically c-commands both pronouns at the point of person valuation. In this case, the person restriction on the lower pronoun is the result of Agree Closest: Agree is only possible between *v* and the structurally closest minimal pronoun. In contrast, the restriction is WEAK when, as shown in (64), the pronouns are valued in Spec*v*P, with the person features on the pronouns acting as probes. In this case, the person restriction on the lower pronoun is the result of the timing of the two person valuations: the first pronoun to move to Spec*v*P can bleed person valuation for the second pronoun to move there, but it is impossible for the first pronoun to move to Spec*v*P and stay unvalued while the second one is valued.

(63) STRONG:



(64) WEAK:

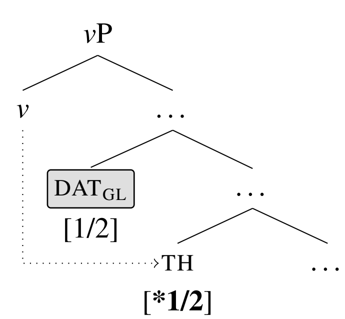


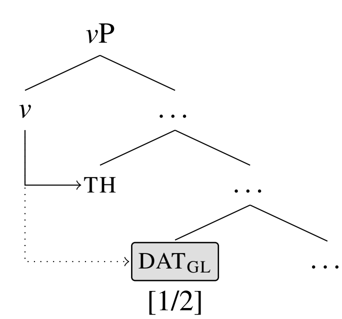
The difference in restriction strength being based on where the pronouns are located at the point of valuation will be the key to deriving the *Strength Generalization* in Chapter 4. There we will see that the attested and unattested combinations of coexisting STRONG and WEAK restrictions follow directly from basic assumptions concerning argument structure.

3.6 Deriving reverse person restrictions

One of the most important aspects of the current analysis is that the person restriction, regardless of whether it is STRONG or WEAK, arises solely due to the structural configuration of the two minimal pronouns. This is unlike many other analyses, which crucially rely on non-structural asymmetries between the two arguments (see Section 2.2 for details).

While morphological and thematic scale approaches face the biggest issue with reverse (IA-IA) restrictions, due to relying exclusively on non-structural asymmetries, most existing syntactic analyses also rely on similar asymmetries. For Béjar and Řezáč (2003) the Goal must bear inherent case and the Theme structural case, and for Anagnostopoulou (2003, 2005) the Goal only has person features but no number features while the Theme only has person features if it is 1/2P. In simplified terms, the result of these non-structural asymmetries is that the Goal pronoun can only induce the restrictions, as in (8), where it blocks Agree with the Theme pronoun, whereas the Goal itself is "protected" from person restrictions (indicated by the gray box). Consequently, even if the c-command relations between the Goal and the Theme are reversed, making the Theme block Agree between *v* and the Goal, as in (9), the Goal can still have 1/2P features.

(65)


(66)


Since the current analysis treats all arguments as equal with respect to person restrictions, this problem does not arise. It can straightforwardly derive canonical restrictions, which we have seen in the previous section, as well as reverse restrictions, which we will see now.

Recall that, with reverse restrictions in the IA-IA domain (reverse PCC), the person restriction applies to the Goal as opposed to the Theme. In Slovenian, the reverse PCC is observed when the Theme clitic precedes the Goal clitic, as shown in (67). For Slovenian speakers with STRONG restrictions, this means that the Goal must always be 3P, whereas for speakers with WEAK restrictions, the Goal must be 3P when the Theme is 3P.

(67) Slovenian [*Slavic*]:

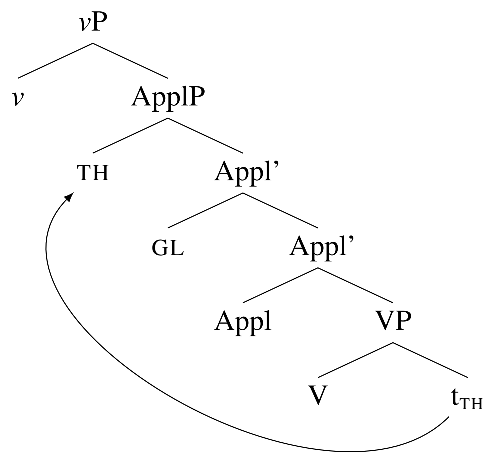
a. Mama { **me** / **te** / **ga** } **mu** bo predstavila. $\boxed{1/2/3.\text{TH} \gg 3.\text{GL}}$
mom { 1.ACC / 2.ACC / 3.M.ACC } 3.M.DAT FUT.3 introduce.F
'Mom will introduce **me/you/him** to **him**.'

b. %Mama **te** **mi** bo predstavila. $\boxed{(\text{✗})2.\text{TH} \gg 1.\text{GL}}$
mom 2.ACC 1.DAT FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

c. *Mama **ga** { **mi** / **ti** } bo predstavila. $\boxed{\text{✗}3.\text{TH} \gg 1/2.\text{GL}}$
mom 3.M.ACC { 1.DAT / 2.DAT } FUT.3 introduce.F
'Mom will introduce **him** to **me/you**.' (Stegovec 2015b, 108–9)

Both STRONG and WEAK reverse restrictions can be derived in the current system as the result of a single structural difference: unlike with the canonical IA-IA restrictions where the Goal asymmetrically c-commands the Theme below *v*, reverse IA-IA restrictions arise when the Theme asymmetrically c-commands the Goal below *v*. We saw in Chapter 2 that the two object clitic orders in Slovenian, which determine whether the restriction will be canonical or reverse, can be analyzed as reflecting different c-command relations between the Goal and the Theme clitic. Here I will argue specifically that the different orders must be established already below *v*. Specifically, I will argue that the reverse IA-IA restriction occurs when the Theme undergoes short A-movement over the Goal, as illustrated in (68).

(68)



Although I will only discuss IA-IA reverse restrictions in this section, and more specifically just the case of Slovenian, I will argue in Chapter 4 that all instances of reverse person restrictions should be analyzed along the same lines.

I will now discuss evidence suggesting that the different object clitic orders in Slovenian double object ditransitives are derived from each other via syntactic movement. The standard tests for A-movement of the kind we need here are unfortunately inapplicable with clitic arguments. However, we can show that when nominal indirect and direct objects in Slovenian reorder, the reordering shows properties associated with short A-movement, specifically short A-scrambling (Mahajan 1990; Saito 1992; Webelhuth 1992; Yatsushiro 2001; Anagnostopoulou 2003; i.a.).

In Slovenian, when the objects in a ditransitive are noun phrases, the Goal either precedes the Theme (cf. (69a)), or the Theme precedes the Goal (cf. (69b)), just as with clitic objects.

(69) a. Pokazal sem zmaju leva. $\boxed{\text{GL»TH}}$
show.M AUX.1 dragon.M.DAT lion.M.ACC

b. Pokazal sem leva zmaju. $\boxed{\text{TH»GL}}$
show.M AUX.1 lion.M.ACC dragon.M.DAT
'I showed the lion to the dragon.'

Crucially, the two orders behave differently with respect to the standard tests for structural height in double object constructions (Barss and Lasnik 1986; Larson 1988, 1990; Aoun and Li 1989; i.a.). For example, when the Goal precedes the Theme, if the Goal is a reciprocal expression it cannot be bound by the Theme, as in (94a); indicating that the Goal asymmetrically c-commands the Theme. However, when we reorder the two objects, the Theme can bind the reciprocal Goal, as in (94b).

(70) a. *Zigi je predstavil [enega drugemu]$_i$ [nova kitarista]$_i$.
Ziggy AUX.3 introduced one.ACC other.DAT new.DU.ACC guitarist.DU.ACC

b. Zigi je predstavil [nova kitarista]$_i$ [enega drugemu]$_i$.
Ziggy AUX.3 introduced new.DU.ACC guitarist.DU.ACC one.ACC other.DAT
'Ziggy introduced the two new guitarists to each other.'

The reordering of the objects can therefore establish new binding relations, a characteristic of A-movement (see Marvin and Stegovec 2012; Stegovec 2012 for more examples). Since the reordering is optional under the right conditions, it also has the hallmarks of scrambling. It therefore exactly fits the type of short A-scrambling described in other languages in double object ditransitives (Mahajan 1990; Saito 1992; Webelhuth 1992; Yatsushiro 2001; Anagnostopoulou 2003; i.a.).

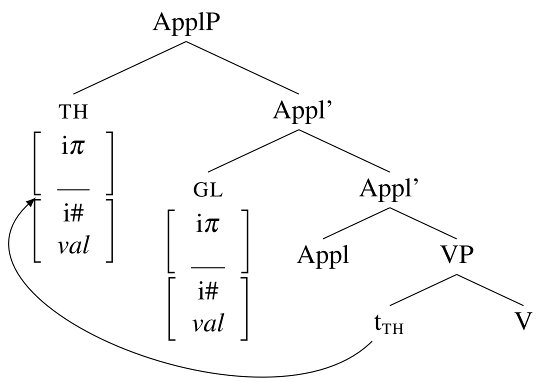
I want to suggest that in Slovenian the exact same movement is possible with double object clitics (see also Stegovec 2019 for more arguments).[13] Within the proposed approach to person restrictions, this short A-movement of the Theme clitic over the Goal clitic below *v* is sufficient to change a canonical PCC pattern into a reverse PCC pattern.

Let us first look at the effects of minimal pronoun scrambling with a STRONG restriction pattern. The derivation is exactly the same as before, except that before *v* enters the derivation the Theme pronoun moves over the Goal pronoun, as shown in (71a). What this means

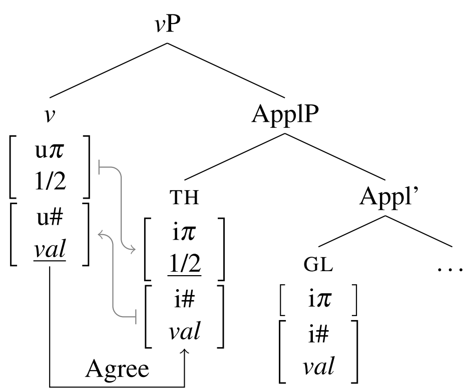
13. It is important to note that not all languages that allow A-scrambling with strong pronouns and noun phrases also allow the same type of reordering with clitic/weak pronouns; for example in Bosnian-Croatian-Serbian, only reordering with strong pronouns and noun phrases is possible (see Stjepanović 1999, 2003). There is thus only a one way correlation between allowing A-scrambling with strong pronouns and noun phrases and allowing it with clitic/weak pronouns (see Section 4.2 in Chapter 4 for more discussion).

is that the Theme pronoun is now the structurally closest argument to *v*, so that once the unvalued [u#] on *v* probes, it finds the Theme's valued [i#] and triggers Agree between them, as shown in (71b). Recall that because of the Maximize Agree principle all the matching features on the two heads in an Agree relation can value each other at this point, which means that the valued [uπ] on *v* can value the Theme's unvalued [iπ].

(71) a.



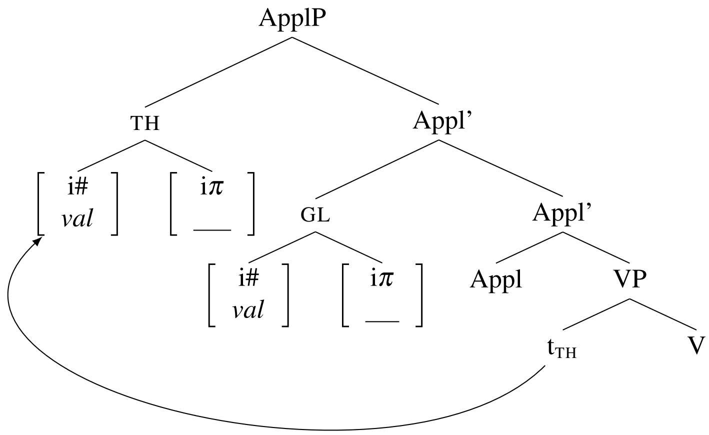
b.



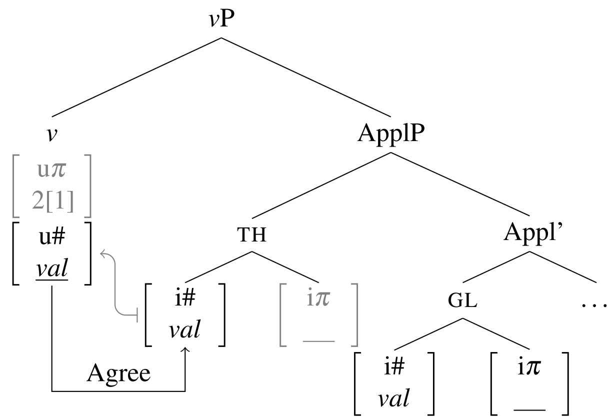
What is impossible here is for *v* to Agree with the Goal, due to Agree Closest. As a result, the Goal can only get a default 3P value, which derives the reverse STRONG restriction.

The consequences of the Theme moving over the Goal below *v* are exactly the same with a WEAK derivation. Recall that the only difference in this derivation is the richer internal structure of the minimal pronouns that makes Maximize Agree inapplicable. Thus, as shown in (72a), just as before, the Theme moves over the Goal before *v* enters the derivation, and *v*'s unvalued [u#] Agrees with the Theme's valued [i#], as shown in (72b).

(72) a.

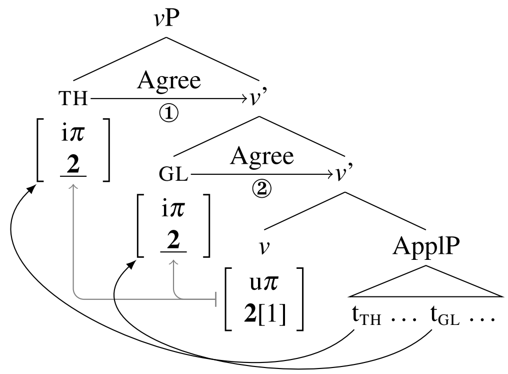


b.



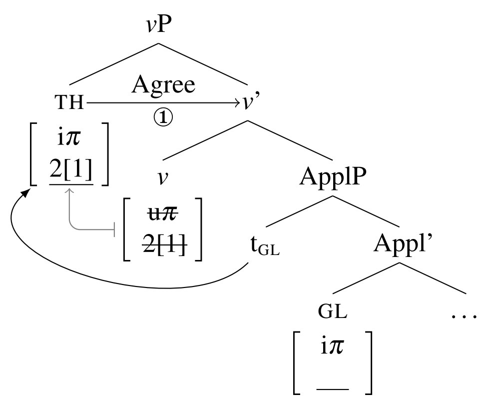
However, since in this case the unvalued [iπ] is on a different head within the pronoun from the valued [i#], there can be no person valuation at this point. Just as with the canonical PCC derivations in the previous section, the Theme pronoun must move to Spec*v*P to be valued first, followed by the Goal also moving to Spec*v*P, tucking in under the Theme. As shown in (73), this allows both pronouns to be valued 1/2P (I do not show the valuation for the speaker '1' feature here, but it proceeds exactly the same way as with the canonical version of this derivation illustrated in the previous section).

(73)

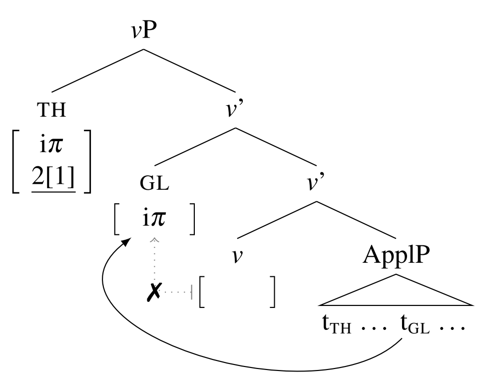


The other option, as discussed in the previous section, is that the first pronoun to move to Spec*v*P—in this case the Theme—bleeds the valuation of the other pronoun. More precisely, when the Theme is valued in Spec*v*P, this results in the deletion of *v*'s valued [uπ], as shown in (74a). This leaves the Goal pronoun without a source of valued person features when it moves to Spec*v*P in (74b), which forces it to receive a default 3P value.

(74) a.



b.



The possible person combinations that can be derived with the reverse WEAK pattern are exactly the same as with the canonical equivalent, with only the roles of the Goal and the Theme reversed. This is the case because the proposed analysis treats both pronouns exactly the same; the reversal in structural height thus simply shifts the restriction from the Theme pronoun to the Goal pronoun. This holds for both STRONG and WEAK restrictions.

What is important regarding this proposal is that the source of the person restriction is totally uniform across canonical and reverse restrictions. All the variation is due to a factor independent from the person restriction itself: the possibility of movement of the Theme minimal pronoun over the Goal minimal pronoun at the right point in the derivation. The nature of the movement that yields this result will be crucial in deducing the *Direction Generalization*; as we will see in the next chapter, there is a strong point to be made in favor of making all reverse restriction patterns just like the Slovenian one, namely, as resulting from short A-scrambling of one pronoun over the other.

To sum up, I have proposed that the source of person restrictions is the limitations imposed on the person valuation of minimal pronouns when more than one such pronoun occurs within the same phase. The motivation for this approach were the observations made in Chapter 2 regarding the special behavior of pronouns sensitive to person restrictions with respect to binding and the obligatory prosodic deficiency of such pronouns. I have attributed both of these properties to severe structural deficiency, which makes the pronouns in question incapable of projecting a phase boundary required for referential as well as prosodic independence. The referential dependence of such pronouns is tied to the proposal that only a phase head may bear valued person features, and since these structurally deficient pronouns may not project their own phase, they then have unvalued person features that must receive a value from an external source—a phase head.

I have shown that in this type of approach, the variation between STRONG and WEAK person restrictions can be derived without resorting to variation in core grammatical operations.

Instead, both types of restrictions can be derived by using a single universal version of the operation Agree—the variation arises solely due to differences in the pronouns themselves. These affect the valuation possibilities for the pronouns: when the minimal pronouns are unstructured heads, they must receive a person value in situ—yielding a STRONG person restriction, whereas when the minimal pronouns have more internal structure, with person features residing on a different head than other φ-features, they must move to the phase head to be valued—yielding a WEAK person restriction. The two restriction patterns are thus tied to the pronouns being positioned differently with respect to the phase head at the point of the valuation. Similarly, it is also a difference in the location of the pronouns at the point of valuation that derives the canonical vs. reverse person restriction variation: depending on which pronoun is closer to the phase head when it enters the derivation, the other pronoun will be the restricted one. In all these cases, the variation in person restrictions is crucially tied to grammatical differences independent from the person restriction itself.

As we will see in the next chapter, the two typological gaps established in Chapter 2, namely the *Strength Generalization* and the *Direction Generalization*, will follow from the current proposal in the same way. The gaps will not be derived by imposing limitations on the grammatical operations driving the person restrictions by brute force. Instead, they will follow from independently required assumptions concerning argument structure and movement operations.

Mind the gap!

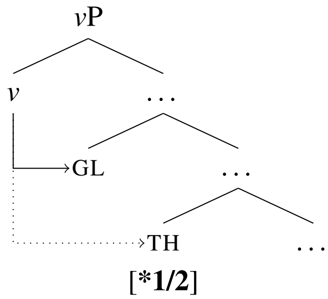
—London Underground

4

Explaining the typological gaps

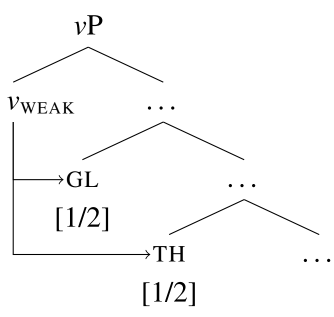
In this chapter, I take the analysis proposed in the previous chapter and show that it can straightforwardly derive the *Strength Generalization* and *Direction Generalization* established in Chapter 2. Before doing that, let us remind ourselves why these two generalizations are problematic to derive in existing syntactic accounts of person restrictions.

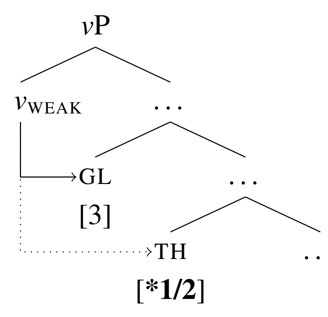
Recall that what drives most syntactic approaches to person restrictions, including mine, is local nature of Agree (*Agree Closest*). That means that in a configuration with one probe and two goals—such as the structure in (1), where *v* is the probe and the Goal and Theme are both matching goals—will ensure that Agree will not be a possibility for both goals.

(1) 

The analysis of STRONG restrictions is not that different in other approaches and the one I proposed in the previous chapter; the impossibility of Agree between *v* and the Theme will result in the Theme only being able to be 3P.

Complications arise with other accounts when it comes to modeling the STRONG VS. WEAK variation. The standard way to derive WEAK restrictions in this line of approaches is to parameterize Agree so that, in some languages, it is allowed between a probe and all the goals in its c-command domain, but only when the goals all have 1/2P features (Anagnostopoulou 2005; Nevins 2007). This is needed to license multiple 1/2P pronouns in configurations like (2), but still block Agree with both goals when these have "mixed" person values, as in (3). Because the Goal in (3) is 3P and the Theme 1/2P, regular Agree is induced and Agree with the Theme is again blocked—prohibiting it from being 1/2P.

(2) 

(3) 

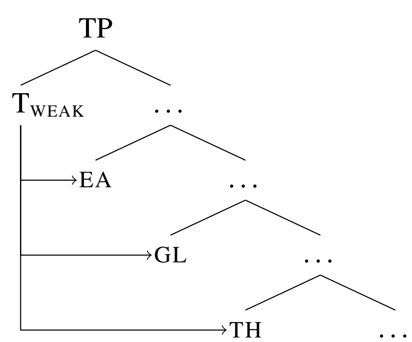
As noted in Chapter 2, this type of approach, where Agree is crosslinguistically not always the same operation, requires the parameterization of a core grammatical operation; there are a number of conceptual arguments to be made against such an approach to parameterization

(Borer 1984; Chomsky 1995, 2005; Biberauer et al. 2014). But the real issue here is an empirical one; this way of approaching the STRONG vs. WEAK variation gets us into trouble with languages where person restrictions exist in both EA-IA and IA-IA domains. In particular, this approach fails to explain the existence of a typological gap concerning possible combinations of STRONG and WEAK restrictions in the two domains, described by the following generalization, established in Chapter 2:

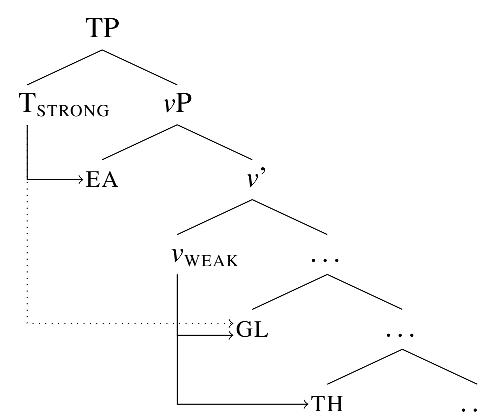
(4) *Strength Generalization*
If a language has person restrictions in both the EA-IA and IA-IA domain, the IA-IA person restriction cannot be weaker than the EA-IA person restriction.

Specifically, if we parameterize Agree uniformly for both domains—for example, by tying both the EA-IA and the IA-IA restriction to the same probe, as in (5)—we can only generate combinations where the two restrictions are either both STRONG or both WEAK (the option shown in (5)). In other words, we under-generate with respect to the attested patterns. Alternatively, if we parameterize the restrictions for each domain separately—for example, by tying the EA-IA and the IA-IA restriction each to a different parameterizable probe, as in (6)—we can generate all the logically possible combinations of restrictions, including STRONG-WEAK (the option shown in (6)), which is unattested crosslinguistically. That is, in this case we over-generate with respect to the attested patterns.

(5)



(6)



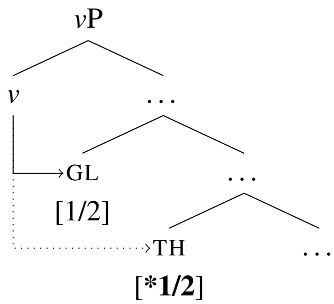
The general issue here is that, even if we can separately parameterize the source of the restrictions in the EA-IA domain and the IA-IA domain, there is no way to capture the dependency of the strength of the IA-IA restriction on the strength of the EA-IA restriction.

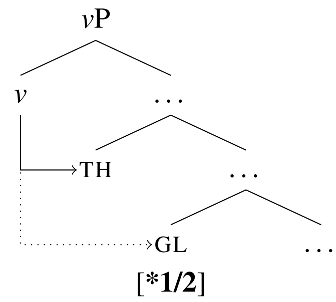
One of the main advantages of the analysis I proposed in the last chapter is that it derives the STRONG vs. WEAK variation without having to parameterize the Agree operation. The trigger for the variation were lexical differences in the pronouns themselves. And, more importantly, the difference between STRONG and WEAK restrictions came from a difference in the position of the minimal pronouns at the point of person valuation; the WEAK pattern arises in the case of IA-IA restrictions because of the movement of the pronouns resulting from a different internal structure of the pronouns. As we will see in this chapter, the connection between where the pronouns are at the point of valuation and the strength of the restriction is the key to explaining the *Strength Generalization*.

Recall also that most existing analyses rely on non-structural asymmetries between the pronouns, which means that they can only derive canonical restrictions, but not reverse ones. We saw that the analysis introduced in Chapter 3 does not have this problem, because the only asymmetry that matters for person restrictions is the structural one; other than the two pronouns being minimal pronouns, no other inherent property of the pronoun matters for deriving the person restriction. However, this still does not resolve the issue of the distribution of reverse person restrictions, which is governed by the generalization in (7).

(7) *Direction Generalization*
If a language has a *reverse* person restriction in the EA-IA or IA-IA domain, it must also have a *canonical* counterpart of the same person restriction.

What we saw in Section 3.6 is that in the proposed analysis the difference between canonical and reverse restrictions amounts to a difference in the asymmetric c-command relation between the pronouns under the phase head. Thus, a canonical IA-IA restriction arises in configurations like (8), while a reverse IA-IA restriction arises in configurations like (9).

(8)


(9)


The issue is that one can imagine a configuration like (9) to in principle be possible even if a language does not independently also allow (8). If that would be possible, we would expect to also find languages with only a reverse person restriction. As we will see in this chapter, he exclusion of languages with only (9) configurations actually follows from assuming that the base positions of all arguments are universally the same and that reverse restriction configurations like (9) can only arise as the result of an optional movement of one minimal pronoun over the other prior to person valuation taking place.

In the continuation I first tackle the deduction of the *Strength Generalization*, which will be based on the similarities and differences with respect to how EA-IA restrictions and IA-IA restrictions arise. Following that, I deduce the *Direction Generalization*, showing that the absence of reverse-only restriction patterns follows from all reverse restrictions arising due to a type of scrambling movement operation. Finally, I discuss some more fine-grained person restriction patterns that I set aside so far and show that those can be accommodated by the current system while still maintaining the ability to deduce the two generalizations.

4.1 Deducing the Strength Generalization

In order to consider the place of the *Strength Generalization* in the current system, we need to look at restrictions beyond the IA-IA domain, which is what we used in the previous chapter to illustrate the basics of the system. Recall that EA-IA restrictions are limited to languages of the pronominal argument type, where what appear to be agreement affixes on the verb, indexing the subject and object(s), are actually the real arguments, as opposed to the DPs/NPs with which they co-vary in φ-features. I hypothesized in Section 3.3.2 that when such languages show an EA-IA restriction, this is because both the EA and IA pronouns are minimal pronouns. In contrast, when they do not have an EA-IA restriction, it is because at least one of the markers is more akin to an agreement marker registering (or licensing) the presence of a phasal *pro*, which does not require person valuation.

It must be noted though that there is disagreement in the literature regarding the proper analysis of pronominal arguments. According to Jelinek (1984, 2006), the markers on the verb are themselves the arguments, while according to Baker (1988, 1996, 2006), the markers on the verb are actually agreement markers that license null pronouns in argument positions. The differences between the two approaches do not matter for the discussion to follow, since they are both compatible with the analysis I propose. As long as the pronouns in question—either Jelinek's markers on the verb or Baker's null pronouns—are minimal pronouns, the predictions regarding person restrictions will not be affected.

However, for the sake of consistency with the derivations in the previous chapter, I will follow Jelinek and treat the markers on the verb as the direct morphological expression of the minimal pronouns. A nice consequence of this approach is that we can also very easily capture the cases where not all the arguments are indexed on the verb and the non-indexed arguments are person restricted. If the markers on the verb are the true arguments—and minimal pronouns—then the non-indexed pronouns are simply phonologically null minimal pronouns. The lack of exponence can be either due to a particular marker always being null, which is often the case with 3P markers in pronominal argument languages, or due to the marker being null only in a particular context, which is also common, for example, with Theme markers in double object constructions. Crucially, the presence of the minimal pronoun in such cases is identified by, among other things, the person restriction.

I will now proceed to show how STRONG and WEAK restrictions arise in the EA-IA domain, which will form the base for discussing how the *Strength Generalization* is deduced in the current system. Most importantly, we will see that the STRONG and WEAK restrictions in the EA-IA domain arise due to the same difference in person valuation as with IA-IA restrictions: STRONG restrictions arise through pure intervention and WEAK restrictions due to the timing of multiple instances of person valuation by the same phase head.

4.1.1 Deriving STRONG EA-IA restrictions

Before looking at how the current system derives the STRONG version of EA-IA person restrictions, let us remind ourselves of how this restriction pattern looks like. An example of this pattern can be observed in Arizona Tewa (Kroskrity 1977, 1985). Recall that Arizona Tewa encodes all the arguments on the verb via prefix pronominal markers—which are pronominal arguments. The prefixes are available in active transitive clauses only for cases where the internal argument is 3P. Consequently, if an EA subject is present, the IA cannot be 1/2P, as shown in (10), which makes this a STRONG person restriction.

(10) Arizona Tewa [*Kiowa-Tanoan*]: **STRONG EA-IA**

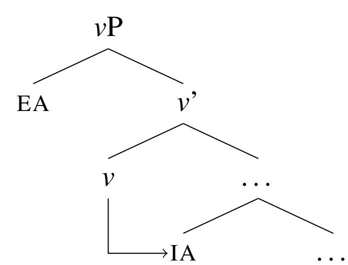
a. *sen* { **dó-** / **ná:-** / **mán-** } mun. 1/2/3.EA ≫ 3.IA
man { 1>3- / 2>3- / 3>3- } see
'**I/You/He** saw **him**, the man.'

b. **na: ų̨* **???-** mun. ✗1.EA ≫ 2.IA
I you 1>2- see
'**I** saw **you**'

c. **sen* **???-** mun. ✗3.EA ≫ 1/2.IA
man 3>1/2- see
'**He** [the man] saw **me/you**.' (Kroskrity 1977, 86, 169, 171)

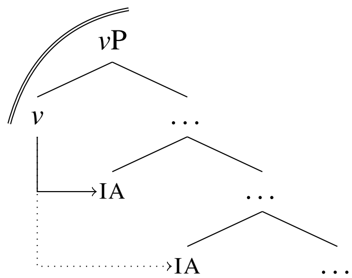
We saw in Chapter 3 that what causes STRONG restrictions in the IA-IA domain is Agree Closest: the probe on the phase head *v*, which bears valued person features, asymmetrically c-commands both internal arguments, but it may only enter into an Agree relation with the argument that is structurally the highest, forcing the lower pronoun to always get default 3P. But note that if the EA is also a minimal pronoun, which is a prerequisite for EA-IA restrictions, it will not enter the derivation in a position where it will structurally intervene between *v* and the structurally lower IAs. This is because the base position of EAs is in Spec*v*P (Chomsky 2000); as illustrated in (11), this means that the EA minimal pronoun will not intervene for Agree between *v* and the closest accessible IA minimal pronoun.

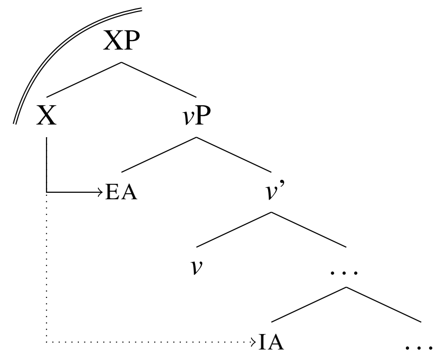
(11)



What is then needed in addition to derive STRONG EA-IA restrictions is that the phase head asymmetrically c-commands both the EA and the IA minimal pronoun. While this is not a possibility in the phase system of (Chomsky 2000, 2001), it is possible in the approach to phases I am adopting (see Bošković 2013), where the size of phases is contextual and phases do not always correspond to *v*P. In fact, it has been argued based on a wide range of syntactic phenomena that phase boundaries must be contextual (Gallego 2005; den Dikken 2007; Wurmbrand 2013, 2017b; Bošković 2013, 2014; Despić 2015; i.a.), allowing phases to be extended beyond their default size under the right conditions.

While in the case of STRONG IA-IA restrictions *v* is the phase head agreeing only with the closest IA, as in (12), I want to argue that in the case of STRONG EA-IA restrictions a head above *v* is the phase head, as in (13). This means the EA becomes the intervener blocking valuation of any IA minimal pronouns lower in the structure.

(12)


(13)


The difference in phase size is also meant to capture here the fact that having a STRONG as opposed to a WEAK EA-IA restriction is comparatively rare: in the survey only 12 languages have a STRONG EA-IA restriction, compared to the 31 languages that have a WEAK EA-IA restriction (we will see in the next section why WEAK restrictions are the default option in the EA-IA domain). We can attribute the smaller number of languages with a STRONG EA-IA restriction to them having two prerequisites: (i) both EA and IA pronouns must be minimal pronouns, and (ii) the phase head must be a head above *v*. On its own, (i) is just a prerequisite for EA-IA restrictions, while the option in (ii) can in principle also yield an IA-IA restriction if there is no EA minimal pronoun in the language. Only when both (i) and (ii) are met in a language do we get a STRONG EA-IA restriction.

In addition to the rarity of STRONG EA-IA restrictions, there is also evidence independent from person restrictions themselves suggesting that the identity of the probe is indeed different with STRONG EA-IA restrictions. This is most obvious when comparing closely related languages that differ in terms of either having or not having a STRONG EA-IA restriction; in such cases, the language with the STRONG EA-IA restriction generally shows signs of the *v* head being deficient in some way (which can be tied to *v*P not being a phase). Additional evidence is also found language internally in cases where only specific constructions in the language show a STRONG EA-IA restriction; in such cases, the construction with the STRONG restriction generally shows either a different indexing pattern or more argument extraction possibilities. I will discuss one case study concerning the relation between a deficient *v* and STRONG EA-IA person restrictions, before going in detail through the derivation of a STRONG EA-IA person restriction. With that in place, the relevance of the language internal variation cases will become clearer, so I will discuss two additional case studies only after presenting the basic derivation.

An important component of the current approach to person restrictions is that there should always be independent correlates to variation in person restrictions. In the case of STRONG EA-IA restrictions, where I argue a syntactic head above *v* is the phase head, we expect that the inability of *v* to be a phase will be reflected beyond the person restriction itself. A very nice illustration of this is observed when we compare Arizona Tewa with the closely related Southern Tiwa.

4.1.1.1 Voice deficiency and phase extension

Southern Tiwa (Allen and Frantz 1978; Rosen 1990) morphologically expresses arguments in the same way as Arizona Tewa; i.e. by way of a pronominal prefix. The main difference is that in Southern Tiwa, the EA-IA restriction is a WEAK one rather than a STRONG one. That means that, unlike in Arizona Tewa, the IA can also be 1/2P, as in (14b), but only when the EA is also 1/2P, as shown by the ungrammaticality of (14c).

(14) Southern Tiwa [*Kiowa-Tanoan*]: **WEAK EA-IA**

a. { **ti-** / **a-** / **Ø-** } mũ-ban.
{ 1>3- / 2>3- / 3>3- } see-PAST
'**I/You/He** saw **him**.' — 1/2/3.EA ≫ 3.IA

b. **i-** mũ-ban.
1>2- see-PAST
'**I** saw **you**' — 1.EA ≫ 2.IA

c. *seuan **???-** mũ-ban.
man.A 3>1/2- see-PAST
'**He** [the man] saw **me/you**.' — ✗3.EA ≫ 1/2.IA

(Allen and Frantz 1978, 11–2)

I discuss how WEAK EA-IA restrictions, such as the one in Southern Tiwa, arise in the next section. What is important for now is that with WEAK EA-IA restrictions *v* is the phase head.

This matters because of how *v* is expressed morphologically in Southern Tiwa in contrast to Arizona Tewa. Since *v* is generally considered the syntactic locus of voice, it is also assumed to be able to realize as active/passive morphology. In the Southern Tiwa passive examples in (15), we can clearly identify a dedicated passive morpheme (underlined).

(15) a. seuanide-ba **te-** mũ-che-ban. — 1.IA.SU + PP.3.EA
man.A-INSTR 1.SU- see-PASS-PAST
'**I** was seen by the man.'

b. seuanide-ba **a-** mũ-che-ban. — 2.IA.SU + PP.3.EA
man.A-INSTR 2.SU- see-PASS-PAST
'**You** were seen by the man.' (Allen and Frantz 1978, 11–2)

Contrast this to the Arizona Tewa passives in (16). Just as in Southern Tiwa, the demoted EA (when expressed) is realized by an oblique/PP element. However, unlike in Southern Tiwa, there is no dedicated passive morpheme. Morphologically, passivization only triggers allomorphy on the pronominal prefix.

(16) Arizona Tewa [*Kiowa-Tanoan*]:

a. *ų̨* na:n-di **wí-** tay. — 2.IA.SU + PP.1.EA
you we-OBL PASS(1>2)- know
'**You** are known (or recognized) by us.'

b. *na:* { sen-en-di / ų̨-di } **dí-** $k^{w}ek^{hw}$ę̨di. — 1.IA.SU + PP.3/2.EA
I { man-PL-OBL / you-OBL } PASS(3/2>1)- shoot
'**I** was shot by the man/by you.' (Kroskrity 1985, 311)

I propose that the lack of the passive morpheme corresponding to *v* is how the deficient nature of *v* is expressed in Arizona Tewa. Importantly, the deficient *v* is also incapable of projecting a phase boundary. I assume that the relevant deficiency on *v* is similar to the deficiency I discussed in relation to minimal pronouns in Chapter 3. Recall that in that case the inability of deficient pronouns to project a phase boundary is the reason why they are minimal pronouns with unvalued person features. The only difference with respect to a deficient *v* is that the next syntactic head up takes up the role of the phase head (in line with the overall approach to phasehood adopted here; see Chapter 3).

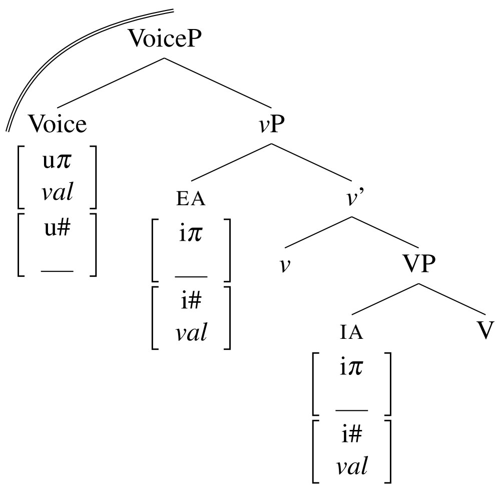
I now turn to the derivation of the STRONG EA-IA restriction in Arizona Tewa. I propose that the deficient nature of *v* results in a higher functional head taking up the role of the locus of voice encoding and, more importantly, the phase head for the verbal domain. I call the head that becomes a phase head under such conditions a *backup* phase head. There are several options for how this can happen. For example, it is possible that the backup phase head is a category not present in languages where *v* is not deficient. Alternatively, the backup phase head could be universally present, but it normally does not play a role in voice alternations—it only does this if *v* is deficient. Either option can be implemented in the current system—where phases correspond to the extended projections of all the major categories—as a change in what counts as the highest head in the extended projection of the verb. With the first option, the deficiency of *v* means that an entirely new verbal head is introduced into the extended projection. With the second option, the deficiency of *v* means that a head which is otherwise outside the extended projection of the verb is co-opted into it; as the new highest head in the extended projection, it counts as a phase head. But the choice between the two options ultimately does not matter for the issue at hand.

What matters is that the extension of the phase takes place when *v* is too deficient to be a phase head. The identity of the backup phase head is also not crucial; it can in principle be any head above *v*P (we will actually see below that different categories may be backup phase heads in different languages). However, I will for completeness sake, in the Arizona Tewa case, label the backup phase head Voice,[1] following Collins (2005), who also adopts a voice encoding head above *v* for independent reasons in his analysis of English passives.

1. Not to be confused with the identically labeled head used, e.g., by Kratzer (1996), Pylkkänen (2002, 2008), and Legate (2014). In current terms, the EA-introducing head they label 'Voice' is what I label *v*.

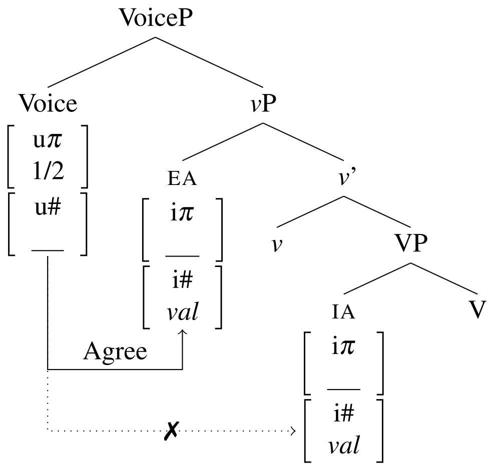
In the current system, this means that Voice, as a phase head, will be able to bear valued [uπ] features, but its [u#] features will be unvalued and active probes. Crucially, Voice asymmetrically c-commands both EA and IA minimal pronouns (just as *v* asymmetrically c-commands both Goal and Theme minimal pronouns with IA-IA restrictions). The VoiceP phase with EA and IA minimal pronouns prior to probing is shown in (17).

(17)

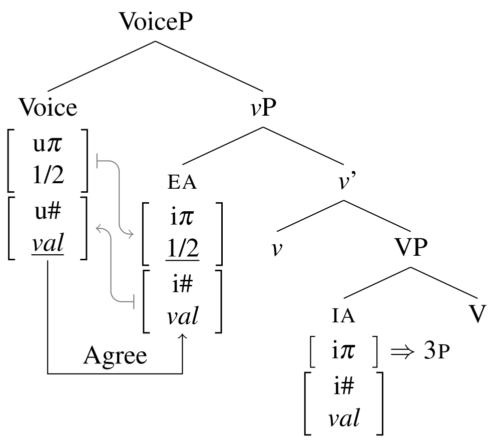


The derivation will proceed in this case exactly like the STRONG IA-IA derivation in the previous chapter, only with the EA minimal pronoun taking up the role of the closest goal. The derivation for when the Voice phase head is specified as 1/2P unfolds as shown in (18). Since Voice also has an unvalued [u#] feature, its [u#] must probe for the closest matching valued features, which reside on the EA pronoun (recall that minimal pronouns have valued number features). The [u#] probe on Voice and the valued [i#] on the EA then initiate Agree between Voice and the EA pronoun, as shown in (18a); note that since the EA asymmetrically c-commands the IA, Agree between Voice and the IA is impossible due to Agree Closest. Once Agree is established, the valued [i#] on the EA may value the [u#] on Voice, and due to the Maximize Agree principle (repeated in (19)) all other matching features on Voice and the EA may value each other. This means that, as shown in (18b), the valuation of the person features on the EA may piggyback on the valuation of number features, resulting in the EA being valued as 1/2P by the valued [uπ] feature on Voice.

(18) a.



b.

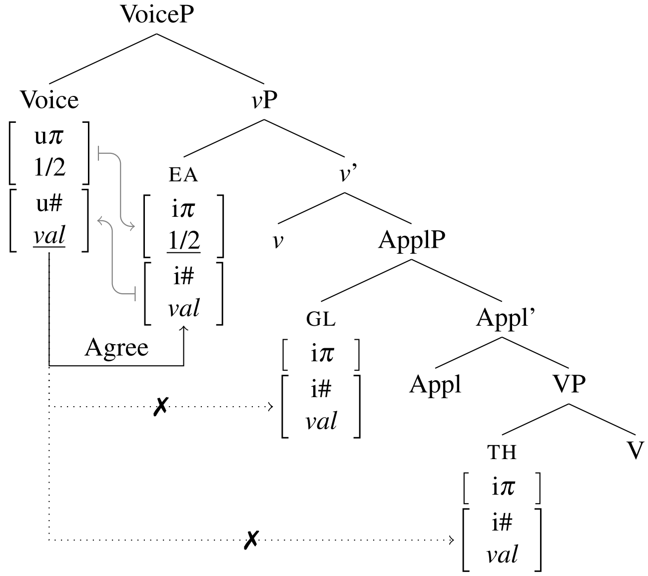


(19) *Maximize Agree*
If a probe matches a goal, all matching features on the probe and goal must be valued at that point in the derivation.

A derivation where Voice's [uπ] is not specified with participant/speaker person features is pretty much the same, the only difference is that the EA receives no person features as a result of Agree with Voice. Most importantly, with both specifications of Voice, the IA pronoun can never enter into an Agree relation with Voice due to Agree Closest. As a result, IA's [iπ] feature cannot be valued. Since a person feature without a value corresponds to 3P (see Section 3.2), the IA's [iπ] must be spelled out as 3P at the interfaces. This is the source of the STRONG EA-IA person restriction: no matter what happens the IA pronoun cannot be valued by Voice, so it will always be 3P whenever an EA pronoun is present and no matter what value the EA receives from Voice.

Note that there is an important additional prediction here: the EA will block the person valuation of any minimal pronouns present below it in the structure. This means that in double object ditransitives both Goal and Theme minimal pronouns will be inaccessible for person valuation, since the EA will always intervene for Agree between Voice and either of them, as shown in (20).

(20)



This is actually borne out in the case of Arizona Tewa. In Arizona Tewa double object ditransitives, even though all three arguments can be expressed in the pronominal prefix (see (21a)), neither the Goal nor the Theme can be 1/2P (see (21b,c)) (Kroskrity 1977, 1985). In fact, even if we passivize such a construction, which lifts the EA-IA restriction and allows the Goal to be 1/2P, the Theme can only be 3P (see (22)). There are no pronominal prefixes in Arizona Tewa, neither active nor passive, that express a 1/2P Theme in the context of a non-PP Goal. And since Arizona Tewa is a pronominal argument language, this means that the Theme can never be 1/2P in such cases.

(21) Arizona Tewa [*Kiowa-Tanoan*]:

a. **dó- béy-** kumɛ.
1>3- 3PL.IO bought
'**I** bought **it** for **them**.' $\boxed{\text{1.EA} \gg \text{3.GL} \gg \text{3.TH}}$

b. **na: ų ko:to* **???-** me:gi
I you bracelet 1>2>3- give
'**I** gave **it** [a bracelet] to **you**.' $\boxed{✗\,\text{1.EA} \gg \text{2.GL} \gg \text{3.TH}}$

c. **he-ʔi sen ų ko:to* **???-** me:gi
that man you bracelet 3>2>3- give
'**He** [the man] gave **it** [a bracelet] to **you**.' $\boxed{✗\,\text{3.EA} \gg \text{2.GL} \gg \text{3.TH}}$

(22) a. *ų ko:to* he-ʔi sen-dí **wó:-** me:gi
you bracelet that man-OBL PASS(3>2)- give
'**You** were given **it** [a bracelet] by that man.' $\boxed{\text{2.GL.SU} \gg \text{3.TH.O} + \text{PP.3.EA}}$

b. **na: ų* **???-** me:gi
I you PASS(3>1>2)- give
'**I** was given **you**.' $\boxed{✗\,\text{1.GL.SU} \gg \text{2.TH.O} + \text{PP.3.EA}}$
(Kroskrity 1977, 80, 168–9)

In other words, the proposed analysis of STRONG EA-IA restrictions automatically extends to languages with a STRONG-STRONG restriction combination (see Section 2.5.1); that is, languages with a STRONG EA-IA as well as a STRONG IA-IA restriction. If a language has a STRONG EA-IA restriction and both objects in a double object ditransitives are minimal pronouns, this will be the pattern that arises. I come back to discuss the implications of this in Section 4.1.3, where I discuss STRONG-STRONG combinations specifically.

Importantly, the alternative phase configuration that gives rise to STRONG EA-IA restrictions does not have to be in place for all types of clauses. In fact, one of the key insights of contextual phase analyses of other syntactic phenomena is that the size of the phase can be different in different grammatical contexts (even within the same language; see Bošković 2013, 2014). In relation to STRONG EA-IA restrictions, this predicts that we may also be able to find languages, where unlike in Arizona Tewa, the STRONG EA-IA restriction is only observed with some constructions. This is actually borne out. In the remainder of this section, I look at two languages where a STRONG EA-IA restriction only arises with some constructions. The two cases I look at here are Christian Barwar [*Neo-Aramaic*] and Kaqchikel [*Quichean-Mamean*], although other cases of this kind exist (see Chapter 6).

4.1.1.2 Aspect-sensitive extended phases

Christian Barwar only has a STRONG EA-IA restriction when the verb has a perfective base (Doron and Khan 2012b; Kalin 2014b; Kalin and van Urk 2015). In those cases, the IA marker on the verb cannot be 1/2P in the presence of a EA marker, as shown in (23).

(23) Christian Barwar [*Neo-Aramaic*]:

a. griš **-a** { **-li** / **-ləx** / **-le** }. $\boxed{\text{1/2/3.EA} \gg \text{3.IA}}$
pull.PERF -3.F.O { -1.SU / -2.F.SU / -3.M.SU }
'**I/you/he** pulled **her**.'

b. *griš **-at** **-li**. $\boxed{\text{✗ 1.EA} \gg \text{2.IA}}$
pull.PERF -2.F.O -1.SU
'**I** pulled **you**.'

c. *griš { **-an** / **-at** } **-le**. $\boxed{\text{✗ 3.EA} \gg \text{1/2.IA}}$
pull.PERF { -1.F.O / -2.F.O } -3.M.SU
'**He** pulled **me/you**.' (Doron and Khan 2012b, 232)

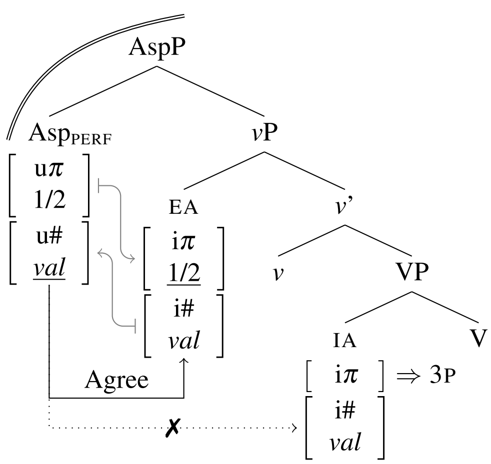
Conversely, with an imperfective base there are no person restrictions, as shown by the contrast between the perfective and imperfective example in (31) (the same pattern is also attested in the related Jewish Zakho, Christian Barwar, Senaya, and Telkepe; see Kalin 2014b). Notice also that the order of argument markers is also reversed between the two aspects: with the perfective base the IA marker is closer to the verb (see (31a)), whereas with the imperfective base the EA marker is closer (see (31b)).

(24) a. *qṭil **-an** **-na**. $\boxed{\text{✗ 3.EA} \gg \text{1.IA}}$
kill.PERF -1.O -3.F.SU
'**She** killed **me**.'

b. qaṭl **-a** **-li**. $\boxed{\text{3.EA} \gg \text{1.IA}}$
kill.IMPF -3.F.SU -1.O
'**She** kills **me**.' (Kalin 2014b, 5)

I follow here the insights of Kalin and van Urk (2015), who argue that the two marker orders are the result of different probe configurations associated with the two aspectual bases; specifically, in the perfective there is only one probe which asymmetrically c-commands both the EA and the IA arguments. In the current system, this insight can be implemented in terms of the *v* in perfective clauses being unable to project a phase, and the Asp(ect) head (the next head up) taking over as the phase head and host of valued person features. The derivation of the STRONG EA-IA restriction in the perfective then proceeds just as above, only with the Asp head being the relevant phase head instead of Voice. As shown in (25), when the unvalued [u#] feature on Asp probes, Agree can only be established between Asp and the EA minimal pronoun. Consequently, the EA minimal pronoun can get its person features valued piggybacking on the number valuation. However, since the EA also blocks Agree between Asp and the IA pronoun, the latter can never be valued through Agree, so it must get a default 3P value.

(25)



Building on Kalin and van Urk's (2015) proposal that the imperfective aspect involves two probes, I suggest that the absence of person restrictions with imperfective aspect is, on the other hand, the result of each of the two arguments being in separate phases valued via Agree with two distinct probes. Kalin and van Urk crucially base their proposal on the analyses of split ergativity of Laka (2016) and Coon (2010), who argue that split ergativity arises because nonperfective aspects have the semantics of locative predicates which can as a result of this function essentially as embedding verbs. I return to discuss this and similar patterns in more detail in Chapter 6, where I show how the order reversal in markers can be tied to a particular configuration of multiple phases in the verbal domain. For now, what is most important, is the contextual nature of the STRONG EA-IA restriction in Christian Barwar, which lends support to the analysis in terms of contextual phases.

4.1.1.3 Person restrictions and extraction asymmetries

I now turn to the other relevant case, namely, Kaqchikel (Preminger 2011, 2014), where the STRONG EA-IA restriction is also not available in all syntactic contexts. For example, the restriction is not observed in plain transitive clauses, where the EA is expressed by an ergative marker and the IA by an absolutive marker on the verb. As the examples in (26) show, the IA can be 2P (also 1P; not shown here) in the presence of an EA.

(26) Kaqchikel [*Quichean-Mamean*]:

a. *ri achin* x- **a-** **r-** ax-aj *rat* — 3.EA ≫ 2.IA
the man CMPL- 2.ABS- 3.ERG- hear-ACT you
'**He** [the man] heard **you**.' (Preminger 2014, 16)

b. ja *röj* x- **ix-** **qa-** tz'et — 1.EA ≫ 2.IA
FOC us CMPL- 2PL.ABS- 1PL.ERG- see
'It was **us** that saw **you(pl.)**.' (Preminger 2014, 23)

The person restriction, which is a STRONG EA-IA restriction, arises in Kaqchikel only in the so called Agent-Focus construction. This is a construction used when the subject of a transitive verb is focused or *wh*-moved (as well as some other cases involving Ā-movement of the subject; see Erlewine 2016) and where the only argument indexed on the verb is expressed via an absolutive marker, as shown in (27a) (the Agent-Focus/AF marker is underlined). Crucially, the argument that is not indexed on the verb, in this case the IA,[2] cannot be 1/2P, as shown in (27b) and (27c).

(27) a. ja *rat* x- **at-** ax-an *ri* *achin* $\boxed{2.\text{EA} \gg (3.\text{IA})}$
FOC you CMPL- 2- hear-AF the man
'It was **you** that heard the man.' (Preminger 2014, 18)

b. *ja *ri* *achin* x- **Ø-** ax-an *rat* $\boxed{✗\,3.\text{EA} \gg (2.\text{IA})}$
FOC the man CMPL- 3.ABS- hear-AF you
'It was **him** [the man] that heard you.'

c. *ja *yïn* x- **in-** ax-an *rat* $\boxed{✗\,1.\text{EA} \gg (2.\text{IA})}$
FOC me CMPL- 1- hear-AF you
'It was **me** that heard you.' (Preminger 2014, 22)

Two things are key here in relation to the Agent-Focus construction, in addition to the person restriction. The first one is the role that Agent-Focus plays in relation to movement. This is important since the primary empirical motivation for phases concerns the locality of movement. In the case of Kaqchikel (and many other Mayan languages), Ā-movement of the EA is prohibited in a plain transitive clause (i.e. a syntactic ergativity effect), as (28a) shows with respect to *wh*-movement. The Agent-Focus construction is required when the EA undergoes Ā-movement as in (28b).

(28) a. **achike* x- **Ø-** **u-** tëj *ri* *wäy*?
who CMPL- 3.ABS- 3.ERG- eat the tortilla

b. *achike* x- **Ø-** tj-ö *ri* *wäy*?
who CMPL- 3.ABS- eat-AF the tortilla
'Who ate the tortilla?' (Erlewine 2016, 430)

The other key aspect of Agent-Focus is the placement of the Agent-Focus morpheme, which must come on top of morphemes expressing that the verb is a derived active transitive verb. In (29a), a baseline transitive clause, the active transitive *-aj* morpheme is attached to the verb *ax* 'hear'. In the Agent-Focus version of the sentence in (29b), where the STRONG

2. Recall from Chapter 2 that Kaqchikel has both canonical and reverse versions of this restriction; here I look only at the canonical version, postponing the discussion of the reverse one until Section 4.2.

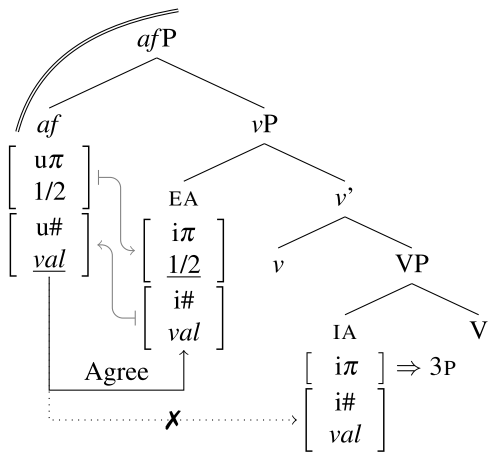
EA-IA restriction occurs, the Agent-Focus morpheme -*an* is attached to the verb in the same place as the active transitive morpheme was before. Crucially, we can tell that -*an* does not simply replace -*aj*, since the form of the Agent-Focus morpheme changes depending on whether the verb requires the active transitive morpheme or not (the Agent-Focus morpheme is -*ö* on bare verbs and -*an* otherwise).

(29) a. *ri achin* x- **a-** **r-** ax-aj *rat* $\boxed{\text{3.EA} \gg \text{2.IA}}$
the man CMPL- 2.ABS- 3.ERG- hear-ACT you
'**He** [the man] heard **you**.' (Preminger 2014, 16)

b. *ja *ri achin* x- **Ø-** ax-<u>an</u> *rat* $\boxed{\text{✗ 3.EA} \gg \text{(2.IA)}}$
FOC the man CMPL- 3.ABS- hear-AF you
'It was **him** [the man] that heard you.' (Preminger 2014, 18)

Recall from the discussion concerning Southern Tiwa and Arizona Tewa passives above that *v* is the syntactic head associated with voice, and here we see the Agent-Focus head merging on top of the morpheme associated with derived *active* transitive verbs. I take this to suggest that the Agent-Focus morpheme corresponds to a syntactic head *af* above *v*P, where the phase boundary extends to in Agent-Focus constructions in order to make EA movement possible (see discussion in Chapter 6). This then gives us the same syntactic configuration of the phase head with respect to EA and IA minimal pronouns that we saw above with Arizona Tewa and Christian Barwar perfective clauses. The derivation of the STRONG EA-IA restriction that arises because of this configuration is illustrated in (30), and proceeds exactly like the derivations discussed above, only with the *af* head playing the role that Voice and Asp played in Arizona Tewa and Christian Barwar respectively.

(30)



Going into why the alternative phase configuration that yields a STRONG EA-IA restriction also lifts the ban on EA-movement would take us too far astray at this point, but I return to discuss this in Chapter 6, where the proposed solution relates to the phase configuration in baseline transitive clauses, where the person restriction is also absent; in fact, it is related to the phase configuration responsible for the absence of person restrictions in imperfective clauses in Christian Barwar, which I also discuss in the same chapter. What is important here is that there is an independent correlate to the alternative Agent-Focus phase configuration aside from the STRONG restriction: the change in EA movement possibilities.[3]

To sum up, just like STRONG IA-IA restrictions, STRONG EA-IA person restrictions arise as the result of a syntactic intervention effect: Agree Closest. A phase head asymmetrically c-commanding both the EA and IA arguments may only establish Agree with the structurally closest one, which is always the EA. Since person valuation is dependent on Agree being established between a minimal pronoun and the relevant phase head, the IA can never be valued by the phase head and can only receive a default 3P value whenever the EA minimal pronoun is present. We have also seen that the exceptional larger phase that yields

3. Erlewine (2016) reduces the ban on EA Ā-movement to *anti-locality*: movement of the EA out of Spec*v*P into the next landing site constitutes movement that is *too short* (see Bošković 1994, 2016a; Abels 2003b; Grohmann 2003). I suggest an alternative implementation of the extraction asymmetry in Chapter 6, which crucially relies on the differences in phase status between Agent-Focus and the baseline transitive construction.

such person restrictions has other syntactic and morphological correlates. Although these correlates will not always be easy to spot, the current account makes the predictions in (31).

(31) Given two minimally different languages/constructions α and β, where α shows a STRONG EA-IA restriction and β does not, we expect to find:
   a. that α has more impoverished voice marking than β; or
   b. that α has a more impoverished argument indexing than β; or
   c. that α has accusative alignment and β ergative alignment; or
   d. that α allows more extraction possibilities for arguments than β.

While not all of these predictions might be clear at this point, we will see in Chapter 6 how they all essentially relate to different phase configurations. The existence of these correlations with the person restriction is relevant also in relation to questions concerning the learnability of different person restriction patterns outlined in Chapter 1. If there were no independent cues for the learner, then acquiring the person restriction pattern would have to be based solely on the grammatical person combinations in the input—I return to discuss the advantages of the current analysis in relation to learnability in Chapter 6 as well.

Another important aspect of the proposed analysis is that the larger phase is a departure from the default *v*P phase, which then fits the fact that STRONG EA-IA restrictions are comparatively rare: if the larger verbal phase is not the normal state of affairs, then it is not surprising that the languages that depart from the default configuration are rarer, and in the case of STRONG EA-IA restrictions, there are two conditions that need to be met: both EA and IA pronouns must be minimal, and the verbal phase must be larger than *v*P. In the next section, we will see what happens with EA-IA restrictions if *v*P is the phase.

4.1.2 Deriving WEAK EA-IA restrictions

Recall that in addition to STRONG restrictions, WEAK restrictions also occur in the EA-IA domain and they are in fact the more common variant. In the previous section, we saw a minimal pair with respect to the strength of the EA-IA restriction with Arizona Tewa (STRONG) and Southern Tiwa (WEAK), where we crucially also saw that the key difference between the two patterns was in the status of *v*, which was deficient and not a phase head in the former case, resulting in the structurally higher Voice head taking up the role of the phase head and consequently yielding a STRONG EA-IA person restriction due to Agree Closest. In this section, I look at the latter option regarding EA-IA restrictions: what happens when *v* is not deficient and hence capable of projecting a phase boundary?

Let us first remind ourselves of what a WEAK EA-IA restriction looks like. The same kind of EA-IA restriction pattern we saw in Southern Tiwa is also attested in many Salish languages, including Lummi (Jelinek and Demers 1983). Salish languages are, just like Kiowa-Tanoan languages, pronominal argument languages—in fact Salish languages are the prototypical case of pronominal argument languages discussed by Jelinek (1984, 2006). That means, in current terms, that all the arguments including the EA are minimal pronouns. The person restriction pattern of Lummi is repeated here in (32): in an active transitive sentence the IA cannot be 1/2P whenever the EA is 3P, and that is the only restriction.

(32) Lummi [*Salish*]: **WEAK EA-IA**

a. x̣či-t **-Ø** { **-sən** / **-sx^{w}** / **-s** } — 1/2/3.EA ≫ 3.IA
know-TR -3.O { -1.SU / -2.SU / -3.SU.ERG }
'**I/you/he** knows **him**.'

b. x̣či-t **-oŋəs -sən** — 1.EA ≫ 2.IA
know-TR -1/2.O -1.SU
'**I** know **you**.'

c. *x̣či-t **-oŋəs -əs** — ✗ 3.EA ≫ 1/2.IA
know-TR -1/2.O -3.SU.ERG
'**He** knows **you/me**.' (Jelinek and Demers 1983, 168)

Just like we saw in Southern Tiwa above, Lummi passives are also characterized by a dedicated passive morpheme on the verb, as seen in the examples in (33) (underlined).[4]

4. The transitive morpheme (TR) is present on all transitive verbs in Lummi, both lexical and derived (see Jelinek 1994; Jelinek and Demers 1994).

(33) a. x̣či-t-ŋ̱ **-sən** $\boxed{\text{1.IA.SU + 3.EA}}$
know-TR-PASS -1.SU
'**I** am known (by him/someone).'

b. x̣či-t-ŋ̱ **-sx$^{\mathbf{w}}$** $\boxed{\text{2.IA.SU + 3.EA}}$
know-TR-PASS -2.SU
'**You** are known (by him/someone).' (Jelinek and Demers 1983, 168)

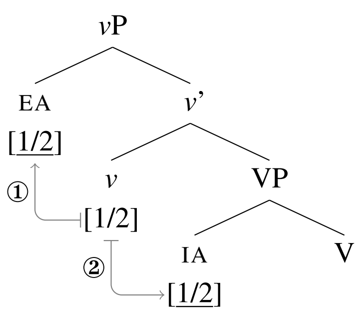
Given the distinction we saw between Arizona Tewa and Southern Tiwa in the previous section, the dedicated passive morpheme indicates that *v* is not deficient and therefore projects a phase boundary. The fact that the same head that projects the phase boundary is also the one introducing the EA will play a key role in how WEAK EA-IA restrictions arise.

4.1.2.1 Minimal pronouns on the edge and the timing of grammatical operations

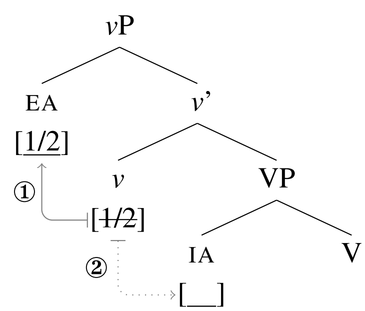
Recall that in the case of WEAK IA-IA restrictions, the person restriction does not arise due to Agree Closest, as it does with STRONG restrictions, but due to the timing of multiple instances of valuation. With WEAK IA-IA restrictions, the minimal pronouns cannot be valued in situ and have to move to Spec*v*P to be valued, which is possible because their unvalued person features act as probes. The object pronoun closest to *v* must move and be valued first, followed by the lower object pronoun, which tucks in as the second Spec*v*P. Crucially, this sequence of valuation steps can never yield the ungrammatical *3P $\gg$ 1/2P pronoun combinations. The possible derivations are: (i) the first pronoun is valued 1/2P and bleeds person valuation of the second pronoun by deleting *v*'s person feature, forcing the second pronoun to get a default 3P value, (ii) the first pronoun is valued 1/2P and does not delete *v*'s valued person feature, which allows the second pronoun to also be valued 1/2P, and (iii) the first pronoun is valued 3P because *v* is not specified with participant/speaker features and the second pronoun is also valued 3P. The two options with respect to deleting *v*'s person feature are the consequence of the person feature's special status as a *valued uninterpretable* feature: it *can* be, but *need not* be, deleted right after it values a matching unvalued feature. The deletion of the person feature is the only way for the pronouns to have mixed values as 1/2P and 3P respectively, which is what derives the WEAK restriction.

I propose that WEAK EA-IA restrictions arise in the same way: due to the timing of multiple instances of person valuation. However, in the EA-IA case, the EA minimal pronoun, which is base generated in Spec*v*P, potentially bleeds the person valuation of the IA minimal pronoun in situ. The basic principle is illustrated in (34)–(35): the EA, which must be valued first for reasons discussed below, can be valued 1/2P without deleting *v*'s person feature, allowing the IA to be valued 1/2P when *v* Agrees with it, as in (34), or the EA can be valued 1/2P and delete *v*'s person feature, forcing the IA to get a default 3P value, as in (35).

(34)



(35)



In the case of WEAK IA-IA restrictions discussed in Chapter 3, it is crucial that the object minimal pronouns cannot be valued in situ. Since their unvalued person features are active probes, this forces them to move to Spec*v*P to be valued by *v*'s valued person feature (see Section 3.5.2). In the case of WEAK EA-IA restrictions, the EA minimal pronoun is crucially base generated Spec*v*P, where it asymmetrically c-commands the valued person feature on *v*. This means that its unvalued person feature can probe and Agree with *v* in its in situ position immediately upon Merge. The basic intuition here is in fact very similar to Chomsky's (2000) analysis of *there*-expletive insertion, where *there*, which only has a person probe, Agrees with T immediately upon Merge with TP (cf. (36a)) and bleeds movement of the indefinite noun phrase to SpecTP, which takes place only in the absence of *there* (cf. (36b)).

(36) a. **there** [$_{TP}$ was-T elected **an unpopular candidate**]
b. **an unpopular candidate** [$_{TP}$ was-T elected *t*]

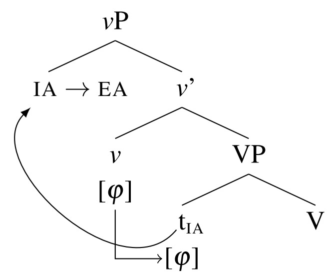
The main point in (36) is that while both *there* and T are probes, it is crucial in the derivational model of Chomsky (2000) that T cannot probe before *there* probes. As a result, *there*-insertion blocks the movement of the indefinite noun phrase because it deletes the feature on T that requires TP to Merge with a specifier ([EPP]); the probing of *there* bleeds the possibility of the indefinite noun phrase checking of the [EPP] feature on T.

The similarity with the derivation of WEAK EA-IA syntactic restrictions, like (34)–(35) above, is that in those cases the EA minimal pronoun introduced in Spec*v*P is a probe (due to its unvalued person feature) and *v* is also a probe (due to its unvalued number feature). The EA probe is abstractly just like the *there* probe and the *v* probe like the T probe. The difference is only that we are not dealing with a *v* that attracts the IA to Spec*v*P, instead we are dealing with a case where the EA is valued by *v* before *v* can Agree with the IA. This means that the way Chomsky (2000) enforces the order of probing in cases like (36), via the *Merge over Move* preference, will not work in our case. Instead, I argue that the solution is in the nature of the two probes: *v* is a theta-role assigner and EA is not.

The difference concerns the timing of probing with respect to the timing of the establishment of theta-relations. I propose that the preference for probes is to probe immediately at Merge, but that this preference is suppressed in cases where the probe is a theta-role assigner, in which case the probing cannot occur until the probe has assigned the theta-role; I call this the *Theta-Probe Condition*. This condition is in place to capture a fundamental asymmetry concerning the establishment of theta-relations with respect to other grammatical dependencies, where the former precedes the latter.

There are two main views on theta-roles with respect to which we can show why something like the Theta-Probe Condition must regulate the timing of operations. The first is the configurational view (see e.g. Hale and Keyser 1998, 2002), where an argument's theta-role follows directly from the structural configuration between the argument and the theta-role assigning head; for example, the EA role is the result of the pronoun or noun phrase being in Spec*v*P (more precisely, in the Spec of a verbal head that takes a VP complement), since *v* is the head associated with the EA role. It is crucial in this type of an approach that only the base positions of arguments are taken into account for determining their theta-role. What has to be excluded in this type of an approach is the possibility of derived theta-roles (since movement into a theta-position is assumed not to be possible); for example, an IA moving into the EA position, as in (37).

(37) ✗



If the movement of the IA is conditioned by Agree, then the impossibility of (37) follows from the Theta-Probe Condition: if *v* cannot probe and Agree with IA until the EA has merged in Spec*v*P, then the IA can never move to Spec*v*P before EA enters the derivation.[5]

The second main approach to theta-roles is feature-based (see e.g. Bošković and Takahashi 1998; Hornstein 1999), where theta-role assigner assigns the theta-role under local feature checking, such as between a head and its specifier. Within such an approach, the timing of grammatical operations must also be constrained. For example, theta-related specifiers must be created before A-specifiers, and the latter must be created before Ā-specifiers (see Bošković 1994; Abels 2007). In a system where the creation of all three

5. Note that given the possibility of multiple specifiers in the current system, the configuration that matters for theta-role assignment in specifiers is the one between the head and the first specifier to be merged—otherwise, we would allow for multiple arguments in the same clause to share a theta-role, going against one of the core insights of the *Theta-Criterion* (Chomsky 1981).

types of specifiers (either via base generation or movement) is driven by feature valuation/checking, a natural way to constrain this is in terms of an obligatory order in which the relevant features are valued/checked: theta-related features ([θ]) must be valued/checked before A-related features (i.e. φ-features and Case features [K]) are valued/checked, and the latter must be valued/checked before Ā-related features (i.e. interrogative features [Q], information-structure features [TOP,FOC], etc.) are valued/checked; which we can state as in (38) (cf. the *UCOOL* principle of Abels 2007).

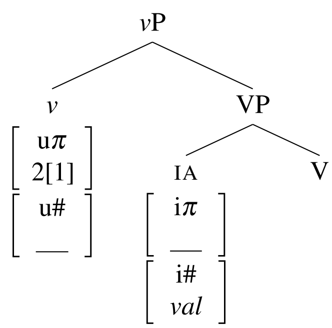
(38) *Order of feature-checking*
Theta-operation [θ] $\succ$ A-operation [φ, K] $\succ$ Ā-operation [Q, FOC, TOP, ...]

This universal ordering in fact deduces the Theta-Probe Condition: if theta-role assignment is accomplished via θ-feature checking, then this operation has to precede any probing for φ-features by the same head (an A-operation). In our case this means that *v* cannot probe until the EA has merged and was assigned the EA theta-role via feature checking with *v*.

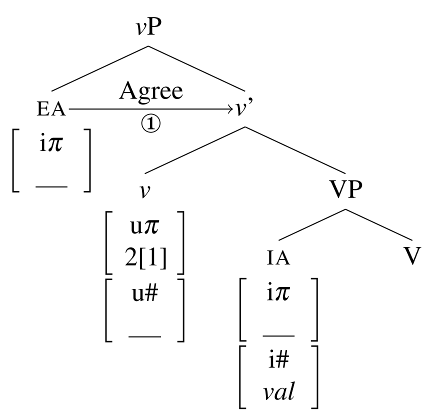
Either characterization of the Theta-Probe Condition will suffice for the issue at hand, although we will see in Section 4.2 that the feature checking approach is to be preferred due to the explanation proposed there for the *Direction Generalization*. What matters is that the Theta-Probe Condition is a way to encode the fundamental asymmetry between theta-role assignment and other operations: the former always has precedence over the latter. In relation to WEAK EA-IA restrictions, what matters is that EA is solely a person probe that probes immediately at Merge, whereas *v* is multi-functional in the sense that it is both a theta-role assigner and a number probe. Because theta-role assignment always takes precedence over φ-probing, the EA has to be merged and person valued before *v* can probe (and that EA and *v* will undergo feature checking before anything else happens).

With all the core assumptions in place, we can now go through the derivation of WEAK EA-IA restrictions in detail. In a language like Lummi, EA and IA are both minimal pronouns and the EA crucially enters the derivation by merging with the phase head *v*. Minimal pronouns have unvalued person features, whereas a phase head has valued person features but unvalued number features. Thus, when *v* enters the structure, which is shown in (39a), its unvalued [u#] feature is an active probe. However, due to the Theta-Probe Condition, *v* cannot probe until EA enters the derivation and is assigned a theta-role. In contrast, when the EA enters the derivation by merging with *v*P, which is shown in (39b), its unvalued [iπ] feature can probe immediately at Merge and find the matching valued [uπ] feature on *v* (which is in this case specified with both a participant '2' and a speaker '1' sub-feature).

(39) a.



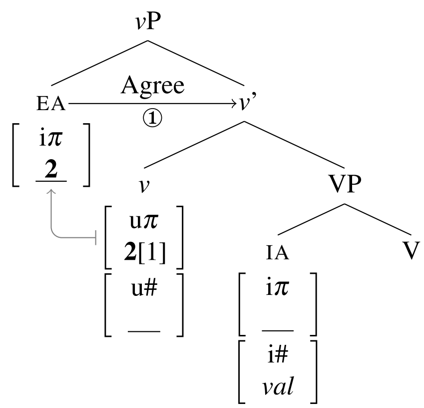
b.



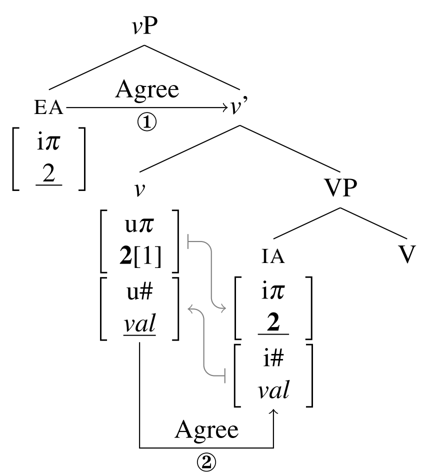
The derivation can go in two directions at this point, just as with WEAK IA-IA restrictions: (i) the valuation of the EA's [iπ] may leave *v*'s [uπ] intact, or (ii) the valuation of the EA's [iπ] deletes *v*'s [uπ]. Recall that these two options are due to [uπ] being a valued uninterpretable feature, which means that it can be, but need not be, deleted right after it values a matching unvalued feature.

Let us first look at the non-deletion option. Once the EA has established Agree with *v*, the pronoun's [iπ] can be valued with a participant '2' feature, as shown in (40a), which in this particular case leaves *v*'s [uπ] intact (I set aside EA's number features for now, coming back to discuss their role below). At this point, after the EA has merged with *v*P (probing and immediately upon Merge), the Theta-Probe Condition is met for *v*, and its unvalued [u#] can probe. As shown in (40b), *v* finds the matching valued [i#] on the IA minimal pronoun and establishes Agree with it. Due to Maximize Agree and since [uπ] on *v* has not been deleted, the valuation of IA's unvalued [iπ] can piggyback on the valuation of [u#].[6]

(40) a.



b.

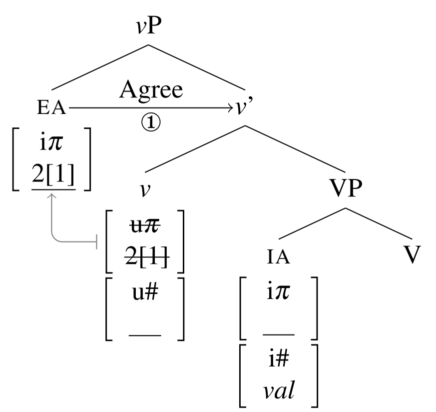


6. This derivation assumes an IA minimal pronoun with no internal structure, but note that the EA-IA restriction pattern would not be affected if the IA had the more complex internal structure that would force it to move to Spec*v*P after its person feature would fail to be valued. In that case, the valuation of the IA in Spec*v*P would still have to follow the valuation of the EA, resulting in the same bleeding configuration that yields a WEAK pattern. To keep things simple, I only show the in-situ valuation option here, although the movement option is needed independently for deriving WEAK-WEAK restriction combinations; see Section 4.1.5.

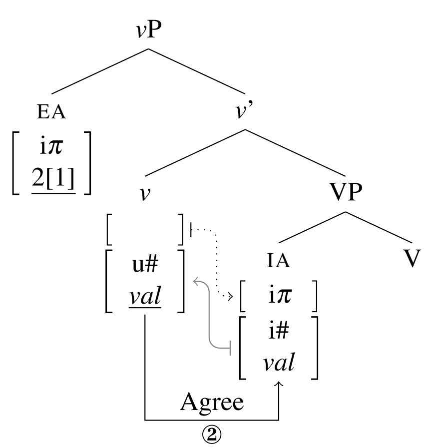
Notice that in (40), both pronouns get their [iπ] valued by *v*, just as with WEAK IA-IA restrictions. The EA is valued in Spec*v*P by probing and Agreeing with *v*, and the IA is valued after *v* probes and Agrees with it. I only show in (40) how participant '2' feature is assigned to both pronouns. Recall that the person feature system forces all valuation for '2' to take place before either of the pronouns can receive a speaker '1' feature. As both EA and IA already stand in an Agree relation with *v*, either can receive '1'. The possibilities are identical to those we saw with WEAK EA-IA restrictions in the previous chapter.

We can now transition to the other direction the derivation can go in after EA Agrees with *v*. Namely, the valuation of the EA's person features by *v*, shown in (41a), can trigger the immediate deletion of *v*'s [uπ] feature, as shown in (41b). Consequently, when *v* subsequently Agrees with the IA, *v* no longer has a [uπ] feature that could value the IA's unvalued [iπ], as shown in (40b), which means that the IA then has to get a default 3P value.

(41) a.

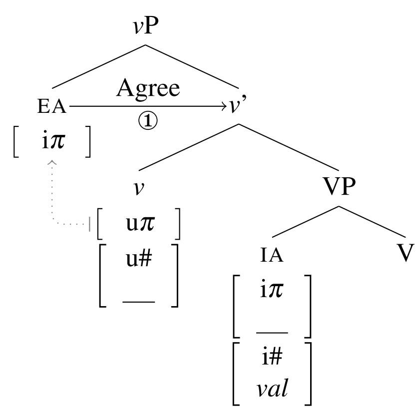


b.

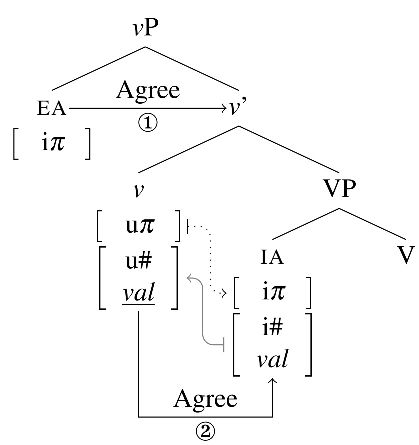


What we see here is that the valuation options for the two pronouns are exactly the same as those we saw in the case of WEAK IA-IA restrictions despite the different position of the arguments in relation to *v*. When *v* has a fully specified [uπ], *v* can either value both the EA and IA as 1/2P, or value just the EA as 1/2P. But the latter is the only possible derivation where the two pronouns can be 1/2P and 3P respectively. The EA can be valued as 3P only when the IA is also valued as 3P. This kind of derivation is shown in (42), where both pronouns are valued as 3P because *v*'s [uπ] is not specified with any participant/speaker features, so it cannot value them as anything else.

(42) a.



b.

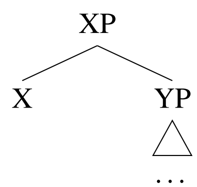


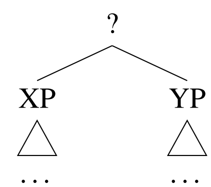
The *3P $\gg$ 1/2P combination is therefore ungrammatical for the same reasons as with the other WEAK restriction: when *v* has a fully specified [uπ], required to value at least one of the pronouns as 1/2P, the first (in this case only) pronoun in Spec*v*P must Agree with *v* and be valued 1/2P. This is because Agree is obligatory: when the conditions for it are met, it has to take place. But not Agreeing in such a case would be the only way for an EA to receive a 3P value while allowing the IA to be valued as 1/2P. Since this is impossible, *3P $\gg$ 1/2P pronoun combinations can never arise.

4.1.2.2 The internal structure of EA *minimal pronouns*

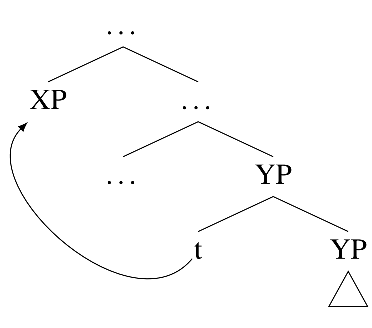
An aspect of EA pronouns that I set aside above was that EA minimal pronouns also have valued number features just like all other minimal pronouns. However, that raises the question of why the valued number on the EA does not value *v*'s unvalued number feature when Agree takes place between them in the same way that the person feature on the IA gets valued by piggybacking on the valuation of number features. Remember that when the pronouns are just unstructured bundles of ϕ-features and therefore heads, Maximize Agree ensures that all matching sets of ϕ-features on the probe and goal value each other regardless of the direction of valuation. The only way this can be counteracted is when the minimal pronoun has more internal structure, with person and number features being on separate heads within the pronoun. This is the driving force behind the movement of the minimal pronouns to Spec*v*P in the case of the WEAK IA-IA restriction. I want to argue that the reason why the EA's valued number features do not value *v*'s unvalued number features under Agree is the same: EA minimal pronouns in WEAK EA-IA languages always have the internal structure of IA minimal pronouns in WEAK IA-IA languages. I furthermore suggest that the reason for this is also related to the syntactic position where the EA enters the derivation, which also has implications for the comparative rarity of EA-IA person restrictions and EA/IA asymmetries in relation to historical changes from strong pronouns to deficient pronouns.

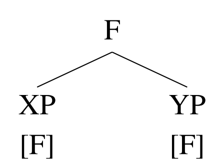
One way to derive the richer structure of EA minimal pronouns in WEAK EA-IA languages is within the approach to projection of Chomsky (2013, 2015), where Chomsky proposes that Merge of two syntactic elements does not always automatically result in one of the two elements projecting; i.e. providing a label for the newly formed constituent. What the label of the constituent created by Merge will be is determined by the *Labeling Algorithm*. In the case where Merge combines a head X with a maximal projection YP, the Labeling Algorithm picks X as the label, resulting in the resulting constituent becoming an XP maximal projection, as shown in (43). But, the labeling of the newly formed constituent is not always guaranteed: when Merge applies to two maximal projections, as in (44), the Labeling Algorithm cannot automatically determine the label of the new object.

(43)


(44)


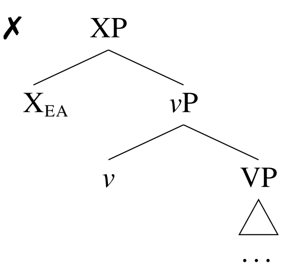
In configurations which present a problem for the Labeling Algorithm, there are two possible ways to bypass the problem. One way is to move one of the problematic maximal projections, as in (45). The Labeling Algorithm ignores traces (the lower ends of a movement chain), which means that the non-moved element then provides the label. A different possibility arises in cases where the two elements share a common syntactic feature, as in (46). In that case, the Labeling Algorithm picks the shared feature as the label (i.e. *feature sharing*).

(45)


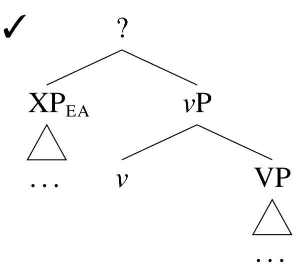
(46)


Consider now, with the Labeling Algorithm in mind, the case of Merge between an EA minimal pronoun and *v*P. If the EA pronoun is a bundle of φ-features without any internal structure—and therefore a head—this is a case of a head (X) and a maximal projection (*v*P) merging. But this would mean that the EA should provide a label for the newly formed constituent, making *v*P its complement, as illustrated in (47). A number of things would go wrong if this were to happen, the main point being that this configuration involves an argument taking a verbal phrase as a complement. Under either of the main approaches to theta-role assignment discussed in Section 4.1.2.1, this would present an issue, since the assignment of a theta-role to the EA must be restricted to Spec-Head configurations; furthermore, the head above *v*P would end up taking the argument as a complement. Alternatively, if the EA is itself a maximal projection (XP), as in (48), the newly established phrase could not receive a label at this point. However, issues concerning labels—unlike the theta-role issue in (47)—can be resolved, for example, by employing one of the strategies in (45)–(46) (see some other options below). Thus, as long as the EA is phrasal, the derivation should have a way of converging to a grammatical result.

(47) ✗



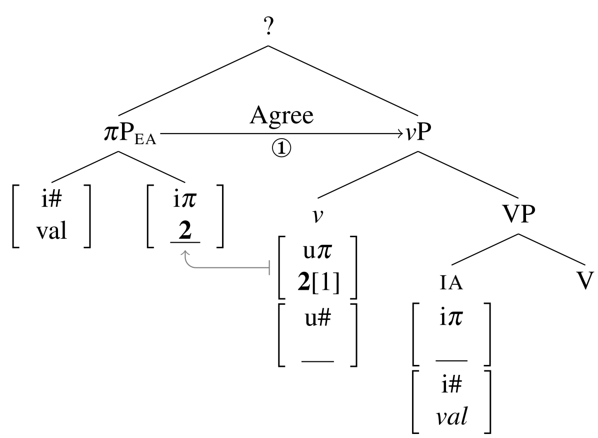
(48) ✓



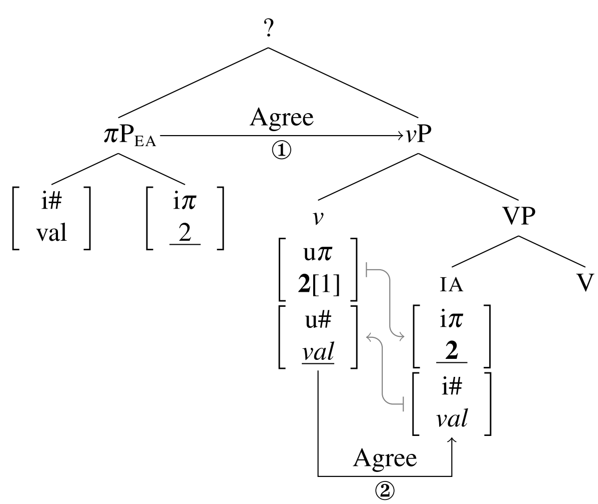
I argue that this is the reason why EA minimal pronouns in languages with WEAK EA-IA restrictions must always have a richer internal structure, like the one IA minimal pronouns have in WEAK IA-IA languages. Although I have left the status of such pronouns open in Chapter 3, I argue here that they are actually phrasal, with the person features taking the other φ-features as a complement, although still deficient enough to not project their own phase boundary and consequently have valued person features.

The consequence of this richer internal structure in relation to Agree between the EA minimal pronoun and *v* is the same as what we saw for richer IA pronouns in the case of WEAK IA-IA restrictions; namely, they make Maximize Agree inapplicable. Thus, when the EA minimal pronoun enters the derivation in Spec*v*P, its unvalued [iπ] feature immediately probes and establishes Agree with the valued [uπ] feature on *v*, as shown in (49a). Since the [iπ] and [i#] features are not bundled on the same head, this results only in the valuation of [iπ] on the EA pronoun, leaving *v*'s [u#] unvalued. Because of this, *v* is still an active probe that can establish Agree with the IA minimal pronoun and value its [iπ] feature, as shown in (49b). Here, person valuation can piggyback on number valuation because both heads involved are bundles of person and number features.

(49) a.



b.



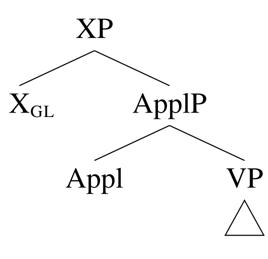
The inability of the EA to deactivate *v*'s number probe is therefore due to its richer internal structure, required to avoid a configuration where the EA would take the *v*P as its complement—which would happen if the EA were a head, given the Labeling Algorithm. But one may ask now, why can pronouns other than the IA be heads; this includes IA pronouns in WEAK EA-IA restriction languages as well as all pronouns subject to STRONG restrictions. I propose that the reason why only the EA is singled out in this respect is essentially the same as the driving force behind the WEAK EA-IA restriction in the first place: it is due to EA's special status as the argument that is base generated at the edge of a phase.

Chomsky (2013) argues that labeling does not have to take place immediately when a constituent is formed, as labels are needed only at the interfaces. Thus, labels are provided only at spell-out, when syntactic structure is sent to the interfaces (LF and PF). But note that phases crucially determine the point of spell-out (Chomsky 2000, 2001). As Bošković (2016a) points out this presents a 'chicken or the egg' type problem: labels are only determined at the point of spell-out, but to determine the point of spell-out, we need phases, which do not really exist prior to labeling (phases are maximal projections like CPs or *v*Ps and determining what is a phrase requires reference to labels), so neither can come first.

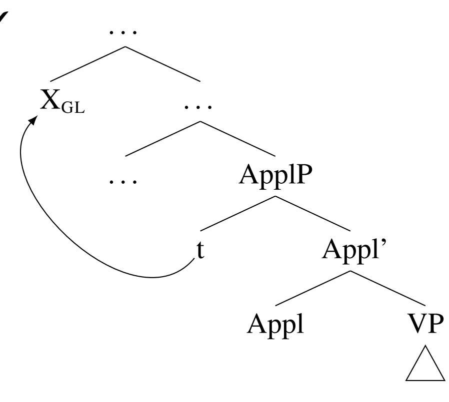
To resolve this 'chicken or the egg' problem, I follow Bošković's reasoning that labeling by the head in head-complement configurations must take place immediately upon Merge because of subcategorization: in order to guarantee selection of the right complement the head must project (i.e. we need to know what is the head is and what is the complement), this is also what resolves the 'chicken or the egg' problem. There are further consequences here, assuming that all labels must be determined at the phase level (i.e. remaining unlabeled elements must be labeled at that point). When two phrases merge, and neither of them contains a phase head, they do not have to immediately label the resulting constituent as there is no problem regarding subcategorization in such cases—there is also still time, before the phase is complete (which is when all labels must be determined), for one of the phrases to move or for the labeling issue to be resolved in some other way. However, if one of the two phrases does contain a phase head (note that the phase would already have a label at this point due to the immediate labeling in head-complement configurations), then this merger occurs at the phasal level. But, in order for the phase to spell out, the projection of the phase must be determined, so such cases also require immediate labeling (recall that everything must be labeled at the phasal level).

Consider with this in mind the case of a Goal pronoun merging with ApplP. If the pronoun is just a head (X), then we encounter the same problem as we did above with the EA: the Labeling Algorithm predicts that the pronoun should label the constituent, as in (50), but this would constitute a problem for theta-role assignment (see discussion above). However, since in this case ApplP is not a phase, there is no need for labeling to take place at this point in the derivation. This means that if the Goal should move out of its base position, as in (51), ApplP could determine the label of the relevant constituent, since traces are ignored by the Labeling Algorithm (cf. (46)).

(50) ✗



(51) ✓

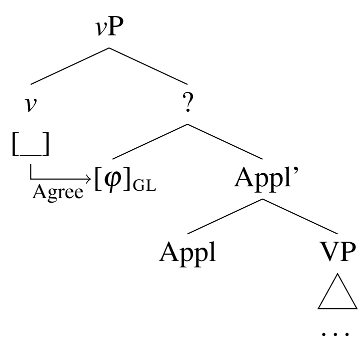


In fact, this has a very nice side effect concerning the syntactic behavior of minimal pronouns. Recall that all minimal pronouns are deficient pronouns, which includes clitic pronouns, traditionally assumed to be heads (i.e. non-branching elements in bare phrase structure). The fact that the Goal pronoun in (50)–(51) cannot stay in situ, or it would cause a problem for theta-role assignment, derives the fact that clitic pronouns are elements that always move out of their base position (see e.g. Cardinaletti and Starke 1994, 1999; Abels 2003a, 2003b).

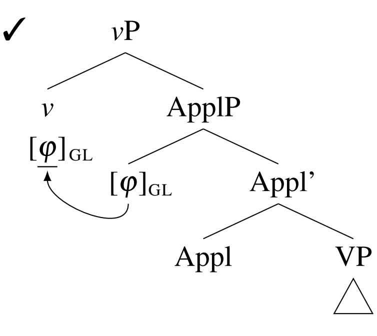
The same result can also be achieved in a framework where cliticization is a direct reflex of Agree and not traditional movement—specifically, the system of Roberts (2010b). In Roberts' analysis of cliticization, if the goal in an Agree relation is a φ-feature bundle containing a subset of the probe's φ-features (a *deficient* goal in his terms), the chain created between the probe and goal counts as equivalent to a movement chain.

When a probe finds such a goal, as shown in (52a), its φ-features are valued by the goal resulting in the features on the probe and goal having identical values (i.e. there are no valued features on the goal that are not also present on the probe). Roberts (2010b) proposes that given the copy theory of movement (Chomsky 1995), the configuration of two identical copies of feature sets (ignoring the difference in interpretability) counts as equivalent to movement, as shown in (52b). But this also means that the lower copy is effectively a trace and therefore ignored by the Labeling Algorithm, just as in (51) above.

(52) a.



b. ✓



Note that this analysis can also be carried over to STRONG EA-IA restrictions, where a phase head asymmetrically c-commanding the EA minimal pronoun enters into Agree with it. The Agree relation between the EA pronoun and the phase head (above *v*P) makes the EA pronoun count as the lower end of a movement chain—a trace—because the features on the probe and the goal become identical after valuation. Since traces are ignored by the Labeling Algorithm, the EA can be a head and not take *v*P as a complement.

The only tough cases for the Labeling Algorithm are those where the EA minimal pronoun merges with a *v*P phase. In such cases, labeling must take place immediately (because of the 'chicken or the egg problem' discussed above), but the EA also has to be phrasal in order to avoid the illicit configuration where it would take *v*P as a complement, since there is no higher heads at that point that the EA could move to or which could establish Agree with the EA. However, even if the EA is a phrase this will still yield a configuration of two phrases, which presents a problem for the Labeling Algorithm.

There are two possibilities for how this can be resolved. One is feature sharing (see (46) above): prominent features shared by *v* and the EA, like their person features, could provide the label. Another possibility is that because a label is required in order to determine the phase boundary and therefore the point of spell-out (see discussion above), *v*'s status as a phase head is sufficient to force the labeling of the constituent in order to prevent a scenario in which *v*P can never get spelled-out (this could be seen as a counterpart to *v* being deficient and unable to project a phase boundary; see Section 4.1.1). I leave for future work the task of establishing whether both of these options are required or if only one of them is operative in the languages with a *v*P phase and EA minimal pronouns.

4.1.2.3 Implications for language variation and change

The analysis in terms of labeling outlined above could also be used to explain the crosslinguistic distribution of EA-IA restrictions: in the current language sample, there are 77 cases of IA-IA restrictions, but only 43 cases of EA-IA restrictions. There seems to be a tendency for languages to not have EA minimal pronouns. This tendency can be tied to the issues regarding the internal structure of EA deficient pronouns discussed above: having an EA pronoun that is deficient to the point of being an unstructured feature bundle causes serious complications for the labeling of the relevant syntactic constituents.

I want to propose that given the issues for labeling that EA minimal pronouns with a head status cause, languages are biased against them. This bias effectively guides language change in directions that steer away from such pronouns. Recall that synchronically the labeling issue can be avoided in the following ways: (i) the EA minimal pronoun can be phrasal (yielding a WEAK EA-IA restriction), or (ii) the phase head is above *v*P, where the EA is introduced (yielding a STRONG EA-IA restriction).[7] But the issue can also be avoided diachronically, by having the EA pronouns gradually become structurally non-deficient pronouns through language change, like the weak object pronouns in Polish or *pro* in canonical null subject languages (see Section 3.3). This is consistent with the findings of studies regarding historical changes in pronoun systems.

There is a tendency for strong pronouns to gradually "shrink" though language change, first becoming weak and clitic pronouns, and eventually true agreement markers; additionally, subject pronouns usually become agreement markers before object pronouns (see e.g. Steele 1977; Fuss 2005). Consider what the change into an agreement marker implies here. There needs to be an agreement controller in such cases, which can be either an overt full DP/NP, strong pronoun, or a *pro*. In relation to the current analysis, this would mean that when EA pronouns shrink to the point of being heads, there are two possible changes that will avoid the problematic state: (i) the pronouns become pure inflectional agreement morphology and hence not pronouns at all—at which point another element must grammaticalize into a

7. To capture the fact that WEAK EA-IA restrictions are more common than STRONG EA-IA restrictions, the first option should be less costly that the second one.

pronoun to serve as a potential controller for agreement, or (ii) the trajectory of the change is reversed and the pronouns become structurally non-deficient. Both options are attested. The option in (i) is what we described above as the gradual "shrinking", which seems to be more common. However, the option in (ii) is also attested—a "bouncing back" change where pronouns go from being heads to being structurally non-deficient (see Haspelmath 2004b on changes from heads to phrases more generally).

Additionally, the subject-object asymmetry, whereby subject pronouns tend to become agreement markers before object pronouns, can also be attributed to the labeling issue. As we saw the labeling issue is most problematic with EA pronouns (which are the subject in the majority of active clauses), because they are introduced at the edge of the phase. All other pronouns have many more options for voiding the labeling issue. Therefore, I suggest that change happens faster with subject pronouns because of there being fewer synchronic strategies to avoid the labeling issue.

4.1.2.4 Summing up

I have argued that WEAK EA-IA person restrictions arise due to the position of the relevant minimal pronouns with respect to the phase head *v*. Its position causes the EA pronoun to be person valued by *v* before the IA pronoun can be person valued via Agree triggered by *v*. Crucially, this also means that the valuation of the EA pronoun may bleed the person valuation of the IA pronoun. This is exactly the timing of valuation operations that we saw with WEAK IA-IA restrictions, which explains why the restriction patterns in the two domains are identical. I have also shown that the ordering of the two valuation steps in this case is related to independent proposals concerning the status of theta-role assignment compared to other grammatical operations and that there are are other ways in which the EA minimal pronoun is singled out with respect to other pronouns due to its base position at the edge of the *v*P phase. But the most important take away message from this section is that differences in the base positions of arguments in relation to the most local phase head are sufficient to give rise to the STRONG VS. WEAK restriction variation. This will be crucial in explaining the attested and unattested combinations of coexisting EA-IA and IA-IA restrictions, which I will go over next. As we will see, those will follow straightforwardly from basic assumptions concerning argument structure.

4.1.3 Deriving STRONG-STRONG combinations

I now turn to the derivation of the combinations of EA-IA and IA-IA restrictions. The first case of an attested combination of EA-IA and IA-IA restrictions I will look at is the case of STRONG-STRONG combinations. Recall from earlier in this chapter that this is the combination found in Arizona Tewa. That means that in double object ditransitives only the EA can be 1/2P, as shown in (53a). The only way for a Goal to be 3P (outside of locatives/PPs) is via passivization, as in (53b), in which case the Theme must still be 3P.

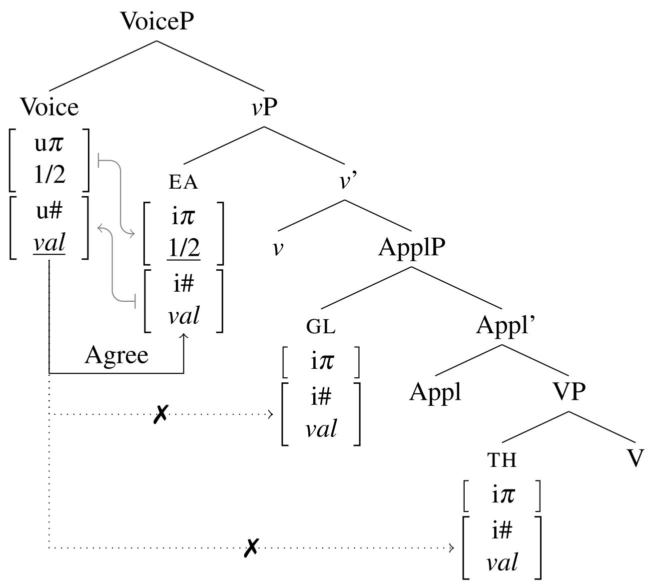
(53) Arizona Tewa [*Kiowa-Tanoan*]:

a. **dó-** **béy-** kumɛ. $\boxed{\text{1.EA} \gg \text{3.GL} \gg \text{3.TH}}$
1>3- 3PL.IO bought
'**I** bought **it** for **them**.'

b. *ų̨* *ko:to* he-ʔi sen-dí **wó:-** me:gi $\boxed{\text{2.GL.SU} \gg \text{3.TH.O} + \text{PP.3.EA}}$
you bracelet that man-OBL PASS(3>2)- give
'**You** were given **it** [a bracelet] by that man.' (Kroskrity 1977, 80, 168–9)

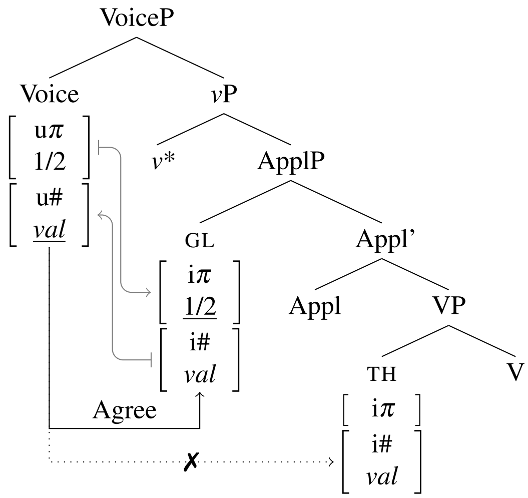
Recall that we already saw the derivation of this combination above. The STRONG-STRONG combination arises automatically when the EA and both IAs are minimal pronouns in double object ditransitives and the phase head is higher than usual—Voice as opposed to *v* in the case of Arizona Tewa—due to the deficient *v* (see Section 4.1.1). Since the phase head then asymmetrically c-commands all three minimal pronouns, the EA minimal pronoun will block the person valuation of both IA minimal pronouns. The derivation of a STRONG-STRONG double object ditransitive is shown again in (54); recall that Voice is an active number probe that can Agree only with the EA pronoun, resulting in the pronoun's person features being valued by piggybacking on the number valuation (i.e. Maximize Agree), while the other two pronouns always get a default 3P value.

(54)



The STRONG IA-IA restriction in passives can be derived just as easily. Since the demoted EA realized as an oblique/PP phrase that is not indexed on the verb, it is natural to assume that there is no EA minimal pronoun in Spec*v*P in those cases. Nonetheless, since the phase head (Voice) asymmetrically c-command all the arguments, the person restriction that results will arise in exactly the same way. As shown in (55), with the EA out of the picture, the Goal is the closest pronoun to Voice, which means that Voice will Agree with it and value its person features. But since the Goal still intervenes structurally between Voice and the Theme pronoun, there is no way for the Theme minimal pronoun to be person valued, hence it can only get a default 3P value no matter what person value the Goal pronoun receives.

(55)



The current analysis also offers a way to capture the rarity of STRONG-STRONG combinations. The reason why STRONG-STRONG combinations are so rare (only 5 of them in my sample), comes down to two factors. First, having a STRONG EA-IA restriction requires a change in the size of the verbal phase, which is a deviation from the norm. Second, the fact that 1/2P Goals and Themes must always be expressed by employing alternative constructions, such as passives and prepositional datives, might put pressure on language change so that the alternative ditransitive constructions, like prepositional datives, become the only ones used in the language. In any case, these restriction combinations can be derived in the current system just like the languages that only have a STRONG IA-IA restriction; the only difference is in the number of minimal pronouns inside the verbal phase.

4.1.4 Deriving WEAK-STRONG combinations

I now turn to WEAK-STRONG combinations of person restrictions. Recall that, unlike STRONG-STRONG combinations, these are fairly common. In fact they seem to be the default option for languages with both EA-IA and IA-IA restrictions, with 16 out of the 23 cases with restrictions in both domains being of this type. As we will see, the fact that these are the most common combination follows naturally from the current system.

I will present the derivation of the WEAK-STRONG language type with the by now familiar case of Southern Tiwa. Recall that Southern Tiwa has a WEAK restriction in the EA-IA domain, which means that when the EA minimal pronoun is 3P, the IA can only be 3P as well. The restriction pattern is shown again in (56).

(56) Southern Tiwa [*Kiowa-Tanoan*]: **WEAK EA-IA**

a. { **ti-** / **a-** / **Ø-** } mũ-ban. $\boxed{\text{1/2/3.EA} \gg \text{3.IA}}$
{ 1>3- / 2>3- / 3>3- } see-PAST
'**I/You/He** saw **him**.'

b. **i-** mũ-ban. $\boxed{\text{1.EA} \gg \text{2.IA}}$
1>2- see-PAST
'**I** saw **you**'

c. *seuan **???-** mũ-ban. $\boxed{✗\ \text{3.EA} \gg \text{1/2.IA}}$
man.A 3>1/2- see-PAST
'**He** [the man] saw **me/you**.' (Allen and Frantz 1978, 11–2)

In double object ditransitives, however, the Theme must always be 3P, just as in Arizona Tewa. But unlike in the latter, the Goal may be 1/2P. The grammatical person combinations in ditransitives are illustrated with the examples in (57).[8] As discussed in Chapter 2, we can see that the IA-IA restriction is truly a strong one when we look at passives, where despite the demoted EA, the Theme can only be 3P in the presence of a Goal, as shown by the contrast between the examples in (58).

(57) Southern Tiwa [*Kiowa-Tanoan*]: **STRONG IA-IA**

a. *'uide* **tow-** wia-ban. $\boxed{\text{1.EA} \gg \text{3.GL} \gg \text{3.TH}}$
child 1>3>3PL- give-PAST
'**I** gave **them** to **him/her** [the child].'

b. **ow-** wia-ban. $\boxed{\text{2.EA} \gg \text{3.GL} \gg \text{3.TH}}$
2>3>3PL- give-PAST
'**You** gave **them** to **him/her**.'

8. In contrast to the WEAK EA-IA restriction in transitive clauses, double object ditransitives show an additional restriction: a non-demoted EA can never be 3P when a Goal is present (see Rosen 1990). This is usually characterized as a ban on the co-occurrence of an ergative and a dative marker, under the assumption that, as is common with person based ergative splits, 3P markers show an ergative-absolutive alignment, whereas 1/2P markers show a nominative-accusative one. I do not discuss this restriction here, since it appears to have more to do with person-based ergative splits than person restrictions.

c. **kam-** musa- wia-ban. $\boxed{1.\text{EA} \gg 2.\text{GL} \gg 3.\text{TH}}$
1>2>3PL- cat(PL)- give-PAST
'**I** gave **them** [the cats] to **you**.'

d. **bow-** wia-ban. $\boxed{2.\text{EA} \gg 1.\text{GL} \gg 3.\text{TH}}$
2>1>3PL- give-PAST
'**You** gave **them** to **me**.' (Rosen 1990, 672–3, 677)

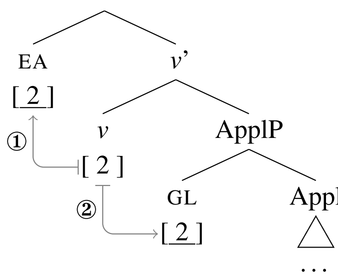
(58) a. liorade-ba **in-** *khwian-* wia-che-ban $\boxed{1.\text{GL.SU} \gg 3.\text{TH.O} + \text{PP}.3.\text{EA}}$
lady-INSTR 1>3- dog- give-PASS-PAST
'**I** was given **him** [the dog] by the lady.'

b. *liorade-ba **???-** wia-che-ban $\boxed{✗\, 1.\text{GL.SU} \gg 2.\text{TH.O} + \text{PP}.3.\text{EA}}$
lady-INSTR 1>2- give-PASS-PAST
'**I** was given **you** by the lady.' (Allen and Frantz 1978, 13–6)

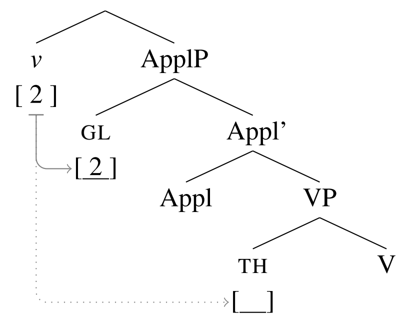
In the current system the WEAK-STRONG restriction combination follows naturally from just combining the derivations of the WEAK EA-IA restriction and the STRONG IA-IA restriction. Recall that the WEAK pattern arises due to the timing of multiple person valuation instances by the same phase head. In the case of the WEAK EA-IA restriction specifically, the EA minimal pronoun must be valued in Spec*v*P first, followed by *v* establishing Agree with the IA. Because the first valuation can either delete or not delete the valued person feature on *v*, the second valuation can result in the IA being valued either 3P or 1/2P; the former option is crucially the only way the two arguments can be 1/2P and 3P respectively—which derives the ungrammaticality of combinations where the EA is 3P and the IA is 1/2P. The STRONG pattern arises on the other hand due to Agree Closest. In the case of the STRONG IA-IA restriction specifically, *v* has an active number probe, which can only Agree with the structurally closest IA, the Goal, which can receive a person value from *v* because the valuation of person features can piggyback on the valuation of number features. However, since Agree is only possible between *v* and the Goal, the Theme can never be valued by *v* and is forced to get a default 3P value. Note that in both cases the relevant phase head is *v* and that in a double object ditransitive the Goal will be the structurally closest IA to *v*. As a result, the WEAK-STRONG combination arises solely from combining the WEAK

EA-IA derivation, which arises in valuation configurations like (59), with the STRONG IA-IA derivation, which arises in valuation configurations like (60).

(59)



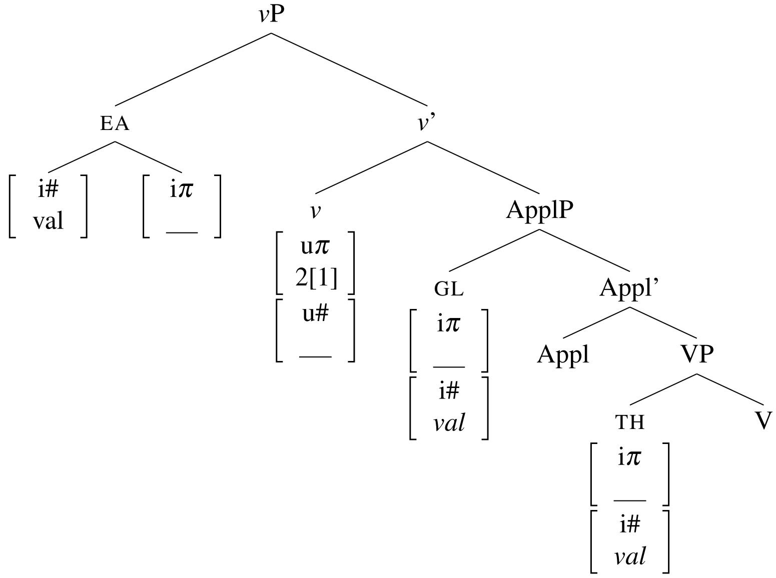
(60)



Crucially, since both restrictions are based around *v* being the phase head, and since *v* is the default phase head for the verbal domain, this approach can explain why this combination of EA-IA and IA-IA restrictions is the most common one (nothing special happens here).

Let us then go through a derivation of this restriction combination in more detail. The initial configuration of the three minimal pronouns in relation to the phase head is presented in (61), which is meant to be a snapshot of the structure at the point when the EA enters the derivation (note the richer structure which isolates its number features from Agree with *v*).

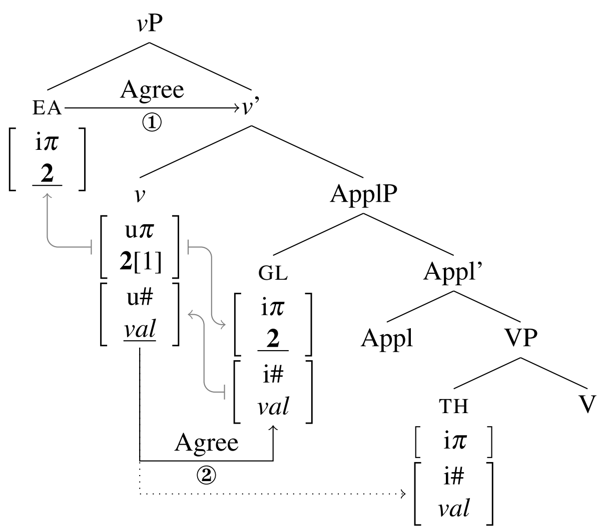
(61)



Recall from Section 4.1.2 that *v*, even though it has an active number probe, cannot probe immediately when it enters the derivation due to the Theta-Probe Condition, which prevents it in (61) from probing before it assigns a theta-role to the EA. In contrast, the EA's unvalued person feature can and does probe immediately upon Merge.

The next step in the derivation is therefore the person valuation of the EA minimal pronoun. As soon as the EA is merged, its unvalued [iπ] feature probes and establishes Agree with its valued counterpart on *v*, as shown in (62) (I ignore the number features on EA from now on, since they have no effect on the derivation). As a result of the successful Agree, the EA can receive a person value from *v*.

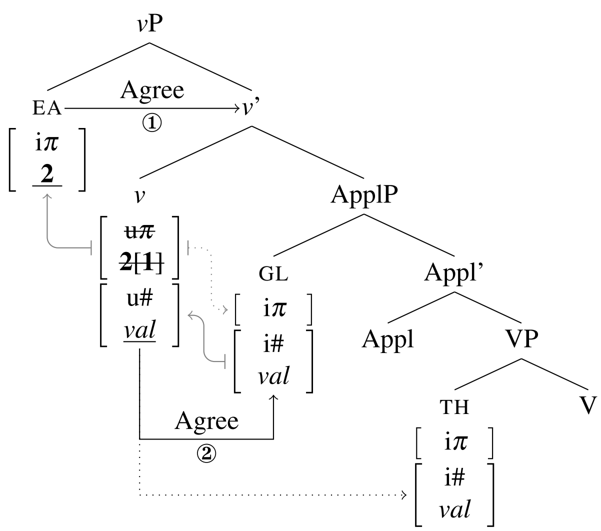
(62)



The derivation in (62) is one where the first Agree does not delete the [uπ] on *v*. As a result, when the unvalued [u#] on *v* probes and finds the valued [i#] feature on the Goal, *v* can Agree with the Goal and the [uπ] on *v* can also value the Goal's [iπ] feature. But unlike in the previous cases of WEAK EA-IA restrictions, there is another IA minimal pronoun in the derivation: the Theme. However, since the Goal structurally intervenes between *v* and the Theme, *v* can never establish Agree with it, and so the Theme is forced to get a default 3P value. Note that I only show the valuation for participant '2' features in (62), but the valuation for the speaker '1' feature proceeds in exactly the same way as with all the WEAK restrictions before. The valuation for participant features must take place for both arguments first, and either of them can receive the speaker '1' feature because they both stand in an Agree relation with *v* (both of them cannot receive '1' as this would make both pronouns 1P, which would make them co-refer and thus violate Binding Principle B).

The other possible derivation for when the [uπ] on *v* is specified with participant/speaker features is one where the person valuation of EA deletes the valued [uπ] feature on *v*, which is shown in (63). As a result, when *v* finally Agrees with the Goal (it could not have done it before due to the Theta-Probe Condition), it no longer has the valued [uπ] feature, which means that the Goal must receive a default 3P value.

(63)



The Theme also cannot be person valued in this derivation; both because of *v* no longer having a valued [uπ] feature and because of being inaccessible to *v* for Agree.

Just as with all the other WEAK person restrictions we saw above, this type of derivation where the first minimal pronoun to be valued bleeds valuation for the second one is the only way to get mixed 1/2P and 3P values on the pronouns. This means that it is impossible to derive pronoun combinations where the EA is 3P and the Goal is 1/2P. The only derivation where the EA can be 3P is the one where the [uπ] feature on *v* has no participant/speaker features, in which case the IA can also only be valued as 3P.

The derivation of the person restriction in ditransitive passives and other constructions without an EA minimal pronoun are just as straightforward. Recall that in such constructions in Southern-Tiwa (as well as other comparable constructions in WEAK-STRONG languages) the person restriction between IAs is also a STRONG; this is shown for passives again in (64) and for the quirky subject EA-less construction in (65).

(64) Southern Tiwa [*Kiowa-Tanoan*]:

a. liorade-ba **in-** khwian- wia-che-ban | $\boxed{\text{1.GL.SU} \gg \text{3.TH.O} + \text{PP}}$
lady-INSTR 1>3- dog- give-PASS-PAST
'**I** was given **him** [the dog] by the lady.'

b. *liorade-ba **???-** wia-che-ban | $\boxed{\text{✗ 1.GL.SU} \gg \text{2.TH.O} + \text{PP}}$
lady-INSTR 1>2- give-PASS-PAST
'**I** was given **you** by the lady.' (Allen and Frantz 1978, 13–6)

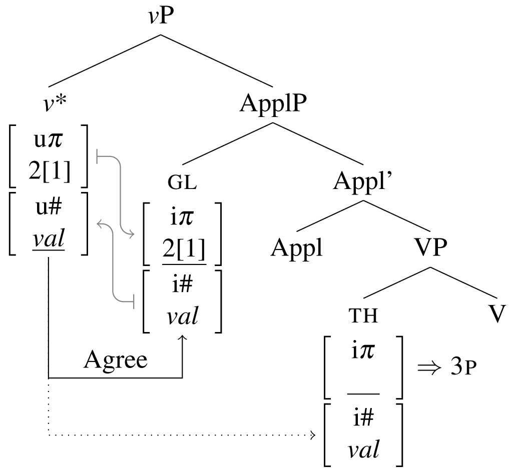
(65) a. { **im-** / **kam-** / **am-** } seuan-wan-ban | $\boxed{\text{1/2/3.GL.SU} \gg \text{3.TH.O}}$
{ 1>3PL- / 2>3PL- / 3>3PL- } man-come-PAST
'**He** [the man] came to **me/you/him**.'

b. ***???-** wan-ban | $\boxed{\text{✗ 1.GL.SU} \gg \text{2.TH.O}}$
1>2- come-PAST
'**You** came to **me**.'

c. **a-** wan-ban na-'ay | $\boxed{\text{2.TH.SU} + \text{PP}}$
2- come-PAST me-to
'**You** came to **me**.' (Rosen 1990, 673–4)

Since the EA minimal pronoun is missing in such constructions (just as in the Arizona Tewa passives we considered in the previous section), the result is exactly like a STRONG IA-IA configuration in languages with only that restriction. The derivation for this restriction is shown in (66). Since *v* does not assign a theta-role to the EA, it can probe immediately at Merge, find the Goal, and value its person feature via Agree. But the Goal still structurally intervenes between *v* and the Theme, so *v* can never Agree with the Theme, which means that the Theme can only receive a default 3P value.

(66)



What is most important about the current system and how it derives the WEAK-STRONG restriction combination is that it does so without having to treat the two restrictions separately.

As we saw in Chapter 2, other intervention-based analyses would have to derive WEAK-STRONG combinations by attributing the two restrictions to two independently parameterizable probes. In this respect, not only does the current system not require the parameterization of probes, but two different person restriction patterns arise only due to the different positions of the arguments in relation to the phase head. As we will see below, the key to deriving the *Strength Generalization* lies exactly in the fact that the strength of person restrictions is merely a reflection of where the minimal pronouns stand in relation to the phase head at the point of valuation.

4.1.5 Deriving WEAK-WEAK combinations

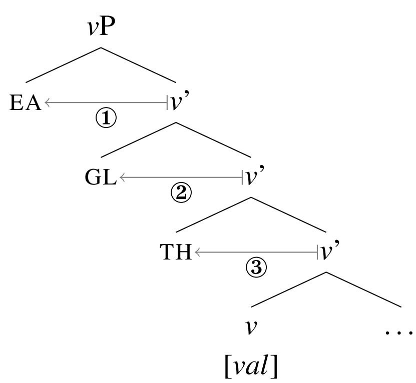
Finally, we turn to the last and rarest person restriction combination: WEAK-WEAK. These combinations can also be derived by the proposed analysis. Let us see how that can be done.

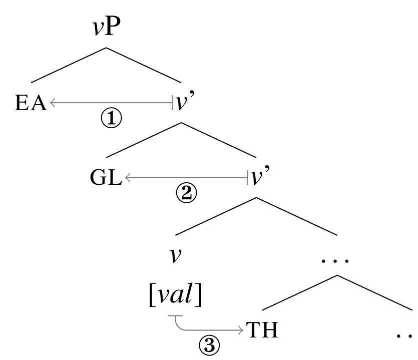
Recall that the key to deriving WEAK restrictions is in having separate instances of person valuation for each of the minimal pronouns, such that the structurally highest pronoun is valued first, followed by the next highest, and so on. The many-to-one relation between the minimal pronouns and the phase head that values them is crucial for deriving the possibility of multiple 1/2P pronouns, whereas the top-to-bottom ordering of operations is crucial for deriving the possibility of bleeding the person valuation of the pronouns lower in the structure, which is required for getting mixed 1/2P and 3P pronoun combinations. The person restriction comes from the fact that such combinations can only be derived with the bleeding option, so mixed sequences can only go from 1/2P to 3P but never from 3P to 1/2P.

So far we only saw WEAK restrictions with two minimal pronouns. With the EA-IA version the EA is valued in Spec*v*P, followed by the valuation of the IA, and with the IA-IA version both Goal and Theme are valued after moving to Spec*v*P one after the other. In the case of WEAK EA-IA restrictions we only looked at cases where the IA pronoun is valued in-situ by piggybacking on the Agree initiated by *v*, which is the type of configuration that derives WEAK-STRONG configurations. But the same EA-IA restriction pattern can arise if the IA pronoun has a richer internal structure that results in it not being valued in-situ, forcing it to move to be valued in Spec*v*P. Here too *v* can only probe after the person feature on EA has probed (due to the Theta-Probe Condition), so the internally complex IA will only move to Spec*v*P after the failure to be person valued via Agree initiated by *v*. This means the EA will still be person valued by *v* before the IA. The result is the same bleeding configuration we get when the IA is valued in-situ, but the difference in where the IA is valued will effect the person restriction pattern if the language also has an IA-IA restriction.

Note that nothing in the current system prevents more than two pronouns to be valued in a bleeding configuration. Recall that the reason why multiple valuations are possible is the special nature of valued uninterpretable features: they *can* be, but *need not* be, deleted right after they value a matching unvalued feature. There is no limit to how many times a valued uninterpretable feature can value a matching unvalued feature and not be deleted.

Because of the lack of a limit on multiple valuations we can combine a WEAK EA-IA restriction with a WEAK IA-IA restriction just as we combined a WEAK EA-IA restriction with a STRONG IA-IA restriction in the previous section. This corresponds to the EA minimal pronoun being introduced in Spec*v*P and both IAs tucking in under it, as in (67). Recall that in the case of WEAK IA-IA restrictions, the order in which the pronouns move to Spec*v*P is Goal first, followed by the Theme tucking in under the Goal due to Shortest Move. Additionally, the object pronouns only move to Spec*v*P *after v* probes and Agrees with the Goal, but fails to value its person feature due to the pronouns' richer internal structure; recall also that *v* can only probe after the EA has been person valued, due to the Theta-Probe Condition. All this means that the three pronouns will always Merge in Spec*v*P in the order: EA first, then Goal, then Theme, and that will also be the sequence in which they will be person valued by *v*. The same effect could in principle be achieved with only the Goal moving to Spec*v*P, as in (68), if we allowed the Goal to move to Spec*v*P before *v* can probe.

(67)


(68)


Because the individual instances of person valuation are ordered in top-to-bottom fashion, the only derivable mixed 1/2P and 3P combinations are those where the 1/2P pronouns are the ones higher in the structure and the 3P pronouns are the ones lower in the structure. These combinations would arise as in all previous cases with the higher valuations bleeding the lower ones. This possibility can thus derive a WEAK-WEAK restriction combination.

Having shown that the attested WEAK-WEAK combination can be derived in the current approach, we should also consider two empirical complications relating to it. The first concerns teasing apart WEAK-WEAK combinations from WEAK-STRONG ones, and the second the extreme rarity of the WEAK-WEAK combination, which is only attested in two related languages in my sample, namely Alutor (Savvina and Mel'čuk 1978; Mel'čuk 1988; Kibrik 2002) and Koryak (Comrie 1980; Abramovitz 2018) [both *Chukotkan*].

Whether or not a language has a WEAK-WEAK combination is going to be pretty much impossible to tell in regular active ditransitives. As I discussed in Section 2.5.1, combinations of more than two 1/2P arguments are problematic for reasons independent from person restrictions. With three 1/2P pronouns in a clause, two of them will inevitably co-refer and induce a Principle B violation: with one 1P and one 2P pronoun there is no issue, but adding another 1/2P pronoun will cause the binding violation, since it will have the same person value as one of the two other pronouns. We therefore need a way to distinguish whether the impossibility of a third 1/2P pronoun is due to a person restriction or the binding restriction.

Making one of the objects plural, could in principle at least marginally improve the grammaticality of such cases; as in English with '?*I* introduced *us* to you' (see footnote 33 in Section 2.5.1), but it is difficult to find the relevant examples for all languages in a typological study of the scale of the current one. Additionally, many pronominal argument languages, unlike English, categorically disallow referential overlap between pronouns (Cysouw and Fernández Landaluce 2012), so such examples would not necessarily disambiguate whether the restriction is due to binding or due to a person restriction. Making the highest pronoun 3P will also not work, since given the WEAK EA-IA restriction, this would force all the IA pronouns to also be 3P. The only way to see whether a language is WEAK-WEAK or WEAK-STRONG is if passivization or a similar kind of argument demotion voids the EA-IA restriction in the language (like in the case of Arizona Tewa and Southern Tiwa above).

Luckily, at least one strategy of this type is available in Alutor and Koryak: the so-called *spurious antipassive* (Halle and Hale 1997; Bobaljik and Branigan 2006). This construction, limited to banned person combinations, has the syntax of a regular (di)transitive construction, but with antipassive morphology on the verb. This means the verb only bears the antipassive morpheme and EA marker, while the presence of the IA is only inferred. For example, the EA can be 1P regardless of the person of the IA, as seen in (69a) and (69b), but the IA cannot be 1P when the EA is 3P if the spurious antipassive paradigm is not used, as shown in (70).[9]

(69) Alutor [*Chukotkan*]: **WEAK EA-IA**

a. **tə-** ta-ktəplə-ŋə **-n** 1.EA ≫ 3.IA
1.SU- POT-hit-PERF -3.O
'**I** probably hit **him/her**.'

b. **tə-** ta-ktəplə **-ɣət** 1.EA ≫ 2.IA
1.SU- POT-hit.PERF -2.O
'**I** probably hit **you**.'

(70) a. ***Ø-** ta-ktəplə **-ɣəm** ✗3.EA ≫ 1.IA
3.SU- POT-hit.PERF -1.O

b. t-ina-ktəpl **-əŋ** 3.EA ≫ (AP.1.IA)
POT-AP(1.O)-hit.PERF -3.SU
'**He/She** probably hit **me**.' (Kibrik 2002, 199–202)

The spurious antipassive allows us to check what the person restriction pattern really is in the IA-IA domain. That is because it neutralizes the EA-IA restriction, allowing us to check whether both the Goal and Theme can be 1/2P, and whether the Theme is person restricted when the Goal is 3P. We can see that this is the case with the examples in (71) (as we saw in Section 2.3, Alutor has both a canonical and reverse IA-IA restriction, here I only look at the former). Note that when the Goal is 1P, the Theme can be either 3P, as in (71a), or 2P, as in (71b), but when the Goal is 3P, the Theme cannot be 1P, as shown in (71c).

(71) Alutor [*Chukotkan*]: **WEAK IA-IA**

a. əlləɣ-a ina-jəl **-i** *ɣəmək-əŋ* ʂininkin ŋavakək 1.GL ≫ (3.TH)
father-ERG AP(1.IO)-give -3.SU me-DAT his.ABS daughter.ABS
'**He** [father] gave his daughter to **me** (as a wife).'

9. I am simplifying the presentation of the pattern here, since the distribution of the spurious antipassive is also sensitive to the number of the EA (Kibrik 2002) and there is an additional 1P vs. 2P asymmetry with respect to the restriction. However, these additional factors do not matter for the point I am making here and I will come back to discuss them in Section 4.3 in the context of other fine-grained person restrictions.

b. əlləɣ-a ina-jəl -i ɣəmək-əŋ ɣəttə | 1.GL ≫ (2.TH)
father-ERG AP(1.IO)-give -3.SU me-DAT you.ABS
'**He** [father] gave you to **me** (as a wife).'

c. *əlləɣ-a **Ø-** jəl **-nin** ənək-əŋ ɣəmmə | ✗ 3.GL ≫ (1.TH)
father-ERG 3.SU- give -3>3.IO him-DAT me.ABS
'**He** [father] gave me (as a wife) to **him**.' (Mel'čuk 1988, 294–5)

Thanks to the spurious antipassive, we can see that the Alutor and Koryak restriction combination is indeed WEAK-WEAK. However, Alutor and Koryak are the only languages in my sample that qualify as having a WEAK-WEAK restriction combination, so there are no other languages of this kind that we can compare them to outside of Chukotkan to help us identify what makes WEAK-WEAK combinations so scarce.[10] Hopefully, a future more extensive survey will reveal more languages of this type, giving us a better chance of determining why they are so rare. What matters for now is that the WEAK-WEAK pattern is attested. As we will see next, regardless of which of the two analyses of WEAK-WEAK combinations suggested above is right for Alutor, the current approach crucially derives only the attested restriction combinations (the ones reviewed in the last three sections), but it can never derive the unattested STRONG-WEAK combination no matter where the pronouns stand in relation to the phase and what the identity of the phase head is.

4.1.6 The impossibility of STRONG-WEAK combinations

In the last three sections, we have seen that the current system can derive all the attested combinations of EA-IA and IA-IA person restrictions. What we have yet to establish is that the current system cannot, under any circumstances, derive the only unattested combination: STRONG-WEAK. In other words, we need to show that the current system can deduce the *Strength Generalization*, which states that the IA-IA restriction can never be weaker than the

10. Also, double object ditransitives are available only with the verb 'give' (Savvina and Mel'čuk 1978; Mel'čuk 1988; Abramovitz 2018), so there are not many contexts where we can examine the IA-IA restriction.

EA-IA restriction in a language that has both. But before we look at how this generalization is deduced, let us consider what a language with a STRONG-WEAK restriction combination would even look like.

Because such a hypothetical language would have a STRONG EA-IA restriction, only the EA would have the possibility of being 1/2P, even in a ditransitive, just like in Arizona Tewa. This pattern is shown in (72) with illustrative English examples. Crucially, unlike in Arizona Tewa, passivization (or an alternative means of neutralizing the EA-IA restriction) in such a language would have to reveal a WEAK IA-IA restriction pattern, illustrated with English examples in (73). In other words, more person combinations of subject and object markers would be available in a ditransitive passive clause (or comparable construction) than in an active transitive clause. No such language was identified by the crosslinguistic survey.

(72) **STRONG EA-IA**

a. **I** showed **them him**. 1.EA ≫ 3.GL ≫ 3.TH

b. ***I** showed **you him**. ✗ 1.EA ≫ 2.GL ≫ 3.TH

c. ***They** showed **me him**. ✗ 3.EA ≫ 1.GL ≫ 3.TH

(73) **WEAK IA-IA**

a. **I** was shown **him**. 1.GL ≫ 3.TH

b. **I** was shown **you**. 1.GL ≫ 2.TH

c. ***They** were shown **me**. ✗ 3.GL ≫ 1.TH

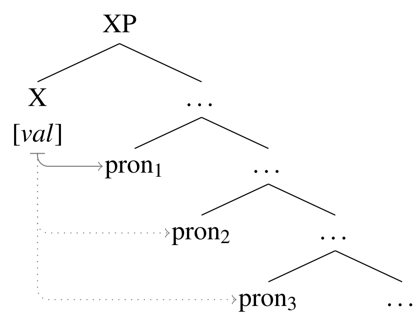
A possible objection here is that perhaps STRONG-WEAK combinations are just very rare, similarly to WEAK-WEAK combinations, and that a more extensive survey might find examples of languages with the restriction combination in question. Perhaps the WEAK restrictions in the IA-IA domain are obscured by the same confound that we saw above in the case of Alutor and Koryak. However, there are good reasons to think that the gap is not just the result of such patterns having gone unnoticed. In virtually all languages with a STRONG EA-IA restriction, voiding the restriction with an alternative construction is always very productive. This is probably due to the fact that, unlike with 1/2P Themes in ditransitives, one finds the need in the discourse to use a sentence with a 1/2P object more often. To put it plainly, researchers and grammarians working on such languages sooner or later have to answer the inevitable question: So how *does* one say 'I see you' or 'I showed you him' in this language? Scholarly works on such languages thus tend to include the answer to this question, which is also the point where we would find the relevant data identifying the language as one with a STRONG-WEAK restriction combination. After all, for a language to be able to express 'I was shown you' but not 'I saw you' is quite striking and probably worth noting. I will therefore proceed to deriving the typological gap in question under the assumption that the impossibility of STRONG-WEAK combinations is indeed real.

4.1.6.1 Deriving the typological gap

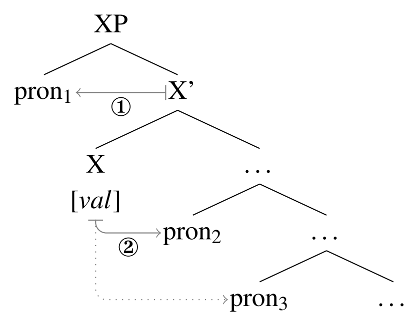
Variation in person restriction patterns is the result of two factors in the current system: (i) the movement of minimal pronouns driven by their richer internal structure, and (ii) the variable position of the phase head in relation to the minimal pronouns. Since these two points of variation are necessary in the system to derive the attested patterns of person restrictions, we need to show that no matter how we vary these two parameters, the current system can never derive the unattested STRONG-WEAK restriction combination. In fact, it can be shown, by abstracting away from any particular argument structure or phase head, that given any three minimal pronouns and a single phase head, all the logically possible valuation configurations allowed by the current system can only derive the attested restriction combinations, thus deducing the *Strength Generalization*.

All the person valuation configurations possible in the current system for three minimal pronouns and a single phase head (X) are given in (74)–(77); this includes the configuration in (74), where the phase head asymmetrically c-commands all three pronouns, the configuration in (75), where one of the pronouns is the specifier of the phase head and the remaining two pronouns are asymmetrically c-commanded by the phase head, the configuration in (76), where two of the pronouns are specifiers of the phase head and the remaining pronoun is asymmetrically c-commanded by the phase head, and finally the configuration in (77), where all three pronouns are specifiers of the phase head.

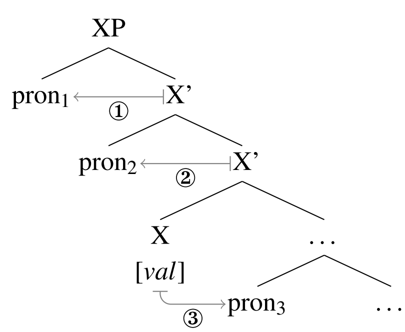
(74) STRONG-STRONG:



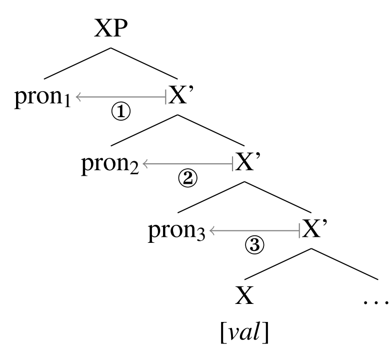
(75) WEAK-STRONG:



(76) WEAK-WEAK:



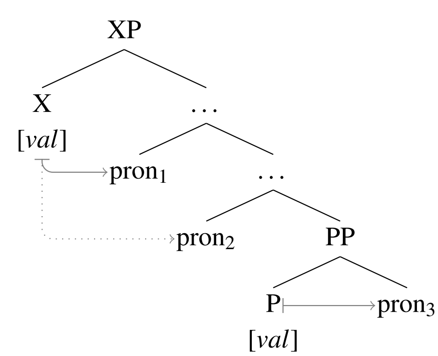
(77) WEAK-WEAK:



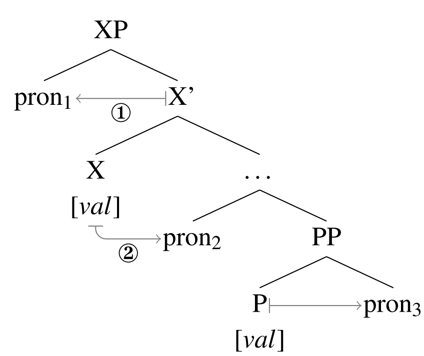
Because of the differences in the positions of the pronouns, differences arise in how the pronouns are person restricted. The configuration in (74) is a pure Agree Closest configuration, where only the top pronoun can be valued by the phase head, so the restriction is STRONG for both bottom pronouns. The configuration in (75) is a combination of a timing effect configuration and an Agree Closest configuration, where the topmost pronoun may or may not bleed the person valuation of the next pronoun down, resulting in a WEAK restriction, but Agree Closest still applies in relation to the two bottom pronouns, resulting in a STRONG restriction. The remaining two configurations in (76) and (77) are both pure timing effect configurations, where the valuations proceed top-to-bottom, either bleeding or not the next valuation, resulting in only WEAK restrictions between all the pronouns. Note that this exhausts all the possibilities for three pronouns and one phase head; if we substitute the three pronouns top-to-bottom with an EA, a Goal, and a Theme minimal pronoun respectively, we get exactly the three crosslinguistically attested restriction combinations: STRONG-STRONG (cf. (74)), WEAK-STRONG (cf. (75)), and WEAK-WEAK (cf. (76),(77)).

The only way we could introduce more valuation possibilities is by adding an additional phase head (which is a general way to void person restrictions), such as a P head, into the mix. However, that would not change the restriction pattern, it would only make one of the two restrictions disappear. For example, if we add a PP projection above the lowest pronoun in (74), we get a STRONG restriction between the top two pronouns and no restriction between the bottom two pronouns, since the P phase head can give the pronoun it takes as its complement any person value, as shown in (78). The same happens if we add the P phase head to (75): we get a WEAK restriction between the top two pronouns and no restriction between the bottom two pronouns, due to the P phase head between them, as shown in (79).

(78) STRONG-∅:



(79) WEAK-∅:



There is thus no way in the current system to derive any other restriction combinations than the ones derived in (74)–(77). Given any single phase, the pronoun pair closest to its edge will be affected by a person restriction of: (a) equal strength, or (b) lesser strength than any pronoun pair further from the phase edge. In other words, the *Strength Generalization*, repeated in (80), is deduced.

(80) *Strength Generalization*
If a language has person restrictions in both the EA-IA and IA-IA domain, the IA-IA person restriction cannot be weaker than the EA-IA person restriction.

Not only is the *Strength Generalization* deduced, in the current system it is actually part of a broader generalization regarding person restrictions: no matter which or how many pairs of pronouns we look at, the restrictions can only stay the same or grow stronger as we go deeper into the structure of a given phase.

Perhaps the most important result of the deduction of the *Strength Generalization* is that it crucially requires reference to syntactic structure and syntactic operations, which more generally provides an additional argument for the syntactic approach to person restrictions. Although this particular argument for a syntactic approach to person restrictions is more abstract than those I provided in Chapter 2, it sheds more light on why person restrictions exist in two versions. Let us take some time to examine why this is so.

Both STRONG and WEAK person restrictions arise due to a failure of the same grammatical operation (Agree), but there are two major ways in which syntax can constrain grammatical operations: locality and timing. Agree Closest, which gives rise to STRONG restrictions, is a locality constraint expressed over syntactic representations. On the other hand, the ordering of separate valuations, which gives rise to WEAK restrictions, follows from a derivational approach to syntax; operations can only take place when the elements that trigger them or condition them enter into the derivation. The two types of person restrictions arise respectively: (i) because different arguments cannot occupy the same syntactic position (one has to always asymmetrically c-command the other) and (ii) because different arguments must be introduced or moved to their respective syntactic positions one at a time. Based on argument structure, movement possibilities, and variations in the size of phases, minimal pronouns can be valued in different syntactic configurations, resulting in the variation in restriction patterns. However, the rigid syntactic constraints that apply to these different configurations severely limit the valuation options. Valuation of minimal pronouns at the phase edge, where new syntactic structure is being built either through base generation or movement, is always going to be constrained in terms of the ordering of separate valuation steps. Conversely, valuation of minimal pronouns lower than the phase edge, where all the syntactic structure is already in place, is going to be constrained in purely representational terms (Agree Closest). That is the key to the *Strength Generalization*: person restrictions can never become stronger the closer the affected pronouns are to the phase edge, they can only grow weaker or remain of the same strength.

4.1.6.2 The significance of the **Strength Generalization**

Another way to appreciate the way the *Strength Generalization* falls out from the proposed syntactic account is to compare it to an approach to person restrictions that makes no reference to syntactic structure, for example, a functionalist approach like that of Haspelmath (2004a), which is an approach that relies on extrinsic person and thematic scales (see also Section 2.2). The core intuition behind such approaches is that in a given language, specific interactions between a person hierarchy, where 1P and 2P outrank 3P, and a thematic hierarchy, where Agent (EA) outranks Goal and both outrank Theme, may be impossible; e.g. an argument low on the thematic scale cannot have a person value higher on the person scale than the person value of a co-occurring argument high on the thematic scale.

Haspelmath's specific implementation of this idea relates to the frequency of arguments having a particular person value and a particular thematic role even in languages without person restrictions, that is: EAs are 1/2P more frequently than Goals, and Goals are 1/2P more frequently than Themes. According to Haspelmath, person restrictions arise due to a *frequency condition on grammaticalization*: the combinations of deficient pronouns that are ungrammatical in a particular language are not ungrammatical due to a failure of any syntactic operation, they are merely combinations that have failed to grammaticalize due to the universal infrequency of specific combinations of person and thematic roles.

The STRONG VS. WEAK variation in this kind of an approach amounts to more or less person and thematic role combinations failing to grammaticalize in a particular language. Thus, in the case of EA-IA restrictions, a WEAK restriction would amount to a failure to grammaticalize only the combinations where the thematic roles are aligned with person values at the opposite ends of the person scale, shown in (81), while a STRONG restriction would amount to a failure to grammaticalize all alignments where an IA is aligned with 1/2P, shown in (82).[11] The STRONG VS. WEAK restriction variation in the IA-IA domain is derived the same way, as shown in (83)–(84), only with the Goal substituting for the EA and the Theme substituting for the IA on the thematic scale.

(81) **WEAK EA-IA**

*EA > IA

1/2 > 3

(82) **STRONG EA-IA**

*EA > IA

1/2 > 3

(83) **WEAK IA-IA**

*GL > TH

1/2 > 3

(84) **STRONG IA-IA**

*GL > TH

1/2 > 3

Setting aside the fact that this type of an approach leaves unexplained, for example, the correlation between STRONG EA-IA restrictions and seemingly unrelated syntactic and morphological phenomena (as discussed in Section 4.1.1) and must treat canonical + reverse

11. Haspelmath (2004a) mainly discusses the variation in IA-IA restrictions, so this is just an extrapolation of what the derivation of the variation would have to look like in the EA-IA domain.

restriction systems as an entirely different phenomenon than other person restrictions, it also cannot derive the *Strength Generalization*.

The problem is essentially the same "too many parameters" problem that was discussed in relation to a syntactic analysis where variation in the EA-IA and IA-IA domain is attributed to two independently parameterized probes: in order to derive the STRONG/WEAK variation in each argument domain independently for languages with only an EA-IA or IA-IA restriction, the restrictions on grammaticalization in each domain must be independent from each other. But that also means that if a language has person restrictions in both domains, the variation in restrictions should also vary independently for the two domains in such cases; allowing for the unattested STRONG-WEAK combination to arise. The same problem would also arise for other analyses in terms of extrinsic person and thematic scales (e.g. Rosen 1990; Aissen 1999b), which crucially do not make reference to syntactic structure, but have to leave room for the EA-IA and IA-IA restrictions to be parameterized independently from each other for the same reasons as the non-syntactic analysis based on Haspelmath 2004a.

In contrast, we saw that an approach to person restrictions that makes reference to syntactic structure and syntactic operations—specifically the one I proposed in Chapter 3 and elaborated on in the sections above—can derive the typological gap quite straightforwardly. That is not to say that the grammaticalization of different person restriction patterns is not an important issue. But the insights of Haspelmath (2004a) are not lost with the proposed analysis. The grammaticalization of person restrictions can be tied in the current approach directly to the tendency of strong pronouns to change into more and more deficient forms through language change (see Steele 1977; Fuss 2005; Section 4.1.2). For example, if combinations of 1/2P Goals and 3P Themes are more frequent even in languages without person restrictions to the point where 1/2P Themes are almost non existent in the primary linguistic data, then grammaticalizing them into deficient pronouns that have to be assigned a value by an external source will not come at that great of a cost: it will still be possible to derive combinations of 1/2P Goals and 3P Themes by valuing only the Goal pronoun. In contrast, the rarer combinations with a 1/2P Theme will be expressed with an alternative construction: a prepositional dative, or with one of the pronouns being strong rather than deficient. Importantly, such alternative constructions also involve "more costly" derivations: they involve more structure and more applications of grammatical operations.

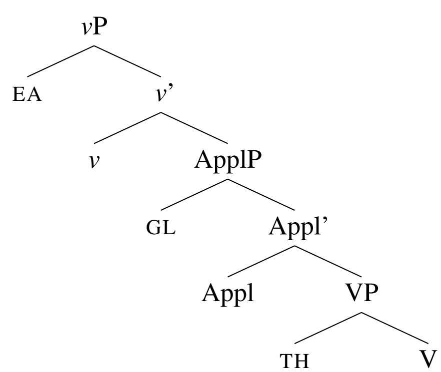
The more important result here is, however, the deduction of the *Strength Generalization* itself and what it means for person restrictions as a syntactic phenomenon. Up until this point, the syntactic nature of person restrictions was a position to be defended in light of alternative analyses in terms of extrinsic scales, morphological constraints, or surface ordering constraints. However, based on the deduction of the *Strength Generalization* we can make a much stronger claim. Person restrictions are not only syntactic, the existence of person restrictions and the limited number of patterns in which they come in show us the need for syntactic structure and syntactic operations in theorizing about language. Without them, it would not be possible to explain the *Strength Generalization* in such a straightforward way. And this argument is not particular to the specific reason that I have offered for why person restrictions arise—the valuation of minimal pronouns—as long as an analysis of person restrictions is based around STRONG restrictions being the result of the strictly local nature of grammatical operations and WEAK restrictions being the result of the timing of grammatical operations, the generalization can be made to follow.

There is one more aspect of the analysis, not strictly required for deriving the *Strength Generalization*, that needs to be highlighted. The current approach captures person restrictions by only employing syntactic elements and operations that were independently proposed to deal with other phenomena, for example: phases, which were originally proposed to deal with a variety of phenomena (e.g. locality of movement and multiple spell-out), and minimal pronouns, which were originally proposed to deal with binding phenomena; it does not introduce new theoretical mechanisms. If our goal is a constrained theory of syntax, then we should strive to reuse what we already have, rather then to constantly create new machinery to deal with new phenomena. That we have been able to capture very complex paradigms regarding person restrictions in a vast number of languages by using only independently proposed syntactic mechanisms is a very nice result from this perspective.

4.2 Deducing the Direction Generalization

I have limited the discussion of person restrictions in this chapter so far to canonical restrictions, which means person restrictions on the IA in the case of EA-IA restrictions and person restrictions on the Theme in the case of IA-IA restrictions.[12] An important result of the proposed analysis is that the derivation of different combinations of EA-IA and IA-IA restrictions does not require the parameterization of Agree. A key part in this was the assumption that the arguments in a clause are always organized in the way shown in (85).

(85)



But this cannot be the only way arguments can be organized in a clause, at least not at the point when person restrictions apply. This is because we have seen that there also exist reverse versions of person restrictions. For example, in Slovenian when the Goal clitic precedes the Theme clitic, the IA-IA restriction pattern is a canonical one, shown in (86) (STRONG version), whereas when the Theme clitic precedes the Goal clitic, the restriction

12. Although in some cases, the languages discussed did also have reverse counterparts to the restriction which I have set aside, to be discussed in this section.

pattern is reversed, shown in (87) (STRONG version); the person restriction applies to the Goal clitic as opposed to the Theme clitic—a mirror image of the canonical restriction.

(86) Slovenian [*Slavic*]:

a. Mama **ti** **ga** bo predstavila. — 2.GL ≫ 3.TH
mom 2.DAT 3.M.ACC FUT.3 introduce.F
'Mom will introduce **him** to **you**.'

b. *Mama **mi** **te** bo predstavila. — ✗ 1.GL ≫ 2.TH
mom 1.DAT 2.ACC FUT.3 introduce.F
'Mom will introduce **you** to **me**.'

c. *Mama **mu** **te** bo predstavila. — ✗ 3.GL ≫ 2.TH
mom 3.M.DAT 2.ACC FUT.3 introduce.F
'Mom will introduce **you** to **him**.'

(87) a. Mama **te** **mu** bo predstavila. — 2.TH ≫ 3.GL
mom 2.ACC 3.M.DAT FUT.3 introduce.F
'Mom will introduce **you** to **him**.'

b. *Mama **me** **ti** bo predstavila. — ✗ 1.TH ≫ 2.GL
mom 1.ACC 2.DAT FUT.3 introduce.F
'Mom will introduce **me** to **you**.'

c. *Mama **ga** **ti** bo predstavila. — ✗ 3.TH ≫ 2.GL
mom 3.M.ACC 2.DAT FUT.3 introduce.F
'Mom will introduce **him** to **you**.' (Stegovec 2015b, 108–9)

Additionally, reverse restrictions of this kind are not limited to the IA-IA domain. For example, Kaqchikel has both canonical and reverse versions of its STRONG EA-IA restriction. This is illustrated by the examples in (88)–(89), which parallel those in Slovenian, except for the arguments involved and the fact that in Kaqchikel the restriction takes the form of person restricted argument indexing: only one argument can be indexed on the verb, resulting in the other one being person restricted. When, as in (88), the EA is indexed, the IA cannot be 1/2P, and when, as in (89), the IA is indexed, the EA cannot be 1/2P.

(88) Kaqchikel [*Quichean-Mamean*]:

a. ja *rat* x- **at-** ax-an *ri* *achin* — 2.EA ≫ (3.IA)
FOC you CMPL- 2- hear-AF the man
'It was **you** that heard the man.'

b. *ja *yïn* x- **in-** ax-an *rat* — ✗ 1.EA ≫ (2.IA)
FOC me CMPL- 1- hear-AF you
'It was **me** that heard you.'

c. *ja *ri achin* x- **Ø-** ax-an *rat* — ✗ 3.EA ≫ (2.IA)
FOC the man CMPL- 3.ABS- hear-AF you
'It was **him** [the man] that heard you.'

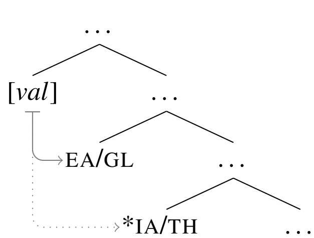
(89) a. ja *ri achin* x- **at-** ax-an *rat* — 2.IA ≫ (3.EA)
FOC the man CMPL- 2- hear-AF you
'It was the man that heard **you**.'

b. *ja *rat* x- **in-** ax-an *yïn* — ✗ 1.IA ≫ (2.EA)
FOC you CMPL- 1- hear-AF me
'It was you that heard **me**.'

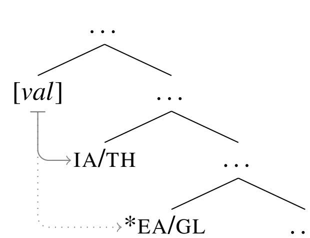
c. *ja *rat* x- **Ø-** ax-an *ri achin* — ✗ 3.IA ≫ (2.EA)
FOC you CMPL- 3.ABS- hear-AF the man
'It was you that heard **him** [the man].' (Preminger 2014, 18, 22)

I have argued in Chapter 2 that the canonical/reverse alternation corresponds to a difference in asymmetric c-command relations between the two relevant pronouns at the point where the person restriction applies. In the current system, this corresponds to two structural configurations of arguments being possible at the point of person valuation. That means that canonical restrictions arise when the position of the arguments at the point of person valuation respects the hierarchy we see above in (85); in the case of a STRONG restriction, this means a configuration like (90). Reverse restrictions, on the other hand, arise when the position of the arguments is reversed with respect to what we see in (85); in the case of a STRONG restriction, this means a configuration like (91).

(90) Canonical:



(91) Reverse:



There are, in principle, two options for how a configuration like (91) could arise. One is that languages simply allow different base generation possibilities for all arguments, but that would force us to abandon all the insights that can be gained from an approach to argument structure that assumes a universal hierarchy of argument base positions (see e.g. Baker 1988, 1997). More importantly for the issue at hand, we would have no way of constraining reverse person restrictions to only occur in languages that already have a canonical counterpart of the same restriction. Recall that these are the only cases in which we find reverse restrictions, as per the *Direction Generalization*. The other possibility is that configurations like (91) are always derived via movement, as with the analysis I suggested for Slovenian in Section 3.6. If the type of movement that can yield a reverse restriction can never be obligatory, then this would explain why languages can never only have reverse restrictions under the assumption that the base argument structure is always (85) in all languages. In this section, I identify the type of movement that has the right characteristics to derive all the reverse person restrictions found crosslinguistically and deduce the *Direction Generalization*.

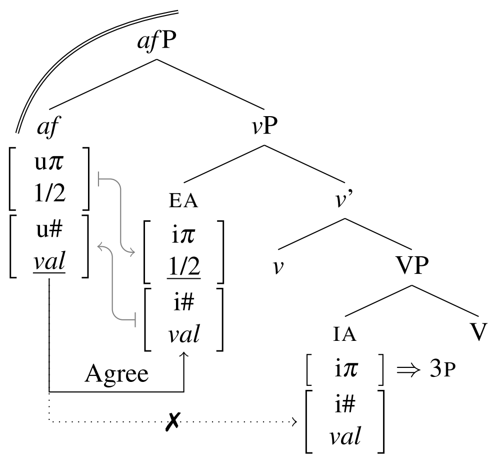
4.2.1 The movement analysis of reverse person restrictions

Before we move on to pinpoint the exact properties of the type of movement required to deduce the *Direction Generalization*, let us remind ourselves of how a movement analysis of reverse restrictions works. Since we already saw in Section 3.6 how this can be done in the case of IA-IA restrictions, I will look at the case of an EA-IA restriction here, which will also show that the movement approach is not limited to the IA-IA domain.

The particular case I derive here is the Kaqchikel pattern we saw in the last section. Recall from Section 4.1.1, that the STRONG EA-IA restriction arises only with Agent-Focus constructions, which I attributed to those constructions not having a regular verbal phase, but an extended one where the phase head (which I labeled *af*) is located just above *v*P.

As shown again in (92), when the arguments are in their base positions below *af*, this phase configuration only allows the EA to be person valued by *af* by piggybacking on the Agree *af* establishes with the EA minimal pronoun due to their matching number features, whereas the IA minimal pronoun can only get a default 3P value.

(92)

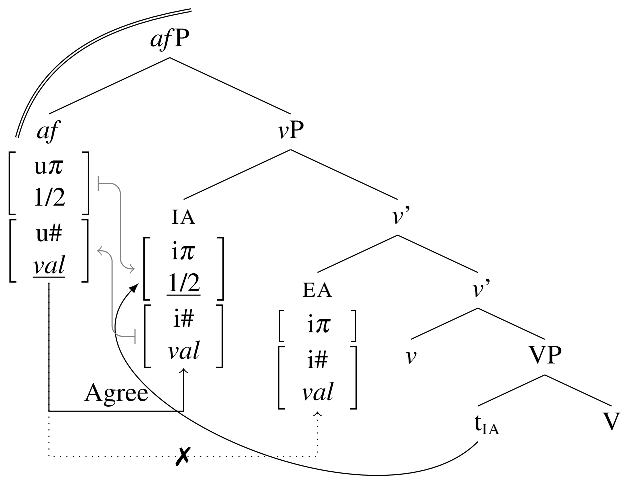


I propose that the reverse STRONG EA-IA restriction occurs when the IA minimal pronoun moves over the EA minimal pronoun below *af*, similarly to how the Theme moves over the Goal below *v* in the analysis I proposed for Slovenian in Section 3.6. What this results in is a reversal of the asymmetric c-command relation between the EA and the IA under *af*, which means that now the IA intervenes for Agree between *af* and the EA.

As shown in (93), the derivation of the person restriction is then essentially the mirror image of the one in (92).[13] The *af* now establishes Agree with the IA minimal pronoun, valuing its person feature as a result, while the EA minimal pronoun, inaccessible for Agree with *af*, can only get a default 3P value.

13. The reason why the IA ends up higher than the EA here in spite of the mechanism of tucking in discussed earlier, will be elaborated on below.

(93)



A movement analysis for reverse restrictions also works for WEAK EA-IA person restrictions, but showing that will be easier once we establish the specific type of movement that can yield reverse person restrictions. At this point, I have been purposefully vague regarding what kind of movement derives the argument reversal we saw in (92)–(93). As we will see next, the type of movement we are looking for, and which can derive all the reverse patterns attested crosslinguistically, must meet the following requirements: (i) it must be able to take place before person valuation, (ii) it must be able to reorder arguments, and most importantly (iii) it must be optional.

4.2.2 Reverse person restrictions require A-scrambling

In fact, exactly the kind of movement that fits the criteria outlined above has been independently proposed to exist; namely, *middle-field scrambling* (also *short-distance scrambling*) (Mahajan 1990; Saito 1992; Webelhuth 1992; Yatsushiro 2001; Anagnostopoulou 2003). Recall from Section 3.6 that I sugegsted this type of movement is what drives the canonical/reverse IA-IA restriction alternation in Slovenian. In Slovenian, it is not only object clitics that can reorder. When the objects in a ditransitive are noun phrases, the Goal either precedes the Theme or the Theme precedes the Goal, and the reordering that yields the latter is optional under the right conditions. In the basic order, where the Goal precedes the Theme, if the Goal is a reciprocal expression it cannot be bound by the Theme, as in (94a). But if the objects change their order, the Theme can bind the reciprocal Goal, as in (94b).

(94) Slovenian [*Slavic*]:

a. *Zigi je pokazal enega drugemu$_i$ [nova kitarista]$_i$. $\boxed{\text{GL} \gg \text{TH}}$
Ziggy AUX.3 showed one other.DAT [new.DU.ACC guitarist.DU.ACC]

b. Zigi je pokazal [nova kitarista]$_i$ enega drugemu$_i$. $\boxed{\text{TH} \gg \text{GL}}$
Ziggy AUX.3 showed [new.DU.ACC guitarist.DU.ACC] one other.DAT
'Ziggy introduced the two new guitarists to each other.'

What this asymmetry between the two orders shows is that the reordering of objects must be syntactic in nature, since if can feed binding relations. While the cases concerning the reordering of full NPs cannot directly confirm that the reordering of clitic objects in Slovenian is the same type of process, there is also no evidence suggesting the reordering of clitics could not be the same type of movement. The most parsimonious assumption here is therefore that the reordering of Goals and Themes in Slovenian, whether they are clitics of other kinds of objects, is always the same kind of grammatical process.

The sensitivity to binding actually also shows that the scrambling movement driving the reordering in (94), and by extension the clitic reordering, must be a type of A-movement. This argument is primarily known from scrambling in Japanese (Saito and Hoji 1983; Saito 1985; see also Nemoto 1999 for an overview), where clause-internal scrambling can also create new binding possibilities, one of the characteristic properties of A-movement (Hoji 1985, 1987; Saito 1992). In fact, the exact same asymmetry we saw above for Slovenian can be replicated in Japanese. As shown in (95), the two object orders found in Japanese ditransitives are associated with different binding possibilities: in (95a), where the Goal precedes the Theme, the Goal can bind the reciprocal Theme, while in (95b), where the Theme precedes the Goal, the reciprocal Theme cannot be bound by the Goal.

(95) Japanese:

a. Masao-ga [Taroo to Hanako]$_i$-ni otegai$_i$-o syookaisita. $\boxed{\text{GL} \gg \text{TH}}$
Masao-NOM [Taroo and Hanako]-DAT each.other-ACC introduced
'Masao introduced, to Taro and Hanako, each other.'

b. *Masao-ga otegai$_i$-o [Taroo to Hanako]$_i$-ni syookaisita. $\boxed{\text{TH} \gg \text{GL}}$
Masao-NOM each.other-ACC [Taroo and Hanako]-DAT introduced
'Masao introduced each other to Taro and Hanako.' (Nemoto 1999, 146)

I propose that all cases of reverse person restrictions, including IA-IA as well as EA-IA restrictions, result from middle-field scrambling of one minimal pronoun over another. Because they occur within *v*P, which is the default verbal phase, and change the asymmetric c-command relation between the two pronouns, they can feed person valuation of the moved minimal pronoun, which is essential for deriving reverse person restrictions. Finally, and most importantly, because scrambling is fundamentally an optional movement operation, we can deduce the *Direction Generalization*: the reason why there are no languages with only reverse person restrictions in either the EA-IA or IA-IA domain is because the type of movement that can yield such restriction patterns is never obligatory.

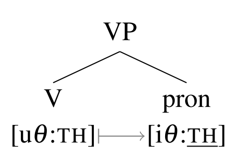
The existence of A-movement that can reorder arguments might seem odd from the perspective of *Relativized Minimality* (Rizzi 1990b), which tells us that movement targeting a position of type α cannot cross another position of type α, where α is either an A-position, Ā-position, or a syntactic head. However, the insensitivity to Relativized Minimality is a well established hallmark of scrambling (Saito and Fukui 1998). A straightforward explanation for this exceptional property of scrambling is possible with a feature based reinterpretation of Relativized Minimality (Rizzi 2001; Starke 2001), where it effectively boils down to Agree Closest (or Shortest Move, when movement is driven by unvalued features on the moving elements à la Bošković 2007). If scrambling is actually not feature driven, as argued by Saito and Fukui (1998), then the reason why scrambling is not sensitive to Relativized Minimality is because Relativized Minimality only applies only to feature driven movement.

Because A-movement scrambling (or just A-scrambling) can apply freely and not because of the need to value any features, it can also occur before φ-feature valuation or Case assignment, but also, in a system where theta-role assignment is also feature based (see Section 4.1.2), before theta-role assignment. I argue, following the insights of Richards (2008b), this is also the reason why A-scrambling is the only type of A-movement that does not result in tucking in configurations, in other words why A-scrambling can reorder arguments.

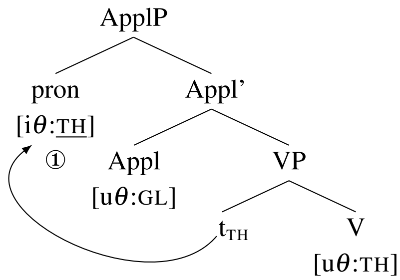
The basic insight of Richards (2008b) is that we can model the argument reordering which results from A-scrambling as movement of an argument to the base generated position of another argument before the latter has even entered the derivation. We can integrate this option in the current system by expanding on the feature based theta-role assignment approach discussed briefly in Section 4.1.2. Specifically, if theta-role assignment involves the theta-role assigner providing a value for unvalued theta-role related features (*θ-features*) of the argument it merges with, then an argument that has already had its θ-features valued will not be θ-valued in the same position. I propose that this is exactly what happens with middle-field scrambling, which involves arguments that have already received a theta-role. Let us look at a sample derivation with a Theme moving across a Goal.

When an argument (here a pronoun) merges with a verb, the verb values the argument's θ-features, making it a Theme, as seen in (96a). Since we are dealing with a double object construction, the newly formed VP will be taken as a complement by an Appl head, which is also a theta-assigner; assigning the Goal theta-role. As scrambling is not driven by any feature valuation, it can apply as soon as possible, which means that when it targets the specifier of ApplP, it will take place immediately after ApplP is formed, as seen in (96b).

(96) a.

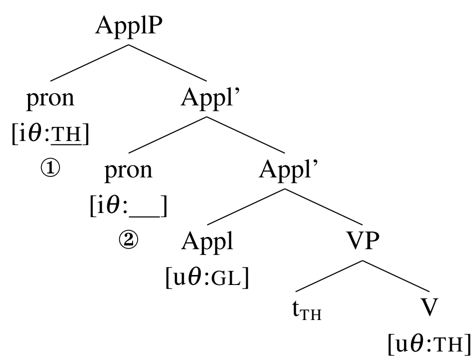


b.

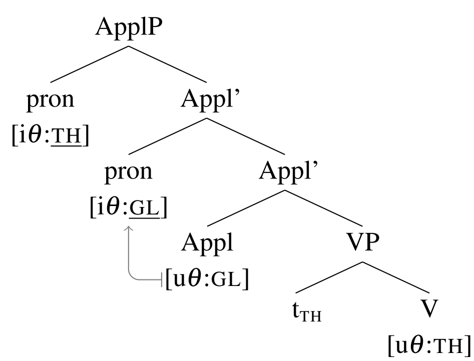


Note that since the argument moving to SpecApplP is already a Theme, the Appl head will not be able to assign it a theta-role. Appl can only assign a theta-role when an argument is base generated in its specifier. When this happens, the base generated argument tucks in under the previously merged Theme, following Richards (2008b), as shown in (97a). At this point, Appl can assign the newly merged argument the Goal theta role, as shown in (97b).

(97) a.



b.

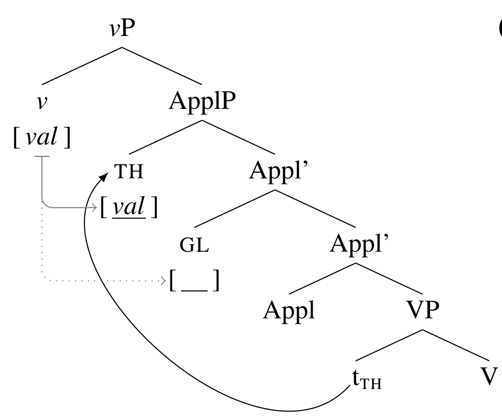


It is important to note that under this approach to scrambling, the ungrammaticality of specific person combinations in the case of reverse person restrictions is not due to a failure of movement. Instead, it is simply the result of the reordering reversing the asymmetric c-command relations between two minimal pronouns. This makes a different pronoun the structurally highest one, which in turn either blocks or bleeds person valuation of the derived lower pronoun. For example, in the case of a reverse IA-IA restriction, the combination of a 3P Theme and a 1/2P Goal is not ungrammatical because a 3P Theme cannot cross a 1/2P Goal (a person configuration ungrammatical with both STRONG and WEAK patterns), but because if movement of the Theme over the Goal were to take place, a 3P Theme would prevent the person valuation of a Goal minimal pronoun and force it to be 3P.

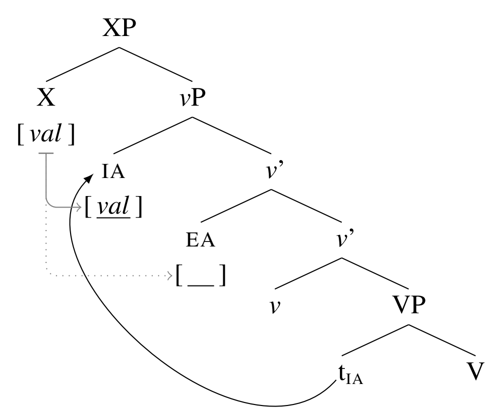
4.2.3 Reverse restrictions across argument domains

The proposed analysis of reverse person restrictions in terms of middle-field scrambling is applicable to both STRONG and WEAK restrictions and both EA-IA and IA-IA restrictions, and to all combinations of the two. In the case of STRONG restrictions, scrambling targets a position below the phase head: with IA-IA restrictions, the Theme moves to SpecApplP, as illustrated in (98), whereas with EA-IA restrictions, where the phase head is above *v*P, the IA scrambles to Spec*v*P, as illustrated in (99).

(98)

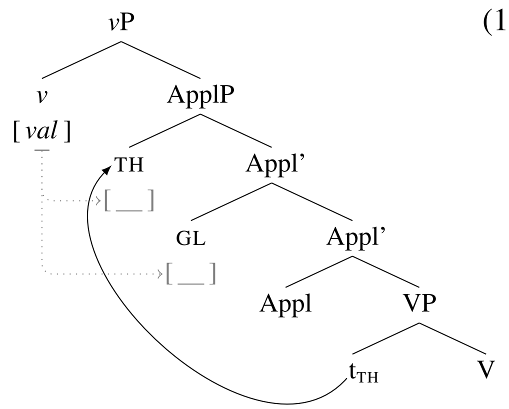


(99)

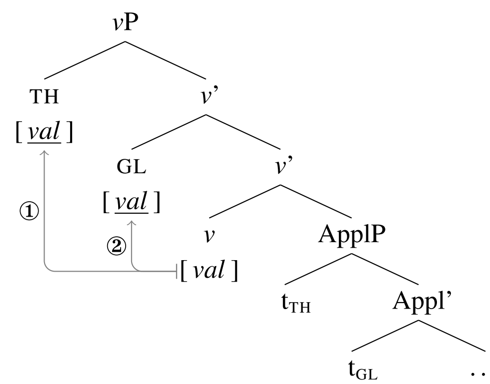


In the case of WEAK IA-IA restrictions, scrambling targets the same position as with their STRONG counterparts; recall that with WEAK IA-IA restrictions the person valuation in situ needs to fail first, as shown in (100), in order for the two pronouns to move to Spec*v*P to be valued. Since that movement is subject to Shortest Move, the order of the two objects is preserved with the movement to Spec*v*P: the valuation of the Theme, which moves to Spec*v*P, takes place first, as shown in (101), potentially bleeding the valuation of the Goal.

(100)



(101)



While WEAK EA-IA restrictions arise in a different valuation configuration, the reverse version of the restriction can also be derived through scrambling, as I will show next.

Let us remind ourselves of what a canonical/reverse alternation looks like with a WEAK EA-IA restriction. This is for example, a pattern pervasive in Algonquian languages; the basic pattern is illustrated in (102)–(103) with examples from Nishnaabemwin (a variety of Ojibwe; Valentine 2001). When the EA marker precedes the IA marker, we get a canonical WEAK EA-IA restriction: the EA and IA can be 1/2P and 3P respectively, as in (102a), both markers can be 1/2P, as in (102b),[14] but the EA cannot be 3P when the IA is 1/2P, as illustrated in (102c). The reverse pattern is observed, on the other hand, when the IA marker

14. As in most Algonquian languages, there is also an asymmetry between 1P and 2P person markers, which will be discussed in Section 4.3 in the context of other related 1P vs. 2P asymmetries.

precedes the EA marker: the IA and EA can be 1/2P and 3P respectively (cf. (103a)), both markers can be 1/2P (cf. (103b)), but the IA cannot be 3P when the EA is 1/2P (cf. (103c)).

(102) Nishnaabemwin/Ojibwe [*Algonquian*]:

a. **n-** waabm -aa **-g** — 1.EA ≫ 3.IA
1- see -EA>3 -3PL
'**I** see **them**.'

b. **g-** waabm **-imin** — 2.EA ≫ 1.IA
2- see -1PL
'**You** see **us**.'

c. ***w-** waabm -igoo (**-waa**) **-???** — ✗ 3.EA ≫ 1/2.IA
3- see -INV.3>IA -PL -1/2
'**They** see **me/you**.'

(103) a. **n-** waabm -igoo **-g** — 1.IA ≫ 3.EA
1- see -INV.3>IA -3PL
'**They** see **me**.'

b. **g-** waabm -in **-imin** — 2.IA ≫ 1.EA
2- see -INV.1>2 -1PL
'**We** see **you**.'

c. ***w-** waabm -aa (**-waa**) **-???** — ✗ 3.IA ≫ 1/2.EA
3- see -EA>3 -PL -1/2
'**We** see **him**.' (Valentine 2001, 274–5, 287)

In addition to the reordering of the pronoun markers, the person combination of the two markers also conditions the appearance of direct or inverse marking on the verb (underlined). I will not discuss the derivation of the direct/inverse marking in detail, but I essentially assume that inverse marking is a reflex of the marker reordering (see e.g. Bruening 2001; Oxford 2019). In fact, the proposal that the reverse restriction pattern in Algonquian is the result of A-movement is also not new (see Oxford 2014 for discussion and references). The A-movement properties of the marker reordering can be observed, just as in Slovenian, when we look at the behavior of the corresponding nominal phrases in terms of binding possibilities; the main difference is that in Algonquian, the minimal pronouns can double (i.e. co-occur with) the nominal phrases in question.

In Algonquian, the alternations with pronoun marker order are also observed with exclusively 3P arguments. In that case, the linearly first marker indexes the more discourse salient argument regardless of whether that corresponds to the EA or the IA; the grammatical role of the two arguments can be read off the direct/inverse marking on the verb, just as with mixed 1/2P and 3P argument combinations. Crucially, in such cases, whether or not one argument can bind into the other is not determined by their respective grammatical roles as subject and object, but based on which one corresponds to the pre-verbal marker and which to the post-verbal marker. This is illustrated with the Nishnaabemwin examples in (104)–(105) (but can be replicated in other Algonquian languages as well; see Oxford 2014 for references and discussion). When the EA is discourse salient and indexed on the verb by the pre-verbal marker, it can bind into the IA, as in (104a), but the IA may not bind into the EA, as shown in (104b). However, when the IA is indexed by the pre-verbal marker, as revealed by the inverse marking on the verb, the situation is reversed: the IA can then bind into the EA, as in (105a), but the EA may not bind into the IA, as shown in (104b).

(104) a. John **w-** gii-waabm <u>-aa</u> **-n** w-gwis-an $\boxed{3.\text{EA} \gg 3'.\text{IA}}$
John 3- PAST-see -EA>3 -3.OBV 3-son.OBV
'John$_i$ saw his$_i$ son.'

b. *w-gwis-an **w-** gii-waabm <u>-aa</u> **-n** John-an $\boxed{3.\text{EA} \gg 3'.\text{IA}}$
3-son-OBV 3- PAST-see -EA>3 -3.OBV John-OBV
'His$_{*i}$ son saw John$_i$.'

(105) a. John **w-** gii-waabm <u>-igoo</u> **-n** w-gwis-an $\boxed{3.\text{IA} \gg 3'.\text{EA}}$
John 3- PAST-see -INV.3>IA -3.OBV 3-son.OBV
'His$_i$ son saw John$_i$.' (Valentine 2001, 632–3)

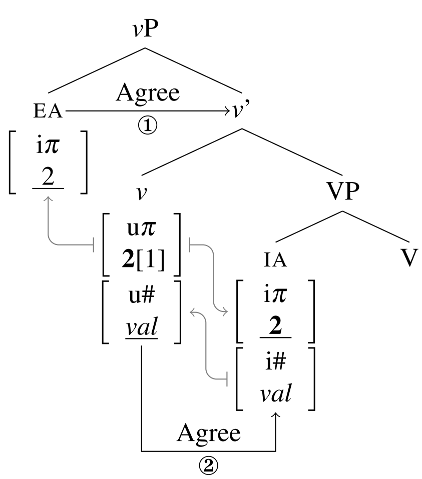
b. **w-** gii-waabm <u>-igoo</u> **-n** w-gwis-an $\boxed{3.\text{IA} \gg 3'.\text{EA}}$
3- PAST-see -INV.3>IA -3.OBV 3-son.OBV
'His$_i$ son sees him$_i$.'
*'He$_i$ sees his$_i$ son.' (Rhodes 1994, 3)

What we see here is the same hallmark of A-movement that we saw in Slovenian and Japanese above. Following Oxford (2019), I assume that when the pre-verbal marker is the

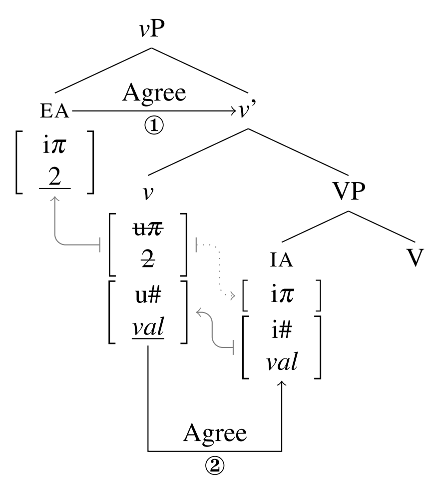
IA this is the result of A-movement of the IA to Spec*v*P over the EA base generated there; I depart from him in making the movement explicitly scrambling, adopting the analysis proposed above, and in the mechanics of how the WEAK restriction arises.

Recall that a canonical WEAK EA-IA restriction arises when an EA minimal pronoun is base generated in Spec*v*P and *v* is the phase head. The restriction arises because of the timing of the person valuation of the EA in Spec*v*P in relation to the valuation of the IA via Agree with *v*. Because of the Theta-Probe Condition, *v* cannot probe (despite its active number probe) until it has assigned the EA a theta-role. This means that the EA minimal pronoun, which can Agree with *v* and be person valued immediately at Merge, will always be person valued before the IA, which is only valued when *v* can finally Agree with is. As shown in (106)–(107), there are two options in such cases when the *v* has a [uπ] specified with participant/speaker features. The first is that the valuation of the EA can leave the valued [uπ] on *v* intact, allowing for the IA to also be valued, as in (106); this option results in both arguments being valued as 1/2P. The second option is that the valuation of the IA can immediately delete the valued [uπ] on *v*, which leaves no valued features for the IA, as in (107). The latter is the only option when it comes to pronoun combinations with a 1/2P and a 3P pronouns, which is why combinations of a 3P EA and a 1/2P IA are impossible.

(106)

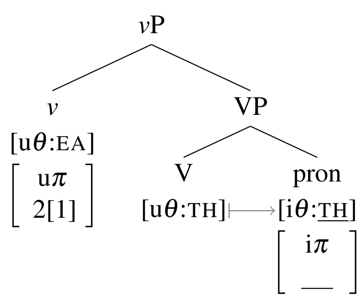


(107)

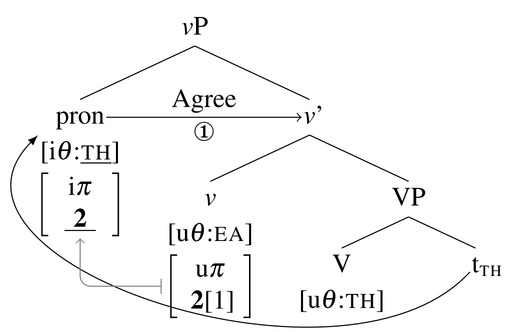


The difference with the reverse version of the derivation is that the IA minimal pronoun can merge with *v*P before the EA minimal pronoun and consequently be person valued first, potentially bleeding the person valuation of the EA. The first step of such a derivation involves a minimal pronoun merging with the verb and the verb assigning the pronoun a Theme theta-role, as in (108a). At this point, even when *v* merges next, no φ-valuation can take place even if *v* has an active probe, due to the Theta-Probe Condition. However, as soon as *v* is merged, the Theme pronoun can undergo A-scrambling to Spec*v*P, as in (108b). Since the Theme pronoun has an unvalued [iπ] feature, it can probe at Merge and Agree with the valued [uπ] on *v*, resulting in person valuation.[15]

(108) a.



b.

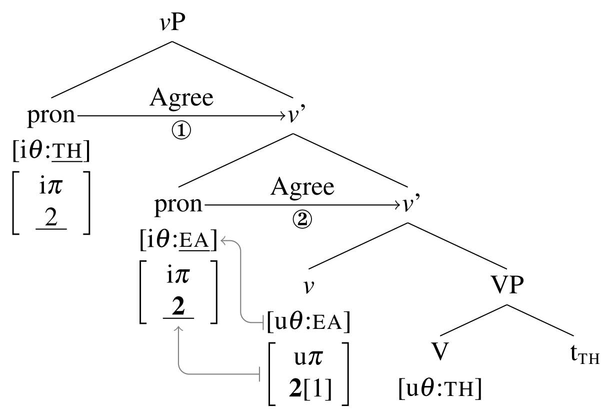


After this step in the derivation, another minimal pronoun is merged with *v*P tucking in under the Theme pronoun as the lower Spec*v*P. At this point the pronoun can be person valued and receive the EA theta-role from *v*. However, since the Theme pronoun has been valued

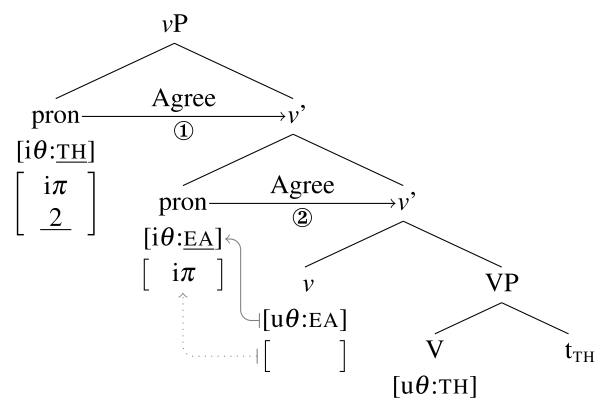
15. Note here in relation to the order of feature valuation and checking ($\theta \succ \mathrm{A} \succ \bar{\mathrm{A}}$) that was used in Section 4.1.2 to provide a possible derivation of the Theta-Probe Condition, that the ordering does not prevent *v* from being a goal for Agree initiated by other elements before it can assign a theta-role. It only prevents *v* from initiating Agree itself; i.e. it prevents it from probing.

by *v* prior to this, we get the same two options for the person valuation of the EA as we got with the person valuation of the Theme in the canonical version of the derivation. If the valuation of the Theme pronoun left the [uπ] on *v* intact, both pronouns can be valued 1/2P, as in (109a). Alternatively, if the valuation of the Theme previously caused the immediate deletion of the [uπ] feature, the EA can only receive a default 3P value, as in (109b).

(109) a.



b.



Note that, due to the Theta-Probe Condition, a *v* with an active number probe (omitted in the derivation) can only probe after the EA has been assigned its theta-role, but in this particular derivation there are also no more matching goals in *v*'s c-command domain. This is one way in which the reverse derivation differs from the canonical one in addition to the argument reordering: due to its timing in relation to *v*'s probing, the movement of the IA bleeds Agree with *v*. This additional asymmetry could be tied to the direct/inverse alternation. Oxford (2014, 2019) argues that the direct/inverse morpheme is in fact the expression of agreement on *v*. Furthermore, he proposes that the reordering of the arguments creates a context for morphological φ-feature impoverishment, yielding a different realization of *v* when the IA is in the specifier above the EA. In current terms, Oxford's proposal can be reinterpreted as syntactic bleeding of agreement rather than post-syntactic impoverishment, although the exact details of the analysis still have to be worked out. Regardless of whether the proposed analysis can also derive direct/inverse morphology, the main result is that the A-scrambling analysis can be applied to WEAK EA-IA restrictions. This means the overarching analysis extends to all person restriction patterns regardless of strength or affected arguments.

More generally, the analysis does more than just derive all the attested reverse restriction patterns and deduce the *Direction Generalization*. Similarly to the deduction of the *Strength Generalization* in Section 4.1, the existence of a typological gap in person restriction patterns under discussion in this section can also be used to make a much broader point about syntax. Namely, the lack of languages with only reverse versions of person restrictions can be used as evidence for some version of UTAH (Baker 1988, 1997).

The need for a universal hierarchy of base argument positions comes from the fact that, when languages only have one type of person restriction in terms of its direction, it arises with an EA $\gg$ Goal $\gg$ Theme argument structure. If there were any other crosslinguistic options for base argument positions, we would expect languages with only reverse restrictions to exist. The fact that reverse restrictions only exist as one part of the paradigm—coexisting with their canonical counterpart—is evidence that what drives them must be an operation that is fundamentally optional. I have argued that the optionality comes from the operation being a particular kind of A-movement, namely A-scrambling, which is crucially not driven by Agree (i.e. the valuation and checking of syntactic features). Therefore, movement driven by Agree exists, Agree independent of movement exists, but movement independent of Agree also exists, which means that the operations of Agree and movement cannot be reduced to being one and the same operation or one being purely the reflex of the other.

4.2.4 A-scrambling interacting with order types of reordering

Before we conclude the discussion of reverse person restrictions, let us briefly consider the implications of the analysis proposed in this section in relation to some of the more complex interactions between pronoun order and person restrictions. We briefly looked at a few such cases in Section 2.5.3, where I used them to argue that a syntactic analysis of canonical + reverse restriction patterns is needed. What I would like to show here is the basics of what such an analysis of these patterns would look like. These are not meant to be fully fledged accounts of the cases in question, but illustrations of how the analysis proposed in this section can be extended to cases which are more complex than the ones discussed above. In addition, these case studies I look at in this section will further confirm the validity of the *Direction Generalization*, in particular with respect to the conclusion made in the last section that only scrambling may yield reverse person restrictions.

The first case study involves Haya, with its canonical + reverse WEAK IA-IA restriction pattern. Recall that what is striking about it is that it is a mirror image of the Slovenian pattern: when the order of the object markers is Theme » Goal, the person restriction is a canonical one—applying to the Theme, whereas when the order of object markers is Goal » Theme, the person restriction is a reverse one—applying to the Goal. On the surface, this results in the basic restriction pattern illustrated in (110): linearly second object marker can be 1/2P, regardless of whether it is a Goal or a Theme marker, as in (110a,b), while the linearly first object marker can be 1/2P, but only when the other marker is also 1/2P, again regardless of whether the former is a Goal or a Theme marker, as in (110c).

(110) Haya [*Bantu*]:

a. a-ka- **mu-** { **n-** / **ku-** } deetela. $\boxed{\text{1/2.GL/TH} \gg \text{3.TH/GL}}$
3.SU-PAST- 3.O- { 1.O- / 2.O } bring.APPL
'He brought **him** to **me/you**.'
'He brought **me/you** to **him**.'

b. a-ka- **ku- n-** deetela. $\boxed{\text{1.GL/TH} \gg \text{2.TH/GL}}$
3.SU-PAST- 2.O- 1.O- bring.APPL
'He brought **you** to **me**.'
'He brought **me** to **you**.'

c. *a-ka- { **n-** / **ku-** } **mu-** leetela. $\boxed{\text{✗ 3.GL/TH} \gg \text{1/2.TH/GL}}$
3.SU-PAST- { 1.O- / 2.O- } 3.O- bring.APPL
'He brought **me/you** to **him**.'
'He brought **him** to **me/you**.' (Duranti 1979, 40)

I argue that this pattern also involves middle-field scrambling of the Theme marker over the Goal marker below *v*. What makes the Haya pattern different from the Slovenian one, where the surface order of clitics transparently reflects their configuration at the point of person valuation, is that it also involves obligatory reordering of markers after person valuation.

Evidence for this second step of reordering comes from a comparison with the related language Sambaa, which only has a canonical WEAK IA-IA restriction, illustrated in (111), with the same surface sequences of object markers we see in Haya. The difference is that in Sambaa, the first marker always corresponds to the Theme and the second to the Goal, where the Theme is the marker to which the person restriction applies to, as seen in (111c).

(111) Sambaa [*Bantu*]:

a. a-za- **m-** { **ni-** / **ku-** } onyesha $\boxed{\text{1/2/3.GL} \gg \text{3.TH}}$
3.SU-PERF- 3.DO- { 1.IO- / 2.IO- } show
'He pointed **her** out to **me/you**.'

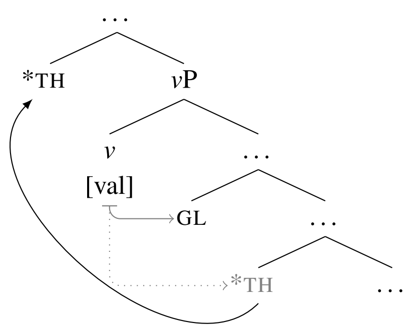
b. a-za- **ku- ni-** onyesha $\boxed{\text{1.GL} \gg \text{2.TH}}$
3.SU-PERF- 2.DO- 1.IO- show
'He pointed **you** out to **me**.'

c. *a-za- { **ni-** / **ku-** } **mu-** onyesha $\boxed{\text{✗ 3.GL} \gg \text{1/2.TH}}$
3.SU-PERF- { 1.DO- / 2.DO- } 3.IO- show
'He pointed **me/you** out to **her**.' (Riedel 2009, 76, 140)

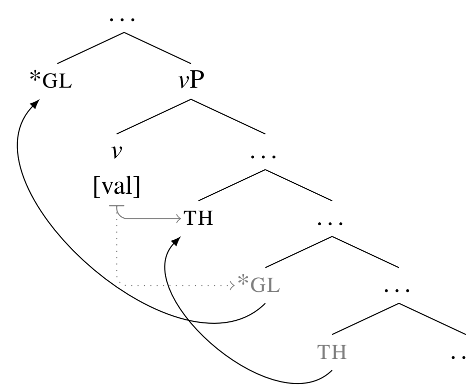
The difference between the two languages can be attributed to just the presence or absence of scrambling below *v*. The canonical restriction in both languages arises when the Goal asymmetrically c-commands the Theme under *v*, which yields a person restriction on the

Theme. Subsequently, the two object markers are reordered, as shown in (112). The option available only in Haya, however, is for A-scrambling to apply below *v*, yielding a reverse restriction pattern due to the Theme asymmetrically c-commanding the Goal under *v*. The same reordering from before can then occur after this, as shown in (113); this reordering makes it seem as if the restriction applies to the linearly first marker, even though the restriction arises before the reordering, when the Theme intervenes between the Goal and *v*.

(112) Canonical (Haya & Sambaa):



(113) Reverse (Haya only):



The landing site or the nature of the second movement is not crucial for deriving this pattern as long as the movement occurs after the person valuation of the two minimal pronouns. In fact, the second reordering can in principle involve low level PF reordering, as that would automatically and straightforwardly place it after person valuation. I leave open what the best analysis of the late reordering is for Haya and Sambaa, pending further evidence that would help determine whether the surface order is the result of syntactic or PF operations.

The derivation of the Haya restriction pattern involves optional movement, A-scrambling, feeding a second obligatory reordering of pronouns. But there is no reason why the reordering of pronouns after person valuation has to be obligatory. We will see next that there are surface patterns of double object deficient pronouns that appear quite random at first glance, but which turn out to be quite unremarkable when we disentangle them as cases where an optional reordering applies after person valuation.

The first example is the complex person restriction pattern of Swiss German. Recall from Section 2.5.3 that on the surface the Swiss German pattern appears as if the IA-IA person restriction occurs only with one of the two available object clitic orders: the Goal » Theme clitic order gives rise to a canonical IA-IA restriction, while the Theme » Goal clitic order appears to not be person restricted at all. We saw that this was a problematic case for non-syntactic accounts. However, the pattern starts making much more sense from the perspective of two different types of clitic reordering interacting with each other. For example, the person combination in (114a) is grammatical no matter what the order of clitics is, which I argue is due to an optional clitic reordering after person valuation. In contrast, the two examples in (115) illustrate a case where a person combination that is ungrammatical with the Goal » Theme clitic order (see (115a)) is grammatical with the Theme » Goal clitic order (see (115b)), which I argue is due to the latter being derived via scrambling under *v*, just as in the derivations of the reverse IA-IA restriction pattern.

(114) Swiss German [*Germanic*]:

a. D' Maria zeigt **mir** **en**. $\boxed{\text{1.GL} \gg \text{3.TH}}$
the Maria shows 1.DAT 3.M.ACC

b. D' Maria zeigt **en** **mir**.
the Maria shows 3.M.ACC 1.DAT
'Maria shows **him** to **me**.'

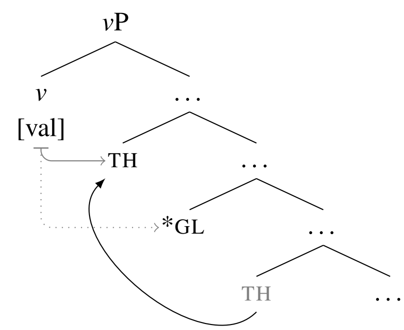
(115) a. *D' Maria zeigt **em** **mich**. $\boxed{\text{✗ 3.GL} \gg \text{1.TH}}$
the Maria shows 3.M.DAT 1.ACC

b. D' Maria zeigt **mi** **em**. $\boxed{\text{1.TH} \gg \text{3.GL}}$
the Maria shows 1.ACC 3.M.DAT
'Maria shows **me** to **him**.' (Bonet 1991, 188n12))

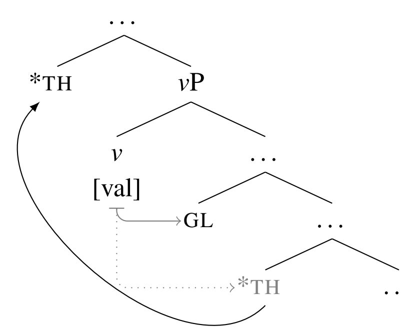
The two different types of reordering can actually be derived by maintaining a scrambling analysis for both. The reason why the optional reordering appears to either effect or not the person restriction pattern is because the Theme clitic can either undergo scrambling to a *v*P internal position or a *v*P external position (which could correspond to AgrOP of

Chomsky 1995, meant to be above the base position of the EA). This is entirely compatible with the analysis of scrambling proposed in the last section. Since scrambling is not feature driven, it can in principle target any position. There are thus two ways of getting a surface Theme » Goal clitic order: one involving movement below *v*, which feeds person valuation of the Theme, as shown in (116), and one involving movement above *v*, which takes place after the person valuation of the Goal minimal pronoun, as shown in (117).

(116) Reverse (Swiss & Zürich):



(117) Canonical (Swiss only):



The appearance of having no person restrictions with the Theme » Goal order is therefore because (116) yields a configuration where the Theme is not person restricted and (117) yields a structural configuration where the Goal is not person restricted. The Goal » Theme clitic order on the other hand always gives rise to a canonical person restriction because it always transparently reflects the base order of the Goal and Theme clitics below *v*. The reason why it is impossible to have a derivation that would yield a Goal » Theme clitic order but no person restriction is because this would require the Theme to first undergo scrambling over the Goal below *v* (allowing the Theme to be person valued by *v*) followed by the Goal scrambling over the Theme above *v*. Such a derivation is impossible because it would constitute a case of string-vacuous scrambling, which is universally prohibited (see Hoji 1985). In other words, not only *can* scrambling result in the reordering of arguments, it *must* result in the reordering of arguments.

Additionally, this analysis can also straightforwardly derive the variation in restriction patterns within Swiss German, where some speakers, specifically speakers of Zürich German (Werner 1999), have the same person restriction pattern as Slovenian, where the clitic order can either be Goal » Theme or Theme » Goal, with a consistent person restriction on the second clitic in the sequence. This is shown in (118)–(119).

(118) Zürich German [*Germanic*]:

a. … das d' **mer** **en** halt morn bringsch.
that 2.NOM 1.DAT 3.M.ACC just tomorrow bring.2 $\boxed{1.\text{GL} \gg 3.\text{TH}}$

b. *… das d' **en** **mer** halt morn bringsch.
that 2.NOM 3.M.ACC 1.DAT just tomorrow bring.2 $\boxed{✗\,3.\text{TH} \gg 1.\text{GL}}$
'… that you just bring **him** to **me** tomorrow then.'

(119) a. De Max hät **mi** **em** voorgschtellt.
the Max has 1.ACC 3.M.DAT introduced.3 $\boxed{1.\text{TH} \gg 3.\text{GL}}$

b. ??De Max hät **em** **mi** voorgschtellt.
the Max has 3.M.DAT 1.ACC introduced.3 $\boxed{✗\,3.\text{GL} \gg 1.\text{TH}}$
'Max has introduced **me** to **him**.' (Werner 1999, 81)

The difference between the two varieties amounts to the availability of a landing site for scrambling above *v*. When it is available, the person restriction will not be observed with the Theme » Goal clitic order, because the Theme will have the option of moving over the Goal after the Goal has been person valued. Conversely, if scrambling is only possible to positions below *v*, the Theme » Goal clitic order will yield a reverse person restriction, because the Theme will only be able to move over the Goal by landing in a position below *v*.

With the Swiss German case it is essential that the two reordering options do not feed each other, which is impossible because that would constitute string-vacuous scrambling. That is the reason why the person restriction is neutralized only with one clitic order. If however, a language would somehow allow the possibility of optional reordering both before and after person valuation, where one would always feed the other, the prediction would be the complete absence of person restrictions. That is because having either a

Theme ≫ Goal or a Goal ≫ Theme configuration under *v*, due to middle-field scrambling, could be completely masked on the surface if an additional optional movement could apply to the output of the first one, but after person valuation. This is in fact a pattern that we do find attested. I have argued in Stegovec 2019 that we observe exactly this kind of movement constellation in Slovenian, but only in a very specific context: in matrix imperatives, if the verb precedes the clitic cluster. As shown in (120)–(121), in that case, there is no person restriction regardless of the surface clitic order.

(120) <u>Slovenian [*Slavic*]:</u>

a. Predstavi **mi** **ga**! $\boxed{\text{1.GL} \gg \text{3.TH}}$
introduce.IMP 1.DAT 3.M.ACC

b. Predstavi **ga** **mi**!
introduce.IMP 3.M.ACC 1.DAT
'Introduce **him** to **me**!'

(121) a. Predstavi **me** **mu**! $\boxed{\text{1.TH} \gg \text{3.GL}}$
introduce.IMP 1.ACC 3.M.DAT

b. Predstavi **mu** **me**!
introduce.IMP 3.M.DAT 1.ACC
'Introduce **me** to **him**!' (Stegovec 2019, 33)

I propose in Stegovec 2019 that these examples show that there are two instances of optional clitic reordering operating in Slovenian: middle-field scrambling and what I will call the *PF-switch*, which can take place at PF when the imperative verb forces the pronunciation of the lower copies of the two clitics (see Stegovec 2019 for details, but also Bošković 2004b and the discussion below regarding the same phenomenon in Greek).

Because the PF-switch occurs at PF, and thus after person valuation, it cannot interact with person restrictions. However, since Slovenian also allows the option of reordering the clitics prior to person valuation via scrambling, the two reordering options can neutralize each other on the surface, giving the appearance of no person restrictions. The reason why one optional reordering can feed the other in this case and not in Swiss German is because the reordering after person valuation in Slovenian does not involve scrambling, so the ban on string-vacuous scrambling does not apply.

The PF-switch is independently needed for cases where it does not interact with other instances of reordering. For example, in Greek, post-verbal double object clitics in imperatives and gerunds can optionally reorder (Warburton 1977; Joseph and Philippaki-Warburton 1987; Terzi 1999; Bošković 2004b). Unlike Slovenian, Greek does not allow middle-field scrambling with clitics, which is why they otherwise have a rigid Goal » Theme order. Because of this, the PF-switch does not make the canonical restriction go away, as evidenced in (122) by the ungrammaticality of the examples with a 3P Goal and 1P Theme clitic combination regardless of the surface clitic order.[16]

(122) Greek [*Hellenic*]:

a. *Danise **tu** **me**! ✗3.GL ≫ 1.TH
lend.IMP 3.M.DAT 1.ACC

b. ??Danise **me** **tu**!
lend.IMP 1.ACC 3.M.DAT
'Lend **me** to **him**!' (Mavrogiorgos 2010, 262)

What is most striking, considering all the complex interactions of pronoun reordering we saw above, is the fact that not all logically possible combinations are attested. Consider the summary of the logically possible combinations of pronoun reordering in Table 4.1, given the two parameters of variation: (i) what is the timing of the movement (where 'above *v*' also encompasses the PF-switch), and (ii) is the movement obligatory or optional.

The only unattested cases are those excluded by the *Direction Generalization*; that is, the restriction patterns that would require obligatory reordering prior to the person valuation of the pronouns. This lends further support to the idea that scrambling is distinct from all

16. '??' indicates, as Mavrogiorgos (2010) reports, that some speakers of Greek feel the restriction is weaker with the Theme » Goal order, depending also on the choice of verb and whether the clitics are case-syncretic (already noted by Bonet 1991, attributing the observation to Sabine Iatridou). Given the influence of verb choice and surface morphological traits, it seems likely that the effect is independent from the syntactic processes that cause person restrictions.

Table 4.1: Possible options for pronoun reordering and IA-IA person restrictions

reordering below v	reordering above v	predicted restriction	example
*	*	canonical	Greek
*	optional	canonical	Greek IMP
*	obligatory	canonical	Sambaa
optional	*	canonical + reverse	Zürich German, Slovenian
optional	optional	no restriction	Swiss German, Slovenian IMP
optional	obligatory	canonical + reverse	Haya
obligatory	*	reverse	**unattested**
obligatory	optional	reverse	**unattested**
obligatory	obligatory	reverse	**unattested**

other types of movement in its ability to take place as soon as possible—a consequence of it not being dependent on Agree. All obligatory movement seem to be limited to occurring much later in the derivation, suggesting that the dependency of grammatical operations on different types of features or their independence from them might be a key factor in determining the timing of when they can take place (see also Sections 4.1.2 and 4.2.2).

4.3 Fine-grained person restrictions

So far, all the person restrictions were categorized as either STRONG or WEAK in terms of their strength; i.e. how many person combinations are allowed between two or more pronouns. However, there is actually further more fine-grained variation within the restrictions I call WEAK. This has been previously acknowledged explicitly by, among others, Nevins (2007, 2011) and Pancheva and Zubizarreta (2018), although they consider fewer patterns than I will look at in this section. In addition to the STRONG pattern and the plain WEAK pattern I have discussed so far, my crosslinguistic survey has revealed 6 additional WEAK patterns; where "WEAK" covers all patterns where the restricted pronoun is not always restricted to 3P. The total 8 restriction patterns are summarized in Table 4.2.[17] Note that this encompasses EA-IA and IA-IA restrictions as well as canonical and reverse restrictions.

Table 4.2: Attested person restriction patterns (shaded = ungrammatical)

STRONG	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2 (shaded)	**2** $\gg$ 1 (shaded)	**3** $\gg$ 2 (shaded)	**3** $\gg$ 1 (shaded)
WEAK (plain)	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2	**2** $\gg$ 1	**3** $\gg$ 2 (shaded)	**3** $\gg$ 1 (shaded)
WEAK *$\gg$ 1	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2	**2** $\gg$ 1 (shaded)	**3** $\gg$ 2 (shaded)	**3** $\gg$ 1 (shaded)
WEAK *$\gg$ 2	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2 (shaded)	**2** $\gg$ 1	**3** $\gg$ 2 (shaded)	**3** $\gg$ 1 (shaded)
ME-FIRST	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2	**2** $\gg$ 1 (shaded)	**3** $\gg$ 2	**3** $\gg$ 1 (shaded)
YOU-FIRST	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2 (shaded)	**2** $\gg$ 1	**3** $\gg$ 2 (shaded)	**3** $\gg$ 1
*2 $\gg$ 1	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2	**2** $\gg$ 1 (shaded)	**3** $\gg$ 2	**3** $\gg$ 1
*3 $\gg$ 2	**1** $\gg$ 3	**2** $\gg$ 3	**3** $\gg$ 3	**1** $\gg$ 2	**2** $\gg$ 1	**3** $\gg$ 2 (shaded)	**3** $\gg$ 1

The additional 6 patterns can be categorized into two groups: (i) WEAK restrictions with an additional 1P vs. 2P asymmetry (abbreviated as WEAK *$\gg$ 1 and WEAK *$\gg$ 2), and (ii) restrictions weaker than the plain WEAK restriction—allowing for more grammatical combinations, but still singling out either 1P or 2P pronouns specifically (see the bottom four rows of the table). Importantly, some of these restriction patterns have been argued to not exist; for example, Nevins (2007) argued for the existence of WEAK *$\gg$ 1 and ME-FIRST restrictions (the term comes from him), but argued against the existence of YOU-FIRST and other 2P-based restrictions. Additionally, the two weakest patterns, *2 $\gg$ 1 and *3 $\gg$ 2, have also not been recognized in previous crosslinguistic surveys of person restrictions.

In this section, I will first take a closer look at the additional 6 restriction patterns, showing that they all still behave like syntactic restrictions. I then proceed to show that all the additional patterns can be derived in the current system and that the additional person

17. I do not include here the restrictions between 3P pronouns that sometimes exist alongside the person restriction; for example, the *spurious se* effect in Spanish (Perlmutter 1971) and comparable effects found in Catalan (Bonet 1991). These tend to be either morphological or arise due to features other than person on the two pronouns (see discussion in Chapter 5). Additionally, their presence seems to be completely independent from the strength of the accompanying person restrictions (see also Pancheva and Zubizarreta 2018, 1310).

valuation possibilities that allow more fine-grained differences between restriction patterns are still within the limits of the *Strength Generalization* and the *Direction Generalization*.

4.3.1 A closer examination of the restriction patterns

The most common out of the more fine-grained WEAK restriction patterns are the WEAK restrictions with an additional 1P vs. 2P asymmetry. This includes WEAK *≫ 1, where on top of the baseline WEAK restriction, the lower pronoun can never be 1P, and WEAK *≫ 2, where on top of the baseline WEAK restriction, the lower pronoun can never be 2P. Let us start by first looking at the more common case, namely the WEAK *≫ 1 restriction.[18]

Different versions of the WEAK *≫ 1 pattern are attested, for example, in Modern Standard Arabic [*Semitic*] (Fassi-Fehri 1988, 1993) in the IA-IA domain, as well as Hungarian [*Finno-Ugric*] (É. Kiss 2013, 2017; Bárány 2015b, 2015a, 2017) and Sahaptin [*Penutian*] (Rude 1994) in the EA-IA domain. In fact, Classical Arabic, which I have been using as an example of a WEAK IA-IA restriction above, has the same restriction pattern as Modern Standard Arabic when we look at the full paradigm. As shown in (123), the baseline pattern is a WEAK restriction, with the possibility of both pronouns being 1/2P, as in (123b). However, in such combinations the Theme can never be 2P, as illustrated by (123c).

(123) Classical Arabic [*Semitic*]: **WEAK *≫ 1 IA-IA**

a. ʔaʕtˀaː { **-niː** / **-ka** / **-haː** } **-hʊ** 1/2/3.GL ≫ 3.TH
gave.3 { -1.IO / -2.IO / -3.F.IO } -3.M.DO
'He gave **him/it** to **me/you/her**.'

b. ʔaʕtˀaː **-niː** **-ka** 1.GL ≫ 2.TH
gave.3 -1.IO -2.DO
'He gave **you** to **me**.'

18. Nevins (2007) calls these *ultrastrong* restrictions, which is a bit misleading since they are technically weaker than STRONG restrictions. Additionally, I want to highlight the parallels with both WEAK restrictions and what I call WEAK *≫ 2 restrictions. Hence, the less catchy term.

c. *ʔaʕtˀaː **-ka** **-niː** ✗2.GL ≫ 1.TH
gave.3 -2.IO -1.DO
'He gave **me** to **you**.'

d. *ʔaʕtˀaː **-huː** { **-niː** / **-ka** } ✗3.GL ≫ 1/2.TH
gave.3 -3.M.IO { -1.DO / -2.DO }
'He gave **me/you** to **him**.' (Walkow 2014, 139–40)

The WEAK *≫ 1 restriction pattern is also found in Alutor, although it is occurs there in both the EA-IA and the IA-IA domain (Savvina and Mel'čuk 1978; Mel'čuk 1988; Kibrik 2002). Let us first look at the EA-IA pattern. The baseline transitive indexing paradigm can be used in Alutor for any person of the subject if the object is 3P, as shown in (124a) with a 3P subject and in (124b) with a 1/2P subject. The same paradigm is also used if the subject is 1P and the object is 2P, as in (124c), showing that we are dealing with a WEAK restriction. It is, however, impossible to index arguments this way with *3 ≫ 1/2 configurations, as shown in (125a), but also with the *2 ≫ 1 configuration, as shown in (125b).

(124) Alutor [*Chukotkan*]: **WEAK *≫ 1 EA-IA**

a. ta-ktəpl-əŋ **-nin** 3.EA ≫ 3.IA
POT-hit-PERF -3>3.O
'**He/She** probably hit **him/her**.'

b. { **tə-** / **Ø-** } ta-ktəplə-ŋə **-n** 1/2.EA ≫ 3.IA
{ 1.SU- / 2.SU- } POT-hit-PERF -3.O
'**I/You** probably hit **him/her**.'

c. **tə-** ta-ktəplə **-ɣət** 1.EA ≫ 2.IA
1.SU- POT-hit.PERF -2.O
'**I** probably hit **you**.'

(125) a. ***Ø-** ta-ktəplə { **-ɣəm** / **-ɣət** } ✗3.EA ≫ 1/2.IA
3.SU- POT-hit.PERF { -1.O / -2.O }
'**He/She** probably hit **me/you**.'

b. ***Ø-** ta-ktəplə **-ɣəm** ✗2.EA ≫ 1.IA
2.SU- POT-hit.PERF -1.O
'**You** probably hit **me**.' (Kibrik 2002, 199–202)

In the cases where the baseline indexing paradigm cannot be used due to the person restriction, an alternative paradigm has to be used. For some combinations that is the spurious antipassive, as shown in (126a) (see Section 4.1.5 for details), where only the EA is indexed. In other cases, like (126b), the alternative paradigm that must be used is what has been termed the "weak inverse" by Comrie (1979b, 1980), where only the IA is indexed on the verb together with the *na-* morpheme (more on this paradigm below).

(126) a. t-<u>ina</u>-ktəpl(ə) { **-ŋ** / **-əŋ** } [2/3.EA ≫ (AP.1.IA)]
POT-AP(1.O)-hit.PERF { -2.SU / -3.SU }
'**You/He/She** probably hit **me**.'

b. <u>na</u>-ta-ktəplə **-ɣət** [(3.EA) ≫ 2.IA]
INV(3.SU)-POT-hit.PERF -2.O
'**He/She** probably hit **you**.' (Kibrik 2002, 199–202)

In Alutor double object constructions, we find that the same person combinations of the Goal and Theme are grammatical and ungrammatical (I only present the canonical IA-IA restriction here to keep things simple; see Section 2.3.3 regarding the reverse pattern in Alutor). For example, the Goal can be 3P, as in (127), or 1P, as in (128a), or 2P, as in (128b),[19] as long as the Theme is 3P. Note that because of the EA-IA restriction, the person-based indexing has to be done via one of the alternative paradigms (see Section 4.1.5 for details). Additionally, the Goal can be 1P and the Theme 2P, as in (128c).

(127) <u>Alutor [*Chukotkan*]:</u> [**WEAK IA-IA**]
əlləɣ-a jəl **-nina-wwi** ənək-əŋ *şininkina-wwi ŋavakka-wwi* [(3.GL) ≫ 3.TH]
father-ERG give -3>3.DO-PL him-DAT his-PL daughter-PL
'**He** [father] gave **them** [his daughters] to him (as a wife).'

(128) a. əlləɣ-a <u>ina</u>-jəl **-i** *ɣəmək-əŋ* şininkin ŋavakək [1.GL ≫ (3.TH)]
father-ERG AP(1.IO)-give -3.SU me-DAT his.ABS daughter.ABS
'**He** [father] gave his daughter to **me** (as a wife).'

b. əlləɣ-a [<u>ne</u>-]jəl **-ɣət** *ɣənək-əŋ* şininkin ŋavakək [2.GL ≫ (3.TH)]
father-ERG INV(3.SU)-give -2.IO you-DAT his.ABS daughter.ABS
'**He** [father] gave his daughter to **you** (as a wife).'

19. The prefix *ne-* is added to the example although is not reported by Mel'čuk (1988, 295). As noted by Bobaljik and Wurmbrand (2002), the available descriptions for the language indicate that the verb should have the prefix in such cases. In a personal communication to Bobaljik and Wurmbrand, I. A. Mel'čuk (11/1998) confirms that the absence of the prefix in this example is anomalous among the data he has collected. An equivalent example with the expected prefix is actually also provided by Mel'čuk (1973, 26). For this reason, I present all the relevant Alutor data with the "non-anomalous" paradigm, but put the prefix in square brackets.

c. əlləɣ-a ina-jəl -i *ɣəmək-əŋ* ɣəttə $\boxed{\text{1.GL} \gg \text{(2.TH)}}$
father-ERG AP(1.IO)-give -3.SU me-DAT you.ABS
'**He** [father] gave you to **me** (as a wife).' (Mel'čuk 1988, 294–5)

The ungrammatical person combinations are the same as in the EA-IA domain: the *3 ≫ 1/2, combinations, shown in (129a), and the *2 ≫ 1 combination, shown in (129b).

(129) a. *əlləɣ-a **Ø-** jəl **-nin** *ənək-əŋ* { ɣəmmə / ɣəttə } $\boxed{\text{✗3.GL} \gg \text{(1/2.TH)}}$
father-ERG 3.SU- give -3>3.IO him-DAT { me.ABS / you.ABS }
'**He** [father] gave me/you (as a wife) to **him**.'

b. *əlləɣ-a [ne-]jəl **-ɣət** *ɣənək-əŋ* ɣəmmə $\boxed{\text{✗2.GL} \gg \text{(1.TH)}}$
father-ERG INV(3.SU)-give -2.IO you-DAT me.ABS
'**He** [father] gave me to **you** (as a wife).' (Mel'čuk 1988, 294–5)

What we see in Alutor is then that both the EA-IA and IA-IA restrictions can be of the WEAK *≫ 1 type. However, there is an important qualification to be made. The status of the "weak inverse" forms as a strategy to void person restrictions in Chukotko-Kamchatkan is controversial (see Bobaljik to appear for discussion); among other things, the *na-* prefix can surface with some combinations of a 3P EA and 3P IA that do not violate any person restriction, and its distribution, also with respect to the spurious antipassive, is partly also determined by the number of the EA (although such indirect sensitivity of person restrictions to number is not an uncommon phenomenon; see Section 6.2.3).

What is important for us is only that even if the "weak inverse" forms really do not reveal a person restriction, this only means that the EA-IA restriction in Alutor is of a different WEAK type: one that only bans *2 ≫ 1 and *3 ≫ 1 combinations; i.e. those combinations that require the spurious antipassive. This would make the EA-IA restriction in Alutor one of the ME-FIRST type (see discussion below). In relation to the *Strength Generalization*, this alternative characterization would only mean that the EA-IA restriction would be weaker than the WEAK *≫ IA-IA restriction, which still complies with the generalization (it is IA-IA restrictions that cannot be weaker than EA-IA restrictions). Either way, Alutor remains a case of a language with a WEAK-WEAK combination.

Another interesting example of a WEAK *≫ 1 restriction comes from Pashto [*Iranian*] (Roberts 2000). The restriction does not occur between two arguments of the verb, but between an ergative EA clitic and a possessor clitic. This is because this is the only configuration where two personal pronoun clitics co-occur in the language. In past tenses, which have an ergative-absolutive alignment, the EA is a second position clitic, while the IA is not—the verb only shows object agreement in this case. In contrast, possessors are second position clitics as well, and crucially, the order of the EA and possessor clitics is only constrained by their person value, usually producing ambiguous parses of the two, as illustrated in (130). Note though that the impossible orders are exactly those expected to be ungrammatical given the WEAK *≫ 1 restriction. The second clitic cannot be 1/2P when the first one is 3P, and when the two are both 1/2P, the 1P clitic cannot be second.

(130) Pashto [*Eastern Iranian*]: **WEAK *≫ 1**

a. topak { **mee** / **dee** } **yee** raaworr-e ✗ 1/2.EA/POSS ≫ 3.POSS/EA
gun.M { 1.OBL / 2.OBL } 3.OBL brought-M.3
'**I/You** brought **his** gun.'
'**He** brought **my/your** gun.'

b. topak **mee** **dee** raaworr-e ✗ 1.EA/POSS ≫ 2.POSS/EA
gun.M 1.OBL 2.OBL brought-M.3
'**I** brought **your** gun.'
'**You** brought **my** gun.'

c. *topak **dee** **mee** raaworr-e ✗ 2.EA/POSS ≫ 1.POSS/EA
gun.M 2.OBL 1.OBL brought-M.3
'**You** brought **my** gun.'
'**I** brought **your** gun.'

d. *topak **yee** { **mee** / **dee** } raaworr-e ✗ 3.EA/POSS ≫ 1/2.POSS/EA
gun.M 3.OBL { 1.OBL / 2.OBL } brought-M.3
'**He** brought **my/your** gun.'
'**I/You** brought **his** gun.' (Roberts 2000, 135–8)

Crucially, all three restriction patterns above are compatible with the syntactic characterization in (131); why two separate conditions are needed will become clearer below.

(131) *WEAK *≫ 1 person restrictions*
  a. If X asymmetrically c-commands Y, and X is 3P, then Y cannot be 1/2P;
  b. If X asymmetrically c-commands Y, then Y cannot be 1P.

Moving to WEAK *≫ 2 restrictions, these are much rarer crosslinguistically, so there are fewer examples to use for illustration. Nonetheless, this is famously the restriction found in pretty much all of the Algonquian languages in the EA-IA domain. So far, I have been counting Algonquian languages as having WEAK EA-IA restrictions with both canonical and reverse versions, but the pattern is actually a bit more constrained (although some speakers of Arapaho seem to have a plain canonical + reverse WEAK pattern; see Cowell and Moss Sr. 2008). The Algonquian pattern has received many treatments over time (see Hockett 1939; 1948; 1966; Goddard 1979; Oxford 2014; 2019; i.a.). I illustrate the restriction pattern here by showing a fuller picture of the Nishnaabemwin paradigm. The canonical part is shown in (132), whereas the reverse part is shown in (133). In addition to the baseline WEAK restriction that we saw when we discussed the language previously, we can now also see that while a 2P marker can precede a 1P marker regardless of which one is the EA and which one the IA (cf. (132b), (133b)), the reverse order is never possible (cf. (132c), (133c)).[20]

(132) Nishnaabemwin/Ojibwe [*Algonquian*]: **WEAK *≫ 2 EA-IA**
  a. **n-** waabm-aa **-g** 1.EA ≫ 3.IA
     1- see-EA>3 -3PL
     '**I** see **them**.'
  b. **g-** waabm **-imin** 2.EA ≫ 1.IA
     2- see -1PL
     '**You** see **us**.'

20. Note that as in previous examples the '???' morpheme and optional plural marker are used in the glosses because the form and appearance of the post-verbal markers is highly contextual in Algonquian and we cannot know for sure the morphological forms the markers would have in the unattested person configurations.

c. ***n-** waabm-in (**-waa**) **-???** $\boxed{✗\,1.\textsc{ea} \gg 2.\textsc{ia}}$
1- see-INV.1>2 -PL -2
'**I** see **you(pl)**.'

d. ***w-** waabm-igoo (**-waa**) **-???** $\boxed{✗\,3.\textsc{ea} \gg 1/2.\textsc{ia}}$
3- see-INV.3>IA -PL -1/2
'**They** see **me/you**.'

(133) a. **n-** waabm-igoo **-g** $\boxed{1.\textsc{ia} \gg 3.\textsc{ea}}$
1- see-INV.3>IA -3PL
'**They** see **me**.'

b. **g-** waabm-in **-imin** $\boxed{2.\textsc{ia} \gg 1.\textsc{ea}}$
2- see-INV.1>2 -1PL
'**We** see **you**.'

c. ***n-** waabm (**-waa**) **-???** $\boxed{✗\,1.\textsc{ia} \gg 2.\textsc{ea}}$
1- see -PL -2
'**You(pl)** see **me**.'

d. ***w-** waabm-aa (**-waa**) **-???** $\boxed{✗\,3.\textsc{ea} \gg 1/2.\textsc{ia}}$
3- see-EA>3 -PL -1/2
'**They** see **me/you**.' (Valentine 2001, 274–5, 287)

As with the WEAK *≫ 1 person restriction pattern, we can also characterize the WEAK *≫ 2 pattern in syntactic terms as in (134). Notice that the only difference from the previous pattern is in the second condition, where the restriction concerns 2P pronouns as opposed to 1P ones.[21] Otherwise the two are entirely parallel.

(134) *WEAK *≫ 2 person restrictions*
a. If X asymmetrically c-commands Y, and X is 3P, then Y cannot be 1/2P;
b. If X asymmetrically c-commands Y, then Y cannot be 2P.

The reason why I am characterizing WEAK *≫ 1 and WEAK *≫ 2 restriction patterns in terms of two conditions comes from languages where we can see the two teased apart. For example, some Italian speakers have a WEAK *≫ 1 IA-IA restriction pattern, but one where the two conditions are clearly acting independently of each other. The baseline WEAK

21. More precisely, 1P inclusive also patterns with 2P in the Algonquian cases, which suggests it is all the person values that include the addressee that are singled out. I will leave out this qualification in the main discussion to help with the exposition. However, the importance of the addressee in such patterns (and other patterns that single out 2P) will be reflected in the analysis offered in Section 4.3.2.

restriction is always a canonical restriction which cannot be voided by reversing the order of the pronominal clitics. This is illustrated in (135).

(135) Italian [*Romance*]:

a. Lui { **me** / **te** } **lo** presentò. $\boxed{\text{1/2.GL} \gg \text{3.TH}}$
he { 1.IO / 2.IO } 3.M.DO introduced.3.M
'He introduced **him** to **me/you**.'

b. *Lui **gli** { **mi** / **ti** } presentò. $\boxed{✗\,\text{3.GL} \gg \text{1/2.TH}}$
he 3.IO { 1.DO / 2.DO } introduced.3.M

c. *Lui { **mi** / **ti** } **gli** presentò.
he { 1.IO / 2.DO } 3.DO introduced.3.M
'He introduced **me/you** to **him**.' (Cardinaletti 2008, 49–50, 77–8)

However, when we look at combinations of two 1/2P object clitics, their order is determined based on their person alone. As shown in (136a), as long as the 1P clitic precedes the 2P one, either clitic can be interpreted as the Goal or the Theme. However, all combinations where the 2P clitic precedes the 1P one are ungrammatical, as shown in (136b).

(136) a. Lui **mi** **ti** presentò. $\boxed{\text{1.GL/TH} \gg \text{2.TH/GL}}$
he 1.IO/DO 2.DO/IO introduced.3.M
'He introduced **me** to **you** / ?**you** to **me**.'

b. *Lui **ti** **mi** presentò. $\boxed{✗\,\text{2.GL/TH} \gg \text{1.TH/GL}}$
he 2.IO/DO 1.DO/IO introduced.3.M
'He introduced **me** to **you** / **you** to **me**.' (Cardinaletti 2008, 49–50)

In other words, the baseline WEAK person restriction is canonical, whereas the 1P vs. 2P asymmetry operates both as a canonical and a reverse restriction.

Spanish, on the order hand, has effectively the WEAK $*\!\gg 2$ version of the same restriction. The baseline WEAK pattern is illustrated in (137), while the additional 1P vs. 2P asymmetry is shown in (138). Notice that unlike in Italian, here the 2P clitic must always precede the 1P one. However, just as in Italian, either clitic can be the Goal or the Theme.

(137) Spanish [*Romance*]:

a. { **Me** / **Te** } **lo** recomendaron. $\boxed{\text{1/2/3.GL} \gg \text{3.TH}}$
{ 1.IO / 2.IO } 3.M.DO recommended.3PL
'They recommended **him** to **me/you**.'

b. *__Le__ { **me** / **te** } recomendaron. $\boxed{✗\,\text{3.GL} \gg \text{1/2.TH}}$
3.IO.DAT { 1.DO / 2.DO } recommended.3PL

c. *{ **Me** / **Te** } **le** recomendaron.
{ 1.DO / 2.DO } 3.IO.DAT recommended.3PL
'They recommended **me/you** to **him/her**.'

(138) a. **Te** **me** recomendaron. $\boxed{\text{2.GL/TH} \gg \text{1.TH/GL}}$
2.IO/DO 1.DO/IO recommended.3PL
'They recommended **me** to **you**.'
'They recommended **you** to **me**.'

b. *__Me__ **te** recomendaron. $\boxed{✗\,\text{1.GL/TH} \gg \text{2.TH/GL}}$
1.IO/DO 2.DO/IO recommended.3PL
'They recommended **you** to **me**.'
'They recommended **me** to **you**.' (Perlmutter 1971, 26n12, 27, 62)

There is an additional indirect argument for decomposing WEAK *≫ 1 and WEAK *≫ 2 into two separate restrictions; namely, that the second component of these restrictions is identical to the full restriction pattern in ME-FIRST and YOU-FIRST restrictions, which I turn to next.

Perhaps the most commonly discussed example of a ME-FIRST restriction is found in Romanian, where this pattern arises with the IA-IA restriction in ditransitives (Farkas and Kazazis 1980; Ciucivara 2006, 2009; Nevins 2007). As illustrated in (139), this restriction pattern excludes only the combination where the Theme clitic is 1P; as in (139b) and (140b).

(139) Romanian [*Romance*]: **ME-FIRST IA-IA**

a. Maria **me-** **te-** a prezentat. $\boxed{\text{1.GL} \gg \text{2.TH}}$
Maria 1.DAT 2.ACC has.3 introduced
'Maria has introduced **you** to **me**.'

b. *Maria **tie-** **m-** a prezentat. $\boxed{✗\,\text{2.GL} \gg \text{1.TH}}$
Maria 2.DAT 1.ACC has.3 introduced
'Maria has introduced **me** to **you**.'

(140) a. Maria **i-** **te-** a prezentat. $\boxed{\text{3.GL} \gg \text{2.TH}}$
Maria 3.F.DAT 2.ACC has.3 introduced
'Maria has introduced **you** to **her**.'

b. *Maria **i-** **m-** a prezentat.
Maria 3.F.DAT 1.ACC has.3 introduced
'Maria has introduced **me** to **her**.' $\boxed{\text{✗ 3.GL} \gg \text{1.TH}}$ (Nevins 2007, 297)

Other notable examples of ME-FIRST IA-IA restrictions are found in Bulgarian [*Slavic*] (Pancheva and Zubizarreta 2018) and Bosnian-Serbian-Croatian [*Slavic*] (Runić 2013b).

The restriction pattern can be characterized in syntactic terms as in (141). Note that this is exactly like an isolated second condition with the WEAK *≫ 1 restriction (cf. (131b)).

(141) *ME-FIRST person restriction*
If X asymmetrically c-commands Y, then Y cannot be 1P.

We could therefore describe ME-FIRST as a degraded version of WEAK *≫ 1; degraded in the sense that it seems to lack the baseline WEAK component (see also Nevins 2007, 2011 for this way of looking at the difference between WEAK, WEAK *≫ 1, and ME-FIRST).

Interestingly, as reported by Ciucivara (2006, 2009), there is another restriction pattern found with some Romanian speakers (including Ciucivara herself). The only difference between that person restriction pattern and the ME-FIRST pattern is that the former also allows for combinations of a 3P Goal and a 1P Theme clitic, as in (142).

(142) Romanian B [*Romance*]:
I- **m-** au recomandat ieri.
3.DAT 1.ACC have recommended yesterday
'They have recommended **me** to **him** yesterday.' $\boxed{\text{3.GL} \gg \text{1.TH}}$ (Ciucivara 2006, 259)

The speakers in question thus only disallow combinations where the Goal is 2P and the Theme is 1P, or *2 ≫ 1. Because only one person combination is ungrammatical here, I will simply adopt *2 ≫ 1 as a label for the restriction. This is the only example of this restriction pattern in my language sample. Nonetheless, it is significant because of its close relationship to the ME-FIRST pattern, which shown that the ME-FIRST pattern should also be viewed as complex, with the grammaticality of *3 ≫ 1 being what changes between the two varieties. The analysis of ME-FIRST and *2 ≫ 1 should therefore seek to reflect this.

In comparison to the ME-FIRST restriction pattern, the YOU-FIRST pattern is much rarer. In my language sample, only Quechua [*Quechuan*] (Myler 2017) seems to have a restriction of this kind, where it takes the form of an ordering constraint between subject and object agreement marking. In Quechua, the canonical restriction pattern is observed when the EA marker is the outermost marker on the verb, while the object marker is closer to the verb, as illustrated by the examples in (143)–(145). In that case, all person combinations are possible except for those where the EA is 1P and the IA is 2P, as in (144b), and those where the EA is 3P and the IA is 2P, as in (145b); in other words, the IA marker cannot be 2P.[22]

(143) Quechua [*Quechuan*] **YOU-FIRST EA-IA**

a. maylla **-Ø** -rqa **-ni** 1.EA ≫ 3.IA
wash -3.O -PAST -1.SU
'**I** washed **him/her**

b. maylla **-wa** -rqa **-n** 3.EA ≫ 1.IA
wash -1.O -PAST -3.SU
'**S/he** washed **me**

(144) a. maylla **-wa** -rqa **-nki** 2.EA ≫ 1.IA
wash -1.O -PAST -2.SU
'**You** washed **me**

b. *maylla -rqa **-Ø** **-ni** ✗ 1.EA ≫ 2.IA
wash -PAST -1>2.O -1.SU
'**I** washed **you**

(145) a. maylla **-Ø** -rqa **-nki** 2.EA ≫ 3.IA
wash -3.O -PAST -2.SU
'**You** washed **him/her**

b. *maylla -rqa **-su** **-n** ✗ 3.EA ≫ 2.IA
wash -PAST -3>2.O -3.SU
'**S/he** washed **you** (Myler 2017, 754, 779)

22. Like in the Algonquian WEAK *≫ 2 pattern (see above), 1P inclusive also patterns with 2P in Quechua. However, just as in that case, I leave it out for the sake of exposition; but see Section 4.3.2 for the analysis of YOU-FIRST where the role of the addressee will be crucial.

The ungrammatical combinations become grammatical with a reversal of the markers, accompanied by a morphological change of the object marker (similar to the inverse marking found in Algonquian). As shown in (146), the IA can be 2P when the order of the markers is reversed (the Quechua pattern is then essentially a canonical/reverse alternation).

(146) a. maylla -rqa **-Ø** **-yki** — $\boxed{2.\text{IA} \gg 1.\text{EA}}$
wash -PAST -1>2.O -2.O
'**I** washed **you**

b. maylla -rqa **-su** **-nki** — $\boxed{2.\text{IA} \gg 3.\text{EA}}$
wash -PAST -3>2.O -2.O
'**S/he** washed **you** (Myler 2017, 754)

The YOU-FIRST pattern found in Quechua can be characterized syntactically as in (147).

(147) *YOU-FIRST person restriction*
If X asymmetrically c-commands Y, then Y cannot be 2P.

Interestingly, just as in the case of the two varieties of Romanian, there also exists a degraded version of YOU-FIRST of sorts, except that in this case it is found in a completely unrelated language family: Salish. The person restriction in question is the also rare $*3 \gg 2$. All the attested cases in my language sample are from the Salish language family: Bella Coola (Forrest 1994), Halkomelem (Jelinek and Demers 1983; Gerdts 1988; Wiltschko 2008), and Squamish (Jelinek and Demers 1983; Jacobs 1994). This person restriction, as should be evident from the label, prohibits only one person combination: $*3 \gg 2$. The restriction in Halkomelem is illustrated by the examples in (148)–(149).

(148) Halkomelem [*Salish*]:

a. máy-t **-∅** **-es** — $\boxed{3.\text{EA} \gg 3.\text{IA}}$
help-TR -3.O -3.SU.ERG
'**S/he** helps **him/her**.'

b. máy-t **-∅** { **-tsel** / **-chexw** } — $\boxed{1/2.\text{EA} \gg 3.\text{IA}}$
help-TR -3.O { -1.SU / -2.SU }
'**I/You** help **him/her**.'

c. máy-th **-óme -tsel** $\boxed{1.\textsc{ea} \gg 2.\textsc{ia}}$
help-TR -2.O -1.SU
'**I** help **you**.'

d. máy-th **-óx -chexw** $\boxed{2.\textsc{ea} \gg 1.\textsc{ia}}$
help-TR -1.O -2.SU
'**You** help **me**.'

(149) a. máy-th **-óx -es** $\boxed{3.\textsc{ea} \gg 1.\textsc{ia}}$
help-TR -1.O -3.SU.ERG
'**He/She** helps **me**.'

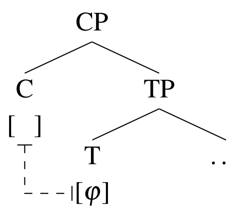
b. *máy-th **-óme -s** $\boxed{✗\,3.\textsc{ea} \gg 2.\textsc{ia}}$
help-TR -2.O -3.SU.ERG
'**He/She** helps **you**.' (Wiltschko 2008, 284–6)

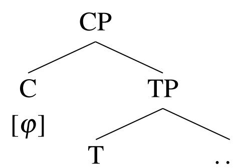
Although there are apparently no closely related languages with a YOU-FIRST pattern, the person restriction variation in Salish is also informative in terms of how the $*3 \gg 2$ pattern relates to other patterns. Other Salish languages either have a plain WEAK pattern (e.g. Lummi) or lack person restrictions entirely (e.g. Lushootseed) (Jelinek and Demers 1983). This variation will be important in the next section where I propose an extension of the current approach to person restrictions that can capture all the attested fine-grained WEAK restriction patterns. The key to deriving them lies in the fact that in one way or another they all seem to be internally complex, such that they often appear to be in a subset-superset relation with each other. In fact, this relationship between them can be viewed as a progressive degradation of the baseline WEAK restriction pattern that comes as a result of introducing new valuation options for the pronouns.

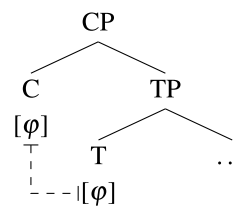
4.3.2 Deriving the fine-grained restriction patterns

I propose that the more limited grammatical person combination possibilities with some WEAK restrictions as well as the 'weaker than WEAK' person restrictions are the result of a single syntactic process interacting with the base WEAK restriction; namely, the possibility of *feature inheritance* (Chomsky 2004, 2005, 2008), where a phase head may transfer its features to the head of the projection it takes as a complement. In particular, I show that the variation we see between different WEAK restrictions can be derived if we adopt specifically the approach to feature inheritance of Ouali (2006).

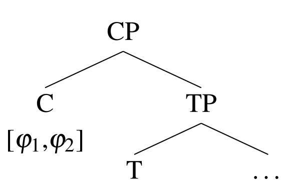
Ouali (2006) suggests three parametric options for feature inheritance. In particular, when the relevant phase head is C and the feature recipient is T, the options are: (i) C transfers its φ-features to T and does not keep a copy of these features, as illustrated in (150), (ii) C does not transfer its φ-features to T at all, as illustrated in (151), and (iii) C transfers its φ-features to T and keeps a copy, as illustrated in (152).

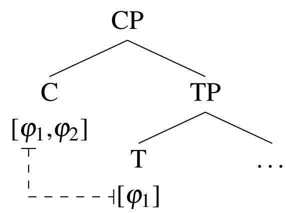
(150)


(151)


(152)


A natural extension of these is the option that C can transfer only a subset of its features; which is actually tacitly assumed by Chomsky (2004, 2005, 2008), since in the cases he discusses, C crucially only transfers its φ-features, even though C may also have other features (e.g. interrogative features). I suggest that the transfer of a subset of the phase head's features can be much more fine-grained; for example, given multiple φ-features, only a specific subset may be transferred, as illustrated in (153). This possibility crucially interacts with (150) and (152) above, since whether or not a copy of the features is kept on C is independent of which features can be transferred.

(153) a.


b. 

The intuition behind the set of features on the lower head ending up being less complex than the original set on the phase head is related to similar asymmetric relationships between higher and lower positions in other phenomena. For example, in relation to information structure, topics and foci may occur in two distinct areas of the clause: a high area (left periphery) and a low area (middle-field) (Belletti 2004). However, more types of topics and foci can occur in the high area than in the low area (see Paul 2002, 2005; Lacerda 2016, in preparation): the left periphery can host, among other things, aboutness, given, and contrastive topics, as well as both new information and contrastive foci, whereas the middle-field can host only contrastive and given topics.

This asymmetry is abstractly just like the one we see with different WEAK restrictions: no matter what the restriction pattern is, the structurally lower pronoun has fewer options regarding the person values it can have compared to the structurally higher pronoun. I propose that this is because the relevant verbal phase head can transfer only some of its person features to the head of its complement. All the variation comes from the parameterization of feature inheritance proposed by Ouali (2006) and variation in the person features. The key to deriving the different WEAK restriction patterns, which often look like they are composed of the same sub-patterns, is in the possibility for the different feature inheritance options to coexist in one language, acting in concert and producing the complex restriction patterns.

In relation to person features, I have been assuming only one person geometry so far, which was sufficient to express the three-way distinction between 1P ([π[2[1]]]), 2P ([π[2]]), and 3P ([π]). However, this type of privative feature geometry is too restricted to capture the crosslinguistic variation in person systems (see e.g. Harley and Ritter 2002; McGinnis

2005; Béjar and Řezáč 2009; Harbour 2016; Ackema and Neeleman 2018), as well as the full fine-grained variation found within WEAK person restrictions. In order to derive all the fine-grained patterns, I expand the person feature system I have been using so far to allow a parameterization along the lines shown in Table 4.3.

Table 4.3: Variation in person feature systems

speaker system	addressee system	inclusive/exclusive system
$\begin{bmatrix}\pi \\ 2[1]\end{bmatrix}$ (1P)	$\begin{bmatrix}\pi \\ 1[2]\end{bmatrix}$ (2P)	$\begin{bmatrix}\pi \\ \mathrm{P}[2]\end{bmatrix}$ (2P) $\begin{bmatrix}\pi \\ \mathrm{P}[1]\end{bmatrix}$ (1P.EXCL) $\begin{bmatrix}\pi \\ \mathrm{P}[2,1]\end{bmatrix}$ (1P.INCL)
$\begin{bmatrix}\pi \\ 2\end{bmatrix}$ (2P)	$\begin{bmatrix}\pi \\ 1\end{bmatrix}$ (1P)	$\begin{bmatrix}\pi \\ \mathrm{P}\end{bmatrix}$ (∅)
$\begin{bmatrix}\pi\end{bmatrix}$ (3P)	$\begin{bmatrix}\pi\end{bmatrix}$ (3P)	$\begin{bmatrix}\pi\end{bmatrix}$ (3P)

In order to distinguish the feature systems, I will call the person system used so far the *speaker system*, because it is the speaker '1' feature that distinguishes 1P from 2P, and I will call the system which uses the addressee '2' feature to distinguish 2P from 1P the *addressee system*. Finally, there is the *inclusive/exclusive system* which makes the distinction between inclusive and exclusive 1P and is essentially a combination of the speaker and the addressee systems.[23] Note that the 'participant' feature, which is common to 1P and 2P, is marked differently in the three systems (2, 1, and P respectively). This notation is mainly adopted to facilitate keeping track of the person of each pronoun in the derivations below.

I propose that the geometry of person features constrains feature inheritance, when it can only occur for a subset of person features in the same way as it constrains feature valuation in the case of the baseline WEAK restriction. Recall that in those cases, whenever only one feature takes part in valuation, it is always the participant feature; e.g. when a phase head

23. Although I will not discuss this in the main text, recall that In Algonquian (see footnote 21) and Quechua (see footnote 22), 2P person patterns with 1P inclusive when it comes to the person restriction. This feature system and how it interacts with person valuation is also designed to capture that.

values two pronouns 2P and 1P respectively, it only gives a participant feature to the former, but a participant and a speaker feature to the latter. That is because speaker/addressee features are dependent on participant features—the former cannot exist in a feature set without the latter. In terms of deriving the different versions of WEAK person restrictions, this will be key in limiting the value of the lower pronoun. Crucially, the possibility for syntactic heads to inherit person features from the phase head is still very constrained due to only being possible between a phase head and the head of its complement. As proposed by Chomsky (2008), transfer is also different from just adding another phase head into the structure for the reason that there is still a dependency between the two heads: if the phase head does not have a specific person feature to transfer, then the head of the complement cannot have that person feature either. In the case of two phase heads, the two are completely independent from each other in terms of what sort of person features they can bear.

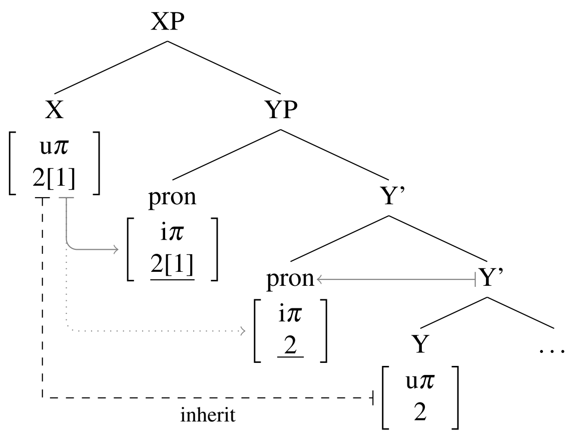
I will discuss the derivations of the different fine-grained person restrictions below in abstract terms, abstracting away from the identity of the pronouns and phase heads. Just as in the case of the deduction of the *Strength Generalization* in Section 4.1.6 above, it is important to show that no matter how we vary the different parametric options, the proposed analysis cannot derive the crosslinguistically unattested patterns.

4.3.2.1 Deriving **WEAK** *restrictions with additional restrictions*

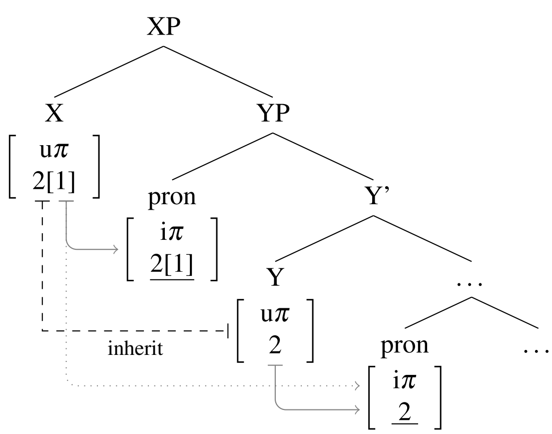
Let us begin by deriving the WEAK patterns with the additional ban on the lower pronoun being 1P or 2P. I will only provide the derivations for the former case, the WEAK $*\gg 1$ pattern, which is the type of WEAK restriction pattern we find in Classical Arabic, but as we will see the other restriction can be derived in exactly the same way, by only using an addressee based person system as opposed to a speaker based one.

This type of restriction occurs when the phase head can only transfer a subset of its person features; specifically, only the participant ('2') feature (which is the only subset that can be transferred due to the feature geometry). In addition, the transfer is of the kind where a copy of the transferred person feature is kept on the phase head. Let us first look at how the 1P vs. 2P asymmetry arises. Given a phase head X and the head of its complement Y, there are three ways in which a WEAK $*\gg 1$ restriction can arise. I will look at two options here, leaving the third one for later, as it crucial also for the derivation of ME-FIRST and YOU-FIRST patterns. The two options are presented in (154) and (155) respectively.

(154)



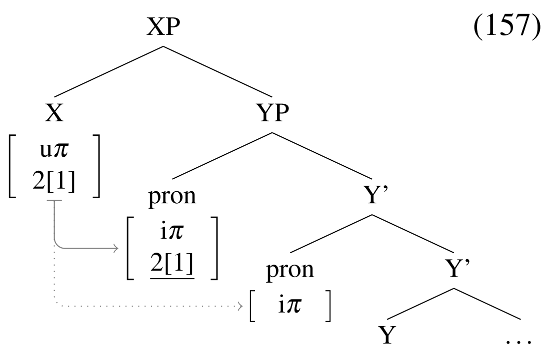
(155)



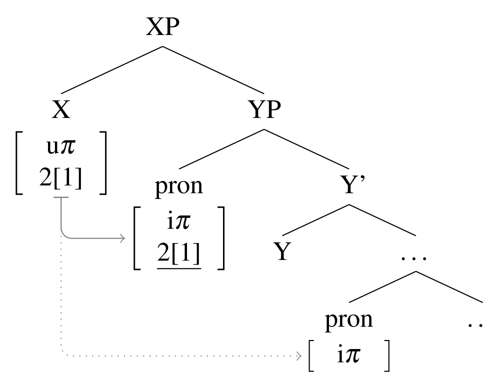
In (154), X transfers the participant '2' feature to Y with two minimal pronouns in SpecYP, while in (155), one minimal pronoun is in SpecYP and the other one below Y. Because of the feature transfer, there is a division of labor between X and Y in terms of person valuation, but their role in it is not equivalent. Note that in both (154) and (155) the higher pronoun is eligible for full person valuation from X, whereas the lower pronoun can only receive a participant feature from Y and be 2P. X cannot value the lower pronoun because the higher pronoun structurally intervenes between X and the lower pronoun. Additionally, in the case of (155), where the lower pronoun is person valued by Y in-situ, Y must value the pronoun by initiating Agree with it (just as in all the cases of in-situ valuation before). However, Y must not probe before X has entered the derivation and transferred its features to Y, or it would not have any person features to value the pronoun with. This ordering can be implemented in the current system in terms of Y having no unvalued features prior to the transfer, so that it gets both its valued person and unvalued number features from X.

Combinations where the higher pronoun is 1/2P and the lower one is 3P arise a bit differently in such languages than in basic WEAK restriction languages. Recall that in those, such combinations require the deletion of the valued person feature on the phase head after the first pronoun is valued, thus bleeding the valuation of the second pronoun. In the present case, such combinations basically arise when X does not transfer any features to Y. As shown in, (156) and (157) (which correspond to the structures in (154) and (155) respectively), the lack of any valued person features on Y means the lower pronoun can only get a 3P value.

(156)

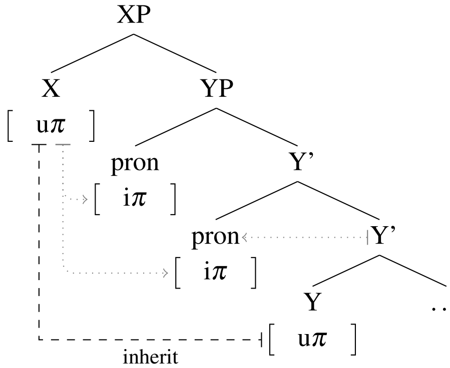


(157)

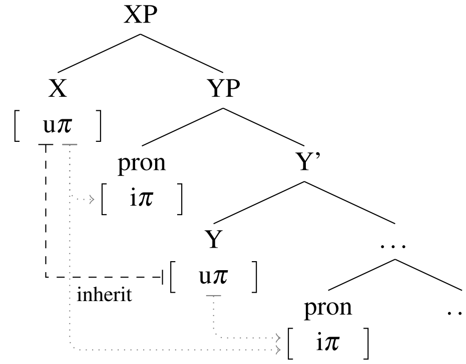


The final option is that X has no participant/speaker features to begin with, in which case whether the transfer takes place or not, there is no source of participant/speaker features in the XP phase, so all the minimal pronouns in the phase can only get a default 3P value. This is illustrated in (158) and (159) (which correspond to (154) and (155) respectively).

(158)



(159)



The reason why 3P ≫ 1/2P combinations cannot arise is because the person features of Y always depend on the person features of X in that the former always has a subset of the person features of the latter. A 3P ≫ 1/2P combination would on the other hand require X to be less specified than Y. Although that is a possibility in other types of languages, which we will look at next, it can never take place when the parameters are set as they are in this case.

Before we move on, consider that the derivation of WEAK *≫ 2 patterns, which are WEAK restrictions with the additional ban against the *1 ≫ 2 combination, can be derived with exactly the same syntactic mechanisms and parameters. The only difference that is required is a difference in the person feature system; an addressee system as opposed to a speaker one. In that case, the participant feature is also the only person feature transferred from X to Y, however, that means that it is the addressee feature which is *not* transferred to Y. Consequently, we get a WEAK restriction where the lower pronoun can only be 1P.

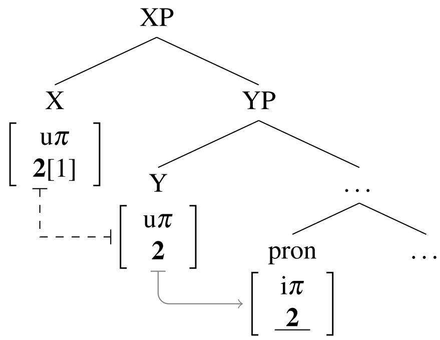
4.3.2.2 Deriving WEAK restrictions with loosened restrictions

At the other extreme of WEAK patterns are the ones that I dubbed 'weaker than WEAK', which allow even more grammatical person combinations than base WEAK restrictions: ME-FIRST, YOU-FIRST, *2 ≫ 1, and *3 ≫ 2. These are all degraded versions of other stronger restrictions, and the analysis I suggest for them below is designed to reflect that.

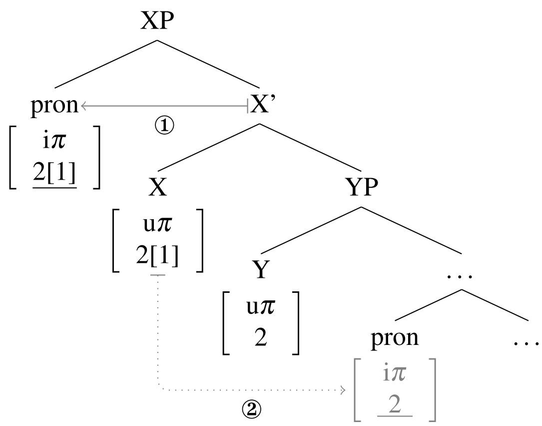
The basic idea behind the analysis of ME-FIRST restrictions—extending to YOU-FIRST restrictions as well—is that they are weakened versions of WEAK $*\gg 1$ restrictions (WEAK $*\gg 2$ in the case of YOU-FIRST), where the weakening is the result of an additional feature transfer option: transferred participant features can either leave behind a copy of themselves on the phase head, as before, or not. The additional option is sufficient to also derive $3 \gg 2$ pronoun combinations ($3 \gg 1$ in the case of YOU-FIRST). I will illustrate the derivation of ME-FIRST restrictions based on the structural configuration I left out before: one of the two pronouns is introduced in the specifier of the phase head. This particular configuration is key for deriving the ME-FIRST restriction because of the timing of feature inheritance, which must occur immediately at the point when the phase head enters the derivation. This contrasts with probing/Agree, which is subject to the Theta-Probe Condition and can only take place after the relevant probe has assigned a theta-role to the argument it introduces.

Let us first look at the part of the derivation that is virtually identical to a WEAK $*\gg 1$ derivation. As shown in (160a), X can transfer its participant '2' feature to Y, which can in turn value the minimal pronoun already present in the derivation; here it does not matter whether the pronoun is below Y and valued when Y probes and Agrees with it, as shown in (160a), or if the pronoun is in SpecYP and it probes and Agrees with Y. At this point X cannot probe because it has not yet introduced an argument in its specifier. Once that happens, as shown in (160b), the pronoun can be valued immediately upon being merged. Since X has a full person feature set, the pronoun can be valued 1P. In contrast, the lower pronoun is at this point inaccessible for Agree with X, which means it can only have the participant feature it received from Y and be 2P.

(160) a.



b.



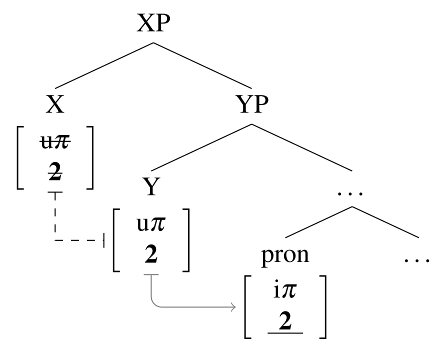
There are many ways in which the inaccessibility of the lower pronoun can be derived. One option is that this is impossible due to the *Activity Condition* (Chomsky 2000, 2001): arguments that have previously been goals cannot be goals for any subsequent Agree operations. Another option is that the features that and up on both X and Y through feature inheritance count as occurrences of the same feature, and thus cannot probe twice once they have already triggered Agree.[24] It is also possible that different conditions apply in different constructions. But since we are dealing with abstract derivations of these person restrictions, it important to note that there are multiple options at our disposal. Crucially, some restriction of this kind must be in place, or the result would simply be a plain WEAK restriction without a 1P vs. 2P asymmetry (see also discussion at the end of this section).

At this point, we have only shown how this configuration provides essentially another way to derive a WEAK $*\gg 1$ pattern (cf. Classical Arabic in (123)). What crucially derives

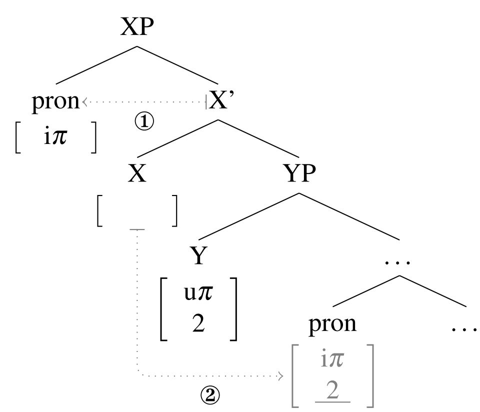
24. This would then also explain why Y does not count as a goal for X once the latter can probe; it would be like a probe probing itself—an option we definitely do not want to allow in any system of Agree.

ME-FIRST restrictions (cf. Romanian in (139)–(140)), where the lower pronoun can be 2P even when the top one is 3P, is specifically the possibility of not keeping a copy of the features on the phase head after they have been transferred. Crucially, since transfer happens before the introduction of the pronoun in SpecXP, the person features on X are removed before the pronoun in question enters the derivation, as shown in (161a).[25]

(161) a.



b.



Since X no longer has any person features when the minimal pronoun enters the derivation in SpecXP, the pronoun can only get a default 3P value. What is important here is that the person features on Y are also inaccessible to the pronoun in SpecXP. One way to ensure this is to assume that Y's person features are deleted after entering into the Agree relation with the lower pronoun. This must happen before the pronoun is merged in SpecXP, because feature inheritance is not conditioned by the Theta-Probe Condition, and if Y has a theta-role

25. Note that X only has a '2' feature in this case essentially for economy reasons: so that it does not redundantly introduce features that will not be used. Crucially, the derivation would not be affected if X had a full set of person features since they would be removed after the partial transfer anyway.

to assign (to an argument in its specifier or complement), it will have done so before X enters the derivation, so the Theta-Probe Condition will also have been met. Thus, Y can initiate probing as soon as possible (i.e. immediately after it gets the features from X), and its person features can be deleted before the minimal pronoun in SpecXP can probe.

Basically, what we see here is the mirror image of the bleeding of valued person features that takes place with plain WEAK restrictions, but because the feature transfer is only partial, affecting only participant features, we can only derive $3 \gg 2$ combinations, and not the ungrammatical $*3 \gg 1$ ones, this way. The partial transfer of person features from X to Y thus still makes it impossible to derive $*2 \gg 1$ and $*3 \gg 1$ person combinations, which are exactly the ones excluded with the ME-FIRST pattern.[26] It should be straightforward for the reader to verify that by only changing the feature system from a speaker based one to an addressee based system will derive the YOU-FIRST pattern the same way.

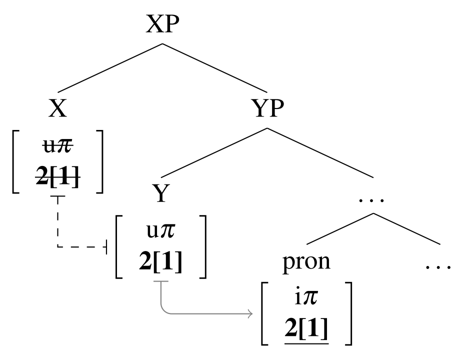
The feature inheritance options we saw so far all involve transferring only a subset of the person features on the phase head, but the system also allows transferring all the features. And indeed these are options that can derive the remaining fine-grained restriction patterns that we have established in the previous section. For example, recall that there are two types of speakers of Romanian: those with a ME-FIRST restriction pattern and those that only exclude $*2 \gg 1$ combinations. Deriving the latter kind in the current system requires only a minor change to the ME-FIRST derivation, which fits nicely with the speaker variation being very idiosyncratic. The only change that is needed is in relation to the feature transfer option that leaves no features behind on the phase head, in particular, feature transfer from X to Y where X does not keep the features. In that case in particular, the transfer option that leaves a copy of the features behind remains just like before—deriving $1 \gg 2$ but not

26. We will see in Section 6.2.2 that there exists another way to derive ME-FIRST patterns in the current system, based on Franks 2016. However, I will suggest there that if both derivations are correct, the one based on Franks 2016 must be limited only to languages with a very specific type of minimal pronouns.

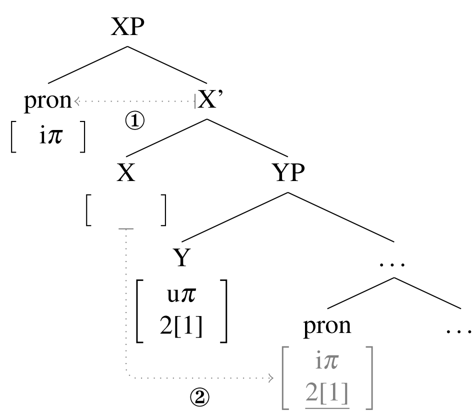
*2 ≫ 1 combinations. The reason why both 3 ≫ 2 and 3 ≫ 1 are now grammatical, though, is because when there is no copy left on X, the features Y inherits from X can be a full set.

The derivation of 3 ≫ 1, which is impossible with ME-FIRST, is shown in (162). What happens here is that X can transfer all its features as soon as it enters the derivation (the Theta-Probe Condition does not apply to inheritance), as in (162a), but then no copy of the features is left on X when the minimal pronoun is merged in SpecXP in (162b).

(162) a.



b.

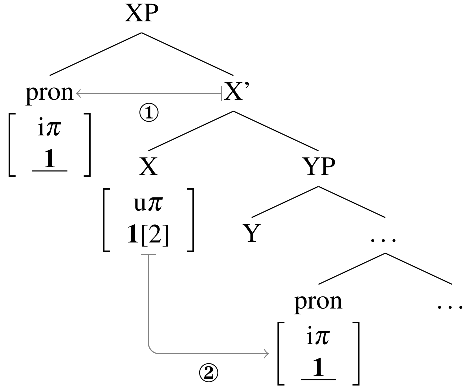


Just as when X only has the option of transferring a subset of its features but leaves no copy of the features behind (required for a ME-FIRST pattern; cf. (141)–(161)), the minimal pronoun entering the derivation in SpecXP in (162b) cannot be valued and must get a default 3P value. The difference compared to a ME-FIRST derivation is that the lower pronoun can be valued either 1P or 2P because X can transfer both speaker and participant features to Y.

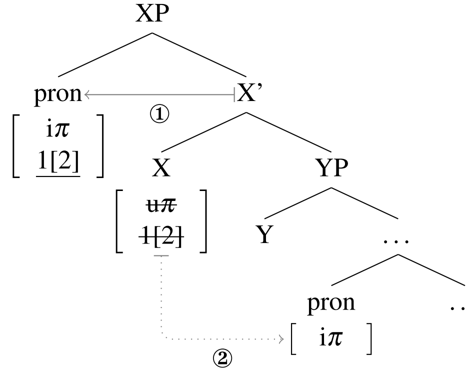
The final parametric possibility for feature transfer is that transfer, whether partial or full, can either take place or not. Here we can nicely illustrate the available options in relation to the variation in person restrictions within the Salish language family (see Jelinek and Demers 1983). There are three types of patterns in Salish, showing a gradual degradation of the plain WEAK restriction. Recall that Lummi is among the Salish languages with a WEAK EA-IA restriction while Halkomelem is one of the languages that only disallows *3 ≫ 2 EA-IA combinations. Finally, Lushootseed does not show any person restrictions.

Let us start by reminding ourselves of the baseline WEAK restriction found in Lummi. I switch here to using an addressee based person system, since the fine-grained restriction found in other Salish languages (*3 ≫ 2) is addressee based, and I know of no evidence against all of Salish having the same person feature system. Recall that in a plain WEAK restriction we can have both pronouns valued by the same person features on the phase head, as illustrated in (163) in relation to the X phase head, which can yield 1/2 ≫ 2/1 person combinations. Additionally, the first pronoun to be valued can delete the valued person feature on X, as in (164); the first pronoun to be valued here is the one merged in SpecXP, due to the Theta-Probe Condition prohibiting X from probing before that.

(163)



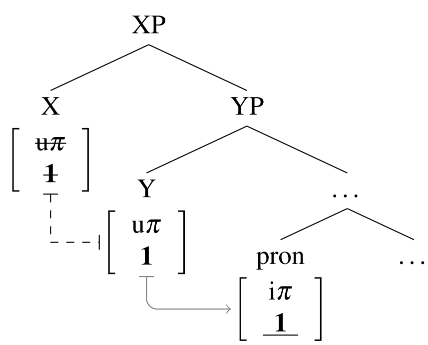
(164)



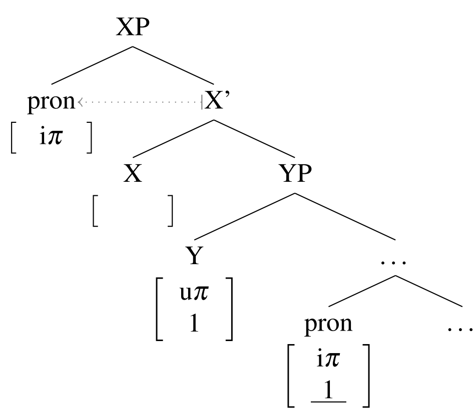
This derivation can yield mixed 1/2 ≫ 3 person combinations. The alternative where the person feature on X is not deleted, yields 1 ≫ 2 or 2 ≫ 1 person combinations. Crucially, the deletion option is the only way to get one pronoun to be 1/2P and the other 3P, which is why it is impossible to get *3 ≫ 1/2 combinations with basic WEAK restrictions.

The Halkomelem pattern, which is weaker in that it only disallows *3 ≫ 2 combinations, can be derived by simply adding to the above derivation the possibility of partial feature transfer that leaves no copy behind. Note that this is the same possibility we saw with ME-FIRST, except that there it was coupled with the possibility of partial transfer with a copy; i.e. transferring participant to Y and leaving both participant and speaker features on X. If leaving behind a copy of the features is impossible, what we are left with is a WEAK pattern with one more grammatical combination. In this case, due to the addressee based system: 3 ≫ 1. The derivation of this additional combination is illustrated in (165). X can transfer the participant feature as soon as it enters the derivation, as in (165a), but then no copy of the features is left on X when the minimal pronoun is merged in SpecXP in (165b).

(165) a.



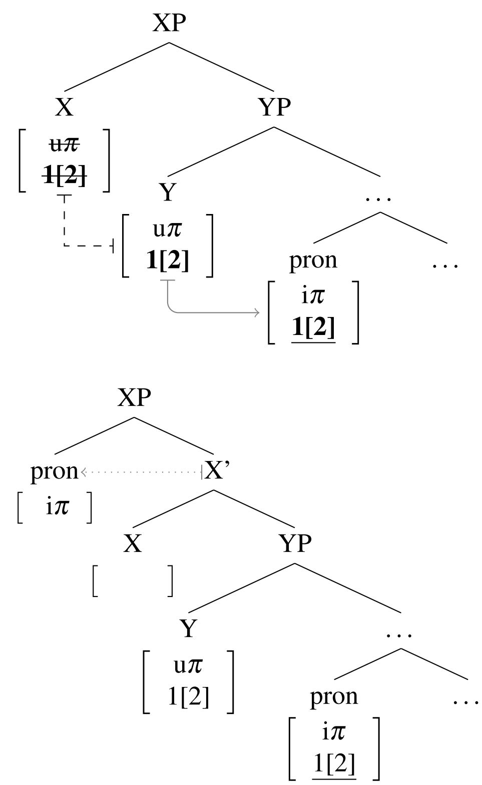
b.



Lastly, by also varying the partial/full transfer parameter which earlier got us the distinction between the two Romanian patterns, we can get the Lushootseed pattern with no person restrictions. This pattern arises because it is possible for both pronouns to be valued 1/2P by X. Because of that it is possible to get a 1/2 ≫ 3 combination through the bleeding derivation. It is also possible to get a 3 ≫ 1/2 combination via full feature transfer between X and Y, if X does not keep a copy of the features, as shown in (166). X can transfer all its features as soon as it enters the derivation, as in (166a), but then no copy of the features is left on X when the minimal pronoun is merged in SpecXP in (166b).

(166) a.

b.



The progressively weaker restrictions in Salish can thus be attributed to the feature transfer first emerging as a possibility and then becoming richer in terms of the amount of person features that can be transferred via feature inheritance.

Having derived the *3 ≫ 2 pattern, all the attested fine-grained versions of WEAK restriction patterns are now accounted for. However, we have not yet exhausted all the parametric possibilities offered by the parameterized feature inheritance model. I will show below that even if we utilize the additional options, it is impossible to derive any unwanted restriction patterns within the proposed system. Before that, let us consider all the possibilities that we have utilized to derive the attested pattern, summarized in Table 4.4.

Table 4.4: Person restrictions derived through feature inheritance

	$*\gg 1$	$*\gg 2$	ME-FIRST	YOU-FIRST	$*2 \gg 1$	$*1 \gg 2$	$*3 \gg 1$	$*3 \gg 2$	∅	∅
π-system	spkr. 2[1]	addr. 1[2]	spkr. 2[1]	addr. 1[2]	spkr. 2[1]	addr. 1[2]	spkr. 2[1]	addr. 1[2]	spkr. 2[1]	addr. 1[2]
partial tr.										
copy	✓	✓	✓	✓	✓	✓	✗	✗	✗	✗
no copy	✗	✗	✓	✓	✗	✗	✓	✓	✗	✗
full tr.										
copy	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
no copy	✗	✗	✗	✗	✓	✓	✗	✗	✓	✓

Note that there are two patterns that this approach predicts that we have not encountered in the language sample (in the gray columns). Person restriction patterns tend to come in symmetric pairs: a speaker based one and an addressee based one. We have, for example, the WEAK $*\gg 1$ and WEAK $*\gg 2$ pair, as well as the ME-FIRST and YOU-FIRST pair. And given that in such cases the role of speaker and addressee features seems interchangeable, it would be striking if the same option did not also exist for other even weaker restrictions: $*3 \gg 2$ is addressee based and $*2 \gg 1$ is speaker based. Admittedly, these patterns are incredibly rare, so it could be that their symmetric doubles, which are missing in the current language sample, can be found in an even larger sample. If, however, such patterns remain illusive, it will be necessary to reevaluate the proposed analysis in terms of its symmetric treatment of speaker and addressee based person systems—perhaps the identity of specific person features will turn out to be relevant for valuation.

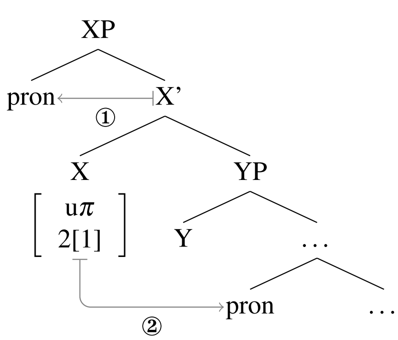
Now we can turn to the parametric options we have not considered above. Some can be excluded a priori; namely, if full transfer is possible given either the copy or no copy option, then also having partial transfer with the same option is entirely redundant; the end result would be no different than allowing just the full transfer, so presumably these parametric options are never exploited. The remaining options are summarized in Table 4.5. Notice that these are either plain WEAK patterns or patterns already present in the previous table.

Table 4.5: Remaining options available through feature inheritance

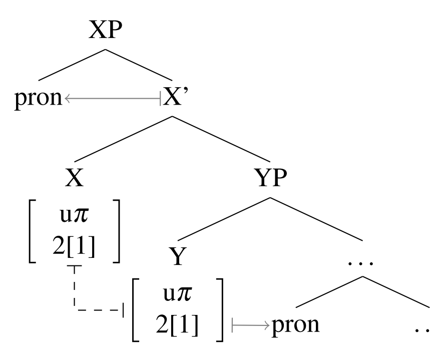
	WEAK	WEAK	$*3 \gg 1$	$*3 \gg 2$
π-system	spkr. 2[1]	addr. 1[2]	spkr. 2[1]	addr. 1[2]
partial tr.				
copy	✗	✗	✗	✗
no copy	✗	✗	✓	✓
full tr.				
copy	✓	✓	✓	✓
no copy	✗	✗	✗	✗

All four options here involve full transfer with no copy, which is relevant because this yields exactly the same valuation possibilities as a single source of valued person features in a basic WEAK configuration. Compare the basic WEAK configuration in (167) with the full feature inheritance version in (168). In both cases both pronouns can receive a value, it is just that with the latter option the valuation is achieved through a division of labor between X and Y. The only difference is that with the basic WEAK derivation, $1/2 \gg 3$ arises through deletion of the person feature on X, while with the inheritance version the same combination arises when there is no feature transfer between X and Y (see (156)–(157) above).

(167)



(168)



The same holds for $*3 \gg 2$ and $*3 \gg 1$, which are, as we saw above, only WEAK restrictions with the option of partial transfer without a copy. Since WEAK restrictions can in principle be derived in two ways, so can $*3 \gg 2$ and $*3 \gg 1$ restrictions.

The are two ways to go about this redundancy. We can assume that the version of these derivations that involves the redundant full feature transfer step never occurs precisely because the extra step is redundant and syntax is constrained by economy of derivation. The other possibility is that there might be independent phenomena that interact with the feature inheritance possibility (maybe regarding features other than person) and both versions of the derivations exist, but have different consequences for those phenomena. The latter would certainly be an interesting possibility given that much of the motivation for the current approach to person restrictions came from finding independent syntactic factors that interact with person restrictions. However, exploring those options will have to be left for another time. For now, the important result is that all the attested fine-grained WEAK restrictions can be derived in the current system, but also that the analysis makes the prediction, to be tested, that all fine-grained WEAK restrictions exist in symmetric speaker/addressee pairs. Hopefully, that prediction can be either verified or falsified in future work.

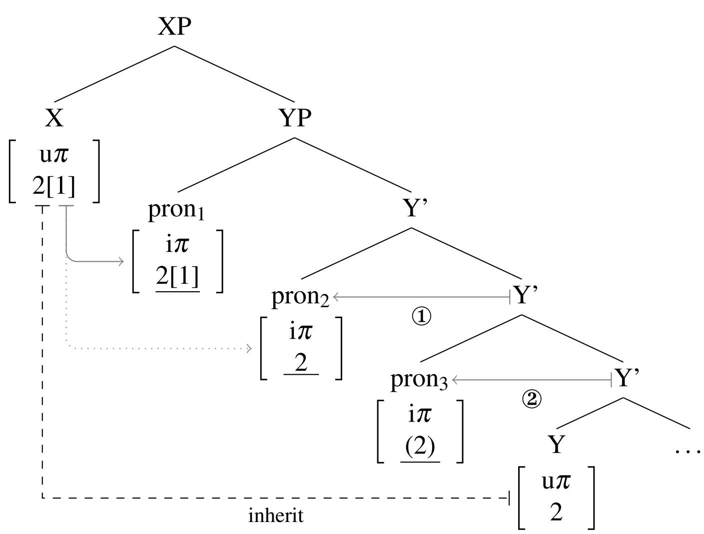
4.3.3 Why the crosslinguistic gaps remain

What remains to be shown in light of the proposed analysis of fine-grained WEAK restrictions is that it does not, despite the additional valuation possibilities it introduces, undermine the deductions of the two crosslinguistic generalizations achieved earlier in this chapter. That this is not the case can be shown pretty easily. One of the main points of invoking feature inheritance was the idea that it is a possibility reserved for phase heads, and that it can only take place between a phase head and the head of its complement. That is the key point: there is no unbounded feature transfer, the operation is extremely local. I will show next how this maintains the two generalizations, starting with the *Strength Generalization*.

The *Strength Generalization* arises because person restrictions between pronouns closer to the phase edge can only be of the same strength or weaker than those between pronouns further away from it. The possibility of feature inheritance between a phase head and its complement maintains this; it can either introduce an additional 1P vs. 2P asymmetry on top of an underlying WEAK restriction or further weaken a WEAK restriction that arises at the phase edge, but it may not weaken a restriction not occurring at the phase edge. Let us look at some examples of relevant derivations to illustrate this point.

We can see for example in (169), a derivation where the partial feature inheritance of participant features from X to Y makes both the restriction between *pron*$_1$ and *pron*$_2$ and the restriction between *pron*$_2$ and *pron*$_3$ of the WEAK $* \gg 1$ type.

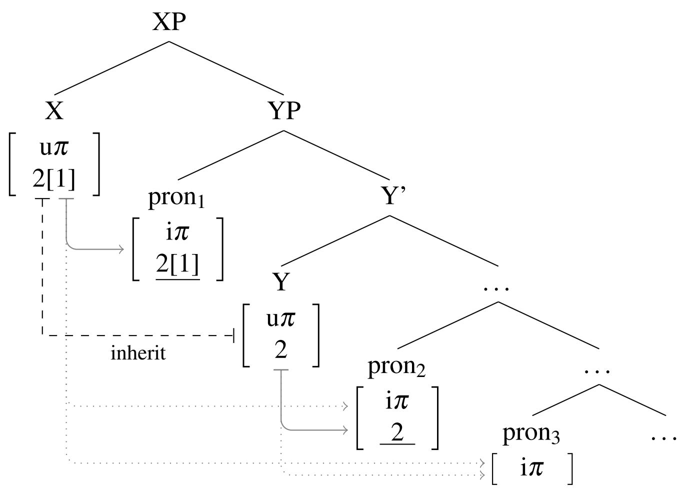
(169)



This is because whichever is the topmost pronoun can be 1P, while the others can only be 2P, while mixed 1/2P and 3P combinations will be possible only with a bleeding derivation—making the derivation of $*3 \gg 1/2$ combinations impossible.

In contrast, the derivation in (170) also involves a partial feature inheritance of participant features from X to Y. But because Y asymmetrically c-commands both *pron*$_2$ and *pron*$_3$, the restriction between them will be STRONG due to Agree Closest, as opposed to the restriction between *pron*$_1$ and *pron*$_2$, which will be WEAK $*\gg 1$ just as in (169) above.

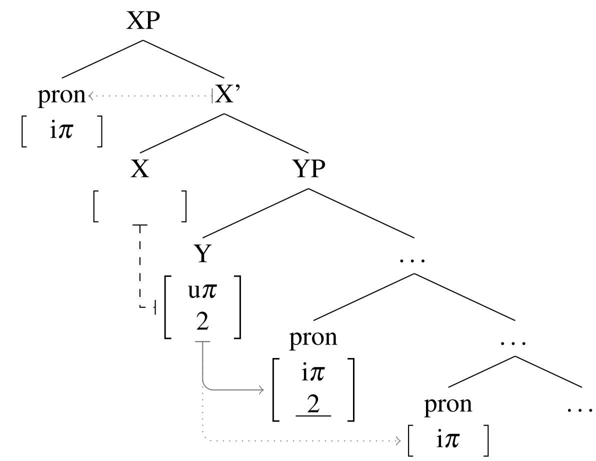
(170)



The restrictions between the two pronoun pairs can thus either both be WEAK $*\gg 1$, as in (169), or WEAK $*\gg 1$ and STRONG respectively, as in (170). In fact, the two derivations fit two attested patterns. The derivation in (169) fits the pattern of Alutor [*Chukotkan*] (Savvina and Mel'čuk 1978; Mel'čuk 1988; Kibrik 2002); under the interpretation where both the EA-IA ad IA-IA restrictions are WEAK $*\gg 1$ (see Section 4.3.1). The derivation in (170), on the other hand, fits the pattern of Sahaptin [*Penutian*] (Rude 1994), where the EA-IA restriction is WEAK $*\gg 1$ and the IA-IA one is STRONG.

We can see the same thing with ME-FIRST derivations, where we can get a configuration where the restriction between the top two pronouns is weakened because the partial feature transfer leaves no copy of the features behind on X. However, in a derivation like (171), Y asymmetrically c-commands both *pron*$_2$ and *pron*$_3$. This means that while the restriction between *pron*$_1$ and *pron*$_2$ is ME-FIRST (see previous section for details; cf. (141)–(161)), the restriction *pron*$_2$ and *pron*$_3$ arises due to Agree Closest and is therefore STRONG.

(171)

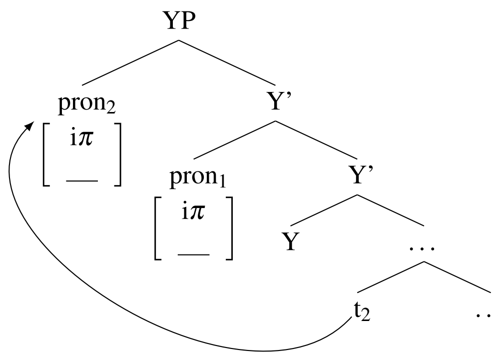


The YOU-FIRST version of the pattern derived in (171) (that is, a version with the addressee feature system rather than a speaker system) is one we find in Quechua (Myler 2017; p.c.).

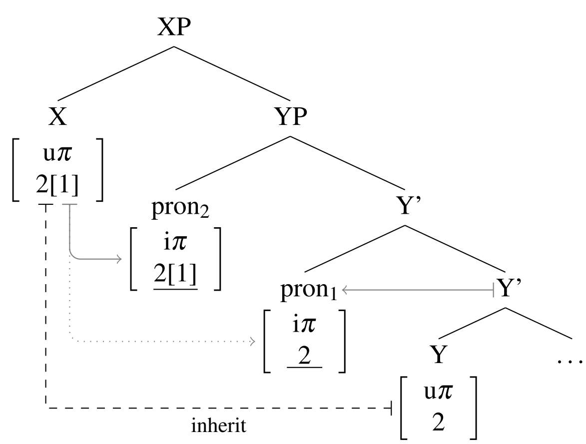
Showing all the logically possible configurations given three minimal pronouns and all the feature inheritance possibilities would take too much time, and it would not necessarily be any more insightful than the derivations we have already seen. So I trust the reader can verify for themselves that due to the extremely local nature of feature inheritance, the features introduced on the phase head cannot be transferred over unbounded intermediate projections; that would be for instance necessary to derive the unattested STRONG-WEAK combination, where the transfer would have to completely bypass the EA-IA domain. Additionally, feature inheritance can never make STRONG restrictions out of WEAK ones; the only strengthening it is capable of is from WEAK to WEAK $*\gg 1$ or WEAK $*\gg 2$. In all other cases, it makes the restriction weaker. Because of that it is impossible to turn a WEAK restriction into a STRONG one via feature inheritance, so STRONG-WEAK combinations cannot be derived this way either. This means the deduction of the *Strength Generalization* is maintained, despite the additional person valuation options that feature inheritance offers.

Turning now to the deduction of the *Direction Generalization*, it is even easier to show that feature inheritance poses no threat for it. Recall that the driving force behind the deduction of the *Direction Generalization* is the nature of the movement that derives reverse person restrictions: scrambling. Since it is optional, we can never get languages with only a reverse restriction in either argument domain. Due to it not being driven by Agree/valuation, scrambling can take place before any person valuation, and crucially before feature inheritance. Thus, minimal pronouns can reorder before the phase head enters the derivation, as in (172a), so that at the point when the phase head X enters the derivation, it proceeds just as it would without the reordering, as in (172b). The only difference is that the pronouns have reordered, so the resulting restriction is a reverse one.

(172) a.

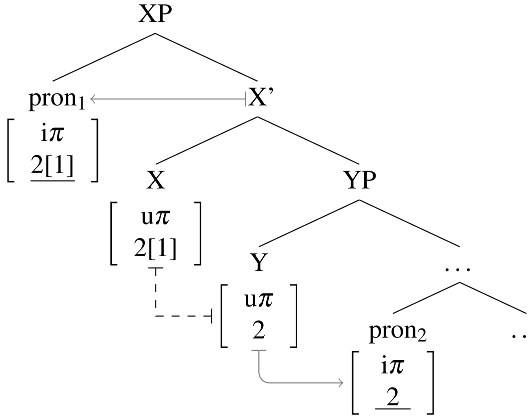


b.

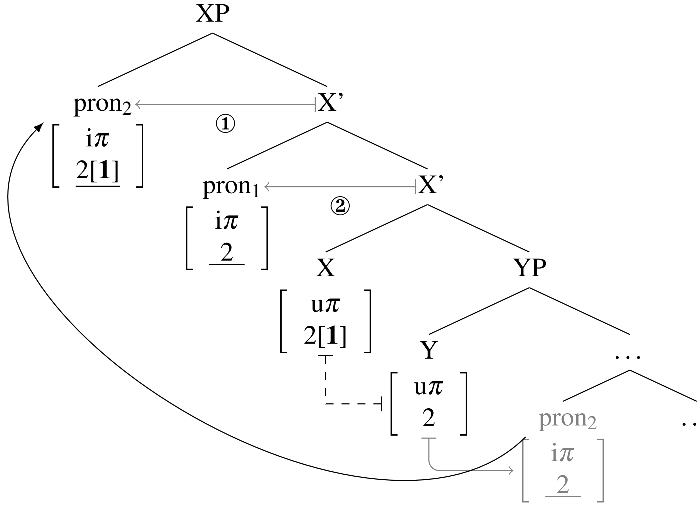


More importantly, because of the dissociation between the 1P vs. 2P asymmetry and the WEAK restriction in how the feature inheritance analysis derives WEAK $*\gg$ 1 and WEAK $*\gg$ 2 patterns, we can derive the patterns of Spanish or Italian, where there is a canonical/reverse alternation with respect to co-occurring 1P and 2P clitic pronouns, but not for the baseline WEAK restriction. For example, in the case of WEAK $*\gg 1$ (Italian), the canonical pattern arises as in (173a), which is the same as the WEAK $*\gg 1$ derivations above. However, there is also the possibility for the low pronoun (pron_2) to scramble to SpecXP after it had received a participant feature from Y, but before the other pronoun (pron_1) even enters the derivation (which is how arguments can reorder through scrambling). As a result pron_2 can be valued as 1P instead of pron_1, as shown in (173b).

(173) a.



b.



This is not the only way in which Spanish/Italian patterns can be derived in the current system, but what is important is that they can be. A closer study of the patterns in question will hopefully help us determine with more accuracy what the actual derivation should be.

In addition, what would have to be explained is why scrambling of clitics is possible with such languages that otherwise do not allow scrambling with full DPs or strong pronouns.

We conclude this section having seen that not only the approach to person restrictions established in Chapter 3 can deduce both the *Strength Generalization* and the *Direction Generalization*, but that even when we expand the approach to derive even more variation in person restriction patterns, the deduction of the two generalizations remains intact. The main reason for this is that valued person features are consistently limited to phase heads. This assumption has been key in deriving all the relevant facts concerning person restrictions in the last two chapters. However, the reason for the existence of this constraint has not yet been explained. I will discuss a possible origin for it in the following chapter.

tea knows precisely why it is tea
it has no desire to be Catalina
—Tomaž Šalamun, "Tea"

5

Why person?

It is crucial for the analysis of person restrictions proposed in Chapter 3 that minimal pronouns have unvalued person features, but also that their other φ-features are valued and that phase heads are the only source of valued person features. This split in valuation status is what drives the all the derivations of person restrictions and what confines the restrictions to person features. However, these assumptions have not yet been fully justified.

The (un)valued status of different φ-features cannot be a matter of arbitrary choice, possibly subject to crosslinguistic variation. If that were the case, we would expect restrictions comparable to person restrictions to also arise with number and gender, which we crucially never find. In other words, there are no *Number-Case Constraint* or *Gender-Case Constraint* effects that would be comparable to the Person-Case Constraint (PCC) effect in double object ditransitives, or equivalent restrictions outside double object ditransitives. However, grammatical categories other than person that can exhibit the same kind of syntactic co-occurrence restrictions do exist. We will see in this chapter that those categories and their semantic similarities to person actually hold the key to identifying what sets apart the φ-features that can be unvalued on pronouns and conversely valued on phase heads.

I will first establish which are the categories other than person that can be subject to the same syntactic co-occurrence restrictions (Section 5.1). After that I will show that these categories, including person, constitute a natural class in terms of their semantic properties (Section 5.2), and propose an account of how the distribution and valuation status of the features that encode these categories in the syntax is constrained by their semantic status (Section 5.3). Finally, I will argue that the proposed analysis, which I show respects the autonomy of syntax, should be preferred over analyses of person restrictions that require semantic considerations to directly constrain syntactic operations (Section 5.4).

5.1 Co-occurrence restrictions beyond person

Outside person, co-occurrence restrictions also arise in relation to: *discourse salience*, *definiteness*, *animacy*, and *deference*. Before we consider what these have in common with person, we must make sure that these restrictions are indeed equivalent to person restrictions.

5.1.1 Obviation (discourse salience)

The most frequently discussed co-occurrence restriction outside person is *obviation*. Famously exhibited in Algonquian languages, but not limited to them (Aissen 1997, 2001), obviation operates on a grammatical opposition that ranks 3P arguments based on what is variously described as: topicality, focus of interest, prominence, viewpoint, or empathy. For current purposes, I simplify the opposition to a relative difference in *discourse salience* (to be elaborated on in Section 5.2.1.1).[1] Following Algonquianist tradition,[2] the 3P arguments most central to the discourse are called *proximate* (PROX) (roughly, those a clause, sentence, or longer discourse unit is about) while all other 3P arguments are *obviative* (OBV).

Obviation is illustrated in (1) with examples from Mapudungun, discussed previously in Chapter 2 as a language with a STRONG EA-IA person restriction: transitive clauses with a 1/2P.IA must be realized as inverse constructions. A similar restriction occurs with two 3P arguments (Grimes 1985; Smeets 1989, 2008; Arnold 1994; Zúñiga 2000): a proximate EA and obviative IA can co-occur in a standard (direct) transitive construction, as in (1a), but the same construction is ungrammatical with an obviative EA and proximate IA, as shown in (1b). In the latter case, an inverse construction like the one in (1c) must be used.

(1) Mapudungun [*Araucanian*]:

a. mütrüm **-fi** -y **-Ø** $\boxed{\text{3PROX.EA} \gg \text{3OBV.IA}}$
call -3.O.OBV -IND -3.SU.PROX
'**He**$_{\text{PROX}}$ called **him**$_{\text{OBV}}$.'

b. *mütrüm **-fi** -y **-Ø** $\boxed{\text{✗ 3OBV.EA} \gg \text{3PROX.IA}}$
call -3.O.PROX -IND -3.SU.OBV
'**He**$_{\text{OBV}}$ called **him**$_{\text{PROX}}$.'

c. mütrüm -e -y **-Ø** **-ew** $\boxed{\text{3OBV.EA} \gg \text{3PROX.IA}}$
call -INV -IND -3.O.PROX -3.SU.OBV.DAT
'**He**$_{\text{OBV}}$ called **him**$_{\text{PROX}}$.' (Smeets 2008, 226–7, 367)

The PROX $\gg$ OBV configuration crucially patterns like a 1/2P $\gg$ 3P configuration, whereas *OBV $\gg$ PROX patterns like *3P $\gg$ 1/2P. In other words, obviative 3P has the distribution that 1/2P has with respect to the person restriction. The main difference between obviation and person restrictions more generally is that the former does not show a crosslinguistic

1. For a discussion of the complexities and parameters of crosslinguistic variation in obviation systems see, among others: Klaiman (1991, 1992), Aissen (1997, 2001), Oshima (2006, 2007), and Zúñiga (2014).

2. See, e.g.: Hockett (1939, 1948, 1966), Bloomfield (1962), Frantz (1966), Wolfart (1973, 1978), Goddard (1979, 1984, 1990), Grafstein (1981, 1984, 1987), Dahlstrom (1986a, 1986b, 1991, 2017), and Rhodes (1990a).

STRONG/WEAK variation: PROX ≫ PROX (the obviation counterpart to 1/2P ≫ 2/1P) is always impossible, unless the arguments co-refer (Aissen 1997, 2001), which makes obviation a STRONG restriction. The behavior of OBV ≫ OBV configurations (usually rare in natural discourse[3]) is also informative: for example, in Plains Cree [*Algonquian*] (Wolfart 1973, 1978; Dahlstrom 1986a, 1991) and Kutenai [*isolate*] (Dryer 1992; Appelbaum 2019) these can surface as either direct and inverse constructions. The possibility of occuring with either construction shows that no restriction occurs in such cases—recall that otherwise the two are in complementary distribution, governed by the person restriction and obviation.[4]

Although obviation is mainly associated with direct/inverse systems, it also manifests crosslinguistically in other ways familiar from the earlier discussions of person restrictions.[5] For example, Aissen (1997, 2001) notes that in both Tzotzil [*Cholan-Tzeltalan*] and Chamorro [*W. Austronesian*] the obviation status of arguments constrains the choice of active/passive voice (cf. the EA-IA person restrictions in Arizona Tewa [*Kiowa-Tanoan*] and Lummi [*Salish*] discussed in Section 2.3.1). Aissen (1999a) also shows that the Agent-Focus construction in Tzotzil patterns with inverse constructions based on the obviation restrictions on the EA and IA (cf. the EA-IA person restriction with Agent-Focus in Kaqchikel [*Quichean-Mamean*] discussed in Section 2.5.3.1). Finally, obviation can also occur in the IA-IA domain:[6] in many Algonquian languages, if the Goal in a double object construction

3. For an example of discourse conditions that may yield multiple obviatives see Mithun (1999, 225–6; using an Odawa Ojibwe text fragment from R. Rhodes). The rarity of OBV ≫ OBV in spontaneous speech might be why some researchers, including Aissen (1997, 2001), assume such configurations should be ruled out. This seems to be tied to an assumption that two 3P arguments with distinct referents are never equal in discourse salience and thus never equal in obviation status. What is often taken as evidence for this is the existence of "second/further obviative" morphemes in some Algonquian languages (Hockett 1939, 1966). However, these are only found on possessum nouns, based on which Wolfart (1978) argues that they do not express a further obviation class, but agreement with the obviative possessor. Genitive constructions are also a context where OBV ≫ OBV configurations (for possessor ≫ possessum) occur freely in Algonquian.

4. See also footnote 6 regarding the order alternations with 3P object clitics in Slovenian.

5. See also Anagnostopoulou (2005, 224–9), Bianchi (2006), Quinn (2006), Béjar and Řezáč (2009, 53n9), Lochbihler (2013), and Oxford (2014, 2019), among others, for minimalist analyses of obviation in Algonquian that treat obviation as being essentially a type of person restriction.

6. In fact, although not previously described in these terms, restrictions on the order of 3P IO and DO clitics in Slovenian are a potential candidate for IA-IA obviation. Recall that Slovenian allows both IO»DO and DO»IO

is 3P the Theme must be obviative (see Ellis 1983, 283–4, 2000, 285 for Moose/Swampy Cree, Rhodes 1990a, 1990b for Ojibwe, and Bruening 2001, Ch. 2 for Passamaquoddy).[7]

5.1.2 Definiteness (and specificity)

Another case of non-person co-occurrence restrictions are those sensitive to definiteness and specificity. A relevant definiteness restriction is found in Nyaturu [*Bantu*] (Hualde 1989; Woolford 2000; Ormazabal and Romero 2007).[8] In Nyaturu, a person restriction arises with double objects, as in (2a), where the combination of a 3P Goal and 2P Theme is ungrammatical (note that only one object marker is realized, just like in Swahili; see Section 2.4). With such combinations, the Goal argument must be realized as a PP, as shown in (2b).

(2) Nyaturu [*Bantu*]:

a. *n-a- { **mU- / kU-** } et-e-aa *mUnca veve*. ✗3.GL ≫ 2.TH
1.SU-PAST- { 3.IO- / 2.DO- } bring-APPL-FV girl you

b. n-a- **kU-** et-aa *veve* kU mUnca. 2.TH + PP.3.GL
1.SU-PAST- 2.DO- bring-FV you for girl
'I brought **you** for **her** [the girl].' (Hualde 1989, 181)

An additional restriction that applies in Nyaturu is that only one of the objects may be definite, so (3) is only grammatical if the Theme is construed as an indefinite.[9]

clitic orders (see Section 2.2.1), which includes combinations of two 3P clitics. Crucially, the order of object clitics in the latter case is governed by their relative discourse salience, where the more salient (proximate) one must precede the less salient (obviative) one (see Stegovec 2019, 16 for the relevant data).

7. These are works where the restriction is explicitly noted, but the restriction seems to be more widespread in Algonquian, as I was not able to find counterexamples. Note also that double object constructions are yet another context where OBV ≫ OBV configurations are easily found (see discussion above and footnote 3).

8. A comparable restriction is also found in Akan [*Kwa*] (Sáàh and Ézè 1997; Haspelmath 2004a, 2007): the Theme in a double object construction cannot be definite (for the relevant data see Sáàh and Ézè 1997, 143–4).

9. An exception to this occurs with 1P.SG and reflexive IO markers, which are exceptional in a number of other ways in Nyaturu and many other Bantu languages (Marlo 2014); e.g. the DO marker can be expressed only alongside these IO markers. Crucially, the examples provided by Marlo (2014, 74) suggest that even the person restriction is either voided or altered in such contexts. There is also evidence from final vowel (FV) alternations that 1P.SG markers fundamentally differ from other object markers in Nyaturu (Marlo 2014, 75–6): object markers on imperative verbs require the *-e* FV, whereas a 1P.SG marker requires the *-a* FV, otherwise used only in the absence of object markers. One way to interpret this is that 1P.SG markers are agreement

(3) n-a- **va-** et-e-aa *anca mUhUmba*. $\boxed{\text{3DEF.GL} \gg \text{3INDEF.TH}}$
1.SU-PAST- 3PL.IO- bring-APPL-FV girls boy
'I brought *a/*the boy* to **them** [the girls].' (Hualde 1989, 181)

In fact, if a proper name is identified with the Theme, as in (4a), the result is always ungrammatical, since then the Theme cannot be indefinite. Just as with the person restriction seen above, the Goal must be realized as a PP in such cases, as shown in (4b).

(4) a. *n-a- **va-** tUm-I-aa *alimu* *Yohana*. $\boxed{\text{✗ 3DEF.GL} \gg \text{3DEF.TH}}$
1.SU-PAST- 3PL.IO- send-APPL-FV teachers Yohana

b. n-a- **mU-** tUm-aa *Yohana* kU alimu. $\boxed{\text{3DEF.TH + PP.3DEF.GL}}$
1.SU-PAST- 3.DO- send-FV Yohana for teachers
'I brought **him** [Yohana] to **them** [the teachers].' (Hualde 1989, 181)

A related restriction occurs in Senaya [*Neo-Aramaic*] (Kalin 2014a, 2014b, 2017, 2018), where perfective verbs disallow specific objects (definite and specific indefinite objects). This is shown in (5a), where a definite—and thereby specific—object is banned (see Kalin 2014b, 2018 for examples showing that the relevant factor is specificity and not definiteness).[10] In contrast, we see in (5b) that a perfective verb with a non-specific object is grammatical.

(5) Senaya [*Neo-Aramaic*]:

a. *Axnī ō ksūta ksū (**-a**) **-lan**. $\boxed{\text{✗ 1.EA} \gg \text{3SPEC.IA}}$
we that book.F write.PFV -3.F.O -1PL.SU
'**We** wrote **it** [that book].'

b. Axnī xa ksūta ksū **-lan**. $\boxed{\text{1.EA} \gg \text{3NSPEC.IA}}$
we a book.F write.PFV -1PL.SU
'**We** wrote a book (non-specific).' (Kalin 2018, 120)

affixes rather than deficient pronouns. As for the other exceptional marker, reflexives have been known to void person restrictions crosslinguistically; see e.g. Harris (1981, 81) on Georgian, Rivero (2004, 500–1) on Bulgarian, and Stegovec (2016, 296–7, 302–3) on Slovenian (see also Section 2.4.1 on reflexives).

10. The optional indexing of the object on the verb is relevant because indexing is obligatory with specific objects when they are allowed (see Kalin 2014a, 2014b, 2017, 2018); e.g. with imperfective aspect, as in the example in (6) below (see also the discussion of differential object marking/indexing below). What (5a) then shows is that the ban on specific objects cannot be voided by not expressing the object marker.

The specificity restriction in (5a) parallels the EA-IA person restriction that also exists in Senaya: the object cannot be 1/2P (recall from Section 4.1.1.2 that this restriction is observed also in a number of other Neo-Aramaic languages). Crucially, both restrictions are voided when the verb is in the imperfective aspect; shown in (6) for the specificity restriction.

(6) Āna ō ksūta kasw **-an** **-ā**. $\boxed{\text{1.EA} \gg \text{3SPEC.IA}}$
I that book.F write.IMPF -1.SU -3.F.O
'**I** (will) write **it** [that book].' (Kalin 2018, 119)

Both co-occurrence restrictions also extend to the IA-IA domain: the Theme in ditransitives must be 3P person unless the Goal is a PP (Kalin 2014a, 206n3), but also the Theme must be non-specific unless a progressive auxiliary is added to the verb, even if the verb is imperfective (Kalin 2014b, 156n11). The relevance of the two restrictions being voided using different strategies will be briefly addressed below in Section 5.2.

Definiteness and specificity restrictions of this type are less frequently discussed then obviation, so the crosslinguistic possibilities relating to the former have not been explored to the same extent. Note that I am referring here specifically to co-occurrence restrictions, where an argument cannot be definite or specific in the context of a structurally higher argument. There are many other syntactic phenomena where definiteness and specificity play a role which have received a great deal more attention, such as *differential object marking* (DOM) and *differential object indexing* (DOI), where, rather than being banned, definite and specific objects either occur with special case morphology/adpositions (DOM) or must be indexed/clitic doubled (DOI).[11] Interestingly, DOM and DOI have been argued by Kalin (2017, 2018) to actually have the same grammatical source as co-occurrence restrictions. If this view is on the right track, then definiteness and specificity co-occurrence

11. See, among many others: Comrie (1979a, 1981), Croft (1988, 1990), Bossong (1991), Enç (1991), Aissen (2003), Rodríguez-Mondoñedo (2007), Dalrymple and Nikolaeva (2011), and Kalin (2017, 2018). Note that DOM/DOI is crosslinguistically not only sensitive to definiteness and specificity, but also to person and animacy—two other categories subject to syntactic co-occurrence restrictions. This parallel lends additional credence to Kalin's (2017, 2018) proposal that the two phenomena should be unified (see below).

restrictions might be more common than previously thought. However, since the status of DOM/DOI is not directly relevant for the discussion at hand, I leave exploring the connection between DOM/DOI and co-occurrence restrictions to future research.

5.1.3 Animacy (and humanness)

Syntactic co-occurrence effects can also apply to animacy. Consider for example the case of Mohawk [*Iroquoian*] (Baker 1988, 1996, 2006). In Mohawk double object constructions, there is a person restriction forcing the Theme to be 3P, as in the examples in (7). The ungrammaticality of Themes of any other person is illustrated with (8a).[12] But the restriction on the Theme is in fact even stricter: the Theme can only be 3P and inanimate, as shown by the ungrammaticality of (8b), where both the 2P Goal and 3P Theme are animate.

(7) Mohawk [*Iroquoian*]:

a. wa- **híy-** u-'. $\boxed{\text{3.GL} \gg \text{3.TH}}$
FACT- 1>3M(>3N)- give-PUNC
'**I** gave **it** to **him**.'

b. wa- **hák-** u-'. $\boxed{\text{1.GL} \gg \text{3.TH}}$
FACT- 3M>1(>3N)- give-PUNC
'**He** gave **it** to **me**.'

(8) a. *érhar ʌ- **kú-** nut-e'. $\boxed{\text{✗ 3.GL.} \gg \text{2.TH}}$
dog FUT- 1(>3Z)>2- feed-PUNC
'I will feed **you** to **it** [the dog].'
(OK as: 'Dog, **I** will feed **it** [something] to **you**.')

b. *ʌ- **kú-** (ya't)-óhare-'s-e' ne owitá'a. $\boxed{\text{✗ 2.GL} \gg \text{3AN.TH}}$
FUT- 1>2(>3F)- wash-BEN-PUNC NE baby
'**I** will wash **her** [the baby] for **you**.' (Baker 1996, 193)

12. The translation of the 3P Goal marker is potentially misleading, as English uses *'it'* to refer to non-human animates, such as *'dog'*, as well as inanimates. Mohawk morphologically distinguishes four 3P "genders": masculine (M), feminine (F; also generic/indefinite), neuter (N; exclusive to inanimates), and *zoic* (Z; exclusive to animals, i.e. non-human animates) (Baker 1996, 36–7n17, 238n1, 518–9). Neuter and zoic markers are often indistinguishable due to being phonologically null, although zoic markers are not systematically null (see Baker 1996, 36–7n16 and Baker 1990 for further morphological distinctions between zoic and neuter).

The animacy restriction is independent from the person of the Goal, as seen with the minimal pair in (9), where both the Goal and Theme are 3P, but only the Goal can be animate.

(9) a. Sak wa- **híy-** u-' ne ashare'. | 3AN.GL ≫ 3INAN.TH
Sak FACT- 1>3M(>3N)- give-PUNC NE knife
'**I** gave **it** [the knife] to **him** [Sak].'
b. *Sak wa- **híy-** u-' ne ashare'. | ✗3AN.GL ≫ 3AN.TH
Sak FACT- 1>3M(>3F)- give-PUNC NE knife
'**I** gave **her** [the baby] to **him** [Sak].' (Baker 2006, 292–3)

Animacy also plays a key role in direct/inverse alternations in Algonquian, in addition to person and obviation restrictions (I will discuss the specifics of that animacy restriction in Section 5.2), as well as other kinds of EA-IA co-occurrence restrictions crosslinguistically (see Aissen 1997 for discussion and references).[13] An interesting example is the EA-IA restriction in Tzotzil, illustrated in (10), where both EA and IA can be animate in an active clause like (10a), but a 3P IA cannot be animate if the EA is inanimate, as in (10b).[14]

(10) Tzotzil [*Cholan-Tzeltalan*]:
a. I- **Ø-** **s-** mil Xun li Petul-e. | 3AN.EA ≫ 3AN.IA
CMPL- 3.ABS- 3.ERG- kill Juan the Pedro-DEF
'**He** [Pedro] killed **him** [Juan].'
b. *I- **Ø-** **s-** mil Xun li ton-e. | ✗3INAN.EA ≫ 3AN.IA
CMPL- 3.ABS- 3.ERG- kill Juan the rock-DEF
'**It** [the rock] killed **him** [Juan].' (Aissen 1997, 724–5)

Just as some EA-IA person restrictions can be lifted with passive voice (see Section 2.3.1), the animate restriction in Tzotzil is also lifted if the clause is passive, as shown in (11).

13. I do not consider here another kind of animacy restriction, where arguments with specific thematic/grammatical roles must be animate. For example, in Southern Tiwa [*Kiowa-Tanoan*], EA subjects of transitives (ergatives) and Goals must be animate (Rosen 1990, 682); see also Wiltschko and Ritter (2015) regarding similar restrictions in Blackfoot [*Algonquian*]. These restrictions are not co-occurrence restrictions: the obligatory animacy of the argument is not tied to the presence of another argument higher in the structure. I assume, following Adger and Harbour (2007) and Wiltschko and Ritter (2015), that these animacy restrictions arise due to selectional requirements by specific functional heads on the content of their specifiers.

14. Interestingly, the restriction does not apply to 1/2P IAs, which are also animate (Aissen 1997, 725–6), which shows that animacy restrictions can be independent from person restrictions (see also Section 5.2). Aissen (1997) draws an analogy between the absence of the animacy restriction in this context and the absence of the 3P proximate/obviative contrast in Algonquian in cases when the other argument is 1/2P.

(11) I- **Ø-** mil-e ta ton li Xun-e. $\boxed{\text{3AN.IA.SU} + \text{3INAN.EA}}$
CMPL- 3.ABS- kill-PASS by rock the Juan-DEF
'**He** [Juan] was killed by the rock.' (Aissen 1997, 727)

By comparing the restrictions in Mohawk and Tzotzil, we can identify another point of variation, beside the former applying in the IA-IA domain and the latter in the EA-IA domain: the Mohawk restriction is an analog of a STRONG person restriction and the Tzotzil restriction is an analog of a WEAK person restriction. In Mohawk, the Theme cannot be animate regardless of the Goal's animacy status, while in Tzotzil the IA cannot be animate only if the EA is inanimate. Thus, not only are the strategies that void animacy restrictions identical to those used with person restrictions, the points of variation are also parallel.

It should be noted that many restrictions described as animacy restrictions are specifically *humanness* restrictions: they single out human animates and do not restrict non-human ones, such as animals (Wiltschko and Ritter 2015). When animacy restrictions are reported, they are often illustrated only for human arguments, making it impossible to determine whether the restriction operates on the animate/inanimate opposition or more specifically on the human/non-human one. Interestingly, languages where both oppositions matter for co-occurrence restrictions also exist, such as Laxopa Zapotec [*Oto-Manguean*] (Toosarvandani 2017; Foley and Toosarvandani 2018; Foley, Kalivoda, and Toosarvandani to appear). In Laxopa, animate (human or animal) IA clitic pronouns are banned in the presence of an EA inanimate clitic pronoun (Toosarvandani 2017, 131). Additionally, the human/animal opposition yields another restriction, shown in (12): with two subject/object clitics where one refers to a human (HUM) and the other to an animal (ANML),[15] the EA must be the human and the IA the animal, as in (12a), but the reverse is impossible, as shown in (12b).[16]

15. Like other Sierra Zapotec languages, Laxopa further distinguishes between respectful and familiar human 3P clitic forms (the latter are used in (12)); see Section 5.1.4 for discussion of this opposition.

16. The two co-occurrence restrictions are independently attested in other Sierra Zapotec varieties: the restriction in Yatzachi (Butler 1980) operates only on the human/non-human opposition, while in Zoogocho (Long and Cruz 2000; Sonnenschein 2004) it operates only on the animate/inanimate oppositions. See Foley and Toosarvandani (2018) and Foley, Kalivoda, and Toosarvandani (to appear) for discussion of this variation.

(12) Laxopa Zapotec [*Oto-Manguean*]:

a. Blen **-ba'** **-b**. $\boxed{\text{3HUM.EA} \gg \text{3ANML.IA}}$
hug.CMPL -3.HUM.SU -3.ANML.O
'**S/he** hugged **it** (an animal).' (Foley and Toosarvandani 2018, 5)

b. *Bdi'in **-ba** **-ba'**. $\boxed{\text{✗ 3ANML.EA} \gg \text{3HUM.IA}}$
bite.CMPL -3.ANML.SU -3.HUM.O
'**It** (an animal) bit **her/him**.' (Toosarvandani 2017, 133)

As we will see in the next section, a related Sierra Zapotec language exhibits an interesting additional co-occurrence restriction on top of the animacy and humanness restrictions.

5.1.4 Deference

The final example of non-person co-occurrence restrictions are those sensitive to *deferential* status (social/familiarity status with respect to the speaker). This type of restriction is attested in Yalálag Zapotec (Avelino Becerra 2004; López and Newberg 2005), where it exists on top of analogous animacy and humanness co-occurrence restrictions—which are the same as those discussed in the previous section for Laxopa Zapotec.

Like other Sierra Zapotec languages, Yalálag has grammaticalized the opposition between inanimate, animal, familiar human, and respected human for 3P entities, and this opposition is morphologically expressed on subject and object clitic pronouns.[17] Familiar 3P forms are used for individuals in a close relationship with the speaker, such that they can be addressed informally or familiarly, and respectful 3P forms are used for individuals who are not intimate with the speaker and deserve a respectful treatment (Avelino Becerra 2004, 50). A grammatical example using clitic pronouns with is given in (13a), where the EA is a

17. The Zapotec animacy and humanness restrictions described in Section 5.1.3 and the deference restriction discussed in this section are together dubbed the *Gender-Case Constraint* by Foley and Toosarvandani (2018), due to the authors treating the inanimate/animal/familiar-human/respected-human opposition as a gender opposition. However, note that this "gender" system is quite unlike the more familiar gender systems that distinguish categories such as feminine, masculine, and neuter, which are interestingly never targeted by co-occurrence restrictions. I will offer a potential explanation for what makes these intuitively different gender systems behave differently with respect to co-occurrence restrictions in Section 5.2.

respected individual (RESP) and the IA someone who the speaker is on familiar terms (FAM). Crucially, the pronouns cannot be reversed: as (13b) illustrates, the IA clitic cannot be one that refers to a respected individual if the EA clitic refers to a familiar individual.

(13) Yalálag Zapotec [*Oto-Manguean*]:

a. W-kwell **-e'** **-be'**. 3RESP.EA ≫ 3FAM.IA
IRR-CAUS.cry -3.RESP.SU -3.FAM.O
'**He** (an elder) will make **him** cry.'

b. *W-kwell **-be'** **-e'**. ✗ 3FAM.EA ≫ 3RESP.IA
IRR-CAUS.cry -3.FAM.SU -3.RESP.O
'**He** will make **him** (an elder) cry.' (Avelino Becerra 2004, 33)

At this point, this is the only case I know of where the co-occurrence of pronouns is restricted based on their deference status.[18] It could be that this is because the necessary conditions for such restrictions—having the relevant type of deficient pronouns and a grammaticalized deference opposition on such pronouns—seldom coexist. However, a more systematic survey is needed before any conclusions can be made about the rarity of deference restrictions.

Nonetheless, the Yalálag Zapotec case alone is very informative regarding the status of the deference restriction. As noted above, Yalálag also exhibits animacy and humanness restrictions of the same type. Furthermore, it exhibits a STRONG EA-IA person restriction (Avelino Becerra 2004, 31–3). What is relevant here is that all three restrictions behave the same in terms of the syntactic environment in which they occur and the strategies that can void them; namely, expressing the IA as a strong pronoun. I take this as evidence that, other than being sensitive to deference status rather than person value, the restriction shown in (13b) is equivalent to the person restrictions discussed in the previous chapters.

18. A potentially related phenomenon is found in Kashaya [*Pomoan*] (Hall 1990), which has a complex DOM pattern sensitive among other things to deference status. However, the Kashaya case is only relevant inasmuch co-occurrence restrictions and DOM are related, as argued by Kalin (2017, 2018) (see Section 5.1.2).

5.2 What can and what cannot be restricted

Grammatical categories beyond person can thus also be subject to co-occurrence restrictions. In fact, the restrictions reviewed in the previous section can also operate independently from person restrictions (I discuss the criteria for independence and why it is a relevant factor below). The restrictions in question are crucially syntactic just like person restrictions: it is always the syntactically lower argument that is restricted; with canonical/reverse restriction alternations this results in being either an order restriction or an indexing restriction just as with the analogous person restrictions (see Sections 2.5.3 and 2.5.4).[19]

Two categories that were conspicuously absent in the previous section are gender and number. Gender and other categories used for noun classification, aside from animacy, are never subject to syntactic co-occurrence restrictions. Number is also never restricted, although the facts are a bit more complicated in that case, as number can interfere with co-occurrence restrictions: for example, in some varieties of Spanish, a WEAK person restriction is observed with two singular object clitics, but a STRONG person restriction is observed with two plural object clitics (Rivero 2008; Alba de la Fuente 2010). What is important in such cases is that number is always an interfering factor rather than the restricted category. Number can affect how other categories are restricted, but it is never directly restricted itself; apparent "number restrictions" are just reflexes of underlying person restrictions. I will suggest in Section 6.2.3 that the number sensitivity is an indirect effect of a difference in pronoun type similar to what is observed in Italian, where plural IO deficient pronouns, which crucially void person restrictions, are of a different pronoun type than their singular counterparts (Cardinaletti and Starke 1994; 1999; see also Section 1.3.2).

19. In some cases there is additional evidence for the syntactic nature of the alternation; e.g. in Algonquian languages, where the order or markers is constrained both by person and obviation status, the different marker orders can be shown to yield different binding possibilities (see Section 4.2.3).

Given the possibility of categories that can interfere with co-occurrence restrictions even if they are not themselves restricted, we must make sure that the categories discussed in the previous section are not of this type—this is where the independence from person restrictions noted in the introduction comes into play. The reasoning is that if obviation, definiteness (i.e. definiteness or specificity), animacy (i.e. animacy or humanness), and deference can each be targeted by co-occurrence restrictions, it should be possible to find languages or constructions: (i) where these categories are restricted in the absence of any person restriction, or (ii) which constitute minimal pairs that differ in the presence of a restriction on one of these categories but have the exact same person restriction.

There are many cases where other co-occurrence restrictions exist in the absence of person restrictions: an obviation restriction can be the sole restriction in a language (e.g. Kutenai [*isolate*]; Dryer 1992; Appelbaum 2019) or construction (e.g. *Agent-Focus* in Tzotzil [*Cholan-Tzeltalan*]; Aissen 1999a) and animacy restrictions can also exist independently of person restrictions (e.g. Tzotzil ergative-absolutive constructions; Aissen 1997, 2001).

In the case of definiteness and deference restrictions, I have so far been unable to find languages where these exist in the absence of any person restrictions.[20] Nonetheless there is evidence that such restrictions are not merely a reflex of the underlying person restriction. Namely, there exist minimal pairs of closely related languages that differ with respect to the presence of a definiteness or deference restriction with no difference in person restriction patterns. For example, Senaya and Telkepe [*Neo-Aramaic*] (Kalin 2014a, 2014b) have the same person restriction pattern, but only Senaya has the specificity restriction. Recall from Section 5.1.2 that the person and specificity restrictions in Senaya also differ in how they can be voided, which makes sense of the two restrictions are independent from each

20. This could just be due to my sample not being large enough combined with the rarity of such restrictions, but it is equally possible that there is a one way implication between having person restrictions and having definiteness and deference restrictions. Note though that this is irrelevant for the issue at hand. What matters is only that these restrictions do not arise indirectly like the “number restrictions” noted above—it only needs to be shown that the relevant categories can be singled out by the syntactic restriction.

other.[21] Similarly, recall from the previous section that Yalálag Zapotec and Laxopa Zapotec [*Oto-Manguean*] differ only in that the former exhibits the deference restriction on top of the same person and animacy restrictions.

Finally, non-person restrictions can also coexist in different combinations. For example, Navajo [*Athabaskan*] (Hale 1973a; Jelinek 1990; Willie 1991; Willie and Jelinek 2000; i.a.) the co-occurrence of pronominal markers is restricted based on obviation, animacy, and definiteness in the absence of any person restrictions.

The existence of cases that reveal a dissociation between obviation, definiteness, animacy, and deference restrictions on one hand and person restrictions on the other suggests that the former are true syntactic co-occurrence restrictions just like the latter. But this raises the question of what makes these categories a natural class?

In traditional descriptive and typological work going back at least to Silverstein (1976, 1986),[22] person, animacy, and definiteness are often grouped together on a scale or hierarchy that is variously called the *Activity Scale* (Moravcsik 1978, 255–6), the *Nominal Hierarchy* (Dixon 1979, 85), the *hierarchy of salience* (Comrie 1986, 94), the *Extended Animacy Hierarchy* (Croft 1990, 128–32), or the *Referential Hierarchy* (Bickel 2008), among other things—the variation in terminology is at least in part due to a lack of agreement on what makes the categories on the scale a natural class. The grammatical phenomena that have been argued to be governed by this scale, which includes co-occurrence restrictions (recall

21. Such close minimal pairs involving the definiteness restriction in Nyaturu [*Bantu*] are harder to find, mainly due to the extent of variation in Bantu object marking in double object ditransitives: (i) regarding the number of realized object markers (Marlo 2014; 2015; Van der Wal 2017; i.a.), (ii) regarding the order of multiple realized object markers (see Section 2.5.4.1 regarding Sambaa and Haya), and (iii) regarding which object is indexed when one object marker is realized (Bresnan and Moshi 1990; Van der Wal 2017; i.a.). To add to the problem, the sources I currently have at my disposal often do not give full paradigms. Nonetheless, near minimal pairs can be identified: Nyaturu has a (WEAK) person restriction and a definiteness restriction, while Swahili (see Section 2.4) has the same person restriction but no definiteness restriction (the minimal pair is not perfect only due to the exceptional status of 1P.SG/reflexive markers in Nyaturu; see footnote 9).

22. A similar notion had been introduced previously by Hawkinson and Hyman (1974) to deal with object marker patterns in Shona [*Bantu*], although their approach was more inspired by the surface templates of Perlmutter (1968, 1970, 1971) (see Section 2.5.4.1). Hawkinson and Hyman's idea later came to be known as the *Topicality Hierarchy* (see Duranti 1979), which is yet another name used for the scales discussed below.

from Section 2.2 that person restrictions have also been given scale-based analyses), do not always show sensitivity to all the categories on the scale—we saw this in relation to the different types of co-occurrence restrictions above. As a result, the scale is often accordingly divided into several sub-scales:[23]

(14) a. *Person Scale*: 1P, 2P > 3P

b. *Animacy Scale*: 3P human > 3P animal/animate > 3P inanimate

c. *Definiteness Scale*: 3P definite > 3P specific indefinite > 3P non-specific

Where: *Person Scale* > *Animacy Scale* > *Definiteness Scale*

However, we also saw that co-occurrence restrictions occur with categories not included in (14). In such cases, such as with obviation restrictions and deference restrictions, additional sub-scales are posited: (15a) with the former case (see Hockett 1966, 60; Frantz 1966, 53; i.a. regarding Algonquian and Aissen 1997, 707–8 for a more general proposal) and (15b) with the later case (Avelino Becerra 2004, 31; López and Newberg 2005, 8).

(15) a. *Obviation scale*: 3P proximate > 3P obviative

b. *Deference scale*: 3P respectful > 3P familiar

As discussed in Section 2.2, scale-based approaches miss several generalizations when it comes to person restrictions. Most importantly, they leave unexplained the sensitivity of the restrictions to syntactic structure. Additionally, these scales do not really address the question of what makes the categories on the scale a natural class. In fact, any category can be added to the scale in order. A practical example of this can be seen with Comrie's (1979; 1980) analysis of the Chukchi [*Chukotkan*] direct/inverse system, where the person restriction interacts with number in a similar way to the Spanish person restriction noted earlier. In order to capture this interaction, Comrie adds number to his Animacy Hierarchy

23. Another sub-scale that is often assumed is the *Referentiality Scale*: pronoun > proper name > common noun (Croft 1990, 130). However, as co-occurrence restriction pertain only to pronouns, it is omitted here.

(specifically: 3P singular > 3P plural). Recall though that number is never directly restricted in co-occurrence restrictions, which is not what we would expect if number had the same status as person in relation to the mechanism responsible for the restrictions.

The scale-based approach is therefore not explanatory in the sense that any arbitrary category can be placed on a scale. This means that even if one can pin down the unifying trait of the relevant grammatical categories, placing them on a scale is no more meaningful than creating a scale from categories with no such unifying characteristic. As a first step towards a more explanatory approach, I will argue over the next few sections that what makes the relevant categories unique with respect to co-occurrence restrictions is their semantics.

5.2.1 Context-dependency and indexicality

The scale-based approaches highlight what makes unifying all the co-occurrence restrictions difficult: the restricted categories are heterogeneous enough to require distinct sub-scales, but also similar enough so that relating the scales to each other makes intuitive sense. It is clear that we need a more explicit theory of what makes the restricted categories a natural class, and I will argue that the solution lies in their context-dependent semantics. Specifically, all the relevant categories are either indexical or closely tied to indexical categories.

Indexicals are expressions like *you* or *today*, which require information from the utterance context to get a semantic value. In simple terms, their meaning amounts to a set of instructions on how to use context to make reference, among other things, to a particular person, time, or place. What this means in practice can be illustrated by considering the simple sentence *"I can see you"*, which contains two indexical expressions: *I* and *you*.

The sentence is true in any context where the person uttering it can see the person or people it is addressed to. In other words, the meaning of *I* is that it must refer to whoever is the speaker in the relevant context and likewise the meaning of *you* is that it must refer to whoever the addressee is in the same context. This context-invariant aspect of the meaning of indexicals is what Kaplan (1978, 1989) calls *character*. However, what makes indexicals special is the aspect of their meaning that can vary from context to context, which Kaplan calls *content*. Consider two different contexts in which the sentence in question can be uttered. In context c_1 the speaker is Buddy and the addressee is Grant, while in context c_2 their roles are reversed. Even though the sentence has the same character in both cases, the truth conditions change depending on the context in the way outlined in (16).

(16) I can see you.
  a. is true in c_1 iff Buddy can see Grant
  b. is true in c_2 iff Grant can see Buddy

The minimally different sentence in (17), where the subject and object are flipped with respect to person value, likewise yields two distinct truth conditions in c_1 and c_2.

(17) You can see me.
  a. is true in c_1 iff Grant can see Buddy
  b. is true in c_2 iff Buddy can see Grant

Crucially, the truth conditions the sentence in (17) has in c_1 are identical to those the sentence in (16) has in c_2, and the truth conditions the sentence in (17) has in c_2 are identical to those the sentence in (16) has in c_1. Indexicals therefore reveal double dissociation between character and content: expressions with the same character may diverge in terms of content or they may converge on the same content despite having different characters. In short, this means that the reference of an indexical pronoun cannot be determined without considering the value of context parameters, such as the identity of the speaker and addressee.

Following Kaplan (1978, 1989), the meaning of indexicals can be formalized roughly along the lines outlined below. The assignment of a denotation to an indexical α can be represented, as is common practice, as $[\![\alpha]\!]$, the output of which is the context-invariant character of α. The denotation of α relative to a particular context c, or content, can be represented as $[\![\alpha]\!]^c$. In the case of a pronoun, the output of this will be the referent of the pronoun in that particular context. To give a practical example, when I address you, the reader, with: *"I would like you to read it"*, the referent for *I* is determined as shown in (18a) and *you* as shown in (18b). Determining the referent for *it* is a bit trickier, as it will require either an explicit pointing gesture or additional context in the preceding discourse: *"This is my thesis. I would like you to read it."* In the latter case, *it* can be identified as referring to the previously established salient inanimate entity roughly along the lines of (18c).

(18)
a. $[\![I]\!]^c = speaker(c) =$ Adrian
b. $[\![you]\!]^c = addressee(c) =$ the reader
c. $[\![it]\!]^c = salient\text{-}inanimate\text{-}entity(c) =$ my thesis

The extra step required in the case of *it* is used by Kaplan (1989) to propose a split between *pure indexicals*, such as *I*, *here*, or *now* on one hand and *true demonstratives*, such as *it*, *that*, *there*, or *then* on the other. The idea is that the content of pure indexicals can be determined solely from their character and basic context parameters like the speaker, addressee, time and place of an utterance, while in the case of true demonstratives additional pragmatic factors and speaker intentions must be considered.[24] I argue that what is relevant for co-occurrence restrictions is indexicality in the broad sense, where the denotation of an expression can only be determined relative to a particular context. As I discuss next, three of the categories sensitive to co-occurrence restrictions in addition to person (obviation, definiteness, and deference) clearly display this kind of context-dependence, and the fourth

24. Other ways to divide indexicals have been proposed. For example, Perry (1997) distinguishes *discretionary indexicals*, which require or allow the speaker to have options in determining the referent, and *automatic indexicals*, which do not. Radulescu (2018), on the other hand, proposes that the relevant criterion is whether the context parameters are fixed independently of the sentence being uttered (*indexicals*) or specifically for that sentence (*demonstratives*). Finally, Mount (2008) makes a case against dividing indexicals into two types regardless of criteria, which is the best fit in terms of which conception of indexicality matters for co-occurrence restrictions (see below). Although, any sub-division of indexicals is compatible with the proposal below, as long as it acknowledges that the two classes are still indexical in a broader sense.

(animacy), although not context-dependent in the same sense, is clearly in some kind of dependency relation with person as well as the other three categories.

5.2.1.1 Obviation

Since Hockett's seminal works on Algonquian morphosyntax (Hockett 1939, 1948, 1966) obviation marking in Algonquian and beyond has received numerous analyses, focusing almost exclusively on the morphological and syntactic sides of the phenomenon.[25] The formal semantic basis for the proximate/obviative opposition has, on the other hand, received almost no attention until very recently (see Oshima 2006, 2007; Mühlbauer 2008; Bittner 2014). The existing proposals on the semantics of obviation marking differ, often significantly, in the details of the analysis, but ultimately they all seek to derive the same two primary functions of obviation marking: proximate singles out arguments the speaker empathizes with, and the proximate/obviative opposition ensures unambiguous reference tracking across clauses as well as longer spans of discourse (see Ch. 4 of Dahlstrom 1986a, 1991 for a detailed discussion of the two functions with examples from Plains Cree).

The indexical nature of proximate arguments that signal speaker empathy is fairly obvious, given that it is intrinsically tied to the speaker of the relevant context. In the case of reference tracking, the indexical status of obviation marking requires a bit more elaboration, so I will first establish the basic pattern and tackle the issue of indexicality afterwards by considering in more detail the relationship between the two functions of obviation marking.

In the most basic cases of reference tracking, obviation can disambiguate the referents of 3P pronouns in embedded clauses. In English, a sentence like *'The man*$_i$ *told his friend*$_j$ *that he*$_{i/j}$ *saw him*$_{j/i}$*'* is ambiguous with respect to the referents of the two bound pronouns in the embedded clause. As shown in (19), this is not the case in an obviation marking language like Blackfoot, where the obviation status of the two pronouns resolves the ambiguity.

25. See footnote 1 for references on crosslinguistic work and footnote 2 for references on Algonquian.

(19) Blackfoot [*Algonquian*]:
om-*a* ninaa-*wa* it-anist-**sii-wa** om-*i* ot-akka-*yi* …
that-3 man-3 then-say-3>OBV-3(SU) that-OBV 3-friend-OBV
'The man$_i$ (PROX) then told his friend$_j$ (OBV) that … '

a. ots-ino-**a**-hs-ayi. (DIRECT)
3-see-3>OBV-CONJ-3
'… he$_i$ (PROX) saw him$_j$ (OBV).'

b. ots-ino-**yi**-ss-ayi. (INVERSE)
3-see-OBV>3.INV-CONJ-3
'… he$_j$ (OBV) saw him$_i$ (PROX).' (Frantz 1971, 42)

The key here is that 'the man' is established in the matrix clause as proximate and 'his friend' as obviative. The direct marking on the verb in (19a) signals that the pronoun referring to the PROX argument is the subject and the pronoun referring to the OBV argument is the object. In contrast, the inverse marking in (19b) is the result of the restriction against *OBV ≫ PROX (see Section 5.1.1) and thus signals that the subject pronoun refers to the OBV argument and the object pronoun refers to the PROX argument from the matrix clause.

The function of obviation marking in (19) is very similar to switch-reference marking (marking same/different reference between subjects/objects across joined clauses; see e.g. Haiman and Munro 1983; McKenzie 2015), however there are key differences between switch-reference and obviation that become apparent when longer spans of discourse are considered. The narrative passage in (20) from Kutenai [*isolate*], a language with a system of obviation marking very closely resembling the one of Algonquian languages (Dryer 1992), nicely illustrates the discourse function of obviation marking. In this passage, 'Chickadee' is first established by the speaker in (20a) as proximate for the relevant span of discourse (to be elaborated on), and in the following sentences (20b–d) 'Chickadee' is understood as the subject—that the proximate argument is the subject in all three is reflected by the absence of obviation marking on the verbs, which are only inflected with the indicative suffix **-(n)i**, as there are no overt verbal pronominal markers for proximate 3P in Kutenai.

(20) Kutenai:

a. taxa-s mi¢̓qaqas$_i$ ¢̓inax-**i**‿¢ ʔat n-ʔaq¢xuʔ-**ni**$_i$ łaǹ$_j$-**[ʔ]is**$_i$
then-OBV chickadee go-IND-and HAB PRED-pound.holes-IND moccasins-3.POSS
'Then Chickadee$_i$ (PROX) went and **he**$_i$ (PROX) would pound holes in **his**$_i$ (PROX) moccasins$_j$ (OBV).'

b. ʔat ła ¢̓inax-**i**$_i$
HAB REV go-IND
'**He**$_i$ (PROX) would return.'

c. ʔat qak-ił-**ni**$_i$ ka·kin-s$_k$ k-ʔumi¢-ik-**i**$_i$ łaǹ$_j$-**[ʔ]is**$_i$
HAB say-TR-IND wolf-OBV SUBR-break-REFL-IND moccasins-3.POSS
'**He**$_i$ (PROX) would tell Wolf$_k$ (OBV) that **he**$_i$ (PROX) wore out **his**$_i$ (PROX) moccasins$_j$ (OBV).'

d. k-qa-tał łaxam$_i$
SUBR-NEG-can arrive
'that **he**$_i$ (PROX) couldn't make it there.' (Dryer 1992, 157)

In the very next sentence in the narrative, the obviative argument 'Wolf' is the subject, and this is reflected by the presence of the inverse morpheme **-aps** (due to *OBV ≫ PROX; in intransitives with an OBV subject, the verb is inflected with the obviative suffix **-s**):

(21) łaʔak̓łak-s ʔat qa·nmił hamat-**ik¢**$_i$-**aps-i**$_k$ łaǹ-s$_l$ (INVERSE)
different-OBV HAB quickly give-IO-INV-IND moccasin-OBV
'**He**$_k$ (OBV) would quickly hand **him**$_i$ (PROX) different moccasins$_l$ (OBV).'
(Dryer 1992, 158)

Note that the same entity is proximate throughout the whole passage above. A span of discourse for which a particular entity is proximate is often referred to as an *obviation span* (Aissen 1997, 2001).[26] A new obviation span begins when the speaker establishes a new referent as proximate, as in (22), which shortly follows the passage in (20)–(21) (see Dryer 1992, 140–5 on the conditions under which this can occur in Kutenai; crucially, the exact conditions may differ slightly from language to language). In these instances of so-called

26. For at least some Algonquian languages, it has been noted that occasionally multiple entities may have proximate status within an obviation span—one typically being more prominent than others; see e.g. Goddard 1984, 1990 regarding Meskwaki/Fox and Dahlstrom 1986a, 1991 regarding Plains Cree. Such cases can be identified when two proximates within a sentence are not coreferential—this can occur when the two are in different clauses within the sentence or when they are in different conjuncts of a conjunction.

proximate shift (Goddard 1984, 1990), the new proximate does not have to be a newly introduced individual—recall that 'Wolf' was obviative in the preceding discourse.

(22) taxa-s ka·kin$_k$ n-ʔin-**i** ¢̓inax-**i**
then-OBV wolf PRED-be-IND go-IND
'Then Wolf$_k$ (PROX) went himself.' (Dryer 1992, 158)

The difference between switch-reference and obviation marking should be clearer now. Within an obviation span, the proximate/obviative status of 3P arguments is used for reference tracking as well as to disambiguate co-reference and disjoint reference between 3P arguments across joined clauses, however the latter is simply a consequence of the asymmetry between the two types of 3P: there is one proximate per obviation span, so two proximates will always be co-referential, whereas a proximate and an obviative will always have disjoint referents within the obviation span. In contrast, switch-reference is exclusively used for marking same/different reference across joined clauses. Additionally, the proximate/obviative also operates clause-internally (e.g. a simple matrix clause may contain both proximate and obviative 3P arguments), whereas switch-reference only operates over clause junctures (see Camacho 2017 for additional discussion).

Recall that reference tracking is not the only function of obviation: proximate status signals that the speaker empathizes with the referent. Oshima (2006, 2007) even argues that empathy marking is the primary function of obviation—reference tracking is only a side effect of having two distinct 3P categories. In a similar analysis, Mühlbauer (2008) characterizes obviative referents as those that from the speaker's point of view lack a perspective on the event under discussion.[27] Mühlbauer also argues that obviative is morphosyntactically not a primitive category: it is constructed from several mechanisms that mark

27. Part of the motivation for this is that in Plains Cree, which Mühlbauer (2008) focuses on, and more generally in Algonquian inanimate entities, which cannot hold a perspective, do not take part in obviation marking. This does not extend to obviation marking in Kutenai though (see Dryer 1992; Appelbaum 2019), although this could also be connected to the absence of grammatical animacy distinctions in the language.

referential dependencies, which accounts for its role in reference tracking. Note though that in both analyses the significance of a proximate/obviative status is directly tied to the perspective of the speaker in the given context. Thus, despite a different implementation, the two analyses converge on treating obviation as indexical. Furthermore, if reference tracking is really just an aspect of empathy marking, then reference tracking via obviation must by extension also be indexical. Although, as we will see next, the indexical nature of reference tracking can be established even without reference to empathy marking.

In Algonquian languages, 1/2P and proximate sometimes form a natural class for morphological and syntactic processes (including person restrictions), which has led some to suggest that proximate is really an additional person value (see e.g. Lochbihler 2013; Oxford 2014).[28] Bittner (2014) notes the same kind of grouping in Kalaallisut [*Inuit*] and develops a semantic analysis that characterizes the natural class based on how the reference of 1/2P and proximate pronouns is determined from context parameters.[29] The core idea of Bittner's proposal is that discourse referents are ranked (building on Dekker 1994) and, crucially, divided into two prominence-based sequences: the center of attention sequence and the background sequence, where discourse referents are put in the first sequence either by the speech act itself (e.g. establishing the speaker and addressee) or by the speaker establishing a new proximate referent. What unifies 1/2P and proximate pronouns under this view is that they both pick out referents from the center of attention sequence.[30]

28. This idea has some precedence in the notational convention sometimes used in Algonquianist circles to distinguish proximate and obviative 3P, where the former is classified as *third person* and the latter as *fourth person* (Uhlenbeck 1938, 29 attributes the terminology to Father van Ginneken; see also Frantz 1966).

29. Bittner (2014) calls proximate pronouns *topic-oriented anaphora*, equating proximate status with topic status and obviative status with background information. Bittner's use of "topic" is somewhat nonstandard (cf. the traditional *topic/comment* and *focus/background* oppositions): she characterizes topics as being the current center of attention, as opposed to background information (see below). Related to this, Dahlstrom (2017) has argued that proximates cannot be equated with topics or foci in the more standard sense.

30. Bittner (2014) takes her argument even a step further and argues for the abandonment of Kaplan's (1978) distinction between the *utterance context*, used to determine reference for pure indexicals, and the *assignment function*, used to determine reference for anaphors, replacing both with the sequences of discourse referents. While this view fits rather well with the analysis I propose in Section 5.3, it is not crucial for its adoption.

What is important here is that Bittner, who concerns herself only with reference-tracking, argues that the way in which contextual information is used to determine a referent with proximate pronouns is the same as with pure indexicals. In other words, a proximate referent is a context parameter for a given obviation span just like speaker and addressee are context parameters for a given utterance—the main difference is that the latter are inherent to an utterance, while the former must be established deliberately. It therefore makes sense to consider the proximate/obviative opposition indexical in the broad sense.

5.2.1.2 Definiteness

Proximate referents are basically a special case of discourse referents, so it is natural to ask in light of the preceding discussion if discourse anaphoric pronouns in general are indexical in the same way as proximate pronouns? Kaplan (1989) originally argued that 3P pronouns are only indexical when used deictically: accompanied by a pointing gesture or if the intended referent is salient enough in the context without pointing. Therefore *discourse anaphora* (also *unbound anaphora* or *pragmatic anaphora*) like *he* and *it* in (23), co-referring with expressions used previously in the discourse, were not considered indexicals.

(23) Obelix$_i$ isn't allowed to drink the potion$_j$. He$_i$ fell into a cauldron of it$_j$ as a boy.

However, with the development of *dynamic semantics* (Kamp 1981; Heim 1982; Kamp and Reyle 1993; i.a.), information derived from previous linguistic discourse is now commonly considered to be part of the context (this is also a key assumption of Bittner's 2014 analysis of proximate pronouns outlined in the previous section). Under this view, discourse anaphoric pronouns become indexical in the same sense as deictic/demonstrative pronouns. Crucially, the nature of the contextual information relevant in determining the referent of a discourse anaphor is intrinsically connected to definiteness and specificity.

A specific indefinite expression can serve as an antecedent to either a discourse anaphoric pronoun or a definite expression, as shown in (24a). However, the reverse is not the case, so *a menhir* in (24b) must either refer to a new menhir distinct from the one in the preceding discourse or be non-specific (meaning "no menhir will fit on the carriage").

(24) a. Obelix brought a menhir$_i$, and $\left\{ \begin{array}{c} \text{it}_i \\ \text{the menhir}_i \end{array} \right\}$ won't fit on the carriage.

b. Obelix brought $\left\{ \begin{array}{c} \text{it}_i \\ \text{the menhir}_i \end{array} \right\}$, and a menhir$_{j,*i}$ won't fit on the carriage.

The behavior of indefinites can be captured in dynamic semantics by characterizing them in terms of their contribution to the context, namely: indefinite expressions must introduce a new discourse referent. When a pronoun or definite expression "refers back" to an indefinite expression, as in (24a), it really refers to a discourse referent, which is a discourse entity and part of the context—the appropriate discourse referent is identified via the referential index it shares with the indefinite expression that introduced it. Just like unbound anaphora, definite expressions are thus indexical because their semantic value is context-dependent.

Note that specific indefinites cannot be indexical in the same sense—their referent is not determined based on the context, instead they introduce a new discourse referent into the context. Crucially, the introduction of a discourse referent also means a new referential index—an index not associated with an existing discourse referent. To ensure this, the indices of old discourse referents in a given context must be accessed when a specific indefinite is interpreted (i.e. when it introduces a discourse referent). The semantic contribution of specific indefinites is thus also context-dependent and indexical in the broad sense.

5.2.1.3 Deference

Context-dependency is also a key characteristic of deference marking (see e.g. Fillmore 1975, Comrie 1976, Levinson 1983, 89–94 also regarding other manifestations of *social* *deixis*) and so-called *referent honorifics* in particular. With referent honorifics, the choice of the appropriate form of a pronoun or noun phrase is governed by the social status of the referent of the pronoun or noun phrase relative to the speaker.[31] A typical example of referent honorifics are Indo-European 2P singular T/V pronouns (e.g. French informal *tu* vs. formal *vous* or Russian informal *ty* vs. formal *vy*). More importantly, the Sierra Zapotec familiar vs. respected 3P pronouns discussed in Section 5.1.4, which form the basis for the deference-based co-occurrence restriction, also exemplify a referent honorific system.

The indexical component is clear with referent honorifics, given that the social relationships that govern the choice of pronoun forms are crucially evaluated with respect to the speaker (although other context parameters may also play a role). A given formal or informal pronoun can thus refer to different individuals in different contexts, but also the same individual may be referred to using an informal or formal pronoun, depending on who the speaker is and what the level of familiarity is between the speaker and referent.

5.2.1.4 Animacy

At first glance, animacy appears to be the exception to the link between co-occurrence restrictions and context-dependence, as it may be used for noun classification in the same way as gender, in which case its semantic contribution is context-invariant. However, animacy is actually intricately linked to all the indexical categories discussed above.

The most obvious link is the dependency between animacy and person: being human is a prerequisite for being a speech act participant (speaker or addressee) and animacy is in turn a prerequisite for being human. Related to this, being human is also a prerequisite for referent honorific marking,[32] since the relevant social relationships can only hold between humans

31. As opposed to, for example, Japanese/Korean-style *addressee honorific* systems, where the choice of honorific forms is always sensitive to social status of the speaker and addressee, regardless of who or what the referent of the honorific-marked element is (Comrie 1976, Levinson 1983, 90).

32. A way to get around this is through personification of inanimate referents. In relation to this strategy, see also the discussion below regarding the grammatical effects of personification.

(recall here the Sierra Zapotec system, where the familiar/respectful opposition is limited to human 3P pronouns). Obviation and definiteness may also depend on animacy, although not always systematically. The proximate/obviative opposition is crosslinguistically often limited to animate 3P nouns and pronouns (see Aissen 1997; Zúñiga 2002; 2006; Oshima 2007; i.a.). Similarly, animacy or humanness may, in some languages and under the right circumstances, force a definite or specific reading or at least be a prerequisite for it.

In most English wh-questions *what* is non-specific (the answer to *what* is not from set of established discourse referents), whereas *who* can be non-specific or specific. Kuroda (1968) shows this by contrasting *what* and *who* with the always specific *which (one)*:

(25) a. You may read *Syntactic structures* or *La nausée*.
b. $\left\{\begin{array}{c}\text{Which one}\\ \text{\#What}\end{array}\right\}$ do you prefer to read?

(26) a. You may see Chomsky or Sartre.
b. $\left\{\begin{array}{c}\text{Which one}\\ \text{Who}\end{array}\right\}$ do you prefer to see? (Kuroda 1968, 252)

In this case, humanness is a prerequisite for a specific interpretation.

A different contrast between *what* and *who* is observed in questions formed with intensional verbs: *what* can be either specific or non-specific, while *who* can only be specific. As noted by Moltmann (1997), the answer to a *what*-question in such cases may be a non-specific indefinite, whereas this is impossible with a corresponding *who*-question:

(27) a. What is John looking for? – A secretary.
b. #Whom is John looking for? – A secretary. (Moltmann 1997, 6)

In this case, using the human *who(m)* forces a specific interpretation.

The contrast can also be observed with syntactic phenomena sensitive to specificity. For example, in Macedonian clitic-doubling occurs with specific and definite DOs; cf. (28).[33]

33. Note that Browne (1970) considers definiteness the relevant factor for clitic-doubling in Macedonian, but later works have shown that indefinites may also be doubled and that the relevant factor is specificity (see Berent 1980, 171–2; Rudin 1997, 242–3; Franks and King 2000, 72–4; i.a.).

(28) Macedonian [*Slavic*]:

a. Vidov eden čovek.
saw.1 a/one man
'I saw a man.'

b. **Go** vidov čovek-ot.
3.M.ACC saw.1 man-the
'I saw **him** [the man].' (Browne 1970, 267)

As discussed by Browne (1970), clitic-doubling in Macedonian is constrained by the choice of wh-phrase: *who*-questions permit clitic-doubling, whereas *what*-questions do not:

(29)

a. Kogo barate?
who.ACC look.for.2
'Who are you looking for?'

b. Kogo **go** barate?
who.ACC 3.M.ACC look.for.2
'Who is **it** you're looking for?'

(30)

a. Što barate?
what.ACC look.for.2
'What are you looking for?'

b. *Što **go** barate?
what.ACC 3.M.ACC look.for.2
'What is **it** you're looking for?'
(Browne 1970, 269)

This is because *kogo* ('whom') may be either specific or non-specific, whereas *što* ('what') can only be non-specific (the clefts in the translations are used to indicate a specific reading).

Another case where the link between animacy and definiteness is observed is with DOM. For example, DOM (generally) occurs in Spanish with objects that are both human and definite (see Bossong 1991; Torrego 1998; Rodríguez-Mondoñedo 2007; i.a.).

In sum, animacy and humanness can directly interact with indexical categories in several different ways: (i) animacy or humanness may be a prerequisite for the expression of all the values of the indexical category, (ii) animacy or humanness may force a particular value of the indexical category, and (iii) animacy or humanness may be required together with a particular value of the indexical category in order to trigger a morpho-syntactic process.

It is also possible to make the stronger claim that animacy and humanness are themselves context-dependent. Wiltschko and Ritter (2015, 899–905) note that lexically inanimate nouns can behave morpho-syntactically like human nouns when personified. For example, DOM in Spanish can exceptionally occur with inanimate specific direct objects when their referents are personified (see Wiltschko and Ritter 2015, 901 regarding a similar effect in Blackfoot [*Algonquian*], interesting in light of its animacy-based noun classification system). I leave open whether or not this is sufficient evidence to claim that animacy is indexical. What is much clearer is that there exist dependency relations between animacy and clearly indexical categories—dependencies that are do not exist in the case of number or gender.

5.2.1.5 Number and gender

It is fairly clear that the semantic contribution of number is context-invariant. There is no dissociation between semantic character and content with number: what counts as plural in one context will count as plural in all other contexts, and the same holds for other number values. Grammatical gender and other categories used for noun classification are context-invariant in a different sense: they are lexically determined, so a given noun is more or less arbitrarily assigned a gender/class value (e.g. "car" is *masculine* in Slovenian, *feminine* in Italian, and *Class 5* in Swahili). Even when a gender/class is exclusive to nouns with shared lexical semantic traits (e.g. the 'edible plant food' class in Dyirbal [*Pama-Nyungan*]; see Dixon 2015, Ch. 2) the gender/class value itself makes no semantic contribution—it only highlights an aspect of their lexical semantics which they all have in common.

In contrast, semantic/natural gender seems to show a kind of context-dependence: the gender of individuals may change from one point in time to another (e.g. due to mistaken identity or actual gender change). However, this is not due to linguistic considerations, as once the gender identity of an individual is extra-linguistically established, (pro)nouns of the appropriate gender can be used to refer to that individual regardless of changes in context.[34]

34. As observed by Stokke (2010, 96–101), number and gender also differ from person in allowing mismatches between pronoun and referent: e.g. if *she* is wrongly used to refer to a male this does not necessarily result in reference failure—we can understand it refers to the male individual. This is not the case if a speaker wrongly uses a 1P pronoun to refer to another person, usually Napoleon. The pronoun cannot be understood as

Number and gender/class are thus semantically context-invariant, unlike the categories discussed previously. Recall that number and gender/class are also singled out with respect to the other categories when it comes to syntactic co-occurrence restrictions, in that their values are never directly restricted in the relevant syntactic contexts.

5.2.1.6 Overview of indexical and non-indexical categories

The motivations for the indexical status of individual categories in the discussion above were quite varied, so it would be useful to go over the relevant categories again and highlight the semantic trait that unifies the categories that can be restricted in co-occurrence restrictions.

Recall that person exhibits a double dissociation between the semantic character and content of 1P and 2P pronouns: a 1P pronoun may refer to different individuals depending on the context (same character—different content), and likewise the same individual may be referred to by either a 1P or a 2P pronoun depending on the context (same content—different character). The same kind of double dissociation is in fact characteristic of all the categories that have been identified as indexical in the previous sections.

For example, in the case of obviation, different individuals may be proximate depending on context parameters—specifically, which discourse referent is currently in the center of attention. This also means that the same individual may be proximate in one context and obviative in another. In the case of deference, a familiar pronoun may be appropriate to refer to different individuals depending on the context—specifically, depending on who the speaker is and what their relationship is with the referent. For example, it may be appropriate for most Gaulish villagers to use a familiar pronoun for the easygoing hero Asterix, but only some of them may be in a close enough relationship with the respected druid Panoramix to do the same for him. This also means that either a familiar or a respectful pronoun can be

referring to Napoleon, it can only mean the speaker thinks they are Napoleon. Sudo (2012, 142–4) takes this to mean that person features are not presupposition triggers, in contrast to other φ-features (cf. Cooper 1983).

used to refer to Panoramix, depending on who the speaker is. Finally, in a given context different entities will be definite, depending on whether they were established as discourse referents or they are otherwise salient in the discourse. As in the previous two cases, this also means that a particular entity will be definite in some contexts but not in others.

Contrast this with animacy, number, and gender specifications. These may all restrict reference to a different subset of entities in different contexts, but the same individual or set of individuals can not correspond to different specifications of those categories in different contexts (barring extra-linguistic considerations—see discussion of semantic/natural gender in the previous section). The relevant difference in character/content dissociation between the two classes of categories is illustrated in Table 5.1.

Table 5.1: Dissociation between semantic character and content

	CHARACTER—CONTENT		CONTENT—CHARACTER	
PERSON	*speaker*	c_1: Buddy c_2: Grant	Buddy	c_1: *speaker* c_2: *addressee*
OBVIATION	*proximate*	c_1: Chickadee c_2: Wolf	Chickadee	c_1: *proximate* c_2: *obviative*
DEFERENCE	*familiar*	c_1: Asterix c_2: Panoramix	Panoramix	c_1: *familiar* c_2: *respected*
DEFINITENESS	*definite*	c_1: potion c_2: menhir	menhir	c_1: *indefinite* c_2: *definite*
ANIMACY	*human*	c_1: druid c_2: legionnaire		
NUMBER	*plural*	c_1: Gauls c_2: Romans		
GENDER	*feminine*	c_1: Cleopatra c_2: Falbala		

What distinguishes the two classes of categories is that only with person, obviation, deference, and definiteness different specifications correspond to different discourse entities or roles. Unlike animacy, number, or gender, the indexical categories have no corresponding concepts outside the realm of discourse. This intrinsic connection to discourse is what I take to be the crucial defining characteristic of the indexical categories.

Animacy is thus the only non-indexical category that can be restricted in co-occurrence restrictions. Recall though that, unlike number and gender, animacy stands in a dependency relation with 1P and 2P: only human, and therefore animate, entities may be speech act participants. Additionally, having an animate or human specification can constrain the specification of other indexical categories. This close relationship with the indexical categories seems to be sufficient to place animacy in the same class as person, obviation, deference, and definiteness when it comes to co-occurrence restrictions.

The question that remains is why should semantics matter for syntactic co-occurrence restrictions? The first step in answering this will be to establish that the indexical vs. non-indexical semantic distinction can have an effect on syntax outside of co-occurrence restrictions, namely in the case of selectional restrictions.

5.2.2 Indexicality and selectional restrictions

A predicate may in addition to determining the lexical category of its arguments (*c-selection*) also determine their semantic content (*s-selection*). In the latter case, the meaning of the predicate restricts the possible arguments to those that are semantically compatible with the predicate; for example, the verb *drink* may only take patients that denote liquids. Along the same lines, predicates may select arguments based on their number or gender.

Collective predicates like the verb *gather* only take plural agents, as shown in (31), while verbs like *collect* only take plural patients, as shown in (32).

(31) a. They gathered on Palm Sunday.
b. *He gathered on Palm Sunday.

(32) a. Kilgore collects stamps.
b. *Kilgore collects the Inverted Jenny.

Examples of gender selection in English, a language without grammatical gender, are not as clear cut and mainly limited to biological/anatomical considerations as they pertain to differences in biological sex rather than gender (e.g. *be pregnant* or *castrate*). Clearer cases can be found in languages with rich systems of noun classification, such as Mawng (Singer 2016). For example, the verb *-waraw* 'light (fire)' may only take objects of the so-called *vegetation gender* (VE), which is the gender used for most trees, plants, types of fires and some wooden artifacts (Singer 2016, 89). An example of this verb taking the object *nganangka* 'clapsticks', a wooden musical instrument, is shown in (33).

(33) Mawng [*Iwaidjan*]:

La	arrkana-waraw-n	mata	nganangka.
And	1PL.INCL>3VE.FUT-light.fire-NPAST	VE	clapsticks

'We will burn the two clapsticks.' (Singer 2016, 90)

Objects not of the vegetation gender are infelicitous with *-waraw*, and can only be used with the underspecified *-wunya* 'burn' instead (Singer 2016, 90). Crucially, even trees that are of masculine gender in Mawng are infelicitous as objects of *-waraw* (Singer 2016, 91).

A way to put this kind of restriction in perspective is to think of it as an issue of availability of lexical items. Mawng has two distinct verbs that correspond to the English *burn*, each limited to specific classes of nouns. Similarly, one can conceive of a language in which the distinction present in English between the collective *gather* and under-specified *show up* does not exist. To a speaker of this language, the *gather* vs. *show up* distinction would seem just as alien as the *-waraw* vs. *-wunya* distinction does to English speakers.

The important point here is that while such lexical distinctions in predicates can form along oppositions in number or gender, they never form along oppositions in person, obviation, deference,[35] and definiteness—the indexical categories.

Perhaps unsurprisingly, animacy is again somewhat of an outlier. There are definitely cases where predicates are incompatible with an animate/human agent, such as with some verbs of sound emission, as shown in (34) for *ring* (the relevant meaning here is one where the agent directly produces the ringing sound). Similarly, there are plenty of verbs that require an animate/human agent, like the psych predicate *afraid* in (35).

(34) a. The phone rang.
b. #John rang. (Folli and Harley 2008, 192)

(35) a. Obelix isn't afraid of anything.
b. #The menhir isn't afraid of anything.

It is not entirely clear that such examples show sensitivity to animacy status rather than some other property that merely aligns with it. Folli and Harley (2008) argue that many apparent cases of selection for animate agents really involve selection for arguments internally capable of generating the entire event described by the predicate (in the case of *ring* in (34) this holds for the phone but not for John). With psych verbs it is also reasonable to say that what is relevant is the ability to have a mental state (being a rock, a menhir is technically not afraid of anything, but what makes (35b) infelicitous is that a menhir can never have a mental state). Pinpointing the status of animacy in selectional restrictions is beyond the scope of the current discussion. What matters is that even if animacy turns out to pattern with number and gender regarding selection, this would not be an issue. This is because animacy, despite not being an indexical category itself, is connected to indexical categories in a way that number and gender are not (a point that I return to in Section 5.3).

35. Recall that what matters here is referent honorification (see discussion in Section 5.2.1.3). There are definitely lexical choices of this kind based on other kinds of deference; e.g. Japanese/Korean style addressee-based honorification, in which case the lexical choices even extend beyond just predicates.

The relevant generalization is then that predicates never select for specific values of indexical categories. The only prima facie potential counterexample are Japanese empathy marking verbs (Kuno and Kaburaki 1977; Kuno 1987),[36] although we will see that these in fact provide further support for the special status of indexical features in relation to selection. The example usually used to illustrate the relevant pattern is the contrast between the two 'give' verbs: *yaru* (*subject-empathy*) and *kureru* (*dative-empathy*). The meaning of the former, used in (36a), roughly corresponds to 'giving that benefits the subject', while the latter, used in (36b), roughly corresponds to 'giving that benefits the indirect object/dative'. The apparent person selection effect here is that *yaru* cannot take a 1P indirect object, as shown in (37a), whereas *kureru* cannot take a 1P subject, as shown in (37b).

(36) Japanese:

a. Taroo-wa Hanako-ni okane-o **yat**-ta. — subject-empathy
Taroo-TOP Hanako-DAT money-ACC give-PAST
'Taroo gave money to Hanako (benefiting Taroo).'

b. Taroo-wa Hanako-ni okane-o **kure**-ta. — dative-empathy
Taroo-TOP Hanako-DAT money-ACC give-PAST
'Taroo gave money to Hanako (benefiting Hanako).'

(37) a. *Taroo-wa boku-ni okane-o **yat**-ta. — ✗ subject-empathy / 1.IO
Taroo-TOP me-DAT money-ACC give-PAST
'Taroo gave money to me (benefiting Taroo).'

b. *Boku-wa Taroo-ni okane-o **kure**-ta. — ✗ dative-empathy / 1.SU
I-TOP Taroo-DAT money-ACC give-PAST
'I gave money to Taroo (benefiting Taroo).' (Kuno and Kaburaki 1977, 630–1)

The examples above seem to suggest that *yaru* selects for a 2/3P indirect object (cf. (36a) vs. (37a)) and *kureru* selects for a 2/3P subject (cf. (36b) vs. (37b)). However, the pattern

36. A superficially similar example is found in Manambu [*Sepik*] (Aikhenvald 2008, 86–9), where the verb 'give' has different forms when it occurs with a 1/2P Goal (*kwatiya-*) and when it occurs with a 3P Goal (*kui-*). However, there is no difference in meaning between the two, and *kui-* is also sometimes used with 1/2P Goals, so it seems more likely that the alternation involves some type of agreement driven suppletion.

changes in embedded clauses; for example, if a clause is embedded under an attitude verb, as in (38), *yaru* can take a 1P indirect object and *kureru* can take a 1P subject.[37]

(38) a. Taroo$_i$-wa [zibun$_i$-ga boku-ni purezento-o **yat**-ta]-to omotte-i-ru.
Taroo-TOP self-NOM me-DAT gift-ACC give-PAST -COMP believe-ASP-PRES
'Taroo$_i$ believes that he$_i$ gave me a gift (benefiting him$_i$).'

b. Taroo$_i$-wa [boku-ga zibun$_i$-ni purezento-o **kure**-ta]-to omotte-i-ru.
Taroo-TOP I-NOM self-DAT gift-ACC give-PAST -COMP believe-ASP-PRES
'Taroo$_i$ believes that I gave him$_i$ a gift (benefiting him$_i$).' (Oshima 2006, 194–5)

This does not mean that there are no restrictions on the arguments of *yaru* and *kureru* in such contexts. For example, while the anaphor *zibun* can be the subject in (38a) and the indirect object in (38b), it cannot be replaced in either case by a bound pronoun (even though bound pronouns are grammatical with other verbs in the same syntactic context):

(39) a. ??Taroo$_i$-wa [kare$_i$-ga boku-ni purezento-o **yat**-ta]-to omotte-i-ru.
Taroo-TOP he-NOM me-DAT gift-ACC give-PAST -COMP believe-ASP-PRES
'Taroo$_i$ believes that he$_i$ gave me a gift (benefiting him$_i$).'

b. ??Taroo$_i$-wa [boku-ga kare$_i$-ni purezento-o **kure**-ta]-to omotte-i-ru.
Taroo-TOP I-NOM him-DAT gift-ACC give-PAST -COMP believe-ASP-PRES
'Taroo$_i$ believes that I gave him$_i$ a gift (benefiting him$_i$).'(Oshima 2006, 194–5)

The restriction is really not about person, but about empathy marking. Recall that if *yaru* and *kureru* are used with a 3P subject and 3P indirect object (cf. (36)), the choice of verb reflects empathy with the subject (*yaru*) vs. the indirect object (*kureru*). The restrictions in (37) and (39) arise because some arguments, such as 1P pronouns and logophors, inherently attract empathy (speakers always empathize the most with themselves), which can lead to a conflict between the locus of empathy signaled by the verb and that signaled by the arguments (Kuno and Kaburaki 1977; Kuno 1987; Oshima 2006). Thus, in (37a) *yaru* signals empathy with

37. The restriction is also different in matrix questions (p.c. Hiroaki Saito, Yuta Tatsumi). Although the pattern of the *yaru/kureru* distribution in questions is quite complex and also appears to show significant speaker variation, so I will not discuss it here. However, the point I make in relation to the more frequently discussed matrix/embedded contrast can also be made in relation to the declarative/interrogative contrast.

the subject, but having a 3P subject and a 1P indirect object conflicts with that. In the same vein, *kureru* in (37b) signals empathy with the indirect object, but having a 3P indirect object and 1P subject conflicts with that. The conflict does not arise with the embedded clauses in (38) because the 3P *zibun* is a logophor in those contexts (Oshima 2006), which means it refers to the attitude holder and is on par with 1P pronouns in terms of empathy. When *zibun* is replaced with a plain bound pronoun, as in (39), the empathy conflict comes back.

What is important here is that neither *yaru* nor *kureru* directly constrains the person of their arguments, they just encode that the subject or indirect object are the arguments we should empathize with. Person is merely one of the factors considered in determining acceptable loci of empathy. Thus, even what looks like a prima facie case of person selection actually also shows that person values are never directly selected by the predicate.

The main take away here is that the distinction between indexical and non-indexical categories matters for selection, and thus there is at least another syntactic phenomenon apart from co-occurrence restrictions where this semantic distinction matters. I will not explore here how the selection facts should be analyzed, but what they illustrate is that the manner in which arguments get their denotation can influence syntax, even though the syntactic derivation precedes semantic interpretation. In the next section, I argue that the special syntactic status of person and other indexical categories can actually be made to follow from their semantics without jeopardizing the autonomy of syntax.

5.3 The person trail

I suggest that the oppositions within each indexical category (proximate/obviative, definite/indefinite, familiar/respected, etc.) reflect different values of interpretable morphosyntactic features. While primarily relevant for semantic behavior, the different specifications can, as we saw, can also give rise differences in morphological expression, which is consistent with them being encoded via features in the syntax. I will refer to these as *indexical features* below, and treat them as a sub-class of φ-features. In line with the analysis of person restrictions introduced in Chapters 3 and 4, I propose that co-occurrence restrictions that arise with other indexical categories are also the result of a failure to value the relevant features on minimal pronouns. Recall that valued person features were argued to be limited to phase heads, and I propose that this holds for all indexical features. Furthermore, the limited distribution of valued indexical features will be argued to be the result of indexical semantics interacting with syntactic principles of economy.

Crucially, I am not proposing that all indexical features on minimal pronouns must be unvalued. Rather, I argue that only indexical features can be unvalued and that having unvalued indexical features of any kind is a characteristic of minimal pronouns. The reason for this is the dissociation between different kinds of co-occurrence restrictions noted in Section 5.2: not only can specific restrictions exist in the absence of others, but there are also minimal pairs of languages/constructions that differ only in the presence of a specific restriction. This is captured straightforwardly by allowing each class of indexical features to be unvalued independently of the valuation status of other indexical features. It is possible that there are typological generalizations yet to be identified regarding the distribution of specific co-occurrence restrictions and therefore the distribution of specific unvalued indexical features. However, identifying them goes beyond the scope of the current survey, which is set up specifically to identify typological gaps in the case of person restrictions.[38]

Of course, animacy is again an outlier, since animacy-based co-occurrence restrictions exist even though the indexical status of animacy is at best uncertain. One way to incorporate animacy with indexical features is to say that animacy status is encoded in the syntax as part

38. A promising line of inquiry would be to determine whether the distribution of (un)valued indexical features is more of a continuum. Franks (2016) suggests some minimal pronouns can have valued participant but unvalued speaker features (see footnote 26 in Section 4.3.2). Something similar could be at play with indexical features more broadly and could be constrained in terms of implicational relationships between them.

of the person feature system (see e.g. Anagnostopoulou 2003, 2005; Adger and Harbour 2007; Richards 2008a), building on the implicational relationship between animacy and person (i.e. being animate is a prerequisite for being 1P or 2P). Another possibility is that there are actually two kinds of animacy, where one relates to noun classification and the other to speech act roles (Wiltschko and Ritter 2015), in which case the latter kind would count as indexical features. Determining the best way to formalize the relationship between animacy and indexical features is not crucial for the issue at hand, so I will simply assume that there is a formal relationship between them which is sufficient to make animacy features count as indexical features when it comes to valuation status.

I argued in Chapter 3 that unvalued features are preferred in syntactic derivations over their valued counterparts for economy reasons, as long as the derivation can still converge. Why must then number and gender be valued on minimal pronouns if unvalued features are always preferred? I propose that this is because of their semantic status. We saw that the primary semantic contribution of non-indexical features like number and gender is context-invariant and that predicates can s-select arguments with a specific number or gender value (e.g. 'collect' requires a plural object), but not a specific indexical value. Even without considering any particular implementation of s-selection, it is clear that the semantic compatibility between the predicate and the argument in such cases is evaluated locally, without reference to other elements in the clause. If a minimal pronoun with no number or gender value were to combine with a predicate that selects for a particular number or gender, this would result in a look-ahead problem: the number or gender features of the pronoun can be valued only after the predicate combines with it, and to guarantee a convergent derivation, the outcome of the valuation has to be known before it takes place. If number and gender features are always valued on minimal pronouns, this kind of problem will never arise.[39]

39. Somewhat related to this, Kratzer (2009, 229–31) argues, based on partial binding data (e.g. 'Only I_i mentioned $\text{our}_{i,k}$ audiovisual capabilities'), that at least number can be valued on minimal pronouns.

In the case of indexical features the issue of s-selection does not arise and so the value can be assigned as late as possible. Furthermore, indexicals only receive a denotation with respect to a context, which occurs at least at the clause level—when a sentence is uttered or when a clause is embedded under an attitude predicate. So, in principle, both valuation and interpretation of indexical features could occur as late as at the CP level. However, this kind of derivation would then be at odds with the cyclic nature of syntax.

5.3.1 Indexical features and derivation by phase

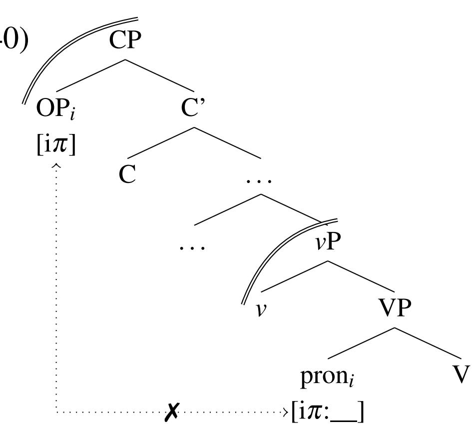
Syntactic derivation is cyclic in the sense that it takes place one phase at the time, spelling-out the individual phases in a bottom up fashion. Once a phase is spelled-out, its contents become inaccessible for further syntactic computation (Uriagereka 1999; Chomsky 2000, 2001). Since most clauses have a phase boundary at the end of the verbal extended projection (*v*P by default), an issue arises for the accessibility of minimal pronouns to elements in the left periphery of the clause when the *v*P phase is spelled-out; note that I assume the entire phase, rather than the complement of the phase head, is the spell-out domain (see Bošković 2016b for empirical support).[40] The spell-out of *v*P occurs when C, the next phase head, enters the derivation (Chomsky 2001). This means that if C were responsible for valuing features on minimal pronouns, they could not receive a value inside the *v*P phase.[41]

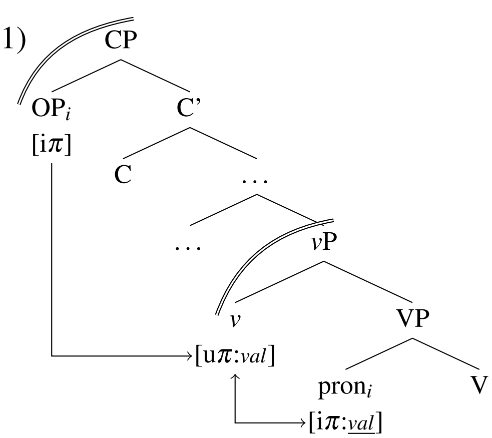
40. This is crucial since the external argument (EA) can also be a minimal pronoun (see Section 3.3.2), even though it is introduced in Spec*v*P, the prototypical escape hatch if the spell-out domain is the complement of *v*. If the entire phase is the spell-out domain, no argument is inherently privileged in this way—any argument can move out of *v*P as long as movement to a non-phasal head below CP is possible (see Bošković 2016b).

41. What must be excluded is minimal pronouns moving successive-cyclically through *v*P, without checking any features in Spec*v*P, to then be valued by C directly (like Ā-movement in the system of Bošković 2007). One possibility is that the Ā/A-distinction is relevant here, and minimal pronouns, perhaps due to being the most minimal expression of arguments, can not bear the features that would allow them to land in Ā-positions. Another possibility is that minimal pronouns in fact can be valued at the CP level, which could be a way to reinterpret in the current system the person restrictions observed with complementizer agreement in Nez Perce [*Sahaptian*] (Deal 2015), although such derivations would then still need to be appropriately constrained. Note that minimal pronouns may (and in fact typically must) move out of *v*P, as long as the movement is not driven by or feeds person valuation. What cannot be allowed is a "look ahead" type of movement, where a feature other than person drives movement out of *v*P only to make person valuation by C possible.

Additionally, it has been argued that the interpretation of 1/2P pronouns is contingent on the establishment of a dependency between the pronouns and speech act operators in the left periphery of the clause, which relate the pronouns to individuals in the context (for different implementations of this idea, see e.g. Speas and Tenny 2003; Baker 2008b; Pearson 2012). Following the unification of all indexical features that I am pursuing, this analysis would extend to all indexicals, possibly with different operators for each type of indexical.

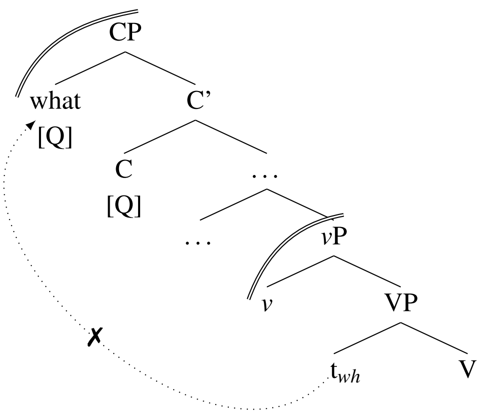
Based on this and the way in which phases constrain syntactic derivations, I propose that valued indexical features on phase heads exist essentially as a bridge to the next phase. I assume that a version of the speech act operator approach to indexicals is correct, but that the phase head, via its valued indexical features, must mediate the dependency between the pronouns and the operators in the left periphery of the clause (the nature of this dependency will be elaborated on in Sections 5.3.2 and 5.3.3). The gist of the idea is illustrated by the simplified derivations without and with valued person features on *v* in (40)–(41). In (40), the phase boundary prevents a dependency between the speech act operator and the minimal pronoun, whereas in (41) the valued person features on *v* provide an "escape hatch". The operator can establish a dependency with *v*, and *v* is in an Agree relation with the minimal pronoun, which means that we have an uninterrupted chain of dependencies from the operator to the pronoun mediated by the valued person features on *v*.

(40)


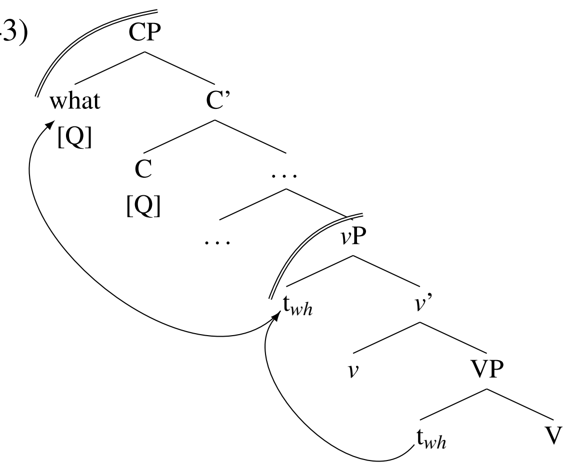
(41)


The logic behind the valued indexical features being present on intermediate phase heads is no different than having intermediate landing sites at the phase edges with successive cyclic movement. With *wh*-movement, movement to SpecCP cannot take place in one fell swoop, as in (42), it needs to proceed through the phase edge first, as in (43).[42]

(42)



(43)



The similarity between the two phenomena is even deeper, since it is the semantics of *wh*-questions that is typically assumed to require the *wh*-phrase to bind its trace from a left peripheral position (see e.g. Dayal 2016), which necessitates the movement of the *wh*-phrase. The intermediate movement step has no semantic motivation, it is imposed solely by the cyclic nature of syntax, it is only the movement to SpecCP as a whole that has a semantic motivation. Nonetheless, if the movement to Spec*v*P would not take place, the movement to SpecCP required to guarantee question semantics would also not be possible. We will see over the next two sections that the relationship between the valued—but uninterpretable—indexical features on the phase head and indexical operators in SpecCP is very similar. Although the uninterpretable features serve no semantic purpose, their absence on the phase head would prevent the establishment of the semantically motivated dependency between the minimal pronouns inside the phase and the indexical operators.

42. Technically, in the approach to phases I am adopting here, it needs to go through a non-phase projection just above the *v*P and below the CP (Bošković 2016b), but I am simplifying here for the sake of exposition.

5.3.2 Bound indexical pronouns

I explore in this section one way to implement the dependency between indexical minimal pronouns and operators in the left periphery, based mainly on insights from Heim (1991, 1994, 2002, 2008), von Stechow (2003, 2004), Schlenker (2003, 2004) (specifically his THEORY II of indexicals), Sigurðsson (2004) and Baker (2008b). The main idea is that 1/2P pronouns are bound variables, bound either by: (i) another 1/2P pronoun—in canonical binding contexts, (ii) an attitude verb—in attitude reports, or (iii) speech act participant operators—in matrix contexts. However, I modify this approach slightly, by proposing that 1/2P pronouns are bound variables only if they are also minimal pronouns, and that the bound variable analysis extends to other pronouns with indexical features.

The key observation that motivates this general approach to 1/2P pronouns is attributed to Heim 1991, who observes that in "Only I did my homework" the possessive pronoun *my*, despite having an interpretable 1P feature, is not interpreted as a 1P pronoun under the bound reading (the reading being: 'Every person *x* such that *x* is different from me didn't do *x*'s homework') (see also Heim 1994, 2002, 2008; Kratzer 1998, 2009; von Stechow 2003, 2004; Schlenker 2003, 2004; i.a.). Indexical pronouns with such readings are now commonly referred to as *fake indexicals*. The existence of fake indexicals tells us that, unlike what one might expect from a standard Kaplanian approach to indexicals, 1/2P pronouns can, just like their 3P counterparts, function as bound variables (cf. "Only he did his homework").

In my implementation of the operator-pronoun dependency, I will adopt specifically von Stechow's (2003, 2004) analysis of fake indexicals, where the relevant interpretable features on the bound pronouns remain uninterpreted because they are deleted at LF (details provided below). Note that von Stechow's proposal and Kratzer's (2009) minimal pronouns have both been used to derive fake indexical readings. Thus, the current system relying on both mechanisms might seem a bit of an overkill. However, I will argue in Section 5.3.3

that a hybrid system using both mechanisms can actually resolve some conceptual issues concerning the status of referential indices in narrow syntax and LF.

Von Stechow (2003, 2004) proposes that the 1P feature on the possessive pronoun in Heim's sentence is not interpreted because it is deleted at LF under semantic binding (i.e. binding through λ-abstraction). Thus, as illustrated in (44a), both the quantified DP subject and the bound possessive pronoun have interpretable 1P features in the syntax. However, the quantified subject moves in LF to resolve a clash in semantic types (i.e. it undergoes Quantifier Raising/QR), creating a co-indexed trace and a λ-operator, as shown in (44b); note that the trace, the λ-operator, and the quantified subject all have a copy of the interpretable person feature (which I return to in Section 5.3.3).

(44) Only I did **my** homework.

a. Syntax: $[[_{\text{DP}}\ \text{only}\ \underset{[\text{i}\pi{:}1\text{P}]}{\mathbf{I}_5}\ ]_8\ \text{did}\ \underset{[\text{i}\pi{:}1\text{P}]}{\mathbf{my}_8}\ \text{homework}\ ]$

b. LF: $[_{\text{DP}}\ \text{only}\ \underset{[\text{i}\pi{:}1\text{P}]}{\text{I}_5}\ ]\ [\ \underset{\cancel{[\text{i}\pi{:}1\text{P}]}}{\boldsymbol{\lambda}_{\mathbf{8}}}\ [\ \underset{\cancel{[\text{i}\pi{:}1\text{P}]}}{t_{\mathbf{8}}}\ \text{did}\ \underset{\cancel{[\text{i}\pi{:}1\text{P}]}}{\mathbf{8}}\text{'s}\ \text{homework}\ ]]$

QR thus induces binding and, as a result, the deletion of all the interpretable 1P features apart from those on the subject. This means the I_5 pronoun inside the subject DP is interpreted as a variable with an interpretable 1P feature, which restricts the denotation of the variable 5 to the actual speaker in the relevant context. Conversely, the binding of the possessive pronoun, translated in LF into the variable 8, results in the deletion of the interpretable 1P feature on the pronoun, leaving it uninterpreted and yielding the desired fake indexical reading: 'Every person x such that x is different from me didn't do x's homework.'

However, the derivation in (44) only accounts for 1/2P pronouns in canonical bound variable contexts. I assume that outside those contexts, they are bound either: (i) by an attitude verb if the relevant CP is embedded under one (see von Stechow 2003, 2004), or (ii) by speech act participant operators. I focus on the latter case here, discussing the former briefly below. The basic idea is that those 1/2P minimal pronouns which are not bound by another argument or by an attitude verb are not actually free, but in fact must be bound via a λ-operator in C by the speech act participants of the actual context, which are instantiated in the syntax by null SPKR and ADDR operators in SpecCP (following Baker 2008b). A sample derivation for a sentence with a 1P and a 3P pronoun is illustrated in (45).

(45) Peter showed **me them**.

a. Syntax: [SPKR [$_{CP}$ C$_5$ [Peter showed **me**$_5$ **them**$_7$]]]
[iπ:1P] [iπ:1P] [iπ:1P] [iπ]

b. LF: [SPKR [$\boldsymbol{\lambda_5}$ [Peter showed **5** 7]]]
[iπ:1P] ~~[iπ:1P]~~ ~~[iπ:1P]~~ [iπ]

The main difference compared to (44) is that instead of movement creating the operator-variable configuration, the C head here functions as a movement-independent λ-operator in LF (see Kratzer 2009), which allows the SPKR operator introduced in SpecCP to bind the 1P pronoun. However, just as in (44), the binder (SPKR), the λ-operator (C), and the bound pronoun (*me*) all initially have an interpretable 1P feature (SPKR has it by virtue of being the syntactic manifestation of the speaker). Crucially, following von Stechow (2003, 2004), the semantic binding configuration results at LF in the feature deletion of all the interpretable 1P features aside from those of SPKR (cf. the quantified subject in (44)). As a result, the bound pronoun is left without its 1P feature and is interpreted as a variable bound by the speaker of the utterance. The 3P pronoun, on the order hand, is a free variable that gets its denotation via an assignment function which maps its index to the appropriate discourse referent.[43]

The derivation in (45) involves a person combination with a 3P direct object that would be grammatical in both STRONG and WEAK person restriction scenarios, but WEAK restrictions also permit combinations where neither pronoun is 3P. This is where I must depart from

43. See Section 5.2.1.2 on how discourse anaphora receive a denotation in a dynamic semantics approach; see also Heim and Kratzer 1998, 239–45 for a brief overview of free variables and discourse anaphora.

Baker (2008b), who argues that in a given clause a single 1/2P pronoun can be bound by a speech act operator, which is in fact Baker way of deriving person restrictions, but crucially limited to only STRONG restrictions. As my goal is to derive all types of person restrictions as well as other similar co-occurrence restrictions, it must be possible to allow more than one indexical pronoun per clause to be bound this way. Thus, I suggest that when more than one minimal indexical pronoun is involved, each indexical pronoun must be bound by an appropriate operator: in the case of a 1P and 2P pronoun, they must respectively be bound by a SPKR and an ADDR operator. Other indexical pronouns must also be bound by operators of the relevant type for each indexical category and value.[44,45]

Another departure from Baker (2008b) and other similar approaches is that I crucially propose that only minimal indexical pronouns must be bound this way, other indexical pronouns are interpreted in the standard Kaplanian fashion (see Section 5.2.1). Recall that the pronouns I analyze as minimal pronouns are distinguishable from other pronouns not only in terms of giving rise to co-occurrence restrictions, but also in terms of their binding behavior (see Section 2.4). In particular, in languages where deficient pronouns (i.e. phonologically weak or null pronouns) are minimal pronouns, the strong counterparts of the pronouns can not be used in bound variable contexts (the Montalbetti Effect; Montalbetti 1984; Despić 2011). I argue, building on Messick (2017), that this extends to differences in the interpretation of indexicals as well. Messick (2017) provides evidence from the behavior of shifted indexicals showing that the deficient vs. strong pronoun distinction matters for the behavior of indexical pronouns in languages that allow indexical shift.

44. I leave open whether each λ-operator corresponds to a different syntactic head or whether C can multitask and bind multiple indexical pronouns—that would imply CP can also host multiple indexical operators or alternatively that indexical operators themselves can multitask in the same way. I am currently not aware of any direct syntactic or semantic evidence that would favor the former or the later approach.

45. It is possible though that there are constraints on the number and combinations of indexical operators in the left periphery. For example, the proximate/obviative 3P distinction in Algonquian is only relevant when there are no 1/2P pronouns in the same clause (see Sections 5.1.1 and 5.2.1.1). This type of complementary distribution could be implemented in terms of a "proximate operator" only being available if SPKR and ADDR operators are absent. I leave exploring this type of approach to the distribution of obviation for future work.

As discussed first by Schlenker (1999, 2003) (but see, among others, also Anand and Nevins 2004; Sudo 2012), there are languages where 1P and 2P pronouns in clauses under attitude verbs do not have to refer to the speaker and addressee of the utterance—they can refer to the speaker and addressee of the reported content. That means that *"Grant said that I was sick"* can be interpreted the same as *"Grant$_i$ said that he$_i$ was sick"*, so that the embedded 1P pronoun has the same denotation it would have in a direct quotation: *"Grant said: 'I am sick"'*, even though the sentences in question do not involve direct quotation. Likewise, *"Grant said to Buddy that you were sick."* can be interpreted the same as *"Grant said to Buddy$_i$ that he$_i$ was sick"*, so that the embedded 2P pronoun is interpreted the same as in: *"Grant said to Buddy: 'You are sick"'*. Interestingly, Messick (2017, Ch. 3) observes that a version of the Montalbetti Effect is operational with indexical shift: in languages that allow indexical shift and have both deficient and strong pronouns, only the former can be shifted and the latter can not. In other words, the deficient pronoun must be chosen over its strong pronoun counterpart in order to be interpreted as a shifted indexical, which parallels the Montalbetti Effect where the deficient pronoun must be chosen over its strong pronoun counterpart in order to be interpreted as a bound variable.

If my proposal is on the right track and minimal pronouns are singled out in terms of how they are interpreted when they are indexicals, then the split identified by Messick (2017) can be seen as another consequence of the proposal. The idea is that in indexical shift scenarios the indexical minimal pronouns are bound by the attitude verbs/predicates which take the embedded CP as a complement (see von Stechow 2003, 2004 for the details of how this type of binding arises). This is in line with how such pronouns are bound by the quantified subject in (44) and the operator in SpecCP in (45). Crucially, just as in the other two binding configurations, the indexical features on the pronouns will be deleted in LF under binding, accounting for why the pronouns look like 1P and 2P pronouns, but are interpreted as 3P pronouns. This analysis thus fits Messick's observation that indexical shift has the same effect on pronoun choice as a bound variable interpretation, as the shifted interpretation is achieved via binding. What is important here is that this type of derivation is not available with strong indexical pronouns, which are not bound pronouns (see above), and have to be interpreted in the standard Kaplanian fashion. Consequently, strong pronouns can only refer to the speaker and addressee of the utterance rather than the embedded context, even in configurations where indexical shift is otherwise possible. Note also that as in the case of person restrictions and the Montalbetti Effect, the proposed correlation between pronoun type and indexical shift is one-directional: it is not the case that all deficient pronouns can undergo indexical shift. As we saw in Section 3.3, pronouns may be deficient at PF even though they syntactically and semantically pattern with strong pronouns.[46]

5.3.3 Co-indexation as a reflex of Agree

In light of the claim that only minimal pronouns are consistently interpreted as bound variables, the apparent redundancy in adopting both minimal pronouns and feature deletion under binding must be addressed. Both mechanisms have been independently proposed to deal with (among other things) the problem of fake indexicals: indexical features on bound pronouns being ignored for semantic interpretation. In approaches to fake indexicals that use minimal pronouns (Heim 1991, 1994, 2002, 2008; Kratzer 1998, 2009; i.a.), the idea is that bound minimal pronouns receive their φ-feature values from the antecedent, but only in the PF branch. The idea is that the pronouns remain minimal in the syntax and in LF, but receive the φ-feature values that are morphologically reflected on the pronouns only in the

46. Recall also from Section 3.3.2, that little *pro* may count as a deficient pronoun with respect to the Montalbetti Effect even though in “classic” null subject languages it does not take part in person restrictions. My proposal would then force a division of minimal pronouns into two classes: ones that can receive an indexical value either via binding or Agree (giving rise to person restrictions) and ones that can only receive a value via binding (little *pro*). This might follow from T rather than *v* being the head that Agrees with *pro* in the languages in question. Since T is normally not a phase head, it cannot provide a value to the minimal pronoun, so it needs to acquire it by other means. I leave the details of this analysis to be worked out in future work.

PF branch—this is why the pronoun can match an antecedent in φ-features morphologically, but only the φ-features on the antecedent are semantically interpreted. This is essentially an inverse version of the von Stechow (2003, 2004) approach to fake indexicals outlined in the previous section, which involves deletion of valued interpretable φ-feature on the pronoun under binding, but crucially the deletion occurs only in the LF branch. The end result is the same in both approaches: the φ-features are visible only in PF and the pronoun is interpreted as a pure variable without interpretable φ-features in LF. Even though both approaches were designed to deal with the same facts, I want to suggest that both are correct, but for entirely different aspects the problem of bound indexicals.

Note that the version of minimal pronouns I have been assuming so far is different from the standard version, as the valuation of φ-features occurs entirely in the syntax—this is a key part of the derivation of person restrictions. That also means that for such pronouns to be fake indexicals, their φ-features must be ignored at LF, which can be achieved, as we saw, via the deletion of the features at LF under binding. But what is then the purpose of minimal pronouns in the proposed system, if not to derive fake indexical readings?

I propose that the purpose of minimal pronouns relevant for semantics is the generation of matching indices on the λ-operator and bound pronoun. This issue is often set aside in works that focus on the semantics of fake indexicals, where matching indices are usually assumed to be arbitrarily generated, presumably in the syntax, and derivations with non-matching indices simply fail to converge semantically. In syntactic works, specifically those considered part of the minimalist program (Chomsky 1995), indices in the syntax are seen as violations of *Inclusiveness*, which "bars introduction of new elements (features) in the course of computation: indices, traces, syntactic categories or bar levels, and so on" (Chomsky 2001, 2–3). However, indices are essential for treating indexical pronouns as bound variables: for the interpretable features on the bound pronoun to be deleted, the pronoun must be semantically bound in the first place, and that by definition requires matching indices between the λ-operator and the bound pronoun. In cases with one bound pronoun, randomly generating matching indices on the λ-operator and the pronoun is manageable. But what about derivations with multiple bound pronouns? Even with an upper limit to the number of indices that can be generated on each element, the combinatorial possibilities would grow exponentially with every additional variable. This is the kind of complexity we should try to avoid if our aim is a computationally optimal theory of language. The idea I entertain here is that indices are neither randomly generated nor present in the syntax. Instead, matching indices are a LF reflex of multiple occurrences/copies of an interpretable φ-feature set.

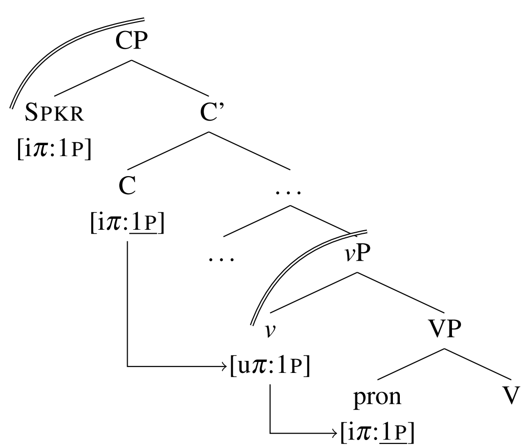
I assume that Agree between distinct *instances* of matching features results in them counting as *occurrences* of the same feature (Pesetsky and Torrego 2007),[47] which distinguishes features that share a value as a result of Agree from features that only coincidentally have the same value. This parallels the almost identical assumption needed with the *copy theory of movement* (Chomsky 1993, 1995), where the in situ copy and the copy generated by movement count as occurrences of the same element (see also Roberts 2010b, 59–61 on this parallelism between Move/Merge and Agree). Crucially, in the case of movement, the resulting chain of copies must convert into a legitimate LF object at LF, such as an operator-variable construction, to ensure LF convergence (Chomsky 1993, 34–6). Operator-variable constructions specifically can be implemented as the tail copy of the chain converting at LF into a variable (trace) bound by the head of the chain via λ-abstraction (Fox 1998, 2002, 2003; Sauerland 1998, 2004), as in the QR derivation in (44). I suggest that in such cases the matching indices on the λ-operator and trace can be analyzed as LF reflexes of the interpretable φ-features on the copies, using something that must be present in the syntax to generate indices, which we do not want to be present in the syntax. More importantly, I argue that individual occurrences of interpretable φ-features resulting from Agree analogously

47. I use the terms *instance* and *occurrence* in a technical sense where the former refers to two identical but distinct elements entering the derivation at different points and the latter to two copies of the same element that result from syntactic operations like movement (Chomsky 1995; see below) or in this case Agree.

also map to matching indices at LF (uninterpretable φ-features are irrelevant in this case as they must be absent at LF; Chomsky 1995, 277–9).

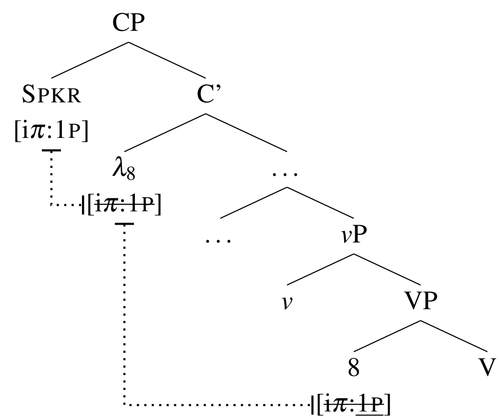
Let's then consider what this means for a derivation with a minimal pronoun. As illustrated in (46a), a 1P minimal pronoun gets its person value from the valued uninterpretable person feature on the phase head *v*, which functions as an escape hatch, allowing the pronoun to indirectly establish a dependency with C outside the spell-out domain (see Section 5.3.1). In line with the proposal above regarding index generation, I assume this dependency between *v* and C to be Agree, which in turn makes all the 1P features on the minimal pronoun, *v*, and C count as occurrences of the same 1P feature.[48] As shown in (46b), at LF the interpretable 1P features on C and the pronoun map to matching indices when the former converts into a λ-operator and the latter into a variable (the uninterpretable features on *v* are not present at LF). The semantic binding between SPKR and the pronoun consequently results in the deletion of all occurrences of the 1P feature apart from the one on SPKR itself.

(46) a. Syntax:



48. Following Kratzer (2009), I assume the φ-features of SPKR and C undergo unification in the Spec-Head configuration analogously to the features of the EA and *v* in reflexive constructions (see Chapter 3). This type of dependency between valued features could also be argued to result in the two sets of features counting as occurrences of the same feature, but this is not crucial for the analysis presented here.

b. LF:



The reason why minimal pronouns must have unvalued indexical features then amounts to ensuring that they are interpreted as bound variables at LF. If their indexical features were not valued by matching counterparts on *v*, and in turn C were not to Agree with *v*, then the minimal pronoun would not have a matching index with the λ-operator in C and consequently it would not be semantically bound. Alternatively, if the pronoun were not a minimal pronoun, *v* would not have to assign it a value, so the trail of dependencies going all the way to C would not be established. But recall that I proposed strong indexical pronouns are not interpreted as bound variables but as regular Kaplanian indexicals.

There are thus two possible derivations that guarantee convergence at LF: (i) a minimal pronoun derivation, which requires the presence of valued indexical features on the phase head, or (ii) a derivation with less deficient pronouns, which not only employs more syntactic structure and valued features, violating *Minimize Structure* (see Section 3.4), but also requires an alternative semantic mechanism to interpret indexical pronouns. The latter, more costly, strategy can only be chosen, given economy considerations, when the former is impossible; for example, due to the syntactic constraints on feature valuation. Additionally, note that the interpretable indexical features on C, which Agree with their valued counterparts on *v*, also serve an important role: they guarantee that the λ-operator has the same index as the pronoun at LF, due to the interpretable φ-features shared between them.

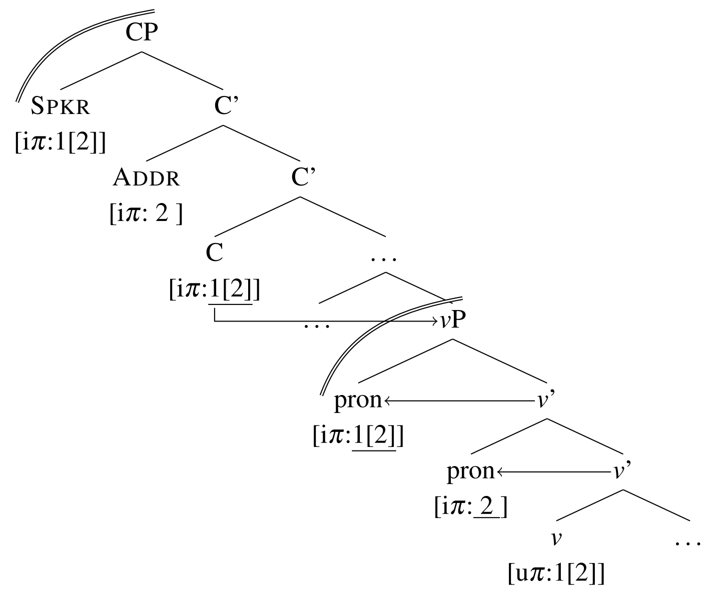
The placement of valued indexical features on the intermediate phase head is thus a minimal cost to be paid to ensure the interpretation of minimal pronouns. It is the trade-off between having minimal pronouns, which are the most economic means of expressing an argument, and ensuring the derivation converges at LF. There is crucially also no need for a special syntactic operation that inserts the valued features on *v*. The *v*'s with or without a valued indexical feature (of the right kind) are simply different lexical items, just like (non-minimal) pronouns with different φ-feature values. Derivations that start with a minimal pronoun and a *v* without valued indexical features either do not converge at LF, because the minimal pronoun cannot obtain—within the *v*P phase—an indexical value matching the operator in SpecCP, or the unvalued feature on the minimal pronoun gets a default value, just as in the derivations where valuation is prevented due to an intervening pronoun.

Some clarification is needed here regarding default values. I have argued that from an LF perspective the purpose of minimal pronouns is to guarantee matching indices on the λ-operator in C and the pronoun, which is a result of Agree between the indexical features of the pronoun, the phase head *v*, and finally C. However, when the minimal pronoun gets a default value for its indexical feature, there is no Agree between it and *v*. Note that the default value is the unmarked value for any of the relevant indexical categories (3P, obviative, inanimate, non-specific, familiar, etc.); in other words, it is the non-indexical value. This means that such pronouns do not require being bound by an indexical operator, and therefore the lack of Agree between them and *v* is not an issue. Nonetheless, default value pronouns still need to be able to pick out a referent, and there are several options for how this can be achieved. I outline here two options that I think are the most promising.

One option is to parallel the assignment of default values in the syntax with default index assignment at LF, essentially a last resort operation applying at LF to lone instances of interpretable φ-features (the index then picks out a referent with respect to the context; see discussion in Section 5.2.1.2). The other option would be to say that default value pronouns actually lack an index at LF. As radical as this may sound, the proposal that only some pronouns have a referential index has a precedent in the work of Cardinaletti and Starke (1994, 1999), who argued that only strong pronouns have a referential index. In the current case, the lack of an index would be a more restricted option, but nonetheless limited to deficient pronouns (recall from Section 3.2 that strong pronouns are always phases and therefore can have valued indexical features). This approach would, however, necessitate a strategy for determining the reference of such pronouns that does not rely on indices; for example, something along the lines of *centering theory* (Grosz, Joshi, and Weinstein 1983, 1995). Either option is in principle compatible with the present account.

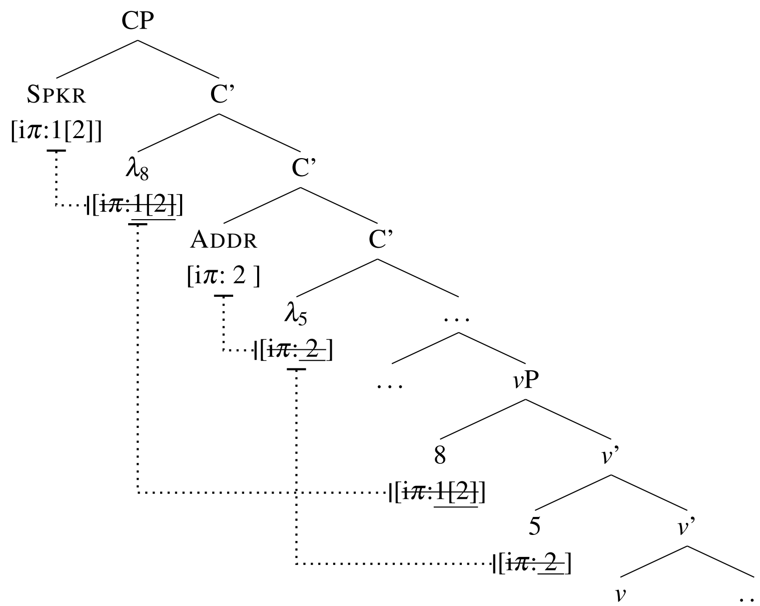
Finally, the proposed indirect interaction between syntax and semantics allows us to confine the variation in restriction type to the syntax (see Chapters 3 and 4 on the relevant parameters). For example, when dealing with two 1/2P pronouns, allowed with WEAK person restrictions, each must be bound by corresponding indexical operators in the left periphery of the clause (see Section 5.3.2). As illustrated in (47), both minimal pronouns are independently valued by the person features on *v* (1[2] = *speaker* [*participant*]) in the syntax and a chain of dependencies ends up linking the operators, via the C head, to *v*, and finally both pronouns (as *v* projects *v*P it is closer to C than any pronoun within *v*P).

(47) Syntax:



Based on the dependencies established in the syntax, each operator can bind the appropriate pronoun in LF. Note that the valued features on *v* are no longer present in LF due to being uninterpretable, however, as shown in (48), the interpretable features aside from those on the operators are mapped to λ-operators and variables with matching indices. Recall that the reason a valued person feature on *v* can value two minimal pronouns 1P and 2P respectively is because a 1P value (1[2]) also contains a 2P value (2). This also allows the interpretable person features on C to map to two λ-operators, one for the SPKR and one for the ADDR operator (but see footnote 44 on the possibility of multiple Cs hosting the operators). Since both pronouns have Agreed with *v*, and C in turn has Agreed with *v*, the person features on the 1P and 2P pronouns and those on the λ-operators count as instances of the same feature. This means that in LF the variables derived from the pronouns and the λ-operators will have matching indices, leading to semantic binding, and finally the deletion of all the interpretable person features other than those on the SPKR and ADDR operators.

(48) LF:



Having one minimal pronoun or two valued by *v* thus does not affect the binding relations in LF, which differs from Baker's (2008) version of the operator-variable analysis of indexical pronouns. The valued indexical features on phase heads perform their escape hatch function just as well with single or multiple operator-variable dependencies.

To sum up, the reason why both minimal pronouns and feature deletion under binding are needed is because they each serve a different but crucial role in the interpretation of indexical pronouns. The former guarantee that the pronoun variable has the appropriate index at LF, and the latter removes unwanted interpretable features once binding is established based on those indices. This is the final piece in establishing the necessity of valued indexical features on phase heads: they are the necessary syntactic link between indexical (minimal) pronouns and the indexical operators in the left periphery. Importantly, the proposed motivation for the distribution of valued and unvalued indexical features respects the autonomy of syntax. Semantics does not directly constrain Agree, valuation, or any other processes in the syntax. Syntax can only fail to establish the conditions necessary for the interpretation for minimal pronouns when there is no uninterrupted trail of syntactic dependencies from C to the minimal pronouns. This trail of dependencies can be likened to a gunpowder trail: it has to be uninterrupted to be useful, otherwise it fizzles out along the way. The apparent interaction between the semantics of a certain class of features and syntax is only indirect. Syntax tries to get away with as few valued features as possible, but in order for convergence to be ensured at LF, syntax needs to spend the minimum extra resources required for convergence, which amounts to having valued indexical features on phase heads.

5.4 Why we want autonomy (of syntax)

In the previous section I outlined an analysis that relates the semantic properties of features that can be subject to co-occurrence restrictions to their special status in the syntax. An important aspect of the analysis is that the link between the syntactic and semantic behavior of the features is only indirect, thus maintaining the autonomy of syntax. I want to conclude this chapter by considering two alternative approaches: *syntactic licensing* and *semantic licensing*. Although these two styles of analysis approach the phenomenon from opposite ends, they are ultimately not that different in terms of implementation, as they both rely on constraining the syntactic operation Agree based on the semantic properties of the features involved. They thus fall short of being purely syntactic and purely semantic respectively in terms of identifying the driving force behind co-occurrence restrictions.

5.4.1 Syntactic licensing

The syntactic licensing approach is probably the most well known of the two, being among the first analyses of person restrictions based on syntactic intervention effects (see Chapters 1 and 2). The idea is that in addition to uninterpretable features, which must Agree with matching features in order to be deleted prior to reaching LF, a subset of interpretable features must also obligatorily take part in Agree. Since these analyses seek to derive person restrictions, the relevant interpretable features are assumed to be person features specifically. One of the earliest proposals of this kind is Béjar and Řezáč's (2003) *Person Licensing Condition (PLC)* (see also Preminger 2019 an updated version):

(49) *Person Licensing Condition (PLC) axiom:*
An interpretable 1st/2nd person feature must be licensed by entering into an Agree relation with a functional category. (Béjar and Řezáč 2003, 53)

The question that naturally arises with this approach is why should the licensing condition be limited to person features? In principle, we could replace person in (49) with any other interpretable φ-feature. In fact, since features other than person can be subject to the same syntactic restrictions, we would have to extend the licensing condition to those features if we wanted a unified analysis.[49] Whether we want to keep the licensing condition limited to person features or indexical features more broadly, the interpetable φ-features it applies to have to be either arbitrarily limited, thus missing an important generalization, or determined based on their semantics. Therefore, somewhat paradoxically, the syntactic licensing approach actually compromises the autonomy of syntax: unless the class of features that requires special licensing is assumed to be purely coincidental, the syntactic licensing approach must allow semantic factors to directly constrain syntactic operations.

5.4.2 Semantic licensing

An alternative approach to person licensing has been proposed by Charnavel and Mateu (2015a, 2015b), Zubizarretta and Pancheva (2017), and Pancheva and Zubizarreta (2018), who point to similarities between person restrictions and restrictions on point-of-view and empathy encoding (cf. Japanese empathy verbs in Section 5.1). Their goal is to unify person restrictions with these phenomena, where the idea is that what requires licensing is

49. In her analysis of DOM/DOI phenomena, Kalin (2017, 2018) in fact adopts a version of (49), extending it to the other interpretable φ-features that crosslinguistically trigger DOM/DOI; see Section 5.1.2 regarding the possible connection between co-occurrence restrictions and DOM/DOI phenomena.

specifically the point-of-view (logophoric) center of the clause. Depending on the specific analysis, both kinds of restrictions are attributed to an intervention effect or to conflicting licensing requirements arising when more than one potential point-of-view center is present. For the main point I make in this section, the exact details of the analyses are not crucial, although I will elaborate on some of the details when necessary in the following section.

Since logophoricity and empathy are at their core semantic notions, this approach to person restrictions is often seen as a semantic licensing approach. As we saw, there are undoubtedly semantic considerations at play with person restrictions, especially in the broader context of co-occurrence restrictions. However, there is a fundamental difference between person restrictions and the restrictions on point-of-view and empathy encoding. The latter revolve around either the speaker or the addressee being the point-of-view center. The classic way of identifying such point-of-view effects involves comparing declarative and interrogative clauses to look for the so called *interrogative flip*. For example, as illustrated in (50), the evidential verbs 'appear' and 'seem' are interpreted with respect to different points-of-view in declarative and interrogative sentences. In the former, they are interpreted with respect to the speaker's point-of-view, whereas in the latter, they are interpreted with respect to the addressee's point-of-view.

(50) a. The bicyclist **appears**$_{\text{spkr}}$ / **seems**$_{\text{spkr}}$ to have escaped injury in the crash. (Tenny 2006, 259)

b. Does the bicyclist **appear**$_{\text{addr}}$ / **seem**$_{\text{addr}}$ to have escaped injury in the crash? (Tenny 2006, 262)

The flip in perspective illustrated in (50) is also observed across other phenomena broadly in the domain of point-of-view and empathy encoding, but crucially not with person restrictions. I was not able to attest any cases where person restrictions, even those that single out 1P or 2P pronouns specifically (see Section 4.3), are sensitive to the interrogative flip. For example, there are no cases where a ME-FIRST restriction (i.e. *3P $\gg$ 1P, *2P $\gg$ 1P) turns into a YOU-FIRST restriction (i.e. *3P $\gg$ 2P, *1P $\gg$ 2P) in questions. The absence of this type of person restriction is unexpected if the two kinds of phenomena are both tied to the grammatical encoding and licensing of point-of-view centers.

There is also another issue with semantic licensing at the level of implementation. Namely, the way a point-of-view center is licensed in the analyses of Charnavel and Mateu (2015a, 2015b), Zubizarretta and Pancheva (2017), and Pancheva and Zubizarreta (2018) is via Agree, a syntactic operation. Thus, in terms of implementation, the semantic licensing approach is not that different from the syntactic licensing approach: an element/category singled out based on its semantic properties must be licensed in the syntax via Agree. The difference is that the semantic licensing approach makes explicit the relevant semantics properties, but it nonetheless inherits the issue concerning the autonomy of syntax.

5.4.3 The relevance of semantic effects in person restrictions

In contrast to the syntactic and semantic licensing approaches, the analysis developed in this thesis does not pose a problem for the autonomy of syntax. Person restrictions arise due to the distribution of unvalued and valued person features, constrained by the tension between syntactic and semantic requirements. The presence of valued person features on phase heads is a solution to the clash between the clause-level semantic requirements of minimal pronouns and the cyclic nature of syntax (valued person features are not present anywhere else due to economy considerations). Note that with this analysis semantics influences syntax only indirectly as a filter: if valued person features would not be present on phase heads, the derivation would not converge at LF. This is crucially not the same as semantics inserting the features on the phase head. The assumption is that there exist two versions of the phase head, with and without valued person features, where the right one must be chosen in order to guarantee LF convergence, as outlined in Section 5.3.

What I show next is that this analysis can also handle the phenomena presented as evidence for the semantic nature of person restrictions. Namely, the amelioration of person restrictions with non-*de se* 1P pronouns, and the parallels between person restrictions and the clitic binding restriction (see also Section 2.4). While I will not offer full analyses here, the tentative proposals outlined below will be sufficient to show that the proposed analysis of person restriction is compatible with these phenomena and that a fully semantic analysis of person restrictions is not necessary. In fact, we will see that even the semantic licensing analyses derive the relevant facts via syntactic, not semantic mechanisms.

Let us first consider the amelioration of person restrictions in contexts where 1P pronouns are not interpreted *de se*; that is, when the speaker does not self-identify with the referent of the 1P pronoun. Charnavel and Mateu (2015a, 2015b) observe this effect in dream reports, where the clause embedded under an attitude verb reports a dream the speaker had, usually dreaming that they were someone other than themselves (see also Lakoff 1975; Percus and Sauerland 2003; Anand 2006, 2007; Arregui 2007). As shown by the French example in (51), the normally ungrammatical combination of a 3P Goal and a 1P Theme clitic becomes grammatical if the 1P pronoun does not refer to the speaker's "dream self" (Marilyn Monroe) but to the speaker's actual self. In other words, in the context of the dream the 1P pronoun does not refer to the individual the speaker self-identifies with in the dream (it is non-*de se*).

(51) French [*Romance*]:
?J$_i$ ai rêvé que j' étais Marilyn Monroe$_m$, que j' étais chez Kennedy$_k$ et
I have dreamed that I was Marilyn Monroe that I was house Kennedy and
... [que *je*$_m$ **me**$_i$ **lui**$_k$ présentais].
[that I 1.DO 3.IO.DAT introduced]
'I$_i$ dreamed that I was Marilyn Monroe$_m$, that I was at Kennedy$_k$'s house and ...
[that *I*$_m$ introduced **me**$_i$ to **him**$_k$].' (Charnavel and Mateu 2015b, 693)

Notice that in such contexts Principle B effects also disappear: the subject and one of the objects in the embedded clause are both 1P pronouns, which would normally also mean co-reference. However, the subject referring to the speaker's dream self and the object referring to the speaker's actual self is sufficient to count as disjoint reference.

The same effect can also be observed outside dream reports. As I noted in previous work (Stegovec 2019), Slovenian allows for 1P pronouns that would otherwise violate the person restriction to be grammatical when they are not interpreted *de se*. For example, in (52), the Theme clitic pronoun is 1P, but it can not be interpreted *de se*. If it were interpreted *de se*, like the other 1P pronoun, it would violate Principle B. Instead, it must refer to someone else's perception of myself, which crucially also voids the person restriction (even for speakers with STRONG person restrictions, where the restricted pronoun has to be 3P).

(52) Slovenian [*Slavic*]:
#Janez **mi** **me** je pojasnil.
Janez 1.DAT 1.ACC is explained.M
'Janez explained **me** (what I am) to **me**.' (Stegovec 2019, 14)

While the distinction between *de se* and non-*de se* 1P at the core of this effect is uncontroversially a semantic distinction, it can not be excluded that the distinction is based on an underlying syntactic distinction. In fact, if we want to keep semantics fully compositional, the semantic distinction has to be a reflection of a preexisting syntactic distinction.

Crucially, the analysis of the amelioration effect that Charnavel and Mateu (2015a, 2015b) offer is fundamentally syntactic. They endow only the pronouns with *de se* readings with a special *logophoric* feature that must be licensed via Agree with a logophoric operator in the left periphery in the clause—failure to do so, due to the syntactic intervention of another pronoun, results in a person restriction. Non-*de se* pronouns are exempt from person restrictions because they lack the logophoric feature and hence do not require licensing via Agree with the logophoric operator. Note that this analysis is effectively identical to the syntactic licensing approach, with logophoric features replacing 1/2P features.

In the current approach there are at least two options for deriving the amelioration effect. One option would be to employ a split in φ-features, where each interpretable (LF) φ-feature set on an argument can be accompanied by an uninterpretable (PF) φ-feature set (see Smith 2015; Messick 2017). In most cases the LF and PF feature sets align, resulting in the interpretation matching the morphological realization of the features. However, the two sets can also not match; see Smith (2015) and Messick (2017) for examples and discussion. Within such a system, it is possible to derive the required mismatch in person features: a pronoun can be 1P at PF, but 3P at LF. This feature combination would amount to a non-*de se* 1P pronoun (see Messick 2017), which due to being 3P in terms of its interpretable features, would have the distribution of 3P minimal pronouns in the current system.

The other option, suggested to me by Richie Kayne, involves treating non-*de se* 1P pronouns as *imposters* in the sense of Collins and Postal (2012). Imposters are nominal expressions that are formally of one person value but function semantically as if they had a different person value, such as *yours truly* (formally 3P, semantically 1P). This approach would be somewhat equivalent to the LF/PF feature mismatch approach discussed above. The basic idea in Collins and Postal's system is that imposters involve a 1/2P pronoun embedded within a larger 3P DP. In order to extend the analysis to non-*de se* pronouns, the possibility of a reverse configuration would have to be introduced: a 3P pronoun embedded within a larger 1/2P DP. In the context of my analysis of person restrictions, this would make non-*de se* pronouns their own phases, just like strong pronouns, and therefore insensitive to person restrictions. What remains a mystery with this analysis is why the additional structural layer present with the 'imposter structure' does not affect the PF character of the relevant pronouns. Recall that a full DP structure and the phase status that comes with it, normally corresponds to prosodic independence (see Section 3.3),[50] so we would expect

50. Unless we are dealing with an analogue of Polish weak pronouns (see Section 3.3), but this does not fully resolve the issue. I argued that Polish weak pronouns are phases in the syntax but prosodically deficient at PF, which accounts for why they are insensitive to person restrictions and behave prosodically as weak pronouns.

these imposter pronouns to be strong pronouns. In any case, what is important is that a syntactic analysis of non-*de se* pronouns is entirely plausible and there are precedents for it with phenomena unrelated to person restrictions. The amelioration of person restrictions with non-*de se* 1P pronouns thus does not force a semantic analysis of person restrictions.

The second phenomenon that is sometimes presented as evidence for person restrictions being semantic is the *clitic binding restriction (CBR)*, which I briefly discussed in Section 2.4. Recall that the restriction occurs essentially in the came contexts as IA-IA person restrictions: when two object clitics co-occur, as shown in (53); the Theme clitic can be bound by the matrix subject only when the animate Goal is not clitic doubled (the observation for Spanish and Catalan was originally made by Roca 1992).

(53) Spanish [*Romance*]:

Mateo$_i$	piensa	que	**se**$_k$	**lo**$_{*i}$	entregaste	*a*	*la policá*$_k$.
Mateo	thinks	that	3.IO	3.DO.ACC	handed.2	to	the police

'Mateo$_i$ thinks that you handed **him**$_i$ over to **her/them**$_k$ [the police].'
(Ormazabal and Romero 2007, 327)

As Pancheva and Zubizarreta (2018) observe, not all languages that exhibit an IA-IA person restriction also exhibit the CBR. For example, Bulgarian has an IA-IA person restriction, just like Spanish, but no CBR, as shown in (54).

(54) Bulgarian [*Slavic*]:

[Scenario: Ivan and Maria are engaged. He is going to dinner to her parents' house tonight. He will meet them for the first time.]

Toj$_i$	misli	če	*tja*	šte	**im**$_k$	**go**$_i$	predstavi	prosto	kato	prijatel,	ne	kato	godenik.
he	thinks	that	she	will	3PL.DAT	3.M.ACC	introduce	just	as	friend	not	as	fiancé

'He$_i$ thinks that she will introduce **him**$_i$ to **them**$_k$ just as a friend and not as a fiancé.'
(Pancheva and Zubizarreta 2018, 1316)

However, as noted in Section 3.3, this special status should also correlate to differences in distribution (e.g. pickiness with respect to appropriate prosodic hosts), which we do not see with *de se* vs. non-*de se* clitic pronouns in Slovenian: they are 2nd position clitics regardless of their interpretation.

Pancheva and Zubizarreta suggest that absence of the CBR is tied to a difference in person restrictions: Spanish has either a STRONG or a WEAK restriction (with different speakers), but Bulgarian has a ME-FIRST restriction (see Section 4.3). They build their analysis on that of Charnavel and Mateu (2015a, 2015b), who argue that person restrictions and the CBR both arise because of a failure to license logophoric centers: a 1/2P clitic in the case of person restrictions and a clitic logophor in the case of the CBR—note that what the CBR excludes is binding by the attitude holder and that logophors are pronouns which must be bound by an attitude holder. Pancheva and Zubizarreta propose that this kind of licensing can be parameterized so that in Spanish 1P, 2P, and certain animate 3P pronouns (which for them form a natural class: [+proximate]) all require licensing via Agree. This parameterization yields a STRONG or a WEAK person restriction (depending on the setting of an additional parameter) and in embedded clauses with two animate 3P clitics specifically it yields the CBR (see their paper for details). In the case of Bulgarian, they propose instead that only 1P pronouns require the same kind of licensing. This parameterization yields a ME-FIRST person restriction, but in embedded clauses like (54) no CBR is observed, since they contain no 1P pronouns and thus no licensing is required.

Note that Pancheva and Zubizarreta's (2018) analysis is, like Charnavel and Mateu's (2015a, 2015b) analysis before it, effectively a syntactic licensing analysis in terms of implementation: a specific class of interpretable φ-features must take part in Agree in order to be licensed. As the class of features is semantically determined, the analysis inherits the issue for the autonomy of syntax present with syntactic licensing analyses.

Another issue concerns the correlation between person restriction strength and the CBR. It is crucial for Pancheva and Zubizarreta (2018) that languages with STRONG and WEAK person restrictions (including those with an extra $*\gg$1P restriction; see Section 4.3) have the CBR, but those with a ME-FIRST person restriction do not. They list Bulgarian and Romanian as instances of the latter, contrasting them with French, Spanish, Catalan, and

Czech, which have both person restrictions and the CBR. However, another language that exhibits the CBR is Bosnian-Croatian-Serbian (Bhatt and Šimík 2009), which has a ME-FIRST person restriction (Runić 2013b).[51] Given the small number of languages considered, it is hard to judge whether this is a true counterexample or an outlier with a principled explanation. Therefore more languages should be examined before we can conclude whether the two phenomena are connected or merely arise in the same syntactic environments.

Setting aside this last issue, let us consider for the sake of argument how one would go about deriving the CBR in the analysis of person restrictions developed in this thesis. The analysis of Charnavel and Mateu (2015b, 2015a), where the CBR is really about the distribution of clitic logophors, can be easily adapted from a licensing analysis into a minimal pronoun analysis. Instead of a logophoric feature requiring licensing, a minimal pronoun needs to have its logophoric feature valued by a corresponding valued feature on a phase head (or the appropriate feature needs to be valued as logophoric), but in the relevant structural configuration the intervening pronoun prevents this and thus prevents the lower pronoun from being a logophor. A logophor requires self-identification by the attitude holder, so the ban on logophors will appear superficially like a ban on binding between the matrix subject (the attitude holder) and the clitic pronoun. This also explains why binding between the matrix subject and the clitic pronoun is possible if the clitic pronoun is not *de se* and thus not a logophor (Bhatt and Šimík 2009; Charnavel and Mateu 2015a, 2015b).

This analysis aligns nicely with the proposal in Section 5.3 that indexical minimal pronouns must be semantically bound. Crucially, von Stechow (2003, 2004), whose approach to semantic binding I adopt, also argues that logophors must be semantically bound by the attitude verb that quantifies over them, resulting in their obligatory *de se* interpretation.

51. Since Bhatt and Šimík (2009) do not provide person restriction data for their consultant, it is a possibility that they do not have a ME-FIRST person restriction, in spite of it being the predominant restriction reported for Bosnian-Croatian-Serbian by Runić (2013b). But the prediction is clear: any micro-variation across speakers with respect to ME-FIRST vs. other person restriction patterns should correlate with the presence of the CBR.

The CBR thus does not necessitate a semantic analysis of person restrictions, as a minimal pronoun analysis of the phenomenon is fairly straightforward. What remains to be established is how deep the connection between the CBR and person restrictions is. The tentative analysis above allows us to be agnostic about it, as it makes the CBR a co-occurrence restriction that is, similarly to the restrictions from Section 5.1, based on something other than person: logophoricity. In terms of implementation, the extent to which person and logophoricity are linked can be adjusted by making person and logophoricity part of the same feature set or not. Establishing on a larger language sample whether or not the CBR and person restriction actually co-vary will be crucial for making the choice.

5.4.4 Summary

The two main alternatives to the minimal pronoun analysis of person restrictions, syntactic licensing and semantic licensing, turn out in fact to be remarkably similar. The basic idea with both is that a specific class of interpretable φ-features must be licensed by taking part in Agree. This kind of analysis presents an issue for the autonomy of syntax, since the class of features must be semantically determined—semantic licensing analyses are a bit more explicit about this than syntactic licensing analyses. Another difference is that in the case of semantic licensing, the direct syntax-semantics interaction is justified by the similarity between person restrictions and phenomena that are clearly tied to semantic notions such as point-of-view and empathy. However, the latter phenomena always show sensitivity to interrogative flip, which person restrictions never do. Since interrogative flip is perhaps the most fundamental test for phenomena sensitive to point-of-view and empathy, its absence casts significant doubt on such a characterization of person restrictions. That said, there are instances where person restrictions interact with semantic factors, but we saw that these too do not force a semantic analysis of person restrictions.

In the first part of this chapter, I offered a possible explanation for why co-occurrence restrictions target a semantically defined class of features, which has to do with the conflicting requirements indexicals impose on semantics and syntax. The analysis I developed based on this idea does not require for semantics to directly influence syntax, and more importantly it can still derive the sensitivity of person restrictions to semantic factors. Because it has the same empirical coverage without compromising the autonomy of syntax, this analysis has an advantage over the alternative options.

If we want to solve a problem that we have never solved before, we must leave the door to the unknown ajar.

—Richard P. Feynman, "The Value of Science"

6

Extensions and open questions

In the course of the previous chapters we saw that, in spite the high degree of crosslinguistic variation, not everything is possible in the realm of person restrictions. Most strikingly, two typological gaps are revealed when a sufficiently large language sample is examined. The first gap concerns languages where EA-IA and IA-IA person restrictions coexist: in such languages the IA-IA restriction is never weaker than the EA-IA restriction—the *Strength Generalization*. The second gap concerns which argument in a given argument pair can be subject to the person restriction (i.e. the direction of the restriction): in a given restriction domain, languages may have only a canonical restriction, both canonical and reverse restrictions, but never only a reverse restriction—the *Direction Generalization*.

Building off of these typological gaps as well as other crosslinguistic generalizations revealed by the survey, I have proposed a new analysis or person restrictions. The most significant result of this analysis is that it makes the typological gaps fall out from grammatical principles independent from the specific mechanisms that cause the person restrictions. On one hand, this gives us a more constrained analysis that does not require the introduction of new operations or constraints specifically to derive the typological gaps, and on the other hand this means that the typological gaps can be seen as evidence reaffirming the validity of those grammatical principles. An additional nice result is that virtually all the crosslinguistic variation can be attributed to small lexical differences with independent grammatical reflexes: (i) the type of pronouns used in the derivation, (ii) the size and number of phases in the clause, and (iii) the movement possibilities afforded to the pronouns.

These minor syntactic differences interacting with each other can result in a domino effect that pushes the person restriction patterns in very different directions. However, since the restrictions still arise through the interactions of very constrained grammatical operations, the variation can never become unbounded. Additionally, none of the points of variation matter exclusively for person restrictions. We saw in several places in earlier chapters how this can help explain many correlations between the variation in person restrictions and several seemingly unrelated phenomena. Based on how the analysis is set up, the expectation is that there are many more such correlations waiting to be discovered.

In this chapter, I explore some possible paths that future explorations within this approach to person restrictions may take. I first review the challenges variation in person restrictions poses for first language acquisition, focusing on the "parameterized Agree" approach to the problem (Section 6.1). I then discuss how the issues that arise with this approach are avoided in the current model, where much of the variation stems from minute differences in the internal structure of pronouns (Section 6.2). Given the relationship between varying phase sizes and person restrictions in the current model, I also discuss how it can derive correlations between person restrictions and extraction asymmetries, as well as another phenomenon that often accompanies them: split alignment (Section 6.3). Finally, I offer some thoughts on the place of person restrictions in grammar, specifically how the proposed analysis relates to a functionalist approach to person restrictions (Section 7.2).

6.1 Person restrictions from a learnability perspective

The mainstream approach to variation in person restriction strength involves parameterization of the Agree operation, so that multiple pronouns of different person values may or may not take part in (Multiple) Agree depending on the parameter setting (Anagnostopoulou 2005; Nevins 2007, 2011; Pancheva and Zubizarreta 2018; Coon and Keine 2018; Foley and Toosarvandani 2018; i.a.). Many of these analyses are designed specifically to handle the more fine-grained variation in person restrictions, such as the different restriction patterns I discuss in Section 4.3. But as I note in that section, no existing account of this type actually derives all the restriction patterns identified through my survey. More importantly, I showed in Section 2.5.2 that analyses where the STRONG vs. WEAK variation is the result of a parameterized Agree cannot explain the *Strength Generalization*. Lastly, as discussed in the same section, there are conceptual arguments against parameterizing Agree, coming from the perspective that it is undesirable to parameterize a core grammatical operation (Borer 1984; Chomsky 1995, 2005; Biberauer et al. 2014). However, I want to argue below that the main challenge for the parameterized Agree approaches is actually learnability.

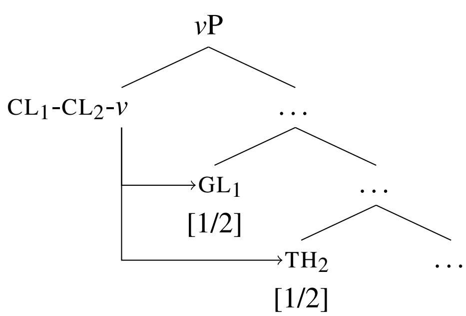
Recall that person restrictions can only arise with deficient pronouns, which is also something that analyses based on a parameterized Agree aim to capture.[1] Specifically,

1. Such analyses generally do not employ a licensing condition on person features (Section 5.4), in contrast to classic intervention analyses like Béjar and Řezáč (2003), where person restrictions are not

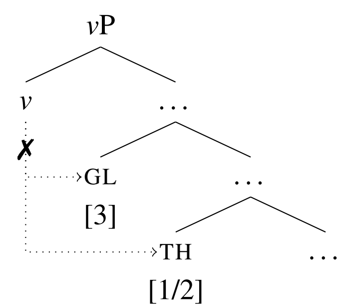
such analyses make the presence of clitics pronouns (henceforth cliticization) contingent on a successful Agree operation with a pronoun. Since Agree, depending on how it is parameterized, may only be successfully established with multiple pronouns when they have specific person values, certain person combinations will render Agree impossible. In this type of approach, the person restriction is effectively a failure of cliticization.

To give a specific example, a WEAK parameterization for the person probe will make possible Agree between the probe (e.g. on *v*) and two 1/2P pronouns, as illustrated in (1). The successful establishment of Agree with both pronouns will then cause them to cliticize. In contrast, if the Goal is 3P and the Theme is 1/2P, the same parameterization must prevent Agree between *v* and the two pronouns, as in (2), making cliticization impossible.[2]

(1)



(2)



The relationship between Agree and cliticization is not crucial here: cliticization may be implemented as a direct reflex of Agree or it may be a two step process fed by movement of the pronoun which is what is really driven or constrained by Agree—the end result is the same. Similarly, there are different ways in which the unwanted person combinations can be

limited to deficient pronouns, drawing a parallel with restrictions on Icelandic nominative objects (see also Anagnostopoulou 2003, 2005); recall that I argued in Section 2.4.3 those are actually a distinct phenomenon. However, nothing prohibits using both parameterized Agree and a (parameterized) licensing condition, as in the case of Pancheva and Zubizarreta (2018). The argument below considers specifically analyses without a special licensing condition, although the learnability issue arises with all analyses that adopt a parameterized Agree approach.

2. Any analysis that uses Multiple Agree in this way must also address why it matters which pronoun is 3P and which 1/2P, since cliticization of both pronouns and therefore Multiple Agree must be possible when the Goal is 1/2P and the Theme is 3P, where the person values are the same as in (2). Different analyses address this differently, but as it does not matter for my argument below, I leave this complication out in the discussion.

excluded, which also does not affect the argument below. Recall that weak pronouns can sometimes also give rise to person restrictions, but that not all deficient pronouns give rise to person restrictions. In order to make this approach to person restriction work with all and only the right pronoun types, Agree must be a prerequisite for the occurrence of only those deficient pronouns that give rise to person restrictions. However, I will keep referring to all the relevant types of deficient pronouns as clitic pronouns for ease of exposition.

Let us now consider the issue of variation in person restrictions from the perspective of first language acquisition (which I touched upon already in Section 1.3.3). The first hurdle for the acquisition of person restrictions is the rarity of relevant examples in the primary linguistic data. In the case of double object distransitives, adult speakers will rarely utter most of the grammatical clitic combinations,[3] like those equivalent to English sentences like "They introduced **me him**", as opposed to prepositional ditransitives like "They introduced **him** to **me**", where the relevant clitic pronoun cluster does not arise. The child learner has to set whatever the relevant parameters are based on very rare utterances. The second hurdle is that the child must set the parameters based only on positive evidence, so they have no way of telling whether the combinations not present in the input are ungrammatical or grammatical but rarely uttered. They could thus in principle end up with a stronger person

3. The rarity of relevant clitic combinations, or even co-occurring object clitics, in adult speech is usually taken for granted, and there are to my knowledge no studies that explicitly provide information on the frequency of specific person combinations in *child-directed speech (CDS)*. To get a better empirical picture, we can look at related studies that indirectly reveal relevant information. For example, Bello (2017) examines a longitudinal corpus of three French acquiring children and their parents (*CHILDES French York Corpus*; MacWhinney 2000; Plunkett 2002) for differences in the production of indirect and direct object clitics. She finds that ditransitive verbs are found in 4–15% of verbal constructions in CDS, which includes all objects, not only clitics, and object omission. Only 50–85% of those contain indirect object clitics, and of those roughly 17–42% also contain a direct object clitic (inferred from the number given for children; the number for CDS is not given, but Bello notes that the two are the same). Given the small sample we cannot draw strong conclusions, but based on these numbers, clitic objects co-occur in CDS in only 0.4–5% of verbal constructions. The person values for the objects are not given, but with such small numbers it is not a stretch to think that a particular parent might not produce all the relevant combinations in the given span of time. It would be interesting to see what counting specific person combinations in CDS would reveal, also for languages like Spanish and Catalan, where adult speakers differ greatly with respect to having STRONG vs. WEAK restrictions.

restriction pattern than the one in the target grammar, should they interpret the gap in the input data as an ungrammatical person combination.

There are two main options here for what we might expect from child learners in such scenarios: (i) they will attempt to generalize the pattern beyond the available data, or (ii) they will act as conservative learners (Snyder 2002, 2007, 2008). In the case of (i), the child will explore different hypotheses even in the absence of evidence, leading them to make errors of commission—producing person combinations ungrammatical in the target grammar—as they gradually get closer to the target grammar. In the case of (ii), the child will not produce a new syntactic structure without first identifying that it is permitted in the target grammar as well as its grammatical basis. Assuming a single parameter regulates the STRONG vs. WEAK variation, the parameter obeys the *Subset Principle* (Berwick 1985; Manzini and Wexler 1987), since the person combinations allowed with a STRONG restriction are a subset of those allowed with a WEAK restriction. So even with a WEAK restriction target grammar, the child will stick to the parameter setting that yields a STRONG restriction at least until they have sufficient evidence that combinations of two 1/2P clitic pronouns are allowed.

Coming back to parameterized Agree analyses, recall that these derive person restrictions as a failure of cliticization, where Agree is a prerequisite for cliticization. Even with a single pronoun, cliticization is possible only if there is Agree between the probe and the pronoun, so acquiring when pronouns can and cannot cliticize entails acquiring when a pronoun can be Agreed with. However, since what needs to be excluded are specific combinations of multiple pronouns, the different parameter settings for Agree only yield different results with multiple pronouns: they regulate when *Multiple* Agree can be established. In other words, a single clitic pronoun must always be a suitable goal for Agree, regardless of its person value. Thus, if all the child hears up to a point are constructions with a single clitic pronoun (by far the most common ones), they could in principle choose any parameter setting for Agree, as they are all compatible with the data in the input so far.

With a learner that generalizes beyond the available data, we then expect a fair number of errors of commission. Namely, they might produce co-occurring clitic pronouns before those occur in the input and even person combinations ungrammatical in the target grammar. When children start producing clitic pronouns, the link between Agree and cliticization must have been acquired, but this also means that a parameter setting for Agree had been chosen. Recall that what the Agree parameter regulates is when Multiple Agree, and therefore multiple cliticization, is possible. Thus, a parameter setting for Agree immediately also gives the child the ability to produce co-occurring clitic pronouns, and even non-target clitic combinations, if the wrong setting is chosen due to limited information. However, studies of the acquisition of clitic pronouns have not identified any such significant errors of commission. In fact, children master the use of clitic pronouns at an adult level surprisingly early and make almost no errors of commission; this has been noted regarding various conditions on clitic distribution and use (see e.g. Snyder, Hyams, and Crisma 1995 for clitic-auxiliary selection in French and Italian, Rodríguez-Mondoñedo, Snyder, and Sugisaki 2006 for clitic-climbing in Spanish, and Powers and Hamann 2000 for a general overview).

The same situation with a conservative learner forces the use of a default parameter setting in order to avoid setting the Agree parameter wrong in the absence of useful data in the input. The default setting in this case has to be one yielding a STRONG restriction, where the setting is changed if person combinations that only arise with weaker restrictions are encountered in the input (see Pancheva and Zubizarreta 2018). However, while this ensures the child will not produce person combinations ungrammatical in the target grammar, it actually does not ensure the child will not produce structures without first identifying that they are permitted in the target grammar. This is again because the parameter regulates only when Multiple Agree can take place: the default setting predicts that any child who has acquired all the clitic forms and the link between Agree and cliticization should be able to immediately produce clitic clusters, as long as they conform to a STRONG person restriction. While this is not an error *per se*, as the clitic combinations that conform to a STRONG person restriction will be grammatical in all possible target grammars that allow co-occurring clitic pronouns, this does mean that children's production is allowed to move ahead of the input, which is not what we generally observe with first language acquisition and in fact goes against the principles behind the idea of conservative learners (see Snyder 2007 for an overview across several distinct phenomena).

In sum, issues arise regardless of the acquisition strategy due the nature of the parameter itself. This is because the parameter is specifically designed to regulate multiple cliticization by way of Multiple Agree. What we expect to see are children whose spontaneous production does not strictly follow what is available in the input: they should be able to produce co-occurring clitic pronouns before these appear in the input. Studies of children's spontaneous production in this domain do not yet exist, so it is entirely possible that children do in fact acquire person restrictions this way. Although, given what has been observed so far in relation to other linguistic phenomena, such a result would be truly surprising.

Another related issue is that person restrictions are observed even in the absence of overt co-occurring clitic pronouns. For example, we saw that in Swahili a WEAK person restriction arises despite there being only one object marker on the verb in double object constructions (see Section 2.4.3). Assuming a conservative learner, the WEAK restriction involves a non-default setting for the Agree parameter, but a child will never receive direct evidence to change the parameter setting from the default STRONG setting to a WEAK setting. That is because 1/2 $\gg$ 2/1 clitic combinations will never occur in the input.

Finally, the rarity of individual person combinations in the input cannot be ignored (see also footnote 3), especially when we consider more fine-grained variation between person restriction patterns (see Section 4.3). There are total a total of 7 person combinations to be considered, where 4 are ungrammatical with a STRONG restriction, 3 are ungrammatical with the two WEAK restrictions that have an additional 1P vs. 2P asymmetry, 2 are ungrammatical with a plain WEAK, ME-FIRST, or YOU-FIRST restriction, and finally a single person combination may be excluded (*$2 \gg 1$ or *$3 \gg 2$). The correct setting of the parameter can thus hinge on a single person combination occurring in the input, and it might take a while for 1 out of the 7 combinations to naturally occur in the input, if it occurs at all.

Note that the potential rarity of the relevant person combinations in the input is an issue even if parameterized Agree is not directly tied to cliticization (Pancheva and Zubizarreta 2018). A version of this poverty of the stimulus problem will arise with all analyses where the parameter regulating the person restriction pattern can only be set by directly observing all the grammatical person combinations in the relevant construction. As I will discuss next, the analysis of person restrictions developed in this thesis allows us to avoid this issue.

6.2 Using external cues to acquire person restrictions

In the analysis I develop in this thesis, whether or not a language has person restrictions is due to the syntactic status of the co-occurring pronouns. The pronouns sensitive to person restrictions are also identifiable via the prosodic and referential properties that reflect their special syntactic status. Thus, person restrictions should be learnable without knowing what all the grammatical person combinations are—other salient cues in the input data should suffice to determine that the pronouns are of the type that gives rise to person restrictions.

Variation in restriction strength in this analysis is also due to differences in pronoun status, specifically in their internal structure. In this section, I review some consequences of this proposal, focusing on its correlates outside person restrictions and micro-variation in restriction strength, the role of number in the structure of pronouns, and contexts where person restrictions arise between co-occurring pronouns of two different types.

6.2.1 Micro-variation in pronoun structure

Recall from Chapter 3 that the source of the STRONG vs. WEAK variation with IA-IA person restrictions lies in the difference in the internal structure of minimal pronouns: those with STRONG restrictions are just a φ-feature bundle, as shown in (3), whereas those with WEAK restrictions have more internal structure, as in (4), where the unvalued person features of the pronoun reside on their own head, separate from other φ-features, like number.

(3)
$$\begin{array}{c} \text{pron} \\ \begin{bmatrix} \text{i}\pi \\ \_\_ \end{bmatrix} \\ \begin{bmatrix} \text{i\#} \\ \text{val} \end{bmatrix} \end{array}$$

(4)
$$\begin{array}{c} \text{pron} \\ \begin{bmatrix} \text{i}\pi \\ \_\_ \end{bmatrix} \quad \begin{bmatrix} \text{i\#} \\ \text{val} \end{bmatrix} \end{array}$$

The reason why (3) vs. (4) is significant for the STRONG vs. WEAK variation concerns how the person feature can be valued. With (3), person can be valued in situ: when the unvalued number feature on the phase head Agrees with the pronoun's valued number feature, the person feature of the pronoun can also be valued by its valued counterpart on the phase head, piggybacking on the Agree already established by the number features. This kind of valuation is allowed due to *Maximize Agree*, see (5), an economy principle that reduces the amount of Agree operations needed to value all the unvalued features in a derivation.

(5) *Maximize Agree*
If a probe matches a goal, all matching features on the probe and goal must be valued at that point in the derivation.

However, *Maximize Agree* is not applicable in the case of the structurally richer (4), because under Agree with the phase head, only number features can be valued—the person features are structurally isolated from them and so do not count as being part of the goal targeted by Agree. Because this leaves the person features unvalued, the minimal pronoun must move to the specifier of the phase head and be valued there (since it c-commands valued person feature there). After that, the lower pronoun can also move to be valued in the same way, which is what makes 1/2 ≫ 2/1 pronoun combinations derivable in WEAK IA-IA restriction languages as opposed to STRONG IA-IA restriction languages.

The question is then, what other correlates are there to this richer internal structure of the pronoun? I suggest that the pronouns in languages with WEAK IA-IA restrictions are generally phonologically stronger and allow a more transparent morphological decomposition. Of course, developing a precise formulation of what this amounts to is just as tricky as developing a solid theory neutral definition of a clitic pronoun. I want to suggest that this is actually an advantage as opposed to a problem. Namely, the learner of a language will in a lot of cases be faced with deficient pronoun forms that can not be straightforwardly parsed as having either the internal structure in (3) or the structure in (4), which predicts a lot of speaker variation in terms of the STRONG VS. WEAK status of the restriction, which is exactly what is generally observed (Perlmutter 1970, 1971; Bonet 1991; Ormazabal and Romero 2007; Anagnostopoulou 2005, 2008; Řezáč 2011; Stegovec 2015b, 2019; i.a.).

Let's consider with this in mind perhaps the best studied examples of clitic pronouns in the generative tradition: Romance clitic object pronouns. In particular, it is useful here to compare French, which is the most stable Romance language in terms of its person restriction (see Řezáč 2011, 150–1 and the references there), and Spanish, which displays a great deal of speaker variation (see e.g. Perlmutter 1970, 1971; Bonet 1991). We would expect, given that only a STRONG IA-IA person restriction is ever found in French, compared to Spanish, that clitic object pronouns will display more fusional inflectional morphology in the former than the later, given the proposal that STRONG restrictions arise with feature bundles and WEAK ones with clitic pronouns with more internal structure.

The two clitic object systems are presented side-by-side in Table 6.1, with the Spanish forms in (a) (I am using the paradigm from Castilian varieties that retain the 2P.PL *-os* clitic) and the French forms in (b). There is a great deal of similarity across the two systems, especially in the 1P and 2P forms, due to their shared origins in their Latin counterparts (see e.g. Wanner 1987; Uriagereka 1988, 346–8), but also across the 3P accusative forms, which have together with definite articles developed from Latin demonstrative pronouns and are morphologically identical to their corresponding definite articles (see e.g. Wanner 1987; Torrego 1988, 1995; Uriagereka 1988, 1995). I suggest that the key differences between the two systems are observed in the dative forms, as well as with how the inflectional morphology corresponds to matching gender-number cases outside the pronominal paradigms.

Table 6.1: Morphology of Spanish versus French clitic object pronouns

(a) Spanish clitic object pronouns

		SG		PL	
		ACC	DAT	ACC	DAT
1P	M F	me		no-s	
2P	M F	te		o-s	
3P	M F	l-o l-a	l-e	l-o-s l-a-s	l-e-s

(b) French clitic object pronouns

		SG		PL	
		ACC	DAT	ACC	DAT
1P	M F	mə		nu(-z)	
2P	M F	tə		vu(-z)	
3P	M F	l-ə l-a	l-ɥi	l-e(-z)	l-œʁ

The dative forms of clitic pronouns are key here, because Spanish and French do not have dative paradigms for strong pronouns and lexical nouns: those are identified as dative by way of the prepositional marker *a/à*. This is important for the issue of acquiring the pronoun forms, because all the evidence for dative inflectional morphology will come from the clitic pronouns themselves, a closed morphological class. This means, the child will have to posit an underlying structure and feature content for dative pronouns based solely on the four paradigmatic cells we get from the two gender and number values (person is ignored here given that 1P and 2P person pronouns never have dedicated dative forms).

Let us consider with this in mind the individual morphemes from the two clitic paradigms. It is important that we identify the dedicated morphemes for each φ-feature or case value, so that we first find as many morphemes that only encode a marked value for a single feature. Note that this way we can easily identify the pronominal stems for 1P, 2P, and 3P for both languages (cf. (6) and (9)), as well as the dedicated inflectional morphemes for plural, dative, masculine, and feminine (cf. (7) and (10)). Additionally, the two languages have the same pattern for the plural 1P and 2P stems (cf. (8) and (11)). French stands out here only in one important way: the presence of fusional inflectional morphology for 3P.PL (*-e*, see (12a)) and DAT.PL (*-œʁ*, see (12b)). In contrast, the 1P and 2P plural stems are the only morphemes in Spanish that encode more than one marked feature value at a time.

(6) a. me (1P)
b. te (2P)
c. l- (3P)

(7) a. -s (PL)
b. -e (DAT)
c. -o (M)
d. -a (F)

(8) a. no- (1P+PL)
b. o- (2P+PL)

(9) a. mə (1P)
b. tə (2P)
c. l- (3P)

(10) a. -z (PL)
b. -ɥi (DAT)
c. -ə (M)
d. -a (F)

(11) a. nu- (1P+PL)
b. vu- (2P+PL)

(12) a. -e (3P+PL)
b. -œʁ (DAT+PL)

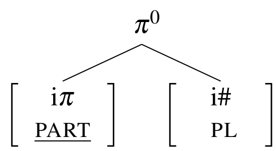
Additionally, Spanish gender and number inflection is the same for 3P accusative clitics, articles, and most nouns and adjectives (Picallo 2007, 2008): masculine *-o*, feminine *-a*, and plural *-s*. This is different in French, where most nouns and adjectives do not have transparent gender and number inflection, and when they do, it typically does not match what is found on clitic object pronouns and articles (Lowenstamm 2012). A child acquiring Spanish clitics has ample evidence for the decomposition of number and gender morphology from both closed and open class categories, which is in principle consistent with a syntactic structure where each φ-feature corresponds to a head (more on this below). In the case of French, however, the input for making generalizations comes only from the clitic object pronouns themselves and the article system—both a closed class. A child acquiring the less transparent inflectional system in French might then posit the segmentation of φ-features and case in clitic object pronouns is largely due to morphological rules of exponence, rather than an underlying articulated syntax of φ-feature heads.

If this line of thought is on the right track, we expect the other Romance languages, which either show WEAK restrictions with some speakers, or only show WEAK restrictions, to align more with Spanish than French or have even more transparent inflectional systems than either. At least for the sample of Romance languages reviewed in my survey, this seems to hold. Catalan, where speakers have STRONG or WEAK restrictions, has a gender and number inflection system very similar to the Spanish one (Bonet 1991; Martín 2012; Walkow 2012), and Italian, where speakers also have STRONG or WEAK restrictions (Seuren 1976; Bianchi 2006; Cardinaletti 2008), has systematic gender and number marking (M: *-o*, F: *-a*, M.PL: *-i*, F.PL: *-e*) in clitic object pronouns (Cardinaletti 2008), as well as articles, and most adjectives and nouns (De Vincenzi and Di Domenico 1999); additionally dative inflection in Italian clitic pronouns even retains the masculine/feminine split. Lastly, Romanian, where only very weak person restrictions like ME-FIRST or $*2 \gg 1$ are attested (Farkas and Kazazis 1980; Ciucivara 2006, 2009; Nevins 2007), is an outlier also in terms of inflection: it has a three-way gender system and marks case distinctions on clitic pronouns, articles, adjectives, and nouns (Daniliuc and Daniliuc 2000). It thus seems that person restrictions weaken as the morphological expression of number, gender, and case inside clitic pronouns becomes more segmentable and transparent, and as the system of number, gender, and case inflection becomes more consistent across closed and open grammatical classes.

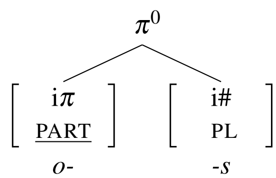
What needs to still be properly addressed though is how the systematicity of inflection in clitic pronouns relates not just to the possibility of having WEAK person restrictions, but specifically to the high degree of speaker variation. I assume here that clear morphological segmentation of morphosyntactic categories does not always entail each segment corresponds to a syntactic head. Specifically, features present in the syntax may still be manipulated in the PF module of grammar (Bonet 1991; Noyer 1992; Halle and Marantz 1993, 1994; Harley and Noyer 1999; i.a.) so that a feature bundle may be broken up into separate features realized by separate morphemes (*fission*), or separate features or sets of features may be realized by a single morpheme (*fusion*, or some other means by which one morpheme can encode grammatical information distributed over distinct syntactic units).[4]

With this type of relationship between syntax and morphology, the child learner has two options when faces with a decomposable clitic pronoun like the Spanish *os*: one with an isomorphic mapping between syntax and morphology, like (13), or one where morphology splits up a feature bundle present in the syntax into multiple morphemes, as in (14).

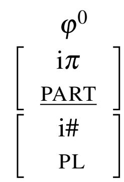
(13) a. Syntax:



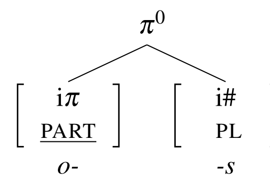
b. PF:



(14) a. Syntax:



b. PF:



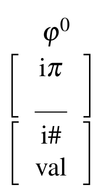
The point I am trying to make in relation to person restrictions is that with the very limited input a child receives concerning the person restriction pattern in a language, and the morphology of clitic pronouns—a closed class, the choice between the two options is close to a coin flip. Choosing one option over another will only be revealed in the rare cases where STRONG VS. WEAK person restrictions can be teased apart with adult speakers

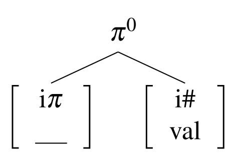
4. For example, contextual allomorphy may also result in a morpheme that encodes both the features of the syntactic terminal where it is realized and the allomorphy trigger. Additionally, several authors have argued that morphemes can be inserted over non-terminal syntactic units (Starke 2009; Caha 2009; Radkevich 2010; Svenonius 2016; i.a.). Importantly, any approach to the morphological realization of syntactic material that can realize information from distinct syntactic terminals on one morpheme will work for the point I make below.

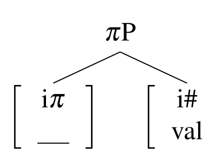
(and hopefully some other grammatical differences). I posit that this is why languages like Spanish display so much fine-grained speaker variation in this domain.

Related to this, we can ask the question of what happens in cases where there is more independent evidence for an articulated internal structure in deficient pronouns. For example, Cardinaletti and Starke (1994, 1999) famously propose that a separate class of deficient pronouns, so called *weak pronouns*, have distinct grammatical behavior from their clitic pronoun counterparts because of having a more articulated internal structure. Most notably, they have a distribution closer to that of phrasal elements than that of heads (see in relation to this the discussion of Polish weak pronouns in Section 3.3.1).

In relation to this, Anagnostopoulou (2008) observes that in the Germanic languages that have person restrictions and unambiguously weak (as opposed to clitic) pronouns, the restriction patterns are always of the WEAK type. I propose, based on this, that there is at least a three-way structural distinction between deficient pronouns that give rise to person restrictions. There are two types of clitic pronouns: those that correspond to feature bundles in the syntax (cf. (15)), and those that are complex heads (cf. (16)). Additionally, weak pronouns have a similar internal structure to the latter, but where the head hosting person features projects a phrase, as illustrated in (17).

(15)


(16)


(17)


The basic idea is that while clitics can either have an internal structure or not, internal structure is a prerequisite for a phrasal status, and thus a weak pronoun status. The result of this is that we correctl predict the one-way correlation between deficient pronoun type and person restriction strength observed by Anagnostopoulou (2008): languages with clitic object pronouns can have both STRONG and WEAK person restrictions, but languages with only weak object pronouns can only have WEAK person restrictions.

In the continuation of this section, I discuss two other ways in which the internal structure and feature content of deficient pronouns may correlate with person restriction type, and offer some tentative suggestions in how this can be captured in the approach to person restrictions I have argued for in the preceding chapters.

6.2.2 Levels of deficiency

Another person restriction pattern that seems to correlate (although not perfectly) with less structurally deficient pronouns is the ME-FIRST variant of WEAK restrictions specifically; recall for example the discussion of Romanian in the previous section. Additionally, the Slavic languages with a ME-FIRST pattern, which includes at least Bosnian-Croatian-Serbian (Runić 2013b) and Bulgarian (Pancheva and Zubizarreta 2018), also have clitic systems that show clear person, number, gender, and case contrasts across the board, and can be largely decomposed into smaller morphemes, unlike most clitics in Romance languages.[5]

Interestingly, Franks (2016) suggests (building on my analysis in Stegovec 2015b, 2019) that ME-FIRST restrictions, which only ban the lower pronoun from being 1P, arise when minimal pronouns are less unspecified in person features. Specifically, although these pronouns have unvalued person features, they can optionally be specified with a participant '2' feature, so they only need to get valued for speaker '1' features if they are to be 1P. When this is impossible, due to an intervening minimal pronoun, they can at most be 2P. In this proposal is on the right track, then we can think of this option as the next step in the progression from minimal pronouns that are just feature bundles (cf. (18)), to more

5. Although having case distinctions for all persons can not be a sufficient factor for having a ME-FIRST pattern, since Greek has a very similar clitic system to Romanian and a consistently STRONG person restriction.

structured minimal pronouns where person is a separate node (cf. (19)), to now structured minimal pronouns with the option of bearing their own participant feature, as in (20).

(18)
$$\begin{array}{c} \text{pron} \\ \begin{bmatrix} \mathrm{i}\pi \\ \_\_ \end{bmatrix} \\ \begin{bmatrix} \mathrm{i\#} \\ \mathrm{val} \end{bmatrix} \end{array}$$

(19)
$$\begin{array}{cc} \multicolumn{2}{c}{\pi \Leftrightarrow \text{pron}} \\ \begin{bmatrix} \mathrm{i}\pi \\ \_\_ \end{bmatrix} & \begin{bmatrix} \mathrm{i\#} \\ \mathrm{val} \end{bmatrix} \end{array}$$

(20)
$$\begin{array}{cc} \multicolumn{2}{c}{\pi \Leftrightarrow \text{pron}} \\ \begin{bmatrix} \mathrm{i}\pi \\ (2)\_\_ \end{bmatrix} & \begin{bmatrix} \mathrm{i\#} \\ \mathrm{val} \end{bmatrix} \end{array}$$

Note that the number feature being on a separate node as person in (20) also means the pronoun has to move to the phase head to be valued just as with plain WEAK restrictions (see above), although just for speaker '1' features in this case. However, person valuation of the first pronoun can bleed the valuation of the next pronoun to move to the phase head, yielding either a $1 \gg 2$ combination, if the second pronoun is specified with the participant feature, or a $1 \gg 3$ combination, if the second pronoun is not specified with the participant feature. Crucially, the first pronoun always has to be valued as 1P when the phase head is specified with a participant feature. Valuing both pronouns as 1P is excluded, just as with the plain WEAK pattern, as it would make the pronouns co-referential and violate Principle B. The only impossible combinations to derive are thus $*2 \gg 1$ and $*3 \gg 1$.

Importantly, if the suggested correlation between richer morphological distinctions on clitic pronouns and ME-FIRST person restrictions proves to be on the right track, it will be at best a one way correlation: minimal pronouns with a particular richer structure would always yield ME-FIRST (or YOU-FIRST) restrictions, but ME-FIRST (or YOU-FIRST) can also arise by alternative means. As I noted in Section 4.3, languages like Quechua exist (Myler 2017), where a YOU-FIRST pattern arises despite there being no independent evidence for the minimal pronouns having a richer internal structure; Quechua is pretty much like any other pronominal argument language in terms of the morphological properties of its deficient pronouns. We therefore need alternative ways of deriving ME-FIRST/YOU-FIRST restrictions, such as the proposal involving feature inheritance I outlined in Section 4.3.

Nonetheless, tying the different levels of deficiency in minimal pronouns to different strengths of person restrictions is an attractive idea that deserves to be explored further. However, since this is not the primary goal of this work, I did not examine the language sample in detail to identify any further correlations between prosodic/morphological deficiency and restriction strength beyond the cases discussed above. But the cases noted above seem to suggest that there is a correlation: the less deficient pronouns get the weaker the restriction gets, until the point where it disappears completely, as in the case of Polish and the Germanic languages without person restrictions (see Section 2.4).

6.2.3 The structure of number

The current system makes yet another interesting prediction concerning minimal pronoun complexity and person restriction patterns. I argued that the main driving force behind WEAK IA-IA restrictions is the inability of person features to be valued by piggybacking on number valuation, resulting in them remaining unvalued and the pronoun having to move. However, the same result would obtain if the minimal pronoun completely lacked number features (and any other φ-features other than person), as illustrated in (21); this would technically make the pronoun even more deficient than the bundled (22). With such pronouns Agree could not even be established between the probe on the phase head and the minimal pronoun, also resulting in them having to move to the phase to be valued.

(21)
$$\begin{array}{c} \text{pron} \\ \begin{bmatrix} \text{i}\pi \\ \text{\_\_} \end{bmatrix} \end{array}$$

(22)
$$\begin{array}{c} \text{pron} \\ \begin{bmatrix} \text{i}\pi \\ \text{\_\_} \end{bmatrix} \\ \begin{bmatrix} \text{i\#} \\ \text{val} \end{bmatrix} \end{array}$$

Interestingly, Kayne (2000, 135–7, 2003) observes that some dialects of different Romance languages have singular 1P and 2P clitics that behave as if they were unspecified for a number value, being singular only by virtue that more specified plural forms exist. Although, I do not have the relevant person restriction data specifically from the dialects discussed by Kayne, there are cases of person restrictions in Romance that show an asymmetry between singular and plural 1P and 2P clitics—and, crucially, a restriction pattern predicted by the current approach under the assumption that singular but not plural clitics lack number features. For example, Rivero (2008) reports that speakers of some varieties of Spanish, including her, have a WEAK restriction with singular object clitic pairs but a STRONG restriction with plural object clitic pairs. The contrast is illustrated in (23): with the singular clitic pair both clitics can be 1/2P, while with the plural clitic pair this is not possible.

(23) Spanish [*Romance*]:

a. **Te** **me** presentaron. — 2SG.GL/TH ≫ 1SG.TH/GL
   2.O 1.O introduced.3PL
   'They introduced **you** to **me**.'
   'They introduced **me** to **you**.'

b. *Ellos **os** **nos** presentaron. — ✗ 2PL.GL/TH ≫ 1PL.TH/GL
   they 2PL.O 1PL.O introduced.3PL
   'They introduced **you(pl)** to **us**.'
   'They introduced **us** to **you(pl)**.' (Rivero 2008, 217, 228)

This is exactly the predicted pattern if singular clitics really lack number features, as in (21), whereas plural clitics have number features bundled together with person, as in (22). In the former case, clitics must move to the phase head to be valued, whereas in the latter case only the top clitic is valued in situ and blocks the person valuation of the lower clitic.

The pattern becomes even more interesting (and complex) when we look at 1P and 2P clitics of mixed number values. As reported by Alba de la Fuente (2010), who also speaks a variety where singular and plural clitics pattern differently, the singular clitic has to be first with such clitic pairs, as shown in (24) (note that the 2P clitics are reflexives, which pattern just like regular 2P clitics in this respect).

(24) a. ¡Huy, que **te** **nos** ensucias! $\boxed{\text{2SG.TH} \gg \text{1PL.GL}}$
oops that 2.DO.(REFL) 1PL.IO stain.2
'You are staining **yourself** on **us**.'

b. *¡Huy, que **os** **me** ensucias! $\boxed{✗\,\text{2PL.TH} \gg \text{1SG.GL}}$
oops that 2PL.DO.(REFL) 1.IO stain.2
'You are staining **yourselves** on **me**.' (Alba de la Fuente 2010, 207)

This pattern is also consistent with the analysis suggested above: only plural clitics completely block valuation of the lower pronoun, by being valued for person at the same time as the phase head establishes Agree with them for number. The assumption is then that in (24a), after the singular pronoun moves, the plural one can follow suit.

There is, however, a complication with this way of interpreting the data. Note that in the examples in (24), the first clitic is always the Theme, but generally with 1P and 2P clitic combinations Spanish allows either clitic to be interpreted as the Goal or Theme, as seen in (23) above. Alba de la Fuente (2010) does not provide sufficient examples to check whether in other cases with mixed singular/plural clitics the order is also rigid. Since not all Spanish speakers have this restriction pattern, I am still looking for speakers that can help determine whether the restriction is based on the order of clitics or on their thematic role, which is crucial in order to determine at which point in the derivation the restriction takes place.

Nonetheless, the important point here is not the correct analysis of this particular Spanish pattern, but the fact that such restriction patterns exist in the first place and that the proposed approach to person restrictions can capture them. Under syntactic analyses where the number features of the intervening pronoun play no role in giving rise to person restrictions—which includes pretty much all the syntactic intervention analyses that I am aware of—we would not expect such asymmetries to exist at all. In order to derive them with those analyses, one would probably have to posit licensing conditions for number features on par with those on person features (see Chapter 5). But then we would expect to find number restrictions comparable to person restrictions crosslinguistically, and those are not attested.

There are no languages with restrictions on exclusively number features, only cases such as (23)–(24), where different number values affect the underlying person restriction pattern. More examples of such languages, limiting ourselves to the IA-IA domain, include Catalan (Alba de la Fuente 2010), Romanian (Ciucivara and Nevins 2008; Alba de la Fuente 2010; Ivan to appear), Bulgarian (Pancheva and Zubizarreta 2018, 1314n24), and Haya (Duranti 1979). Although the restriction patterns themselves can be different in their specifics, the general picture is that plural clitics are subject to stronger restrictions than singular clitics. What is important to consider about these cases, especially in relation to the discussion in Chapter 5, is that the influence of number on person restrictions is not due to number being somehow part of a different (semantic) class of features in these languages. The change in the restriction patterns is the consequence of unvalued person features having to share space on minimal pronouns with other φ-features; if these features are for whatever reason exceptional in a language in terms of how they affect, for example, the successful application of Agree, then this will also affect how the person features will be valued.

6.2.4 Intervening strong pronouns

I key assumption made in the chapter so far is that person restrictions only affect deficient pronouns, clitic and weak pronouns, and the agreement markers in pronominal argument languages. In fact, I argued in Section 2.4.3 that apparent cases of person restrictions arising with strong pronouns or lexical noun phrases, such as with nominative objects in Icelandic, are not in fact the same kind of phenomenon as what I have been calling person restrictions. However, there exist person restrictions which arise between pronouns where one of them is unambiguously a strong pronoun, and which cannot be excluded along the lines of the Icelandic restriction and other similar restrictions discussed in Section 2.4.3.

We established already in Chapter 1 that French has a STRONG person restriction between indirect and direct object clitics. However, a similar restriction is also observed with the *faire-infinitif* causative construction (see, among many others, Kayne 1975, Ch. 3 & 4, 2005 for discussion). As illustrated in (25), when both the dative Causee and the DO from the infinitival clause are clitic pronouns, the DO must be 3P.

(25) French [*Romance*]:

a. Marcel **la** **lui** a fait dessiner. $\boxed{\text{3.DAT} \gg \text{3.DO}}$
Marcel 3F.DO 3.DAT has made draw.INF
'Marcel made **her/him** draw **her**'.

b. *Marcel **vous** **lui** a fait dessiner. $\boxed{\text{✗ 3.DAT} \gg \text{2.DO}}$
Marcel 2(PL).DO 3.DAT has made draw.INF
'Marcel made **her/him** draw **you**'. (Řezáč 2011, 128)

However, what is most important for the issue at hand is that even when the dative Causee is a lexical noun phrase or string pronoun, the clitic DO must still be 3P, as illustrated in (26).

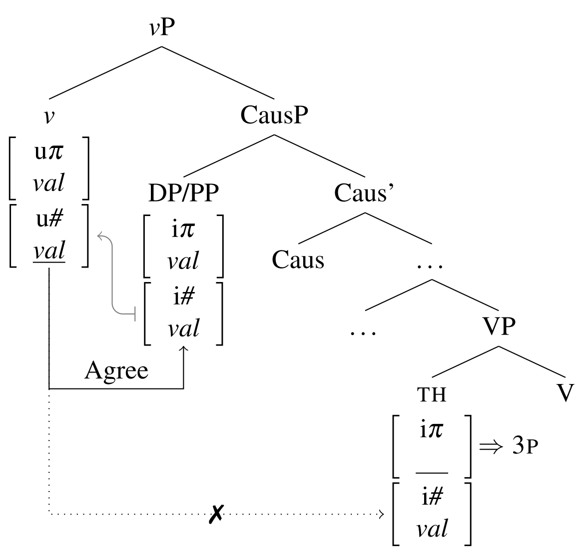
(26) a. Marcel **l'** a fait dessiner *à* *Ilse* . $\boxed{\text{DP.DAT} \gg \text{3.DO}}$
Marcel 3F.DO has made draw.INF to Ilse
'Marcel made *Ilsa* draw **her**'.

b. *Marcel **vous** a fait dessiner *á* { *Ilse* / *nous* }. $\boxed{\text{✗ DP.DAT} \gg \text{2.DO}}$
Marcel 2(PL).DO has made draw.INF to { Ilsa / us }
'Marcel made *Ilsa/us* draw **you**'. (Řezáč 2011, 128)

This pattern, already noted by Kayne (1975, 241) (see also Řezáč 2011, 124–131 for detailed discussion and additional references), was first discussed at length by Postal (1989), who dubbed it the *Fancy Constraint* (FC). What I would like to draw attention to here is that unlike with the restrictions from Section 2.4.3, what is restricted in person value is a clitic pronoun. The Causee—clitic or not—is merely the intervener that creates the syntactic configuration where the person restriction can arise (see Řezáč 2011, 129 on why the Causee is structurally higher than the DO in the relevant point in the derivation).

Although I so far only discussed derivations of IA-IA person restrictions where both the IO and DO are deficient pronouns, the analysis I proposed in this work (and its predecessors presented in Stegovec 2015a, 2015b, 2016, 2019; i.a.) does not actually require the intervening element to be a deficient pronoun as well. This is because the first step of the relevant derivation always involves Agree established between a *v* with unvalued number features and matching valued number features on the closest argument. As illustrated in the simplified derivation in (27), the argument closest to *v* is an intervener for Agree between *v* and any minimal pronoun lower in the structure, regardless of whether the former has valued or unvalued person features. In both cases, the impossibility of Agree between *v* and the minimal pronoun will result in it getting a default 3P value as a last resort, just as it would in a regular STRONG IA-IA person restriction derivation in this system.

(27)



What is particularly interesting about the Fancy Constraint is that it also seemingly bypasses the competition effects that Perlmutter (1970, 1971) observed: strong pronouns have marked/contrastive interpretations when a clitic counterpart exists, but lose such interpretations when the use of the clitic counterpart would lead to ungrammaticaily—such as with a person restriction. In Section 3.4, I attributed such effects to a *Minimize Structure* principle. In all the cases I discussed so far, the avoidance strategy (or repair) involves a prepositional dative construction where the PP is structurally lower than the minimal pronouns or a structure where the strong pronoun is the structurally lower pronoun (i.e. the one that would otherwise be restricted). I direct the reader to Řezáč (2011, 124–131) for a detailed discussion of the (non-)availability of the usual person restriction repair structures in this context. An interesting future direction to explore in relation to this issue would be a broader crosslinguistic examination of non-deficient interveners.

6.2.5 One last open question about the deficiency of pronouns

Before I conclude this section, I want to draw attention to one more open question about different levels of pronoun deficiency. I argued that pronouns can at one extreme be just bundles of features, and at the other extreme develop sufficient structure to count as their own phase, thus becoming immune to person restrictions. But what remains unclear is the deciding factor that tips the scales so that a deficient pronoun no longer counts as deficient. One possibility is that what makes pronouns count as phases is that none of the features they can have (apart from maybe case) are unvalued. Franks (2016) suggests that some pronouns are less deficient in that they can have valued participant features, and I also suggested in Chapter 5 that indexical features other than person can be unvalued, but do not have to be unvalued in all languages. A natural extension of these two ideas is that languages vary in how deficient their indexical feature set is, to the point that all can be unvalued or none of them can be unvalued. If a pronoun has only valued indexical features, it is a phase.

Another possibility is to look for a solution in terms of why minimal pronouns are not phases in the first place. Recall that I assumed that the extended projections of all major lexical categories are phases (following Bošković 2013, 2014). But where does this leave pronouns? It is a pretty common assumption that pronouns are in fact not a lexical category, but functional elements (see e.g. Postal 1969; Longobardi 1994; Moskal 2015a, 2015b). In that case, the way pronouns can count as phases must be different from how lexical categories do. The latter are phases by default, no matter how big their extended projection is (the problem seems related to how CP ends up counting as a phase in this type of model; see Bošković 2014 for discussion). Thus, if we do not want to say that strong pronouns, unlike deficient pronouns, simply are nouns,[6] we must assume that there is a backup strategy for establishing phase-hood. A possibility here is that at some point, as the extended projection of pronouns becomes large enough, it becomes indistinguishable from the extended projection of nouns, even though the two categories differ in terms of their lexical/functional status. That could be sufficient to "trick" whatever mechanism determines the phase-hood of syntactic objects into treating both as the same. Although I leave open here what is the right way to tackle the phasal status of strong pronouns, the issue more or less comes down to the question of what is the key factor that makes any syntactic projection a phase in the first place. While we may be closer to a definitive answer to that question than we were when phases were first proposed, we are still far from having one.

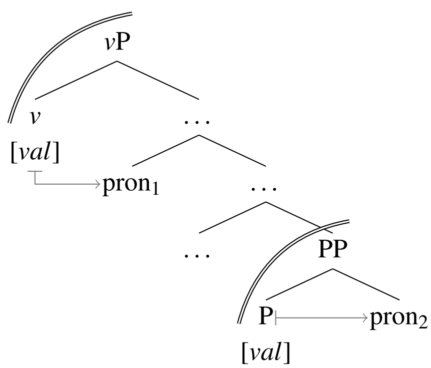
6.3 A few notes on phases (within phases)

In this section, I explore the other factor responsible for variation in person restrictions: phases. Although I will not make direct reference to the problem of language acquisition, as in the previous section, the main point is similar: a theory where person restriction patterns correlate with independent grammatical phenomena should be preferred due to providing cues for language acquisition outside the grammatical pronoun combinations themselves.

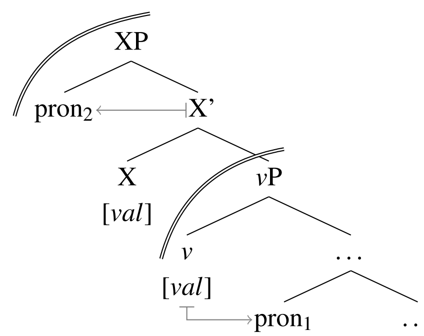
6. In which case we also have to be careful in how to distinguish strong pronouns from regular nouns as to not lose all the insights from the works which suggest that all pronouns are functional categories.

So far, when I considered cases where two or more minimal pronouns are valued in separate phases, I only considered cases where the two phase configuration involved another extended projection within the verbal extended projection, such as a PP within *v*P, as in (28). However, it is at least a logical possibility for the verbal phase to be embedded within another extended projection, as in (29) (in addition to the obvious case of the CP phase, in which case there is presumably always more material between them).

(28)



(29)



In fact, (29) looks very much like what has been proposed in relation to aspect-based split ergativity by Laka (2016) and Coon (2010, 2013), who argue that split ergativity arises because nonperfective aspects have the semantics of locative predicates that can consequently be expressed as embedding verbs. Either because of the aspect head counting as an embedding verb, or because the projection it embeds must be nominalized (see Laka 2016; Coon 2010, 2013), this means that the aspect projection above the verb will count as a separate extended projection, and therefore its own phase in the system of Bošković (2013, 2014). Recall that the possibility of aspect projecting a phase was employed in the analysis of the Christian Barwar [*Neo-Aramaic*] STRONG EA-IA restriction in Chapter 4, where I also suggested that the lack of person restrictions with imperfective aspect is due to the two minimal pronouns being in separate phases. Here I come back to explore the consequences of this configuration independent from the person restriction itself.

In Christian Barwar, the STRONG EA-IA restriction only arises with the perfective aspect (Doron and Khan 2012b; Kalin 2014b; Kalin and van Urk 2015), as illustrated in (30).

(30) Christian Barwar [*Neo-Aramaic*]:

a. griš **-a** { **-li** / **-ləx** / **-le** }. — 1/2/3.EA ≫ 3.IA
pull.PERF -3.F.O { -1.SU / -2.F.SU / -3.M.SU }
'**I/you/he** pulled **her**.'

b. *griš **-at** **-li**. — ✗ 1.EA ≫ 2.IA
pull.PERF -2.F.O -1.SU
'**I** pulled **you**.'

c. *griš { **-an** / **-at** } **-le**. — ✗ 3.EA ≫ 1/2.IA
pull.PERF { -1.F.O / -2.F.O } -3.M.SU
'**He** pulled **me/you**.' (Doron and Khan 2012b, 232)

However, the IA marker is completely unrestricted with imperfective aspect. This is shown by the contrast between the perfective (31a) and the imperfective (31b) (the same pattern is also attested in the related Jewish Zakho, Christian Barwar, Senaya, and Telkepe; see Kalin 2014b). Note also that not only is the person restriction absent in the imperfective example, but that the markers reverse their order between the two aspects: in the perfective (31a), the marker closer to the verb expresses the IA and the one farther from it expresses the EA. In the imperfective (31b), the order is flipped and the marker closer to the verb expresses the EA while the one farther from it expresses the IA.

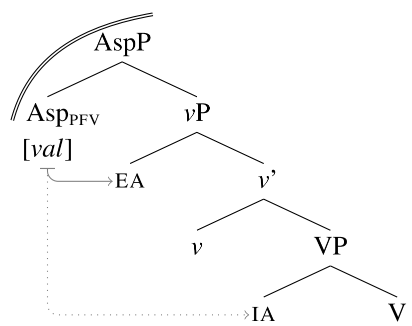
(31) a. *qṭil **-an** **-na**. — ✗ 3.EA ≫ 1.IA
kill.PERF -1.O -3.F.SU
'**She** killed **me**.'

b. qaṭl **-a** **-li**. — 3.EA ≫ 1.IA
kill.IMPF -3.F.SU -1.O
'**She** kills **me**.' (Kalin 2014b, 5)

I would like to suggest that the reversal is actually a direct result of the alternative phase configuration responsible for the absence of person restrictions in (31b). This is inspired by Kalin and van Urk (2015), who attribute the reversal to the presence of an extra probe. Here I will argue specifically that the extra probe is also an additional phase head.

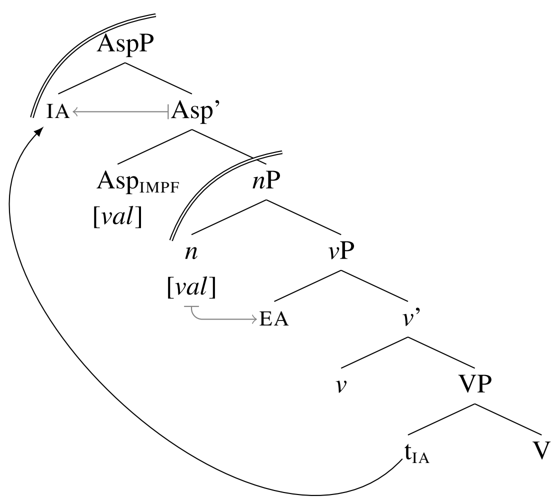
I argued STRONG EA-IA restrictions are the result of *v*P not projecting a phase (due to being too deficient to do so), so in a case like this the perfective aspect head inherits its person valuation duties as the backup phase head (see Section 4.1.1 for discussion of other cases when this takes place). As shown in (32), this means the EA minimal pronoun blocks the person valuation of the IA pronoun, forcing the latter to be 3P.

(32)



Based on insights from Laka (2016) and Coon (2010, 2013), I assume imperfective aspect involves the nominalization of *v*P, where *v* is still unable to project a phase, but the nominalizing head *n* can. This results in the structure in (33) (discussed below).

(33)



Since the *n* head is a phase head, it can person value the EA, but not the IA. However, I assume that the imperfective Asp head, which is essentially an embedding light verb (Laka 2016; Coon 2010, 2013) projects its own extended projection and therefore a phase. As such, the Asp head may also person value a minimal pronoun. And since the EA is already in an Agree relation with *v*, but it nonetheless intervenes between Asp and the IA, the IA can only be person valued by Asp if it moves to it. I crucially assume here (following Rackowski and Richards 2005; den Dikken 2009; Bošković 2016a, 2016b) that movement out of phases does not proceed through the specifier of the phase, but targets the projection just above the phase. In the resulting configuration, not only can both minimal pronouns receive a person value independently from each other, we also get the reordering effect with the IA marker ending up being the one further from the verb (the suggested analysis also has similarities to Borer 1994, where aspect is responsible for assigning case to the IA).[7]

I want to now turn to the other case that was discussed in Section 4.1.1 in relation to a variable STRONG EA-IA restriction: Kaqchikel. Recall that in Kaqchikel the EA-IA restriction is not present with the baseline transitive construction, which has an ergative-absolutive alignment, but only with the so-called Agent-Focus construction, which is employed whenever the EA is Ā-moved. The contrast is illustrated again in (34).

(34) Kaqchikel [*Quichean-Mamean*]:

a. *ri achin* x- **a-** **r-** ax-aj *rat* — $\boxed{3.\text{EA} \gg 2.\text{IA}}$
the man CMPL- 2.ABS- 3.ERG- hear-ACT you
'**He** [the man] heard **you**.' (Preminger 2014, 16)

b. *ja *ri achin* x- **Ø-** ax-<u>an</u> *rat* — $\boxed{\text{✗}\, 3.\text{EA} \gg (2.\text{IA})}$
FOC the man CMPL- 3.ABS- hear-AF you
'It was **him** [the man] that heard you.' (Preminger 2014, 18)

Crucially, the change in alignment (and presence of person restrictions) is also accompanied by a change in extraction possibilities for the EA: the EA can not undergo Ā-movement with

7. Interestingly, other Neo-Aramaic languages like Senaya (Kalin 2014b; Kalin and van Urk 2015), only show a *partial reversal* of the markers: the inner marker expresses the EA and the outer marker the IA with imperfective aspect, while in the perfective aspect there is only a marker referencing the EA. In the current system, this can be correlated to the fact that the IA never Agrees with Asp in the perfective aspect (cf. (32)). The overtness of the IA marker could therefore be attributed to different levels of deficiency of *v*: if it has no φ-features at all, then the IA is null, but if it at least has some unvalued φ-features, it can Agree with the IA and license it to be expressed overtly. Note that since *v* is not a phase head, it can never have valued person features, so this Agree relation would leave the IA restricted to 3P.

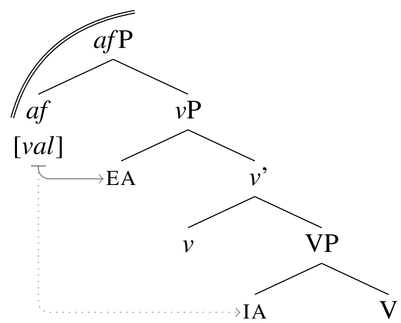
the baseline ergative-absolutive clause, as shown in (35a), which is what the Agent-Focus construction is employed for, as in (35b).

(35) a. **achike* x- **Ø-** **u-** tëj *ri* *wäy*?
who CMPL- 3.ABS- 3.ERG- eat the tortilla

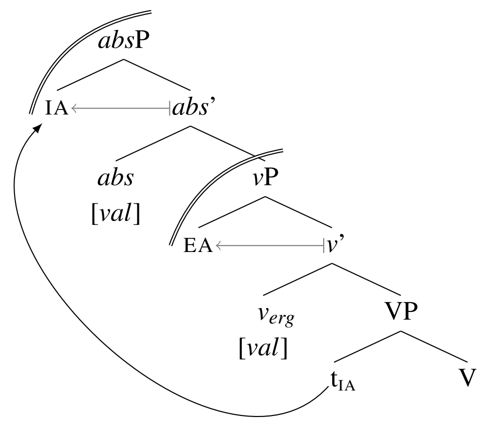
b. *achike* x- **Ø-** tj-ö *ri* *wäy*?
who CMPL- 3.ABS- eat-AF the tortilla
'Who ate the tortilla?' (Erlewine 2016, 430)

I propose that the two constructions are related to each other in basically the same way as the perfective and imperfective constructions are in Christian Barwar. As I already suggested in Section 4.1.1, the Agent-Focus construction involves a deficient *v* that can not project a phase boundary, so the Agent-Focus head (*af*) above *v*P must serve as the backup phase head, as shown in (36). This is what causes the STRONG EA-IA restriction (recall also that Kaqchikel also has a canonical/reverse restriction alternation, where the IA A-scrambles above the EA, reversing the restriction pattern). In contrast, the ergative-absolutive construction involves a different kind of *v*, which can assign ergative case to the EA (see Woolford 1997, 2006; Legate 2008 on ergative case assigned in this type of configuration) and can project a phase boundary. However, I crucially assume that this type of *v* cannot Agree with the IA (perhaps because it only has valued person features), which means that the IA has to move to a special absolutive (*abs*) head (potentially a voice-related head), which is also a phase head with valued person features (comparable to the imperfective aspect head in Christian Barwar). As shown in (37), this allows both the EA and the IA to be person valued and correctly predicts the absence of person restrictions in the baseline ergative-absolutive transitives.

(36)



(37)

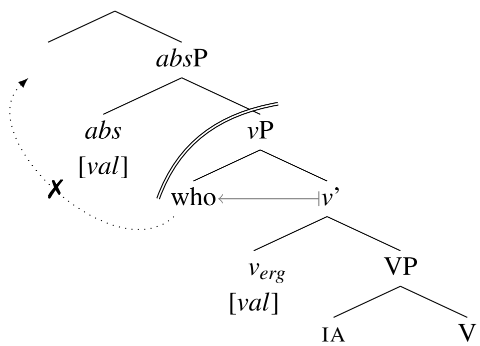


Note that just as in Christian Barwar, the absolutive marker is further from the verb stem than the ergative marker (cf. (34a) vs. (35a)), and there is additional evidence from Kaqchikel (and related Mayan languages) suggesting that it is a 'high absolutive' language, where the absolutive IA asymmetrically c-commands the EA in their final A-positions (see Coon, Baier, and Levin 2019 for discussion). In the current analysis, this follows from the phase configuration required to person value the two minimal pronouns and assign them ergative and absolutive case. Additionally, this phase configuration can also explain the extraction asymmetries between ergative-absolutive constructions and Agent-Focus, as I show next.

An influential analysis of the Kaqchikel extraction facts is that it results from an anti-locality violation: a ban on movement that is too short (see Bošković 1994; Abels 2003b; Grohmann 2003). This type of approach is adopted, among others, by Erlewine (2016), where the gist is that Ā-movement from a subject position in SpecTP to SpecCP is too short in the ergative-absolutive construction, but there is an additional projection between them in the Agent-Focus construction that makes the Ā-movement possible. However, this type of analysis of the extraction asymmetry has been recently challenged by Henderson and Coon (2018) on empirical grounds. The alternative analysis I propose is instead tied to the 'high absolutive' construction (following in that sense Coon, Baier, and Levin 2019) in conjunction with the phase stacking of the *abs*P phase over the *v*P phase.

In addition, I assume following Bošković (2016a) that Ā-movement of the EAs can proceed directly from its base position to SpecCP in some languages (see Bošković 2016a for data and additional references), and that in fact in Kaqchikel it always proceeds this way—which is when Agent-Focus morphology is observed. This means, among other things, that Ā-movement of the EA will never proceed through *abs*P, as shown in (38).

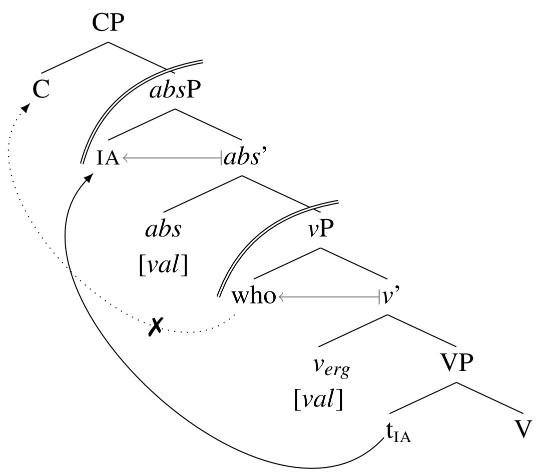
(38)



However, recall that both *abs*P and *v*P are phases. Let us assume than that Chomsky (2001) is correct in that the spell-out for a given phase happens as soon as the next phase up is complete. In addition, what is sent to spell-out is not the complement of the phase head, but the entire phase (see Bošković 2016b for arguments for this type of approach). Note that, as discussed by Bošković (2016b), this is very much in line with what I have already been assuming, namely that movement out of phases does not proceed through the phase edge (as in Chomsky 2001), but that it targets a position just above the phase (following Rackowski and Richards 2005; den Dikken 2009; Bošković 2016a, 2016b). A consequence of this approach is that when a phase (XP) is immediately contained inside another projection that is also a phase (YP), XP becomes an island for syntactic extraction as soon as the YP phase is completed. This is because of how spell-out is timed in this system: once the YP phase is complete, the XP phase is spelled-out, and all syntactic elements still inside XP become frozen in place. An "escape hatch" is available only for elements inside XP that move into YP before the latter is complete (e.g. movement of the complement of XP into SpecYP).

I propose that this is exactly what happens in Kaqchikel, specifically in a configuration like (39), which is forced when the EA in ergative-absolutive clauses should move to SpecCP: the *abs*P phase is complete as soon as the IA merges with *abs*P, which then traps the EA in Spec*v*P by spelling-out the *v*P phase before the EA can move above *abs*P.

(39)



What is important here is that the IA can move to the specifier of *abs*P, because at that point *abs*P is not yet complete—in fact, the IA merging with *abs*P and forming its specifier is what completes the *abs*P phase and triggers the spell-out of *v*P, trapping the EA inside.

The difference with Agent-Focus derivations is that *v*P is not a phase—as I argued in Section 4.1.1, so when *af*P is complete, the EA is not trapped in *v*P and can still move to SpecCP, which I assume is just above *af*P. Since the CP phase is only complete once its specifiers are merged, the *af*P phase is not yet spelled-out at the point at which the EA moves to it. The phase stacking analysis therefore not only captures the relevant person restriction asymmetry, but also the extraction asymmetry. This is completely in line with the goals of the current analysis of person restrictions whose aim it is to find independent cues for the structural configurations that either yield or ameliorate person restrictions.

Person restriction patterns with this kind of ‘split alignment’ asymmetry are actually quite common crosslinguistically, even though they do not all necessarily look exactly the same on the surface. Another interesting case is Tangut [*Qiangic*] (Kepping 1979), where the pattern is abstractly very much like the one in Kaqchikel: a STRONG EA-IA restriction with a canonical/reverse alternation in one construction, and no EA-IA restriction in an alternative ergative-absolutive construction. The main difference is that it is the former which is the default transitive construction, and the latter only serves as a strategy to avoid person restrictions. As shown in (40): either the EA or IA can be the sole argument indexed on the verb, and the one that is indexed is the 1/2P one if the other one is 3P (just as in Kaqchikel). This essentially forms the basis of the canonical/reverse alternation in Tangut.

(40) Tangut [*Qiangic*]:

a. *ni*^{2} *pha*^{1} *ngi*^{2}*-mbɪn*^{2} ndɪ^{2}-śi̯ei^{1} **-na**^{2} — $\boxed{2.\text{EA} \gg (3.\text{IA})}$
you other ∅-wife ∅-choose -2
‘**You** choose **her**, another wife!’

b. *Mei*^{1} *Swen*^{1} ma^{1}-nə *na*^{2} khê^{1} **-na**^{2} … — $\boxed{2.\text{IA} \gg (3.\text{EA})}$
M. S. formerly-∅ you hate -2
‘**Meng Sun** formerly hated **you**, … ’ (Kepping 1979, 273)

The person restriction is observed when both arguments are 1/2P. In that case, the regular transitive alignment is unavailable. Instead, it is always the same argument that is indexed on the verb: the IA This is shown in the examples in (41) (as Tangut is a dead language, the ungrammaticality of two 1/2P arguments is actually inferred by Kepping from the obligatory use of the alternative construction in such cases in the existing written records).

(41) a. … *nga*^{2}*-mi*^{2} ngi̯u^{2} **-na** — $\boxed{(1.\text{EA}) \gg 2.\text{IA}}$
me-PL save -2
‘… **we** will save **you**.’

b. *ni*^{2} tɪn^{1} *nga*^{2}*-ɪn*^{1} ldi̯ə̣^{1} thɪ^{1} **-nga**^{2} … — $\boxed{(2.\text{EA}) \gg 1.\text{IA}}$
you if I-ACC indeed chase -1
‘If indeed **you** are chasing **me**, … ’

c. *nga²-in śâ¹-men¹ kwi̯e¹* ndɪ²-khi̯on **-nga** $\boxed{(2.\text{EA}) \gg 1.\text{GL} \gg (3.\text{TH})}$
me-DAT śramana fruit give(.IMP)-∅ -1
'[you] Give **it**, the fruit of the śramana, to **me**!' (Kepping 1979, 268–74)

DeLancey (1981) suggests that this pattern is likely a historical remnant of an earlier less degraded ergative-absolutive pattern. Independently of whether that is correct, the data are consistent with a characterization on terms of an alternation between two constructions comparable to the Agent-Focus and the ergative-absolutive constructions in Kaqchikel.

Specifically, the pattern in (40) is consistent with an analysis in terms of a structural configuration where the default phase head relevant for the person valuation of both EA and IA is a head above *v*P (resulting in the STRONG restriction; cf. the Kaqchikel derivation in (36)): only one of the arguments can be person valued, acting as an intervener for the person valuation of the other. In contrast, the pattern in (41) is consistent with a configuration where the EA is person valued in the Spec*v*P of an "ergative" *v* (as in the Kaqchikel derivation in (37)), while the IA moves to a phase head just above it—comparable to *abs*P in Kaqchikel—to be person valued. Since each of the arguments can be person valued by a different phase head, the result is the absence of person restriction with the "ergative" indexing pattern.

In addition to such alternations with STRONG EA-IA restrictions, comparable alternations where the EA-IA restriction is a WEAK one also exist. This is for example the case in Kashmiri [*Indo-Aryan*] (Wali and Koul 1997, 2006; Bárány 2015a, 2017) and Pashto [*Eastern Iranian*] (Roberts 2000), where the WEAK EA-IA restriction is observed with some tenses/aspects and no restriction occurs in others. Although a more careful consideration of those types of languages is needed to offer a complete analysis of the alternation in those languages, these are consistent with an approach comparable to the one I outlined above for Christian Barwar, Kaqchikel, and Tangut. That is, the person restriction arises with a single phase in the verbal extended projection, while the lack of person restrictions is the result of phase stacking.

The difference is only in that the WEAK restriction comes from a regular *v*P phase (see Section 4.1.2), where the person restriction arises due to the timing of the person valuation of the EA minimal pronoun in Spec*v*P in relation to the person valuation of the IA minimal pronoun inside the complement of *v*P: the former valuation can bleed the other. In contrast, the alternative alignment where person restrictions are not observed is also the result of a phase stacking, where tense/aspect projections above *v*P project a phase for some tense and aspect values (comparable to what I suggest for Christian Barwar in (33)). The key point is that even though we need to consider each of these languages on a case-to-case basis, the proposed analysis of person restrictions offers a new set of tools for approaching the analysis of alignment splits that interact with person restrictions. This is a welcome result since the exact reason why such alternations exist is still largely a mystery.

6.4 Summary and concluding remarks

In this chapter I primarily discussed the acquisition problem posed by person restrictions, and offered some tentative solutions from the perspective of the analysis of person restrictions introduced in Chapters 3–5. The main claim was that, unlike other analyses, my analysis offers a way of bypassing the acquisition problem by shifting the issue to independent cues in the primary linguistic input. The main cases I discussed were the morphology of pronouns themselves and degrees of pronominal deficiency. Another potentially relevant case that was brought up at the end of the chapter was other syntactic phenomena sensitive to phases, such as movement of pronouns and extraction asymmetries.

A related issue that was tackled was the propensity for person restrictions to display a great deal of fine-grained speaker variation, especially in the IA-IA domain. I argued that this too is related to the acquisition problem, and the difficulty of identifying whether or not morphological decomposition is interpreted as a transparent reflection of the internal structure of deficient pronouns. I suggested that children may interpret the same morphological patterns as corresponding to different types of structures, which yield the fine-grained STRONG vs. WEAK variation in adult grammars. In a sense, this problem can be related to a similar study of syntactic patterns being affected by the morphology of pronouns: the work of Neeleman and Szendrői (2007) on *radical pro-drop*. They identify a crosslinguistic correlation between agglutinating morphology for case, number, or other nominal features on pronouns and the availability of radical pro-drop, such that only languages with clear agglutinating morphology in pronouns allow Chinese-style radical pro-drop. What is particularly relevant here for the related issue with the relationship between the morphology of pronouns and person restrictions is Neeleman and Szendrői's discussion of cases where morphological decomposition not straightforward, but also not completely out of the question:

> *A rule-based analysis can be devised for any paradigm, as long as one allows oneself an unlimited number of allomorphy rules for both stems and affixes. But if the complexity of the rule system is comparable to (or exceeds) the complexity of the actual paradigm, it seems likely that native speakers simply list unanalyzed forms* [. . .] (Neeleman and Szendrői 2007, 693)

What I suggested in this chapter is that there is a third option, at least in the case of the relationship between the internal syntax of pronouns and the variation in person restriction types. Namely, due to the closed class nature of deficient pronoun paradigms, child learners may reach different conclusions regarding the internal syntax of deficient pronouns from the limited data available to them in the input. The hope is that future studies can establish with more precision what exactly is the cut-off point between cases that can be interpreted ambiguously and those cases that are clear-cut enough to not pose the same problem for children acquiring the language.

. . . but the Universe is an awfully big place.
There is room enough for an awful lot of people
to be right about things and still not agree.

—Kurt Vonnegut, *The Sirens of Titan*

7

Conclusion

I began this thesis by suggesting that not only is combining the insights of the generative linguistics enterprise with linguistic typology possible, it can also be applied to an empirical problem as complex as syntactic person restrictions. I hope to have shown the viability of this style of linguistics over the course of the thesis. Guided by generative syntactic theory, I have identified two typological gaps in the domain of person restrictions (the *Strength Generalization* and the *Direction Generalization*), and put forth an analysis of person restrictions that can derive the two gaps in question.

Additionally, I have shown how the proposed analysis can explain the correlation between seemingly independent grammatical properties of pronouns that are constrained by person restrictions crosslinguistically. Namely, these pronouns are characterized on one hand by their distinct deficient prosodic behavior, and on the other hand by distinct referential and binding behavior. I argued that this follows from the internal structure of such pronouns: they are both structurally deficient and featurally deficient—lacking a person value, which restricts what they can be mapped to at the interfaces with syntax. In the case of the PF interface, they can only be mapped to deficient prosodic units unable to project a prosodic word or phrase. In the case of the LF interface, they can only be mapped to simple variables that only receive a full interpretation when bound by speech act operators. The unvalued person features of minimal pronouns—the pronouns that give rise to person restrictions—are key for establishing this binding relation, where the Agree operation required to value the person features of the pronouns is also required to establish the binding relation with the speech act operators at LF and in the semantics.

The locality and timing constraints on the application of Agree are also the reason why person restrictions arise: given more than one minimal pronoun in a phase, only one can be closest to the phase head—the source of valued person features. This means that the valuation of the person features of other minimal pronouns further away from the phase head will either be blocked by syntactic intervention (Agree Closest), or constrained by the bleeding configuration of multiple instances of Agree applying top to bottom. Although this analysis ultimately derives person restrictions as a syntactic phenomenon, since they arise when there is a one-to-many syntactic configuration between a phase head and deficient pronouns of a certain kind, I argued that the reason why the restrictions apply to person specifically are ultimately due to the special semantic status of person.

I will spend the remainder of the conclusion on a brief review of the implications of this analysis, divided into three parts. First, I will consider in a little bit more detail the implications of the proposed analysis on syntactic theory in a broader sense (Section 7.1). Then, I will discuss how the proposed analysis relates to a more functionalist approach to the phenomenon of person restrictions (Section 7.2). Finally, I will summarize the answer the proposed analysis offers to the question: Why do person restrictions exist? (Section 7.3).

7.1 Implications for syntactic theory

A syntactic analysis of person restrictions, such as the one I put forth in this thesis, has to provide answers for at least two questions: (i) what characterizes the syntactic environments that give rise to person restrictions in contrast to those that do not, and (ii) what is the source of crosslinguistic variation in person restrictions?

The answer I provided for the first question is broadly speaking in line with what has been the mainstream syntactic analysis of person restrictions since Anagnostopoulou (2003, 2005) and Béjar and Řezáč (2003): person restrictions arise when two or more pronouns are in the domain of a single functional head, such that one of the pronouns prevents the establishment of an Agree dependency between the functional head and the other pronoun, or at least constrains the outcome of subsequent applications of Agree. The difference in my proposal is in the direction of the valuation of person features. That is, instead of Agree resulting in the valuation of uninterpretable person features ([uπ]) on the functional head, successful Agree between the functional head—specifically a phase head—and the minimal pronoun, results in the valuation of the pronoun's interpretable person feature ([iπ]). A person restriction arises when the minimal pronoun closer to the phase head interferes with the valuation of the person features of a minimal pronoun further away from the phase head.

Minimal pronouns, which are pronouns with unvalued φ-features, have been previously invoked, for example, in the domain of binding phenomena (see e.g. Kratzer 2009), and in the domain of the interpretation of imperative/jussive subjects (see e.g. Zanuttini 2008;

Zanuttini, Pak, and Portner 2012). I proposed that minimal pronouns are available outside of the aforementioned restricted uses, and that when they are used as regular arguments, the presence of more than one minimal pronoun will result in person restrictions. In addition, the success of the proposed analysis of person restrictions also serves as additional evidence for the double dissociation of interpretability and valuation status of formal features (see Pesetsky and Torrego 2007; Bošković 2007, 2011).

I turn now to the second question, which concerns crosslinguistic variation in person restrictions. I will focus here on three main points of variation: (i) variation in excluded person combinations (*restriction strength*), (ii) variation in the argument pairs affected by the person restriction (*restriction domain*), and (iii) variation in which of the two arguments is affected by the person restriction (*restriction direction*).

In relation to the variation in restriction strength, the mainstream approach involves a parameterization of the Agree operation, so that it can be established between a functional head and multiple arguments depending on the person feature values of the latter (Anagnostopoulou 2005; Nevins 2007, 2011; Pancheva and Zubizarreta 2018; Coon and Keine 2018; i.a.). I argued in Section 3.4 based on conceptual grounds, and the fine-grained speaker variation with respect to STRONG vs. WEAK restriction, that such variation should not be derived via a parameterization of a core grammatical operation. The alternative analysis I proposed, shifts the burden instead onto lexical variation: specifically the internal structure of minimal pronouns. This way, the STRONG vs. WEAK variation (at least between internal argument pronouns) boils down to how the internal structure of pronouns affects the application of Agree needed to value the person features of minimal pronouns.

In contrast, variation in the domain of person restrictions is typically handled in terms of the position of the functional head initiating Agree in relation to the pronouns involved in the restriction. For example, *v* yields person restrictions between co-occurring internal arguments (IA-IA), and heads higher in the clause, like T, yield restrictions between external and internal arguments (EA-IA) (see e.g. Nevins 2011). I proposed a refinement of this approach, where what matters for which functional heads yield person restrictions is their status as phase heads. What is important here is that *v* is a phase head by default, and verbal heads between *v* and C may only be phase heads under a restricted set of conditions (in line with contextual approaches to phase-hood, such as: Gallego 2005; den Dikken 2007; Wurmbrand 2013, 2017b; Bošković 2013, 2014; Despić 2015; i.a.). Even more importantly, the position of the phase head relative to the minimal pronouns determines more than just the domain of the person restriction.

Whether or not a person restriction arises in a particular domain is determined mainly based on what minimal pronouns a language has in its inventory. Specifically, what matters is whether the co-occurrence of minimal pronouns is possible in the EA-IA or IA-IA domain, or both. Additionally, the position of the phase head in relation fo the co-occurring minimal pronouns also determines the strength of the person restriction. In short, what matters is whether at least one of the pronouns is valued in a specifier of a phase head—which yields a WEAK restriction—or both are valued in the complement of a phase head. Therefore, with a *v* phase head, an EA minimal pronoun is introduced in Spec*v*P, which will yield a WEAK restriction in the EA-IA domain, due to the configuration of the two minimal pronouns. In contrast, a phase head higher than *v* will result in a STRONG restriction in the same argument domain. In the IA-IA domain, the STRONG vs. WEAK variation is the result of the internal structure of the pronouns (see discussion above) requiring valuation in-situ (STRONG) or in the specifiers of *v*P after they move there one after the other (WEAK).

This approach is also what ultimately derives the *Strength Generalization*: co-occurring minimal pronouns closer to the phase edge will always yield person restrictions weaker or the same strength as the person restrictions on pairs further from the phase edge. Recall that the generalization states that in a language with both EA-IA and IA-IA restrictions, the IA-IA person restriction can never be weaker than the EA-IA person restriction. This follows from how external and internal arguments are introduced into a clause, following a strict universal structural hierarchy (see e.g. Baker 1988, 1997).

The existence of such a universal argument structure hierarchy is also behind the derivation of the *Direction Generalization*: no language can have a reverse person restriction without also having its canonical counterpart. Recall that canonical restrictions target the IA with EA-IA restrictions and the Theme argument with IA-IA restrictions, whereas with reverse restrictions the EA is targeted with EA-IA restrictions and the Goal argument is targeted with IA-IA restrictions. I proposed that reverse person restrictions always involve optional middle-field scrambling (see e.g. Mahajan 1990; Saito 1992; Webelhuth 1992; Yatsushiro 2001; Anagnostopoulou 2003) before the phase head enters the derivation. Given a universal argument structure where the EA asymmetrically c-commands the Goal, which in turn asymmetrically c-commands the Theme, a language can either only have a canonical person restriction—because middle-field scrambling is unavailable, or both canonical and reverse person restrictions of the same kind, where the reverse restriction is derived via middle-field scrambling. This result then validates not only a universal argument structure hierarchy (i.e. some version of Baker's UTAH), but also the extension of optional movement operations to derivations involving clitic/weak and other deficient pronouns.

7.2 Finding the middle ground

An important aspect of person restrictions is that languages generally have a way to bypass them. That is, the person combinations of arguments that are excluded can be expressed by alternative grammatical means, such as: an alternative ditransitive construction with IA-IA restrictions, or with an alternative grammatical voice with EA-IA restrictions, or just by the use of one or more strong pronouns in place of the restricted deficient pronouns. What any formal analysis of person restrictions has to explain then is why the constructions that are used as "repairs" for the person restrictions are not simply always used. In other words, why is there complementary distribution between the restricted construction and the repair construction? One could just as easily imagine grammars avoiding the problem altogether by never using the constructions that yield person restrictions.

The complementary distribution of the two types of constructions in a language with person restrictions is one of the main concerns of the functionalist approach to person restrictions of Haspelmath (2004a, 2007). The gist of this approach is that the grammaticalization of clitic/affixal pronouns is guided by "harmonic" alignments/associations between a person scale (see Section 5.2) and a thematic/grammatical role scale for arguments (see Sections 2.2 and 4.1.6.2 for more details). In this context, grammaticalization refers to the historical process by which strong pronouns become clitics/affixes, in which case particular combinations of the pronouns may become fixed grammatical structures. Although the exact nature of these grammatical structures is by design not explicitly stated, they can be imagined as a version of the templates discussed in Section 2.5.4.1. The key idea is that more frequent person-role associations (e.g. a 1/2P subject and a 3P object) are also the more harmonious ones, and thus more likely to be grammaticalized. The complementary distribution noted above follows from the idea that more harmonious associations are expressed by "simpler constructions" and less harmonious associations by "more complex constructions". Grammaticalized clitic/affixal pronoun combinations crucially count as simpler, more efficient ways of expressing the arguments of a verb, while the alternative "repair" constructions count as more complex, less efficient ways of expressing the arguments.

Haspelmath's claim is that such an approach has a clear advantage over formalist and specifically generative analyses like the one I put forth in this thesis, because it does not, among other things, make use of *"any highly specific formal metalanguage for describing the relevant grammatical structures"* (see Haspelmath 2007, 98–9). However, the radically informal nature of this approach is also one of the reasons why it misses some additional generalizations, among them the *Strength Generalization* and the *Direction Generalization*. This was discussed in detail in Sections 2.5.2 and 4.1.6.2 in relation to the *Strength Generalization* and in Section 2.5.4.1 in relation to the *Direction Generalization*. In short, the *Strength Generalization* is missed because there is no way to relate EA-IA and IA-IA person restrictions to each other—in my proposed analysis they are related because of where arguments are introduced within the structure of a clause, a consequence of the specific formalism used for encoding thematic relations between the verb and its arguments. And with respect to the *Direction Generalization*, the issue is that grammatical templates are too powerful as a tool for encoding possible and impossible combinations of clitic/affixal pronouns: any format for templates that can describe the attested patterns can also describe the unattested ones. This is in sharp contrast with the proposed derivation of the *Direction Generalization*, which is that it also follows from how thematic relations are encoded in the structure of the clause in conjunction with the specific type of syntactic movement operation that yields reverse person restrictions (which I return to below).

It will probably not surprise anyone that I think using “highly specific formal metalanguage” is not a drawback but a virtue of my analysis, especially given the position I outlined in the introduction chapter. However, let me try to make more precise why I think this is so, before discussing why my analysis is in many ways similar to Haspelmath’s, and how there is no reason for formalist and functionalist analyses to be seen as diametrically opposed.

The biggest difference between an analysis like Haspelmath’s and a highly specific formal analysis like the one I proposed is in what they tell us about exceptions. When a grammatical pattern deviates from a functionalist universal, all we can say about it is that it is an exception to the rule, since the universal is only meant to guide grammaticalization. In contrast, an exception to a formal analysis will tell us what aspect of the analysis must either be abandoned or adjusted. This is in fact how I arrived at my analysis, by reevaluating existing formal analyses based on new data that did not fit them. But let us consider a specific example here to illustrate the role of exceptions in the two approaches: languages which I described as involving an alternation between a canonical person restriction and its reverse counterpart, such as Slovenian [*Slavic*] and Haya [*Bantu*].

What we see in Slovenian and Haya is that even the person-role associations that are very infrequent, such as a 1/2P Theme co-occurring with a 3P Goal (see Haspelmath 2004a, 32–6) have grammaticalized into combinations of clitic/affixal pronouns. While this does not contradict Haspelmath's analysis, it also tells us nothing about how such patterns arise and why the high frequency of more harmonious person-role associations was ignored by the grammaticalization process that yielded them. On the other hand, we saw in Sections 2.5.3 and 4.2 what one can conclude based on such patterns, and how they fit within the broader analysis of person restrictions. Moreover, one of the other virtues of formalist generative analyses, as noted in the introduction chapter, is their usefulness in identifying non-obvious implicational relationships between phenomena, and there are many non-obvious correlations that were identified over the course of this thesis between person restrictions and, for example, the interpretation of pronouns or even syntactic displacement phenomena. Most importantly, the proposed analysis will sooner or later also be challenged by new data—because it is specifically designed to be—and we will learn something from this.

I spite of these differences, Haspelmath's (2004, 2007) analysis and my analysis are similar in more ways than a superficial comparison would suggest. First, note that Haspelmath's analysis is not completely free of formal devices, as this is what person and thematic role scales as well as the concept of person-role associations are. In fact, as discussed in Section 2.2, there are several formalist approaches to person restrictions and related phenomena that adopt the same formalism. I want to highlight here in particular Aissen (1997, 1999b, 1999a, 2001, 2003), who formalizes the notion of complementary distribution between less and more complex constructions within *Optimality Theory* (Prince and Smolensky 1993/2004).

Second, it has not been true for a while that generative, or specifically minimalist, do not take into consideration efficient coding of information or domain general computational principles. Following Chomsky (2005), it has become commonplace to consider so-called *third-factor* considerations, or non-domain-specific cognitive optimization principles, and even before that *economy* considerations were a key part of syntactic theorizing (Chomsky 1991, 1993, 1995). This is also where my analysis relates most to Haspelmath's.

Recall that I have made use in my analysis of principles like *Minimize Structure* (see (1)), which is an economy condition on syntactic derivations.

(1) *Minimize Structure*
Given two extended projections of the same lexical item: α and β, if α has fewer syntactic nodes than β, β is used iff α is independently ruled out.
(see also Cardinaletti and Starke 1999, 47)

The role of (1) in particular is key for explaining the complementary distribution between deficient pronouns and strong pronouns when it comes to person restrictions. The use of the former "simpler" versions of pronouns is enforced unless they are independently ruled out: such as in the case when a minimal pronoun cannot be 1/2P person because it cannot be valued as such, so a strong pronoun counterpart, inherently valued for 1/2P person, will be used instead. The same principle can be generalized to other alternative constructions, such as prepositional datives, or passives, or inverse constructions, among many others. Note that all of these constructions involve additional structure or additional syntactic movements in the proposed analysis. Their presence is normally blocked by economy considerations, but allowed if the syntactic derivation would not converge without them.

Generally speaking, a version of *Minimize Structure* that applies not only to syntactic nodes but to features is responsible for person restrictions in the first place. Recall that I proposed that valued person features can only be present on phase heads. By way of Agree, a single value can be spread over multiple matching features, such as person features. In terms of economy this means that instead of introducing several instances of valued features, only one valued feature can be introduced per phase (the unit of structure that defines a syntactic cycle) and via Agree that value can distributed over several syntactic elements within a phase. This can be considered a more economical derivation than one where each element enters the derivation already specified with a person value. More importantly, this is where we can make a comparison between the “efficient coding” assumption driving Haspelmath’s (2004, 2007) analysis and the formal syntactic analysis I proposed.

The idea is that for the more frequent person combinations of arguments, a language with the appropriate minimal pronouns in its inventory can use more economical derivations that still ultimately converge. For example, with a combination of a 1/2P Goal pronoun and a 3P Theme pronoun, a derivation exists where both pronouns enter the derivation without a person value and the Goal is valued as 1/2P and the Theme gets a default 3P value. The fact that the application of Agree is constrained in a way that prevents a minimal pronoun Theme from being valued as 1/2P is not an issue, since convergence is guaranteed for derivations corresponding to the more frequent 1/2P $\gg$ 3P person combination. In contrast, a core complex syntactic derivation, where the Goal or Theme is a strong pronoun or the construction is a prepositional dative construction, will have to be employed for the less frequent person combinations like 3P $\gg$ 1/2P person combination. Additionally, in a language like Slovenian, where the less frequent person-role associations can be expressed with clitic pronouns, are distinguished by allowing a the syntactic reordering of the minimal pronoun arguments, which can be considered a more economical way to express those combinations in that it does not involve the addition of extra syntactic structure or features.

In relation to the grammaticalization of strong pronoun forms into deficient pronoun forms, this can also be captured in the proposed analysis. The process of historical change by which strong pronouns turn into weak, clitic, and affixal forms can be explained in terms of the loss of internal structure (along the lines suggested in Section 4.1.2.3), which ultimately results in minimal pronouns, whose person values are constrained based on where and how they can establish Agree with phase heads, yielding person restrictions. In sum, the relationship between the frequency of certain person-role associations and person restrictions, as well as how those relate to the grammaticalization of strong pronouns into deficient pronoun forms can be explained in a formal generative analysis such as mine just as they can in Haspelmath's (2004, 2007) functionalist analysis.

I began this thesis by pointing out that the methodology of looking at large samples of languages, which typically associated with functionalist typological research, is not incompatible with a generative approach to language, and that in fact generative theoretical assumptions can help us find typological generalizations that would be missed otherwise. In this section, I focused on a related but similar point. In spite of the differences in theoretical assumptions, functionalist and formalist analyses can converge on similar analyses. There is no reason why these two approaches to language should not inform each other.

7.3 Why do person restrictions exist?

I have to disappoint those who were expecting a very simple answer. If anything has become clear over the course of this thesis is that one cannot pin the existence of person restrictions to just syntax, just morphology, or just semantics. One has to approach the problem in a holistic way, and unsurprisingly the answer I have reached is in a sense also holistic.

The simplest way in which I can phrase the answer is that person restrictions arise due to a tension between the cyclic, local, and economic nature of syntactic operations on one hand and the indexical semantics of person on the other. Starting from the latter, determining the referent of a 1/2P pronoun necessarily makes reference to the linguistic context (who is the speaker or addressee), which means it is determined at the level of the utterance, and thus at least a whole clause. However, arguments are introduced very deep in a clause's structure, and in addition the periphery of the clause is separated from the argument domain by a phase—the syntactic projection that delimits a syntactic cycle and the point of spell-out to the interfaces with syntax. This means, that establishing a formal dependency between a 1/2P pronoun with the periphery of a clause is not a straightforward task.

I argued that pronouns without an inherent person value, or minimal pronouns, are the most efficient way for resolving this tension. They receive a person value via Agree with a phase head, which in turn ends up in a dependency with speech act operators in the periphery of the clause. Thus, the referent of 1/2P minimal pronouns can be determined via an operator-variable relation, where Agree is key in establishing a link between the outermost and innermost parts of the clause. The reason why the person of the lower minimal pronoun is restricted when minimal pronouns co-occur is because Agree is a local operation, so a pronoun can completely block or constrain the application of Agree between the phase head and another pronoun, restricting the possible person values it can receive.

Finally, the reason why this type of restriction only applies to pronouns also characterized in terms of their morphological size and inability to be independent words has to do with the mapping between narrow syntax and morphology/prosody. The internal structure of minimal pronouns is much smaller compared to the structure of independent words, which determines what kinds of morphological and prosodic units they can be mapped to.

This is why it is so difficult to place the source of person restrictions in a single module of grammar: they are ultimately a very intricate interface phenomenon involving syntax and both sides of the interfaces with syntax.

Appendix: Overview of language sample

I present here the language sample used for the typological survey together with the relevant references. I present only the data points relevant for person restrictions—setting aside the additional restrictions discussed in Chapter 5 and number asymmetries discussed briefly in Chapter 6—unless the information somehow relates to the main data points. The conventions for describing the relevant data points are:

1. *Restriction strength:*

   - Gray EA-IA or EA-IA row in table signifies no person restriction for that domain;
   - (%) next to multiple options signifies speaker variation with respect to the pattern or conflicting reports in the cited literature (possibly due to speaker variation);
   - Sub-types of WEAK restrictions:

     WEAK $*\gg 1$: WEAK restriction with additional ban on $*2 \gg 1$, where $*\gg 1$ may be independent with respect to allowing a reverse restriction;

     WEAK $*\gg 2$: WEAK restriction with additional ban on $*1 \gg 2$, where $*\gg 2$ may be independent with respect to allowing a reverse restriction;

     WEAK – ME-FIRST: WEAK restriction where $x \gg 2$ is grammatical;

WEAK – YOU-FIRST: WEAK restriction where $x \gg 1$ is grammatical;

WEAK – $*2 \gg 1$: WEAK restriction where only $*2 \gg 1$ is ungrammatical;

WEAK – $*3 \gg 2$: WEAK restriction where only $*3 \gg 2$ is ungrammatical.

2. *Restriction direction:*

   - canonical
   - canonical + reverse
   - canonical + %reverse (variation regarding acceptance of reverse variant)

3. *Other:*

   - (%) next to only restriction in the domain signifies speaker variation with respect to having the restriction at all or conflicting reports in the cited literature regarding the existence of the restriction (possibly due to speaker variation);
   - Salient additional constraints on the restriction are stated if present;
   - Languages from the sample are grouped according to family;
   - † marks a dead language.

Language list

[1] Indo-European

[1] **French** (Perlmutter 1971; Kayne 1975; Postal 1981, 1990; Bonet 1991, 1994; Laenzlinger 1993; Béjar and Řezáč 2003; Nicol 2005; Řezáč 2011; Raynaud 2018)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

2 **Spanish** (Perlmutter 1971; Seuren 1976; Bonet 1991; Laenzlinger 1993; Albizu 1997a; Ormazabal and Romero 2007; Rivero 2008; Alba de la Fuente 2010)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG (%)	canonical
	WEAK (%)	canonical
	*≫ 2	canonical + reverse

3 **Catalan** (Bonet 1991, 1994)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG (%)	canonical
	WEAK (%)	canonical
	*≫ 2	canonical + %reverse

4 **Italian** (Seuren 1976; Wanner 1977; Bonet 1991; Laenzlinger 1993; Cardinaletti and Starke 1994, 1999; Cardinaletti 2008)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG (%)	canonical
	WEAK (%)	canonical
	*≫ 1	canonical + %reverse

5 **Romanian** (Farkas and Kazazis 1980; Ciucivara 2006, 2009; Ciucivara and Nevins 2008; Alba de la Fuente 2010; Ivan to appear)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK – ME-FIRST (%)	canonical
	WEAK – *2 ≫ 1 (%)	canonical

6 **German (standard)** (Cardinaletti 1999; Haspelmath 2004a; Anagnostopoulou 2008; Coon, Keine, and Wagner 2017; Coon and Keine 2018)

Domain	Strength	Direction
EA-IA	agreement restriction only	
IA-IA	WEAK (%)	canonical + %reverse

7 **Swiss German** (Bonet 1991)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK restriction with GL » TH order no restriction with TH » GL order	canonical

8 **Zürich German** (Werner 1999)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK	canonical + reverse

9 **Dutch** (Cardinaletti 1999; Anagnostopoulou 2008)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK (%) restriction with GL » TH order no restriction with TH » GL order	canonical

10 **Swedish** (Hellan and Platzack 1999)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK (%) restriction with TH » GL order no restriction with GL » TH order	canonical

11 **English** (Bonet 1991; Richards 2008a)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG (%)	canonical

12 **Icelandic** (Sigurðsson 1990–1991, 1996; Taraldsen 1994, 1995; Maling and Jónsson 1995; Boeckx 2000, 2008; Burzio 2000; Schütze 2003; Holmberg and Hróarsdóttir 2004; Sigurðsson and Holmberg 2008; Bobaljik 2008; Ussery 2013)

Domain	Strength	Direction
EA-IA	agreement restriction only	
IA-IA		

13 **Faroese** (Schütze 2003)

Domain	Strength	Direction
EA-IA		
IA-IA		

14 **Slovenian** (Stegovec 2015b, 2016, 2019)

Domain	Strength	Direction
EA-IA	agreement restriction only	
IA-IA	STRONG (%)	canonical + reverse
	WEAK (%)	canonical + reverse

15 **Bosnian-Croatian-Serbian** (Migdalski 2006; Runić 2013b)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK – ME-FIRST (%)	canonical

16 **Czech** (Sturgeon et al. 2012)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK *≫ 1	canonical + %reverse

17 **Polish** (Cetnarowska 2003; Migdalski 2006; Bondaruk 2012)

Domain	Strength	Direction
EA-IA	agreement restriction only	
IA-IA		

18 **Bulgarian** (Rudin 1997; Hauge 1999; Franks and King 2000; Harizanov 2014a, 2014b; Pancheva and Zubizarreta 2018)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG (%)	canonical
	WEAK – ME-FIRST (%)	canonical

19 **Macedonian** (Migdalski 2006; Tomić 2008)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG (%)	canonical
	WEAK (%)	canonical

20 **Greek** (Bonet 1991; Anagnostopoulou 2003, 2005; Mavrogiorgos 2010)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

21 **Albanian** (Buchholz and Fiedler 1987)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

22 **Kurdish** (Karimi 2010)

Domain	Strength	Direction
EA-IA	STRONG with ergative alignment	canonical
IA-IA		

23 **Pashto** (Roberts 2000)

Domain	Strength	Direction
EA-IA	WEAK $*\gg 1$ subject + possessor clitic with ergative alignment	canonical + reverse
IA-IA		

24 **Iron Ossetic** (Erschler 2014, 2015)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK *≫ 1 with dative + another clitic	canonical
	WEAK with allative + accusative	canonical
	STRONG with superessive + ablative clitic	canonical
	STRONG with allative + ablative clitic	canonical

25 **Digor Ossetic** (Erschler 2014, 2015)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK *≫ 1 with dative + another clitic	canonical
	WEAK *≫ 1 with allative + ablative	canonical

26 **Kashmiri** (Wali and Koul 1997, 2006; Nichols 1997, 2001; Del Bon 2002; Verbeke 2017)

Domain	Strength	Direction
EA-IA	WEAK *≫ 1	canonical
IA-IA	STRONG	canonical

⇒ Person restrictions (in both domains) only observed with non-past tenses where the clitic pronouns display a nominative-accusative alignment;

⇒ Evidence for STRONG IA-IA restriction based on absence of 1/2P Themes in relevant examples—needs to be verified with speakers.

[2] Uralic

27 **Finnish** (Kiparsky 2001; Řezáč 2006)

Domain	Strength	Direction
EA-IA	agreement restriction only	
IA-IA		

28 **Hungarian** (É. Kiss 2013, 2017; Bárány 2015a, 2015b, 2017)

Domain	Strength	Direction
EA-IA	WEAK $* \gg 1$	canonical
IA-IA		

29 **Eastern Mansi** (Virtanen 2012, 2014; É. Kiss 2017)

Domain	Strength	Direction
EA-IA	STRONG	canonical
IA-IA		

30 **Khanty/Ostyak** (Nikolaeva 1999, 2001; Dalrymple and Nikolaeva 2011)

Domain	Strength	Direction
EA-IA	agreement restriction only	
IA-IA		

31 **Tundra Nenets** (Dalrymple and Nikolaeva 2011; Nikolaeva 2014)

Domain	Strength	Direction
EA-IA	STRONG	canonical
IA-IA		

[3] Afro-Asiatic

32 **Modern Standard Arabic** (Fassi-Fehri 1988, 1993; Bonet 1991)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK $*\gg 1$	canonical

33 **Classical Arabic** (Sibawayh 1881; Bonet 1991; Walkow 2014)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK $*\gg 1$	canonical

34 **Cairene Arabic** (Broselow 1983)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

35 **Maltese** (Borg and Azzopardi-Alexander 1997)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

36 **Senaya (Neo-Aramaic)** (Kalin and McPherson 2012; Kalin 2014a, 2014b)

Domain	Strength	Direction
EA-IA	STRONG with perfective aspect	canonical
IA-IA	STRONG	canonical

37 **Christian Barwar (Neo-Aramaic)** (Khan 2008; Doron and Khan 2012a, 2012b; Kalin 2014b; Kalin and van Urk 2015)

Domain	Strength	Direction
EA-IA	STRONG with perfective aspect	canonical
IA-IA		

38 **Telkepe (Neo-Aramaic)** (Coghill 2010, 2018; Kalin 2014b)

Domain	Strength	Direction
EA-IA	STRONG with perfective aspect	canonical
IA-IA	STRONG	canonical

39 **Migama** (Jungraithmayr and Adams 1992)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

40 **Baraïn** (Lovestrand 2012)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[4] Nilo-Saharan

41 **Maasai/Maa** (Payne, Hamaya, and Jacobs 1994; Lamoureaux 2004)

Domain	Strength	Direction
EA-IA	WEAK $*\gg 1$ direct/inverse	canonical
IA-IA	STRONG person-based indexing	canonical + reverse

[5] Niger-Congo

42 **Sambaa/Shambala** (Duranti 1979; Riedel 2009)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG (%)	canonical
	WEAK (%)	canonical

43 **Haya** (Duranti 1979; Riedel 2009)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK $*\gg 1$	canonical + reverse

44 **Swahili** (Riedel 2009)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK	canonical

45 **KiRimi/Nyaturu** (Hualde 1989; Woolford 2000; Ormazabal and Romero 2007)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

46 **Limbum** (Riedel 2009)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK	canonical

[6] Kartvelian

47 **Georgian** (Harris 1981; Boeder 1999; Amiridze and Leuschner 2002)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[7] Northwest Caucasian

48 Abhkaz (Hewitt 1979, 1989; Mel'čuk 1988)

Domain	Strength	Direction
EA-IA		
IA-IA		

⇒ Although all person combinations are attested, Mark Baker (p.c.) notes that Abkhaz has a peculiar pronoun marking paradigm where the IO marker is ergative—as we saw in many cases above, person restrictions are often voided with ergative-absolutive alignment in languages with ergative splits.

[8] Sino-Tibetan

49 **Hakha Chin** (Peterson 1998)

Domain	Strength	Direction
EA-IA		
IA-IA	WEAK	canonical

50 **Chepang** (DeLancey 1981)

Domain	Strength	Direction
EA-IA	WEAK $*\gg 1$ direct/inverse	canonical
IA-IA		

51 **Jyarong** (DeLancey 1981)

Domain	Strength	Direction
EA-IA	WEAK $*\gg 1$ direct/inverse	canonical
IA-IA		

52 **Nocte** (Das Gupta 1971; DeLancey 1981; Thompson 1994)

Domain	Strength	Direction
EA-IA	WEAK $*\gg 1$ direct/inverse	canonical
IA-IA		

53 **Tangut**† (Kepping 1979; DeLancey 1981; Comrie 2003)

Domain	Strength	Direction
EA-IA	STRONG person-based indexing	canonical + reverse
IA-IA	STRONG	canonical

[9] Austronesian

54 **Kambera** (Klamer 1997)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

55 **Manam** (Lichtenberg 1983)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

56 **Tagalog** (Richards 2005)

Domain	Strength	Direction
EA-IA		
IA-IA		

⇒ While there are no traditional person restrictions in Tagalog, there is a person-based constraint on extraction out of embedded clauses; see Richards 2005.

[10] Sepik-Ramu

57 **Yimas** (Foley 1991; Phillips 1993)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

58 **Manambu** (Aikhenvald 2008)

Domain	Strength	Direction
EA-IA		
IA-IA		

[11] Toricelli/Monumbo

59 **Monumbo** (Vormann and Scharfenberger 1914)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[12] Pama-Nyungan

60 **Djaru** (Tsunoda 1981b, 1981a)

Domain	Strength	Direction
EA-IA	WEAK $^*\!\gg 1$ (?)	canonical + reverse
IA-IA	WEAK $^*\!\gg 1$	canonical + reverse

⇒ Possible candidate for a WEAK-WEAK combination; however, a closer inspection of the complex EA-IA marking pattern is required to eliminate the possibility of a surface constraint (the IA-IA restriction is much simpler).

61 **Warlpiri** (Hale 1973b, 1983; Simpson 1991; Laughren and Hoogenraad 1996)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[13] Chukotko-Kamchatkan

62 **Chukchi** (Comrie 1979b, 1980; Dunn 1999; Bobaljik and Branigan 2006)

Domain	Strength	Direction
EA-IA	WEAK $^*\!\gg 1$ direct/inverse	canonical
IA-IA	STRONG	canonical

63 **Koryak** (Comrie 1980; Abramovitz 2018)

Domain	Strength	Direction
EA-IA	WEAK *≫ 1 direct/inverse	canonical
IA-IA		

⇒ No accessible data regarding any IA-IA restriction.

64 **Alutor** (Mel'čuk 1973, 1988; Savvina and Mel'čuk 1978; Kibrik 2002)

Domain	Strength	Direction
EA-IA	WEAK *≫ 1 direct/inverse	canonical
IA-IA	WEAK *≫ 1 person-based indexing	canonical + reverse

65 **Itelmen** (Comrie 1980; Bobaljik and Wurmbrand 2002; Bobaljik to appear)

Domain	Strength	Direction
EA-IA IA-IA		

⇒ Only under Comrie's (1980) interpretation of the EA-IA marking paradigm, Itelmen has a WEAK EA-IA restriction, but only with plural 3P EAs.

[14] Penutian

66 **Sahaptin** (Rude 1994, 2009)

Domain	Strength	Direction
EA-IA	WEAK *≫ 1 direct/inverse	canonical
IA-IA	STRONG	canonical

67 **Takelma** (Sapir 1922)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[15] Algic

68 **Algonquin/Algonkin** (Henderson 1971; Grafstein 1980, 1984; Klaiman 1991)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

69 **Arapaho** (Cowell and Moss Sr. 2008; Wagner, Cowell, and Hwang 2016)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2 (%) direct/inverse	canonical + reverse
	WEAK (%) direct/inverse	canonical + reverse
IA-IA		

70 **Blackfoot** (Frantz 1966, 1971, 1991; Bliss 2013)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

71 **Plains Cree** (Dahlstrom 1986a, 1991, 2013; Campana 1989)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

72 **Delaware** (Goddard 1969, 1979)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

73 **Meskwaki/Fox** (Goddard 1984; Dahlstrom 2015)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

74 **Mi'kmaq** (Pacifique, Francis, and Hewson 1990; Coon and Bale 2014; Hamilton 2015)

Domain	Strength	Direction
EA-IA	WEAK person-based indexing	canonical + reverse
IA-IA	STRONG	canonical

75 **Maniwaki** (Oxford 2014, 2019)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

76 **Ojibwe** (Rhodes 1990b; Jelinek 1990; Valentine 2001)

Domain	Strength	Direction
EA-IA	WEAK $*\gg$ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

77 **Passamaquoddy** (Leavitt 1996; Bruening 2001)

Domain	Strength	Direction
EA-IA	WEAK $*\gg$ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

78 **Potawatomi** (Hockett 1939, 1948, 1966; Erickson 1965)

Domain	Strength	Direction
EA-IA	WEAK $*\gg$ 2 direct/inverse	canonical + reverse
IA-IA	STRONG	canonical

[16] Kiowa-Tanoan

79 **Southern Tiwa** (Allen and Frantz 1978, 1986; Allen et al. 1990; Rosen 1990)

Domain	Strength	Direction
EA-IA	WEAK	canonical
IA-IA	STRONG	canonical

80 **Picurís** (Zaharlick 1977; Nichols 2001)

Domain	Strength	Direction
EA-IA	WEAK	canonical
IA-IA	STRONG	canonical

81 **Arizona Tewa** (Kroskrity 1977, 1985; Kroskrity 2010)

Domain	Strength	Direction
EA-IA	STRONG	canonical
IA-IA	STRONG	canonical

82 **Kiowa** (Watkins 1984; Adger and Harbour 2007; Adger, Harbour, and Watkins 2009)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[17] Iroquoian

83 **Cherokee** (Scancarelli 1987; Dukes 1996; Montgomery-Anderson 2008)

Domain	Strength	Direction
EA-IA	WEAK direct/inverse	canonical
IA-IA	STRONG	canonical

84 **Mohawk** (Baker 1985, 1988, 1991, 2006)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[18] Uto-Aztecan

85 **Tetelcingo Nahuatl** (Tuggy 1977)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

86 **Classical Nahuatl**† (Andrews 1975)

Domain	Strength	Direction
EA-IA		
IA-IA		

⇒ All person combinations are listed in the grammar, but Andrews notes that some object combinations are hypothetical and not attested in original sources;

⇒ This raises the question if the gaps are due to a person restriction—unfortunately, the grammar does not single out the hypothetical person combinations.

87 **O'odham/Papago** (Hale 1959; Zepeda 1983; Langacker 1977)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

[19] Mayan

88 **Tzotzil** (Aissen 1987)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

89 **Kaqchikel** (Preminger 2011, 2014)

Domain	Strength	Direction
EA-IA	STRONG person-based indexing with Agent-Focus construction	canonical + reverse
IA-IA		

[20] Oto-Manguean

90 **Oaxaca Zapotec** (Foley and Toosarvandani 2018; Foley, Kalivoda, and Toosarvandani to appear)

Domain	Strength	Direction
EA-IA	STRONG	canonical
IA-IA	STRONG	canonical

[21] Quechuan

91 **Quechua** (Myler 2017, p.c.)

Domain	Strength	Direction
EA-IA	WEAK – YOU-FIRST	canonical + reverse
IA-IA	STRONG	canonical + %reverse

[22] Salish

92 **Bella Coola** (Forrest 1994; Jacobs 1994)

Domain	Strength	Direction
EA-IA	WEAK – $*3 \gg 2$	canonical
IA-IA		

93 **Clallam** (Jelinek and Demers 1983)

Domain	Strength	Direction
EA-IA	WEAK	canonical
IA-IA		

94 **Lummi** (Jelinek and Demers 1983; Jelinek 1994)

Domain	Strength	Direction
EA-IA	WEAK	canonical
IA-IA		

95 **Halkomelem** (Jelinek and Demers 1983; Gerdts 1988; Galloway 1993; Wiltschko 2008)

Domain	Strength	Direction
EA-IA	WEAK – $*3 \gg 2$	canonical
IA-IA		

96 **Squamish** (Jelinek and Demers 1983; Jacobs 1994)

Domain	Strength	Direction
EA-IA	WEAK – $*3 \gg 2$	canonical
IA-IA		

97 **Lushootseed** (Jelinek and Demers 1983)

Domain	Strength	Direction
EA-IA		
IA-IA		

[23] Dené-Yeniseian

98 **Koyukon** (Thompson 1994)

Domain	Strength	Direction
EA-IA		
IA-IA		

99 **Navajo** (Jelinek 1990; Fountain 1998)

Domain	Strength	Direction
EA-IA		
IA-IA		

[24] Eskimo-Aleut

100 **Labrador Inuttut** (Johns 1996; Johns and Kučerova 2017; Compton 2018; Yuan 2018)

Domain	Strength	Direction
EA-IA	STRONG (%) with participial, indicative mood	canonical
IA-IA		

101 **Inuktitut (South Baffin)** (Yuan 2018)

Domain	Strength	Direction
EA-IA		
IA-IA		

[25] Araucanian

102 **Mapudungun** (Grimes 1985; Arnold 1994; Smeets 2008; Golluscio 2010)

Domain	Strength	Direction
EA-IA	STRONG direct/inverse	canonical
IA-IA		

⇒ Smeets 2008 notes that given two objects, it is "the animate, more agentive or more definite object" which is indexed on the verb. This looks a lot like a canonical/inverse alternation based on non-person indexical values;

⇒ There is also a possibility of an IA-IA restriction and thus a STRONG-STRONG combination. As Golluscio 2010 discusses, a 1/2P Theme can affect the choice of ditransitive verb: requiring 'give away' (*wïl*) rather than 'give' (*elu*), where with the former the Goal is only implicit. However, more research is needed to show that this restriction is really systematic.

[26] Tupian

103 **Paraguayan Guaraní** (Payne 1994; Zubizarretta and Pancheva 2017; Roessler 2019)

Domain	Strength	Direction
EA-IA	WEAK *≫ 1 direct/inverse	canonical
IA-IA	STRONG	canonical

⇒ The evidence for the IA-IA restriction is so far indirect; Roessler 2019 shows that in double object constructions, a 1/2P Theme is not indexed on the verb or alternatively a prepositional dative construction must be used.

Isolate languages

104 **Basque** (Bonet 1991, 1994; Laka 1993; Albizu 1997a; Řezáč 2008b)

Domain	Strength	Direction
EA-IA		
IA-IA	STRONG	canonical

105 **Zuni** (Nichols 1997)

Domain	Strength	Direction
EA-IA	WEAK *≫ 2	canonical + reverse
IA-IA		

⇒ Goal interacts with the EA-IA restriction in ditransitives, but so far insufficient evidence for presence of an IA-IA restriction.

106 **Kutenai** (Dryer 1992)

Domain	Strength	Direction
EA-IA		
IA-IA		

Bibliography

Abels, Klaus. 2003a. *[P clitic]!—Why? In *Proceedings of FDSL 4, Potsdam 2001,* edited by Peter Kosta, Joanna Blaszczak, Jens Frasek, Ludmila Geist, and Marzena Zygis, 443–460. Frankfurt am Main: Linguistik International, Peter Lang.

———. 2003b. Successive cyclicity, anti-locality, and adposition stranding. PhD diss., UConn.

———. 2007. Towards a restrictive theory of (remnant) movement! *Linguistic Variation Yearbook* 7:53–120.

Abels, Klaus and Ad Neeleman. 2012. Linear asymmetries and the LCA. *Syntax* 15 (1): 25–74.

Abramovitz, Rafael. 2018. *Successive-cyclic wh-movement feeds dependent case competition.* Poster presented at NELS 49, Cornell University, October 6th 2018.

Ackema, Peter and Ad Neeleman. 2018. *Features of person: From the inventory of persons to their morphological realization.* Cambridge, MA: MIT Press.

Adger, David. 2003. *Core syntax: A minimalist approach.* Oxford: Oxford University Press.

Adger, David and Daniel Harbour. 2007. Syntax and syncretism of the Person Case Constraint. *Syntax* 10:2–37.

Adger, David, Daniel Harbour, and Laurel J. Watkins. 2009. *Mirrors and microparameters: Phrase structure beyond free word order.* Cambridge: Cambridge University Press.

Aikhenvald, Alexandra Y. 2008. *The Manambu language of East Sepik, Papua New Guinea.* Oxford: Oxford University Press.

Aissen, Judith. 1997. On the syntax of obviation. *Language* 73 (4): 705–750.

———. 1999a. Agent Focus and inverse in Tzotzil. *Language* 7 (3): 451–485.

———. 1999b. Markedness and subject choice in optimality theory. *Natural Language & Linguistic Theory* 17:673–711.

Aissen, Judith. 2001. The obviation hierarchy and morphosyntactic markedness. *Linguistica atlantica* 23:1–34.

———. 2003. Differential object marking: Iconicity vs. economy. *Natural Language & Linguistic Theory* 21 (3): 435–483.

Aissen, Judith L. 1987. *Tzotzil clause structure.* Dordrecht: D. Riedel.

Alba de la Fuente, Anahí. 2010. More on the clitic combination puzzle: Evidence from Spanish, Catalan and Romanian. In *Romance linguistics 2009: Selected papers from the 39th linguistic symposium on Romance languages (LSRL), Tuscon, Arizona, March 2009,* edited by Sonia Colina, Antxon Olarrea, and Ana Maria Carvalho, 203–216. Amsterdam/Philadelphia: John Benjamins.

Albizu, Pablo. 1997a. Generalized person-case constraint: A case for a syntax driven inflectional morphology. In *Theoretical issues on the morphology-syntax interface,* edited by Myriam Utribe-Extebarria and Amaya Mendikoetxea, 1–33. Donostia: UPV/EHU.

———. 1997b. The syntax of person agreement. PhD diss. draft, USC, Los Angeles.

Allen, Barbara J. and Donald G. Frantz. 1978. Verb agreement in Southern Tiwa. In *Proceedings of the 4th annual meeting of the Berkeley Linguistics Society,* edited by Jeri J. Jager et. al., 11–17. Berkeley, CA: BLS.

———. 1986. Goal advancement in Southern Tiwa. *International Journal of American Linguistics* 52 (4): 388–403.

Allen, Barbara J., Donald G. Frantz, Donna B. Gardiner, and David M. Perlmutter. 1990. Verb agreement, possessor ascension, and multistratal representation in Southern Tiwa. In *Studies in relational grammar 3,* edited by Paul M. Postal and Brian D. Joseph, 321–383. Chicago: University of Chicago Press.

Amiridze, Nino and Torsten Leuschner. 2002. Body-part nouns as a source of reflexives: towards a grammaticalization account of Georgian *tav-* ‘head’. *Sprachtypologie und Universalienforschung* 55.3:259–276.

Anagnostopoulou, Elena. 1999. *Person restrictions.* Paper presented at the 21st GLOW Colloquium, ZAS-Berlin, March 29, 1999. [*Glow Newsletter* 42: 12-13].

———. 2003. *The syntax of ditransitives.* Berlin: De Gruyter.

———. 2005. Strong and weak person restrictions: A feature checking analysis. In *Clitic and affix combinations,* edited by Lorie Heggie and Francisco Ordoñez, 199–235. Amsterdam/Philadelphia: John Benjamins.

———. 2008. Notes on the Person-Case Constraint in Germanic. In *Agreement restrictions,* edited by Roberta D'Alessandro, Susann Fischer, and Gunnar Hrafn Hrafnjargarson, 15–47. Berlin: De Gruyter.

———. 2017. The Person Case Constraint. In *The Wiley-Blackwell companion to syntax,* 2nd ed., edited by Martin Everaert and Henk van Riemsdijk. Hoboken, NJ: Wiley.

Anand, Pranav. 2006. De de se. PhD diss., MIT.

———. 2007. Dream report pronouns, local binding, and attitudes 'de se'. In *Proceedings of SALT 17,* edited by Tova Friedman and Masayuki Gibson, 1–18. Ithaca, NY: CLC Publications.

Anand, Pranav and Andrew Nevins. 2004. Shifty operators in changing contexts. In *Proceedings of SALT 14,* edited by Robert B. Young. Ithaca, NY: CLC Publications.

Andrews, J. Richard. 1975. *Introduction to classical nahuatl.* Norman: University of Oklahoma Press.

Aoun, Joseph and Audrey Yun-Hui Li. 1989. Scope and constituency. *Linguistic Inquiry* 20 (2): 141–172.

Appelbaum, Irene. 2019. Double-obviatives and direction-marking in Kutenai. In *Proceedings of the 30th Western Conference on Linguistics,* edited by Trevor Driscoll, 18–24. Fresno, CA: Department of Linguistics, CSU Fresno.

Arnold, Jennifer. 1994. Inverse voice marking in Mapudungun. In *Proceedings of the twentieth annual meeting of the Berkeley Linguistics Society: General session dedicated to the contributions of Charles J. Fillmore,* edited by Susanne Gahl, Andy Dolbey, and Christopher Johnson, 28–41. Berkeley, CA: BLS.

Arregui, Ana. 2007. Being 'me', being 'you': Pronoun puzzles in modal contexts. In *Proceedings of Sinn und Bedeutung 11,* edited by Louise McNally and Estela Puig-Waldmüller, 31–45. Barcelona: Universitat Pompeu Fabra.

Authier, J.Marc and Lisa Reed. 1992. On the syntactic status of French affected datives. *The Linguistic Review* 9 (4): 295–312.

Avelino Becerra, Heriberto. 2004. Topics in Yalálag, with particular reference to its phonetic structures. PhD diss., UCLA.

Baker, C. L. 1979. Syntactic theory and the projection problem. *Linguistic Inquiry* 10:533–581.

Baker, C. L. and John McCarthy, eds. 1981. *The logical problem of language acquisition.* Campridge, MA: MIT Press.

Baker, Mark C. 1985. Incorporation. PhD diss., MIT.

Baker, Mark C. 1988. *Incorporation.* Chicago: University of Chicago Press.

———. 1990. Pronominal inflection and the morphology-syntax interface. In *Proceedings of the 26th regional meeting of the Chicago Linguistic Society,* 25–48. Chicago, IL: CLS.

———. 1991. On some subject/object non-asymmetries in Mohawk. *Natural Language & Linguistic Theory* 9 (4): 537–576.

———. 1996. *The polysynthesis parameter.* Oxford: Oxford University Press.

———. 1997. Thematic roles and syntactic structure. In *Elements of grammar: Handbook in generative syntax,* edited by Liliane Haegeman, 73–137. Dordrecht: Springer.

———. 2003a. Agreement, dislocation, and partial configurationality. In *Formal approaches to function in grammar: In honor of Eloise Jelinek,* edited by Andrew Carnie, Heidi Harley, and Mary Ann Willie, 107–132. Amsterdam & Philadelphia: John Benjamins.

———. 2003b. *Lexical categories: Verbs, nouns and adjectives.* Cambridge: Cambridge University Press.

———. 2006. On zero agreement and polysynthesis. In *Arguments and agreement,* edited by Peter Ackema, Partick Brandt, Maaike Schoorlemmer, and Fred Weerman, 289–320. Oxford: Oxford University Press.

———. 2008a. The macroparameter in a microparametric world. In *The limits of variation,* edited by Theresa Biberauer. Amsterdam: John Benjamins.

———. 2008b. *The syntax of agreement and concord.* Cambridge: Cambridge University Press.

———. 2010. Formal generative typology. In *The Oxford handbook of linguistic analysis,* edited by Bernd Heine and Heiko Narrog, 285–312. Oxford: OUP.

———. 2011. When agreement is for number and gender but not person. *Natural Language & Linguistic Theory* 29 (4): 875–915.

———. 2015. *Case: Its principles and its parameters.* Cambridge: Cambridge University Press.

Baker, Mark C. and Jonathan David Bobaljik. 2017. On inherent and dependent theories of ergative case. In *The Oxford handbook of ergativity,* edited by Jessica Coon, Diane Massam, and Lisa DeMena Travis, 111–134. New York, NY: Oxford University Press.

Baker, Mark C. and Jim McCloskey. 2007. On the relationship of typology to theoretical syntax. *Linguistic Typology* 11:285–296.

Bárány, András. 2015a. Differential object marking in Hungarian and the morphosyntax of case and agreement. PhD diss., University of Cambridge.

———. 2015b. Inverse agreement and Hungarian verb paradgims. In *Approaches to Hungarian. Vol. 14: Papers from the 2013 Piliscsaba conference,* edited by Katalin É. Kiss, Balázs Surányi, and Éva Dékány, 37–65. Amsterdam: John Benjamins.

———. 2017. *Person, case, and agreement: The morphosyntax of inverse agreement and global case splits.* Oxford: Oxford University Press.

Barss, Andrew and Howard Lasnik. 1986. A note on anaphora and double objects. *Linguistic Inquiry* 17:347–354.

Béjar, Susana and Milan Řezáč. 2003. Person licensing and the derivation of PCC effects. In *Romance linguistics: Theory and acquisition,* edited by Ana Teresa Pérez-Leroux and Yves Roberge, 49–62. Amsterdam/Philadelphia: John Benjamins.

———. 2009. Cyclic Agree. *Linguistic Inquiry* 40:35–73.

Belletti, Adriana. 2004. Aspects of the low IP area. In *The structure of CP and IP,* edited by Luigi Rizzi, 16–51. Oxford: Oxford University Press.

Bello, Sophia. 2017. Prolegomenon to the study of French indirect objects in first language acquisition. PhD diss., University of Toronto.

Benincà, Paola. 1983. Il clitico 'a' nel dialetto padovano. In *Scritti linguistici in onore di Giovan Battista Pellegrini,* edited by Paola Benincá, Manilio Cortelazzo, Aldo Prosdocimi, Laura Vanelli, and Alberto Zamboni, 25–32. Pisa: Pacini.

Berent, Gerald P. 1980. On the realization of trace: Macedonian clitic pronouns. In *Morphosyntax in Slavic,* edited by Catherine V. Chvany and Richard D. Brecht, 150–186. Columbus, OH: Slavica Publishers.

Berwick, Robert. 1985. *The acquisition of syntactic knowledge.* Cambridge, MA: MIT Press.

Bhatt, Rajesh and Radek Šimík. 2009. *Variable binding and the person-case constraint.* Talk given at *IATL 25*. Ben Gurion, University of the Negev.

Bianchi, Valentina. 2006. On the syntax of person arguments. *Lingua* 16 (12): 2023–2067.

Biberauer, Theresa, Anders Holmberg, and Ian Roberts. 2014. A syntactic universal and its consequences. *Linguistic Inquiry* 45 (2): 169–225.

Biberauer, Theresa, Anders Holmberg, Ian Roberts, and Michelle Sheehan. 2014. Complexity in comparative syntax: The view from modern parametric theory. In *Measuring grammatical complexity,* edited by Laurel B. Preston and Friderick Newmeyer, 103–127. Oxford: Oxford University Press.

Bickel, Balthasar. 2008. On the scope of the referential hierarchy in the typologyof grammatical relations. In *Case and grammatical relations: Studies in honor of Bernard Comrie,* edited by Greville Corbett and Michal Noonan, 191–211. Amsterdam/Philadelphia: John Benjamins.

Bittner, Maria. 2014. Perspectival discourse referents for indexicals. In *Proceedings of the seventh meeting of the Semantics of Under-represented Languages in the Americas,* edited by Hannah Greene, 1–22. Amherst, MA: GLSA.

Bliss, Heather. 2013. The Blackfoot configurationality conspiracy. PhD diss., University of British Columbia.

Bloomfield, Leonard. 1962. *The Menomini language.* New Haven, CT: Yale University Press.

Bobaljik, Jonathan David. 1995. Morphosyntax: The syntax of verbal inflection. PhD diss., MIT.

———. 2007. *On comparative suppletion.* Manuscript: UConn. https://ling.auf.net/lingbuzz/000443.

———. 2008. Where's Phi? Agreement as a post-syntactic operation. In *Phi-theory: Phi features across interfaces and modules,* edited by Daniel Harbour, David Adger, and Susana Béjar, 295–328. Oxford: Oxford University Press.

———. 2012. *Universals in comparative morphology: Suppletion, superlatives, and the structure of words.* Cambridge, MA: MIT Press.

———. to appear. The Chukotkan "inverse" from an Itelmen perspective. *Acta Linguistica Petropolitana* 13:1–9.

Bobaljik, Jonathan David and Philip Branigan. 2006. Eccentric agreement and multiple case checking. In *Ergativity,* edited by Alana Johns, Diane Massam, and Juvenal Ndayiragije, 47–77. Dordrecht: Springer.

Bobaljik, Jonathan David and Susi Wurmbrand. 2002. Notes on agreement in Itelmen. [online journal], *Linguistic Discovery* 1:1–28.

Boeckx, Cedric. 2000. Quirky agreement. *Studia Linguistica* 54:451–480.

———. 2008. *Aspects of the syntax of agreement.* London: Routledge.

Boeder, Winfried. 1999. A slot-filling constraint in the Georgian verb and its syntactic corollary. *Sprachtypologie und Universalienforschung* 52.3 (4): 241–254.

Bondaruk, Anna. 2012. Person case constraint effect in Polish copular constructions. *Acta Linguistica Hungarica* 59 (1–2): 49–84.

Bonet, Eulalia. 1991. Morphology after syntax: Pronominal clitics in Romance languages. PhD diss., MIT.

———. 1994. The Person-Case Constraint. *MIT Working papers in Linguistics* 22:33–52.

———. 1995. Feature structure of Romance clitics. *Natural Language & Linguistic Theory* 13:607–647.

Borer, Hagit. 1984. *Parametric syntax: Case studies in Semitic and Romance languages.* Dordrecht: Foris.

———. 1994. The projection of arguments. In *Functional projections (UMass working papers in linguistics 17),* edited by Elena Benedicto and Jeffrey Runner, 19–47. Amherst, MA: GLSA.

Borer, Hagit and Yosef Grodzinsky. 1986. Syntactic cliticization and lexical cliticization: The case of Hebrew dative clitics. *Syntax and Semantics (The Syntax of Pronominal Clitics)* 19:175–217.

Borg, Albert and Marie Azzopardi-Alexander. 1997. *Maltese.* Routlege Descriptive Grammar. London: Routledge.

Bošković, Željko. 1994. D-structure, θ-criterion, and movement into θ-positions. *Language Analysis* 24:247–286.

———. 1997a. On certain violations of the superiority condition, AgrO, and Economy of Derivation. *Journal of Linguistics* 33 (2): 227–254.

———. 1997b. *The syntax of nonfinite complementation: An economy approach.* Cambridge, MA: MIT Press.

———. 1999. On multiple feature checking: multiple *Wh*-fronting and multiple head movement. Chap. 7 in *Working minimalism,* edited by Samuel David Epstein and Norbert Hornstein, 159–187. Cambridge, MA: MIT Press.

———. 2001. *On the nature of the syntax-phonology interface: Cliticization and related phenomena.* Amsterdam: Elsevier.

———. 2003. Agree, phases, and intervention effects. *Linguistic Analysis* 33:54–96.

———. 2004a. Be careful where you float your quantifiers. *Natural Language & Linguistic Theory* 22 (4): 681–742.

———. 2004b. On the clitic switch in Greek imperatives. In *Balkan syntax and semantics,* edited by Olga Mišeska Tomić, 269–291. Amsterdam: John Benjamins.

———. 2005. On the locality of left branch extraction and the structure of NP. *Studia Linguistica* 59:1–45.

———. 2007. On the locality and motivation of Move and Agree: An even more minimal theory. *Linguistic Inquiry* 38 (4): 589–644.

———. 2008. What will you have, DP or NP? In *Proceedings of NELS 37,* edited by Emily Elfner and Martin Walkow, 101–114. Amherst, MA: GLSA.

Bošković, Željko. 2011. On unvalued uninterpretable features. In *Proceedings of NELS 39,* edited by Suzi Lima, Kevin Mullin, and Brian Smith, 109–120. Amherst, MA: GLSA.

———. 2012. On NPs and clauses. In *Discourse and grammar: From sentence types to lexical categories,* edited by Günther Grewendorf and Thomas Ede Zimmermann, 179–242. Berlin: De Gruyter.

———. 2013. Phases beyond clauses. In *Nominal constructions in Slavic and beyond,* edited by Lilia Schürcks, Anastasia Giannakidou, and Urtzi Etxeberria, 75–128. Berlin: De Gruyter.

———. 2014. Now I'm a phase, now I'm not a phase. *Linguistic Inquiry* 45:27–89.

———. 2015. From the complex NP constraint to everything: On deep extraction across categories. *The Linguistic Review* 32:603–669.

———. 2016a. On the timing of labeling: deducing comp-trace effects, the subject condition, the adjunct condition, and tucking in from labeling. *The Linguistic Review* 33:17–66.

———. 2016b. What is sent to spell-out is phases, not phasal complements. *Linguistica* LVI (Current Trends in Generative Linguistics):25–66.

———. 2018. On clitic doubling and argument ellipsis: Argument ellipsis as predicate ellipsis. Edited by Myriam Dali, Eric Mathieu, and Gita Zareikar. *English Linguistics* 35:1–37.

Bošković, Željko and Troy Messick. to appear. Derivation economy in syntax and semantics. In *Oxford research of linguistics,* edited by Mark Aronoff. [manuscript version 2017]. Oxford: Oxford University Press.

Bošković, Željko and Daiko Takahashi. 1998. Scrambling and last resort. *Linguistic Inquiry* 29 (3): 347–366.

Bossong, Georg. 1991. Differential object merking in Romance and beyond. In *New analyses in Romance linguistics: Selected papers from the Linguistic Symposium on Romance Languages XVIII, Urbana-Champaign, april 7–9, 1988,* edited by Dieter Wanner and Douglas A. Kibbee, 143–170. Amsterdam: John Benjamins.

Bresnan, Joan and Sam A. McHombo. 1987. Topic, pronoun, and agreement in Chicheŵa. *Language* 63 (4): 741–782.

Bresnan, Joan and Lioba Moshi. 1990. Object asymmetries in comparative Bantu syntax. *Linguistic Inquiry* 21 (2): 147–185.

Broselow, Ellen. 1983. A lexical treatment of Cairene Arabic object clitics. In *Current approaches to African linguistics,* edited by Ivan R. Dihoff, vol. 1. Foris.

Browne, Wayles. 1970. Noun phrase definiteness in relatives and questions: Evidence from Macedonian. *Linguistic Inquiry* 1 (2): 267–270.

Bruening, Benjamin. 2001. Syntax at the edge: Cross-clausal phenomena and the syntax of Passamaquoddy. PhD diss., MIT.

Buchholz, Oda and Wilfried Fiedler. 1987. *Albanische grammatik.* Leipzig: Enzyclopädie.

Burzio, Luigi. 2000. Anatomy of a generalization. In *Arguments and case: Explaining Burzio's Generalization,* edited by Eric Reuland, 195–240. Amsterdam: John Benjamins.

Butler, Inez M. 1980. *Gramática Zapoteca: Zapoteco de Yatzachi el Bajo.* Mexico City: Instituto Lingüístico de Verano.

Caha, Pavel. 2009. The nanosyntax of case. PhD diss., University of Tromsø.

Camacho, José. 2017. Switch reference and obviation. In *The Wiley Blackwell companion to syntax,* 2nd, edited by Martin Evaraert and Henk van Riemsdijk, 1–30. Malden, MA: Wiley-Blackwell.

Campana, Mark. 1989. Algonquian and the absolutive case hypothesis. In *Actes du vingtiéme congrés des algonquinistes,* 70–78.

Cardinaletti, Anna. 1999. Pronouns in Germanic and Romance languages: An overview. In *Clitics in the languages of Europe,* edited by Henk van Riemsdijk, 33–82. New York: Mouton de Gruyter.

———. 2008. On different types of clitic clusters. In *The Bantu–Romance connection,* edited by Cécile De Cat and Katherine Demuth, 41–82. Amsterdam: John Benjamins.

Cardinaletti, Anna and Michal Starke. 1994. The typology of structural deficiency: On the three grammatical classes. University of Venice, *Working Papers in Linguistics* 4 (2): 41–109.

———. 1999. The typology of structural deficiency: A case study of the three classes of pronouns. In *Clitics in the languages of Europe,* edited by Henk van Riemsdijk, 145–233. New York: Mouton de Gruyter.

Cetnarowska, Božena. 2003. On pronominal clusters in Polish. In *Investigations into formal Slavic linguistics: Contributions of the fourth European conference on Formal Description of Slavic Languages (FDSL IV) held at Potsdam University, November 28-30, 2001,* edited by Peter Kosta, Joanna Blaszczak, Jens Frasek, Ljudmila Geist, and Marzena Zygis, 1:13–30. Frankfurt am Main: Peter Lang.

Charnavel, Isabelle and Victoria Mateu. 2015a. Antilogophoricity in clitic clusters. In *Proceedings of WCCFL 32,* edited by Ulrike Steindl et. al., 1–10. Somerville, MA: Cascadilla Proceedings Project.

———. 2015b. The clitic binding restriction revisited: Evidence for antilogophoricity. *The Linguistic Review* 32 (4): 671–701.

Cheng, Lisa Lai Shen and Hamida Demirdash. 1990. Superiority violations. *MIT Working papers in Linguistics (Papers on Wh-movement)* 13:27–46.

Chomsky, Noam. 1957. *Syntactic structures.* The Hague: Mouton.

———. 1965. *Aspects of the theory of syntax.* Cambridge: MIT Press.

———. 1973. Conditions on tranformations. In *A festschrift for Morris Halle,* edited by Stephen R. Anderson and Paul Kiparsky, 232–286. New York: Holt, Rinehart / Winston.

———. 1980. On cognitive structures and their development: A reply to Piaget. In *Language and learning: The debate between Jean Piaget and Noam Chomsky,* edited by Massimo Piattelli-Palmarini, 35–54. Cambridge, MA: Harvard University Press.

———. 1981. *Lectures on government and binding.* Dordrecht: Foris.

———. 1986. *Knowledge of language, its nature, origin, and use.* New York: Praeger.

———. 1991. Some notes on economy of derivation and representation. In *Principles and parameters in comparative grammar,* edited by Robert Friedin, 417–454. MIT Press.

———. 1993. A minimalist program for linguistic theory. In *MIT occasional papers in linguistics,* edited by Kenneth Hale and Samuel Jay Keyser, vol. 1. Cambridge, MA: Department of Linguistics / Philosophy, MIT.

———. 1995. *The minimalist program.* Cambridge: MIT Press.

———. 2000. Minimalist inquiries: The framework. In *Step by step: Essays on minimalism in honor of Howard Lasnik,* edited by Roger Martin, David Michaels, Juan Uriagereka, and Samuel Jay Keyser., 89–155. Cambridge: MIT Press.

———. 2001. Derivation by phase. In *Ken Hale: A life in language,* edited by Michael Kenstowicz, 1–52. Cambridge, MA: MIT Press.

———. 2004. Beyond explanatory adequacy. In *The cartography of syntactic structures, vol. 3, Structures and beyond,* edited by Adriana Belletti, 104–131. Oxford: Oxford University Press.

———. 2005. Three factors in language design. *Linguistic Inquiry* 36:1–22.

———. 2008. On phases. In *Foundational issues in linguistic theory: Essays in honor of Jean-Roger Vergnaud,* edited by Robert Freidin, Carlos P. Otero, and Maria Luisa Zubizarreta, 133–165. Cambridge, MA: MIT Press.

———. 2013. Problems of projection. *Lingua* 130 (0): 33–49.

———. 2015. Problems of projection: Extensions. In *Structures, strategies and beyond: Studies in honor of Adriana Belletti,* edited by Elisa Di Domenico, Cornelia Hamann, and Simona Matteini, 1–16. Amsterdam: John Benjamins.

Chomsky, Noam and Howard Lasnik. 1993. The theory of principles and parameters. In *Syntax: An international handbook of contemporary research,* edited by Joachim Jacobs, Arnim von Stechow, Wolfgang Sternfield, and Theo Vennemann, 506–569. Berlin: de Gruyter.

Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective.* Oxford: Oxford University Press.

———. 2005. Deriving Greenberg's Universal 20 and its exceptions. *Linguistic Inquiry* 36 (3): 315–332.

———. 2007. A note on linguistic theory and typology. *Linguistic Typology* 11:93–106.

———. 2013a. Cognition, universal grammar, and typological generalizations. *Lingua* 130:50–65.

———. 2013b. *Typological studies: Word order and relative clauses.* New York: Routledge.

Ciucivara, Oana Săvescu. 2006. Challenging the Person Case Constraint: Evidence from Romanian. In *Selected papers from the 36th LSRL, New Brunswick, March-April 2006,* edited by María José Cabrera, José Camacho, Viviane Déprez, Nydia Flores-Ferrán, and Liliana Sanchez, 255–269. Amsterdam/Philadelphia: John Benjamins.

———. 2009. A syntactic analysis of pronominal clitic clusters in Romance: The view from Romanian. PhD diss., NYU.

Ciucivara, Oana Săvescu and Andrew Nevins. 2008. An apparent Number Case Constraint in Romanian: The role of syncretism. In *Proceedings of the 38th LSRL,* edited by Karlos Arregi, Zsuzsanna Fagyal, Silvina Montrul, and Annie Tremblay, 185–203. Amsterdam/Philadelphia: John Benjamins.

Coghill, Eleanor. 2010. Ditransitive constructions in the Neo-Aramaic dialect of Telkepe. In *Studies in ditransitive constructions: A comparative handbook,* edited by Andrej Malchukov, Martin Haspelmath, and Bernard Comrie, 221–242. Berlin/New York: Mouton de Gruyter.

———. 2018. The Neo-Aramaic dialect of Telkepe. In *Semitic linguistics and manuscripts: a Liber Discipulorum in honour of Professor Geoffrey Khan,* edited by Nadia Vidro, Ronny Vollandt, Esther-Miriam Wagner, and Judith Olszowy-Schlanger, 234–271. Uppsala: Acta Universitatis Upsaliensis.

Collins, Chris. 1995. Toward a theory of optimal derivation. *MIT Working papers in Linguistics* 27:65–104.

———. 2005. A smuggling approach to the passive in English. *Syntax* 8 (2): 81–120.

Collins, Chris and Paul M. Postal. 2012. *Imposters: A study of pronominal agreement.* Cambridge, MA: MIT Press.

Compton, Richard. 2018. *Spurious Person-Case Constraints: Evidence from Inuktitut.* Poster presentation at WCCFL 36.

Compton, Richard and Christine Pittman. 2010. Word-formation by phase in Inuit. *Lingua* 120 (9): 2167–2192.

Comrie, Bernard. 1976. Linguistic politeness axes: Speaker-addressee, speaker-reference, speaker-bystander. *Pragmatics Microfiche* 1 (7): A3–B1.

———. 1979a. Definite and animate direct objects. *Linguistica Silesiana* 3:13–21.

———. 1979b. The animacy hierarchy in Chukchee. In *The elements: A parasession on linguistic units and levels,* edited by Paul R. Clyne, William F. Hanks, and Carol L. Hofbauer, 322–329. Chicago: Chicago Linguistic Society.

———. 1980. Inverse verb forms in Siberia: Evidence from Chukchee, Koryak and Kamchadal. *Folia Linguistica* 1:61–74.

———. 1981. *Language universals and linguistic typology: Syntax and morphology.* Chicago, IL: University of Chicago Press.

———. 1986. Markedness, grammar, people and the world. In *Markedness,* edited by Fred R. Eckman, Edith A. Moravcsik, and Jessica R. Wirth, 85–106. New York, NY: Plenum Press.

———. 2003. When agreement gets trigger-happy. *Transactions of the Pholological Society* 101 (2): 313–337.

Coon, Jessica. 2010. Rethinking split ergativity in Chol. *International Journal of American Linguistics* 76 (2): 207–253.

———. 2013. *Aspects of split ergativity.* Oxford: Oxford University Press.

Coon, Jessica, Nico Baier, and Theodore Levin. 2019. *Mayan Agent Focus and the ergative extraction constraint: Facts and fictions revisited.* Ms. McGill. Https://ling.auf.net/lingbuzz/004545.

Coon, Jessica and Alan Bale. 2014. The interaction of person and number in Mi'gmaq. Eds. Martin Krämer, Sandra-Iulia Ronai and Peter Svenonious, *Nordlyd: Special issue on Features* 40 (1): 85–101.

Coon, Jessica and Stefan Keine. 2018. *Feature gluttony.* Manuscript: McGill. https://ling.auf.net/lingbuzz/004224.

Coon, Jessica, Stefan Keine, and Michael Wagner. 2017. Hierarchy effects in copular constructions: The PCC corner of German. In *Proceedings of NELS 47,* edited by Andrew Lamont and Katerina Tetzloff, 205–214. Amherst, MA: GLSA.

Cooper, Robin. 1983. *Quantification and semantic theory.* Dordrecht: Riedel.

Cowell, Andrew and Alonzo Moss Sr. 2008. *The Arapaho language.* Boulder, CO: University Press of Colorado.

Croft, William. 1988. Agreement vs. case marking and direct objects. In *Agreement in natural language: Approaches, theories, descriptions,* 159–179. Stanford, CA: CSLI.

———. 1990. *Typology and universals.* Cambridge: Cambridge University Press.

Cysouw, Michael. 2003. *The paradigmatic structure of person marking.* Oxford: Oxford University Press.

Cysouw, Michael and Javier Fernández Landaluce. 2012. On the (im)possibility of partial argument coreference. *Linguistics* 50 (4): 765–782.

Dahlstrom, Amy. 1986a. Plains Cree morphosyntax. PhD diss., Berkeley.

———. 1986b. Weak crossover and obviation. In *Proceedings of the 12th annual meeting of the Berkeley Linguistics Society,* edited by Vassiliki Nikiforidou, Mary VanClay, Mary Niepokuj, and Deborah Feder, 51–60. Berkeley, CA: BLS.

———. 1991. *Plains Cree morphosyntax.* New York: Garland.

———. 2013. Argument structure of quirky Algonquian verbs. In *From quirky case to representing space: Papers in honor of Annie Zaenen,* edited by Tracy Holloway King and Valeria de Paiva, 61–71. Stanford, CA: CSLI Publications.

———. 2015. *Meskwaki syntax book.* Ms. University of Chicago [accessed Dec 19th, 2015].

———. 2017. Obviation and information structure in Meskwaki. In *Papers of the 46th Algonquian conference,* edited by Monica Macaulay and Margaret Noodin, 39–54. East Lansing, MI: Michigan State University Press.

Dalrymple, Mary and Irina Nikolaeva. 2011. *Objects and information structure.* Camridge: Cambridge University Press.

Daniliuc, Radu and Laura Daniliuc. 2000. *A descriptive grammar of Romanian: An outline.* Munich: LINCOM.

Das Gupta, Kamalesh. 1971. *An introduction to the nocte language.* Shillong: North East Frontier Agency.

Dayal, Veneeta. 2016. *Questions.* Oxford: Oxford University Press.

De Vincenzi, Marica and Elisa Di Domenico. 1999. A distinction among ϕ-features: The role of gender and number in the retrieval of pronoun antecedents. *Rivista di linguistica* 11 (1): 41–74.

Deal, Amy Rose. 2015. Interaction and satisfaction in ϕ-agreement. In *Proceedings of nels 45,* edited by Thuy Bui and Deniz Ozyildiz, 1:179–192. Amherst, MA: GLSA.

Déchaine, Rose-Marie and Martina Wiltschko. 2002. Decomposing pronouns. *Linguistic Inquiry* 33 (3): 409–442.

———. 2017. A formal typology of reflexives. *Studia Linguistica* 71 (1-2): 60–106.

Dekker, Paul. 1994. Predicate logic with anaphora. In *Proceedings of SALT IV,* edited by Mandy Harvey and Lynn Samelman, 79–95. Ithaca, NY: CLC Publications.

Del Bon, Estella. 2002. Personal inflection and order of clitics in Kashmiri. In *Topics in Kashmiri,* edited by Omkar N. Koul and Kashi Wali, 129–142. New Delhi: Creative Books.

DeLancey, Scott. 1981. The category of direction in Tibeto-Burman. *Linguistics of the Tibeto-Burman Area* 6 (1): 83–101.

———. 2017. Hierarchical and accusative alignment of verbal person marking in Trans-Himalayan. *Journal of South Asian Languages and Linguistics (Special Issue: Typological profiles of language families of South Asia)* 4 (1): 85–105.

———. 2018. Deictic and sociopragmatic effects in Tibeto-Burman SAP indexation. In *Typological hierarchies in synchrony and diachrony,* edited by Sonia Cristofaro and Fernando Zúñiga, 343–376. John Benjamins.

Demuth, Katherine and Jeffrey Gruber. 1995. Constraining XP-sequences. *Niger-Congo Syntax & Semantics* 6:3–30.

Despić, Miloje. 2011. Syntax in the absence of determiner phrase. PhD diss., UConn.

———. 2015. Phases, reflexives, and definiteness. *Syntax* 18 (3): 201–234.

den Dikken, Marcel. 2007. Phase extension. Contours of a theory of the role of head movement in phrasal extraction. *Theoretical Linguistics* 33:1–41.

———. 2009. Arguments for successive-cyclic movement through SpecCP. *Linguistic Variation Yearbook* 9:89–126.

Dixon, R. M. W. 1979. Ergativity. *Language* 55:59–138.

———. 2015. *Edible gender, mother-in-law style, and other grammatical wonders: Studies in Dyirbal, Yidiñ, and Warrgamay.* Oxford: Oxford University Press.

Doron, Edit and Geoffrey Khan. 2012a. PCC and ergative case: Evidence from Neo-Aramaic. In *Proceedings of the 29th West Coast Conference on Formal Linguistics,* edited by Jaehoon et al. Choi. Somerville, MA: Cascadilla Proceedings Project.

———. 2012b. The typology of morphological ergativity in Neo-Aramaic. *Lingua* 122 (3): 225–240.

Dryer, Matthew S. 1992. A comparison of the obviation systems of Kutenai and Algonquian. In *Papers on the 23rd Algonquian conference,* 119–163. Ottawa: Carleton University.

Dukes, Michael. 1996. Animacy and agreement in Cherokee. In *Cherokee papers from UCLA,* edited by Pamela Munro, 16:75–96. UCLA Occasional Papers in Linguistics. Los Angeles: UCLA Department of Linguistics.

Dunn, Michael John. 1999. A grammar of Chukchi. PhD diss., The Australian National University.

Duranti, Alessandro. 1979. Object clitic pronouns and the Topicality Hierarchy. *Studies in African Linguistics* 10:31–45.

Ellis, C. Douglas. 1983. *Spoken Cree: West Coast of James Bay (2nd edition).* Edmonton: The University of Alberta Press.

———. 2000. *Spoken Cree: ê-ililîmonâniwahk. level I. West Coast of James Bay.* Edmonton: The University of Alberta Press.

Emonds, Joseph. 1975. A transformational analysis of French clitics without positive output constraints. *Linguistic Analysis* 1:3–24.

Enç, Mürvet. 1991. The semantics of specificity. *Linguistic Inquiry* 22 (1): 1–25.

Erickson, Barbara E. 1965. Patterns of person-number reference in Potawatomi. *International Journal of American Linguistics* 31 (3): 226–236.

Erlewine, Michael Yoshitaka. 2016. Anti-locality and optimality in Kaqchikel Agent Focus. *Natural Language & Linguistic Theory* 34 (2): 429–479.

Erschler, David. 2014. *Person case constraints beyond the dative and accusative: Evidence from Ossetic.* Ms. Max Planck Institut/Tübinger Zentrum für Linguistik.

———. 2015. *Variation in Person Case Constraints: Evidence from Digor and Iron Ossetic.* Ms. UMass, Amherst.

Faltz, Leonard M. 1977. Reflexivization: A study in universal syntax. PhD diss., UC Berkeley.

———. 1985. *Reflexivization: A study in universal syntax.* New York, NY: Garland.

Farghaly, Ali. 1982. Subject pronoun deletion rule in Egyptian Arabic. In *Discourse analysis: Theory and application proceedings of the second national symposium on linguistics* *and English language teaching,* edited by S. Gamal and R. Bower, 60–69. Cairo: Center for Developing English Language Teaching, Ain Shams University.

Farkas, Donca and Kostas Kazazis. 1980. Clitic pronouns and topicality in Rumanian. In *Proceedings of the 16th regional meeting of the Chicago Linguistic Society,* 75–82. Chicago, IL: CLS.

Fassi-Fehri, Abdelkader. 1988. Agreement in Arabic, binding and coherence. In *Agreement in natural language,* edited by Michael Barlow and Charles A. Ferguson, 107–158. Stanford: CSLI Publications.

———. 1993. *Issues in the structure of Arabic clauses and words.* Vol. 29. Dordrecht: Kluwer Academic.

Fenger, Paula. to appear. Size matters: Auxiliary formation in the morpho-syntax and morpho-phonology. In *Proceedings of NELS 49.* Amherst, MA: GLSA.

———. in preparation. Words within words. PhD diss., UConn.

Fillmore, Charles J. 1975. *Santa Cruz lectures on deixis, 1971.* Bloomington, IN: Indiana University Linguistics Club.

Foley, Steven, Nick Kalivoda, and Maziar Toosarvandani. to appear. Gender-case constraints in Zapotec. In *Proceedings of the 22nd WSCLA 22.*

Foley, Steven and Maziar Toosarvandani. 2018. *Pronoun movement and probe generosity.* Ms. UCSC. Https://ling.auf.net/lingbuzz/004267.

Foley, William. 1991. *The Yimas language of New Guinea.* Stanford: Stanford University Press.

Folli, Raffaella and Heidi Harley. 2008. Teleology and animacy in external arguments. *Lingua* 118:190–202.

Forrest, Linda B. 1994. The de-transitive clauses in Bella Coola. In *Voice and inversion,* edited by Talmy Givón, 147–168. Amsterdam: John Benjamins.

Fountain, Amy Velita. 1998. An optimality theoretic account of Navajo prefixal syllables. PhD diss., University of Arizona.

Fox, Danny. 1998. Economy and semantic interpretation – a study of scope and variable binding. PhD diss., MIT.

———. 2002. Antecedent-contained deletion and the copy theory of movement. *Linguistic Inquiry* 33 (1): 63–96.

———. 2003. On logical form. In *Minimalist syntax,* 82–123. Oxford: Blackwell Publishers.

Franks, Steven. 1995. *Parameters of Slavic morphosyntax.* New York: OUP.

———. 2013. Orphans, doubling, coordination, and phases: On nominal structure in Slovenian. *Slovene Linguistic Studies* 9:55–92.

———. 2016. Clitics are/become minimal(ist). In *Formal studies in Slovenian syntax. In honor of Janez Orešnik,* edited by Lanko Marušič and Rok Žaucer, 91–128. Amsterdam: John Benjamins.

———. 1998/2010. *Clitics in Slavic.* Paper presented at the Comparative Slavic Morphosyntax Workshop, Spencer, Indiana. [Updated version published online in *Glossos 10: Contemporary Issues in Slavic Linguistics*].

Franks, Steven and Tracy Holloway King. 2000. *Handbook of Slavic clitics.* New York: Oxford University Press.

Frantz, Donald G. 1966. Person indexing in Blackfoot. *International Journal of American Linguistics* 32 (1): 50–58.

———. 1971. *Toward a generative grammar of Blackfoot (with particular attention to selected stem formation processes).* Norman, OK: Summer Institute of Linguistics of the University of Oklahoma.

———. 1976. Unspecified-subject phenomena in Algonquian. In *Papers of the 7th Algonquian conference 1975,* edited by William Cowan, 197–216. Ottawa: Carleton University.

———. 1978. Copying from complements in blackfoot. In *Linguistic studies of native Canada,* edited by Eung-Do Cook and Jonathan Kaye, 89–110. Vancouver: University of British Columbia Press.

———. 1991. *Blackfoot grammar.* Toronto: University of Toronto Press.

Fuss, Eric. 2005. *The rise of agreement: A formal approach to the syntax and grammaticalization of verbal inflection.* Amsterdam/Philadephia: John Benjamins.

Gallego, Ángel J. 2005. *Phase sliding.* Manuscript: UAB/UMD.

Galloway, Brent D. 1993. *A grammar of Upriver Halkomelem.* Berkeley, CA: University of California Press.

Georgala, Effi. 2011. Applicatives in their structural and thematic function: a minimalist account of multitransitivity. PhD diss., Cornell Unversity.

Gerdts, Donna B. 1988. *Object and absolutive in Halkomelem Salish.* New York: Routledge.

Givón, Talmy. 1994. *Voice and inversion.* Amsterdam: John Benjamins.

Goddard, Ives. 1969. Delaware verbal morphology: A descriptive and comparative study. PhD diss., Harvard.

———. 1979. *Delaware verbal morphology.* Garland.

Goddard, Ives. 1984. The obviative in Fox narrative discourse. In *Papers on the 15th Algonquian conference,* edited by William Cowan and José Mailhot, 273–286. Ottawa: Carleton University.

———. 1990. Aspects of the topic structure of Fox narratives: proximate shifts and the use of overt and inflectional NPs. *International Journal of American Linguistics* 56 (3): 317–340.

Golluscio, Lucía A. 2010. Ditransitives in Mapudungun. In *Studies in ditransitive constructions: a comparative handbook,* edited by Andrej Malchukov, Martin Haspelmath, and Bernard Comrie, 710–756. Berlin/New York: Mouton de Gruyter.

Grafstein, Ann. 1980. Some properties of Ojibwa verb stem formation: A lexical analysis. In *Papers of the 11th Algonquian conference,* edited by William Cowan, 83–95. Ottawa: Carleton University.

———. 1981. Obviation in Ojibwa. *Recherches linguistiques à Montréal / Montreal working papers in linguistics* 16:83–134.

———. 1984. Argument structure and the syntax of a non-configurational language. PhD diss., McGill.

———. 1987. The Algonquian obviative and the Binding Theory. In *Proceedings of NELS 18,* edited by James Blevins and Juli Carter. Amherst, MA: GLSA.

Greenberg, Joseph H. 1957. The nature and uses of linguistic typologies. *International Journal of American Linguistics* 23 (2): 68–77.

———. 1963. *Universals of language.* Cambridge, MA: MIT Press.

Grimes, Joseph E. 1985. Topic inflection in Mapudungun verbs. *International Journal of American Linguistics* 52 (2): 141–163.

Grohmann, Kleanthes K. 2003. *Prolific domains: On the anti-locality of movement dependencies.* Amsterdam/Philadelphia: John Benjamins.

Grosz, Barbara J., Aravind K. Joshi, and Scott Weinstein. 1983. Providing a unified account of definite noun phrases in discourse. In *Proceedings, 21st annual meeting of the Association of Computational Linguistics,* 44–50. Cambridge, MA: Association for Computational Linguistics.

———. 1995. Centering: A framework for modeling the local coherence of discourse. *Computational Linguistics* 21 (2): 203–225.

Haiman, John and Pamela Munro, eds. 1983. *Switch reference and universal grammar: proceedings of a symposium on switch reference and universal grammar, Winnipeg, May 1981.* Amsterdam/Philadelphia: John benjamins.

Hale, Kenneth L. 1973a. A note on subject-object inversion in Navajo. In *Issues in linguistics: Papers in honor of Henry and Reneée Kahane.* Edited by Braj B. Kachru, Robert B. Lees, Yakov Malkiel, Angelina Pietrangeli, and Sol Saporta, 300–309. Chicago, IL: University of Illinois Press.

Hale, Kenneth Locke. 1959. A Papago grammar. PhD diss., Indiana University.

———. 1973b. Person marking in Walbiri. In *A festschrift for Morris Halle,* edited by Stephen R. Anderson and Paul Kiparsky. New York: Holt, Rinehart / Winston.

———. 1983. Warlpiri and the grammar of non-configurational languages. *Natural Language & Linguistic Theory* 1 (1): 5–47.

Hale, Kenneth Locke and Samuel Jay Keyser. 1998. The basic elements of argument structure. *MIT Working papers in Linguistics (Papers from the UPenn/MIT Roundtable on Argument Structure and Aspect)* 32:73–118.

———. 2002. *Prolegomenon to a theory of argument structure.* Cambridge, MA: MIT Press.

Hall, Kira. 1990. Agentivily and the animacy hierarchy in Kashaya. In *Papers from the 1990 Hokan-Penutian languages workshop (Occasional Papers in Linguistics 15),* edited by James E. Redden, 118–135. Carbondale, IL: Department of Linguistics, Southern Illinois University.

Halle, Morris and Kenneth Locke Hale. 1997. *Chukchi transitive and antipassive constructions.* Manuscript: MIT.

Halle, Morris and Alec Marantz. 1993. Distributed morphology and the pieces of inflection. In *The view from Building 20,* edited by Ken L. Hale and Samuel Jay Keyser, 111–176. Cambridge, MA: MIT Press.

———. 1994. Some key features of distributed morphology. *MIT Working papers in Linguistics: Papers on Phonology and Morphology* 21:275–288.

Hamilton, Michael David. 2015. The syntax of Mi'gmaq: A configurational account. PhD diss., McGill.

Harbour, Daniel. 2006. *Morphosemantic number: From Kiowa noun classes to UG number features.* Dordrecht: Springer.

———. 2016. *Impossible persons.* Cambridge, MA: MIT Press.

Harizanov, Boris. 2014a. Clitic doubling at the syntax-morphophonology interface: A-movement and morphological merger in Bulgarian. *Natural Language & Linguistic Theory* 32 (4): 1033–1088.

Harizanov, Boris. 2014b. On the mapping from syntax to morphophonology. PhD diss., University of California, Santa Cruz.

Harley, Heidi and Rolf Noyer. 1999. State-of-the-Article: Distributed Morphology. *GLOT International* 4 (4): 3–9.

Harley, Heidi and Elizabeth Ritter. 2002. Person and number in pronouns. *Language* 78 (3): 482–526.

Harris, Alice. 1981. *Georgian syntax: A study in relational grammar.* Cambridge: Cambridge University Press.

Haspelmath, Martin. 2004a. Explaining the ditransitive person-role constraint: A usage-based approach. Free online journal, University of Düsseldorf, *Constructions* 2:49.

———. 2004b. On directional in language change with particular reference to grammaticalization. In *Up and down the cline: The nature of grammaticalization,* edited by Olga Fischer, Muriel Norde, and Harry Perridon, 17–44. Amsterdam/Philadelphia: John Benjamins.

———. 2007. Ditransitive alignment splits and inverse alignment. In *Ditransitivity,* edited by Anna Siewierska and Willem B. Hollmann, vol. 14:1, 79–102. Functions of Language. Amsterdam/Philadelphia: John Benjamins.

Hauge, Kjetil Rå. 1999. The word order of predicate clitics in Bulgarian. *Journal of Slavic Linguistics* 7 (1): 89–137.

Hawkinson, Annie K. and Larry M. Hyman. 1974. Hierarchies of natural topic in Shona. *Studies in African Linguistics* 5 (2): 147–170.

Heim, Irene. 1982. The semantics of definite and indefinite noun phrases. PhD diss., University of Massachusetts, Amherst.

———. 1991. *'Control' (Seminar taught by Heim & Higginbotham, spring 1991): "Interpretation of PRO" (February 22nd), "The first person" (March 8th), and "Connectedness" (April 12th).* [class handouts].

———. 1994. *Puzzling reflexive pronouns in de se reports.* Handout of talk presented at Bielefeld, March 1994.

———. 2002. *Features of pronouns in semantics and morphology.* Handout of talk given at USC on January 31, 2002.

———. 2008. Features on bound pronouns. In *Phi theory: Phi-features across modules and interfaces,* edited by Daniel Harbour, David Adger, and Susana Béjar, 36–56. Oxford: Oxford University Press.

Heim, Irene and Angelika Kratzer. 1998. *Semantics in generative grammar.* Oxford UK: Blackwell Publishing.

Hellan, Lars and Christer Platzack. 1999. Pronouns in Scandinavian languages. An overview. In *Clitics in the languages of Europe,* edited by Henk van Riemsdijk, 123–142. Berlin/New York: Mouton de Gruyter.

Henderson, Robert and Jessica Coon. 2018. Adverbs and variability in Kaqchikel Agent Focus. *Natural Language & Linguistic Theory* 36 (1): 149–173.

Henderson, T. S. T. 1971. Participant reference in Algonkin. *Cahiers linguistiques d'Ottawa* 1:27–49.

Hewitt, B. George. 1979. *Abkhaz.* Vol. 2. Lingua Descriptive Studies. Amsterdam: North-Holland.

———. 1989. *Abkhaz.* Croom Helm descriptive grammars. London: Routledge.

Heycock, Caroline. 2012. Specification, equation, and agreement in copular sentences. *Canadian Journal of Linguistics* 57 (2): 209–240.

Hiraiwa, Ken. 2001. Multiple Agree and the defective intervention constraint in Japanese. *MIT working papers in linguistics* 40:67–80.

———. 2005. Dimensions of symmetry in syntax: Agreement and causal architecture. PhD diss., MIT.

Hockett, Charles. 1939. Potawatomi syntax. *Language* 15 (4): 235–248.

———. 1948. Potawatomi III: The verb complex. *International Journal of American Linguistics* 14 (3): 139–149.

———. 1966. What Algonquian is really like. *International Journal of American Linguistics* 32 (1): 59–73.

Hoji, Hajime. 1985. Logical Form constraints and configurational structures in Japanese. PhD diss., University of Washington.

———. 1987. Weak Crossover and Japanese phrase structure. In *Issues in Japanese linguistics,* edited by Takashi Imai and Mamoru Saito, 163–201. Dordrecht: Foris Publications.

Holmberg, Anders. 2005. Is there a little pro? Evidence from Finnish. *Linguistic Inquiry* 36 (4): 533–564.

Holmberg, Anders and Thorbjörg Hróarsdóttir. 2004. Agreement and movement in Icelandic raising constructions. *Lingua* 114:651–673.

Hornstein, Norbert. 1999. Movement and control. *Linguistic Inquiry* 30:69–96.

Hornstein, Norbert and David Lightfoot, eds. 1981. *Explanation in linguistics: The logical problem of language acquisition.* London: Longman.

Hualde, José. 1989. Double object constructions in KiRimi. In *Current approaches to African linguistics,* edited by Paul Newman and Robert D. Botne, 5:179–189. Dordrecht: Foris.

Ivan, Rudmila-Rodica. to appear. Romanian loves Me: clitic clusters, ethics & cyclic AGREE. In *Proceedings of NELS 49,* edited by Maggie Baird, Duygu Goksu, and Johnathan Pesetsky. Amherst, MA: GLSA.

Jacobs, Peter. 1994. The inverse in Squamish. In *Voice and inversion,* edited by Talmy Givón, 121–146. Amsterdam: John Benjamins.

Jacques, Guillaume and Anton Antonov. 2014. Direct/inverse systems. *Language and Linguistics Compass* 8 (7): 301–318.

Jelinek, Eloise. 1983. Person-subject marking in AUX in Egyptian Arabic. In *Linguistic categories: auxiliaries and related puzzles: Volume One: Categories,* edited by Frank Heny and Barry Richards, 21–46. Dordrecht: Springer Netherlands.

———. 1984. Empty categories, case, and configurationality. *Natural Language & Linguistic Theory* 2 (1): 39–76.

———. 1990. Grammatical relations and coindexing in inverse systems. In *Grammatical relations: A cross-theoretical perspective,* edited by Katarzyna Dziwirek, Patrick Farrell, and Errapel Mejías Bikandi. Stanford, CA: CSLI Publications.

———. 1994. Transitivity and voice in Lummi. In *29th international conference on Salish and neighboring languages, Salish Kootenai College, Pablo, Montana, August 1994,* 1–10.

———. 2002. Agreement, clitics and focus in Egyptian Arabic. In *Themes in Arabic and Hebrew syntax,* edited by Jamal Ouhalla and Ur Shlonsky, 71–105. Dordrecht: Springer Netherlands.

———. 2006. The pronominal argument parameter. In *Arguments and agreement,* edited by Peter Ackema, Partick Brandt, Maaike Schoorlemmer, and Fred Weerman, 261–288. Oxford: Oxford University Press.

Jelinek, Eloise and Richard A. Demers. 1983. The agent hierarchy and voice in some Coast Salish languages. *International Journal of American Linguistics* 49 (2): 167–185.

———. 1994. Predicates and pronominal arguments in Straits Salish. *Language* 70 (4): 697–736.

Johns, Alana. 1996. The occasional absence of anaphoric agreement in Labrador Inuttut. In *Micro-parametric syntax and dialect variation,* edited by James R. Black and Virginia Motapanyane, 121–143. Amsterdam & Pholapelphia: John Benjamins.

Johns, Alana and Ivona Kučerova. 2017. On the morphosyntactic reflexes of information structure in the ergative patterning of Inuit language. In *The Oxford handbook of ergativity,* edited by Jessica Coon, Diane Massam, and Lisa DeMena Travis, 397–418. New York, NY: Oxford University Press.

Joseph, Brian D. and Irene Philippaki-Warburton. 1987. *Modern Greek.* London: Croom Helm.

Julien, Marit. 2002. *Syntactic heads and word formation.* Oxford: Oxford University Press.

Jungraithmayr, Herrmann and Abkar Adams. 1992. *Lexique migama: Migama-français et français-migama (Guéra, Tchad); avec une introduction grammaticale.* Berlin: Reimer.

Kalin, Laura. 2014a. Apparent last resort agreement in Senaya ditransitives. In *Proceedings of NELS 43,* edited by Hsin-Lun Huang, Ethan Poole, and Amanda Rysling, 203–214. Amherst, MA: GLSA.

———. 2014b. Aspect and argument licensing in Neo-Aramaic. PhD diss., UCLA.

———. 2017. Dropping the F-Bomb: An argument for valued features as derivational time bombs. In *Proceedings of NELS 47,* edited by Andrew Lamont and Katerina Tetzloff, 119–132. Amherst, MA: GLSA.

———. 2018. Licensing and differential object marking: The view from Neo-Aramaic. *Syntax* 21 (2): 112–159.

Kalin, Laura and Laura McPherson. 2012. Senaya (Neo-Aramaic): Structural person case constraint effects in progressives. In *Proceedings of the 30th West Coast Conference on Formal Linguistics,* edited by Nathan Arnett and Ryan Bennett, 173–183. Somerville, MA: Cascadilla Proceedings Project.

Kalin, Laura and Coppe van Urk. 2015. Aspect splits without ergativity: Agreement asymmetries in Neo-Aramaic. *Natural Language & Linguistic Theory* 33 (2): 659–702.

Kamp, Hans. 1981. A theory of truth and semantic representation. In *Formal methods in the study of language, Proceedings of the Third Amsterdam Colloquium,* edited by Jeroen Groenendijk, Theo M. V. Janssen, and Martin Stokhof, 277–322. Amsterdam: Mathematisch Centrum.

Kamp, Hans and Uwe Reyle. 1993. *From discourse to logic: Introduction to modeltheoretic semantics of natural language, formal logic and discourse representation theory.* Dordrecht: Kluwer.

Kaplan, David. 1978. On the logic of demonstratives. *Journal of Philosophical Logic* 8:81–98.

Kaplan, David. 1989. Demonstratives. In *Themes from Kaplan,* edited by Joseph Almog, John Perry, and Howard Wettstein. Reprinted Manuscript from 1977, UCLA. Oxford: Oxford University Press.

Karimi, Yadgar. 2010. Unaccusative transitives and the person-case constraint effects in Kurdish. *Lingua* 120:693–716.

Kayne, Richard S. 1969. The transformational cycle in French syntax. PhD diss., MIT.

———. 1972. Subject inversion in French interrogatives. In *Generative studies in Romance languages,* edited by Jean Casagrande and Bohdan Saciuk, 70–126. Rowley, MA: Newbury House.

———. 1975. *French syntax: The transformational cycle.* Cambridge: MIT Press.

———. 1983. Chains, categories external to S, and French complex inversion. *Natural Language & Linguistic Theory* 1:107–139.

———. 1990. Romance clitics and PRO. In *Proceedings of NELS 20,* edited by Juli Carter, Rose-Marie Dechaine, Bill Philip, and Tim Sherer, 255–302. Amherst, MA: GLSA.

———. 1991. Romance clitics, verb movement, and PRO. *Linguistic Inquiry* 22 (4): 647–686.

———. 1994. *The antisymmetry of syntax.* Cambridge: MIT Press.

———. 2000. *Parameters and universals.* Oxford: Oxford University Press.

———. 2003. Person morphemes and reflexives in italian, french, and related languages. In *The syntax of italian dialects,* edited by C. Tortora, 102–136. Oxford: Oxford University Press.

———. 2005a. Prepositions as probes. In *Movement and silence,* 85–104. Oxford: Oxford University Press.

———. 2005b. Some notes on comparative syntax, with special reference to English and French. In *The Oxford handbook of comparative syntax,* edited by Guglielmo Cinque and Richard S. Kayne, 3–69. Oxford: Oxford University Press.

Kenstowicz, Michael. 1989. The null subject parameter in Modern Arabic dialects. In *The null subject parameter,* edited by Osvaldo A. Jaeggli and Kenneth J. Safir, 263–275. Dordrecht: Springer Netherlands.

Kepping, Ksenia B. 1979. Elements of ergativity and nominativity in Tangut. In *Towards a theory of grammatical relations,* edited by Frans Plank, 263–277. London: Academic Press.

Khan, Geoffrey. 2008. *The Neo-Aramaic dialect of Barwar.* Leiden: Brill.

Kibrik, A. E. 2002. ‘Anomalies’ of cross-reference marking: The Alutor case. In *Morphology 2000: Selected papers from the 9th morphology meeting, Vienna, 24-28 february 2000,* edited by Sabrina Bendjaballah, Wolfgang U. Dressler, Oskar E. Pfeiffer, and Maria D. Voeikova, 199–212. Amsterdam/Philadelphia: John Benjamins.

Kiparsky, Paul. 2001. Structural case in Finnish. *Lingua* 111:315–376.

Kiparsky, Paul and Morris Halle. 1977. Towards a reconstruction of the Indo-European accent. In *Studies in stress and accent,* edited by Larry M. Hyman, 4:209–238. Southern California Occasional Papers in Linguistics. Los Angeles, CA: Department of Linguistics, University of Southern California.

É. Kiss, Katalin. 2013. The inverse agreement constraint in Uralic languages. *Finno-Ugric Languages and Linguistics* 2 (1): 2–21.

———. 2017. The person-case constraint and the inverse agreement constraint are manifestations of the same inverse topicality constraint. *The Linguistic Review* 34 (2): 365–395.

Klaiman, M. H. 1991. *Grammatical voice.* Cambridge: Cambridge University Press.

———. 1992. Inverse languages. *Lingua* 88 (3/4): 227–261.

Klamer, Marian. 1997. Spelling out clitics in Kambera. *Linguistics* 35 (5): 895–927.

Kratzer, Angelika. 1996. Severing the external argument from its verb. In *Phrase structure and the lexicon,* edited by Johan Rooryck and Laurie Zaring, 109–136. Dordrecht: Kluwer Academic Publishers.

———. 1998. More structural analogies between pronouns and tenses. In *Proceedings of SALT VIII,* edited by Devon Strolovich and Aaron Lawson, 92–110. Ithaca, NY: CLC Publications.

———. 2009. Making a pronoun. *Linguistic Inquiry* 40 (2): 187–237.

Kroskrity, Paul V. 2010. The art of voice: Understanding the Arizona Tewa inverse in its grammatical, narrative, and language-ideological contexts. *Anthropological Linguistics* 52 (1): 49–79.

Kroskrity, Paul Vincent. 1977. Aspects of Arizona Tewa language structure and language use. PhD diss., Indiana University.

———. 1985. A holistic understanding of Arizona Tewa passives. *Language* 61 (2): 306–328.

Kuno, Susumu. 1987. *Functional syntax. anaphora, discourse and empathy.* Chicago: University of Chicago Press.

Kuno, Susumu and Etsuko Kaburaki. 1977. Empathy and syntax. *Linguistic Inquiry* 8 (4): 627–672.

Kuroda, Sige-Yuki. 1968. English relativization and certain related problems. *Language* 44 (2): 244–266.

Lacerda, Renato. 2016. Topicalization as predication: The syntax-semanticsinterface of low topics in Brazilian Portuguese. In *Proceedings of the 33rd West Coast Conference on Formal Linguistics,* edited by Kyeong-min Kim, Pocholo Umbal, Trevor Block, Queenie Chan, Tanie Cheng, Kelli Finney, Mara Katz, Sophie Nickel-Thompson, and Lisa Shorten, 256–265. Somerville, MA: Cascadilla Proceedings Project.

———. in preparation. Middle-field syntax and information structure in Brazilian Portuguese. PhD diss., UConn.

Laenzlinger, Christopher. 1993. A syntactic view of Romance pronominal sequences. *Probus* 5:241–270.

Laka, Itziar. 1993. The structure of inflection: A case study in X^0 syntax. In *Generative studies in Basque linguistics,* edited by José Ignacio Hualde and Jon Ortiz de Urbina, 21–70. Amsterdam/Philadelphia: John Benjamins.

———. 2016. Deriving split ergativity in the progressive: The case of Basque. In *Ergativity,* edited by Alana Johns, Diane Massam, and Juvanal Ndayiragije, 173–96. Dordrecht: Springer.

Lakoff, G. 1975. Pragmatics in natural logic. In *Formal semantics of natural language,* edited by Ed Keenan, 253–286. Reprinted in Rogers et. al (1977). Cambridge: Cambridge University Press.

Lamoureaux, Siri van Dorn. 2004. *Applicative constructions in Maasai.* MA Thesis.

Langacker, Ronald W. 1977. *Studies in Uto-Aztecan grammar (An overview of Uto-Aztecan grammar 1.)* Dallas: Summer Institute of Linguistics.

Larson, Richard K. 1988. On the double object construction. *Linguistic Inquiry* 19 (3): 335–392.

———. 1990. Double objects revisited: Reply to Jackendoff. *Linguistic Inquiry* 21 (4): 589–632.

Lasnik, Howard. 1995. Case and expletives revisited: On Greed and other human failings. *Linguistic Inquiry* 26 (4): 615–633.

Lasnik, Howard and Mamoru Saito. 1992. *Move α: Conditions on its application and output.* Cambridge, MA: MIT Press.

Laughren, Mary and Robert Hoogenraad. 1996. *A learner's guide to Warlpiri.* Alice Springs: Iad Press.

Law, P. 1991. Effects of head movement on theories of subjacency and proper government. PhD diss., MIT.

Leavitt, Robert M. 1996. *Passamaquoddy-maliseet.* (LW/M, 27). Munich: Lincom Europa.

Legate, Julie Anne. 2008. Morphological and abstract case. *Linguistic Inquiry* 39:55–101.

Legate, Julie-Anne. 2014. *Voice and v: Lessons from acehnese.* Cambridge, MA: MIT Press.

Levinson, Stephen C. 1983. *Pragmatics.* Cambridge: Cambridge University Press.

Lichtenberg, Frantisek. 1983. *A grammar of Manam.* Honolulu: University of Hawaii Press.

Lochbihler, Bethany. 2013. Aspects of argument licensing. PhD diss., McGill.

Long, Rebecca and Sofronio Cruz. 2000. *Diccionaria Zapoteco de San Bartolomé Zoogocho Oaxaca.* Mexico City: Instituto Lingüístico de Verano.

Longobardi, Giuseppe. 1994. Reference and proper names: A theory of N-movement in syntax and Logical Form. *Linguistic Inquiry* 25 (4): 609–665.

López, Filemón and Ronaldo Newberg. 2005. *La conjugación del verbo Zapoteco: Zapoteco de Yalálag.* Mexico City: Instituto Lingüístico de Verano.

Lovestrand, Joseph. 2012. *The linguistic structure of Baraïn (Chadic).* MA thesis, GIAL.

Lowenstamm, Jean. 2012. Feminine and gender, or why the 'feminine' profile of French nouns has nothing to do with gender. In *Linguistic inspirations. Edmund Gussmann in memoriam,* edited by Eugeniusz Cyran, Henryk Kardela, and Bogdan Szymanek, 371–406. Lublin: Wydawnictwo Katolicki Uniwersytet Lubelski.

MacWhinney, Brian. 2000. *The CHILDES project: Tools for analyzing talk.* 3rd. Mahwah, NJ: Lawrence Erlbaum Associates.

Mahajan, Anoop. 1990. The A/A-bar distinction and movement theory. PhD diss., MIT.

Maling, Joan and Jóhannes Gísli Jónsson. 1995. On nominative objects in Icelandic and the feature [+human]. *Working Papers in Scandinavian Syntax* 56:71–79.

Manzini, M. Rita and Kenneth Wexler. 1987. Parameters, binding theory, and learnability. *Linguistic Inquiry* 18 (3): 413–444.

Marantz, Alec. 1993. Implications of asymmetries in double object constructions. In *Theoretical aspects of Bantu grammar 1,* edited by Sam A. Mchombo, 113–151. Stanford, CA: CSLI Publications.

Marlo, Michael R. 2014. Exceptional patterns of object marking in Bantu. *Studies in African Linguistics* 43 (1&2): 73–112.

Marlo, Michael R. 2015. On the number of object markers in Bantu languages. *Journal ofAfrican Languages and Linguistics* 36:1–65.

Martín, Francisco Jesús. 2012. Deconstructing Catalan object clitics. PhD diss., NYU.

Marvin, Tatjana and Adrian Stegovec. 2012. On the syntax of ditransitive sentences in slovenian. *Acta Linguistica Hungarica* 59 (1–2): 177–203.

Mavrogiorgos, Marios. 2010. *Clitics in Greek: A minimalist account of proclisis and enclisis.* Amsterdam/Philadelphia: John Benjamins.

McGinnis, Martha. 2001. Variation in the phase structure of applicatives. In *Linguistic variations yearbook,* edited by J. Rooryck and P. Pica. Amsterdam: John Benjamins.

McGinnis, Martha Jo. 2005. On markedness asymmetries in person and number. *Language* 81 (3): 699–718.

McKenzie, Andrew. 2015. A survey of switch-reference in North America. *International Journal of American Linguistics* 81 (3): 409–448.

Mel'čuk, Igor A. 1973. *Model' sprjazenija v aljutorskom jazyke [a model for Alutor conjugation].* Moscow: Institut russkogo jazyka, Akademija Nauk.

———. 1988. *Dependency syntax: Theory and practice.* Albany, NY: SUNY Press.

Messick, Troy. 2017. The morphosyntax of self-ascription: A cross-linguistic study. PhD diss., UConn.

Meyer-Lübke, Wilhelm. 1899. *Grammatik der Romanischen Sprachen.* Leipzig: O.R. Reisland.

Migdalski, Krzysztof. 2006. The syntax of compound tenses in Slavic. PhD diss., Tilburg University.

Mithun, Marianne. 1999. *The languages of Native North America.* Cambridge: Cambridge University Press.

Moltmann, Friederike. 1997. Intensional verbs and quantifiers. *Natural Language Semantics* 5:1–52.

Montalbetti, Mario. 1984. After binding: On the interpretation of pronouns. PhD diss., MIT.

Montgomery-Anderson, Brad. 2008. A reference grammar of Oklahoma Cherokee. PhD diss., University of Kansas.

Moravcsik, Edith A. 1978. On the distribution of ergative and accusative patterns. *Lingua* 45 (3-4): 233–279.

Moskal, Beata. 2015a. Domains at the border: Between morphology and phonology. PhD diss., UConn.

———. 2015b. Limits on allomorphy: A case study in nominal suppletion. *Linguistic Inquiry* 46 (2): 363–376.

Mount, Allyson. 2008. The impurity of "pure" indexicals. *Philosophical Studies* 138 (2): 193–209.

Mühlbauer, Jeffrey Thomas. 2008. The representation of intentionality in Plains Cree. PhD diss., University of British Columbia.

Myler, Neil. 2017. Cliticization feeds agreement: A view from Quechua. *Natural Language & Linguistic Theory* 35:751–800.

Neeleman, Ad and Kriszta Szendrői. 2007. Radical pro drop and the morphology of pronouns. *Linguistic Inquiry* 38 (4): 671–714.

Nemoto, Naoko. 1999. Scrambling. In *The handbook of Japanese linguistics,* edited by Natsuko Tsujimura, 121–153. Oxford: Blackwell.

Nespor, Marina and Irene Vogel. 1986. *Prosodic phonology.* Dordrecht: Foris.

Nevins, Andrew. 2007. The representation of third person and its consequences for person-case effects. *Natural Language & Linguistic Theory* 25 (2): 273–313.

———. 2011. Multiple Agree with clitics. *Natural Language & Linguistic Theory* 29 (4): 939–971.

Nichols, Johanna. 1986. Head-marking and dependent-marking grammar. *Language* 62 (1): 56–119.

———. 1992. *Linguistic diversity in space and time.* Chicago, IL: Chicago University Press.

Nichols, Lynn. 1997. Topics in Zuni syntax. PhD diss., Harvard.

———. 2001. The syntactic basis of referential hierarchy phenomena: Clues from languages with and without morphological case. *Lingua* 111:515–537.

Nicol, Fabrice. 2005. Romance clitic clusters: On diachronic changes and cross-linguistic contrasts. In *Clitic and affix combinations,* edited by Lorie Heggie and Francisco Ordoñez, 141–199. Amsterdam/Philadelphia: John Benjamins.

Nikolaeva, Irina. 1999. *Ostyak.* München: Lincom Europa.

———. 2001. Secondary topic as a relation in information structure. *Linguistics* 39:1–49.

———. 2014. *A grammar of Tundra Nenets.* Berlin: De Gruyter.

Noyer, Rolf. 1992. Features, positions and affixes in autonomous morphological structure. PhD diss., MIT.

Operstein, Natalie. 2003. Personal pronouns in Zapotec and Zapotecan. *International Journal of American Linguistics* 69 (2): 154–185.

Ordóñez, Francisco. 2002. Some clitic combinations in the syntax of Romance. *Catalan Journal of Linguistics* 1:201–224.

Ormazabal, Javier and Juan Romero. 2007. The object agreement constraint. *Natural Language & Linguistic Theory* 25:315–347.

Oshima, David Y. 2006. Perspectives in reported discourse. PhD diss., Stanford University.

———. 2007. Syntactic direction and obviation as empathy-based phenomena: a typological approach. *Linguistics: An Interdisciplinary Journal of the Language Sciences* 45 (4): 727–764.

Ouali, Hamid. 2006. Unifying agreement relations: A minimalist analysis of Berber. PhD diss., The University of Michigan.

Oxford, Will. 2014. Microparameters of agreement: A diachronic perspective on Algonquian verb inflection. PhD diss., University of Toronto.

———. 2019. Inverse marking and Multiple Agree in Algonquin: Complementarity and variability. *Natural Language & Linguistic Theory* 37 (3): 955–996.

Pacifique, Pére, Bernard Francis, and John Hewson. 1990. *The Micmac grammar of Father Pacifique.* [English translation of: Pacifique, P. (1939). *Lecons grammaticales théoriques et pratiques de la langue micmaque*. Global Language Press, Toronto.] Winnipeg: Algonquian / Iroquoian Linguistics.

Pancheva, Roumyana and Maria Luisa Zubizarreta. 2018. The Person Case Constraint. *Natural Language & Linguistic Theory* 36 (4): 1291–1337.

Patel-Grosz, Pritty and Pagrtrick G. Grosz. 2017. Revisiting pronominal typology. *Linguistic Inquiry* 48 (2): 259–297.

Paul, Waltraud. 2002. Sentence-internal topics in Mandarin Chinese: The case of object preposing. *Language and Linguistics* 3 (4): 695–714.

———. 2005. Low IP area and left periphery in Mandarin Chinese. *Recherches linguistiques de Vincennes* 33:111–134.

Payne, Doris, Mitsuyo Hamaya, and Peter Jacobs. 1994. Active, inverse and passive in Maasai. In *Voice and inversion,* edited by Talmy Givón, 283–316. Amsterdam: John Benjamins.

Payne, Doris L. 1994. The Tupí-Guaraní inverse. In *Voice: Form and function,* edited by Barbara A. Fox and Paul J. Hopper, 313–340. Amsterdam/Philadelphia: John Benjamins.

Pearson, Hazel. 2012. The sense of self: Topics in the semantics of de se expressions. PhD diss., Harvard.

Percus, Orin and Uli Sauerland. 2003. Pronoun binding in dream reports. In *Proceedings of the NELS 33,* edited by Shigeto Kawahara and Makoto Kadowaki, 265–283. Amherst, MA: GLSA.

Perlmutter, David M. 1968. Deep and surface structure constraints in syntax. PhD diss., MIT.

———. 1970. Surface structure constraints in syntax. *Linguistic Inquiry* 1 (2): 187–255.

———. 1971. *Deep and surface structure constraints in syntax.* New York: Holt, Rinehart / Winston.

Perry, John. 1997. Indexicals and demonstratives. In *A companion to philosophy of language,* edited by Bob Hale and Crispin Wright, 586–612. Oxford: Blackwell.

Pesetsky, David. 1982. Paths and categories. PhD diss., MIT.

Pesetsky, David and Esther Torrego. 2007. The syntax of valuation and the interpretability of features. In *Phrasal and clausal architecture: Syntactic derivation and interpretation. In honor of Joseph E. Emonds.* Edited by Simin Karimi, Vida Samiian, and Wendy K. Wilkins, 262–294. Amsterdam: John Benjamins.

Peterson, David M. 1998. The morpho-syntax of transitivization in Lai (Haka Chin). *Linguistics of the Tibeto-Burman Area* 22 (1): 87–153.

Phillips, Collin. 1993. Conditions on agreement in Yimas. *MIT Working papers in Linguistics* 18:173–213.

Picallo, M. Carme. 2007. *On gender and number.* Ms. Departament de Filologia Catalana - UAB. Https://clt.uab.cat/publicacions_clt/reports/pdf/GGT-07-08.pdf.

———. 2008. Gender and number in Romance. *Lingue e linguaggio* VII (1): 47–66.

Pinker, Steven. 1979. Formal models of language learning. *Cognition* 7:217–283.

———. 1989. *Learnability and cognition: The acquisition of argument structure.* Cambridge, MA: MIT Press.

Plunkett, Bernadette. 2002. Null subjects in child French interrogatives: A view from the York Corpus. In *Romanistische Korpuslinguistik: Korpora und gesproche Sprache,* 441–452. Tübingen: Narr.

Poletto, Cecilia. 2000. *The higher functional field: evidence from Northern Italian Dialects.* Oxford: Oxford University Press.

Poletto, Cecilia and Christina Tortora. 2016. Subject clitics: syntax. Chap. 47 in *The oxford guide to the romance languages,* edited by Adam Ledgeway and Martin Maiden, 772–785. Oxford: Oxford University Press.

Postal, Paul M. 1969. On so-called "pronouns" in English. In *Modern studies in english,* edited by David Reibel and Sanford Schane, 201–224. Englewood Cliffs, NJ: Prentice-Hall.

———. 1981. A failed analysis of the French cohesive infinitive construction. *Linguistic Analysis* 8:281–323.

———. 1989. *Masked inversion in French.* Chicago, IL: Chicago University Press.

———. 1990. French indirect object demotion. In *Studies in relational grammar 3,* edited by Paul M. Postal and Brian D. Joseph, 104–200. Chicago: University of Chicago Press.

Powers, Susan M. and Cornelia Hamann. 2000. Introduction: The acquisition of clause-internal rules. In *The acquisition of scrambling and cliticization,* edited by Susan M. Powers and Cornelia Hamann, 1–18. Dordrecht: Kluwer.

Preminger, Omer. 2011. Agreement as a fallible operation. PhD diss., MIT.

———. 2014. *Agreement and its failures.* Cambridge: MIT Press.

———. 2019. What the PCC tells us about "abstract" agreement, head movement, and locality. *Glossa* 4 (1): 13.

Prince, Alan and Paul Smolensky. 1993/2004. *Optimality theory: Constraint interaction in generative grammar.* [Rutgers Optimality Archive, ROA 537, http://roa.rutgers.edu]. Malden, MA: Blackwell.

Pylkkänen, Liina. 2002. Introducing arguments. PhD diss., MIT.

———. 2008. *Introducing arguments.* Cambridge, MA: MIT Press.

Quinn, Conor. 2006. Referential-access dependency in Penobscot. PhD diss., Harvard University.

Rackowski, Andrea and Norvin Richards. 2005. Phase edge and extraction. *Linguistic Inquiry* 36:565–599.

Radkevich, Nina. 2010. On location: The structure of case and adpositions. PhD diss., UConn.

Radulescu, Alexandru. 2018. The difference between indexicals and demonstratives. *Synthese* 195 (7): 3173–3196.

Raynaud, Louise. 2018. Reflexives and participants: A natural class. In *ConSOLE XXVI: Proceedings of the 26th Conference of the Student Organisation of Linguistics in Europe (14-16 January 2018, University College London),* edited by Cora Pots, 367–391. Leiden: Leiden University Centre for Linguistics.

Renzi, Lorenzo and Laura Vanelli. 1983. I pronomi soggetto in alcune varietà romanze. In *Scritti linguistici in onore di Giovan Battista Pellegrini,* edited by Paola Benincá, Manilio Cortelazzo, Aldo Prosdocimi, Laura Vanelli, and Alberto Zamboni, 121–145. Pisa: Pacini.

Řezáč, Milan. 2003. The fine structure of Cyclic Agree. *Syntax* 6 (2): 156–182.

———. 2004. The EPP in Breton: An uninterpretable categorial feature. In *Triggers,* edited by Henk van Riemsdijk and Anne Breitbarth, 451–492. Berlin: De Gruyter.

———. 2006. Escaping the person case constraint: Referential computation in the phi-system. *Linguistic Variation Yearbook* 6:97–138.

———. 2008a. Phi-Agree and Theta-related Case. In *Phi-theory: Phi features across interfaces and modules,* edited by Daniel Harbour, David Adger, and Susana Béjar, 83–129. Oxford: Oxford University Press.

———. 2008b. The syntax of eccentric agreement: The Person Case Constraint and absolutive displacement in Basque. *Natural Language & Linguistic Theory* 26:61–106.

———. 2011. *Phi-features and the modular architecture of language.* Dordrecht: Springer.

Řezáč, Milan and Mélanie Jouitteau. 2008. The French ethical dative: 13 syntactic tests. *Bucharest Working Papers in Linguistics* 9:97–108.

Rhodes, Richard A. 1990a. Obviation, inversion, and topic rank in Ojibwa. In *Proceedings of the 16th annual meeting of the Berkeley Linguistics Society: Special session on general topics in American indian linguistics,* edited by David J. Costa, 101–115. Berkeley, CA: BLS.

———. 1990b. Ojibwa secondary objects. In *Grammatical relations: A cross-theoretical perspective,* edited by Katarzyna Dziwirek, Patrick Farrell, and Errapel Mejías Bikandi, 401–414. Stanford: CSLI Publications.

———. 1994. Agency, inversion, and thematic alignment in Ojibwe. In *Proceedings of the 20th annual meeting of the Berkeley Linguistics Society,* edited by Susanne Gahl, Andy Dolbey, and Christopher Johnson, 431–446. BLS.

Richards, Marc. 2008a. Defective Agree, case alternations, and the prominence of person. In *Scales,* edited by Marc Richards and Andrej L. Malchukov, 137–161. Leipzig: Universität Leipzig.

Richards, Norvin. 1997. What moves where when in which language? PhD diss., MIT.

———. 2001. *Movement in language: Interactions and architectures.* 326. Oxford: Oxford University Press.

Richards, Norvin. 2005. Person-case effect in Tagalog and the nature of long-distance extraction. In *Proceedings of AFLA XII: UCLA Working Papers in Linguistics,* edited by Jeffrey Heinz and Dimitris Ntelitheos, 12:383–394.

———. 2008b. *Can A-scrambling reorder DPs?* Ms. MIT http://web.mit.edu/norvin/www/papers/Ascrambling.pdf.

———. 2016. *Contiguity theory.* Cambridge, MA: MIT Press.

Riedel, Kristina. 2009. The syntax of object marking in Sambaa. PhD diss., Universiteit Leiden.

Rivero, María Luisa. 2004. Spanish quirky subjects, person restrictions, and the Person-Case Constraint. *Linguistic Inquiry* 35 (3): 494–502.

———. 2008. Oblique subjects and person restrictions in Spanish. In *Agreement restrictions,* edited by Roberta D'Alessandro, Susann Fischer, and Gunnar Hrafn Hrafnjargarson, 215–250. Berlin/New York: Mouton de Gruyter.

Rizzi, Luigi. 1982. *Issues in Italian syntax.* Foris: Foris.

———. 1986a. Null objects in Italian and the theory of *pro*. *Linguistic Inquiry* 17 (3): 501–557.

———. 1986b. On the status of subject clitics in Romance. In *Studies in romance linguistics,* edited by Osvaldo Jaeggli and Carmen Silva-Corvalán, 391–419. Dordrecht: Foris.

———. 1990a. On the anaphor-agreement effect. *Rivista di Linguistica* 2:27–42.

———. 1990b. *Relativized minimality.* Cambridge: MIT Press.

———. 2001. Relativized minimality effects. In *The handbook of contemporary syntactic theory,* edited by Mark Baltin and Chris Collins, 89–110. Blackwell Publishers Ltd.

Rizzi, Luigi and Ian Roberts. 1989. Complex inversion in French. *Probus* 1:1–30.

Roberts, Ian. 2007. *Diachronic syntax.* Oxford: Oxford University Press.

———. 2010a. A deletion analysis of null subjects. In *Parametric variation,* edited by Theresa Biberauer, Anders Holmberg, Ian G. Roberts, and Michelle Sheehan, 58–87. Campridge: Cambridge University Press.

———. 2010b. *Agreement and head movement: Clitics, incorporation, and defective goals.* Cambridge, MA: MIT Press.

Roberts, Taylor. 2000. Clitics and agreement. PhD diss., MIT.

Roca, Francesc. 1992. On the licensing of pronominal clitics. The properties of object clitics in Spanish and Catalan. Master's thesis, Universitat Autónoma de Barcelona.

Rodríguez-Mondoñedo, Miguel. 2007. The syntax of objects: Agree and Differential Object Marking. PhD diss., UConn.

Rodríguez-Mondoñedo, Miguel, William Snyder, and Koji Sugisaki. 2006. The parameter of clitic-climbing: The view from child Spanish. In *The Proceedings of the inaugural conference on Generative Approaches to Language Acquisition–North America (GALANA), honolulu, hi,* edited by Kamil Ud Deen, Jun Nomura, Barbara Schulz, and Bonnie Schwarz, 2:241–248. University of Connecticut Occasional Papers in Linguistics. Cambridge, MA: MITWPL.

Roessler, Eva-Maria. 2019. Differential object marking and object scrambling in the Guaraní language cluster. *Linguisticæ Investigationes* 42:31–55.

Rosen, Carol. 1990. Rethinking Southern Tiwa. *Language* 66 (4): 669–713.

Ross, John Robert 'Haj'. 1967. Constraints on variables in syntax. PhD diss., MIT.

Rouveret, Alain and Jean-Roger Vergnaud. 1980. Specifying reference to the subject: French causatives and conditions on representations. *Linguistic Inquiry* 11:97–202.

Rude, Noel. 1994. Direct, inverse and passive in Northwest Sahaptin. In *Voice and inversion,* edited by Talmy Givón, 101–120. Amsterdam/Philadelphia: John Benjamins.

———. 2009. Transitivity in sahaptin. *Northwest Journal of Linguistics* 3 (3): 1–37.

Rudin, Cahtherine. 1988. On multiple questions and multiple *wh*-fronting. *Natural Language & Linguistic Theory* 6:445–501.

Rudin, Catherine. 1997. AgrO and Bulgarian pronominal clitics. In *Proceedings of FASL V: The Indiana meeting,* edited by Martin Lindseth and Steven Franks, 224–253. Ann Arbor, MI: Michigan Slavic Publications.

Runić, Jelena. 2013a. A new look at clitics: Evidence from Slavic. In *Formal Approaches to Slavic Linguistics: The third Indiana meeting 2012,* edited by Steven Franks, Markus Dickerson, George Fowler, Melissa Witcombe, and Ksenia Zanon, 275–288. Ann Arbor, MI: Michigan Slavic Publications.

———. 2013b. *The Person-Case Constraint.* Poster presented at the 87th Meeting of the LSA.

———. 2014. A new look at clitics, clitic doubling, and argument ellipsis. PhD diss., UConn.

Sáàh, Kòfí K. and Éjiké Ézè. 1997. Double objects in àkán and ìgbo. In *Object positions in Benue-Kwa,* edited by Rose-Marie Déchaine and Victor Manfredi, 139–151. The Hague: Academic Graphics.

Safir, Ken. 1993. Perception, selection, and structural economy. *Natural Language Semantics* 2:47–70.

Saito, Mamoru. 1985. Some asymmetries in Japanese and their theoretical consequences. PhD diss., MIT.

———. 1992. Long distance scrambling in Japanese. *Journal of East Asian Linguistics* 1:69–118.

Saito, Mamoru and Naoki Fukui. 1998. Order in phrase structure and movement. *Linguistic Inquiry* 29 (3): 439–474.

Saito, Mamoru and Hajime Hoji. 1983. Weak crossover and move α in Japanese. *Natural Language & Linguistic Theory* 1 (2): 245–259.

Sapir, Edward. 1922. The Takelma language of southwestern Oregon. In *Handbook of American Indian language,* edited by Franz Boas, 1–296. Washington: Government Printing Office.

Sauerland, Uli. 1998. On the making and meaning of chains. PhD diss., MIT.

———. 2004. The interpretation of traces. *Natural Language Semantics* 12 (1): 63–127.

Savvina, Elena N. and Igor. Mel'čuk. 1978. *Toward a formal model of Alutor surface syntax: Nominative and ergative constructions.* Bloomington, IN: Indiana University Linguistics Club.

Scancarelli, Janine. 1987. Grammatical relations and verb agreement in Cherokee. PhD diss., UCLA.

Schlenker, Philippe. 1999. Propositional attitudes and indexicality: A cross-categorial approach. PhD diss., MIT.

———. 2003. A plea for monsters. *Linguistics and Philosophy* 26:29–120.

———. 2004. Person and binding (a partial survey). *Rivista di Linguistica* 16 (1): 155–218.

———. 2005. Minimize Restrictors! (notes on definite descriptions, Condition C and epiteths). In *Proceedings of SuB 9,* edited by Emar Maier, Corien Bary, and Janneke Huitink, 385–416. Nijmegen: Nijmegen Centre of Semantics.

Schneider-Zioga, Patricia. 2000. Anti-agreement and the fine structure of the left edge. In *University of California Irvine Working Papers in Linguistics,* edited by Ruixi Ai, Francesca Del Gobbo, Makie Irie, and Hajime Ono, 6:94–114. Irvine, CA: University of California Irvine.

———. 2007. Anti-agreement, anti-locality and minimality. *Natural Language & Linguistic Theory* 25 (2): 403–446.

Schütze, Carson T. 2003. Syncretism and double agreement with Icelandic nominative objects. In *Grammar in focus: festschrift for Christer Platzack,* edited by L. O. Delsing, C. Falk, G. Josefsson, and H. Á. Sigurðsson, 2:295–303. Lund: Department of Scandinavian Languages.

Selkirk, Elisabeth. 1978. On prosodic structure and its relation to syntactic structure. In *Nordic prosody 2,* edited by Thorstein Frenheim, 111–140. Trondheim: TAPIR.

Selkirk, Elizabeth. 1980. Prosodic domains in phonology: Sanskrit revisited. In *Juncture,* edited by Mark Aronoff and Mary-Louise Kean, 7:107–129. Saratoga: Anma Libri.

———. 1984. *Phonology and syntax: The relation between sound and structure.* Cambridge, MA: MIT Press.

———. 1996. The prosodic structure of function words. *Phonology Yearbook* 3:371–405.

———. 2011. The syntax-phonology interface. In *The handbook of phonological theory,* 2nd, edited by John Goldsmith, Jason Riggle, and Alan Yu, 435–484. Oxford: Wiley-Blackwell.

Seuren, Pieter. 1976. Clitic pronoun clusters. *Italian Linguistics* 2:17–36.

Sibawayh, ’Amr ibn ’Uthmān. 1881. *Le livre de sîbawaihi, traité de grammaire arabe.* Vol. 1. [French translation of the original 8th century grammar]. Paris: Imprimerie Nationale.

Siewierska, Anna. 2004. *Person.* Cambridge: Cambridge University Press.

Sigurðsson, Halldór Ármann. 1996. Icelandic finite verb agreement. *Working Papers in Scandinavian Syntax* 57:1–46.

———. 2004. The syntax of person, tense, and speech features. *Rivista di Linguistica* 16 (1): 219–251.

———. 1990–1991. Beygingarsamræmi. *Íslenskt mál og almenn málfræði* 12-13:31–77.

Sigurðsson, Halldór Ármann and Anders Holmberg. 2008. Icelandic dative intervention: Person and number are separate probes. In *Agreement restrictions,* edited by Roberta D’Alessandro, Susann Fischer, and Gunnar Hrafn Hrafnjargarson, 251–280. Berlin: de Gruyter.

Silverstein, Michael. 1976. Hierarchy of features and ergativity. In *Grammatical categories in australian languages,* edited by Robert M. W. Dixon, 112–171. Canberra: Australian Institutes of Aboriginal Studies.

———. 1986. Hierarchy of features and ergativity. In *Features and projections,* edited by Pieter Muysken and Henk van Riemsdijk, 163–232. Dordrecht: Foris.

Simpson, Jane Helen. 1991. *Warlpiri morpho-syntax: A lexicalist approach.* Springer: Dordrecht.

Singer, Ruth. 2016. *The dynamics of nominal classification: Productive and lexicalised uses of gender agreement in Mawng.* Boston/Berlin: De Gruyter Mouton.

Smeets, Ineke. 1989. A Mapuche grammar. PhD diss., Leiden University.

———. 2008. *A grammar of Mapuche.* Berlin: Mouton de Gruyter.

Smith, Peter W. 2015. Feature mismatches: Consequences for syntax, morphology and semantics. PhD diss., UConn.

Smith, Peter W., Beata Moskal, Ting Xu, Jungmin Kang, and Jonathan David Bobaljik. 2018. Case and number suppletion in pronouns. *Natural Language & Linguistic Theory* 37 (3): 1029–1101.

Snyder, William. 2002. Parameters: The view from child language. In *Proceedings of the 3rd Tokyo Conference on Psycholinguistics,* edited by Yukio Otsu, 27–44. Tokyo: Hituzi Shobo.

———. 2007. *Child language: The parametric approach.* New York: OUP.

———. 2008. Children's grammatical conservatism: Implications for linguistic theory. In *An enterprise in the cognitive science of language: A festschrift for Yukio Otsu,* edited by Tetsuya Sano, Mika Endo, Miwa Isobe, Koichi Otaki, Koji Sugisaki, and Takeru Suzuki, 41–51. Tokyo: Hituzi Syobo.

Snyder, William, Nina Hyams, and Paola Crisma. 1995. Romance auxiliary selection with reflexive clitics: evidence for early knowledge of unaccusativity. In *Proceedings of the twenty-sixth annual child language research forum,* edited by Eve Clark, 127–136. Stanford, CA: CSLI Publications.

Sonnenschein, Aaron. 2004. A descriptive grammar of San Bartolomé Zoogocho Zapotec. PhD diss., USC.

Speas, Peggy. 1994. Null arguments in a theory of economy of projection. In *University of Massachusetts occasional papers 17: Functional projections,* edited by E. Banedicto and J. Runner, 179–208. Amherst, MA: GLSA.

Speas, Peggy and Carol L. Tenny. 2003. Configurational properties of point of view roles. In *Asymmetry in grammar,* edited by Anna Maria Di Sciulo, 315–343. Amsterdam: John Benjamins.

Sportiche, Dominique. 1999. Subject clitics in French and Romance complex inversion and clitic doubling. In *Beyond principles and parameters: Essays in memory of Osvaldo Jaeggli,* edited by Kyle Johnson and Ian G. Roberts, 189–222. Dordrecht: Kluwer.

Starke, Michal. 2001. Move reduces to Merge: A theory of locality. PhD diss., University of Geneva.

———. 2009. A short primer to a new approach to language. University of Tromsø, *Nordlyd: Special issue on Nanosyntax* 36 (1): Peter Svenonius, Gillian Ramchand, Michal Starke, and Knut Tarald Taraldsen.

von Stechow, Arnim. 2003. Feature deletion under semantic binding: Tense, person, and mood under verbal quantifiers. In *Proceedings of NELS 33,* edited by M. Kadowaki and S. Kawahara. Amherst, MA: GLSA.

———. 2004. Binding by verbs: Tense, person and mood under attitudes. In *The syntax and semantics of the left periphery,* edited by H. Lohnstein and S. Trissler, 431–488. Berlin: de Gruyter.

Steele, Susan. 1977. Clisis and diachrony. In *Mechanisms of syntactic change,* edited by Charles N. Li, 539–579. Austin, TX: University of Texas Press.

Stegovec, Adrian. 2012. *Ditransitives in Slovenian: Evidence for two separate ditransitive constructions.* BA Thesis, University of Ljubljana. University of Ljubljana.

———. 2015a. It's not Case, it's personal! Reassessing the PCC and clitic restrictions in O'odham and Warlpiri. In *Proceedings of the poster session of the 33rd WCCFL,* edited by Pocholo Umbal, 5:131–140. SFU Working Papers in Linguistics.

———. 2015b. Now you PCC me, now you don't: Slovenian clitic-switch as a repair for person-case effects. In *Proceeding of NELS 45,* edited by Thuy Bui and Deniz Özyıldız, 3:107–117. Amherst, MA: GLSA.

———. 2016. Not two sides of one coin: Clitic person restrictions and Icelandic quirky agreement. In *Formal studies in Slovenian syntax. In honor of Janez Orešnik,* edited by Lanko Marušič and Rok Žaucer, 283–312. Amsterdam/Pholadelphia: John Benjamins.

———. 2019. Taking case out of the person-case constraint. *Natural Language & Linguistic Theory [online only]*:1–52.

Stjepanović, Sandra. 1999. What do second-position cliticization, scrambling and multiple wh-fronting have in common? PhD diss., UConn.

———. 2003. A word order paradox resolved by copy deletion at PF. *Linguistic Variation Yearbook* 3 (1): 139–177.

Stokke, Andreas. 2010. Indexicality and presupposition: Explorations beyond truth-conditional information. PhD diss., University of St Andrews.

Sturgeon, Anne, Boris Harizanov, Maria Polinsky, Ekaterina Kravtchenko, Carlos Gómez Gallo, Lucie Medová, and Václav Koula. 2012. Revisiting the Person Case Constraint in Czech. In *Proceedings of FASL 19,* edited by J. Bailyn, E. Dunbar, Y. Kronrod, and C. LaTerza, 116–130. Ann Arbor, MI: Michigan Slavic Publications.

Sudo, Yasutada. 2012. On the semantics of phi features on pronouns. PhD diss., MIT.

Svenonius, Peter. 2004. On the edge. In *Peripheries: Syntactic edges and their effects,* edited by David Adger, Cecile De Cat, and George Tsoulas, 259–287. Dordrecht: Kluwer.

———. 2016. Spans and words. In *Morphological metatheory,* edited by Daniel Siddiqi and Heidi Harley, 201–222. Amsterdam/Philadelphia: John Benjamins.

Taraldsen, Knut Tarald. 1994. Reflexives, pronouns and subject/verb agreement in Icelandic and Faroese. *Working Papers in Scandinavian Syntax* 54 (December): 43–58.

———. 1995. On agreement and nominative objects in Icelandic. In *Studies in comparative Germanic syntax,* edited by Hubert Haider, Susan Olsen, and Sten Vikner, 307–327. Dordrecht: Kluwer Academic.

Tenny, Carol L. 2006. Evidentiality, experiencers, and the syntax of sentience in Japanese. *Journal of East Asian Linguistics* 15:245–288.

Terzi, Arhonto. 1999. Clitic combinations, their hosts and their ordering. *Natural Language & Linguistic Theory* 17:85–121.

Thompson, Chad L. 1994. Passive and inverse constructions. In *Voice and inversion,* edited by Talmy Givón, 47–64. Amsterdam: John Benjamins.

Tomić, Olga Mišeska. 2008. Towards grammaticalization of clitic doubling: Clitic doubling in Macedonian and neighbouring languages. In *Clitic doubling in the Balkan languages,* edited by Dalina Kallulli and Liliane Tasmowski, 65–88. Amsterdam: John Benjamins.

Toosarvandani, Maziar. 2017. On reaching agreement early (and late). In *Asking the right questions: Essays in honor of Sandra Chung,* edited by Jason Ostrove, Ruth Kramer, and Joseph Sabbagh, 127–141. Santa Cruz, CA: Department of Linguistics, UCSC.

Torrego, Esther. 1988. *Pronouns and determiners: A DP analysis of Spanish nominals.* Ms. University of Massechusetts, Boston.

———. 1995. On the nature of clitic doubling. In *Evolution and revolution in linguistic theory: Studies in honor of Carlos P. Otero,* edited by Héctor Campos and Paula Kempchinsky, 399–418. Washington, D.C.: Georgetown University Press.

———. 1998. *The dependencies of objects.* Cambridge, MA: MIT Press.

Tsunoda, Tasaku. 1981a. Interaction of phonological, grammatical and semantic factors: An Australian example. *Oceanic Linguistics* 20 (1): 45–92.

———. 1981b. *The Djaru language of Kimberley, Western Australia.* Canberra: Australian National University.

Tuggy, David H. 1977. Tetelcingo Nahuatl. In *Modern Aztec grammatical sketches,* edited by Ronald W. Langacker. Arlington: Summer Institute of Linguistics.

Tvica, Seid. 2017. Agreement and verb movement: the rich agreement hypothesis from a typological perspective. PhD diss., University of Amsterdam.

Uhlenbeck, Christianus Cornelius. 1938. *A concise Blackfoot grammar: Based on material from the Southern Peigans.* Amsterdam: Noord-Hollandsche Uitgevers-Maatschappij.

Ura, Hiroyuki. 1996. Multiple feature checking: A theory of grammatical function splitting. PhD diss., MIT.

Uriagereka, Juan. 1988. On government. PhD diss., UConn.

———. 1995. Aspects of the syntax of clitic placement in Western Romance. *Linguistic Inquiry* 26:79–123.

———. 1999. Multiple spell-out. In *Working minimalism,* edited by Samuel David Epstein and Norbert Hornstein, 251–282. Cambridge, MA: MIT Press.

Ussery, Cherlon. 2013. *The syntax of optimal agreement in Icelandic.* Ms. Carleton College.

Valentine, Randolph J. 2001. *Nishnaabemwin reference grammar.* Toronto: University of Toronto Press.

Van der Wal, Jenneke. 2017. Flexibility in symmetry: An implicational relation in Bantu double object constructions. In *Order and structurein syntax II: Subjecthood and argument structure,* edited by Michelle Sheehan and Laura R. Bailey, 115–152. Berlin: Language Science Press.

Verbeke, Saartje. 2017. *Argument structure in Kashmiri: Form and function of pronominal suffixation.* Leiden: Brill.

Virtanen, Susanna. 2012. Variation in three-participant constructions in Eastern Mansi. *Linguistica Uralica* 48 (2): 120–130.

———. 2014. Pragmatic direct object marking in Eastern Mansi. *Linguistics* 52 (2): 391–413.

Vormann, Franz and Wilhelm Scharfenberger. 1914. *Die Monumbo-Sprache: Grammatik und Wörterverzeichnis.* Wien: Mechitharisten-Buchdruckerei.

Wagner, Irina, Andrew Cowell, and Jena D. Hwang. 2016. Applying universal dependency to the Arapaho language. In *Proceedings of the 10th Linguistic Annotation Workshop held in conjunction with ACL 2016 (LAW-x 2016),* 171–179. Berlin: Association for Computational Linguistics. https://www.aclweb.org/anthology/W16-1719.

Wali, Kashi and Omkar N. Koul. 1997. *Kashmiri.* London: Routledge.

———. 2006. *Modern Kashmiri grammar.* Springfield, VA: Dunwoody Press.

Walkow, Martin. 2012. Goals, big and small. PhD diss., UMass.

Walkow, Martin. 2014. Cyclic AGREE derives restrictions on cliticization in classical Arabic. In *Perspectives on Arabic linguistics XXVI: Papers from the annual symposium on Arabic linguistics. New York, 2012,* edited by Reem Khamis-Dakwar and Karen Froud, 2:135–160. Studies in Arabic Linguistics. Amsterdam: John Benjamins.

Wanner, Dieter. 1977. On the order of clitics in Italian. *Lingua* 43:101–128.

———. 1987. *The development of Romance clitic pronouns: From Latin to Old Romance.* New York/Amsterdam: Mouton de Gruyter.

Warburton, Irene. 1977. Modern Greek clitic pronouns and the 'surface structure constraint' hypothesis. *Journal of Linguistics* 13:259–181.

Watkins, Laurel J. 1984. *A grammar of Kiowa.* Lincoln, NE: University of Nebraska Press.

Webelhuth, Gert. 1992. *Principles and parameters of syntactic saturation.* Oxford: Oxford University Press.

Werner, Ingegerd. 1999. *Die Personalpronomen im Zürichdeutschen.* Stockholm: Almqvist & Wiksell.

Willie, MaryAnn. 1991. Navajo pronouns and obviation. PhD diss., University of Arizona.

Willie, MaryAnn and Eloise Jelinek. 2000. Navajo as a discourse configurational language. In *The Athabaskan languages: Perspectives on a Native American language family,* edited by Theodore B. Fernald and Paul R. Platero, 252–287. Oxford: Oxford University Press.

Wiltschko, Martina. 2008. Person-hierarchy effects without a person-hierarchy. In *Agreement restrictions,* edited by Roberta D'Alessandro, Susann Fischer, and Gunnar Hrafn Hrafnjargarson, 281–313. Berlin: De Gruyter.

Wiltschko, Martina and Elizabeth Ritter. 2015. Animating the narrow syntax. *The Linguistic Review* 32 (4): 869–908.

Wolfart, Hans Christoph. 1973. Plains Cree: A grammatical study. *Transactions of the American Philosophical Society* 63 (5): 1–90.

———. 1978. How many obviatives: Sense and reference in a Cree verb paradigm. In *Linguistic studies of Native Canada,* edited by Eung-Do Cook and Jonathan Kaye, 255–272. Vancouver: UBC Press.

Woolford, Ellen. 1997. Four-way Case systems: Ergative, nominative, objective, and accusative. *Natural Language & Linguistic Theory* 15:181–227.

———. 1999. More on the Anaphor Agreement Effect. *Linguistic Inquiry* 30 (2): 257–287.

———. 2000. Agreement in disguise. In *Advances in African linguistics,* edited by Vicki M. Carstens, 103–117. Trenton, NJ: Africa World Press.

———. 2006. Lexical case, inherent case, and argument structure. *Linguistic Inquiry* 37:111–30.

Wurmbrand, Susi. 2013. QR and selection: Covert evidence for phasehood. In *Proceedings of the NELS 42,* edited by Stefan Keine and Shayne Sloggett, 619–632. Amherst, MA: GLSA.

———. 2017a. Feature sharing or How i value my son. In *The Pesky Set: Papers for David Pesetsky,* edited by Claire Halpert, Hadas Korek, and Coppe van Urk, 173–182. Cambridge, MA: MIT Working Papers in Linguistics.

———. 2017b. Stripping and topless complements. *Linguistic Inquiry* 48 (2): 341–366.

Yatsushiro, Kazuko. 2001. Case licensing and VP structure. PhD diss., UConn.

Yuan, Michelle. 2018. Dimensions of ergativity in Inuit: Theory and microvariation. PhD diss., MIT.

Zaharlick, Ann Marie. 1977. Picurís syntax. PhD diss., The American University.

Zanuttini, Rafaela. 2008. Encoding the addresse in the syntax: Evidence from English imperative subjects. *Natural Language & Linguistic Theory* 26:185–218.

Zanuttini, Rafaela, Miok Pak, and Paul Portner. 2012. A syntactic analysis of interpretive restrictions on imperative, promissive, and exhortative subjects. *Natural Language & Linguistic Theory* 30:1231–1274.

Zepeda, Ofelia. 1983. *A Tohono O'Odham grammar.* Tuscon, AZ: University of Arizona Press.

Zubizarretta, Maria Luisa and Roumyana Pancheva. 2017. A formal characterization of person-based alignment: The case of Paraguayan Guaraní. *Natural Language & Linguistic Theory* 35 (4): 1161–1204.

Zúñiga, Fernando. 2000. *Mapudungun.* München: LINCOM EUROPA.

———. 2002. Inverse systems in indigenous languages of the Americas. PhD diss., University of Zurich.

———. 2006. *Deixis and alignment: Inverse systems in indigenous languages of the Americas.* Amsterdam/Philadelphia: John Benjamins.

———. 2014. Inversion, obviation, and animacy in native languages of the Americas: Elements for a cross-linguistic survey. *Anthropological Linguistics* 56 (3/4): 334–355.

———. 2016. Selected semitransitive constructions in Algonquian. *Lingua Posnanensis* 58 (2): 207–225.

Zwicky, Arnold M. 1977. *On clitics.* Bloomington, IN: Indiana Linguistics Club.

Zwicky, Arnold M. and Geoffrey K. Pullum. 1983. Cliticization vs. inflection: English N'T. *Language* 59 (3): 502–513.